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Carl Friedrich Gauss

Chapter - 1

Applications of Matrices and Determinants

EXERCISE 1.1



Video class

Concept Corner

- The determinant of submatrix is called **minor** of the element a_{ij} . It is denoted by M_{ij} .
- The product of M_{ij} and $(-1)^{i+j}$ is called **cofactor** of the element a_{ij} . It is denoted by A_{ij} . Thus **the cofactor of a_{ij} is $A_{ij} = (-1)^{i+j}M_{ij}$.**
- Let A be a square matrix of order n . Then the matrix of cofactors of A is defined as the matrix obtained by replacing each element a_{ij} of A with the corresponding cofactor A_{ij} .
- The **adjoint matrix of A** is defined as the transpose of the matrix of cofactors of A . It is denoted by **adj A** .
- Let A be a square matrix of order n . If there exists a square matrix B of order n such that $AB = BA = I_n$, then the matrix B is called an **inverse of A** .
- A square matrix A is called **orthogonal** if $AA^T = A^T A = I$.
(A is **orthogonal** if and only if A is non-singular and $A^{-1} = A^T$)

Theorems:

1. For every square matrix A of order n , $A(\text{adj } A) = (\text{adj } A)A = |A|I_n$.
2. If a square matrix has an inverse, then it is unique.
3. Let A be square matrix of order n . Then, A^{-1} exists if and only if A is non-singular.
4. If A is non-singular, then
 - (i) $|A^{-1}| = \frac{1}{|A|}$ (ii) $(A^T)^{-1} = (A^{-1})^T$ (iii) $(\lambda A)^{-1} = \frac{1}{\lambda}A^{-1}$, where λ is a non-zero scalar.
5. **Left Cancellation Law**
Let A, B , and C be square matrices of order n . If A is non-singular and $AB = AC$, then $B = C$.
6. **Right Cancellation Law**
Let A, B and C be square matrices of order n . If A is non-singular and $BA = CA$, then $B = C$.
7. **Reversal Law for Inverses**
If A and B are non-singular matrices of the same order, then the product AB is also non-singular and $(AB)^{-1} = B^{-1}A^{-1}$.
8. **Law of Double Inverse:** If A is non-singular, then A^{-1} is also non-singular and $(A^{-1})^{-1} = A$.
9. If A is a non-singular square matrix of order n , then
 - (i) $(\text{adj } A)^{-1} = \text{adj}(A^{-1}) = \frac{1}{|A|}A$ (ii) $|\text{adj } A| = |A|^{n-1}$
 - (iii) $\text{adj}(\text{adj } A) = |A|^{n-2}A$ (iv) $\text{adj}(\lambda A) = \lambda^{n-1}\text{adj}(A)$, λ is a nonzero scalar
 - (v) $|\text{adj}(\text{adj } A)| = |A|^{(n-1)^2}$ (vi) $(\text{adj } A)^T = \text{adj}(A^T)$
10. If A and B are any two non-singular square matrices of order n , then
 $\text{adj}(AB) = (\text{adj } B)(\text{adj } A)$.

1. Find the adjoint of the following: (i) $\begin{bmatrix} -3 & 4 \\ 6 & 2 \end{bmatrix}$ (ii) $\begin{bmatrix} 2 & 3 & 1 \\ 3 & 4 & 1 \\ 3 & 7 & 2 \end{bmatrix}$ (iii) $\frac{1}{3} \begin{bmatrix} 2 & 2 & 1 \\ -2 & 1 & 2 \\ 1 & -2 & 2 \end{bmatrix}$

(i) Let $A = \begin{bmatrix} -3 & 4 \\ 6 & 2 \end{bmatrix}$ (2M)

Co-factor of $A = \begin{bmatrix} 2 & -6 \\ -4 & -3 \end{bmatrix}$

$\text{adj} A = \begin{bmatrix} 2 & -4 \\ -6 & -3 \end{bmatrix}$

Note:

$A = \begin{bmatrix} -3 & 4 \\ 6 & 2 \end{bmatrix}$

Change sign Interchange

(ii) Let $A = \begin{bmatrix} 2 & 3 & 1 \\ 3 & 4 & 1 \\ 3 & 7 & 2 \end{bmatrix}$ (3M)

$\text{adj} A = \begin{bmatrix} \begin{vmatrix} 4 & 1 \\ 7 & 2 \end{vmatrix} & -\begin{vmatrix} 3 & 1 \\ 3 & 2 \end{vmatrix} & \begin{vmatrix} 3 & 4 \\ 3 & 7 \end{vmatrix} \\ -\begin{vmatrix} 3 & 1 \\ 7 & 2 \end{vmatrix} & \begin{vmatrix} 2 & 1 \\ 3 & 2 \end{vmatrix} & -\begin{vmatrix} 2 & 3 \\ 3 & 7 \end{vmatrix} \\ \begin{vmatrix} 3 & 1 \\ 4 & 1 \end{vmatrix} & -\begin{vmatrix} 2 & 1 \\ 3 & 1 \end{vmatrix} & \begin{vmatrix} 2 & 3 \\ 3 & 4 \end{vmatrix} \end{bmatrix}^T$

$= \begin{bmatrix} (8-7) & -(6-3) & (21-12) \\ -(6-7) & (4-3) & -(14-9) \\ (3-4) & -(2-3) & (8-9) \end{bmatrix}^T$

$= \begin{bmatrix} 1 & -3 & 9 \\ 1 & 1 & -5 \\ -1 & 1 & -1 \end{bmatrix}^T$

$\therefore \text{adj} A = \begin{bmatrix} 1 & 1 & -1 \\ -3 & 1 & 1 \\ 9 & -5 & -1 \end{bmatrix}$

(iii) Let $B = \frac{1}{3} \begin{bmatrix} 2 & 2 & 1 \\ -2 & 1 & 2 \\ 1 & -2 & 2 \end{bmatrix}$ (3M)

$A = \begin{bmatrix} 2 & 2 & 1 \\ -2 & 1 & 2 \\ 1 & -2 & 2 \end{bmatrix} \therefore B = \left(\frac{1}{3}A\right)$

$\text{adj} A = \begin{bmatrix} \begin{vmatrix} 1 & 2 \\ -2 & 2 \end{vmatrix} & -\begin{vmatrix} -2 & 2 \\ 1 & 2 \end{vmatrix} & \begin{vmatrix} -2 & 1 \\ 1 & -2 \end{vmatrix} \\ -\begin{vmatrix} 2 & 1 \\ -2 & 2 \end{vmatrix} & \begin{vmatrix} 2 & 1 \\ 1 & 2 \end{vmatrix} & -\begin{vmatrix} 2 & 2 \\ 1 & -2 \end{vmatrix} \\ \begin{vmatrix} 2 & 1 \\ 1 & 2 \end{vmatrix} & -\begin{vmatrix} 2 & 1 \\ -2 & 2 \end{vmatrix} & \begin{vmatrix} 2 & 2 \\ -2 & 1 \end{vmatrix} \end{bmatrix}^T$

$= \begin{bmatrix} (2+4) & -(-4-2) & (4-1) \\ -(4+2) & (4-1) & -(-4-2) \\ (4-1) & -(4+2) & (2+4) \end{bmatrix}^T$

$= \begin{bmatrix} 6 & 6 & 3 \\ -6 & 3 & 6 \\ 3 & -6 & 6 \end{bmatrix}^T$

$\text{adj} A = \begin{bmatrix} 6 & -6 & 3 \\ 6 & 3 & -6 \\ 3 & 6 & 6 \end{bmatrix}$

$\text{adj} B = \text{adj} \left(\frac{1}{3}A\right) = \left(\frac{1}{3}\right)^{3-1} \text{adj} A$

[$\because$ If A is a matrix of order n , then

$\text{adj}(\lambda A) = \lambda^{n-1} \text{adj} A$]

$= \frac{1}{3^2} \begin{bmatrix} 6 & -6 & 3 \\ 6 & 3 & -6 \\ 3 & 6 & 6 \end{bmatrix} = \frac{1}{9} \times 3 \begin{bmatrix} 2 & -2 & 1 \\ 2 & 1 & -2 \\ 1 & 2 & 2 \end{bmatrix}$

$\therefore \text{adj} B = \frac{1}{3} \begin{bmatrix} 2 & -2 & 1 \\ 2 & 1 & -2 \\ 1 & 2 & 2 \end{bmatrix}$

2. Find the inverse (if it exists) of the following (i) $\begin{bmatrix} -2 & 4 \\ 1 & -3 \end{bmatrix}$ (ii) $\begin{bmatrix} 5 & 1 & 1 \\ 1 & 5 & 1 \\ 1 & 1 & 5 \end{bmatrix}$ (iii) $\begin{bmatrix} 2 & 3 & 1 \\ 3 & 4 & 1 \\ 3 & 7 & 2 \end{bmatrix}$

(i) Let $A = \begin{bmatrix} -2 & 4 \\ 1 & -3 \end{bmatrix}$ (2M)

$|A| = 6 - 4 = 2 \neq 0.$

$\therefore A^{-1}$ exists

$A^{-1} = \frac{1}{|A|} \text{adj} A$

$\text{adj} A = \begin{bmatrix} -3 & -4 \\ -1 & -2 \end{bmatrix}$

$\therefore A^{-1} = \frac{1}{2} \begin{bmatrix} -3 & -4 \\ -1 & -2 \end{bmatrix}$

(ii) Let $A = \begin{bmatrix} 5 & 1 & 1 \\ 1 & 5 & 1 \\ 1 & 1 & 5 \end{bmatrix}$ (3M)

$$|A| = 5(25 - 1) - 1(5 - 1) + 1(1 - 5)$$

$$= 5(24) - 1(4) + 1(-4)$$

$$= 120 - 4 - 4 = 112 \neq 0$$

$\therefore A^{-1}$ exists.

$$\text{adj}A = \begin{bmatrix} \begin{vmatrix} 5 & 1 \\ 1 & 5 \end{vmatrix} & -\begin{vmatrix} 1 & 1 \\ 1 & 5 \end{vmatrix} & \begin{vmatrix} 1 & 5 \\ 1 & 1 \end{vmatrix} \\ -\begin{vmatrix} 1 & 1 \\ 1 & 5 \end{vmatrix} & \begin{vmatrix} 5 & 1 \\ 1 & 5 \end{vmatrix} & -\begin{vmatrix} 5 & 1 \\ 1 & 1 \end{vmatrix} \\ \begin{vmatrix} 1 & 1 \\ 5 & 1 \end{vmatrix} & -\begin{vmatrix} 5 & 1 \\ 1 & 1 \end{vmatrix} & \begin{vmatrix} 5 & 1 \\ 1 & 5 \end{vmatrix} \end{bmatrix}^T$$

$$= \begin{bmatrix} (25 - 1) & -(5 - 1) & (1 - 5) \\ -(5 - 1) & (25 - 1) & -(5 - 1) \\ (1 - 5) & -(5 - 1) & (25 - 1) \end{bmatrix}^T$$

$$= \begin{bmatrix} 24 & -4 & -4 \\ -4 & 24 & -4 \\ -4 & -4 & 24 \end{bmatrix}^T = \begin{bmatrix} 24 & -4 & -4 \\ -4 & 24 & -4 \\ -4 & -4 & 24 \end{bmatrix}$$

$$A^{-1} = \frac{1}{|A|} \text{adj} A = \frac{1}{112} \begin{bmatrix} 24 & -4 & -4 \\ -4 & 24 & -4 \\ -4 & -4 & 24 \end{bmatrix}$$

$$= \frac{1}{112} \times 4 \begin{bmatrix} 6 & -1 & -1 \\ -1 & 6 & -1 \\ -1 & -1 & 6 \end{bmatrix}$$

$$\therefore A^{-1} = \frac{1}{28} \begin{bmatrix} 6 & -1 & -1 \\ -1 & 6 & -1 \\ -1 & -1 & 6 \end{bmatrix}$$

(iii) Let $A = \begin{bmatrix} 2 & 3 & 1 \\ 3 & 4 & 1 \\ 3 & 7 & 2 \end{bmatrix}$ (3M)

$$|A| = 2(8 - 7) - 3(6 - 3) + 1(21 - 12)$$

$$= 2(1) - 3(3) + 1(9)$$

$$= 2 - 9 + 9$$

$$= 2 \neq 0$$

$\therefore A^{-1}$ exists

$$\text{adj}A = \begin{bmatrix} \begin{vmatrix} 4 & 1 \\ 7 & 2 \end{vmatrix} & -\begin{vmatrix} 3 & 1 \\ 3 & 2 \end{vmatrix} & \begin{vmatrix} 3 & 4 \\ 3 & 7 \end{vmatrix} \\ -\begin{vmatrix} 3 & 1 \\ 7 & 2 \end{vmatrix} & \begin{vmatrix} 2 & 1 \\ 3 & 2 \end{vmatrix} & -\begin{vmatrix} 2 & 3 \\ 3 & 7 \end{vmatrix} \\ \begin{vmatrix} 3 & 1 \\ 4 & 1 \end{vmatrix} & -\begin{vmatrix} 2 & 1 \\ 3 & 1 \end{vmatrix} & \begin{vmatrix} 2 & 3 \\ 3 & 4 \end{vmatrix} \end{bmatrix}^T$$

$$= \begin{bmatrix} (8 - 7) & -(6 - 3) & (21 - 12) \\ -(6 - 7) & (4 - 3) & -(14 - 9) \\ (3 - 4) & -(2 - 3) & (8 - 9) \end{bmatrix}^T$$

$$= \begin{bmatrix} 1 & -3 & 9 \\ 1 & 1 & -5 \\ -1 & 1 & -1 \end{bmatrix}^T$$

$$\therefore \text{adj} A = \begin{bmatrix} 1 & 1 & -1 \\ -3 & 1 & 1 \\ 9 & -5 & -1 \end{bmatrix}$$

$$A^{-1} = \frac{1}{|A|} \text{adj} A$$

$$A^{-1} = \frac{1}{2} \begin{bmatrix} 1 & 1 & -1 \\ -3 & 1 & 1 \\ 9 & -5 & -1 \end{bmatrix}$$

3. If $F(\alpha) = \begin{bmatrix} \cos \alpha & 0 & \sin \alpha \\ 0 & 1 & 0 \\ -\sin \alpha & 0 & \cos \alpha \end{bmatrix}$, show that $[F(\alpha)]^{-1} = F(-\alpha)$. (APR-23) (5M)

$$F(\alpha) = \begin{bmatrix} \cos \alpha & 0 & \sin \alpha \\ 0 & 1 & 0 \\ -\sin \alpha & 0 & \cos \alpha \end{bmatrix}$$

LHS: $[F(\alpha)]^{-1}$: $[F(\alpha)]^{-1} = \frac{1}{|F(\alpha)|} \text{adj} [F(\alpha)]$

$$|F(\alpha)| = \cos \alpha [\cos \alpha - 0] - 0 + \sin \alpha [0 + \sin \alpha]$$

$$= \cos^2 \alpha + \sin^2 \alpha = 1 \neq 0$$

$\therefore [F(\alpha)]^{-1}$ exists.

$$\text{adj}[F(\alpha)] = \begin{bmatrix} \begin{vmatrix} 0 & 0 \\ 0 & \cos \alpha \end{vmatrix} & -\begin{vmatrix} 0 & \sin \alpha \\ -\sin \alpha & \cos \alpha \end{vmatrix} & \begin{vmatrix} 0 & 1 \\ -\sin \alpha & 0 \end{vmatrix} \\ -\begin{vmatrix} 0 & \sin \alpha \\ 0 & \cos \alpha \end{vmatrix} & \begin{vmatrix} \cos \alpha & \sin \alpha \\ -\sin \alpha & \cos \alpha \end{vmatrix} & -\begin{vmatrix} \cos \alpha & 0 \\ -\sin \alpha & 0 \end{vmatrix} \\ \begin{vmatrix} 0 & +\sin \alpha \\ 1 & 0 \end{vmatrix} & -\begin{vmatrix} \cos \alpha & +\sin \alpha \\ 0 & 0 \end{vmatrix} & \begin{vmatrix} \cos \alpha & 0 \\ 0 & 1 \end{vmatrix} \end{bmatrix}^T$$

$$= \begin{bmatrix} (\cos \alpha - 0) & -(0 - 0) & (0 + \sin \alpha) \\ -(0 - 0) & (\cos^2 \alpha + \sin^2 \alpha) & -(0 - 0) \\ (0 - \sin \alpha) & -(0 - 0) & (\cos \alpha - 0) \end{bmatrix}^T$$

$$= \begin{bmatrix} \cos \alpha & 0 & \sin \alpha \\ 0 & 1 & 0 \\ -\sin \alpha & 0 & \cos \alpha \end{bmatrix}^T = \begin{bmatrix} \cos \alpha & 0 & -\sin \alpha \\ 0 & 1 & 0 \\ \sin \alpha & 0 & \cos \alpha \end{bmatrix}$$

Now, $[F(\alpha)]^{-1} = \frac{1}{1} \begin{bmatrix} \cos \alpha & 0 & -\sin \alpha \\ 0 & 1 & 0 \\ \sin \alpha & 0 & \cos \alpha \end{bmatrix}$

$$= \begin{bmatrix} \cos \alpha & 0 & -\sin \alpha \\ 0 & 1 & 0 \\ \sin \alpha & 0 & \cos \alpha \end{bmatrix}$$

RHS: $F(-\alpha) = \begin{bmatrix} \cos(-\alpha) & 0 & \sin(-\alpha) \\ 0 & 1 & 0 \\ -\sin(-\alpha) & 0 & \cos(-\alpha) \end{bmatrix}$

$[\because \sin(-\alpha) = -\sin \alpha] [\cos(-\alpha) = \cos \alpha]$

$$= \begin{bmatrix} \cos \alpha & 0 & -\sin \alpha \\ 0 & 1 & 0 \\ \sin \alpha & 0 & \cos \alpha \end{bmatrix}$$

LHS = RHS $\Rightarrow [F(\alpha)]^{-1} = F(-\alpha)$ Hence proved.

4. If $A = \begin{bmatrix} 5 & 3 \\ -1 & -2 \end{bmatrix}$, show that $A^2 - 3A - 7I_2 = O_2$. Hence find A^{-1} .

5M

$$A^2 = A \cdot A = \begin{bmatrix} 5 & 3 \\ -1 & -2 \end{bmatrix} \begin{bmatrix} 5 & 3 \\ -1 & -2 \end{bmatrix} = \begin{bmatrix} 25-3 & 15-6 \\ -5+2 & -3+4 \end{bmatrix} = \begin{bmatrix} 22 & 9 \\ -3 & 1 \end{bmatrix}$$

$$3A = 3 \begin{bmatrix} 5 & 3 \\ -1 & -2 \end{bmatrix} = \begin{bmatrix} 15 & 9 \\ -3 & -6 \end{bmatrix}$$

$$7I_2 = 7 \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} = \begin{bmatrix} 7 & 0 \\ 0 & 7 \end{bmatrix}$$

$$\begin{aligned} A^2 - 3A - 7I_2 &= \begin{bmatrix} 22 & 9 \\ -3 & 1 \end{bmatrix} - \begin{bmatrix} 15 & 9 \\ -3 & -6 \end{bmatrix} - \begin{bmatrix} 7 & 0 \\ 0 & 7 \end{bmatrix} \\ &= \begin{bmatrix} 22-15-7 & 9-9-0 \\ -3+3-0 & 1+6-7 \end{bmatrix} = \begin{bmatrix} 0 & 0 \\ 0 & 0 \end{bmatrix} \end{aligned}$$

$$\therefore A^2 - 3A - 7I_2 = O$$

To find A^{-1} : Post-multiplying by A^{-1} , we get

$$A - 3I - 7A^{-1} = O$$

$$7A^{-1} = A - 3I$$

$$= \begin{bmatrix} 5 & 3 \\ -1 & -2 \end{bmatrix} - 3 \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$$

$$= \begin{bmatrix} 5 & 3 \\ -1 & -2 \end{bmatrix} + \begin{bmatrix} -3 & 0 \\ 0 & -3 \end{bmatrix}$$

$$= \begin{bmatrix} 2 & 3 \\ -1 & -5 \end{bmatrix}$$

$$A^{-1} = \frac{1}{7} \begin{bmatrix} 2 & 3 \\ -1 & -5 \end{bmatrix}$$

$$\therefore A^{-1}A^{+2} = (A^{-1}A)A = IA = A$$

$$\therefore A^{-1}3A = 3(A^{-1}A) = 3I$$

$$\therefore A^{-1}I_2 = A^{-1}$$

5. If $A = \frac{1}{9} \begin{bmatrix} -8 & 1 & 4 \\ 4 & 4 & 7 \\ 1 & -8 & 4 \end{bmatrix}$, prove that $A^{-1} = A^T$.

3M

$$A = \frac{1}{9} \begin{bmatrix} -8 & 1 & 4 \\ 4 & 4 & 7 \\ 1 & -8 & 4 \end{bmatrix}; A^T = \frac{1}{9} \begin{bmatrix} -8 & 4 & 1 \\ 1 & 4 & -8 \\ 4 & 7 & 4 \end{bmatrix}$$

To prove : $A^{-1} = A^T$

$$AA^{-1} = AA^T$$

Enough to prove: $AA^T = A^T A = I$

$$AA^T = \frac{1}{9} \begin{bmatrix} -8 & 1 & 4 \\ 4 & 4 & 7 \\ 1 & -8 & 4 \end{bmatrix} \frac{1}{9} \begin{bmatrix} -8 & 4 & 1 \\ 1 & 4 & -8 \\ 4 & 7 & 4 \end{bmatrix} = \frac{1}{81} \begin{bmatrix} 81 & 0 & 0 \\ 0 & 81 & 0 \\ 0 & 0 & 81 \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} = I \quad \dots\dots (1)$$

$$A^T A = \frac{1}{9} \begin{bmatrix} -8 & 4 & 1 \\ 1 & 4 & -8 \\ 4 & 7 & 4 \end{bmatrix} \frac{1}{9} \begin{bmatrix} -8 & 1 & 4 \\ 4 & 4 & 7 \\ 1 & -8 & 4 \end{bmatrix} = \frac{1}{81} \begin{bmatrix} 81 & 0 & 0 \\ 0 & 81 & 0 \\ 0 & 0 & 81 \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} = I \quad \dots\dots (2)$$

From (1) and (2), A is called **orthogonal** if $AA^T = A^T A = I$

$$AA^{-1} = A^{-1}A = I$$

$$\therefore A^{-1} = A^T \quad \text{Hence Proved.}$$

6. If $A = \begin{bmatrix} 8 & -4 \\ -5 & 3 \end{bmatrix}$, verify that JUL-20,21

$$A(\text{adj} A) = (\text{adj} A)A = |A|I_2. \quad \text{3M}$$

$$A = \begin{bmatrix} 8 & -4 \\ -5 & 3 \end{bmatrix}$$

$$\text{adj} A = \begin{bmatrix} 3 & 4 \\ 5 & 8 \end{bmatrix}$$

$$\begin{aligned} A(\text{adj} A) &= \begin{bmatrix} 8 & -4 \\ -5 & 3 \end{bmatrix} \begin{bmatrix} 3 & 4 \\ 5 & 8 \end{bmatrix} \\ &= \begin{bmatrix} 24 - 20 & 32 - 32 \\ -15 + 15 & -20 + 24 \end{bmatrix} \\ &= \begin{bmatrix} 4 & 0 \\ 0 & 4 \end{bmatrix} \end{aligned}$$

$$= 4 \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} \dots\dots(1)$$

$$\begin{aligned} (\text{adj} A)A &= \begin{bmatrix} 3 & 4 \\ 5 & 8 \end{bmatrix} \begin{bmatrix} 8 & -4 \\ -5 & 3 \end{bmatrix} \\ &= \begin{bmatrix} 24 - 20 & -12 + 12 \\ 40 - 40 & -20 + 24 \end{bmatrix} \\ &= \begin{bmatrix} 4 & 0 \\ 0 & 4 \end{bmatrix} \end{aligned}$$

$$= 4 \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} \dots\dots(2)$$

$$|A| = 24 - 20 = 4$$

$$|A|I_2 = 4 \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} = \begin{bmatrix} 4 & 0 \\ 0 & 4 \end{bmatrix} \dots\dots(3)$$

From (1), (2) and (3), we get

$$A(\text{adj} A) = (\text{adj} A)A = |A|I_2$$

8. If $\text{adj}(A) = \begin{bmatrix} 2 & -4 & 2 \\ -3 & 12 & -7 \\ -2 & 0 & 2 \end{bmatrix}$, find A .

$$|\text{adj} A| = 2[24 - 0] + 4[-6 - 14] + 2[0 + 24] = 48 - 80 + 48 = 16$$

$$\text{adj}(\text{adj} A) = \begin{bmatrix} \begin{vmatrix} 12 & -7 \\ 0 & 2 \end{vmatrix} & -\begin{vmatrix} -3 & -7 \\ -2 & 2 \end{vmatrix} & \begin{vmatrix} -3 & 12 \\ -2 & 0 \end{vmatrix} \\ -\begin{vmatrix} -4 & 2 \\ 0 & 2 \end{vmatrix} & \begin{vmatrix} 2 & 2 \\ -2 & 2 \end{vmatrix} & -\begin{vmatrix} 2 & -4 \\ -2 & 0 \end{vmatrix} \\ \begin{vmatrix} -4 & 2 \\ 12 & -7 \end{vmatrix} & -\begin{vmatrix} 2 & 2 \\ -3 & -7 \end{vmatrix} & \begin{vmatrix} 2 & -4 \\ -3 & 12 \end{vmatrix} \end{bmatrix}^T$$

$$= \begin{bmatrix} (24 - 0) & -(-6 - 14) & (0 + 24) \\ -(-8 - 0) & (4 + 4) & -(0 - 8) \\ (28 - 24) & -(-14 + 6) & (24 - 12) \end{bmatrix}^T$$

$$= \begin{bmatrix} 24 & 20 & 24 \\ 8 & 8 & 8 \\ 4 & 8 & 12 \end{bmatrix}^T = \begin{bmatrix} 24 & 8 & 4 \\ 20 & 8 & 8 \\ 24 & 8 & 12 \end{bmatrix} = 4 \begin{bmatrix} 6 & 2 & 1 \\ 5 & 2 & 2 \\ 6 & 2 & 3 \end{bmatrix}$$

$$A = \pm \frac{1}{\sqrt{|\text{adj} A|}} \text{adj}(\text{adj} A) = \pm \frac{1}{\sqrt{16}} \times 4 \begin{bmatrix} 6 & 2 & 1 \\ 5 & 2 & 2 \\ 6 & 2 & 3 \end{bmatrix} = \pm \begin{bmatrix} 6 & 2 & 1 \\ 5 & 2 & 2 \\ 6 & 2 & 3 \end{bmatrix}$$

7. If $A = \begin{bmatrix} 3 & 2 \\ 7 & 5 \end{bmatrix}$ and $B = \begin{bmatrix} -1 & -3 \\ 5 & 2 \end{bmatrix}$, verify that

$$(\text{adj} AB)^{-1} = B^{-1}A^{-1} \quad \text{3M}$$

$$\begin{aligned} AB &= \begin{bmatrix} 3 & 2 \\ 7 & 5 \end{bmatrix} \begin{bmatrix} -1 & -3 \\ 5 & 2 \end{bmatrix} \\ &= \begin{bmatrix} -3 + 10 & -9 + 4 \\ -7 + 25 & -21 + 10 \end{bmatrix} = \begin{bmatrix} 7 & -5 \\ +18 & -11 \end{bmatrix} \end{aligned}$$

$$|AB| = \begin{vmatrix} 7 & -5 \\ +18 & -11 \end{vmatrix} = -77 + 90 = 13 \neq 0$$

$(AB)^{-1}$ exists.

$$\text{adj}(AB) = \begin{bmatrix} -11 & 5 \\ -18 & 7 \end{bmatrix}$$

$$(AB)^{-1} = \frac{1}{|AB|} \text{adj}(AB) = \frac{1}{13} \begin{bmatrix} -11 & 5 \\ -18 & 7 \end{bmatrix} \dots\dots(1)$$

$$A = \begin{bmatrix} 3 & 2 \\ 7 & 5 \end{bmatrix}$$

$$B = \begin{bmatrix} -1 & -3 \\ 5 & 2 \end{bmatrix}$$

$$|A| = 15 - 14 = 1 \neq 0$$

$$|B| = -2 + 15 = 13 \neq 0$$

$\therefore A^{-1}$ exists

$\therefore B^{-1}$ exists

$$\text{adj} A = \begin{bmatrix} 5 & -2 \\ -7 & 3 \end{bmatrix}$$

$$\text{adj} B = \begin{bmatrix} 2 & 3 \\ -5 & -1 \end{bmatrix}$$

$$A^{-1} = \frac{1}{|A|} \text{adj} A$$

$$B^{-1} = \frac{1}{|B|} \text{adj} B$$

$$= \frac{1}{1} \begin{bmatrix} 5 & -2 \\ -7 & 3 \end{bmatrix}$$

$$= \frac{1}{13} \begin{bmatrix} 2 & 3 \\ -5 & -1 \end{bmatrix}$$

$$B^{-1}A^{-1} = \frac{1}{13} \begin{bmatrix} 2 & 3 \\ -5 & -1 \end{bmatrix} \begin{bmatrix} 5 & -2 \\ -7 & 3 \end{bmatrix}$$

$$= \frac{1}{13} \begin{bmatrix} 10 - 21 & -4 + 9 \\ -25 + 7 & 10 - 3 \end{bmatrix} = \frac{1}{13} \begin{bmatrix} -11 & 5 \\ -18 & 7 \end{bmatrix} \dots\dots(2)$$

From (1) and (2), we get $(AB)^{-1} = B^{-1}A^{-1}$

3M

9. If $\text{adj}(A) = \begin{bmatrix} 0 & -2 & 0 \\ 6 & 2 & -6 \\ -3 & 0 & 6 \end{bmatrix}$, find A^{-1} .

$\text{adj}(A) = \begin{bmatrix} 0 & -2 & 0 \\ 6 & 2 & -6 \\ -3 & 0 & 6 \end{bmatrix}$ PTA-6, JUL-23,24 2M

$$|\text{adj}A| = 0 + 2(36 - 18) + 0 = 36$$

$$A^{-1} = \pm \frac{1}{\sqrt{|\text{adj}A|}} (\text{adj}A)$$

$$= \pm \frac{1}{\sqrt{36}} \begin{bmatrix} 0 & -2 & 0 \\ 6 & 2 & -6 \\ -3 & 0 & 6 \end{bmatrix}$$

$$= \pm \frac{1}{\sqrt{6 \times 6}} \begin{bmatrix} 0 & -2 & 0 \\ 6 & 2 & -6 \\ -3 & 0 & 6 \end{bmatrix}$$

$$A^{-1} = \pm \frac{1}{6} \begin{bmatrix} 0 & -2 & 0 \\ 6 & 2 & -6 \\ -3 & 0 & 6 \end{bmatrix}$$

10. Find $\text{adj}(\text{adj}(A))$ if $\text{adj}A = \begin{bmatrix} 1 & 0 & 1 \\ 0 & 2 & 0 \\ -1 & 0 & 1 \end{bmatrix}$ 3M

$$\text{adj}A = \begin{bmatrix} 1 & 0 & 1 \\ 0 & 2 & 0 \\ -1 & 0 & 1 \end{bmatrix}$$

$$\text{adj}(\text{adj}A) = \begin{bmatrix} \begin{vmatrix} 2 & 0 \\ 0 & 1 \end{vmatrix} & -\begin{vmatrix} 0 & 0 \\ -1 & 1 \end{vmatrix} & \begin{vmatrix} 0 & 2 \\ -1 & 0 \end{vmatrix} \\ -\begin{vmatrix} 0 & 1 \\ 0 & 1 \end{vmatrix} & \begin{vmatrix} 1 & 1 \\ -1 & 1 \end{vmatrix} & -\begin{vmatrix} 1 & 0 \\ -1 & 0 \end{vmatrix} \\ \begin{vmatrix} 0 & 1 \\ 2 & 0 \end{vmatrix} & -\begin{vmatrix} 1 & 1 \\ 0 & 0 \end{vmatrix} & \begin{vmatrix} 1 & 0 \\ 0 & 2 \end{vmatrix} \end{bmatrix}^T$$

$$= \begin{bmatrix} (2-0) & -(0+0) & (0+2) \\ -(0-0) & (1+1) & -(0+0) \\ (0-2) & -(0-0) & (2-0) \end{bmatrix}^T$$

$$= \begin{bmatrix} 2 & 0 & 2 \\ 0 & 2 & 0 \\ -2 & 0 & 2 \end{bmatrix}^T$$

$$\therefore \text{adj}(\text{adj}A) = \begin{bmatrix} 2 & 0 & -2 \\ 0 & 2 & 0 \\ 2 & 0 & 2 \end{bmatrix}$$

11. $A = \begin{bmatrix} 1 & \tan x \\ -\tan x & 1 \end{bmatrix}$, show that $A^T A^{-1} = \begin{bmatrix} \cos 2x & -\sin 2x \\ \sin 2x & \cos 2x \end{bmatrix}$ 3M

$$A = \begin{bmatrix} 1 & \tan x \\ -\tan x & 1 \end{bmatrix}; \quad A^T = \begin{bmatrix} 1 & -\tan x \\ \tan x & 1 \end{bmatrix}$$

$$A^{-1} = \frac{1}{|A|} \text{adj}A$$

$$|A| = 1 + \tan^2 x = \sec^2 x; \quad \text{adj}A = \begin{bmatrix} 1 & -\tan x \\ \tan x & 1 \end{bmatrix}$$

$$A^{-1} = \frac{1}{\sec^2 x} \begin{bmatrix} 1 & -\tan x \\ \tan x & 1 \end{bmatrix}$$

$$A^T A^{-1} = \begin{bmatrix} 1 & -\tan x \\ \tan x & 1 \end{bmatrix} \frac{1}{\sec^2 x} \begin{bmatrix} 1 & -\tan x \\ \tan x & 1 \end{bmatrix}$$

$$= \frac{1}{\sec^2 x} \begin{bmatrix} 1 - \tan^2 x & -2 \tan x \\ 2 \tan x & 1 - \tan^2 x \end{bmatrix}$$

$$= \cos^2 x \begin{bmatrix} 1 - \frac{\sin^2 x}{\cos^2 x} & -\frac{2 \sin x}{\cos x} \\ 2 \frac{\sin x}{\cos x} & 1 - \frac{\sin^2 x}{\cos^2 x} \end{bmatrix}$$

$$= \begin{bmatrix} \frac{\cos^2 x - \sin^2 x}{\cos^2 x} \cancel{\cos^2 x} & \frac{-2 \sin x \cos^2 x}{\cancel{\cos x}} \\ \frac{2 \sin x}{\cancel{\cos x}} \cancel{\cos^2 x} & \frac{\cos^2 x - \sin^2 x}{\cos^2 x} \cancel{\cos^2 x} \end{bmatrix}$$

$$= \begin{bmatrix} \cos^2 x - \sin^2 x & -2 \sin x \cos x \\ 2 \sin x \cos x & \cos^2 x - \sin^2 x \end{bmatrix}$$

$$\left[\begin{array}{l} \because \cos^2 x - \sin^2 x = \cos 2x \\ 2 \sin x \cos x = \sin 2x \end{array} \right]$$

$$A^T A^{-1} = \begin{bmatrix} \cos 2x & -\sin 2x \\ \sin 2x & \cos 2x \end{bmatrix} \quad \text{Hence proved.}$$

12. Find the matrix A for which

3M

$$A \begin{bmatrix} 5 & 3 \\ -1 & -2 \end{bmatrix} = \begin{bmatrix} 14 & 7 \\ 7 & 7 \end{bmatrix}.$$

$$\text{Let } B = \begin{bmatrix} 5 & 3 \\ -1 & -2 \end{bmatrix} \text{ and } C = \begin{bmatrix} 14 & 7 \\ 7 & 7 \end{bmatrix}$$

$$\text{Then } AB = C$$

Post multiplying by B^{-1} , we get

$$A(BB^{-1}) = CB^{-1}$$

$$AI = CB^{-1}$$

$$A = CB^{-1} \dots\dots\dots(1)$$

$$|B| = \begin{vmatrix} 5 & 3 \\ -1 & -2 \end{vmatrix} = -10 + 3 = -7 \neq 0$$

 $\therefore B^{-1}$ exists

$$\text{adj } B = \begin{bmatrix} -2 & -3 \\ 1 & 5 \end{bmatrix}$$

$$B^{-1} = \frac{1}{|B|} \text{adj } B$$

$$= \frac{1}{-7} \begin{bmatrix} -2 & -3 \\ 1 & 5 \end{bmatrix} = \frac{1}{7} \begin{bmatrix} 2 & 3 \\ -1 & -5 \end{bmatrix}$$

$$(1) \Rightarrow A = \begin{bmatrix} 14 & 7 \\ 7 & 7 \end{bmatrix} \frac{1}{7} \begin{bmatrix} 2 & 3 \\ -1 & -5 \end{bmatrix}$$

$$= \frac{1}{7} \times 7 \begin{bmatrix} 2 & 1 \\ 1 & 1 \end{bmatrix} \begin{bmatrix} 2 & 3 \\ -1 & -5 \end{bmatrix}$$

$$= \begin{bmatrix} 4-1 & 6-5 \\ 2-1 & 3-5 \end{bmatrix}$$

$$= \begin{bmatrix} 3 & 1 \\ 1 & -2 \end{bmatrix}$$

13. Given $A = \begin{bmatrix} 1 & -1 \\ 2 & 0 \end{bmatrix}$, $B = \begin{bmatrix} 3 & -2 \\ 1 & 1 \end{bmatrix}$ & $C = \begin{bmatrix} 1 & 1 \\ 2 & 2 \end{bmatrix}$,find a matrix X such that $AXB = C$.

3M

Given: $AXB = C$ Pre-multiplying by A^{-1} , we get $(A^{-1}A)XB = A^{-1}C$

$$(IX)B = A^{-1}C$$

$$XB = A^{-1}C$$

Post-multiplying by B^{-1} , we get $X(BB^{-1}) = A^{-1}CB^{-1}$

$$X = A^{-1}CB^{-1}$$

$$A = \begin{bmatrix} 1 & -1 \\ 2 & 0 \end{bmatrix}$$

$$|A| = 0 + 2 = 2 \neq 0$$

 $\therefore A^{-1}$ exists

$$\text{adj } A = \begin{bmatrix} 0 & 1 \\ -2 & 1 \end{bmatrix}$$

$$A^{-1} = \frac{1}{|A|} \text{adj } A$$

$$= \frac{1}{2} \begin{bmatrix} 0 & 1 \\ -2 & 1 \end{bmatrix}$$

Now, $X = A^{-1}(CB^{-1})$

$$X = \frac{1}{2} \begin{bmatrix} 0 & 1 \\ -2 & 1 \end{bmatrix} \begin{bmatrix} 1 & 1 \\ 2 & 2 \end{bmatrix} \frac{1}{5} \begin{bmatrix} 1 & 2 \\ -1 & 3 \end{bmatrix}$$

$$= \frac{1}{10} \begin{bmatrix} 0+2 & 0+2 \\ -2+2 & -2+2 \end{bmatrix} \begin{bmatrix} 1 & 2 \\ -1 & 3 \end{bmatrix}$$

$$= \frac{1}{10} \begin{bmatrix} 2 & 2 \\ 0 & 0 \end{bmatrix} \begin{bmatrix} 1 & 2 \\ -1 & 3 \end{bmatrix}$$

$$X = \frac{1}{10} \begin{bmatrix} 2-2 & 4+6 \\ 0+0 & 0+0 \end{bmatrix} = \frac{1}{10} \begin{bmatrix} 0 & 10 \\ 0 & 0 \end{bmatrix} = \begin{bmatrix} 0 & 1 \\ 0 & 0 \end{bmatrix}$$

14. If $A = \begin{bmatrix} 0 & 1 & 1 \\ 1 & 0 & 1 \\ 1 & 1 & 0 \end{bmatrix}$, show that $A^{-1} = \frac{1}{2}(A^2 - 3I)$.

3M

To prove: $A^{-1} = \frac{1}{2}(A^2 - 3I) \Rightarrow 2A^{-1} = A^2 - 3I$ Pre-multiplying by A , $2AA^{-1} = AA^2 - 3A$

$$\therefore AA^{-1} = A^{-1}A = I$$

Enough to prove: $2I = A^3 - 3A$

$$A^2 = A.A = \begin{bmatrix} 0 & 1 & 1 \\ 1 & 0 & 1 \\ 1 & 1 & 0 \end{bmatrix} \begin{bmatrix} 0 & 1 & 1 \\ 1 & 0 & 1 \\ 1 & 1 & 0 \end{bmatrix} = \begin{bmatrix} 0+1+1 & 0+0+1 & 0+1+0 \\ 0+0+1 & 1+0+1 & 1+0+0 \\ 0+1+0 & 1+0+0 & 1+1+0 \end{bmatrix} = \begin{bmatrix} 2 & 1 & 1 \\ 1 & 2 & 1 \\ 1 & 1 & 2 \end{bmatrix}$$

$$A^3 = \begin{bmatrix} 0 & 1 & 1 \\ 1 & 0 & 1 \\ 1 & 1 & 0 \end{bmatrix} \begin{bmatrix} 2 & 1 & 1 \\ 1 & 2 & 1 \\ 1 & 1 & 2 \end{bmatrix} = \begin{bmatrix} 0+1+1 & 0+2+1 & 0+1+2 \\ 2+0+1 & 1+0+1 & 1+0+2 \\ 2+1+0 & 1+2+0 & 1+1+0 \end{bmatrix} = \begin{bmatrix} 2 & 3 & 3 \\ 3 & 2 & 3 \\ 3 & 3 & 2 \end{bmatrix}$$

$$3A = 3 \begin{bmatrix} 0 & 1 & 1 \\ 1 & 0 & 1 \\ 1 & 1 & 0 \end{bmatrix} = \begin{bmatrix} 0 & 3 & 3 \\ 3 & 0 & 3 \\ 3 & 3 & 0 \end{bmatrix}$$

$$A^3 - 3A = \begin{bmatrix} 2 & 3 & 3 \\ 3 & 2 & 3 \\ 3 & 3 & 2 \end{bmatrix} - \begin{bmatrix} 0 & 3 & 3 \\ 3 & 0 & 3 \\ 3 & 3 & 0 \end{bmatrix} = \begin{bmatrix} 2-0 & 3-3 & 3-3 \\ 3-3 & 2-0 & 3-3 \\ 3-3 & 3-3 & 2-0 \end{bmatrix} = \begin{bmatrix} 2 & 0 & 0 \\ 0 & 2 & 0 \\ 0 & 0 & 2 \end{bmatrix} = 2 \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} = 2I$$

$$A^{-1} = \frac{1}{2}(A^2 - 3I) \text{ Hence Proved.}$$

15. Decrypt the received encoded message $[2 \ -3][20 \ 4]$ with the encryption matrix $\begin{bmatrix} -1 & -1 \\ 2 & 1 \end{bmatrix}$ and the decryption matrix as its inverse, where the system of codes are described by the numbers 1-26 to the letters A – Z respectively, and the number 0 to a blank space.

3M

$$\begin{array}{l|l} \text{Let the encoding matrix be } A = \begin{bmatrix} -1 & -1 \\ 2 & 1 \end{bmatrix} & A^{-1} = \frac{1}{|A|} \text{adj } A \\ |A| = -1 + 2 = 1 \neq 0 \quad \therefore A^{-1} \text{ exists} & = \frac{1}{1} \begin{bmatrix} 1 & 1 \\ -2 & -1 \end{bmatrix} \\ \text{adj } A = \begin{bmatrix} 1 & 1 \\ -2 & -1 \end{bmatrix} & A^{-1} = \begin{bmatrix} 1 & 1 \\ -2 & -1 \end{bmatrix} \\ & \therefore \text{Decoding matrix} = A^{-1} = \begin{bmatrix} 1 & 1 \\ -2 & -1 \end{bmatrix} \end{array}$$

Coded row matrix	decoding matrix	decoded row matrix
$[2 \ -3]$	$\begin{bmatrix} 1 & 1 \\ -2 & -1 \end{bmatrix}$	$= [2 + 6 \ 2 + 3] = [8 \ 5]$
$[20 \ 4]$	$\begin{bmatrix} 1 & 1 \\ -2 & -1 \end{bmatrix}$	$= [20 - 8 \ 20 - 4] = [12 \ 16]$

So, the sequence of decoded row matrix $= [8 \ 5][12 \ 16] = \text{H, E, L, P}$

Thus, the message received is "HELP".

EXERCISE 1.2



Video class

Concept Corner

Elementary Transformations of a matrix:

A matrix can be transformed to another matrix by certain operations called elementary row operations and elementary column operations.

Elementary Row and Column Operations: Elementary row operations and elementary column operations on a matrix are known as **elementary transformations**.

we use the following notations for elementary row transformations.

- Interchanging of i^{th} and j^{th} rows is denoted by $R_i \leftrightarrow R_j$.
- The multiplication of each element of i^{th} row by a non-zero constant λ is denoted by $R_i \rightarrow \lambda R_i$.
- Addition to i^{th} row, a non-zero constant λ multiple of j^{th} row is denoted by $R_i \rightarrow R_i + \lambda R_j$.

Two matrices A and B of same order are said to be **equivalent** to one another if one can be obtained from the other by the applications of elementary transformations. Symbolically, we write $A \sim B$ to mean that the matrix A is equivalent to the matrix B .

Row-echelon form:

A non-zero matrix E is said to be in a **row-echelon form** if.

- All zero rows of E occur below every non-zero row of E .
- The first non-zero element in any row i of E occurs in the j^{th} column of E , then all other entries in the j^{th} column of E below the first non-zero element of row i are zeros.
- The first non-zero entry in the i^{th} row of E lies to the left of the first non-zero entry in $(i + 1)^{\text{th}}$ row of E .

Note : A non-zero matrix is in a row-echelon form if all zero rows occur as bottom rows of the matrix, and if the first non-zero element in any lower row occurs to the right of the first non-zero entry in the higher row.

Rank of a matrix:

The **rank of a matrix** A is defined as the order of a highest order non-vanishing minor of the matrix A . It is denoted by the symbol $\rho(A)$. The rank of a zero matrix is defined to be 0.

Note: (i) If a matrix contains at-least one non-zero element, then $\rho(A) \geq 1$.

(ii) The rank of the identity matrix I_n is n .

(iii) If the rank of a matrix A is r , then there exists at-least one minor of A of order r which does not vanish and every minor of A of order $r + 1$ and higher order (if any) vanishes.

(iv) If A is an $m \times n$ matrix, then $\rho(A) \leq \min\{m, n\}$ = minimum of m, n .

(v) A square matrix A of order n is invertible if and only if $\rho(A) = n$.

An **elementary matrix** is defined as a matrix which is obtained from an identity matrix by applying only one elementary transformation.

Theorems:

- The rank of the matrix in row echelon form is the number of non-zero rows in it.
- The rank of a non-zero matrix is equal to the number of non-zero rows in a row echelon form of the matrix.
- Every non-singular matrix can be transformed to an identity matrix, by a sequence of elementary row operations.

Gauss-Jordan method:

Transforming a non-singular matrix A to the form I_n by applying elementary row operations, is called **Gauss-Jordan method**. The steps in finding A^{-1} by Gauss-Jordan method are given below.

Step 1 - Augment the identity matrix I_n on the right-side of A to get the matrix $[A|I_n]$.

Step 2 - Obtain elementary matrices (row operations) $E_1, E_2, \dots, E_k$ such that $(E_k \dots E_2 E_1)A = I_n$.

Apply $E_1, E_2 \dots E_k$ on $[A|I_n]$. Then $[(E_k \dots E_2 E_1)A | (E_k \dots E_2 E_1)I_n]$. that is, $[I_n | A^{-1}]$.

1. Find the rank of the following matrices by minor method.

(i) $\begin{bmatrix} 2 & -4 \\ -1 & 2 \end{bmatrix}$ (ii) $\begin{bmatrix} -1 & 3 \\ 4 & -7 \\ 3 & -4 \end{bmatrix}$ (iii) $\begin{bmatrix} 1 & -2 & -1 & 0 \\ 3 & -6 & -3 & 1 \end{bmatrix}$ (iv) $\begin{bmatrix} 1 & -2 & 3 \\ 2 & 4 & -6 \\ 5 & 1 & -1 \end{bmatrix}$ (v) $\begin{bmatrix} 0 & 1 & 2 & 1 \\ 0 & 2 & 4 & 3 \\ 8 & 1 & 0 & 2 \end{bmatrix}$

(i) Let $A = \begin{bmatrix} 2 & -4 \\ -1 & 2 \end{bmatrix}$ **2M**

Order of the matrix A is 2×2

$$\therefore \rho(A) \leq \min\{2, 2\} = 2$$

$$|A| = 4 - 4 = 0$$

$$\therefore \rho(A) \neq 2$$

$$\text{Thus } \rho(A) = 1$$

(ii) Let $A = \begin{bmatrix} -1 & 3 \\ 4 & -7 \\ 3 & -4 \end{bmatrix}$ **2M**

Order of the matrix A is 3×2

$$\therefore \rho(A) \leq \min\{3, 2\} = 2$$

We find that the 2nd order minor

$$\begin{vmatrix} -1 & 3 \\ 4 & -7 \end{vmatrix} = 7 - 12 = -5 \neq 0.$$

$$\therefore \rho(A) = 2.$$

APR-24

(iii) Let $A = \begin{bmatrix} 1 & -2 & -1 & 0 \\ 3 & -6 & -3 & 1 \end{bmatrix}$

Order of the matrix A is 2×4

$$\therefore \rho(A) \leq \min\{2, 4\} = 2$$

2M

We find that the 2nd order minor

$$\begin{vmatrix} -1 & 0 \\ -3 & 1 \end{vmatrix} = -1 + 0 = -1 \neq 0.$$

$$\therefore \rho(A) = 2$$

PTA-5

(iv) Let $A = \begin{bmatrix} 1 & -2 & 3 \\ 2 & 4 & -6 \\ 5 & 1 & -1 \end{bmatrix}$ **JUL-22** **3M**

Order of the matrix A is 3×3

$$\therefore \rho(A) \leq \min\{3, 3\} = 3$$

There is only one third order minor of A

$$|A| = 1(-4 + 6) + 2(-2 + 30) + 3(2 - 20)$$

$$= 1(2) + 2(28) + 3(-18)$$

$$= 2 + 56 - 54 = 4 \neq 0.$$

$$\therefore \rho(A) = 3$$

(v) Let $A = \begin{bmatrix} 0 & 1 & 2 & 1 \\ 0 & 2 & 4 & 3 \\ 8 & 1 & 0 & 2 \end{bmatrix}$ **JUL-21** **2M**

Order of the matrix A is 3×4

$$\therefore \rho(A) \leq \min\{3, 4\} = 3$$

We find that the third order minor

$$\begin{vmatrix} 0 & 1 & 1 \\ 0 & 2 & 3 \\ 8 & 1 & 2 \end{vmatrix} = 0 - 0 + 8(3 - 2)$$

$$= 8(1) = 8 \neq 0$$

$$\therefore \rho(A) = 3$$

2. Find the rank of the following matrices by row reduction method:

(i) $\begin{bmatrix} 1 & 1 & 1 & 3 \\ 2 & -1 & 3 & 4 \\ 5 & -1 & 7 & 11 \end{bmatrix}$

(ii) $\begin{bmatrix} 1 & 2 & -1 \\ 3 & -1 & 2 \\ 1 & -2 & 3 \\ 1 & -1 & 1 \end{bmatrix}$

(iii) $\begin{bmatrix} 3 & -8 & 5 & 2 \\ 2 & -5 & 1 & 4 \\ -1 & 2 & 3 & -2 \end{bmatrix}$

(i) Let $A = \begin{bmatrix} 1 & 1 & 1 & 3 \\ 2 & -1 & 3 & 4 \\ 5 & -1 & 7 & 11 \end{bmatrix}$ 3M

$$\sim \begin{bmatrix} 1 & 1 & 1 & 3 \\ 0 & -3 & 1 & -2 \\ 0 & -6 & 2 & -4 \end{bmatrix} \begin{array}{l} R_2 \rightarrow R_2 - 2R_1 \\ R_3 \rightarrow R_3 - 5R_1 \end{array}$$

$$\sim \begin{bmatrix} 1 & 1 & 1 & 3 \\ 0 & -3 & 1 & -2 \\ 0 & 0 & 0 & 0 \end{bmatrix} \begin{array}{l} R_3 \rightarrow R_3 - 2R_2 \end{array}$$

The last equivalent matrix is in row-echelon form.

It has two non-zero rows.

$$\therefore \rho(A) = 2$$

(iii) Let $A = \begin{bmatrix} 3 & -8 & 5 & 2 \\ 2 & -5 & 1 & 4 \\ -1 & 2 & 3 & -2 \end{bmatrix}$ JUL-24 3M

$$\sim \begin{bmatrix} 3 & -8 & 5 & 2 \\ 0 & 1 & -7 & 8 \\ 0 & -2 & 14 & -4 \end{bmatrix} \begin{array}{l} R_2 \rightarrow 3R_2 - 2R_1 \\ R_3 \rightarrow 3R_3 + R_1 \end{array}$$

$$\sim \begin{bmatrix} 3 & -8 & 5 & 2 \\ 0 & 1 & -7 & 8 \\ 0 & 0 & 0 & 12 \end{bmatrix} \begin{array}{l} R_3 \rightarrow R_3 + 2R_2 \end{array}$$

The last equivalent matrix is in row-echelon form.

It has three non-zero rows.

$$\therefore \rho(A) = 3$$

(ii) Let $A = \begin{bmatrix} 1 & 2 & -1 \\ 3 & -1 & 2 \\ 1 & -2 & 3 \\ 1 & -1 & 1 \end{bmatrix}$ APR-22 3M

$$\sim \begin{bmatrix} 1 & 2 & -1 \\ 0 & -7 & 5 \\ 0 & -4 & 4 \\ 0 & -3 & 2 \end{bmatrix} \begin{array}{l} R_2 \rightarrow R_2 - 3R_1 \\ R_3 \rightarrow R_3 - R_1 \\ R_4 \rightarrow R_4 - R_1 \end{array}$$

$$\sim \begin{bmatrix} 1 & 2 & -1 \\ 0 & -7 & 5 \\ 0 & 0 & 8 \\ 0 & 0 & -4 \end{bmatrix} \begin{array}{l} R_3 \rightarrow 7R_3 - 4R_2 \\ R_4 \rightarrow 4R_4 - 3R_3 \end{array}$$

$$\sim \begin{bmatrix} 1 & 2 & -1 \\ 0 & -7 & 5 \\ 0 & 0 & 8 \\ 0 & 0 & 0 \end{bmatrix} \begin{array}{l} R_4 \rightarrow 2R_4 + R_3 \end{array}$$

The last equivalent matrix is in row-echelon form.

It has three non-zero rows.

$$\therefore \rho(A) = 3$$

3. Find the inverse of each of the following by Gauss-Jordan method:

(i) $\begin{bmatrix} 2 & -1 \\ 5 & -2 \end{bmatrix}$ (ii) $\begin{bmatrix} 1 & -1 & 0 \\ 1 & 0 & -1 \\ 6 & -2 & -3 \end{bmatrix}$ (iii) $\begin{bmatrix} 1 & 2 & 3 \\ 2 & 5 & 3 \\ 1 & 0 & 8 \end{bmatrix}$

(i) Let $A = \begin{bmatrix} 2 & -1 \\ 5 & -2 \end{bmatrix}$, applying Gauss - Jordan method, we get

$$[A|I] = \begin{bmatrix} 2 & -1 & 1 & 0 \\ 5 & -2 & 0 & 1 \end{bmatrix}$$

$$\sim \begin{bmatrix} 1 & -\frac{1}{2} & \frac{1}{2} & 0 \\ 5 & -2 & 0 & 1 \end{bmatrix} \begin{array}{l} R_1 \rightarrow \frac{1}{2}R_1 \end{array}$$

$$\sim \begin{bmatrix} 1 & -\frac{1}{2} & \frac{1}{2} & 0 \\ 0 & \frac{1}{2} & -\frac{5}{2} & 1 \end{bmatrix} \begin{array}{l} R_2 \rightarrow R_2 - 5R_1 \end{array}$$

$$\sim \begin{bmatrix} 1 & 0 & -2 & 1 \\ 0 & \frac{1}{2} & -\frac{5}{2} & 1 \end{bmatrix} \begin{array}{l} R_1 \rightarrow R_1 + R_2 \end{array}$$

$$\sim \begin{bmatrix} 1 & 0 & -2 & 1 \\ 0 & 1 & -5 & 2 \end{bmatrix} \begin{array}{l} R_2 \rightarrow 2R_2 \end{array}$$

$$\therefore A^{-1} = \begin{bmatrix} -2 & 1 \\ -5 & 2 \end{bmatrix}$$

3M

MDL

(ii) Let $A = \begin{bmatrix} 1 & -1 & 0 \\ 1 & 0 & -1 \\ 6 & -2 & -3 \end{bmatrix}$ applying

5M

Gauss - Jordan method, we get

$$[A|I] = \left[\begin{array}{ccc|ccc} 1 & -1 & 0 & 1 & 0 & 0 \\ 1 & 0 & -1 & 0 & 1 & 0 \\ 6 & -2 & -3 & 0 & 0 & 1 \end{array} \right]$$

$$\sim \left[\begin{array}{ccc|ccc} 1 & -1 & 0 & 1 & 0 & 0 \\ 0 & 1 & -1 & -1 & 1 & 0 \\ 0 & 4 & -3 & -6 & 0 & 1 \end{array} \right] \begin{array}{l} R_2 \rightarrow R_2 - R_1 \\ R_3 \rightarrow R_3 - 6R_1 \end{array}$$

$$\sim \left[\begin{array}{ccc|ccc} 1 & -1 & 0 & 1 & 0 & 0 \\ 0 & 1 & -1 & -1 & 1 & 0 \\ 0 & 0 & 1 & -2 & -4 & 1 \end{array} \right] R_3 \rightarrow R_3 - 4R_2$$

$$\sim \left[\begin{array}{ccc|ccc} 1 & -1 & 0 & 1 & 0 & 0 \\ 0 & 1 & 0 & -3 & -3 & 1 \\ 0 & 0 & 1 & -2 & -4 & 1 \end{array} \right] R_2 \rightarrow R_2 + R_3$$

$$\sim \left[\begin{array}{ccc|ccc} 1 & 0 & 0 & -2 & -3 & 1 \\ 0 & 1 & 0 & -3 & -3 & 1 \\ 0 & 0 & 1 & -2 & -4 & 1 \end{array} \right] R_1 \rightarrow R_1 + R_2$$

$$\therefore A^{-1} = \begin{bmatrix} -2 & -3 & 1 \\ -3 & -3 & 1 \\ -2 & -4 & 1 \end{bmatrix}$$

(iii) Let $A = \begin{bmatrix} 1 & 2 & 3 \\ 2 & 5 & 3 \\ 1 & 0 & 8 \end{bmatrix}$ applying

5M

Gauss-Jordan method, we get

$$[A|I] = \left[\begin{array}{ccc|ccc} 1 & 2 & 3 & 1 & 0 & 0 \\ 2 & 5 & 3 & 0 & 1 & 0 \\ 1 & 0 & 8 & 0 & 0 & 1 \end{array} \right]$$

$$\sim \left[\begin{array}{ccc|ccc} 1 & 2 & 3 & 1 & 0 & 0 \\ 0 & 1 & -3 & -2 & 1 & 0 \\ 0 & -2 & 5 & -1 & 0 & 1 \end{array} \right] \begin{array}{l} R_2 \rightarrow R_2 - 2R_1 \\ R_3 \rightarrow R_3 - R_1 \end{array}$$

$$\sim \left[\begin{array}{ccc|ccc} 1 & 2 & 3 & 1 & 0 & 0 \\ 0 & 1 & -3 & -2 & 1 & 0 \\ 0 & 0 & -1 & -5 & 2 & 1 \end{array} \right] R_3 \rightarrow R_3 + 2R_2$$

$$\sim \left[\begin{array}{ccc|ccc} 1 & 2 & 3 & 1 & 0 & 0 \\ 0 & 1 & -3 & -2 & 1 & 0 \\ 0 & 0 & 1 & 5 & -2 & -1 \end{array} \right] R_3 \rightarrow (-1)R_3$$

$$\sim \left[\begin{array}{ccc|ccc} 1 & 2 & 0 & -14 & 6 & 3 \\ 0 & 1 & 0 & 13 & -5 & -3 \\ 0 & 0 & 1 & 5 & -2 & -1 \end{array} \right] \begin{array}{l} R_1 \rightarrow R_1 - 3R_3 \\ R_2 \rightarrow R_2 + 3R_3 \end{array}$$

$$\sim \left[\begin{array}{ccc|ccc} 1 & 0 & 0 & -40 & 16 & 9 \\ 0 & 1 & 0 & 13 & -5 & -3 \\ 0 & 0 & 1 & 5 & -2 & -1 \end{array} \right] R_1 \rightarrow R_1 - 2R_2$$

$$\therefore A^{-1} = \begin{bmatrix} -40 & 16 & 9 \\ 13 & -5 & -3 \\ 5 & -2 & -1 \end{bmatrix}$$

EXERCISE 1.3

Concept Corner

➤ A system of m linear equations in n unknowns is of the following form:

$$a_{11}x_1 + a_{12}x_2 + \cdots + a_{1n}x_n = b_1,$$

$$a_{21}x_1 + a_{22}x_2 + \cdots + a_{2n}x_n = b_2,$$

.....

$$a_{m1}x_1 + a_{m2}x_2 + \cdots + a_{mn}x_n = b_m,$$

where the coefficients $a_{ij}, i = 1, 2, \dots, m; j = 1, 2, \dots, n$ and

$b_k, k = 1, 2, \dots, m$ are constants.

If all the b'_k 's are zeros, then the above system is called a **homogeneous system** of linear equations.

On the other hand, if at least one of the b'_k 's is non-zero, then the above system is called a **non-homogeneous system** of linear equations.

A system of linear equations having at least one solution is said to be **consistent**.

A system of linear equations having no solution is said to be **inconsistent**.



Video class

➤ **Matrix Inversion Method**

This method can be applied only when the coefficient matrix is a square matrix & non-singular.

Consider the matrix equation $AX = B$ (1)

Where A is a square matrix and non-singular.

Since A is non-singular, A^{-1} exists and $A^{-1}A = AA^{-1} = I$.

Pre-multiplying both sides of (1) by A^{-1} , we get

$$A^{-1}(AX) = A^{-1}B.$$

$$\text{That is, } (A^{-1}A)X = A^{-1}B.$$

$$\text{Hence, we get } X = A^{-1}B.$$

1. Solve the following system of linear equations by matrix inversion method.

(i) $2x + 5y = -2, x + 2y = -3$

(ii) $2x - y = 8, 3x + 2y = -2$

(iii) $2x + 3y - z = 9, x + y + z = 9, 3x - y - z = -1$

(iv) $x + y + z - 2 = 0, 6x - 4y + 5z - 31 = 0, 5x + 2y + 2z = 13$

(i) $2x + 5y = -2, x + 2y = -3$

3M

The matrix form of the system is

$$AX = B \quad \text{APR-25,JUL-23}$$

$$X = A^{-1}B$$

$$A = \begin{bmatrix} 2 & 5 \\ 1 & 2 \end{bmatrix}, \quad X = \begin{bmatrix} x \\ y \end{bmatrix}, \quad B = \begin{bmatrix} -2 \\ -3 \end{bmatrix}$$

$$|A| = 4 - 5 = -1 \neq 0$$

$$\therefore A^{-1} \text{ exists}$$

$$\text{adj}A = \begin{bmatrix} 2 & -5 \\ -1 & 2 \end{bmatrix}$$

$$A^{-1} = \frac{1}{-1} \begin{bmatrix} 2 & -5 \\ -1 & 2 \end{bmatrix}$$

$$= \begin{bmatrix} -2 & 5 \\ 1 & -2 \end{bmatrix}$$

$$AX = B \Rightarrow X = A^{-1}B$$

$$A^{-1} = \frac{1}{|A|} \text{adj}A$$

$$\begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} -2 & 5 \\ 1 & -2 \end{bmatrix} \begin{bmatrix} -2 \\ -3 \end{bmatrix}$$

$$= \begin{bmatrix} 4 - 15 \\ -2 + 6 \end{bmatrix}$$

$$\begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} -11 \\ 4 \end{bmatrix}$$

The solution is $(x = -11, y = 4)$

(ii) $2x - y = 8, 3x + 2y = -2$

3M

The matrix form of the system is

$$AX = B$$

PTA-3

$$X = A^{-1}B$$

$$A = \begin{bmatrix} 2 & -1 \\ 3 & 2 \end{bmatrix}, \quad X = \begin{bmatrix} x \\ y \end{bmatrix}, \quad B = \begin{bmatrix} 8 \\ -2 \end{bmatrix}$$

$$|A| = 4 + 3 = 7 \neq 0$$

$$\therefore A^{-1} \text{ exists}$$

$$\text{adj}A = \begin{bmatrix} 2 & 1 \\ -3 & 2 \end{bmatrix}$$

$$= \frac{1}{7} \begin{bmatrix} 2 & 1 \\ -3 & 2 \end{bmatrix}$$

$$AX = B \Rightarrow X = A^{-1}B$$

$$A^{-1} = \frac{1}{|A|} \text{adj}A$$

$$= \frac{1}{7} \begin{bmatrix} 2 & 1 \\ -3 & 2 \end{bmatrix} \begin{bmatrix} 8 \\ -2 \end{bmatrix}$$

$$\begin{bmatrix} x \\ y \end{bmatrix} = \frac{1}{7} \begin{bmatrix} 16 - 2 \\ -24 - 4 \end{bmatrix}$$

$$= \frac{1}{7} \begin{bmatrix} 14 \\ -28 \end{bmatrix}$$

$$\begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} 2 \\ -4 \end{bmatrix}$$

The solution is $(x = 2, y = -4)$

(iii) $2x + 3y - z = 9, x + y + z = 9,$
 $3x - y - z = -1$

JUL-24

The matrix form of the system is

5M

$$AX = B \Rightarrow X = A^{-1}B$$

$$A = \begin{bmatrix} 2 & 3 & -1 \\ 1 & 1 & 1 \\ 3 & -1 & -1 \end{bmatrix}, X = \begin{bmatrix} x \\ y \\ z \end{bmatrix}, B = \begin{bmatrix} 9 \\ 9 \\ -1 \end{bmatrix}$$

$$|A| = 2(-1 + 1) - 3(-1 - 3) - 1(-1 - 3) = 16 \neq 0.$$

 $\therefore A^{-1}$ exists

$$\text{adj}A = \begin{bmatrix} \begin{vmatrix} 1 & 1 \\ -1 & -1 \end{vmatrix} & -\begin{vmatrix} 1 & 1 \\ 3 & -1 \end{vmatrix} & \begin{vmatrix} 1 & 1 \\ 3 & -1 \end{vmatrix} \\ -\begin{vmatrix} 3 & -1 \\ -1 & -1 \end{vmatrix} & \begin{vmatrix} 2 & -1 \\ 3 & -1 \end{vmatrix} & -\begin{vmatrix} 2 & 3 \\ 3 & -1 \end{vmatrix} \\ \begin{vmatrix} 3 & -1 \\ 1 & 1 \end{vmatrix} & -\begin{vmatrix} 2 & -1 \\ 1 & 1 \end{vmatrix} & \begin{vmatrix} 2 & 3 \\ 1 & 1 \end{vmatrix} \end{bmatrix}^T$$

$$= \begin{bmatrix} (-1 + 1) & -(-1 - 3) & (-1 - 3) \\ -(-3 - 1) & (-2 + 3) & -(-2 - 9) \\ (3 + 1) & -(2 + 1) & (2 - 3) \end{bmatrix}^T$$

$$= \begin{bmatrix} 0 & 4 & -4 \\ 4 & 1 & 11 \\ 4 & -3 & -1 \end{bmatrix}^T$$

$$= \begin{bmatrix} 0 & 4 & 4 \\ 4 & 1 & -3 \\ -4 & 11 & -1 \end{bmatrix}$$

$$A^{-1} = \frac{1}{|A|} \text{adj}A$$

$$= \frac{1}{16} \begin{bmatrix} 0 & 4 & 4 \\ 4 & 1 & -3 \\ -4 & 11 & -1 \end{bmatrix}$$

$$AX = B \Rightarrow X = A^{-1}B$$

$$\Rightarrow \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \frac{1}{16} \begin{bmatrix} 0 & 4 & 4 \\ 4 & 1 & -3 \\ -4 & 11 & -1 \end{bmatrix} \begin{bmatrix} 9 \\ 9 \\ -1 \end{bmatrix}$$

$$= \frac{1}{16} \begin{bmatrix} 0 + 36 - 4 \\ 36 + 9 + 3 \\ -36 + 99 + 1 \end{bmatrix}$$

$$\begin{bmatrix} x \\ y \\ z \end{bmatrix} = \frac{1}{16} \begin{bmatrix} 32 \\ 48 \\ 64 \end{bmatrix}$$

$$\begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} 2 \\ 3 \\ 4 \end{bmatrix}$$

 $\therefore$ The solution is $(x = 2, y = 3, z = 4)$

(iv) $x + y + z - 2 = 0, 6x - 4y + 5z - 31 = 0,$

$$5x + 2y + 2z = 13$$

5M

The matrix form of the system

$$\text{is } AX = B \Rightarrow X = A^{-1}B$$

$$A = \begin{bmatrix} 1 & 1 & 1 \\ 6 & -4 & 5 \\ 5 & 2 & 2 \end{bmatrix}, X = \begin{bmatrix} x \\ y \\ z \end{bmatrix}, B = \begin{bmatrix} 2 \\ 31 \\ 13 \end{bmatrix}$$

$$|A| = 1(-8 - 10) - 1(12 - 25) + 1(12 + 20) = 27 \neq 0$$

 $\therefore A^{-1}$ exists.

$$\text{adj}A = \begin{bmatrix} (-8 - 10) & -(12 - 25) & (12 + 20) \\ -(2 - 2) & (2 - 5) & -(2 - 5) \\ (5 + 4) & -(5 - 6) & (-4 - 6) \end{bmatrix}^T$$

$$= \begin{bmatrix} -18 & 13 & 32 \\ 0 & -3 & 3 \\ 9 & 1 & -10 \end{bmatrix}^T$$

$$= \begin{bmatrix} -18 & 0 & 9 \\ 13 & -3 & 1 \\ 32 & 3 & -10 \end{bmatrix}$$

$$A^{-1} = \frac{1}{|A|} \text{adj}A$$

$$= \frac{1}{27} \begin{bmatrix} -18 & 0 & 9 \\ 13 & -3 & 1 \\ 32 & 3 & -10 \end{bmatrix}$$

$$AX = B \Rightarrow X = A^{-1}B$$

$$= \frac{1}{27} \begin{bmatrix} -18 & 0 & 9 \\ 13 & -3 & 1 \\ 32 & 3 & -10 \end{bmatrix} \begin{bmatrix} 2 \\ 31 \\ 13 \end{bmatrix}$$

$$\begin{bmatrix} x \\ y \\ z \end{bmatrix} = \frac{1}{27} \begin{bmatrix} -36 + 0 + 117 \\ 26 - 93 + 13 \\ 64 + 93 - 130 \end{bmatrix}$$

$$= \frac{1}{27} \begin{bmatrix} 81 \\ -54 \\ 27 \end{bmatrix}$$

$$\begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} 3 \\ -2 \\ 1 \end{bmatrix}$$

 $\therefore$ The solution is $(x = 3, y = -2, z = 1)$

2. If $A = \begin{bmatrix} -5 & 1 & 3 \\ 7 & 1 & -5 \\ 1 & -1 & 1 \end{bmatrix}$ and $B = \begin{bmatrix} 1 & 1 & 2 \\ 3 & 2 & 1 \\ 2 & 1 & 3 \end{bmatrix}$, find the products AB and BA and hence solve the system of equations $x + y + 2z = 1$, $3x + 2y + z = 7$, $2x + y + 3z = 2$. 5M

$$\begin{aligned}
 AB &= \begin{bmatrix} -5 & 1 & 3 \\ 7 & 1 & -5 \\ 1 & -1 & 1 \end{bmatrix} \begin{bmatrix} 1 & 1 & 2 \\ 3 & 2 & 1 \\ 2 & 1 & 3 \end{bmatrix} \\
 &= \begin{bmatrix} -5+3+6 & -5+2+3 & -10+1+9 \\ 7+3-10 & 7+2-5 & 14+1-15 \\ 1-3+2 & 1-2+1 & 2-1+3 \end{bmatrix} \\
 &= \begin{bmatrix} 4 & 0 & 0 \\ 0 & 4 & 0 \\ 0 & 0 & 4 \end{bmatrix} \\
 &= 4 \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} \\
 \therefore AB &= 4I_3
 \end{aligned}$$

$$\begin{aligned}
 &[\because AB = BA = I \\
 &B^{-1} = A(\text{or}) A^{-1} = B]
 \end{aligned}$$

$$\begin{aligned}
 BA &= \begin{bmatrix} 1 & 1 & 2 \\ 3 & 2 & 1 \\ 2 & 1 & 3 \end{bmatrix} \begin{bmatrix} -5 & 1 & 3 \\ 7 & 1 & -5 \\ 1 & -1 & 1 \end{bmatrix} \\
 &= \begin{bmatrix} -5+7+2 & 1+1-2 & 3-5+2 \\ -15+14+1 & 3+2-1 & 9-10+1 \\ -10+7+3 & 2+1-3 & 6-5+3 \end{bmatrix} \\
 &= \begin{bmatrix} 4 & 0 & 0 \\ 0 & 4 & 0 \\ 0 & 0 & 4 \end{bmatrix} \\
 &= 4 \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} \\
 \therefore BA &= 4I_3
 \end{aligned}$$

So, we get $AB = BA = 4I_3$

$$\Rightarrow \left(\frac{1}{4}A\right)B = B\left(\frac{1}{4}A\right) = I_3$$

$$B^{-1} = \frac{1}{4}A$$

$$= \frac{1}{4} \begin{bmatrix} -5 & 1 & 3 \\ 7 & 1 & -5 \\ 1 & -1 & 1 \end{bmatrix}$$

Matrix form of the given system of equations: $\begin{bmatrix} 1 & 1 & 2 \\ 3 & 2 & 1 \\ 2 & 1 & 3 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} 1 \\ 7 \\ 2 \end{bmatrix}$

$$BX = C$$

$$B = \begin{bmatrix} 1 & 1 & 2 \\ 3 & 2 & 1 \\ 2 & 1 & 3 \end{bmatrix}, C = \begin{bmatrix} 1 \\ 7 \\ 2 \end{bmatrix}, X = \begin{bmatrix} x \\ y \\ z \end{bmatrix}$$

$$X = B^{-1}C$$

$$\begin{bmatrix} x \\ y \\ z \end{bmatrix} = \frac{1}{4} \begin{bmatrix} -5 & 1 & 3 \\ 7 & 1 & -5 \\ 1 & -1 & 1 \end{bmatrix} \begin{bmatrix} 1 \\ 7 \\ 2 \end{bmatrix}$$

$$= \frac{1}{4} \begin{bmatrix} -5+7+6 \\ 7+7-10 \\ 1-7+2 \end{bmatrix}$$

$$= \frac{1}{4} \begin{bmatrix} 8 \\ 4 \\ -4 \end{bmatrix}$$

$$\begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} 2 \\ 1 \\ -1 \end{bmatrix}$$

$\therefore$ The solution is $(x = 2, y = 1, z = -1)$

3. A man is appointed in a job with a monthly salary of certain amount & a fixed amount of annual increment. If his salary was ₹ 19,800 per month at the end of the first month after 3 years of service and ₹ 23,400 per month at the end of the first month after 9 years of service, find his starting salary and his annual increment. (Use matrix inversion method to solve the problem.) 3M

Let the monthly salary = ₹ x

Fixed Annual increment = ₹ y

$$\text{Given: } x + 3y = 19800$$

$$x + 9y = 23400$$

$$\begin{bmatrix} 1 & 3 \\ 1 & 9 \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} 19800 \\ 23400 \end{bmatrix}$$

where $A = \begin{bmatrix} 1 & 3 \\ 1 & 9 \end{bmatrix}$, $X = \begin{bmatrix} x \\ y \end{bmatrix}$ and $B = \begin{bmatrix} 19800 \\ 23400 \end{bmatrix}$

$$AX = B \Rightarrow X = A^{-1}B$$

$$|A| = \begin{vmatrix} 1 & 3 \\ 1 & 9 \end{vmatrix} = 9 - 3 = 6 \neq 0$$

$\therefore A^{-1}$ exists

$$\text{adj}A = \begin{bmatrix} 9 & -3 \\ -1 & 1 \end{bmatrix}$$

$$A^{-1} = \frac{1}{|A|} \text{adj}A$$

$$= \frac{1}{6} \begin{bmatrix} 9 & -3 \\ -1 & 1 \end{bmatrix}$$

$$AX = B$$

$$X = A^{-1}B$$

$$= \frac{1}{6} \begin{bmatrix} 9 & -3 \\ -1 & 1 \end{bmatrix} \begin{bmatrix} 19800 \\ 23400 \end{bmatrix}$$

$$X = \frac{1}{6} \begin{bmatrix} 178200 - 70200 \\ -19800 + 23400 \end{bmatrix}$$

$$\begin{bmatrix} x \\ y \end{bmatrix} = \frac{1}{6} \begin{bmatrix} 108000 \\ 3600 \end{bmatrix}$$

$$= \begin{bmatrix} 18000 \\ 600 \end{bmatrix}$$

$$\begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} 18000 \\ 600 \end{bmatrix}$$

The solution is $x = 18000$, $y = 600$

$\therefore$ Monthly salary = ₹ 18,000

Fixed Annual increment = ₹ 600

4. Four men and 4 women can finish a piece of work jointly in 3 days while 2 men and 5 women can finish the same work jointly in 4 days. Find the time taken by one man alone and that of one woman alone to finish the same work by using matrix inversion method.

If a man takes x days to finish a job, then in one day he will complete $\frac{1}{x}$ of the job. 3M

If a man takes y days to finish a job, then in one day he will complete $\frac{1}{y}$ of the job.

$$\text{Given: } \frac{4}{x} + \frac{4}{y} = \frac{1}{3}$$

$$\frac{2}{x} + \frac{5}{y} = \frac{1}{4}$$

The matrix form is $AX = B$ where

$$A = \begin{bmatrix} 4 & 4 \\ 2 & 5 \end{bmatrix}, X = \begin{bmatrix} \frac{1}{x} \\ \frac{1}{y} \end{bmatrix}, B = \begin{bmatrix} \frac{1}{3} \\ \frac{1}{4} \end{bmatrix}$$

$$|A| = 20 - 8 = 12 \neq 0$$

$\therefore A^{-1}$ exists

$$A^{-1} = \frac{1}{|A|} \text{adj}A$$

$$= \frac{1}{12} \begin{bmatrix} 5 & -4 \\ -2 & 4 \end{bmatrix}$$

$$AX = B \Rightarrow X = A^{-1}B$$

$$X = \frac{1}{12} \begin{bmatrix} 5 & -4 \\ -2 & 4 \end{bmatrix} \begin{bmatrix} \frac{1}{3} \\ \frac{1}{4} \end{bmatrix} = \begin{bmatrix} \frac{5}{3} - \frac{4}{4} \\ -\frac{2}{3} + \frac{4}{4} \end{bmatrix}$$

$$X = \frac{1}{12} \begin{bmatrix} \frac{5}{3} - 1 \\ -\frac{2}{3} + 1 \end{bmatrix}$$

$$\begin{bmatrix} \frac{1}{x} \\ \frac{1}{y} \end{bmatrix} = \frac{1}{12} \begin{bmatrix} \frac{2}{3} \\ \frac{1}{3} \end{bmatrix} = \begin{bmatrix} \frac{1}{12} \times \frac{2}{3} \\ \frac{1}{12} \times \frac{1}{3} \end{bmatrix}$$

$$\begin{bmatrix} \frac{1}{x} \\ \frac{1}{y} \end{bmatrix} = \begin{bmatrix} \frac{1}{18} \\ \frac{1}{36} \end{bmatrix}$$

$$\frac{1}{x} = \frac{1}{18} \Rightarrow x = 18$$

$$\frac{1}{y} = \frac{1}{36} \Rightarrow y = 36$$

$\therefore$ Number of days taken by a man to complete the work = **18 days**

Number of days taken by a woman to complete the work = **36 days**

5. The prices of three commodities A, B and C are ₹ x, y and z per units respectively. A person P purchases 4 units of B and sells two units of A and 5 units of C . Person Q purchases 2 units of C and sells 3 units of A and one unit of B . Person R purchases one unit of A and sells 3 units of B and one unit of C . In the process, P, Q and R earn ₹ 15,000, ₹ 1,000 and ₹ 4,000 respectively. Find the prices per unit of A, B and C . (Use matrix inversion method to solve the problem.)

Let the price per unit of

Commodity $A = ₹ x$, Commodity $B = ₹ y$, Commodity $C = ₹ z$

Given: $2x - 4y + 5z = 15000$

$3x + y - 2z = 1000$

$-x + 3y + z = 4000$

The matrix form is $AX = B$ where

$$A = \begin{bmatrix} 2 & -4 & 5 \\ 3 & 1 & -2 \\ -1 & 3 & 1 \end{bmatrix}, X = \begin{bmatrix} x \\ y \\ z \end{bmatrix}, B = \begin{bmatrix} 15000 \\ 1000 \\ 4000 \end{bmatrix}$$

$$|A| = 2(1 + 6) + 4(3 - 2) + 5(9 + 1) = 68 \neq 0$$

$\therefore A^{-1}$ exists

$$\text{adj}A = \begin{bmatrix} (1 + 6) & -(3 - 2) & (9 + 1) \\ -(-4 - 15) & (2 + 5) & -(6 - 4) \\ (8 - 5) & -(-4 - 15) & (2 + 12) \end{bmatrix}^T$$

$$\text{adj}A = \begin{bmatrix} 7 & -1 & 10 \\ 19 & 7 & -2 \\ 3 & 19 & 14 \end{bmatrix}^T = \begin{bmatrix} 7 & 19 & 3 \\ -1 & 7 & 19 \\ 10 & -2 & 14 \end{bmatrix}$$

$$A^{-1} = \frac{1}{|A|} \text{adj}A = \frac{1}{68} \begin{bmatrix} 7 & 19 & 3 \\ -1 & 7 & 19 \\ 10 & -2 & 14 \end{bmatrix}$$

$$AX = B \Rightarrow X = A^{-1}B$$

$$X = \frac{1}{68} \begin{bmatrix} 7 & 19 & 3 \\ -1 & 7 & 19 \\ 10 & -2 & 14 \end{bmatrix} \begin{bmatrix} 15000 \\ 1000 \\ 4000 \end{bmatrix}$$

$$= \frac{1}{68} \begin{bmatrix} 105000 + 19000 + 12000 \\ -15000 + 7000 + 76000 \\ 150000 - 2000 + 56000 \end{bmatrix}$$

$$= \frac{1}{68} \begin{bmatrix} 136000 \\ 68000 \\ 204000 \end{bmatrix}$$

$$\begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} 2000 \\ 1000 \\ 3000 \end{bmatrix}$$

$\therefore$ The solution is $x = 2000, y = 1000, z = 3000$

Price per unit of $A = ₹ 2000$ & Price per unit of $B = ₹ 1000$ & Price per unit of $C = ₹ 3000$

5M

Note: 1. The amount spent on purchasing the commodity is negative
2. The amount earned by selling the commodity is positive.

EXERCISE 1.4



Video class

Concept Corner

Cramer's rule: When $\Delta \neq 0$, $x_1 = \frac{\Delta_1}{\Delta}$, $x_2 = \frac{\Delta_2}{\Delta}$, $x_3 = \frac{\Delta_3}{\Delta}$

Where $\Delta = \begin{vmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{vmatrix}$, $\Delta_1 = \begin{vmatrix} b_1 & a_{12} & a_{13} \\ b_2 & a_{22} & a_{23} \\ b_3 & a_{32} & a_{33} \end{vmatrix}$, $\Delta_2 = \begin{vmatrix} a_{11} & b_1 & a_{13} \\ a_{21} & b_2 & a_{23} \\ a_{31} & b_3 & a_{33} \end{vmatrix}$, $\Delta_3 = \begin{vmatrix} a_{11} & a_{12} & b_1 \\ a_{21} & a_{22} & b_2 \\ a_{31} & a_{32} & b_3 \end{vmatrix}$

1. Solve the following systems of linear equations by Cramer's rule:

(i) $5x - 2y + 16 = 0, x + 3y - 7 = 0$

(ii) $\frac{3}{x} + 2y = 12, \frac{2}{x} + 3y = 13$

(iii) $3x + 3y - z = 11, 2x - y + 2z = 9, 4x + 3y + 2z = 25$

(iv) $\frac{3}{x} - \frac{4}{y} - \frac{2}{z} - 1 = 0, \frac{1}{x} + \frac{2}{y} + \frac{1}{z} - 2 = 0, \frac{2}{x} - \frac{5}{y} - \frac{4}{z} + 1 = 0$

(i) $5x - 2y + 16 = 0, x + 3y - 7 = 0$

$$5x - 2y = -16, x + 3y = 7$$

$$\Delta = \begin{vmatrix} 5 & -2 \\ 1 & 3 \end{vmatrix} = 15 + 2 = 17 \quad (3M)$$

$$\Delta_1 = \begin{vmatrix} -16 & -2 \\ 7 & 3 \end{vmatrix} = -48 + 14 = -34$$

$$\Delta_2 = \begin{vmatrix} 5 & -16 \\ 1 & 7 \end{vmatrix} = 35 + 16 = 51$$

By Cramer's rule, we get

$$x = \frac{\Delta_1}{\Delta} = -\frac{34}{17} = -2$$

$$y = \frac{\Delta_2}{\Delta} = \frac{51}{17} = 3$$

$\therefore$ The solution is $(x = -2, y = 3)$.

(ii) $\frac{3}{x} + 2y = 12, \frac{2}{x} + 3y = 13 \quad (3M)$

$$\Delta = \begin{vmatrix} 3 & 2 \\ 2 & 3 \end{vmatrix} = 9 - 4 = 5$$

$$\Delta_1 = \begin{vmatrix} 12 & 2 \\ 13 & 3 \end{vmatrix} = 36 - 26 = 10$$

$$\Delta_2 = \begin{vmatrix} 3 & 12 \\ 2 & 13 \end{vmatrix} = 39 - 24 = 15$$

$$\frac{1}{x} = \frac{\Delta_1}{\Delta} = \frac{10}{5} = 2$$

By Cramer's Rule, we get

$$x = \frac{1}{2}$$

$$y = \frac{\Delta_2}{\Delta} = \frac{15}{5} = 3$$

$\therefore$ The solution is $(x = \frac{1}{2}, y = 3)$

(iii) $3x + 3y - z = 11, 2x - y + 2z = 9,$

$$4x + 3y + 2z = 25$$

$$\Delta = \begin{vmatrix} 3 & 3 & -1 \\ 2 & -1 & 2 \\ 4 & 3 & 2 \end{vmatrix}$$

$$= 3[-2 - 6] - 3[4 - 8] - 1[6 + 4]$$

$$= -24 + 12 - 10 = -22$$

$$\Delta_1 = \begin{vmatrix} 11 & 3 & -1 \\ 9 & -1 & 2 \\ 25 & 3 & 2 \end{vmatrix}$$

$$= 11(-2 - 6) - 3(18 - 50) - 1(27 + 25)$$

$$= 11(-8) - 3(-32) - 1(52)$$

$$= -88 + 96 - 52 = -44$$

$$\Delta_2 = \begin{vmatrix} 3 & 11 & -1 \\ 2 & 9 & 2 \\ 4 & 25 & 2 \end{vmatrix}$$

$$= 3(18 - 50) - 11(4 - 8) - 1(50 - 36)$$

$$= 3(-32) - 11(-4) - 1(14)$$

$$= -96 + 44 - 14 = -66$$

$$\Delta_3 = \begin{vmatrix} 3 & 3 & 11 \\ 2 & -1 & 9 \\ 4 & 3 & 25 \end{vmatrix}$$

$$= 3(-25 - 27) - 3(50 - 36) + 11(6 + 4)$$

$$= 3(-52) - 3(14) + 11(10)$$

$$= -156 - 42 + 110 = -88$$

$$x = \frac{\Delta_1}{\Delta} = \frac{-44}{-22} = 2 ; y = \frac{\Delta_2}{\Delta} = \frac{-66}{-22} = 3 ; z = \frac{\Delta_3}{\Delta} = \frac{-88}{-22} = 4$$

$\therefore$ The solution is $(x = 2, y = 3, z = 4)$

JUL-22

5M

$$(iv) \frac{3}{x} - \frac{4}{y} - \frac{2}{z} - 1 = 0 \Rightarrow \frac{3}{x} - \frac{4}{y} - \frac{2}{z} = 1$$

$$\frac{1}{x} + \frac{2}{y} + \frac{1}{z} - 2 = 0 \Rightarrow \frac{1}{x} + \frac{2}{y} + \frac{1}{z} = 2$$

$$\frac{2}{x} - \frac{5}{y} - \frac{4}{z} - 1 = 0 \Rightarrow \frac{2}{x} - \frac{5}{y} - \frac{4}{z} = 1$$

$$X = \frac{1}{x}; \quad Y = \frac{1}{y}; \quad Z = \frac{1}{z}$$

$$3X - 4Y - 2Z = 1$$

$$X + 2Y + Z = 2$$

$$2X - 5Y - 4Z = -1$$

$$\Delta = \begin{vmatrix} 3 & -4 & -2 \\ 1 & 2 & 1 \\ 2 & -5 & -4 \end{vmatrix}$$

$$= 3(-8 + 5) + 4(-4 - 2) - 2(-5 - 4)$$

$$= 3(-3) + 4(-6) - 2(-9)$$

$$= -9 - 24 + 18 = -15$$

$$\Delta_1 = \begin{vmatrix} 1 & -4 & -2 \\ 2 & 2 & 1 \\ -1 & -5 & -4 \end{vmatrix}$$

$$= 1(-8 + 5) + 4(-8 + 1) - 2(-10 + 2)$$

$$= 1(-3) + 4(-7) - 2(-8)$$

$$= -3 - 28 + 16 = -15$$

$$\Delta_2 = \begin{vmatrix} 3 & 1 & -2 \\ 1 & 2 & 1 \\ 2 & -1 & -4 \end{vmatrix}$$

$$= 3(-8 + 1) - 1(-4 - 2) - 2(-1 - 4)$$

$$= 3(-7) - 1(-6) - 2(-5)$$

$$= -21 + 6 + 10 = -5$$

$$\Delta_2 = -5$$

$$\Delta_3 = \begin{vmatrix} 3 & -4 & 1 \\ 1 & 2 & 2 \\ 2 & -5 & -1 \end{vmatrix}$$

$$= 3(-2 + 10) + 4(-1 - 4) + 1(-5 - 4)$$

$$= 3(8) + 4(-5) + 1(-9)$$

$$= 24 - 20 - 9 = -5$$

$$\Delta_3 = -5$$

$$\frac{1}{x} = \frac{\Delta_1}{\Delta} = \frac{-15}{-15} = 1 \Rightarrow x = 1$$

$$\frac{1}{y} = \frac{\Delta_2}{\Delta} = \frac{-5}{-15} = \frac{1}{3} \Rightarrow y = 3$$

$$\frac{1}{z} = \frac{\Delta_3}{\Delta} = \frac{-5}{-15} = \frac{1}{3} \Rightarrow z = 3$$

$\therefore$ The solution is $(x = 1, y = 3, z = 3)$.

2. In a competitive examination, one mark is awarded for every correct answer while $\frac{1}{4}$ mark is deducted for every wrong answer. A student answered 100 questions and got 80 marks. How many questions did he answer correctly? (Use Cramer's rule to solve the problem).

Let the number of question Answered correctly = x

The number of question Answered wrongly = y

$$\therefore x + y = 100 \quad \dots\dots\dots(1)$$

Marks awarded for one, Correct answer = 1, Wrong answer = $-\frac{1}{4}$

$$\therefore (1 \times x) + \left(-\frac{1}{4} \times y\right) = 80$$

$$x - \frac{1}{4}y = 80$$

Multiplying by 4, we get $4x - y = 320 \quad \dots\dots\dots(2)$

$$\text{From (1) and (2), } \Delta = \begin{vmatrix} 1 & 1 \\ 4 & -1 \end{vmatrix} = -1 - 4 = -5$$

$$\Delta_1 = \begin{vmatrix} 100 & 1 \\ 320 & -1 \end{vmatrix} = -100 - 320 = -420 \Rightarrow \Delta_1 = -420$$

$$\Delta_2 = \begin{vmatrix} 1 & 100 \\ 4 & 320 \end{vmatrix} = 320 - 400 = -80 \Rightarrow \Delta_2 = -80$$

$$x = \frac{\Delta_1}{\Delta} = \frac{-420}{-5} = 84; \quad y = \frac{\Delta_2}{\Delta} = \frac{-80}{-5} = 16$$

$\therefore$ The student answered **84** questions correctly.

3. A chemist has one solution which is 50% acid and another solution which is 25% acid. How much each should be mixed to make 10 litres of a 40% acid solution? (Use Cramer's rule to solve the problem).

3M

Let x and y be the amount of solution containing 50% and 25% acid respectively.

$$x + y = 10 \quad \text{..... (1)}$$

$$50\% \text{ of } x + 25\% \text{ of } y = 40\% \text{ of } (x + y)$$

$$50\% \text{ of } x + 25\% \text{ of } y = 40\% \text{ of } 10$$

$$\frac{50}{100}x + \frac{25}{100}y = \frac{40}{100}(10)$$

$$50x + 25y = 400$$

$$\div 25, \quad 2x + y = 16 \quad \text{..... (2)}$$

$$x + y = 10$$

$$2x + y = 16$$

$$\Delta = \begin{vmatrix} 1 & 1 \\ 2 & 1 \end{vmatrix} = 1 - 2 = -1$$

$$\Delta_x = \begin{vmatrix} 10 & 1 \\ 16 & 1 \end{vmatrix} = 10 - 16 = -6$$

$$\Delta_y = \begin{vmatrix} 1 & 10 \\ 2 & 16 \end{vmatrix} = 16 - 20 = -4$$

$$x = \frac{\Delta_x}{\Delta} = \frac{-6}{-1} = 6$$

$$y = \frac{\Delta_y}{\Delta} = \frac{-4}{-1} = 4$$

$$x = 6 \text{ litres, } y = 4 \text{ litres}$$

6 litres of solution containing 50% acid and 4 litres of solution containing 25% acid must be mixed to make 10 litres 40% acid solution.

4. A fish tank can be filled in 10 minutes using both pumps A and B simultaneously. However, pump B can pump water in or out at the same rate. If pump B is inadvertently run in reverse, then the tank will be filled in 30 minutes. How long would it take each pump to fill the tank by itself? (use Cramer's rule to solve the problem).

3M

Let, The time taken by pump A to fill the tank by itself = x minutes

The time taken by pump B to fill the tank by itself = y minutes

$$\text{So, the part of the tank filled by pump A in 1 minute} = \frac{1}{x}$$

$$\text{The part of the tank filled by pump B in 1 minute} = \frac{1}{y}$$

$$\text{The part of the tank filled by both pumps A \& B in 1 minute} = \frac{1}{10}$$

$$\therefore \frac{1}{x} + \frac{1}{y} = \frac{1}{10}$$

If pump B runs in reverse, then the tank will be filled by both pumps in 30 minutes.

$$\text{In this case, the part of the tank filled by both pumps A \& B in 1 minute} = \frac{1}{30}$$

$$\therefore \frac{1}{x} - \frac{1}{y} = \frac{1}{30}$$

$$\text{Let } a = \frac{1}{x} \text{ and } b = \frac{1}{y}$$

$$a + b = \frac{1}{10}$$

$$10a + 10b = 1$$

$$a - b = \frac{1}{30}$$

$$30a - 30b = 1$$

$$\Delta = \begin{vmatrix} 10 & 10 \\ 30 & -30 \end{vmatrix} = -300 - 300 = -600$$

$$\Delta_1 = \begin{vmatrix} 1 & 10 \\ 1 & -30 \end{vmatrix} = -30 - 10 = -40$$

$$\Delta_2 = \begin{vmatrix} 10 & 1 \\ 30 & 1 \end{vmatrix} = 10 - 30 = -20$$

$$a = \frac{\Delta_1}{\Delta} = \frac{-40}{-600} = \frac{1}{15}$$

$$b = \frac{\Delta_2}{\Delta} = \frac{-20}{-600} = +\frac{1}{30}$$

$$a = \frac{1}{x} = \frac{1}{15} \Rightarrow x = 15$$

$$b = \frac{1}{y} = \frac{1}{30} \Rightarrow y = 30.$$

Pump A will take 15 minutes to fill the tank by itself.

Pump B will take 30 minutes to fill the tank by itself.

5. A family of 3 people went out for dinner in a restaurant. The cost of two dosai, three idlies and two vadais is ₹ 150. The cost of the two dosai, two idlies and four vadais is ₹ 200. The cost of five dosai, four idlies and two vadais is ₹ 250. The family has ₹ 350 in hand and they ate 3 dosai and six idlies and six vadais. Will they be able to manage to pay the bill within the amount they had?

5M

Let, The cost of one dosai = ₹ x

The cost of one idly = ₹ y

The cost of one vadai = ₹ z

According to the given information, we get

$$2x + 3y + 2z = 150$$

$$2x + 2y + 4z = 200$$

$$5x + 4y + 2z = 250$$

$$\begin{aligned}\Delta &= \begin{vmatrix} 2 & 3 & 2 \\ 2 & 2 & 4 \\ 5 & 4 & 2 \end{vmatrix} \\ &= 2(4 - 16) - 3(4 - 20) + 2(8 - 10) \\ &= 2(-12) - 3(-16) + 2(-2) \\ &= -24 + 48 - 4 = 20\end{aligned}$$

$$\Delta = 20$$

$$\begin{aligned}\Delta_1 &= \begin{vmatrix} 150 & 3 & 2 \\ 200 & 2 & 4 \\ 250 & 4 & 2 \end{vmatrix} \\ &= 150(4 - 16) - 3(400 - 1000) \\ &\quad + 2(800 - 500) \\ &= 150(-12) - 3(-600) + 2(300) \\ &= -1800 + 1800 + 600 = 600\end{aligned}$$

$$\therefore \Delta_1 = 600$$

$$\begin{aligned}\Delta_2 &= \begin{vmatrix} 2 & 150 & 2 \\ 2 & 200 & 4 \\ 5 & 250 & 2 \end{vmatrix} \\ &= 2(400 - 1000) - 150(4 - 20) \\ &\quad + 2(500 - 1000) \\ &= 2(-600) - 150(-16) + 2(-500) \\ &= -1200 + 2400 - 1000 = 200\end{aligned}$$

$$\therefore \Delta_2 = 200$$

$$\begin{aligned}\Delta_3 &= \begin{vmatrix} 2 & 3 & 150 \\ 2 & 2 & 200 \\ 5 & 4 & 250 \end{vmatrix} \\ &= 2(500 - 800) - 3(500 - 1000) \\ &\quad + 150(8 - 10) \\ &= 2(-300) - 3(-500) + 150(-2) \\ &= -600 + 1500 - 300 = 600\end{aligned}$$

$$\Delta_3 = 600$$

By Cramer's Rule, we get

$$x = \frac{\Delta_1}{\Delta} = \frac{600}{20} = 30 \Rightarrow \text{The cost of one dosai} = \text{₹ } 30$$

$$y = \frac{\Delta_2}{\Delta} = \frac{200}{20} = 10 \Rightarrow \text{The cost of one idly} = \text{₹ } 10$$

$$z = \frac{\Delta_3}{\Delta} = \frac{600}{20} = 30 \Rightarrow \text{The cost of one vadai} = \text{₹ } 30$$

$$\begin{aligned}\text{The cost of 3 dosai and six idly and six vadai} &= 3(30) + 6(10) + 6(30) \\ &= 90 + 60 + 180 \\ &= \text{₹ } 330 < \text{₹ } 350\end{aligned}$$

Since the family has ₹ 350 in hand, **Yes**, they will be able to manage to pay the bill.

EXERCISE 1.5



1. Solve the following systems of linear equations by Gaussian elimination method.

(i) $2x - 2y + 3z = 2, x + 2y - z = 3,$

5M

$$3x - y + 2z = 1$$

The augmented matrix is

$$[A|B] = \left[\begin{array}{ccc|c} 2 & -2 & 3 & 2 \\ 1 & 2 & -1 & 3 \\ 3 & -1 & 2 & 1 \end{array} \right]$$

$$\sim \left[\begin{array}{ccc|c} 1 & 2 & -1 & 3 \\ 2 & -2 & 3 & 2 \\ 3 & -1 & 2 & 1 \end{array} \right] R_1 \leftrightarrow R_2$$

$$\sim \left[\begin{array}{ccc|c} 1 & 2 & -1 & 3 \\ 0 & -6 & 5 & -4 \\ 0 & -7 & 5 & -8 \end{array} \right] \begin{array}{l} R_2 \rightarrow R_2 - 2R_1 \\ R_3 \rightarrow R_3 - 3R_1 \end{array}$$

$$\sim \left[\begin{array}{ccc|c} 1 & 2 & -1 & 3 \\ 0 & -6 & 5 & -4 \\ 0 & 0 & -5 & -20 \end{array} \right] R_3 \rightarrow 6R_3 - 7R_2$$

Writing the equivalent system of equation from the row-echelon matrix, we get

$$x + 2y - z = 3 \dots\dots\dots(1)$$

$$-6y + 5z = -4 \dots\dots\dots(2)$$

$$-5z = -20 \dots\dots\dots(3)$$

$$(3) \Rightarrow z = \frac{20}{5} = 4 \Rightarrow z = 4$$

Substituting $z = 4$ in (2), we get

$$-6y + 5(4) = -4$$

$$-6y = -4 - 20 = -24$$

$$y = \frac{-24}{-6} = 4$$

$$y = 4$$

Substituting $y = 4$ and $z = 4$ in (1), we get

$$x + 2(4) - 4 = 3$$

$$x + 4 = 3$$

$$x = 3 - 4 = -1$$

$$x = -1$$

$\therefore$ The solution is $(x = -1, y = 4, z = 4)$

(ii) $2x + 4y + 6z = 22,$

$$3x + 8y + 5z = 27,$$

$$-x + y + 2z = 2$$

The augmented matrix is

5M

$$[A|B] = \left[\begin{array}{ccc|c} 2 & 4 & 6 & 22 \\ 3 & 8 & 5 & 27 \\ -1 & 1 & 2 & 2 \end{array} \right]$$

$$\sim \left[\begin{array}{ccc|c} 1 & 2 & 3 & 11 \\ 3 & 8 & 5 & 27 \\ -1 & 1 & 2 & 2 \end{array} \right] R_1 \rightarrow \frac{1}{2}R_1$$

$$\sim \left[\begin{array}{ccc|c} 1 & 2 & 3 & 11 \\ 0 & 2 & -4 & -6 \\ 0 & 3 & 5 & 13 \end{array} \right] \begin{array}{l} R_2 \rightarrow R_2 - 3R_1 \\ R_3 \rightarrow R_3 + R_1 \end{array}$$

$$\sim \left[\begin{array}{ccc|c} 1 & 2 & 3 & 11 \\ 0 & 1 & -2 & -3 \\ 0 & 3 & 5 & 13 \end{array} \right] R_2 \rightarrow \frac{1}{2}R_2$$

$$\sim \left[\begin{array}{ccc|c} 1 & 2 & 3 & 11 \\ 0 & 1 & -2 & -3 \\ 0 & 0 & 11 & 22 \end{array} \right] R_3 \rightarrow R_3 - 3R_2$$

Writing the equivalent system of equations from the row - echelon matrix, we get

$$x + 2y + 3z = 11 \dots\dots\dots(1)$$

$$y - 2z = -3 \dots\dots\dots(2)$$

$$11z = 22 \dots\dots\dots(3)$$

$$(3) \Rightarrow z = \frac{22}{11} = 2$$

$$z = 2$$

Substituting $z = 2$ in (2), we get

$$y - 2(2) = -3$$

$$y - 4 = -3$$

$$y = -3 + 4 = 1$$

$$y = 1$$

Substituting $y = 1$ & $z = 2$ in (1), we get

$$x + 2(1) + 3(2) = 11$$

$$x + 2 + 6 = 11 = 11 - 8$$

$$x = 3$$

$\therefore$ The solution is $(x = 3, y = 1, z = 2)$

2. If $ax^2 + bx + c$ is divided by $x + 3$, $x - 5$, and $x - 1$, the remainders are 21, 61 and 9 respectively. Find a , b and c . (Use Gaussian elimination method.)

PTA-3

5M

$$\text{Let } P(x) = ax^2 + bx + c$$

Given: $P(x) \div (x + 3)$ and leaves the remainder 21

$$\therefore P(-3) = a(-3)^2 + b(-3) + c = 21$$

$$9a - 3b + c = 21$$

Given: $P(x) \div (x - 5)$ and leaves the remainder 61

$$\therefore P(5) = a(5)^2 + b(5) + c = 61$$

$$25a + 5b + c = 61$$

Given: $P(x) \div (x - 1)$ and leave the remainder 9

$$\therefore P(+1) = a(+1)^2 + b(+1) + c = 9$$

$$a + b + c = 9$$

$\therefore$ The system of linear equations: $9a - 3b + c = 21$

$$25a + 5b + c = 61$$

$$a + b + c = 9$$

The augmented matrix is $[A|B] = \left[\begin{array}{ccc|c} 9 & -3 & 1 & 21 \\ 25 & 5 & 1 & 61 \\ 1 & 1 & 1 & 9 \end{array} \right]$

$$\sim \left[\begin{array}{ccc|c} 1 & 1 & 1 & 9 \\ 25 & 5 & 1 & 61 \\ 9 & -3 & 1 & 21 \end{array} \right] R_1 \leftrightarrow R_3$$

$$\sim \left[\begin{array}{ccc|c} 1 & 1 & 1 & 9 \\ 0 & -20 & -24 & -164 \\ 0 & -12 & -8 & -60 \end{array} \right] \begin{array}{l} R_2 \rightarrow R_2 - 25R_1 \\ R_3 \rightarrow R_3 - 9R_1 \end{array}$$

$$\sim \left[\begin{array}{ccc|c} 1 & 1 & 1 & 9 \\ 0 & 5 & 6 & 41 \\ 0 & 3 & 2 & 15 \end{array} \right] \begin{array}{l} R_2 \rightarrow \frac{R_2}{5} \\ R_3 \rightarrow \frac{R_3}{1} \end{array}$$

$$\sim \left[\begin{array}{ccc|c} 1 & 1 & 1 & 9 \\ 0 & 5 & 6 & 41 \\ 0 & 0 & -8 & -48 \end{array} \right] R_3 \rightarrow 5R_3 - 3R_2$$

Writing the equivalent equation from the row - echelon matrix, we get

$$a + b + c = 9 \dots\dots\dots(1)$$

$$5b + 6c = 41 \dots\dots(2)$$

$$-8c = -48 \dots\dots\dots(3)$$

(3) $\Rightarrow$

$$c = \frac{-48}{-8}$$

$$= 6$$

$$c = 6$$

Substituting $c = 6$ in (2), we get

$$5b + 6(6) = 41$$

$$5b + 36 = 41$$

$$5b = 41 - 36 = 5$$

$$b = \frac{5}{5} = 1$$

Substituting $b = 1, c = 6$ in (1), we get

$$a + 1 + 6 = 9$$

$$a = 9 - 7 = 2$$

$$a = 2$$

$\therefore$ The solution is ($a = 2, b = 1, c = 6$)

3. An amount of ₹ 65,000 is invested in three bonds at the rates of 6%, 8% and 9% per annum respectively. The total annual income is ₹ 4,800. The income from the third bond is ₹ 600 more than that from the second bond. Determine the price of each bond. (Use Gaussian elimination method.)

5M

Let the price of 6%, 8%, 9% bond be ₹ x , ₹ y , ₹ z respectively.

From the given data, $x + y + z = 65000$ (1)

$$(6\% x) + (8\% y) + (9\% z) = 4800$$

$$x \times \frac{6}{100} + y \times \frac{8}{100} + z \times \frac{9}{100} = 4800$$

$$6x + 8y + 9z = 480000 \quad \text{..... (2)}$$

$$9\% z = 600 + 8\% y$$

$$\frac{9}{100} z = 600 + y \times \frac{8}{100}$$

$$9z = 60000 + 8y$$

$$-8y + 9z = 60000 \quad \text{..... (3)}$$

From (1), (2) and (3)

The augmented matrix is

$$[A|B] = \left[\begin{array}{ccc|c} 1 & 1 & 1 & 65000 \\ 6 & 8 & 9 & 480000 \\ 0 & -8 & 9 & 60000 \end{array} \right]$$

$$\sim \left[\begin{array}{ccc|c} 1 & 1 & 1 & 65000 \\ 0 & 2 & 3 & 90000 \\ 0 & -8 & 9 & 60000 \end{array} \right] R_2 \rightarrow R_2 - 6R_1$$

$$\sim \left[\begin{array}{ccc|c} 1 & 1 & 1 & 65000 \\ 0 & 2 & 3 & 90000 \\ 0 & 0 & 21 & 42000 \end{array} \right] R_3 \rightarrow R_3 + 4R_2$$

$$x + y + z = 65000 \quad \text{.....(4)}$$

$$2y + 3z = 90000 \quad \text{.....(5)}$$

$$21z = 42000$$

$$z = \frac{42000}{21} = 20,000$$

$$(5) \Rightarrow 2y + 3(20000) = 90000$$

$$2y = 90000 - 60000 = 30000$$

$$y = 15000$$

$$(4) \Rightarrow x + 15000 + 20000 = 65000$$

$$x = 65000 - 35000 = 30,000$$

$\therefore$ The price of 6% bond = ₹ 30,000

The price of 8% bond = ₹ 15,000

The price of 9% bond = ₹ 20,000

4. A boy is walking along the path $y = ax^2 + bx + c$ through the points $(-6, 8)$, $(-2, -12)$ and $(3, 8)$. He wants to meet his friend at $P(7, 60)$. Will he meet his friend? (Use Gaussian elimination method.)

5M

APR-23

The path $y = ax^2 + bx + c$ passes through the points $(-6, 8)$, $(-2, -12)$ and $(3, 8)$

So, we get the system of equations

$$8 = a(-6)^2 + b(-6) + c \Rightarrow 36a - 6b + c = 8$$

$$-12 = a(-2)^2 + b(-2) + c \Rightarrow 4a - 2b + c = -12$$

$$8 = a(3)^2 + b(3) + c \Rightarrow 9a + 3b + c = 8$$

The augmented matrix is

$$[A|B] = \left[\begin{array}{ccc|c} 36 & -6 & 1 & 8 \\ 4 & -2 & 1 & -12 \\ 9 & 3 & 1 & 8 \end{array} \right]$$

$$\sim \left[\begin{array}{ccc|c} 36 & -6 & 1 & 8 \\ 0 & -12 & 8 & -116 \\ 0 & 18 & 3 & 24 \end{array} \right] \begin{array}{l} R_2 \rightarrow 9R_2 - R_1 \\ R_3 \rightarrow 4R_3 - R_1 \end{array}$$

$$\sim \left[\begin{array}{ccc|c} 36 & -6 & 1 & 8 \\ 0 & -3 & 2 & -29 \\ 0 & 6 & 1 & 8 \end{array} \right] \begin{array}{l} R_2 \rightarrow \frac{1}{4}R_2 \\ R_3 \rightarrow \frac{1}{3}R_3 \end{array}$$

$$\sim \left[\begin{array}{ccc|c} 36 & -6 & 1 & 8 \\ 0 & -3 & 2 & -29 \\ 0 & 0 & 5 & -50 \end{array} \right] R_3 \rightarrow R_3 + 2R_2$$

Writing the equivalent equations from the row-echelon matrix, we get

$$36a - 6b + c = 8 \quad \dots\dots\dots (1)$$

$$-3b + 2c = -29 \quad \dots\dots\dots (2)$$

$$5c = -50 \quad \dots\dots\dots (3)$$

$$(3) \Rightarrow c = -\frac{50}{5} = -10 \quad \Rightarrow \mathbf{c = -10}$$

$$(2) \Rightarrow -3b + 2(-10) = -29$$

$$-3b = -29 + 20 = -9$$

$$b = \frac{-9}{-3} = 3 \quad \Rightarrow \mathbf{b = 3}$$

$$(1) \Rightarrow 36a - 6(3) - 10 = 8$$

$$36a - 28 = 8$$

$$36a = 8 + 28 = 36$$

$$a = \frac{36}{36} = 1 \quad \Rightarrow \mathbf{a = 1}$$

$$\therefore \text{Equation of the path is } y = x^2 + 3x - 10$$

Substituting $x = 7$, we get

$$\begin{aligned} y &= 7^2 + 3(7) - 10 \\ &= 49 + 21 - 10 = 60 \end{aligned}$$

Thus, the path passes through the point $P(7, 60)$.

Yes, the boy will meet his friend.

Video class

A system of linear equations, written in the form as $AX = B$, is consistent if and only if the rank of the coefficient matrix is equal to the rank of the augmented matrix; that is, $\rho(A) = \rho([A|B])$.

```

graph TD
    A[ ] --> B["ρ(A) = ρ([A|B]) = number of unknowns"]
    A --> C["ρ(A) = ρ([A|B]) < number of unknowns"]
    A --> D["ρ(A) ≠ ρ([A|B])"]
    B --> E["The system is consistent and has unique solution."]
    C --> F["The system is consistent and has infinite number of solution."]
    D --> G["The system is inconsistent and has no solution."]
    style A width:0px,height:0px
  
```

The diagram illustrates the relationship between the rank of the coefficient matrix $\rho(A)$ and the augmented matrix $\rho([A|B])$ and the number of unknowns. It branches into three cases:

- Case 1:** If $\rho(A) = \rho([A|B]) = \text{number of unknowns}$, the system is consistent and has a unique solution.
- Case 2:** If $\rho(A) = \rho([A|B]) < \text{number of unknowns}$, the system is consistent and has an infinite number of solutions.
- Case 3:** If $\rho(A) \neq \rho([A|B])$, the system is inconsistent and has no solution.

- 5M

$$\text{Also } A = \begin{bmatrix} 1 & -1 & 2 \\ 0 & 3 & 0 \\ 0 & 0 & -7 \end{bmatrix}$$

The solution is $(x = 1, y = 1, z = 1)$

(ii) $3x + y + z = 2, x - 3y + 2z = 1, 7x - y + 4z = 5$

5M

The matrix form of the system is $AX = B$,

where $A = \begin{bmatrix} 3 & 1 & 1 \\ 1 & -3 & 2 \\ 7 & -1 & 4 \end{bmatrix}, X = \begin{bmatrix} x \\ y \\ z \end{bmatrix}, B = \begin{bmatrix} 2 \\ 1 \\ 5 \end{bmatrix}$

The augmented matrix is

$$\begin{aligned} [A, B] &= \left[\begin{array}{ccc|c} 3 & 1 & 1 & 2 \\ 1 & -3 & 2 & 1 \\ 7 & -1 & 4 & 5 \end{array} \right] \\ &\sim \left[\begin{array}{ccc|c} 1 & -3 & 2 & 1 \\ 3 & 1 & 1 & 2 \\ 7 & -1 & 4 & 5 \end{array} \right] R_1 \leftrightarrow R_2 \\ &\sim \left[\begin{array}{ccc|c} 1 & -3 & 2 & 1 \\ 0 & 10 & -5 & -1 \\ 0 & 20 & -10 & -2 \end{array} \right] R_2 \rightarrow R_2 - 3R_1 \\ &\quad R_3 \rightarrow R_3 - 7R_1 \\ &\sim \left[\begin{array}{ccc|c} 1 & -3 & 2 & 1 \\ 0 & 10 & -5 & -1 \\ 0 & 0 & 0 & 0 \end{array} \right] R_3 \rightarrow R_3 - 2R_2 \end{aligned}$$

The last equivalent matrix is in row-echelon form and has two non-zero rows.

$$\therefore \rho[A|B] = 2$$

$$\text{Also } A = \begin{bmatrix} 1 & -3 & 2 \\ 0 & 10 & -5 \\ 0 & 0 & 0 \end{bmatrix}$$

$$\text{It has two non-zero rows} \quad \therefore \rho(A) = 2$$

Since $\rho(A) = \rho[A|B] = 2 < \text{Number of unknowns}$, the given system is consistent and has infinitely many solutions.

The equivalent system of equations:

$$x - 3y + 2z = 1 \quad \dots\dots\dots (1)$$

$$10y - 5z = -1 \quad \dots\dots\dots (2)$$

Let $z = t$, where $t \in R$

$$(2) \Rightarrow 10y - 5t = -1$$

$$\Rightarrow 10y = 5t - 1 \Rightarrow y = \frac{5t-1}{10}$$

$$(1) \Rightarrow x - \frac{3}{10}(5t - 1) + 2t = 1$$

$$x + \frac{-15t+3+20t}{10} = 1$$

$$x + \frac{5t+3}{10} = 1$$

$$x = \frac{10-5t-3}{10} = \frac{7-5t}{10}$$

$$\therefore \text{The solution is } \left(x = \frac{1}{10}(7 - 5t), y = \frac{1}{10}(5t - 1), z = t \right), t \in R$$

(iii) $2x + 2y + z = 5, x - y + z = 1, 3x + y + 2z = 4$

JUL-20

5M

The matrix form of the system is $AX = B$,

where $A = \begin{bmatrix} 2 & 2 & 1 \\ 1 & -1 & 1 \\ 3 & 1 & 2 \end{bmatrix}, X = \begin{bmatrix} x \\ y \\ z \end{bmatrix}, B = \begin{bmatrix} 5 \\ 1 \\ 4 \end{bmatrix}$

The augmented matrix is

$$\begin{aligned}
 [A, B] &= \begin{bmatrix} 2 & 2 & 1 & 5 \\ 1 & -1 & 1 & 1 \\ 3 & 1 & 2 & 4 \end{bmatrix} \\
 &\sim \begin{bmatrix} 1 & -1 & 1 & 1 \\ 2 & 2 & 1 & 5 \\ 3 & 1 & 2 & 4 \end{bmatrix} R_1 \leftrightarrow R_2 \\
 &\sim \begin{bmatrix} 1 & -1 & 1 & 1 \\ 0 & 4 & -1 & 3 \\ 0 & 4 & -1 & 1 \end{bmatrix} R_2 \rightarrow R_2 - 2R_1 \\
 &\quad R_3 \rightarrow R_3 - 3R_1 \\
 &\sim \begin{bmatrix} 1 & -1 & 1 & 1 \\ 0 & 4 & -1 & 3 \\ 0 & 0 & 0 & -2 \end{bmatrix} R_3 \rightarrow R_3 - R_1
 \end{aligned}$$

The last equivalent matrix is in row-echelon form and it has three non-zero rows.

$$\therefore \rho[A|B] = 3$$

Also $A = \begin{bmatrix} 1 & -1 & 1 \\ 0 & 4 & -1 \\ 0 & 0 & 0 \end{bmatrix}$

It is in row - echelon form and it has two non-zero rows

$$\therefore \rho(A) = 2$$

Since $\rho(A) \neq \rho[A|B]$ the given system is inconsistent and has no solutions.

(iv) $2x - y + z = 2, 6x - 3y + 3z = 6, 4x - 2y + 2z = 4$

PTA-5

5M

The matrix form of the system is $AX = B$,

where $A = \begin{bmatrix} 2 & -1 & 1 \\ 6 & -3 & 3 \\ 4 & -2 & 2 \end{bmatrix}, X = \begin{bmatrix} x \\ y \\ z \end{bmatrix}, B = \begin{bmatrix} 2 \\ 6 \\ 4 \end{bmatrix}$

The augmented matrix is

$$\begin{aligned}
 [A|B] &= \begin{bmatrix} 2 & -1 & 1 & 2 \\ 6 & -3 & 3 & 6 \\ 4 & -2 & 2 & 4 \end{bmatrix} \\
 &\sim \begin{bmatrix} 2 & -1 & 1 & 2 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix} R_2 \rightarrow R_2 - 3R_1 \\
 &\quad R_3 \rightarrow R_3 - 2R_1
 \end{aligned}$$

The last equivalent matrix is in row-echelon form and has one non-zero row.

$$\therefore \rho[A|B] = 1$$

Also $A = \begin{bmatrix} 2 & -1 & 1 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}$

A is in row - echelon form and it has one non-zero row.

$$\therefore \rho(A) = 1$$

Since $\rho(A) = \rho[A|B] = 1 < \text{Number of unknowns}$, the given system is consistent and it has infinitely many solutions.

The equivalent system of equations:

$$2x - y + z = 2$$

Let $y = s$ and $z = t$ where $s, t \in R$

$$2x - s + t = 2$$

$$2x = 2 + s - t$$

$$x = \frac{1}{2}(2 + s - t)$$

$$\therefore \text{The solution is } x = \frac{1}{2}(s - t + 2),$$

$$y = s, z = t, s, t \in R$$

2. Find the value of k for which the equations $kx - 2y + z = 1, x - 2ky + z = -2, x - 2y + kz = 1$

have (i) no solution (ii) unique solution (iii) infinitely many solution

5M

$$kx - 2y + z = 1$$

$$x - 2ky + z = -2$$

$$x - 2y + kz = 1$$

The matrix form of the system is $AX = B$, where

$$A = \begin{bmatrix} k & -2 & 1 \\ 1 & -2k & 1 \\ 1 & -2 & k \end{bmatrix}, X = \begin{bmatrix} x \\ y \\ z \end{bmatrix}, B = \begin{bmatrix} 1 \\ -2 \\ 1 \end{bmatrix}$$

The augmented matrix is

$$\begin{aligned} [A|B] &= \left[\begin{array}{ccc|c} k & -2 & 1 & 1 \\ 1 & -2k & 1 & -2 \\ 1 & -2 & k & 1 \end{array} \right] \\ &\sim \left[\begin{array}{ccc|c} 1 & -2 & k & 1 \\ 1 & -2k & 1 & -2 \\ k & -2 & 1 & 1 \end{array} \right] R_1 \leftrightarrow R_3 \\ &\sim \left[\begin{array}{ccc|c} 1 & -2 & k & 1 \\ 0 & 2-2k & 1-k & -3 \\ 0 & 2k-2 & 1-k^2 & 1-k \end{array} \right] R_2 \rightarrow R_2 - R_1 \\ &\quad R_3 \rightarrow R_3 - kR_1 \\ &\sim \left[\begin{array}{ccc|c} 1 & -2 & k & 1 \\ 0 & 2-2k & 1-k & -3 \\ 0 & 0 & 2-k-k^2 & -2-k \end{array} \right] R_3 \rightarrow R_3 + R_2 \\ &\sim \left[\begin{array}{ccc|c} 1 & -2 & k & 1 \\ 0 & 2(1-k) & 1-k & -3 \\ 0 & 0 & (2+k)(1-k) & -(2+k) \end{array} \right] \end{aligned}$$

Case (i) If $k = 1$

then $\rho[A|B] = 3$ and $\rho(A) = 1$

Since $\rho(A) \neq \rho[A|B]$, the given system of equations is inconsistent and has no solution.

Case (ii) If $k \neq -2, k \neq 1$,

then $\rho[A|B] = 3$ and $\rho(A) = 3$

Since $\rho[A] = \rho[A|B] = 3 = \text{number of unknowns}$, the given system is consistent and has a unique solution

Case (iii) If $k = -2$,

then $\rho[A|B] = 2$ and $\rho(A) = 2$

Since $\rho[A|B] = \rho(A) = 2 < \text{number of unknowns}$, the given system of equations is consistent and has infinitely many solution.

3. Investigate the values of λ and μ the system of linear equations $2x + 3y + 5z = 9$, $7x + 3y - 5z = 8$, $2x + 3y + \lambda z = \mu$, have (i) no solution (ii) a unique solution (iii) an infinite number of solutions.

5M

$$2x + 3y + 5z = 9$$

$$7x + 3y - 5z = 8$$

$$2x + 3y + \lambda z = \mu$$

The matrix form of the system is $AX = B$, where

$$A = \begin{bmatrix} 2 & 3 & 5 \\ 7 & 3 & -5 \\ 2 & 3 & \lambda \end{bmatrix}, X = \begin{bmatrix} x \\ y \\ z \end{bmatrix}, B = \begin{bmatrix} 9 \\ 8 \\ \mu \end{bmatrix}$$

The augmented matrix is

$$[A|B] = \left[\begin{array}{ccc|c} 2 & 3 & 5 & 9 \\ 7 & 3 & -5 & 8 \\ 2 & 3 & \lambda & \mu \end{array} \right] \\ \sim \left[\begin{array}{ccc|c} 2 & 3 & 5 & 9 \\ 0 & -15 & -45 & -47 \\ 0 & 0 & \lambda - 5 & \mu - 9 \end{array} \right] \begin{array}{l} R_2 \rightarrow 2R_2 - 7R_1 \\ R_3 \rightarrow R_3 - R_1 \end{array}$$

Case (i)

If $\lambda = 5$ and $\mu \neq 9$, then $\rho(A) = 2$ and $\rho[A|B] = 3$

Since $\rho(A) \neq \rho[A|B]$, the given system of equations is inconsistent and has no solution.

Case (ii)

If $\lambda \neq 5$ and $\mu \in R$, then

Since $\rho(A) = \rho[A|B] = 3 = \text{no. of. unknowns}$, the given system of equations is consistent and has a unique solution.

Case (iii)

If $\lambda = 5$ and $\mu = 9$ then $\rho(A) = 2$ and $\rho[A|B] = 2 < \text{No. of unknowns}$ the given system of equations is consistent and has infinitely many solutions.

EXERCISE 1.7



Video class

Concept Corner

The homogeneous system of linear equation $AX = O$

where $A = [a_{ij}]_{n \times n}$ and $X = [x_{ij}]_{n \times 1}$ and $O = [o]_{n \times 1}$

(i) has the trivial solution if $|A| \neq 0$ (ii) has a non trivial solution if $|A| = 0$

System of Homogeneous Equation

$|A| \neq 0$ and
 $\rho(A) = \rho([A|O]) = \text{number of unknowns}$

The system has trivial solution. $x = y = z = 0$

$|A| = 0$ and
 $\rho(A) = \rho([A|O]) < \text{number of unknowns}$

The system has non-trivial solution

1. Solve the following system of homogenous equations.

(i) $3x + 2y + 7z = 0$, $4x - 3y - 2z = 0$, $5x + 9y + 23z = 0$

5M

The matrix form of the system is

$AX = O$, where

$$A = \begin{bmatrix} 3 & 2 & 7 \\ 4 & -3 & -2 \\ 5 & 9 & 23 \end{bmatrix}, X = \begin{bmatrix} x \\ y \\ z \end{bmatrix}, O = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$$

The augmented matrix is

$$[A|O] = \begin{bmatrix} 3 & 2 & 7 & | & 0 \\ 4 & -3 & -2 & | & 0 \\ 5 & 9 & 23 & | & 0 \end{bmatrix}$$

$$\sim \begin{bmatrix} 3 & 2 & 7 & | & 0 \\ 0 & -17 & -34 & | & 0 \\ 0 & 17 & 34 & | & 0 \end{bmatrix} \begin{matrix} R_3 \rightarrow 3R_2 - 4R_1 \\ R_3 \rightarrow 3R_3 - 5R_1 \end{matrix}$$

$$\sim \begin{bmatrix} 3 & 2 & 7 & | & 0 \\ 0 & -17 & -34 & | & 0 \\ 0 & 0 & 0 & | & 0 \end{bmatrix} R_3 \rightarrow R_3 + R_1$$

$$\sim \begin{bmatrix} 3 & 2 & 7 & | & 0 \\ 0 & 1 & 2 & | & 0 \\ 0 & 0 & 0 & | & 0 \end{bmatrix} R_2 \rightarrow -\frac{1}{17}R_2$$

$$\therefore \rho(A) = \rho([A|O]) = 2 < \text{No. of unknowns.}$$

$\therefore$ The system is consistent and has infinite number of (non-trivial) solution.

The equivalent system of equation is

$$3x + 2y + 7z = 0 \quad \dots\dots\dots (1)$$

$$y + 2z = 0 \quad \dots\dots\dots (2)$$

$$\text{Let } z = t, t \in R$$

$$(2) \Rightarrow y + 2t = 0$$

$$\Rightarrow y = -2t$$

$$(1) \Rightarrow 3x + 2(-2t) + 7(t) = 0$$

$$3x - 4t + 7t = 0$$

$$3x + 3t = 0$$

$$x + t = 0$$

$$x = -t$$

$\therefore$ The solution is $(x = -t, y = -2t, z = t), t \in R$

(ii) $2x + 3y - z = 0$, $x - y - 2z = 0$, $3x + y + 3z = 0$

2M

The matrix form of the system is $AX = O$, where

$$A = \begin{bmatrix} 2 & 3 & -1 \\ 1 & -1 & -2 \\ 3 & 1 & 3 \end{bmatrix}, X = \begin{bmatrix} x \\ y \\ z \end{bmatrix}, O = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$$

The augmented matrix is

$$[A|O] = \begin{bmatrix} 2 & 3 & -1 & | & 0 \\ 1 & -1 & -2 & | & 0 \\ 3 & 1 & 3 & | & 0 \end{bmatrix}$$

$$\sim \begin{bmatrix} 1 & -1 & -2 & | & 0 \\ 2 & 3 & -1 & | & 0 \\ 3 & 1 & 3 & | & 0 \end{bmatrix} R_1 \leftrightarrow R_2$$

$$\sim \begin{bmatrix} 1 & -1 & -2 & | & 0 \\ 0 & 5 & 3 & | & 0 \\ 0 & 4 & 9 & | & 0 \end{bmatrix} \begin{matrix} R_2 \rightarrow R_2 - 2R_1 \\ R_3 \rightarrow R_3 - 3R_1 \end{matrix}$$

$$\sim \begin{bmatrix} 1 & -1 & -2 & | & 0 \\ 0 & 5 & 3 & | & 0 \\ 0 & 0 & 33 & | & 0 \end{bmatrix} R_3 \rightarrow 5R_3 - 4R_2$$

$$\rho(A) = \rho([A|O]) = 3 = \text{number of unknowns.}$$

$\therefore$ The given system is consistent and has a trivial solution.

$\therefore$ The trivial solution is $(x = 0, y = 0, z = 0)$

2. Determine the values of λ for which the following system of equations $x + y + 3z = 0$, $4x + 3y + \lambda z = 0$, $2x + y + 2z = 0$ has (i) a unique solution (ii) a non-trivial solution.

$$x + y + 3z = 0, \quad 4x + 3y + \lambda z = 0, \quad 2x + y + 2z = 0$$

PTA-4

5M

The matrix form of the system is $AX = O$, where $A = \begin{bmatrix} 1 & 1 & 3 \\ 4 & 3 & \lambda \\ 2 & 1 & 2 \end{bmatrix}$, $X = \begin{bmatrix} x \\ y \\ z \end{bmatrix}$, $O = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$

The augmented matrix is $[A|O] = \left[\begin{array}{ccc|c} 1 & 1 & 3 & 0 \\ 4 & 3 & \lambda & 0 \\ 2 & 1 & 2 & 0 \end{array} \right]$

$$\sim \left[\begin{array}{ccc|c} 1 & 1 & 3 & 0 \\ 2 & 1 & 2 & 0 \\ 4 & 3 & \lambda & 0 \end{array} \right] R_2 \leftrightarrow R_3$$

$$\sim \left[\begin{array}{ccc|c} 1 & 1 & 3 & 0 \\ 0 & -1 & -4 & 0 \\ 0 & -1 & \lambda - 12 & 0 \end{array} \right] \begin{array}{l} R_2 \rightarrow R_2 - 2R_1 \\ R_3 \rightarrow R_3 - 4R_1 \end{array}$$

$$\sim \left[\begin{array}{ccc|c} 1 & 1 & 3 & 0 \\ 0 & -1 & -4 & 0 \\ 0 & 0 & \lambda - 8 & 0 \end{array} \right] R_3 \rightarrow R_3 - R_2$$

Case (i) : If $\lambda \neq 8$, then $\rho(A) = \rho([A|O]) = 3 = \text{number of unknowns}$

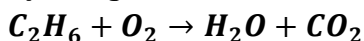
$\therefore$ The given system of equations is consistent and has a unique solution or trivial solution.

Case (ii) : If $\lambda = 8$, then $\rho(A) = \rho([A|O]) = 2 < \text{number of unknowns}$

$\therefore$ The given system is consistent and has a non-trivial solution.

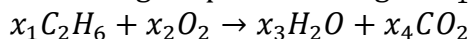
3. By using Gaussian elimination method, balance the chemical reaction equation:

5M



We are searching for positive integers x_1, x_2, x_3 and x_4 such that

MDL



LHS : No. of atoms (Carbon, Hydrogen, Oxygen) = RHS : No. of atoms (Carbon, Hydrogen, Oxygen)

$$2x_1 = x_4 \dots\dots (1)$$

$$6x_1 = 2x_3 \dots\dots (2)$$

$$2x_2 = x_3 + 2x_4 \dots\dots(3)$$

Let $x_1 = t, t \in R$

$$\begin{array}{ccc} (1) \Rightarrow 2x_1 = x_4 & (2) \Rightarrow 6x_1 = 2x_3 & (3) \Rightarrow 2x_2 = x_3 + 2x_4 \\ x_4 = 2t & x_3 = \frac{6t}{2} = 3t & = 3t + 2(2t) = 7t \\ & & x_2 = \frac{7t}{2} \end{array}$$

Since, all x_i are positive integers, let the minimum value of t be taken as 2.

Put $t = 2 \Rightarrow x_1 = 2, \quad x_2 = \frac{7 \times 2}{2} = 7, \quad x_3 = 3(2) = 6, \quad x_4 = 2(2) = 4$

So, the balanced equation is $2C_2H_6 + 7O_2 \rightarrow 6H_2O + 4CO_2$

EXERCISE 1.8



Online Test

Choose the correct answer:

1. If
- $|adj(adjA)| = |A|^9$
- , then the order of the square matrix
- A
- is

(1) 3 (2) 4 (3) 2 (4) 5

$$|adj(adjA)| = |A|^{(n-1)^2}$$

$$|A|^{(n-1)^2} = |A|^9$$

$$(n-1)^2 = 9$$

$$\Rightarrow n-1 = 3$$

$$\therefore n = 4$$

 $\therefore$ order of the square matrix A is 4

2. If
- A
- is a
- 3×3
- non-singular matrix such that
- $AA^T = A^T A$
- and
- $B = A^{-1}A^T$
- , then
- $BB^T =$

(1) A (2) B (3) I_3 (4) B^T

$$BB^T = [A^{-1}A^T][A^{-1}A^T]^T$$

$$= [A^{-1}A^T][(A^T)^T(A^{-1})^T]$$

$$[\because (AB)^T = B^T A^T]$$

$$= (A^{-1}A^T)(A(A^{-1})^T)$$

$$[\because (A^T)^T = A, (A^{-1})^T = (A^T)^{-1}]$$

$$= (A^{-1}A)[A^T(A^T)^{-1}]$$

$$[AA^T = A^T A]$$

$$\therefore BB^T = I_3$$

3. If
- $A = \begin{bmatrix} 3 & 5 \\ 1 & 2 \end{bmatrix}$
- ,
- $B = adjA$
- and
- $C = 3A$
- , then
- $\frac{|adj B|}{|C|} =$

(1) $\frac{1}{3}$ (2) $\frac{1}{9}$ (3) $\frac{1}{4}$ (4) 1

$$\frac{|adj B|}{|C|} = \frac{|adj(adjA)|}{|3A|}$$

$$= \frac{|A|^{(2-1)^2}}{3^2|A|}$$

$$[\because |adj(adjA)| = |A|^{(n-1)^2}, \text{ here } n = 2]$$

$$= \frac{|A|}{9|A|} = \frac{1}{9}$$

$$[|KA| = K^n|A| \text{ Here } n = 2]$$

4. If
- $A \begin{bmatrix} 1 & -2 \\ 1 & 4 \end{bmatrix} = \begin{bmatrix} 6 & 0 \\ 0 & 6 \end{bmatrix}$
- , then
- $A =$

(1) $\begin{bmatrix} 1 & -2 \\ 1 & 4 \end{bmatrix}$ (2) $\begin{bmatrix} 1 & 2 \\ -1 & 4 \end{bmatrix}$ (3) $\begin{bmatrix} 4 & 2 \\ -1 & 1 \end{bmatrix}$ (4) $\begin{bmatrix} 4 & -1 \\ 2 & 1 \end{bmatrix}$

$$A \begin{bmatrix} 1 & -2 \\ 1 & 4 \end{bmatrix} = \begin{bmatrix} 6 & 0 \\ 0 & 6 \end{bmatrix}$$

$$AX = B \Rightarrow A = X^{-1}B$$

$$|X| = \begin{vmatrix} 1 & -2 \\ 1 & 4 \end{vmatrix} = 4 + 2 = 6; adj(x) = \begin{bmatrix} 4 & 2 \\ -1 & 1 \end{bmatrix}$$

$$A = X^{-1}B = \frac{1}{6} \begin{bmatrix} 4 & 2 \\ -1 & 1 \end{bmatrix}$$

$$= \begin{bmatrix} 6 & 0 \\ 0 & 6 \end{bmatrix} \frac{1}{6} \begin{bmatrix} 4 & 2 \\ -1 & 1 \end{bmatrix}$$

$$A = 6 \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} \frac{1}{6} \begin{bmatrix} 4 & 2 \\ -1 & 1 \end{bmatrix} = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} \begin{bmatrix} 4 & 2 \\ -1 & 1 \end{bmatrix} = \begin{bmatrix} 4 & 2 \\ -1 & 1 \end{bmatrix}$$

5. If $A = \begin{bmatrix} 7 & 3 \\ 4 & 2 \end{bmatrix}$, then $9I_2 - A =$

- (1) A^{-1} (2) $\frac{A^{-1}}{2}$ (3) $3A^{-1}$ (4) $2A^{-1}$

$$A = \begin{bmatrix} 7 & 3 \\ 4 & 2 \end{bmatrix}, \quad |A| = 14 - 12 = 2, \quad \text{adj } A = \begin{bmatrix} 2 & -3 \\ -4 & 7 \end{bmatrix}$$

$$A^{-1} = \frac{1}{|A|} \text{adj } A = \frac{1}{2} \begin{bmatrix} 2 & -3 \\ -4 & 7 \end{bmatrix}$$

$$\begin{aligned} 9I - A &= 9 \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} - \begin{bmatrix} 7 & 3 \\ 4 & 2 \end{bmatrix} \\ &= \begin{bmatrix} 9-7 & 0-3 \\ 0-4 & 9-2 \end{bmatrix} \\ &= \begin{bmatrix} 2 & -3 \\ -4 & 7 \end{bmatrix} \end{aligned}$$

$$9I - A = 2A^{-1}$$

6. If $A = \begin{bmatrix} 2 & 0 \\ 1 & 5 \end{bmatrix}$ and $B = \begin{bmatrix} 1 & 4 \\ 2 & 0 \end{bmatrix}$ then $|\text{adj}(AB)| =$

- (1) -40 (2) -80 (3) -60 (4) -20

$$|A| = 10 - 0 = 10; \quad |B| = 0 - 8 = -8$$

$$|\text{adj}(AB)| = |(\text{adj } B)(\text{adj } A)| \quad [\because \text{adj}(AB) = (\text{adj } B)(\text{adj } A)]$$

$$= |\text{adj } B| |\text{adj } A| \quad [\because |AB| = |A||B|]$$

$$= |B|^{2-1} |A|^{2-1} \quad [|\text{adj } A| = |A|^{n-1}]$$

$$= |B||A|$$

$$|\text{adj}(AB)| = (-8)(10) = -80$$

7. If $P = \begin{bmatrix} 1 & x & 0 \\ 1 & 3 & 0 \\ 2 & 4 & -2 \end{bmatrix}$ is the adjoint of 3×3 matrix A and $|A| = 4$, then x is

- (1) 15 (2) 12 (3) 14 (4) 11

$$P = \text{adj } A = \begin{bmatrix} 1 & x & 0 \\ 1 & 3 & 0 \\ 2 & 4 & -2 \end{bmatrix}$$

$$\text{We know that } |\text{adj } A| = |A|^{n-1} \quad (\text{Here } n = 3)$$

$$\Rightarrow 1(-6-0) - x(-2-0) + 0 = (4)^{3-1}$$

$$-6 + 2x = 16$$

$$2x = 16 + 6 = 22$$

$$x = \frac{22}{2} = 11$$

8. If $A = \begin{bmatrix} 3 & 1 & -1 \\ 2 & -2 & 0 \\ 1 & 2 & -1 \end{bmatrix}$ and $A^{-1} = \begin{bmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{bmatrix}$ then the value of a_{23} is

PTA-5

- (1) 0 (2) -2 (3) -3 (4) -1

$$|A| = 3(2-0) - 1(-2-0) - 1(4+2) = 6 + 2 - 6 = 2$$

$$\text{Co-factor of 2 in } A \text{ is } = - \begin{vmatrix} 3 & -1 \\ 2 & 0 \end{vmatrix} = -(0+2) = -2$$

$$\begin{aligned} \text{The value of } a_{23} \text{ in } A^{-1} &= \frac{1}{|A|} (\text{co-factor of 2 in } a_{32} \text{ in } A) \\ &= \frac{1}{2} (-2) = -1 \end{aligned}$$

9. If A, B and C are invertible matrices of some order, then which one of the following is not true?

(1) $\text{adj}A = |A|A^{-1}$

(2) $\text{adj}(AB) = (\text{adj}A)(\text{adj}B)$

(3) $\det A^{-1} = (\det A)^{-1}$

(4) $(ABC)^{-1} = C^{-1}B^{-1}A^{-1}$

APR-24, JUL-21

We know that $\text{adj}(AB) = (\text{adj}B)(\text{adj}A)$

Thus option (2) is not true.

10. If $(AB)^{-1} = \begin{bmatrix} 12 & -17 \\ -19 & 27 \end{bmatrix}$ and $A^{-1} = \begin{bmatrix} 1 & -1 \\ -2 & 3 \end{bmatrix}$, then $B^{-1} =$

APR-20

(1) $\begin{bmatrix} 2 & -5 \\ -3 & 8 \end{bmatrix}$

(2) $\begin{bmatrix} 8 & 5 \\ 3 & 2 \end{bmatrix}$

(3) $\begin{bmatrix} 3 & 1 \\ 2 & 1 \end{bmatrix}$

(4) $\begin{bmatrix} 8 & -5 \\ -3 & 2 \end{bmatrix}$

$$(AB)^{-1} = B^{-1}A^{-1} = \begin{bmatrix} 12 & -17 \\ -19 & 27 \end{bmatrix}$$

$$B^{-1}(A^{-1}A) = \begin{bmatrix} 12 & -17 \\ -19 & 27 \end{bmatrix} A$$

$$B^{-1} = \begin{bmatrix} 12 & -17 \\ -19 & 27 \end{bmatrix} (A^{-1})^{-1}$$

$$\left[(A^{-1})^{-1} = \frac{1}{|A^{-1}|} \text{adj}(A^{-1}) \right]$$

$$\begin{aligned} B^{-1} &= \begin{bmatrix} 12 & -17 \\ -19 & 27 \end{bmatrix} \frac{1}{1} \begin{bmatrix} 3 & 1 \\ 2 & 1 \end{bmatrix} \\ &= \begin{bmatrix} 36 & -34 & 12 & -17 \\ -57 & 54 & -19 & 27 \end{bmatrix} \\ &= \begin{bmatrix} 2 & -5 \\ -3 & 8 \end{bmatrix} \end{aligned}$$

11. If $A^T A^{-1}$ is symmetric, then $A^2 =$

JUL-20, 23, 24

(1) A^{-1}

(2) $(A^T)^2$

(3) A^T

(4) $(A^{-1})^2$

Given that $A^T A^{-1}$ is symmetric

$$\therefore (A^T A^{-1})^T = A^T A^{-1}$$

$$(A^{-1})^T (A^T)^T = A^T A^{-1}$$

$$(A^{-1})^T (A) = A^T A^{-1}$$

$$(A^T)^{-1} (A) (A) = A^T (A^{-1} A)$$

$$(A^T)^{-1} A^2 = A^T (I) = A^T$$

$$(A^T) (A^T)^{-1} A^2 = (A^T) (A^T)$$

$$I A^2 = (A^T)^2$$

$$\therefore A^2 = (A^T)^2$$

12. If A is a non-singular matrix such that $A^{-1} = \begin{bmatrix} 5 & 3 \\ -2 & -1 \end{bmatrix}$, then $(A^T)^{-1} =$

(1) $\begin{bmatrix} -5 & 3 \\ 2 & 1 \end{bmatrix}$

(2) $\begin{bmatrix} 5 & 3 \\ -2 & -1 \end{bmatrix}$

(3) $\begin{bmatrix} -1 & -3 \\ 2 & 5 \end{bmatrix}$

(4) $\begin{bmatrix} 5 & -2 \\ 3 & -1 \end{bmatrix}$

$$A^{-1} = \begin{bmatrix} 5 & 3 \\ -2 & -1 \end{bmatrix}$$

$$(A^T)^{-1} = (A^{-1})^T$$

$$= \left(\begin{bmatrix} 5 & 3 \\ -2 & -1 \end{bmatrix} \right)^T = \begin{bmatrix} 5 & -2 \\ 3 & -1 \end{bmatrix}$$

APR-24, PTA-1, JUL-22

JUL-22

13. If $A = \begin{bmatrix} \frac{3}{5} & \frac{4}{5} \\ x & \frac{3}{5} \end{bmatrix}$ and $A^T = A^{-1}$, then the value of x is

- (1) $-\frac{4}{5}$ (2) $-\frac{3}{5}$ (3) $\frac{3}{5}$ (4) $\frac{4}{5}$

$$AA^T = I$$

$$\frac{1}{5} \begin{bmatrix} 3 & 4 \\ 5x & 3 \end{bmatrix} \frac{1}{5} \begin{bmatrix} 3 & 5x \\ 4 & 3 \end{bmatrix} = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$$

Multiply and equating a_{12}

$$\frac{1}{25} [15x + 12] = 0$$

$$15x = -12$$

$$x = -\frac{4}{5}$$

PTA-3

14. If $A = \begin{bmatrix} 1 & \tan \frac{\theta}{2} \\ -\tan \frac{\theta}{2} & 1 \end{bmatrix}$ and $AB = I_2$, then $B =$

- (1) $\left(\cos^2 \frac{\theta}{2}\right) A$ (2) $\left(\cos^2 \frac{\theta}{2}\right) A^T$ (3) $(\cos^2 \theta) I$ (4) $\left(\sin^2 \frac{\theta}{2}\right) A$

$$AB = I \Rightarrow B = A^{-1} = \frac{1}{|A|} \text{adj } A$$

$$B = \frac{1}{1 + \tan^2 \frac{\theta}{2}} \begin{bmatrix} 1 & -\tan \frac{\theta}{2} \\ \tan \frac{\theta}{2} & 1 \end{bmatrix}$$

$$= \frac{1}{\sec^2 \frac{\theta}{2}} A^T = \left(\cos^2 \frac{\theta}{2}\right) A^T$$

15. If $A = \begin{bmatrix} \cos \theta & \sin \theta \\ -\sin \theta & \cos \theta \end{bmatrix}$ and $A(\text{adj } A) = \begin{bmatrix} k & 0 \\ 0 & k \end{bmatrix}$, then $k =$

- (1) 0 (2) $\sin \theta$ (3) $\cos \theta$ (4) 1

$$A(\text{adj } A) = \begin{bmatrix} k & 0 \\ 0 & k \end{bmatrix} = kI = |A|I$$

$$|A| = k$$

$$\cos^2 \theta + \sin^2 \theta = k$$

$$\therefore k = 1$$

16. If $A = \begin{bmatrix} 2 & 3 \\ 5 & -2 \end{bmatrix}$ be such that $\lambda A^{-1} = A$, then λ is

APR-25, 22

- (1) 17 (2) 14 (3) 19 (4) 21

$$\lambda A^{-1} = A$$

$$\lambda(AA^{-1}) = (A)A$$

$$|\lambda I| = A^2$$

$$\lambda^2 = |A|^2$$

$$= \begin{vmatrix} 2 & 3 \\ 5 & -2 \end{vmatrix}^2 = (-4 - 15)^2 = (-19)^2$$

$$\lambda = 19$$

17. If $\text{adj } A = \begin{bmatrix} 2 & 3 \\ 4 & -1 \end{bmatrix}$ and $\text{adj } B = \begin{bmatrix} 1 & -2 \\ -3 & 1 \end{bmatrix}$ then $\text{adj}(AB)$ is

- (1) $\begin{bmatrix} -7 & -1 \\ 7 & -9 \end{bmatrix}$ (2) $\begin{bmatrix} -6 & 5 \\ -2 & -10 \end{bmatrix}$ (3) $\begin{bmatrix} -7 & 7 \\ -1 & -9 \end{bmatrix}$ (4) $\begin{bmatrix} -6 & -2 \\ 5 & -10 \end{bmatrix}$

$$\begin{aligned} \text{adj}(AB) &= (\text{adj } B)(\text{adj } A) = \begin{bmatrix} 1 & -2 \\ -3 & 1 \end{bmatrix} \begin{bmatrix} 2 & 3 \\ 4 & -1 \end{bmatrix} \\ &= \begin{bmatrix} 2-8 & 3+2 \\ -6+4 & -9-1 \end{bmatrix} = \begin{bmatrix} -6 & 5 \\ -2 & -10 \end{bmatrix} \end{aligned}$$

18. The rank of the matrix $\begin{bmatrix} 1 & 2 & 3 & 4 \\ 2 & 4 & 6 & 8 \\ -1 & -2 & -3 & -4 \end{bmatrix}$ is

JUL-23

- (1) 1 (2) 2 (3) 4 (4) 3

$$\begin{aligned} A &= \begin{bmatrix} 1 & 2 & 3 & 4 \\ 2 & 4 & 6 & 8 \\ -1 & -2 & -3 & -4 \end{bmatrix} \\ &\sim \begin{bmatrix} 1 & 2 & 3 & 4 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix} \begin{matrix} R_2 \rightarrow R_2 - 2R_1 \\ R_3 \rightarrow R_3 + R_1 \end{matrix} \\ \therefore \rho(A) &= 1 \end{aligned}$$

19. If $x^a y^b = e^m$, $x^c y^d = e^n$, $\Delta_1 = \begin{vmatrix} m & b \\ n & d \end{vmatrix}$, $\Delta_2 = \begin{vmatrix} a & m \\ c & n \end{vmatrix}$, $\Delta_3 = \begin{vmatrix} a & b \\ c & d \end{vmatrix}$, then the values of x and y are respectively.

PTA-3, JUL-20

- (1) $e^{\left(\frac{\Delta_2}{\Delta_1}\right)}, e^{\left(\frac{\Delta_3}{\Delta_1}\right)}$ (2) $\log(\Delta_1/\Delta_3), \log(\Delta_2/\Delta_3)$
(3) $\log(\Delta_2/\Delta_1), \log(\Delta_3/\Delta_1)$ (4) $e^{\left(\frac{\Delta_1}{\Delta_3}\right)}, e^{\left(\frac{\Delta_2}{\Delta_3}\right)}$

$x^a y^b = e^m$ $\log(x^a y^b) = \log e^m$ $a \log x + b \log y = m \dots\dots\dots (1)$	$x^c y^d = e^n$ $\log(x^c y^d) = \log e^n$ $c \log x + d \log y = n \dots\dots\dots (2)$
By Crammer's rule, we get $\Delta = \begin{vmatrix} a & b \\ c & d \end{vmatrix}$, $\Delta_1 = \begin{vmatrix} m & b \\ n & d \end{vmatrix}$, $\Delta_2 = \begin{vmatrix} a & m \\ c & n \end{vmatrix}$, $\Delta_3 = \begin{vmatrix} a & b \\ c & d \end{vmatrix}$ $\therefore \Delta = \Delta_3$ So, $\log_e x = \frac{\Delta_1}{\Delta} = \frac{\Delta_1}{\Delta_3} \Rightarrow x = e^{\left(\frac{\Delta_1}{\Delta_3}\right)}$; $\log_e y = \frac{\Delta_2}{\Delta} = \frac{\Delta_2}{\Delta_3} \Rightarrow y = e^{\left(\frac{\Delta_2}{\Delta_3}\right)}$	

20. Which of the following is / are correct?

APR-22

- (i) Adjoint of a symmetric matrix is also a symmetric matrix.
 (ii) Adjoint of a diagonal matrix is also a diagonal matrix.
 (iii) If A is a square matrix of order n and λ is a scalar, then $\text{adj}(\lambda A) = \lambda^n \text{adj}(A)$.
 (iv) $A(\text{adj } A) = (\text{adj } A)A = |A|I$

- (1) Only (i) (2) (ii) and (iii) (3) (iii) and (iv) (4) (i), (ii) and (iv)

$$\begin{aligned} \text{adj}(\lambda A) &= \lambda^{n-1} \text{adj}(A) \quad \text{But given that } \text{adj}(\lambda A) = \lambda^n \text{adj}(A) \\ \text{So, (iii) only a wrong statement.} \end{aligned}$$

21. If $\rho(A) = \rho([A|B])$, then the system $AX = B$ of linear equations is

APR-20, PTA-6

- (1) consistent and has a unique solution (2) consistent
 (3) consistent and has infinitely many solution (4) inconsistent

22. If $0 \leq \theta \leq \pi$ and the system of equations $x + (\sin \theta)y - (\cos \theta)z = 0$,
 $(\cos \theta)x - y + z = 0$, $(\sin \theta)x + y - z = 0$ has a non-trivial solution then θ is

(1) $\frac{2\pi}{3}$

(2) $\frac{3\pi}{4}$

(3) $\frac{5\pi}{6}$

(4) $\frac{\pi}{4}$

The system has a non-trivial solution. $\therefore \Delta = \begin{vmatrix} 1 & \sin \theta & -\cos \theta \\ \cos \theta & -1 & 1 \\ \sin \theta & 1 & -1 \end{vmatrix} = 0$

$$\Rightarrow 1(1 - 1) - \sin \theta(-\cos \theta - \sin \theta) - \cos \theta(\cos \theta + \sin \theta) = 0$$

$$\Rightarrow \cancel{\sin \theta \cos \theta} + \sin^2 \theta - \cos^2 \theta - \cancel{\sin \theta \cos \theta} = 0$$

$$\sin^2 \theta - \cos^2 \theta = 0$$

$$\frac{\sin^2 \theta}{\cos^2 \theta} - \frac{\cos^2 \theta}{\cos^2 \theta} = 0$$

$$\tan^2 \theta = 1 \Rightarrow \tan \theta = 1$$

$$\Rightarrow \theta = \frac{\pi}{4} [\because 0 \leq \theta \leq \pi]$$

23. The augmented matrix of a system of linear equations is $\begin{bmatrix} 1 & 2 & 7 & 3 \\ 0 & 1 & 4 & 6 \\ 0 & 0 & \lambda - 7 & \mu + 5 \end{bmatrix}$. The system has infinitely many solutions if

(1) $\lambda = 7, \mu \neq -5$

(2) $\lambda = -7, \mu = 5$

(3) $\lambda \neq 7, \mu \neq -5$

(4) $\lambda = 7, \mu = -5$

Let $A = \begin{bmatrix} 1 & 2 & 7 & 3 \\ 0 & 1 & 4 & 6 \\ 0 & 0 & \lambda - 7 & \mu + 5 \end{bmatrix}$

If $\lambda = 7, \mu = -5$ then $\rho(A) = 2, \rho([A|B]) = 2$

$\therefore \rho(A) = \rho([A|B]) = 2 < \text{no. of unknowns}$

Thus, the given system is consistent and has infinitely many solutions if $\lambda = 7, \mu = -5$

24. Let $A = \begin{bmatrix} 2 & -1 & 1 \\ -1 & 2 & -1 \\ 1 & -1 & 2 \end{bmatrix}$ and $4B = \begin{bmatrix} 3 & 1 & -1 \\ 1 & 3 & x \\ -1 & 1 & 3 \end{bmatrix}$. If B is the inverse of A , then the value of x is

(1) 2

(2) 4

(3) 3

(4) 1

$$4AB = I \Rightarrow AB = \frac{1}{4}I$$

Evaluate $a_{23} = 0$ only

$$1 + 2x - 3 = 0 \Rightarrow 2x = 2 \Rightarrow x = 1$$

25. If $A = \begin{bmatrix} 3 & -3 & 4 \\ 2 & -3 & 4 \\ 0 & -1 & 1 \end{bmatrix}$, then $\text{adj}(\text{adj}A)$ is

PTA-2

(1) $\begin{bmatrix} 3 & -3 & 4 \\ 2 & -3 & 4 \\ 0 & -1 & 1 \end{bmatrix}$

(2) $\begin{bmatrix} 6 & -6 & 8 \\ 4 & -6 & 8 \\ 0 & -2 & 2 \end{bmatrix}$

(3) $\begin{bmatrix} -3 & 3 & -4 \\ -2 & 3 & -4 \\ 0 & 1 & -1 \end{bmatrix}$

(4) $\begin{bmatrix} 3 & -3 & 4 \\ 0 & -1 & 1 \\ 2 & -3 & 4 \end{bmatrix}$

$$A = \begin{bmatrix} 3 & -3 & 4 \\ 2 & -3 & 4 \\ 0 & -1 & 1 \end{bmatrix}$$

$$|A| = 3[-3 + 4] + 3[2 - 0] + 4[-2 - 0] = 3(1) + 3(2) + 4(-2) = 3 + 6 - 8 = 1$$

$$\text{adj}(\text{adj}A) = |A|^{n-2}A = (1)^{3-2} \begin{bmatrix} 3 & -3 & 4 \\ 2 & -3 & 4 \\ 0 & -1 & 1 \end{bmatrix} = \begin{bmatrix} 3 & -3 & 4 \\ 2 & -3 & 4 \\ 0 & -1 & 1 \end{bmatrix}$$



Rafeel Bombelli

Chapter - 2

Complex Numbers

EXERCISE 2.1



Video class

Concept Corner

1. Powers of imaginary unit i

$i^0 = 1, i^1 = i$	$i^2 = -1$	$i^3 = -i$	$i^4 = 1$
$(i)^{-1} = \frac{1}{i} = -i$	$(i)^{-2} = \frac{1}{i^2} = -1$	$i^{-3} = i$	$i^{-4} = 1$

2. For any integer n ,

$$\begin{aligned}
 i^{4n} &= 1 \\
 i^{4n+1} &= i^1 = i \\
 i^{4n+2} &= i^2 = -1 \\
 i^{4n+3} &= i^3 = -i
 \end{aligned}$$

3. $\sqrt{ab} = \sqrt{a}\sqrt{b}$ Valid only if at least one of a, b is non-negative.4. $i^0 + i^1 + i^2 + i^3 = 0$; $i^1 + i^2 + i^3 + i^4 = 0$

$$i^{-4} + i^{-3} + i^{-2} + i^{-1} = 0 \text{ in general for any } n \in \mathbb{Z}$$

$$i^n + i^{n+1} + i^{n+2} + i^{n+3} = 0$$

Simplify the following:

2M

1. $i^{1947} + i^{1950}$

$$\begin{aligned}
 i^{1947} + i^{1950} &= i^{4(486)+3} + i^{4(487)+2} \\
 &= (1)i^3 + (1)i^2 \\
 &= i^3 + i^2 \\
 &= -i - 1 \\
 &= -1 - i
 \end{aligned}$$

2. $i^{1948} - i^{-1869}$

$$\begin{aligned}
 &= i^{4(487)+0} - i^{4(467)+1} \\
 &= i^0 - i^{-1} \\
 &= 1 - (-i) \\
 &= 1 + i
 \end{aligned}$$

5. $i i^2 i^3 \dots i^{2000}$

$$\begin{aligned}
 &= i^{1+2+\dots+2000} \\
 &= i^{\left[\frac{1+2000}{2} \times 2000 \right]} \\
 &= i^{\frac{2000 \times 2001}{2}} \\
 &= (i^{1000})^{2001} \\
 &= (1)^{2001} \\
 &= 1
 \end{aligned}$$

3. $\sum_{n=1}^{12} i^n$

$$\begin{aligned}
 &= (i^1 + i^2 + i^3 + i^4) + (i^5 + i^6 + i^7 + i^8) \\
 &\quad + (i^9 + i^{10} + i^{11} + i^{12}) \\
 &= 0 + 0 + 0 \\
 \sum_{n=1}^{12} i^n &= 0
 \end{aligned}$$

APR-24

$$\begin{aligned}
 &\text{For any } n \in \mathbb{Z} \\
 &i^n + i^{n+1} + i^{n+2} + i^{n+3} = 0
 \end{aligned}$$

4. $i^{59} + \frac{1}{i^{59}}$

$$\begin{aligned}
 &= i^{4(14)+3} + \frac{1}{i^{4(14)+3}} \\
 &= i^3 + \frac{1}{i^3} \\
 &= i^3 + i^{-3} = -i + i \\
 i^{59} + \frac{1}{i^{59}} &= 0
 \end{aligned}$$

6. $\sum_{n=1}^{10} i^{n+50}$

$$\begin{aligned}
 &= (i^{51} + i^{52} + i^{53} + i^{54}) + (i^{55} + i^{56} + i^{57} + i^{58}) \\
 &\quad + i^{59} + i^{60} \\
 &= 0 + 0 + i^{4(14)+3} + i^{4(15)+0} \\
 &= i^3 + i^0 \\
 &= -i + 1 \\
 &= 1 - i
 \end{aligned}$$

EXERCISE 2.2



Video class

Concept Corner

1. Rectangular form of a complex number

A complex number is of the form $z = x + iy$, where $x, y \in R$.

Here, the real part of $z = Re(z) = x$; the imaginary part of $z = Im(z) = y$

2. Purely imaginary and real numbers

Let $z = x + iy$

(i) If $Re(z) = 0$ (ie) $x = 0$, then z is said to be purely imaginary

(ii) If $Im(z) = 0$ (ie) $y = 0$, then z is said to be real.

3. Equality of two complex numbers

Two complex numbers $z_1 = x_1 + iy_1$ & $z_2 = x_2 + iy_2$ are equal if and only if $x_1 = x_2, y_1 = y_2$.
i.e., $Re(z_1) = Re(z_2)$ and $Im(z_1) = Im(z_2)$.

4. Algebraic operations on complex numbers

(i) Scalar multiplication of complex numbers :

If $z = x + iy$ and $k \in R$, then
we define $kz = (kx) + (ky)i$.

(iii) Subtraction of complex numbers :

If $z_1 = x_1 + iy_1$ and $z_2 = x_2 + iy_2$
Where x_1, x_2, y_1 and $y_2 \in R$ then
 $z_1 - z_2 = (x_1 - x_2) + i(y_1 - y_2)$

(v) Multiplication of complex number z by i :

If $z = x + iy$ then $iz = i(x + iy) = -y + ix$

$\therefore$ The complex number iz is a rotation of z by 90° (or) $\frac{\pi}{2}$ radians in the counter clockwise direction about the origin.

(ii) Addition of complex numbers :

If $z_1 = x_1 + iy_1$ and $z_2 = x_2 + iy_2$

Where x_1, x_2, y_1 and $y_2 \in R$ then

$$z_1 + z_2 = (x_1 + x_2) + i(y_1 + y_2)$$

(iv) Multiplication of complex numbers :

If $z_1 = x_1 + iy_1$ and $z_2 = x_2 + iy_2$

Where x_1, x_2, y_1 and $y_2 \in R$ then

$$z_1 z_2 = (x_1 x_2 - y_1 y_2) + i(x_1 y_2 + x_2 y_1)$$

1. Evaluate the following if $z = 5 - 2i$ and $w = -1 + 3i$

(i) $z + w$

(ii) $z - iw$

(iii) $2z + 3w$

(iv) zw

(v) $z^2 + 2zw + w^2$

(vi) $(z + w)^2$

(i) $z + w$

$$z + w = 5 - 2i - 1 + 3i = 4 + i$$

(ii) $z - iw$

$$z - iw = 5 - 2i + i - 3i^2$$

$$= 5 - i - 3(-1)$$

$$= 5 - i + 3 = 8 - i$$

(iii) $2z + 3w$

$$2z + 3w = 10 - 4i - 3 + 9i = 7 + 5i$$

(iv) zw

$$zw = (5 - 2i)(-1 + 3i)$$

$$= -5 + 15i + 2i - 6i^2$$

$$= -5 + 17i - 6(-1)$$

$$= -5 + 17i + 6 = 1 + 17i$$

(v) $z^2 + 2zw + w^2$

$$z^2 + 2zw + w^2 = (z + w)^2 = [5 - 2i - 1 + 3i]^2$$

$$= (4 + i)^2$$

$$= 16 + 2(4)i + i^2$$

$$= 16 + 8i - 1$$

$$(z + w)^2 = 15 + 8i$$

(vi) $(z + w)^2$

$$(z + w)^2 = [5 - 2i - 1 + 3i]^2$$

$$= (4 + i)^2$$

$$= 16 + 2(4)i + i^2$$

$$= 16 + 8i - 1$$

$$(z + w)^2 = 15 + 8i$$

2. Given the complex number $z = 2 + 3i$, represent the complex numbers in Argand diagram

(i) z , iz and $z + iz$

(ii) z , $-iz$ and $z - iz$

3M

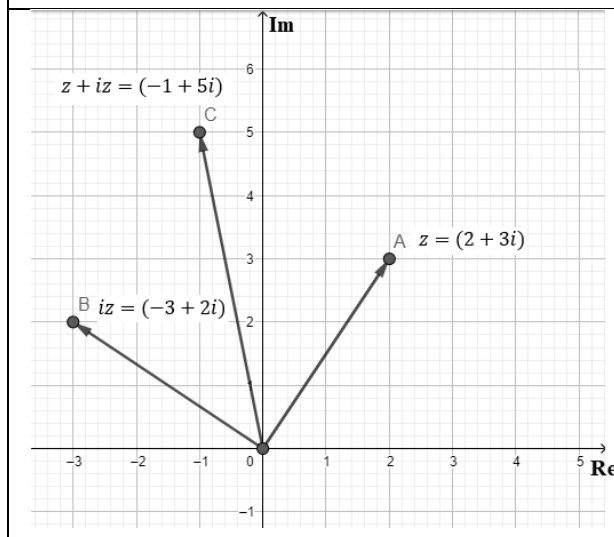
(i) z , iz and $z + iz$

$$z = 2 + 3i$$

$$iz = i(2 + 3i)$$

$$= 2i + 3i^2 = 2i + 3(-1) = -3 + 2i$$

$$z + iz = 2 + 3i - 3 + 2i + iz = -1 + 5i$$



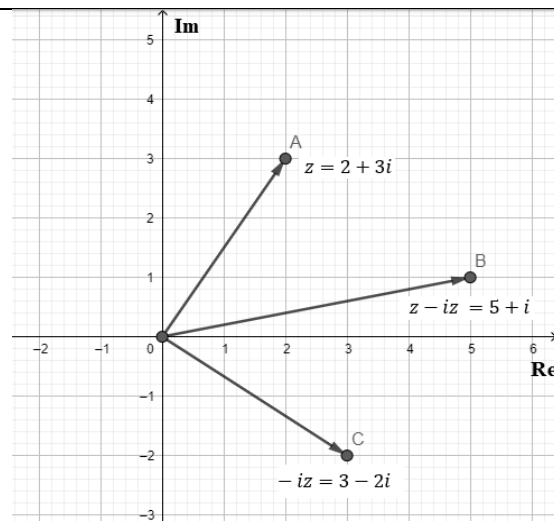
(ii) z , $-iz$ and $z - iz$

$$z = 2 + 3i$$

$$-iz = -i(2 + 3i)$$

$$= -2i - 3i^2 = -2i - 3(-1) = 3 - 2i$$

$$z - iz = 2 + 3i + 3 - 2i = 5 + i$$



3. Find the values of the real numbers x and y , if the complex numbers

3M

$(3 - i)x - (2 - i)y + 2i + 5$ and $2x + (-1 + 2i)y + 3 + 2i$ are equal.

Given that $(3 - i)x - (2 - i)y + 2i + 5 = 2x + (-1 + 2i)y + 3 + 2i$

$$\Rightarrow 3x - ix - 2y + iy + 2i + 5 = 2x - y + i2y + 3 + 2i$$

$$\Rightarrow (3x - 2y + 5) + i(-x + y + 2) = (2x - y + 3) + i(2y + 2)$$

Equating real and imaginary parts separately, we get

$$3x - 2y + 5 = 2x - y + 3$$

$$\Rightarrow 3x - 2y - 2x + y = 3 - 5$$

$$x - y = -2 \quad \dots\dots\dots (1)$$

$$-x + y + 2 = 2y + 2$$

$$-x + y - 2y = 2 - 2$$

$$x + y = 0 \quad \dots\dots\dots (2)$$

$$x - y = -2$$

$$x + y = 0$$

$$(1) + (2) \Rightarrow 2x = -2$$

$$\Rightarrow x = \frac{-2}{2} = -1$$

$$x = -1$$

Substituting $x = -1$ in (2), we get $-1 + y = 0$

$$y = 1$$

$\therefore$ The solution is $x = -1$ and $y = 1$.

EXERCISE 2.3



Video class

Concept Corner

Basic Algebraic Properties of Complex Numbers

The complex numbers satisfy the following properties under addition.	The complex numbers satisfy the following properties under multiplication.
(i) Closure property For any two complex numbers z_1 and z_2 , the sum $z_1 + z_2$ is also a complex number.	(i) Closure property For any two complex numbers z_1 and z_2 the product $z_1 z_2$ is also a complex number.
(ii) The commutative property For any two complex numbers z_1 and z_2 $z_1 + z_2 = z_2 + z_1$.	(ii) The commutative property For any two complex numbers z_1 and z_2 $z_1 z_2 = z_2 z_1$.
(iii) The associative property For any three complex numbers z_1, z_2 & z_3 $(z_1 + z_2) + z_3 = z_1 + (z_2 + z_3)$.	(iii) The associative property For any three complex numbers z_1, z_2 & z_3 $(z_1 z_2) z_3 = z_1 (z_2 z_3)$
(iv) The additive identity There exists a complex number $0 = 0 + 0i$ such that, for every complex number z , $z + 0 = 0 + z = z$ The complex number $0 = 0 + 0i$ is known as additive identity	(iv) The Multiplicative identity There exists a complex number $1 = 1 + 0i$ such that, for every complex number z , $z1 = 1z = z$ The complex number $1 = 1 + 0i$ is known as multiplicative identity.
(v) The additive inverse For every complex number z there exists a complex number $-z$ such that, $z + (-z) = (-z) + z = 0$. $-z$ is called the additive inverse of z .	(v) The Multiplicative inverse For any nonzero complex number z , there exists a complex number w such that, $zw = wz = 1$ w is called the multiplicative inverse of z . w is denoted by z^{-1}
(vi) Distributive property (multiplication distributes over addition) For any three complex numbers z_1, z_2 and z_3 then, $z_1(z_2 + z_3) = z_1 z_2 + z_1 z_3$ and $(z_1 + z_2)z_3 = z_1 z_3 + z_2 z_3$.	
Commutative property under addition: $z_1 = x_1 + iy_1; z_2 = x_2 + iy_2 \Rightarrow z_1 + z_2 = z_2 + z_1$	
Inverse property under addition: $z^{-1} = \frac{1}{z}; z = x + iy \Rightarrow z^{-1} = \frac{x}{x^2+y^2} + i\left(\frac{-y}{x^2+y^2}\right)$	
Complex numbers obey the laws of indices:	
(i) $z^m z^n = z^{m+n}$ (ii) $\frac{z^m}{z^n} = z^{m-n}, z \neq 0$ (iii) $(z^m)^n = z^{mn}$ (iv) $(z_1 z_2)^m = z_1^m z_2^m$	

1. If $z_1 = 1 - 3i, z_2 = -4i$ and $z_3 = 5$, show that

2M

(i) $(z_1 + z_2) + z_3 = z_1 + (z_2 + z_3)$
 (i) $(z_1 + z_2) + z_3 = z_1 + (z_2 + z_3)$
 L. H. S: $(z_1 + z_2) + z_3 = (1 - 3i - 4i) + 5$
 $= 1 - 7i + 5 = 6 - 7i$
 R. H. S: $z_1 + (z_2 + z_3) = 1 - 3i + (-4i + 5)$
 $= 1 - 3i - 4i + 5 = 6 - 7i$
 L. H. S = R. H. S
 $\therefore (z_1 + z_2) + z_3 = z_1 + (z_2 + z_3)$

(ii) $(z_1 z_2) z_3 = z_1 (z_2 z_3)$
 (ii) $(z_1 z_2) z_3 = z_1 (z_2 z_3)$
 L. H. S: $(z_1 z_2) z_3 = [(1 - 3i)(-4i)](5)$
 $= [-4i + 12i^2](5)$
 $= [-4i - 12]5 = -60 - 20i$
 R. H. S: $z_1 (z_2 z_3) = (1 - 3i)[(-4i)(5)]$
 $= (1 - 3i)(-20i)$
 $= -20i + 60i^2 = -60 - 20i$
 L. H. S = R. H. S $\Rightarrow \therefore (z_1 z_2) z_3 = z_1 (z_2 z_3)$

2. If $z_1 = 3$, $z_2 = -7i$, and $z_3 = 5 + 4i$, show that

2M

(i) $z_1(z_2 + z_3) = z_1z_2 + z_1z_3$

(ii) $(z_1 + z_2)z_3 = z_1z_3 + z_2z_3$

(i) $z_1(z_2 + z_3) = z_1z_2 + z_1z_3$

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(ii) $(z_1 + z_2)z_3 = z_1z_3 + z_2z_3$

L. H. S: $z_1(z_2 + z_3) = 3(-7i + 5 + 4i)$

L. H. S $(z_1 + z_2)z_3 = (3 - 7i)(5 + 4i)$

$$= 3(5 - 3i)$$

$$= 15 + 12i - 35i - 28i^2$$

$$= 15 - 9i$$

$$= 15 - 23i + 28$$

$$= 43 - 23i$$

R. H. S: $z_1z_2 + z_1z_3 = 3(-7i) + 3(5 + 4i)$

R. H. S $z_1z_3 + z_2z_3 = (3)(5 + 4i) - 7i(5 + 4i)$

$$= -21i + 15 + 12i$$

$$= 15 + 12i - 35i - 28i^2$$

$$= 15 - 9i$$

$$= 15 - 23i + 28$$

$$= 43 - 23i$$

L. H. S = R. H. S

$$\therefore z_1(z_2 + z_3) = z_1z_2 + z_1z_3$$

L. H. S = R. H. S $\Rightarrow \therefore (z_1 + z_2)z_3 = z_1z_3 + z_2z_3$

3. If $z_1 = 2 + 5i$, $z_2 = -3 - 4i$, and $z_3 = 1 + i$, find the additive and multiplicative inverse of z_1 , z_2 , and z_3 .

3M

(i) $z_1 = 2 + 5i$

The additive inverse of

$$z_1 = -z_1$$

$$= -(2 + 5i) = -2 - 5i$$

The multiplication inverse of

$$z_1 = z_1^{-1}$$

$$= \frac{2}{2^2+5^2} + i \frac{-5}{2^2+5^2}$$

$$= \frac{2}{29} - \frac{5}{29}i = \frac{1}{29}(2 - 5i)$$

(ii) $z_2 = -3 - 4i$

The additive inverse of

$$z_2 = -z_2$$

$$= -(-3 - 4i) = 3 + 4i$$

The multiplication inverse of

$$z_2 = z_2^{-1}$$

$$= \frac{-3}{(-3)^2+(-4)^2} + i \frac{-(-4)}{(-3)^2+(-4)^2}$$

$$= \frac{-3}{9+16} + i \frac{4}{9+16}$$

$$= \frac{-3}{25} + \frac{4}{25}i = \frac{1}{25}(-3 + 4i)$$

(iii) $z_3 = 1 + i$

The additive inverse of

$$z_3 = -z_3$$

$$= -(1 + i) = -1 - i$$

The multiplication inverse of

$$z_3 = z_3^{-1}$$

$$= \frac{1}{1^2+1^2} + i \frac{-1}{1^2+1^2}$$

$$= \frac{1}{2} - \frac{1}{2}i = \frac{1}{2}(1 - i)$$

EXERCISE 2.4



Video class

Concept Corner

Conjugate of a Complex Number

The conjugate of the complex number $z = x + iy$ is denoted by $\bar{z}$ and is defined by $\bar{z} = x - iy$

Properties of complex conjugates

1. $\overline{z_1 + z_2} = \bar{z}_1 + \bar{z}_2$

2. $\overline{z_1 - z_2} = \bar{z}_1 - \bar{z}_2$

3. $\overline{z_1 z_2} = \bar{z}_1 \bar{z}_2$

4. $\overline{\left(\frac{z_1}{z_2}\right)} = \frac{\bar{z}_1}{\bar{z}_2}, z_2 \neq 0$

5. $\bar{\bar{z}} = z$

5. $Re(z) = \frac{z+\bar{z}}{2}$

6. $Im(z) = \frac{z-\bar{z}}{2i}$

8. $(\bar{z}^n) = (\bar{z})^n$, where n is an integer

9. z is real if and only if $z = \bar{z}$

10. z is purely imaginary if and only if $z = -\bar{z}$

1. Write the following in the rectangular form:

2M

$$(i) \overline{(5 + 9i)} + \overline{(2 - 4i)} = \overline{(5 + 9i)} + \overline{(2 - 4i)} \quad [\because \text{By the property, } \overline{z_1 + z_2} = \overline{z_1} + \overline{z_2}]$$

$$= 5 - 9i + 2 + 4i = 7 - 5i$$

$$(ii) \frac{10-5i}{6+2i}$$

$$\frac{10-5i}{6+2i} = \frac{10-5i}{6+2i} \times \frac{6-2i}{6-2i} \quad \boxed{(a+ib)(a-ib) = a^2 + b^2}$$

$$= \frac{60-20i-30i+10i^2}{6^2+2^2}$$

$$= \frac{60-50i-10}{36+4}$$

$$= \frac{50-50i}{40} = \frac{50}{40}(1-i) = \frac{5}{4}(1-i)$$

$$(iii) \overline{3i} + \frac{1}{2-i}$$

$$\overline{3i} + \frac{1}{2-i} = -3i + \frac{1}{2-i} \times \frac{2+i}{2+i}$$

$$= -3i + \frac{2+i}{4+1}$$

$$= -3i + \frac{2}{5} + \frac{1}{5}i$$

$$= \frac{2}{5} + \left(-3 + \frac{1}{5}\right)i = \frac{2}{5} - \frac{14}{5}i$$

2. If $z = x + iy$, find the following in rectangular form.

2M

$$(i) \operatorname{Re}\left(\frac{1}{z}\right)$$

APR-25

$$\text{let } z = x + iy$$

$$\operatorname{Re}\left(\frac{1}{z}\right) = \operatorname{Re}\left(\frac{1}{x+iy} \times \frac{x-iy}{x-iy}\right) = \operatorname{Re}\left(\frac{x-iy}{x^2+y^2}\right)$$

$$= \operatorname{Re}\left(\frac{x}{x^2+y^2} + i\frac{-y}{x^2+y^2}\right) = \frac{x}{x^2+y^2}$$

$$(ii) \operatorname{Re}(i\bar{z})$$

$$\operatorname{Re}(i\bar{z}) = \operatorname{Re}(i(x - iy))$$

$$= \operatorname{Re}(ix - i^2y)$$

$$= \operatorname{Re}(ix + y) = y$$

$$(iii) \operatorname{Im}(3z + 4\bar{z} - 4i)$$

$$\text{let } z = x + iy$$

$$\bar{z} = (x - iy)$$

$$\operatorname{Im}(3z + 4\bar{z} - 4i) = \operatorname{Im}(3(x + iy) + 4(x - iy) - 4i)$$

$$= \operatorname{Im}(3x + i3y + 4x - i4y - 4i)$$

$$= \operatorname{Im}(7x - iy - 4i)$$

$$= \operatorname{Im}(7x + i(-y - 4))$$

$$\operatorname{Im}(3z + 4\bar{z} - 4i) = -y - 4$$

3. If $z_1 = 2 - i$ and $z_2 = -4 + 3i$, find the inverse of $z_1 z_2$ and $\frac{z_1}{z_2}$.

3M

$$(i) z_1 z_2 = (2 - i)(-4 + 3i)$$

$$= -8 + 6i + 4i - 3i^2$$

$$= -8 + 10i + 3 = -5 + 10i$$

$$\left(\frac{z_1}{z_2}\right)^{-1} = \frac{1}{\left(\frac{z_1}{z_2}\right)}$$

$$= \frac{z_2}{z_1} = \frac{1}{-5+10i} = \frac{1}{-5+10i} \times \frac{-5-10i}{-5-10i} = \frac{-5-10i}{25+100}$$

$$= \frac{-5}{125} + i\frac{-10}{125}$$

$$= \frac{-1}{25} + i\frac{-2}{25}$$

$$\therefore \text{The inverse of } z_1 z_2 = \frac{1}{25}(-1 - 2i)$$

$$(ii) \left(\frac{z_1}{z_2}\right)^{-1} = \frac{1}{\left(\frac{z_1}{z_2}\right)} = \frac{z_2}{z_1}$$

$$= \frac{-4+3i}{2-i}$$

$$= \frac{-4+3i}{2-i} \times \frac{2+i}{2+i}$$

$$= \frac{-8-4i+6i+3i^2}{4+1}$$

$$= \frac{-8+2i-3}{5}$$

$$= \frac{-11}{5} + i\frac{2}{5}$$

$$\therefore \text{The inverse of } \frac{z_1}{z_2} = \frac{1}{5}(-11 + 2i)$$

4. The complex numbers u , v , and w are related by $\frac{1}{u} = \frac{1}{v} + \frac{1}{w}$. If $v = 3 - 4i$ and $w = 4 + 3i$, find u in rectangular form.

3M

$$\frac{1}{u} = \frac{1}{v} + \frac{1}{w} = \frac{1}{3-4i} + \frac{1}{4+3i}$$

$$\frac{1}{u} = \frac{3+4i}{(3-4i)(3+4i)} + \frac{4-3i}{(4+3i)(4-3i)} = \frac{3+4i}{9+16} + \frac{4-3i}{16+9} = \frac{3+4i+4-3i}{25} = \frac{7+i}{25}$$

$$\Rightarrow u = \frac{25}{7+i} \times \frac{7-i}{7-i} = \frac{25(7-i)}{49+1} = \frac{25(7-i)}{50} = \frac{1}{2}(7 - i)$$

5. Prove the following properties:

3M

(i) z is real if and only if $z = \bar{z}$ (i) z is real number $\Leftrightarrow z = \bar{z}$

$$z = x + iy \text{ then } \bar{z} = x - iy$$

$$z = \bar{z} \Leftrightarrow x + iy = x - iy$$

$$\Leftrightarrow \cancel{x} + iy - \cancel{x} + iy = 0$$

$$\Leftrightarrow i2y = 0$$

$$\Leftrightarrow y = 0$$

$$\Leftrightarrow z \text{ is real number}$$

(ii) $\operatorname{Re}(z) = \frac{z+\bar{z}}{2}$ and $\operatorname{Im}(z) = \frac{z-\bar{z}}{2i}$ (ii) $\operatorname{Re}(z) = \frac{z+\bar{z}}{2}$ and $\operatorname{Im}(z) = \frac{z-\bar{z}}{2i}$

$$z = x + iy \Rightarrow \operatorname{Re}(z) = x, \operatorname{Im}(z) = y$$

$$z + \bar{z} = x + \cancel{iy} + x - \cancel{iy}$$

$$z + \bar{z} = 2x$$

$$\frac{z+\bar{z}}{2} = x$$

$$\frac{z+\bar{z}}{2} = \operatorname{Re}(z)$$

$$z - \bar{z} = \cancel{x} + iy - \cancel{x} - iy$$

$$z - \bar{z} = 2iy$$

$$\frac{z-\bar{z}}{2i} = y$$

$$\frac{z-\bar{z}}{2i} = \operatorname{Im}(z)$$

6. Find the least value of the positive integer n for which $(\sqrt{3} + i)^n$ (i) real (ii) purely imaginary.

$$\begin{aligned}
 (\sqrt{3} + i)^3 &= \sqrt{3}^3 + 3(\sqrt{3})^2 i + 3(\sqrt{3})i^2 + i^3 & [\because (a+b)^3 = a^3 + 3a^2b + 3ab^2 + b^3] \\
 &= \sqrt[3]{3} + 9i - 3\sqrt{3} - i = 8i
 \end{aligned}$$

3M

$$(\sqrt{3} + i)^6 = [(\sqrt{3} + i)^3]^2 = (8i)^2 = 64i^2 = -64$$

(i) $(\sqrt{3} + i)^n$ is purely real when $n = 6$ (ii) $(\sqrt{3} + i)^n$ is purely imaginary when $n = 3$ 7. Show that (i) $(2 + i\sqrt{3})^{10} - (2 - i\sqrt{3})^{10}$ is purely imaginary (ii) $\left(\frac{19-7i}{9+i}\right)^{12} + \left(\frac{20-5i}{7-6i}\right)^{12}$ is real. 5M(i) Let $z = (2 + i\sqrt{3})^{10} - (2 - i\sqrt{3})^{10}$

JUL-22

$$[\because \overline{z_1 - z_2} = \bar{z}_1 - \bar{z}_2]$$

$$[\because (\bar{z}^n) = (\bar{z})^n]$$

$$\bar{z} = \overline{(2 + i\sqrt{3})^{10} - (2 - i\sqrt{3})^{10}}$$

$$= \overline{(2 + i\sqrt{3})^{10}} - \overline{(2 - i\sqrt{3})^{10}}$$

$$= (\overline{2 + i\sqrt{3}})^{10} - (\overline{2 - i\sqrt{3}})^{10}$$

$$= (2 - i\sqrt{3})^{10} - (2 + i\sqrt{3})^{10}$$

$$= -[(2 + i\sqrt{3})^{10} - (2 - i\sqrt{3})^{10}]$$

$$= -z$$

$$\bar{z} = -z$$

 $\Rightarrow z$ is purely imaginary(ii) $\frac{19-7i}{9+i} = \frac{19-7i}{9+i} \times \frac{9-i}{9-i}$

$$= \frac{171-19i-63i+7i^2}{81+1}$$

$$= \frac{171-82i-7}{82} = \frac{164}{82} - \frac{82}{82}i = 2 - i$$

$$\frac{20-5i}{7-6i} = \frac{20-5i}{7-6i} \times \frac{7+6i}{7+6i}$$

$$= \frac{140+120i-35i-30i^2}{49+36}$$

$$= \frac{140+85i+30}{85} = \frac{170}{85} + \frac{85}{85}i = 2 + i$$

$$\text{Let } z = \left(\frac{19-7i}{9+i}\right)^{12} + \left(\frac{20-5i}{7-6i}\right)^{12}$$

$$z = (2 - i)^{12} + (2 + i)^{12}$$

$$\bar{z} = \overline{(2 - i)^{12} + (2 + i)^{12}}$$

$$= \overline{(2 - i)^{12}} + \overline{(2 + i)^{12}}$$

$$= (\overline{2 - i})^{12} + (\overline{2 + i})^{12}$$

$$= (2 + i)^{12} + (2 - i)^{12}$$

$$= z$$

 $\bar{z} = z \Rightarrow z$ is real.

EXERCISE 2.5



Video class

Concept Corner

Modulus of a complex numbers

If $z = x + iy$, then the modulus of z is $\sqrt{x^2 + y^2}$. It is denoted by $|z|$.

If $z = x + iy$, then $\bar{z} = x - iy$ then $z\bar{z} = (x + iy)(x - iy) = x^2 + y^2 = |z|^2$. $\therefore |z^2| = z\bar{z}$

Properties of modulus of a complex number

- | | |
|--|---|
| 1. $ z = \bar{z} $ | 5. $ z_1 + z_2 \leq z_1 + z_2 $ (Triangle inequality) |
| 2. $ z^n = z ^n$, where n is an integer | 6. $ z_1 - z_2 \geq z_1 - z_2 $ |
| 3. $ z_1 z_2 = z_1 z_2 $ | 7. $\operatorname{Re}(z) \leq z $ |
| 4. $\left \frac{z_1}{z_2} \right = \frac{ z_1 }{ z_2 }$, $z_2 \neq 0$ | 8. $\operatorname{Im}(z) \leq z $ |

9. $||z_1| - |z_2|| \leq |z_1 + z_2| \leq |z_1| + |z_2|$ and $||z_1| - |z_2|| \leq |z_1 - z_2| \leq |z_1| + |z_2|$
10. $|z_1 + z_2 + z_3 + \dots + z_n| \leq |z_1| + |z_2| + |z_3| + \dots + |z_n|$ for $n = 2, 3, \dots$
11. $|z_1 z_2 z_3 \dots z_n| = |z_1| |z_2| |z_3| \dots |z_n|$
12. $|z_1 - z_2|$ = Distance between the two points z_1 and z_2 in the complex plane
- $$|z_1 - z_2| = \sqrt{(x_1 - x_2)^2 + (y_1 - y_2)^2}$$
13. $|z| = 1$ if and only if $\bar{z} = \frac{1}{z}$
14. $|z_1 + z_2|^2 + |z_1 - z_2|^2 = 2[|z_1|^2 + |z_2|^2]$
15. $|z_1 + z_2|^2 - |z_1 - z_2|^2 = 4\operatorname{Re}(z_1 \bar{z}_2)$

1. Find the modulus of the following complex numbers

2M

(i) $\frac{2i}{3+4i}$

(ii) $\frac{2-i}{1+i} + \frac{1-2i}{1-i}$

(iii) $(1-i)^{10}$

(iv) $2i(3-4i)(4-3i)$

(i) Let $z = \frac{2i}{3+4i}$

$|z| = \left| \frac{2i}{3+4i} \right|$

$= \frac{|2i|}{|3+4i|}$

$= \frac{\sqrt{0^2+2^2}}{\sqrt{3^2+4^2}}$

$= \frac{\sqrt{4}}{\sqrt{9+16}} = \frac{\sqrt{4}}{\sqrt{25}}$

$|z| = \frac{2}{5}$

(ii) Let $z = \frac{2-i}{1+i} + \frac{1-2i}{1-i}$

$= \frac{(2-i)(1-i) + (1-2i)(1+i)}{(1+i)(1-i)}$

$= \frac{2-2i-1+2i+2}{1+1}$

$= \frac{4-4i}{2}$

$z = 2 - 2i$

$|z| = |2 - 2i| = \sqrt{4+4}$

$= \sqrt{8} = \sqrt{4 \times 2}$

$|z| = 2\sqrt{2}$

(iii) Let $z = (1-i)^{10}$

$|z| = |(1-i)^{10}|$

$= |1-i|^{10}$

$[\because |z^n| = |z|^n]$

$= (\sqrt{1+1})^{10}$

$= \left(\sqrt{2} \right)^{10} = 2^5 = 32$

$\therefore |z| = 32$

(iv) Let $z = 2i(3-4i)(4-3i)$

$|z| = |2i(3-4i)(4-3i)|$

$= |2i| |3-4i| |4-3i|$

$= (\sqrt{0+4})(\sqrt{9+16})(\sqrt{16+9})$

$= (\sqrt{4})(\sqrt{25})(\sqrt{25})$

$= (2)(5)(5)$

$|z| = 50$

2. For any two complex numbers z_1 and z_2 , such that $|z_1| = |z_2| = 1$ and $z_1 z_2 \neq -1$, then show that $\frac{z_1 + z_2}{1 + z_1 z_2}$ is a real number.

Given that $|z_1| = |z_2| = 1$

3M

$$|z_1| = 1 \Rightarrow |z_1|^2 = 1$$

$$[\because |z|^2 = z\bar{z}]$$

$$z_1 \bar{z}_1 = 1$$

$$\bar{z}_1 = \frac{1}{z_1}$$

$$|z_2| = 1 \Rightarrow |z_2|^2 = 1$$

$$z_2 \bar{z}_2 = 1$$

$$\bar{z}_2 = \frac{1}{z_2}$$

$$\text{Let } z = \frac{z_1 + z_2}{1 + z_1 z_2} \Rightarrow \bar{z} = \frac{\overline{z_1 + z_2}}{\overline{1 + z_1 z_2}}$$

$$= \frac{\bar{z}_1 + \bar{z}_2}{1 + \bar{z}_1 \bar{z}_2} = \frac{\frac{1}{z_1} + \frac{1}{z_2}}{1 + \left(\frac{1}{z_1}\right)\left(\frac{1}{z_2}\right)}$$

$$= \frac{\frac{z_2 + z_1}{z_1 z_2}}{\frac{z_1 z_2 + 1}{z_1 z_2}} = \frac{z_1 + z_2}{1 + z_1 z_2} = z$$

$$\bar{z} = z \quad \text{Thus, } \frac{z_1 + z_2}{1 + z_1 z_2} \text{ is a real number.}$$

3. Which one of the points $10 - 8i$, $11 + 6i$ is closest to $1 + i$.

Let A , B and C be the points representing the complex numbers $10 - 8i$, $11 + 6i$ & $1 + i$

$$AC = |(10 - 8i) - (1 + i)|$$

3M

$$= |10 - 8i - 1 - i|$$

$$= |9 - 9i|$$

APR-22

$$= \sqrt{9^2 + (-9)^2}$$

$$= \sqrt{162} \text{ units}$$

$$BC = |(11 + 6i) - (1 + i)|$$

$$= |11 + 6i - 1 - i|$$

$$= |10 + 5i|$$

$$= \sqrt{10^2 + 5^2}$$

$$= \sqrt{100 + 25}$$

$$= \sqrt{125} \text{ Units.}$$

Since $AC > BC$, B is closest to C .

$\therefore$ The point $11 + 6i$ is closest to $1 + i$.

4. If $|z| = 3$, show that $7 \leq |z + 6 - 8i| \leq 13$.

3M

Let $z_1 = z$ and $z_2 = 6 - 8i$

$$|z_1| = |z| = 3 \text{ and } |z_2| = |6 - 8i| = \sqrt{36 + 64} = \sqrt{100} = 10$$

$$\text{We know that, } ||z_1| - |z_2|| \leq |z_1 + z_2| \leq |z_1| + |z_2|$$

$$\Rightarrow |3 - 10| \leq |z + 6 - 8i| \leq 3 + 10$$

$$\Rightarrow |-7| \leq |z + 6 - 8i| \leq 13$$

$$\Rightarrow 7 \leq |z + 6 - 8i| \leq 13$$

5. If $|z| = 1$, show that $2 \leq |z^2 - 3| \leq 4$.

3M

Let $z_1 = z^2$ and $z_2 = -3$

$$|z_1| = |z|^2 = 1^2 = 1 \text{ and } |z_2| = |-3| = 3$$

$$\text{We know that, } |z_1| - |z_2| \leq |z_1 + z_2| \leq |z_1| + |z_2|$$

$$\Rightarrow |1 - 3| \leq |z^2 - 3| \leq 1 + 3$$

$$\Rightarrow |-2| \leq |z^2 - 3| \leq 4$$

$$\Rightarrow 2 \leq |z^2 - 3| \leq 4$$

6. If $|z| = 2$, show that $8 \leq |z + 6 + 8i| \leq 12$.

APR-25

3M

Let $z_1 = z$ and $z_2 = 6 + 8i$

$$\Rightarrow |z_1| = |z| = 2 \text{ and } |z_2| = |6 + 8i| = \sqrt{36 + 64} = \sqrt{100} = 10$$

$$\text{We know that, } ||z_1| - |z_2|| \leq |z_1 + z_2| \leq |z_1| + |z_2|$$

$$\Rightarrow |2 - 10| \leq |z + 6 + 8i| \leq 2 + 10$$

$$\Rightarrow |-8| \leq |z + 6 + 8i| \leq 12$$

$$\Rightarrow 8 \leq |z + 6 + 8i| \leq 12$$

7. If z_1, z_2 , and z_3 are three complex numbers such that $|z_1| = 1, |z_2| = 2, |z_3| = 3$ and $|z_1 + z_2 + z_3| = 1$, show that $|9z_1z_2 + 4z_1z_3 + z_2z_3| = 6$.

5M

Given that $|z_1| = 1, |z_2| = 2, |z_3| = 3$

$$\Rightarrow |z_1|^2 = 1, |z_2|^2 = 4, |z_3|^2 = 9$$

$$\Rightarrow z_1\bar{z}_1 = 1, z_2\bar{z}_2 = 4, z_3\bar{z}_3 = 9$$

$$[\because |z|^2 = z\bar{z}]$$

$$\Rightarrow z_1 = \frac{1}{\bar{z}_1}, z_2 = \frac{4}{\bar{z}_2}, z_3 = \frac{9}{\bar{z}_3}$$

$$z_1 + z_2 + z_3 = \frac{1}{\bar{z}_1} + \frac{4}{\bar{z}_2} + \frac{9}{\bar{z}_3}$$

$$= \frac{z_2\bar{z}_3 + 4z_1\bar{z}_3 + 9z_1\bar{z}_2}{z_1z_2z_3} = \frac{z_2\bar{z}_3 + 4z_1\bar{z}_3 + 9z_1\bar{z}_2}{z_1z_2z_3} = \frac{9z_1z_2 + 4z_1z_3 + z_2z_3}{z_1z_2z_3}$$

$$z_1 + z_2 + z_3 = \left[\frac{9z_1z_2 + 4z_1z_3 + z_2z_3}{z_1z_2z_3} \right]$$

$$|z_1 + z_2 + z_3| = \left| \left[\frac{9z_1z_2 + 4z_1z_3 + z_2z_3}{z_1z_2z_3} \right] \right|$$

$$1 = \left| \frac{9z_1z_2 + 4z_1z_3 + z_2z_3}{z_1z_2z_3} \right|$$

$$[\because |\bar{z}| = |z|]$$

$$1 = \frac{|9z_1z_2 + 4z_1z_3 + z_2z_3|}{|z_1||z_2||z_3|}$$

$$1 = \frac{|9z_1z_2 + 4z_1z_3 + z_2z_3|}{(1)(2)(3)}$$

$$1 = \frac{|9z_1z_2 + 4z_1z_3 + z_2z_3|}{6}$$

$$\therefore |9z_1z_2 + 4z_1z_3 + z_2z_3| = 6$$

8. If the area of the triangle formed by the vertices z, iz , and $z + iz$ is 50 square units, find the value of $|z|$.

2M

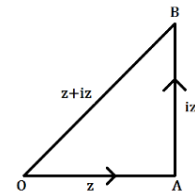
$$\text{Area of the triangle} = \frac{1}{2}|z||iz| = \frac{1}{2}|z||z| = \frac{1}{2}|z|^2 \quad (\because |i| = 1)$$

But, Area = 50 square units

$$\therefore \frac{1}{2}|z|^2 = 50$$

$$|z|^2 = 100$$

$$|z| = 10$$



9. Show that the equation $z^3 + 2\bar{z} = 0$ has five solutions.

PTA-1

3M

$$z^3 + 2\bar{z} = 0 \quad \dots\dots\dots (1)$$

$$z^3 = -2\bar{z}$$

$$|z^3| = |-2||\bar{z}| = 2|z|$$

$$|z|^3 - 2|z| = 0$$

$$[\because z\bar{z} = |z|^2]$$

$$|z|(|z|^2 - 2) = 0$$

$$|z| = 0 \text{ (or) } |z|^2 - 2 = 0$$

$$|z| = 0 \Rightarrow z = 0 \text{ is a solution.}$$

$$|z|^2 - 2 = 0 \Rightarrow |z|^2 = 2 \Rightarrow z\bar{z} = 2 \Rightarrow \bar{z} = \frac{2}{z}$$

$$(1) \Rightarrow z^3 + 2\left(\frac{2}{z}\right) = 0 \Rightarrow z^4 + 4 = 0 \quad \text{It has 4 non-zero solution.}$$

Hence including zero solution, there are five solutions.

Square roots of a complex number:

$$\sqrt{a+ib} = \pm \left[\sqrt{\frac{|z|+a}{2}} + i \frac{b}{|b|} \sqrt{\frac{|z|-a}{2}} \right] \quad \text{Where } z = a+ib \text{ and } b \neq 0$$

Note : If b is negative, $\frac{b}{|b|} = -1$, x and y have different signs.

If b is positive, $\frac{b}{|b|} = 1$, x and y have same sign.

10. Find the square roots of (i) $4+3i$ (ii) $-6+8i$ (iii) $-5-12i$

2M

(i) Let $4+3i$

$$|z| = \sqrt{16+9} = \sqrt{25} = 5$$

Here $a = 4$ and $b = 3$

Since b is positive, $\frac{b}{|b|} = \frac{3}{|3|} = 1$

$$\begin{aligned} \sqrt{4+3i} &= \pm \left[\sqrt{\frac{5+4}{2}} + i(1) \sqrt{\frac{5-4}{2}} \right] \\ &= \pm \left[\sqrt{\frac{9}{2}} + i \sqrt{\frac{1}{2}} \right] = \pm \left[\frac{3}{\sqrt{2}} + i \frac{1}{\sqrt{2}} \right] \end{aligned}$$

(ii) Let $z = -6+8i$

PTA-2

$$|z| = \sqrt{36+64} = \sqrt{100} = 10$$

JUL-24

Here $a = -6$ and $b = 8$

Since b is positive, $\frac{b}{|b|} = \frac{8}{|8|} = 1$

$$\begin{aligned} \sqrt{z} &= \pm \left[\sqrt{\frac{|z|+a}{2}} + i \frac{b}{|b|} \sqrt{\frac{|z|-a}{2}} \right] \\ \sqrt{-6+8i} &= \pm \left[\sqrt{\frac{10-6}{2}} + i(1) \sqrt{\frac{10+6}{2}} \right] \\ &= \pm \left[\sqrt{\frac{4}{2}} + i \sqrt{\frac{16}{2}} \right] \\ &= \pm [\sqrt{2} + i\sqrt{8}] \\ &= \pm [\sqrt{2} + i2\sqrt{2}] = \pm \sqrt{2} [1 + 2i] \end{aligned}$$

(iii) Let $z = -5-12i$

$$|z| = \sqrt{25+144} = \sqrt{169} = 13$$

Here $a = -5$ and $b = -12$

Since b is negative, $\frac{b}{|b|} = \frac{-12}{|-12|} = -1$

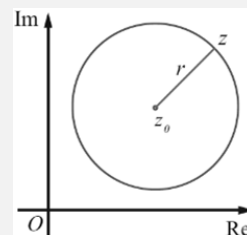
$$\begin{aligned} \sqrt{z} &= \pm \left[\sqrt{\frac{|z|+a}{2}} + i \frac{b}{|b|} \sqrt{\frac{|z|-a}{2}} \right] \\ \sqrt{-5-12i} &= \pm \left[\sqrt{\frac{13-5}{2}} + i(-1) \sqrt{\frac{13+5}{2}} \right] \\ &= \pm \left[\sqrt{\frac{8}{2}} - i \sqrt{\frac{18}{2}} \right] \\ &= \pm [\sqrt{4} - \sqrt{9}i] \\ &= \pm [2 - 3i] \end{aligned}$$

EXERCISE 2.6

Video class

Concept Corner**Geometry and locus of a complex numbers**

- ❖ Equation of complex form of a circle is $|z - z_0| = r$
where z_0 is the fixed (centre) complex number and r is the fixed positive real number (radius of the circle).
- ❖ $|z - z_0| < r$ represents the points interior of the circle.
- ❖ $|z - z_0| > r$ represents the points exterior of the circle
- ❖ If $z = \frac{a+ib}{c+id}$ then (i) $Re(z) = \frac{ac+bd}{c^2+d^2}$ (ii) $Im(z) = \frac{bc-ad}{c^2+d^2}$



1. If $z = x + iy$ is a complex number such that

$$\left| \frac{z-4i}{z+4i} \right| = 1 \text{ show that the locus of } z \text{ is real axis.}$$

$$\left| \frac{z-4i}{z+4i} \right| = 1 \quad \text{JUL-21}$$

$$|z - 4i| = |z + 4i| \quad \text{3M}$$

$$|x + iy - 4i| = |x + iy + 4i|$$

$$|x + i(y - 4)| = |x + i(y + 4)|$$

$$\sqrt{x^2 + (y - 4)^2} = \sqrt{x^2 + (y + 4)^2}$$

Squaring on both sides, we get

$$\cancel{x^2} + (y - 4)^2 = \cancel{x^2} + (y + 4)^2$$

$$[\because (a + b)^2 - (a - b)^2 = 4ab]$$

$$(y - 4)^2 - (y + 4)^2 = 0$$

$$4y(4) = 0$$

$$16y = 0$$

$$y = 0$$

$\therefore$ The locus of z is real axis.

2. If $z = x + iy$ is a complex number such that

$$\operatorname{Im} \left(\frac{2z+1}{iz+1} \right) = 0, \text{ show that the locus of } z \text{ is}$$

$$2x^2 + 2y^2 + x - 2y = 0. \quad \text{5M}$$

$$\frac{2z+1}{iz+1} = \frac{2(x+iy)+1}{i(x+iy)+1}$$

$$= \frac{2x+i2y+1}{ix+i^2y+1}$$

$$= \frac{(2x+1)+i2y}{(1-y)+ix}$$

$$= \frac{(2x+1)+i2y}{(1-y)+ix} \times \frac{(1-y)-ix}{(1-y)-ix}$$

$$= \text{Realpart} + i \left(\frac{2y(1-y)+(2x+1)(-x)}{x^2+(1-y)^2} \right) = 0$$

$$\text{Given: } \operatorname{Im} \left(\frac{2z+1}{iz+1} \right) = 0$$

$$\frac{2y(1-y)-(2x+1)x}{x^2+(1-y)^2} = 0$$

$$2y - 2y^2 - 2x^2 - x = 0$$

$$2x^2 + 2y^2 + x - 2y = 0$$

$\therefore$ The locus of z is $2x^2 + 2y^2 + x - 2y = 0$

3. Obtain the Cartesian form of the locus of $z = x + iy$ in each of the following cases:

(i) $[Re(iz)]^2 = 3$ (ii) $\operatorname{Im}[(1-i)z + 1] = 0$ (iii) $|z + i| = |z - 1|$ (iv) $\bar{z} = z^{-1}$

(i) $z = x + iy$ 2M

$$[Re(i(x + iy))]^2 = 3$$

$$[Re(-y + ix)]^2 = 3$$

$$(-y)^2 = 3$$

$$y^2 = 3$$

$\therefore$ The locus of z is $y^2 = 3$

(iii) $|z + i| = |z - 1|$ APR-22, JUL-20

$$\Rightarrow |x + iy + i| = |x + iy - 1| \quad \text{5M}$$

$$\Rightarrow |x + i(y + 1)| = |(x - 1) + iy|$$

$$\Rightarrow \sqrt{x^2 + (y + 1)^2} = \sqrt{(x - 1)^2 + y^2}$$

Squaring on both sides, we get

$$x^2 + y^2 + 2y + 1 = x^2 - 2x + 1 + y^2$$

$$x^2 + y^2 + 2y + 1 - x^2 + 2x - 1 - y^2 = 0$$

$$2x + 2y = 0$$

$$\div \text{ by } 2 \quad x + y = 0$$

$\therefore$ The locus of z is $x + y = 0$

(ii) $(1 - i)z + 1 = (1 - i)(x + iy) + 1$ 3M

$$= x + iy - ix - i^2y + 1$$

$$= (x + y + 1) + i(y - x)$$

$$\operatorname{Im}[(1 - i)z + 1] = y - x$$

Given:

$$\operatorname{Im}[(1 - i)z + 1] = 0$$

$$\Rightarrow y - x = 0 \text{ (or) } x - y = 0$$

$\therefore$ The locus of z is $x - y = 0$

(iv) $\bar{z} = z^{-1}$ 2M

$$z = x + iy; \bar{z} = x - iy$$

$$\bar{z} = \frac{1}{z}$$

$$x - iy = \frac{1}{x + iy}$$

$$(x - iy)(x + iy) = 1$$

$$x^2 + iyx - iyx - i^2y^2 = 1$$

$$x^2 + y^2 = 1$$

$\therefore$ The locus of z is $x^2 + y^2 = 1$

EXERCISE 2.9

PTA-1 JUL-22



Online Test

1. $i^n + i^{n+1} + i^{n+2} + i^{n+3}$ is

(1) 0

(2) 1

(3) -1

(4) i

$$i^n + i^{n+1} + i^{n+2} + i^{n+3} = i^n[1 + i + i^2 + i^3]$$

$$= i^n[1 + i - 1 - i]$$

$$= i^n(0)$$

$$i^n + i^{n+1} + i^{n+2} + i^{n+3} = 0$$

2. The value of $\sum_{n=1}^{13}(i^n + i^{n-1})$ is

(1) $1 + i$ (2) i

(3) 1

(4) 0

APR-25, 20

$$\sum_{n=1}^{13}(i^n + i^{n-1}) = \sum_{n=1}^{13}i^n + \sum_{n=1}^{13}i^{n-1}$$

$$= \sum_{n=1}^{12}i^n + i^{13} + \sum_{n=0}^{12}i^n$$

$$= \sum_{n=1}^{12}i^n + i^{4(3)+1} + i^0 + \sum_{n=1}^{12}i^n$$

$$= 0 + i^1 + 1 + 0$$

$$= 1 + i \quad [\because \sum_{n=1}^{12}i^n = i^1 + i^2 + i^3 + i^4 \dots \dots + i^{12} = 0]$$

3. The area of the triangle formed by the complex numbers z, iz and $z + iz$ in the Argand's diagram is

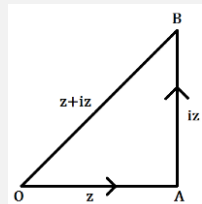
(1) $\frac{1}{2}|z|^2$ (2) $|z|^2$ (3) $\frac{3}{2}|z|^2$ (4) $2|z|^2$

PTA-2

$$\text{Area of the Triangle} = \frac{1}{2} \times OA \times OB = \frac{1}{2} \times |z| \times |iz|$$

$$= \frac{1}{2} \times |z| \times |z|$$

$$= \frac{1}{2}|z|^2$$



4. The conjugate of a complex number is $\frac{1}{i-2}$ then the complex number is

(1) $\frac{1}{i+2}$ (2) $\frac{-1}{i+2}$ (3) $\frac{-1}{i-2}$ (4) $\frac{1}{i-2}$

$$\bar{z} = \frac{1}{i-2}$$

$$z = \bar{\bar{z}} = \left[\frac{1}{-2+i} \right] = \frac{1}{-2-i} = \frac{-1}{2+i}$$

$$\therefore \text{The complex number is } z = \frac{-1}{2+i}$$

5. If $z = \frac{(\sqrt{3}+i)^3(3i+4)^2}{(8+6i)^2}$, then $|z|$ is equal to

(1) 0

(2) 1

(3) 2

(4) 3

JUL-24,PTA-6

$$|z| = \left| \frac{(\sqrt{3}+i)^3(3i+4)^2}{(8+6i)^2} \right|$$

$$= \frac{|\sqrt{3}+i|^3 |3i+4|^2}{|8+6i|^2}$$

$$= \frac{(\sqrt{4})^3 (\sqrt{25})^2}{(\sqrt{100})^2} = \frac{(2)^3 (5)^2}{(10)^2} = \frac{8 \times 25}{100} = 2$$

6. If z is a non zero complex number, such that $2iz^2 = \bar{z}$ then $|z|$ is

JUL-20

- (1) $\frac{1}{2}$ (2) 1 (3) 2 (4) 3

Given: $2iz^2 = \bar{z}$
 $|2iz^2| = |\bar{z}|$
 $|2i||z^2| = |z|$
 $2|z^2| = |z|$
 $2|z| = 1$
 $|z| = \frac{1}{2}$

7. If $|z - 2 + i| \leq 2$, then the greatest value of $|z|$ is

JUL-23

- (1) $\sqrt{3} - 2$ (2) $\sqrt{3} + 2$ (3) $\sqrt{5} - 2$ (4) $\sqrt{5} + 2$

$|z - 2 + i| \leq 2 \Rightarrow 2$ Represents all points inside or on the circle with centre at $C(2, -1)$ and radius 2 units and $|z|$ is the distance of such points from the origin.

$|z| \leq OC + r$
 $\Rightarrow |z| \leq |(2 - i) - (0 - i0)| + 2$
 $\Rightarrow |z| \leq |2 - i| + 2$
 $\Rightarrow |z| \leq \sqrt{5} + 2$

$\therefore$ The greatest value of $|z|$ is $\sqrt{5} + 2$

8. If $\left|z - \frac{3}{z}\right| = 2$, then the least value of $|z|$ is

- (1) 1 (2) 2 (3) 3 (4) 5

$\left|z - \frac{3}{z}\right| = 2$
 $\Rightarrow -2 \leq |z| - \left|\frac{3}{z}\right| \leq 2$
 $\Rightarrow -2 \leq |z| - \frac{3}{|z|} \leq 2$
 $\Rightarrow -2|z| \leq |z|^2 - 3 \leq 2|z|$
 From $-2|z| \leq |z|^2 - 3$, we get
 $0 \leq |z|^2 + 2|z| - 3$
 $\Rightarrow (|z| + 3)(|z| - 1) \geq 0$
 $\Rightarrow |z| \geq 1$ (or) $|z| \geq -3$

Since $|z|$ is a non-negative real number $|z|$ cannot be less than -3 .

Hence we must have $|z| \geq 1$

$\therefore$ The least value of $|z|$ is 1

9. If $|z| = 1$, then the value of $\frac{1+z}{1+\bar{z}}$ is

JUL-21, APR-23

- (1) z (2) $\bar{z}$ (3) $\frac{1}{z}$ (4) 1

Given: $|z| = 1$
 $\Rightarrow z\bar{z} = 1 \Rightarrow \bar{z} = \frac{1}{z}$
 $\frac{1+z}{\left(\frac{1+\bar{z}}{z}\right)} = (1+z) \times \frac{z}{1+\bar{z}} = z$

10. The solution of the equation $|z| - z = 1 + 2i$ is

- (1) $\frac{3}{2} - 2i$ (2) $-\frac{3}{2} + 2i$ (3) $2 - \frac{3}{2}i$ (4) $2 + \frac{3}{2}i$

Let $z = x + iy$

$$|z| = \sqrt{x^2 + y^2}$$

$$|z| - z = 1 + 2i$$

$$\Rightarrow \sqrt{x^2 + y^2} - (x + iy) = 1 + 2i$$

Equating real and imaginary parts, we get

$$\sqrt{x^2 + y^2} - x = 1 \quad \dots\dots\dots (1)$$

$$-y = 2 \Rightarrow y = -2$$

$$(1) \Rightarrow \sqrt{x^2 + y^2} = 1 + x \quad \text{On squaring, we get}$$

$$x^2 + 4 = (1 + x)^2$$

$$x^2 + 4 = 1 + 2x + x^2$$

$$4 - 1 = 2x$$

$$x = \frac{3}{2}$$

$$\therefore z = \frac{3}{2} - 2i$$

11. If $|z_1| = 1, |z_2| = 2, |z_3| = 3$ and $|9z_1z_2 + 4z_1z_3 + z_2z_3| = 12$, then the value of $|z_1 + z_2 + z_3|$ is

- (1) 1 (2) 2 (3) 3 (4) 4

$$|z_1| = 1 \Rightarrow |z_1|^2 = 1 \Rightarrow z_1\bar{z}_1 = 1 \Rightarrow z_1 = \frac{1}{\bar{z}_1}$$

$$|z_2| = 2 \Rightarrow |z_2|^2 = 4 \Rightarrow z_2\bar{z}_2 = 4 \Rightarrow z_2 = \frac{4}{\bar{z}_2}$$

$$|z_3| = 3 \Rightarrow |z_3|^2 = 9 \Rightarrow z_3\bar{z}_3 = 9 \Rightarrow z_3 = \frac{9}{\bar{z}_3}$$

$$\begin{aligned} z_1 + z_2 + z_3 &= \frac{1}{\bar{z}_1} + \frac{4}{\bar{z}_2} + \frac{9}{\bar{z}_3} \\ &= \frac{\bar{z}_2\bar{z}_3 + 4\bar{z}_1\bar{z}_3 + 9\bar{z}_1\bar{z}_2}{\bar{z}_1\bar{z}_2\bar{z}_3} \\ &= \frac{9\bar{z}_1\bar{z}_2 + 4\bar{z}_1\bar{z}_3 + \bar{z}_2\bar{z}_3}{\bar{z}_1\bar{z}_2\bar{z}_3} \\ &= \frac{(9z_1z_2 + 4z_1z_3 + z_2z_3)}{\bar{z}_1\bar{z}_2\bar{z}_3} \end{aligned}$$

$$z_1 + z_2 + z_3 = \left[\frac{9z_1z_2 + 4z_1z_3 + z_2z_3}{z_1z_2z_3} \right]$$

$$|z_1 + z_2 + z_3| = \left| \left[\frac{9z_1z_2 + 4z_1z_3 + z_2z_3}{z_1z_2z_3} \right] \right|$$

$$= \left| \frac{9z_1z_2 + 4z_1z_3 + z_2z_3}{z_1z_2z_3} \right|$$

$$= \frac{|9z_1z_2 + 4z_1z_3 + z_2z_3|}{|z_1||z_2||z_3|}$$

$$|z_1 + z_2 + z_3| = \frac{12}{1 \times 2 \times 3} = 2$$

APR-24, JUL-23

12. If z is a complex number such that $z \in \mathbb{C} \setminus \mathbb{R}$ and $z + \frac{1}{z} \in \mathbb{R}$, then $|z|$ is

- (1) 0 (2) 1 (3) 2 (4) 3

$\frac{1}{z}$ is conjugate of z

$\bar{z}$ is conjugate of z

$$\Rightarrow z\bar{z} = 1$$

$$\Rightarrow |z|^2 = 1$$

$$\therefore |z| = 1$$

13. z_1, z_2 and z_3 are complex numbers such that $z_1 + z_2 + z_3 = 0$ and $|z_1| = |z_2| = |z_3| = 1$ then $z_1^2 + z_2^2 + z_3^2$ is

PTA-5

- (1) 3 (2) 2 (3) 1 (4) 0

$$|z_1| = 1 \Rightarrow |z_1|^2 \Rightarrow z_1 \bar{z}_1 = 1 \Rightarrow z_1 = \frac{1}{\bar{z}_1}$$

Similarly $|z_2| = 1 \Rightarrow z_2 = \frac{1}{\bar{z}_2}$, $|z_3| = 1 \Rightarrow z_3 = \frac{1}{\bar{z}_3}$

$$z_1 + z_2 + z_3 = 0 \Rightarrow \frac{1}{\bar{z}_1} + \frac{1}{\bar{z}_2} + \frac{1}{\bar{z}_3} = 0$$

$$\Rightarrow \frac{\bar{z}_2 \bar{z}_3 + \bar{z}_1 \bar{z}_3 + \bar{z}_1 \bar{z}_2}{\bar{z}_1 \bar{z}_2 \bar{z}_3} = 0$$

$$\Rightarrow \frac{\bar{z}_1 \bar{z}_2 + \bar{z}_2 \bar{z}_3 + \bar{z}_1 \bar{z}_3}{\bar{z}_1 \bar{z}_2 \bar{z}_3} = 0$$

$$\Rightarrow z_1 z_2 + z_2 z_3 + z_1 z_3 = 0$$

[If $\bar{z} = 0$ then $z = 0$]

Given $z_1 + z_2 + z_3 = 0$

$$(z_1 + z_2 + z_3)^2 = 0$$

$$\Rightarrow z_1^2 + z_2^2 + z_3^2 + 2(z_1 z_2 + z_2 z_3 + z_1 z_3) = 0$$

$$\Rightarrow z_1^2 + z_2^2 + z_3^2 + 2(0) = 0$$

$$\therefore z_1^2 + z_2^2 + z_3^2 = 0$$

14. If $\frac{z-1}{z+1}$ is purely imaginary, then $|z|$ is

PTA-5

- (1) $\frac{1}{2}$ (2) 1 (3) 2 (4) 3

If z is purely imaginary, then $\bar{z} = -z$

Given that $\frac{z-1}{z+1}$ is purely imaginary,

$$\therefore \left(\frac{z-1}{z+1} \right) = - \frac{z-1}{z+1}$$

$$\frac{\bar{z}-1}{\bar{z}+1} = \frac{1-z}{1+z}$$

$$(\bar{z} - 1)(1 + z) = (1 - z)(\bar{z} + 1)$$

$$\bar{z} + z\bar{z} - 1 - z = \bar{z} + 1 - z\bar{z} - z$$

$$z\bar{z} + z\bar{z} = 1 + 1$$

$$2z\bar{z} = 2$$

$$z\bar{z} = 1$$

$$|z|^2 = 1$$

$$\therefore |z| = 1$$

15. If $z = x + iy$ is a complex number such that $|z + 2| = |z - 2|$, then the locus of z is

PTA-3

JUL-22

- (1) real axis (2) imaginary axis (3) ellipse (4) circle

$$|z + 2| = |z - 2|$$

$$|x + iy + 2|^2 = |x + iy - 2|^2$$

$$(x + 2)^2 + y^2 = (x - 2)^2 + y^2$$

$$x^2 + 4x + 4 + y^2 = x^2 - 4x + 4 + y^2$$

$$4x + 4x = 4 - 4$$

$$8x = 0$$

$$x = 0$$

$\therefore$ The locus of z is imaginary axis.

16. The principal argument of $\frac{3}{-1+i}$ is

JUL-24,MDL

- (1) $-\frac{5\pi}{6}$ (2) $-\frac{2\pi}{3}$ (3) $-\frac{3\pi}{4}$ (4) $-\frac{\pi}{2}$

$$\arg\left(\frac{3}{-1+i}\right) = \arg 3 - \arg(-1 + i) \text{ But } \frac{3}{-1+i} = -\frac{3}{2} - \frac{3}{2}i \text{ lies in the 3rd quadrant}$$

$$= 0 - \left(\pi - \frac{\pi}{4}\right) = -\frac{3\pi}{4}$$

17. The principal argument of $(\sin 40^\circ + i \cos 40^\circ)^5$ is

- (1) -110° (2) -70° (3) 70° (4) 110°

$$\begin{aligned}(\sin 40 + i \cos 40)^5 &= [\sin(90 - 50) + i \cos(90 - 40)]^5 \\&= (\cos 50 + i \sin 50)^5 \\&= \cos 250 + i \sin 250 \\&= \cos(360 - 110) + i \sin(360 - 110) \\&= \cos 110 - i \sin(110^\circ) = \cos(-110^\circ) + i \sin(-110^\circ)\end{aligned}$$

$\therefore$ The principal argument of $(\sin 40 + i \cos 40)^5$ is -110°

18. If $(1 + i)(1 + 2i)(1 + 3i) \dots (1 + ni) = x + iy$ then $2.5.10 \dots (1 + n^2)$ is

- (1) 1 (2) i (3) $x^2 + y^2$ (4) $1 + n^2$ APR-24, 22, 20

$$\begin{aligned}|(1 + i)(1 + 2i)(1 + 3i) \dots (1 + ni)| &= |x + iy| \\|1 + i||1 + 2i||1 + 3i| \dots |1 + ni| &= |x + iy| \\\sqrt{1+1}\sqrt{1+4}\sqrt{1+9} \dots \sqrt{1+n^2} &= \sqrt{x^2 + y^2} \\\sqrt{2}\sqrt{5}\sqrt{10} \dots \sqrt{1+n^2} &= \sqrt{x^2 + y^2}\end{aligned}$$

Squaring on both sides, we get, $2.5.10 \dots (1 + n^2) = x^2 + y^2$

19. If $\omega \neq 1$ is a cubic root of unity and $(1 + \omega)^7 = A + B\omega$, then (A, B) equals

- (1) (1,0) (2) $(-1,1)$ (3) (0,1) (4) (1, 1) PTA-4

$$\begin{aligned}(1 + \omega)^7 &= A + B\omega \\\Rightarrow (-\omega^2)^7 &= A + B\omega \\\Rightarrow -\omega^{14} &= A + B\omega \\\Rightarrow -\omega^{4(3)+2} &= A + B\omega \\\Rightarrow -\omega^2 &= A + B\omega \\\Rightarrow 1 + \omega &= A + B\omega\end{aligned}$$

Since ω is a cubic root of unity

$$\begin{aligned}1 + \omega + \omega^2 &= 0 \\\Rightarrow 1 + \omega &= -\omega^2\end{aligned}$$

Equating the corresponding $A = 1$ and $B = 1$

$\therefore (A, B) = (1, 1)$

20. The principal argument of the complex number $\frac{(1+i\sqrt{3})^2}{4i(1-i\sqrt{3})}$ is

- (1) $\frac{2\pi}{3}$ (2) $\frac{\pi}{6}$ (3) $\frac{5\pi}{6}$ (4) $\frac{\pi}{2}$ PTA-3

Let $z_1 = 1 + i\sqrt{3}, z_2 = 4i, z_3 = 1 - i\sqrt{3}$

$$\arg z_1 = \tan^{-1} \left| \frac{\sqrt{3}}{1} \right| = \tan^{-1}(\sqrt{3}) = \frac{\pi}{3}$$

$$\arg z_2 = \tan^{-1} \left| \frac{4}{0} \right| = \tan^{-1}(\infty) = \frac{\pi}{2}$$

$$\arg z_3 = -\tan^{-1} \left| \frac{-\sqrt{3}}{1} \right| = -\tan^{-1}(\sqrt{3})$$

$$\arg z_3 = -\frac{\pi}{3}$$

$[\because 1 - i\sqrt{3}$ lies in the 4th quadrant $\theta = -\alpha]$

$$\arg z = \arg \left[\frac{z_1^2}{z_2 z_3} \right]$$

$$= 2\arg z_1 - \arg z_2 - \arg z_3$$

$$= 2\left(\frac{\pi}{3}\right) - \frac{\pi}{2} - \left(-\frac{\pi}{3}\right) = \pi - \frac{\pi}{2} = \frac{\pi}{2}$$

21. If α and β are the roots of $x^2 + x + 1 = 0$, then $\alpha^{2020} + \beta^{2020}$ is

(1) -2

(2) -1

(3) 1

(4) 2

$$x = \frac{-1 \pm \sqrt{1-4}}{2} = \frac{-1 \pm i\sqrt{3}}{2}$$

$$\alpha = \omega = \frac{-1+i\sqrt{3}}{2} \text{ and } \beta = \omega^2 = \frac{-1-i\sqrt{3}}{2}$$

$$\alpha^{2020} + \beta^{2020} = \omega^{2020} + (\omega^2)^{2020} = \omega^{2020} + \omega^{4040} = \omega + \omega^2 = -1$$

$$\left[\begin{array}{l} \because 1 + \omega + \omega^2 = 0 \\ \Rightarrow \omega + \omega^2 = -1 \end{array} \right]$$

22. The product of all four values of $\left(\cos \frac{\pi}{3} + i \sin \frac{\pi}{3}\right)^{\frac{3}{4}}$ is

PTA-1, 6

(1) -2

(2) -1

(3) 1

(4) 2

$$\text{let } x = \left(\cos \frac{\pi}{3} + i \sin \frac{\pi}{3}\right)^{\frac{3}{4}}$$

$$= \left(\cos \frac{3\pi}{3} + i \sin \frac{3\pi}{3}\right)^{\frac{1}{4}}$$

$$= (\cos \pi + i \sin \pi)^{\frac{1}{4}}$$

$$= \left(\cos \left(\frac{2k+1}{4}\pi\right) + i \sin \left(\frac{2k+1}{4}\pi\right)\right), k = 0, 1, 2, 3$$

$$\text{product of values} = \text{cis} \left(\frac{\pi}{4} + \frac{3\pi}{4} + \frac{5\pi}{4} + \frac{7\pi}{4}\right) = \text{cis} \left(\frac{16\pi}{4}\right) = \text{cis} 4\pi = 1$$

23. If $\omega \neq 1$ is a cubic root of unity and $\begin{vmatrix} 1 & 1 & 1 \\ 1 & -\omega^2 - 1 & \omega^2 \\ 1 & \omega^2 & \omega^7 \end{vmatrix} = 3k$, then k is equal to

(1) 1

(2) -1

(3) $\sqrt{3}i$ (4) $-\sqrt{3}i$

$$\begin{vmatrix} 1 & 1 & 1 \\ 1 & -\omega^2 - 1 & \omega^2 \\ 1 & \omega^2 & \omega^7 \end{vmatrix} = 3k$$

$$\begin{vmatrix} 1 & 1 & 1 \\ 1 & -\omega^2 - 1 & \omega^2 \\ 3 & 0 & 0 \end{vmatrix} R_3 \Rightarrow R_1 + R_2 + R_3$$

$$\Rightarrow 3(1 + \omega^2 + \omega^2) = 3k$$

$$k \Rightarrow 1 + \omega^2 + \omega^2 = -\omega + \omega^2 = -\omega - 1 - \omega = -1 - 2\omega = -1 - 2\left(\frac{-1}{2} + i\frac{\sqrt{3}}{2}\right) = -\sqrt{3}i$$

24. The value of $\left(\frac{1+\sqrt{3}i}{1-\sqrt{3}i}\right)^{10}$ is

PTA-4

(1) $\text{cis} \frac{2\pi}{3}$ (2) $\text{cis} \frac{4\pi}{3}$ (3) $-\text{cis} \frac{2\pi}{3}$ (4) $-\text{cis} \frac{4\pi}{3}$

$$\text{Let } 1 + \sqrt{3}i = r[\cos \theta + i \sin \theta]$$

$$r = \sqrt{1+3} = \sqrt{4} = 2$$

$$\alpha = \tan^{-1} \left| \frac{\sqrt{3}}{1} \right| = \frac{\pi}{3}$$

$1 + \sqrt{3}i$ lies in the 1st quadrant

$$\therefore \theta = \alpha = \frac{\pi}{3}$$

$$\therefore 1 + \sqrt{3}i = 2 \left[\cos \frac{\pi}{3} + i \sin \frac{\pi}{3} \right]$$

$$\text{Similarly } 1 - \sqrt{3}i = 2 \left[\cos \frac{\pi}{3} - i \sin \frac{\pi}{3} \right]$$

$$= 2 \left[\cos \left(\frac{-\pi}{3}\right) + i \sin \left(\frac{-\pi}{3}\right) \right]$$

$$\frac{1+\sqrt{3}i}{1-\sqrt{3}i} = \frac{2 \left[\cos \frac{\pi}{3} + i \sin \frac{\pi}{3} \right]}{2 \left[\cos \left(\frac{-\pi}{3}\right) + i \sin \left(\frac{-\pi}{3}\right) \right]}$$

$$= \cos \left(\frac{\pi}{3} + \frac{\pi}{3}\right) + i \sin \left(\frac{\pi}{3} + \frac{\pi}{3}\right)$$

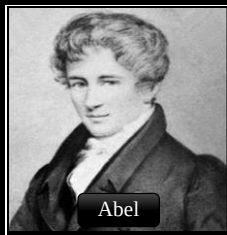
$$= \text{cis} \frac{2\pi}{3}$$

$$\left(\frac{1+\sqrt{3}i}{1-\sqrt{3}i}\right)^{10} = \left(\text{cis} \frac{2\pi}{3} + i \sin \frac{2\pi}{3}\right)^{10}$$

$$= \cos \frac{20\pi}{3} + i \sin \frac{20\pi}{3}$$

$$= \cos \left(6\pi + \frac{2\pi}{3}\right) + i \sin \left(6\pi + \frac{2\pi}{3}\right)$$

$$= \cos \frac{2\pi}{3} + i \sin \frac{2\pi}{3} = \text{cis} \frac{2\pi}{3}$$



Abel

Chapter - 3

Theory of Equations

EXERCISE 3.1



Video class

Concept Corner

Different types of polynomial equation

Definition:

$P \equiv P(x) = a_n x^n + a_{n-1} x^{n-1} + \dots a_1 x + a_0$, where n is positive integer, $a_n, a_{n-1} \dots \dots a_1, a_0$ are constants is a polynomial of degree n in one variable x .

If the coefficients of P are real. Then

- ❖ P is a polynomial over R
- ❖ P is a polynomial over C
- ❖ dP is a polynomial over Q
- ❖ P is a polynomial over Z

Polynomial function: $P(x) = a_n x^n + a_{n-1} x^{n-1} + \dots a_1 x + a_0$

Polynomial equation: $a_n x^n + a_{n-1} x^{n-1} + \dots a_1 x + a_0 = 0$

Note:

- ❖ a_n is called leading coefficient.
- ❖ $a_n, a_{n-1}, \dots \dots a_1, a_0$ are coefficients or constants.
- ❖ $a_n x^n$ is called leading term.

Degree of a polynomial:

The exponent of the highest degree term in a polynomial is known as its degree.

Example:

- (i) $f(x) = 3x + \frac{1}{2}$ is a polynomial degree 1 in the variable x .
- (ii) $g(y) = 2y^2 - \frac{3}{2}y + 7$ is a polynomial degree 2 in the variable y .
- (iii) $h(z) = 8z^3 - 4z^2 + 2z - \frac{5}{\sqrt{2}}$ is a polynomial of degree 3 in the variable z .

Constant polynomial: A polynomial of degree zero 0 is called constant polynomial.

Example: $f(x) = 8$, $g(y) = -\frac{7}{3}$, $p(t) = 1$

Zero polynomial:

The constant 0 or $f(x) = 0$ is called zero polynomial. The degree is not defined (∞).

Example: $f(x) = 0$, $g(x) = 0x$, $p(y) = 0y^2$, $q(z) = 0z^{15}$

Linear polynomial: A polynomial of degree 1 is called linear polynomial.

Example: $p(x) = 4x - 3$; $q(y) = 3y$; $f(t) = \sqrt{3}t + 5$

A linear polynomial may be monomial or binomial

Example: $p(x) = 4x - 3$ is a Binomial. & $q(y) = 3y$ is a monomial.

Quadratic polynomial: A polynomial of degree 2 is called Quadratic polynomial.

Example: $f(x) = 5x^2 - 3x + 2$ [trinomial]

$$g(y) = -\frac{4}{3}y^2 + 3 \text{ [Binomial]}$$

$$h(t) = -\frac{1}{2}t^2 \text{ [Monomial]}$$

Degree 2

Cubic polynomial: A polynomial of degree 3 is called Cubic polynomial.

$$\left. \begin{array}{l} \text{Example : } f(x) = \frac{9}{7}x^3 - 5x^2 + 4x \text{ [trinomial]} \\ h(t) = -5t^3 + t^2 \text{ [Binomial]} \\ \phi(t) = 4.5z^3 \text{ [Monomial]} \end{array} \right\} \text{Degree 3}$$

Quartic polynomial: A polynomial of degree 4 is called Quartic polynomial.

Example: $f(x) = 5x^4 - 3x^3 + x^2$, $h(x) = x^4 + x^3 - 2$

Note:

- ❖ The degree of a polynomial is non-negative integer.
- ❖ $-2.5, \frac{4}{3}, \sqrt{3} \dots$ are not degree of any polynomial.
- ❖ So, $3x^{-2} + 1$, $5x^{\frac{1}{2}} + 1$, $\sqrt{x} + 5$, $\dots$ are not polynomial.
- ❖ $\sin^2 x + \cos^2 x = 1$ is an Identity. But $\sin x + \cos x = 1$ & $\sin^3 x + \cos^3 x = 1$ are equations.

Quadratic equations: Standard form: $ax^2 + bx + c = 0$: $a \neq 0$.

$\Delta = b^2 - 4ac$ is called discriminant in any Quadratic equation, we have 3 types of roots.

Discriminant	Roots
1. $\Delta = 0$	The roots are real and equal
2. $\Delta > 0$ (+ve)	The roots are real and distinct
3. $\Delta < 0$ (-ve)	The roots are not real (i.e.) no real roots

Vieta's formula for Quadratic Equation:

Given two roots: Let α and β be the roots of the quadratic equation $ax^2 + bx + c = 0$, $a \neq 0$.

Then i) Sum of the roots $= \alpha + \beta = -\frac{b}{a}$ ii) Product of the roots $= \alpha\beta = \frac{c}{a}$

Formation of Quadratic equation:

If α & β are the roots then the form of Quadratic equation is $x^2 - (\alpha + \beta)x + \alpha\beta = 0$

Vieta's formula for cubic equation:

Let α, β & γ be the roots of the equation $ax^3 + bx^2 + cx + d = 0$, $a \neq 0$.

Then, Sum of the roots : $\alpha + \beta + \gamma = -\frac{b}{a}$

Sum of the roots taken two at a time: $\alpha\beta + \beta\gamma + \gamma\alpha = \frac{c}{a}$

Product of the roots: $\alpha\beta\gamma = -\frac{d}{a}$

If a cubic polynomial has roots α, β , & γ

Then coefficient of $x^2 = -(\alpha + \beta + \gamma)$

coefficient of $x = \alpha\beta + \beta\gamma + \gamma\alpha$

constant term $= -\alpha\beta\gamma$.

Vieta's Formula for Polynomial equation of degree > 3

If a monic polynomial equation of degree n has roots $\alpha_1, \alpha_2, \dots, \alpha_n$, then

Coefficient of $x^{n-1} = \sum_1 = -\sum \alpha_1$

Coefficient of $x^{n-2} = \sum_2 = \sum \alpha_1 \alpha_2$

Coefficient of $x^{n-3} = \sum_3 = -\sum \alpha_1 \alpha_2 \alpha_3$

Coefficient of $x = \sum_{n-1} = (-1)^{n-1} \sum \alpha_1 \alpha_2 \dots \alpha_{n-1}$

Coefficient of $x^0 = \text{constant term} = \sum_n = (-1)^n \alpha_1 \alpha_2 \dots \alpha_n$

The Fundamental Theorem of Algebra :

Every polynomial equation of degree $n \geq 1$ has atleast one root in $\mathbb{C}$.

1. If the sides of a cubic box are increased by 1, 2, 3 units respectively to form a cuboid, then the volume is increased by 52 cubic units. Find the volume of the cuboid. JUL-21 3M

Let x be the side of a cube.

Increased sides are $x + 1, x + 2, x + 3$

$$\text{volume of cuboid} = x^3 + 52$$

$$(x + 1)(x + 2)(x + 3) = x^3 + 52$$

$$\cancel{x^3} + 6x^2 + 11x + 6 = \cancel{x^3} + 52$$

$$6x^2 + 11x + 6 - 52 = 0$$

$$6x^2 + 11x - 46 = 0$$

$$\left(x + \frac{23}{6}\right)(x - 2) = 0$$

$$x = -\frac{23}{6}; x = 2$$

$$\therefore x = 2$$

$$\therefore \text{volume of cube} = x^3 = 2^3 = 8$$

$$\text{volume of cuboid} = 8 + 52 = \mathbf{60}$$

2. Construct a cubic equation with roots (i) 1, 2, and 3 (ii) 1, 1, and -2 (iii) $2, \frac{1}{2}$ and 1. 2M

(i) Given roots 1, 2, and 3

$$\Sigma 1 = 1 + 2 + 3 = 6$$

$$\Sigma 2 = 1 \times 2 + 2 \times 3 + 3 \times 1 = 2 + 6 + 3 = 11$$

$$\Sigma 3 = 1 \times 2 \times 3 = 6$$

$\therefore$ Required cubic equation is

$$x^3 - x^2(\Sigma 1) + x(\Sigma 2) - \Sigma 3 = 0$$

$$x^3 - x^2(6) + x(11) - 6 = 0$$

$$\mathbf{x^3 - 6x^2 + 11x - 6 = 0}$$

(ii) Given roots 1, 1, and -2

$$\Sigma 1 = 1 + 1 - 2 = 0$$

$$\begin{aligned}\Sigma 2 &= 1 \times 1 + 1 \times (-2) + (-2) \times 1 \\ &= 1 - 2 - 2 = -3\end{aligned}$$

$$\Sigma 3 = 1 \times 1 \times (-2) = -2$$

$\therefore$ Required cubic equation is

$$x^3 - x^2(\Sigma 1) + x(\Sigma 2) - \Sigma 3 = 0$$

$$x^3 - x^2(0) + x(-3) - (-2) = 0$$

$$x^3 - 0 - 3x + 2 = 0$$

$$\mathbf{x^3 - 3x + 2 = 0}$$

(iii) Given roots $2, \frac{1}{2}$ and 1.

$$\Sigma 1 = 2 + \frac{1}{2} + 1 = \frac{7}{2}$$

$$\Sigma 2 = 2 \times \left(\frac{1}{2}\right) + \left(\frac{1}{2}\right) \times 1 + 1 \times 2 = \frac{7}{2}$$

$$\Sigma 3 = 2 \times \left(\frac{1}{2}\right) \times 1 = 1$$

$\therefore$ Required cubic equation is

$$x^3 - x^2(\Sigma 1) + x(\Sigma 2) - \Sigma 3 = 0$$

$$x^3 - x^2\left(\frac{7}{2}\right) + x\left(\frac{7}{2}\right) - (1) = 0$$

$$\mathbf{2x^3 - 7x^2 + 7x - 2 = 0}$$

3. If α, β and γ are the roots of the cubic equation $x^3 + 2x^2 + 3x + 4 = 0$, form a cubic equation whose roots are (i) $2\alpha, 2\beta, 2\gamma$ (ii) $\frac{1}{\alpha}, \frac{1}{\beta}, \frac{1}{\gamma}$ (iii) $-\alpha, -\beta - \gamma$ 3M

Given equation $x^3 + 2x^2 + 3x + 4 = 0$ Where $a = 1, b = 2, c = 3, d = 4$

let α, β & γ be the roots $\sum 1 = \alpha + \beta + \gamma = -\frac{b}{a} = -2$

$$\sum 2 = \alpha\beta + \beta\gamma + \gamma\alpha = \frac{c}{a} = 3$$

$$\sum 3 = \alpha\beta\gamma = -\frac{d}{a} = -4$$

(i) The new roots are $2\alpha, 2\beta, 2\gamma$

$$\sum 1 = 2\alpha + 2\beta + 2\gamma = 2(\alpha + \beta + \gamma) = 2(-2) = -4$$

$$\sum 2 = 2\alpha \cdot 2\beta + 2\beta \cdot 2\gamma + 2\gamma \cdot 2\alpha = 4(\alpha\beta + \beta\gamma + \gamma\alpha) = 4(3) = 12$$

$$\sum 3 = 2\alpha \cdot 2\beta \cdot 2\gamma = 8(\alpha\beta\gamma) = -32$$

$\therefore$ Required cubic equation is $x^3 - x^2(\sum 1) + x(\sum 2) - \sum 3 = 0$

$$x^3 - x^2(-4) + x(12) - (-32) = 0$$

$$x^3 + 4x^2 + 12x + 32 = 0$$

(ii) The new roots are $\frac{1}{\alpha}, \frac{1}{\beta}, \frac{1}{\gamma}$

$$\sum 1 = \frac{1}{\alpha} + \frac{1}{\beta} + \frac{1}{\gamma} = \frac{\alpha\beta + \beta\gamma + \gamma\alpha}{\alpha\beta\gamma} = \frac{3}{-4} = -\frac{3}{4}$$

$$\sum 2 = \frac{1}{\alpha} \cdot \frac{1}{\beta} + \frac{1}{\beta} \cdot \frac{1}{\gamma} + \frac{1}{\gamma} \cdot \frac{1}{\alpha} = \frac{1}{\alpha\beta} + \frac{1}{\beta\gamma} + \frac{1}{\gamma\alpha} = \frac{\alpha + \beta + \gamma}{\alpha\beta\gamma} = \frac{-2}{-4} = \frac{1}{2}$$

$$\sum 3 = \frac{1}{\alpha} \cdot \frac{1}{\beta} \cdot \frac{1}{\gamma} = \frac{1}{\alpha\beta\gamma} = \frac{1}{-4} = -\frac{1}{4}$$

$\therefore$ Required cubic equation is $x^3 - x^2(\sum 1) + x(\sum 2) - \sum 3 = 0$

$$x^3 + x^2\left(\frac{3}{4}\right) + x\left(\frac{1}{2}\right) - \left(-\frac{1}{4}\right) = 0$$

$$x^3 + \frac{3}{4}x^2 + \frac{1}{2}x + \frac{1}{4} = 0$$

$$\frac{4x^3 + 3x^2 + 2x + 1}{4} = 0$$

$$4x^3 + 3x^2 + 2x + 1 = 0$$

(iii) The new roots are $-\alpha, -\beta - \gamma$

$$\sum 1 = -\alpha - \beta - \gamma = -(\alpha + \beta + \gamma) = 2$$

$$\sum 2 = \alpha\beta + \beta\gamma + \gamma\alpha = 3$$

$$\sum 3 = -\alpha\beta\gamma = 4$$

$\therefore$ Required cubic equation is $x^3 - x^2(\sum 1) + x(\sum 2) - \sum 3 = 0$

$$x^3 - x^2(2) + x(3) - (4) = 0$$

$$x^3 - 2x^2 + 3x - 4 = 0$$

4. Solve the equation $3x^3 - 16x^2 + 23x - 6 = 0$ if the product of two roots is 1

Given equation $3x^3 - 16x^2 + 23x - 6 = 0$

Where $a = 3, b = -16, c = 23, d = -6$ 5M

Let the roots be $\alpha, \frac{1}{\alpha}, \beta$

$$\begin{aligned}\Sigma 1 &= \alpha + \frac{1}{\alpha} + \beta = -\frac{b}{a} = \frac{16}{3} \\ &= \alpha + \frac{1}{\alpha} + \beta = \frac{16}{3} \dots\dots\dots (1)\end{aligned}$$

$$\begin{aligned}\Sigma 2 &= \alpha \frac{1}{\alpha} + \frac{1}{\alpha} \beta + \beta \alpha = \frac{c}{a} = \frac{23}{3} \\ &= 1 + \frac{1}{\alpha} \beta + \beta \alpha = \frac{23}{3} \dots\dots\dots (2)\end{aligned}$$

$$\begin{aligned}\Sigma 3 &= \alpha \cdot \frac{1}{\alpha} \cdot \beta = -\frac{d}{a} = \frac{6}{3} = 2 \\ \beta &= 2\end{aligned}$$

$$\begin{aligned}\text{Sub in (1)} \quad \alpha + \frac{1}{\alpha} + 2 &= \frac{16}{3} \\ \alpha + \frac{1}{\alpha} &= \frac{16}{3} - 2 = \frac{16-6}{3} \\ \frac{\alpha^2+1}{\alpha} &= \frac{10}{3}\end{aligned}$$

$$\begin{aligned}3\alpha^2 + 3 &= 10\alpha \\ 3\alpha^2 - 10\alpha + 3 &= 0\end{aligned}$$

$$\begin{aligned}\left(\alpha - \frac{1}{3}\right)(\alpha - 3) &= 0 \\ \alpha &= \frac{1}{3}, \alpha = 3\end{aligned}$$

$\therefore$ The roots are $2, 3, \frac{1}{3}$

5. Find the sum of squares of roots of the equation $2x^4 - 8x^3 + 6x^2 - 3 = 0$ 3M

Given: $2x^4 - 8x^3 + 6x^2 - 3 = 0$ PTA-2

Let α, β, γ , & δ be the roots,

$$a = 2, b = -8, c = 6, d = 0, e = -3$$

$$\Sigma 1 = \alpha + \beta + \gamma + \delta = -\frac{b}{a} = \frac{8}{2} = 4$$

$$\begin{aligned}\Sigma 2 &= \alpha\beta + \alpha\gamma + \alpha\delta + \beta\gamma + \beta\delta + \gamma\delta \\ &= \frac{c}{a} = \frac{6}{2} = 3\end{aligned}$$

$$\begin{aligned}\Sigma 3 &= \alpha\beta\gamma + \alpha\beta\delta + \beta\gamma\delta + \alpha\gamma\delta \\ &= -\frac{d}{a} = 0\end{aligned}$$

$$\Sigma 4 = \alpha\beta\gamma\delta = \frac{e}{a} = -\frac{3}{2}$$

To find $\alpha^2 + \beta^2 + \gamma^2 + \delta^2$

$$\begin{aligned}\alpha^2 + \beta^2 + \gamma^2 + \delta^2 &= (\alpha + \beta + \gamma + \delta)^2 - \\ &\quad 2(\alpha\beta + \alpha\gamma + \alpha\delta + \beta\gamma + \beta\delta + \gamma\delta) \\ &= (4)^2 - 2(3) = 16 - 6\end{aligned}$$

$$\alpha^2 + \beta^2 + \gamma^2 + \delta^2 = 10$$

6. Solve the equation 5M

$x^3 - 9x^2 + 14x + 24 = 0$ if it is given that two of its roots are in the ratio 3: 2

Given: $x^3 - 9x^2 + 14x + 24 = 0$

Let roots be $3\alpha, 2\alpha, \beta$

$$3\alpha + 2\alpha + \beta = 9$$

$$5\alpha + \beta = 9 \dots\dots\dots (1)$$

$$3\alpha \cdot 2\alpha + 3\alpha\beta + 2\alpha\beta = 14$$

$$6\alpha^2 + 5\alpha\beta = 14 \dots\dots (2)$$

$$3\alpha \cdot 2\alpha \cdot \beta = -24$$

$$\alpha^2\beta = -4 \dots\dots\dots (3)$$

$$(1) \Rightarrow \beta = 9 - 5\alpha$$

Sub in (2), we get

$$6\alpha^2 + 5\alpha(9 - 5\alpha) = 14$$

$$19\alpha^2 - 45\alpha + 14 = 0$$

Solving we get

$$\alpha = 2 \text{ or } \frac{7}{19}$$

- (i) Sub $\alpha = \frac{7}{19}$ in (1)

$$\text{we get } \beta = \frac{136}{19}$$

$$\therefore 3\alpha = 3 \times \frac{7}{19} = \frac{21}{19}$$

$$2\alpha = 2 \times \frac{7}{19} = \frac{14}{19}$$

$$\beta = \frac{136}{19}$$

$$\therefore \text{The roots are } \frac{21}{19}, \frac{14}{19}, \frac{136}{19}$$

- (ii) $\alpha = 2$

$$(1) \Rightarrow \beta = 9 - 10$$

$$\beta = -1$$

Roots are $3\alpha, 2\alpha, \beta$

$$6, 4, -1$$

7. If α, β and γ are the roots of the polynomial equation $ax^3 + bx^2 + cx + d = 0$, find the value of $\sum \frac{\alpha}{\beta\gamma}$ in terms of the coefficients.

3M

Given: $ax^3 + bx^2 + cx + d = 0$

Let α, β , & γ be the roots

$$\sum 1 = \alpha + \beta + \gamma$$

$$= -\frac{b}{a}$$

$$\sum 2 = \alpha\beta + \beta\gamma + \gamma\alpha$$

$$= \frac{c}{a}$$

$$\sum 3 = \alpha\beta\gamma$$

$$= -\frac{d}{a}$$

To find $\sum \frac{\alpha}{\beta\gamma}$

$$= \frac{\alpha}{\beta\gamma} + \frac{\beta}{\alpha\gamma} + \frac{\gamma}{\alpha\beta}$$

$$= \frac{\alpha^2 + \beta^2 + \gamma^2}{\alpha\beta\gamma}$$

$$\sum \frac{\alpha}{\beta\gamma} = \frac{(\alpha + \beta + \gamma)^2 - 2(\alpha\beta + \beta\gamma + \gamma\alpha)}{\alpha\beta\gamma}$$

$$= \frac{\left(-\frac{b}{a}\right)^2 - 2\left(\frac{c}{a}\right)}{-\frac{d}{a}}$$

$$= \frac{\frac{b^2}{a^2} - \frac{2c}{a}}{-\frac{d}{a}}$$

$$= -\left(\frac{b^2 - 2ac}{a^2}\right) \cdot \frac{a}{d}$$

$$= -\left(\frac{b^2 - 2ac}{ad}\right)$$

$$\sum \frac{\alpha}{\beta\gamma} = \frac{2ac - b^2}{ad}$$

8. If α, β, γ & δ are the roots of the polynomial equation $2x^4 + 5x^3 - 7x^2 + 8 = 0$, find a quadratic equation with integer coefficients whose roots are $\alpha + \beta + \gamma + \delta$ and $\alpha\beta\gamma\delta$.

Given $2x^4 + 5x^3 - 7x^2 + 8 = 0$

2M

Let α, β, γ , & δ be the roots,

JUL-20

Here $a = 2, b = 5, c = -7, d = 0, e = 0$

$$\sum 1 = \alpha + \beta + \gamma + \delta = -\frac{b}{a} = -\frac{5}{2}$$

$$\sum 4 = \alpha\beta\gamma\delta = \frac{e}{a} = \frac{8}{2} = 4$$

The new roots are $\alpha + \beta + \gamma + \delta$, $\alpha\beta\gamma\delta$

$$\sum 1 = \alpha + \beta + \gamma + \delta + \alpha\beta\gamma\delta$$

$$= -\frac{5}{2} + 4 = \frac{-5+8}{2} = \frac{3}{2}$$

$$\sum 2 = (\alpha + \beta + \gamma + \delta)(\alpha\beta\gamma\delta)$$

$$= -\frac{5}{2}(4) = -10$$

$\therefore$ The required Quadratic equation:

$$x^2 - x(\sum 1) + \sum 2 = 0$$

$$x^2 - x\left(\frac{3}{2}\right) - 10 = 0$$

$$x^2 - \frac{3}{2}x - 10 = 0$$

$$2x^2 - 3x - 20 = 0$$

10. If the equations $x^2 + px + q = 0$ and $x^2 + p'x + q' = 0$ have a common root, show that it must be equal to $\frac{pq' - p'q}{q - q'}$ or $\frac{q - q'}{p' - p}$

Given: $x^2 + px + q = 0$, $x^2 + p'x + q' = 0$

By cross multiplication,

3M

$$\begin{array}{cccc} x^2 & x & 1 & \\ p & q & 1 & p \\ p' & q' & 1 & p' \end{array}$$

$$\frac{p}{p'} \quad \frac{q}{q'} \quad \frac{1}{1} \quad \frac{p}{p'}$$

$$\frac{x^2}{pq' - qp'} = \frac{x}{q - q'} = \frac{1}{p' - p}$$

(i) (ii) (iii)

From (i) & (ii)

$$\frac{x^2}{pq' - qp'} = \frac{x}{q - q'}$$

$$x = \frac{pq' - qp'}{q - q'}$$

$$\therefore x = \frac{pq' - qp'}{q - q'}$$

From (ii) & (iii)

$$\frac{x}{q - q'} = \frac{1}{p' - p}$$

$$x = \frac{q - q'}{p' - p}$$

$$x = \frac{q - q'}{p' - p}$$

11. A 12 metre tall tree was broken into two parts. It was found that the height of the part which was left standing was the cube root of the length of the part that was cut away. Formulate this into a mathematical problem to find the height of the part which was left standing. 2M

Let the height of the part which was left standing is x & $12 - x$

The cube root of the length of the part $x^3 = 12 - x$

$$x^3 + x - 12 = 0$$

EXERCISE 3.2



Video class

Concept Corner

Nature of Roots and Nature of Coefficients of Polynomial Equations

Theorem : Complex Conjugate Root Theorem

If a complex number z_0 is a root of a polynomial equation with real coefficients, then its complex conjugate $\bar{z}_0$ is also a root.

Complex roots occurs in pairs:

Example 1: $z = \alpha + i\beta$ is a root. Then $\bar{z} = \alpha - i\beta$ is also a root of a polynomial equation.

Example 2: $z = 2 + i3$ is a root. Then $\bar{z} = 2 - i3$ is also a root.

Irrational Roots:

If $\sqrt{\Delta}$ is rational, then both roots are rational.

If $\sqrt{\Delta}$ is irrational, then both roots are irrational.

Theorem :

Let p and q be rational numbers such that $\sqrt{q}$ is irrational. If $p + \sqrt{q}$ is a root of a quadratic equation with rational coefficients, then $p - \sqrt{q}$ is also a root of the same equation.

Rational roots:

- If $\Delta > 0$ and a perfect square, then the roots are real, rational and distinct.
- If $\Delta > 0$ and not a perfect square, then the roots are real, irrational and distinct.

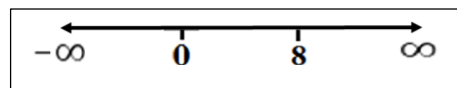
1. If k is real, discuss the nature of the roots of the polynomial equation $2x^2 + kx + k = 0$, in terms of k . 3M

Given $2x^2 + kx + k = 0$

$$a = 2, b = k, c = k$$

$$\Delta = b^2 - 4ac$$

$$= k^2 - 4(2)(k) = k^2 - 8k = k(k - 8) \Rightarrow k = 0, 8$$



- (i) When $k < 0$

$$\text{Say } k = -2$$

$$\Delta = -2(-2 - 8)$$

$$= -2(-10)$$

$$= 20 > 0$$

When $k < 0$, the polynomial has real roots.

- (ii) When $k = 0$ or $k = 8$

$$\Delta = 0$$

When $k = 0$ (or) $k = 8$, the roots are real and equal.

- (iii) When $0 < k < 8$

$$\text{Say } k = 2$$

$$\Delta = 2(2 - 8)$$

$$= 2(-6)$$

$$= -12 < 0$$

When $0 < k < 8$, the roots are imaginary.

- (iv) When $k > 8$

$$\text{Say } k = 9$$

$$\Delta = 9(9 - 8)$$

$$= 9 > 0$$

When $k > 8$, the roots are real and distinct.

2. Find a polynomial equation of minimum degree with rational coefficients, having $2 + \sqrt{3}i$ as a root.

Given one root is $2 + \sqrt{3}i$ $\therefore$ Another root is $2 - \sqrt{3}i$

2M

Sum of the roots (S.O.R) = $2 + \sqrt{3}i + 2 - \sqrt{3}i = 4$

Product of the roots (P.O.R) = $(2 + \sqrt{3}i)(2 - \sqrt{3}i) = 4 + 3 = 7$

$\therefore$ Required equation: $x^2 - x(\text{Sum of roots}) + (\text{Product of roots}) = 0$

$$x^2 - x(4) + 7 = 0$$

$$x^2 - 4x + 7 = 0$$

3. Find a polynomial equation of minimum degree with rational coefficients, having $2i + 3$ as a root.

Given one root is $2i + 3$ $\therefore$ Another root is $-2i + 3$

JUL-24

2M

Sum of the roots (S.O.R) = $2i + 3 - 2i + 3 = 6$

Product of the roots (P.O.R) = $(2i + 3)(-2i + 3)$
 $= 2^2 + 3^2 = 13$

$\therefore$ Required equation: $x^2 - x(\text{S. O. R}) + (\text{P. O. R}) = 0$

$$x^2 - 6x + 13 = 0$$

4. Find a polynomial equation of minimum degree with rational coefficients, having $\sqrt{5} - \sqrt{3}$ as a root.

PTA-5

5M

Given root is $\sqrt{5} - \sqrt{3}$

$$x = \sqrt{5} - \sqrt{3}$$

$$x^2 = (\sqrt{5} - \sqrt{3})^2$$

$$x^2 = 5 + 3 - 2\sqrt{5}\sqrt{3}$$

$$x^2 = 8 - 2\sqrt{15}$$

$$x^2 - 8 = -2\sqrt{15}$$

Squaring on both sides,

$$x^4 + 64 - 2x^2(8) = 4(15)$$

$$x^4 - 16x^2 + 64 = 60$$

$$x^4 - 16x^2 + 4 = 0$$

5. Prove that a straight line and parabola cannot intersect at more than two points.

3M

Let the equation of straight line be

$$y = mx + c \dots\dots\dots (1)$$

And parabola $y^2 = 4ax \dots\dots\dots (2)$

Sub (1) in (2)

$$\Rightarrow (mx + c)^2 = 4ax$$

$$\Rightarrow m^2x^2 + c^2 + 2mcx - 4ax = 0$$

$$\Rightarrow m^2x^2 + (2mc - 4a)x + c^2 = 0$$

This is a Quadratic equation

So, It must have only two solutions.

$\therefore$ A straight line and parabola cannot intersect more than two points.

EXERCISE 3.3



Video class

Concept Corner

Radical solution:

A solution of a polynomial equation written only using its coefficients, the four basic arithmetic operators (addition, multiplication, subtraction and division), and rational exponentiation (power to a rational number, such as square, cube, square roots, cube roots and so on) is called a radical solution.

Roots of higher degree polynomial equations:

1. Every polynomial in one variable is a continuous function from $\mathbb{R}$ to $\mathbb{R}$
2. For a polynomial equation $P(x) = 0$ of even degree, $P(x) \rightarrow \infty$ as $(x) \rightarrow \pm\infty$. Thus the graph of an even degree polynomial starts from left top and ends at right top.
3. Every polynomial is differentiable any number of times.
4. The real roots of a polynomial equation $P(x) = 0$ are the points on the x -axis where the graph of $P(x) = 0$ cuts the x - axis.

Polynomial equations with Even Powers Only:

The polynomial equation $P(x)$ of degree $2n$, having only even powers of x .

1. Solve the cubic equation:

$2x^3 - x^2 - 18x + 9 = 0$ if sum of two of its roots vanishes. 3M

Given: $2x^3 - x^2 - 18x + 9 = 0$

Let the roots be α, β, γ

where $a = 2, b = -1, c = -18, d = 9$

Applying Vieta's formula

$$\sum 1 = \alpha + \beta + \gamma = -\frac{b}{a}$$

$$\alpha + \beta + \gamma = \frac{1}{2}$$

Sum of two roots = 0

$$\alpha + \beta = 0$$

$$0 + \gamma = \frac{1}{2}$$

$$\gamma = \frac{1}{2}$$

$$\sum 3 = \alpha\beta\gamma = -\frac{d}{a}$$

$$\alpha\beta\left(\frac{1}{2}\right) = -\frac{9}{2}$$

$$\alpha + \beta = 0$$

$$\alpha\beta = -9$$

$$\Rightarrow \alpha = -\beta$$

$$-\beta \cdot \beta = -9$$

$$-\beta^2 = -9$$

$$\beta = \pm 3$$

$$\beta = 3$$

$$\Rightarrow \alpha = -3$$

$$\therefore \text{Solutions } -3, 3, \frac{1}{2}$$

2. Solve the equation: $9x^3 - 36x^2 + 44x - 16 = 0$ if the roots form an arithmetic progression. 3M

Given $9x^3 - 36x^2 + 44x - 16 = 0$

Also given the roots are in A.P.

Where $a = 9, b = -36, c = 44, d = -16$

$\therefore$ Let it $a - d, a, a + d$ are the roots (1)

By applying vieta's formula,

$$\sum 1 = a - d + a + a + d = -\frac{b}{a}$$

$$3a = \frac{36}{9}$$

$$3a = 4 \Rightarrow a = \frac{4}{3}$$

$$\sum 3 = (a - d)a(a + d) = -\frac{d}{a}$$

$$a(a^2 - d^2) = \frac{16}{9}$$

$$\text{Put } a = \frac{4}{3} \Rightarrow \frac{4}{3}\left(\frac{16}{9} - d^2\right) = \frac{16}{9}$$

$$\frac{16}{9} - d^2 = \frac{16}{9} \times \frac{3}{4}$$

$$\frac{16}{9} - d^2 = \frac{4}{3}$$

$$d^2 = \frac{16}{9} - \frac{4}{3} = \frac{16-12}{9}$$

$$d^2 = \frac{4}{9} \Rightarrow d = \pm \frac{2}{3}$$

$$\therefore d = \frac{2}{3} \text{ or } d = -\frac{2}{3}$$

Sub in (1)

$$a = \frac{4}{3}; d = \frac{2}{3}$$

$$a - d = \frac{4}{3} - \frac{2}{3} = \frac{2}{3}$$

$$a + d = \frac{4}{3} + \frac{2}{3} = \frac{6}{3} = 2$$

$$\therefore \text{solution } \frac{2}{3}, \frac{4}{3}, 2$$

Sub in (1)

$$a = \frac{4}{3}; d = -\frac{2}{3}$$

$$a - d = \frac{4}{3} + \frac{2}{3} = \frac{6}{3} = 2$$

$$a + d = \frac{4}{3} - \frac{2}{3} = \frac{2}{3}$$

3. Solve the equation:

$3x^3 - 26x^2 + 52x - 24 = 0$ if its roots form a geometric progression.

3M

Given: $3x^3 - 26x^2 + 52x - 24 = 0$

Also given the roots are in G.P.

Where $a = 3, b = -26,$

$c = 52, d = -24$

Let it $\frac{a}{r}, a, ar$ are the roots (1)

By applying vieta's formula,

$$\sum 1 = \frac{a}{r} + a + ar = -\frac{b}{a}$$

$$\frac{a}{r} + a + ar = \frac{26}{3} \quad \text{..... (2)}$$

$$\sum 3 = \frac{a}{r} \cdot a \cdot ar = -\frac{d}{a}$$

$$a^3 = \frac{24}{3}$$

$$a^3 = 8$$

$$a = 2$$

Sub $a = 2$ in (2)

$$\frac{a}{r} + a + ar = \frac{26}{3}$$

$$a \left[\frac{1}{r} + 1 + r \right] = \frac{26}{3}$$

$$2 \left[\frac{1}{r} + 1 + r \right] = \frac{26}{3}$$

$$\frac{1+r+r^2}{r} = \frac{26}{3} \times \frac{1}{2}$$

$$\frac{1+r+r^2}{r} = \frac{13}{3}$$

By cross multiply

$$3 + 3r + 3r^2 = 13r$$

$$3r^2 + 3r - 13r + 3 = 0$$

$$3r^2 - 10r + 3 = 0$$

$$\left(r - \frac{1}{3}\right)(r - 3) = 0$$

$$r = \frac{1}{3} \text{ or } 3$$

Sub in (1)

$$a = 2; r = \frac{1}{3}$$

$$\frac{a}{r} = 2 \times 3 = 6$$

$$a = 2$$

$$ar = 2 \times \frac{1}{3} = \frac{2}{3}$$

$$\therefore \text{solution } \frac{2}{3}, 2, 6$$

$$a = 2; r = 3$$

$$\frac{a}{r} = \frac{2}{3}$$

$$a = 2$$

$$ar = 2 \times 3 = 6$$

4. Determine k and solve the equation $2x^3 - 6x^2 + 3x + k = 0$ if one of its roots is twice the sum of the other two roots.

5M

Given: $2x^3 - 6x^2 + 3x + k = 0$

Let α, β, γ be the roots.

Where $a = 2, b = -6,$

$c = 3, d = k$

And also given $\alpha = 2(\beta + \gamma)$ (1)

Applying Vieta's formula

$$\sum 1 = \alpha + \beta + \gamma = -\frac{b}{a}$$

$$\alpha + \frac{\alpha}{2} = 3$$

$$\frac{2\alpha + \alpha}{2} = 3$$

$$3\alpha = 6$$

$$\alpha = 2 \text{ is a root}$$

$$\therefore p(x): 2x^3 - 6x^2 + 3x + k$$

2 is a root

$$\therefore p(x) = 2(2)^3 - 6(2)^2 + 3(2) + k = 0$$

$$16 - 24 + 6 + k = 0$$

$$-2 + k = 0$$

$$k = 2$$

$$\therefore \text{The equation: } 2x^3 - 6x^2 + 3x + 2$$

$$\begin{array}{r|rrrr} 2 & 2 & -6 & 3 & 2 \\ & 0 & 4 & -4 & -2 \\ \hline & 2 & -2 & -1 & 0 \end{array}$$

$(x - 2)(2x^2 - 2x - 1)$ is a factor

$$x - 2 = 0 \Rightarrow x = 2$$

$$2x^2 - 2x - 1 = 0$$

$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

$$= \frac{2 \pm \sqrt{4 - 4(2)(-1)}}{4}$$

$$= \frac{2 \pm \sqrt{12}}{4}$$

$$= \frac{2 \pm 2\sqrt{3}}{4}$$

$$= \frac{2(1 \pm \sqrt{3})}{4}$$

$$= \frac{1 \pm \sqrt{3}}{2}$$

$$\therefore \text{Solutions are } 2, 2, \frac{1 \pm \sqrt{3}}{2}$$

5. Find all zeros of the polynomial $x^6 - 3x^5 - 5x^4 + 22x^3 - 39x^2 - 39x + 135$, if it is known that $1 + 2i$ and $\sqrt{3}$ are two of its zeros.

5M

Given polynomial: $x^6 - 3x^5 - 5x^4 + 22x^3 - 39x^2 - 39x + 135$

Also given that $1 + 2i$ and $\sqrt{3}$ are zeros,

Given: $1 + 2i$ is a zero. $\Rightarrow 1 - 2i$ is also a zero.

Given: $\sqrt{3}$ is a zero. $\Rightarrow -\sqrt{3}$ is also a zero.

$$\begin{aligned}
 &\Rightarrow [x - (1 + 2i)][x - (1 - 2i)][x - \sqrt{3}][x + \sqrt{3}] \\
 &= ((x - 1) - 2i)((x - 1) + 2i)(x - \sqrt{3})(x + \sqrt{3}) \\
 &= ((x - 1)^2 + 2^2)(x^2 - 3) \\
 &= (x^2 + 1 - 2x + 4)(x^2 - 3) \\
 &= x^4 - 3x^2 - 2x^3 + 6x + 5x^2 - 15 \\
 &= x^4 - 2x^3 + 2x^2 + 6x - 15 \text{ is a factor of given polynomial.}
 \end{aligned}$$

$x^4 - 2x^3 + 2x^2 + 6x - 15$	$x^2 - x - 9$
	$x^6 - 3x^5 - 5x^4 + 22x^3 - 39x^2 - 39x + 135$
	$\cancel{x^6} - 2x^5 + 2x^4 + 6x^3 - 15x^2$
	(-) (+) (-) (-) (+)
	$0 - \cancel{x^5} - 7x^4 + 16x^3 - 24x^2 - 39x$
	$- \cancel{x^5} + 2x^4 - 2x^3 - 6x^2 - 15x$
	(+)(-)(+)(+)(+)
	$0 - 9\cancel{x^4} + 18\cancel{x^3} - 18\cancel{x^2} - 54x + 135$
	$-9\cancel{x^4} + 18\cancel{x^3} - 18\cancel{x^2} - 54x + 135$
	(+)(-)(+)(+)(+)
	0

Remaining factor: $x^2 - x - 9$

$$x = \frac{1 \pm \sqrt{1+36}}{2}$$

$$x = \frac{1 \pm \sqrt{37}}{2}$$

$\therefore$ The required zeros are

$$1 + 2i, 1 - 2i, \sqrt{3}, -\sqrt{3}, \frac{1+\sqrt{37}}{2}, \frac{1-\sqrt{37}}{2}$$

6. Solve the cubic equations: (i) $2x^3 - 9x^2 + 10x = 3$, (ii) $8x^3 - 2x^2 - 7x + 3 = 0$

3M

(i) $2x^3 - 9x^2 + 10x = 3$

APR-22

Given: $2x^3 - 9x^2 + 10x = 3$

$$\Rightarrow 2x^3 - 9x^2 + 10x - 3 = 0$$

Sum of the coefficients $2 - 9 + 10 - 3 = 0$

$\therefore x - 1$ is a factor

$$\begin{array}{r|rrrr} 1 & 2 & -9 & 10 & -3 \\ & 0 & 2 & -7 & 3 \end{array}$$

$$\begin{array}{r|rrrr} 3 & 2 & -7 & 3 & 0 \\ & 0 & 6 & -3 & \end{array}$$

$$\begin{array}{r|rr} & 2 & -1 & 0 \end{array}$$

$\therefore$ factor $(x - 1)(x - 3)(2x - 1)$

$$\begin{array}{l|l|l} x - 1 = 0 & x - 3 = 0 & 2x - 1 = 0 \\ x = 1 & x = 3 & x = \frac{1}{2} \end{array}$$

$$\Rightarrow x = 1, 3, \frac{1}{2}$$

$\therefore$ Roots are $1, \frac{1}{2}, 3$

(ii) $8x^3 - 2x^2 - 7x + 3 = 0$

Given: $8x^3 - 2x^2 - 7x + 3 = 0$

Sum of the coefficients $8 - 2 - 7 + 3 = 0$

Sum of x^3 , x terms

= Sum of x^2 , constant terms

$$8 - 7 = -2 + 3$$

$$1 = 1$$

$\therefore x + 1$ is a factor

$$\begin{array}{r|rrrr} -1 & 8 & -2 & -7 & 3 \\ & 0 & -8 & 10 & -3 \end{array}$$

$$\begin{array}{r|rr} & 8 & -10 & 3 & 0 \end{array}$$

$\therefore$ factor $(x + 1)(8x^2 - 10x + 3)$

$$(x + 1)(4x - 3)(2x - 1) = 0$$

$$\Rightarrow x = -1, \frac{3}{4}, \frac{1}{2}$$

$\therefore$ Roots are $-1, \frac{1}{2}, \frac{3}{4}$

7. Solve the equation: $x^4 - 14x^2 + 45 = 0$

2M

Given equation: $x^4 - 14x^2 + 45 = 0$

Put $x^2 = y$

$\therefore$ The equation becomes $y^2 - 14y + 45 = 0$

$$(y - 9)(y - 5) = 0$$

$$y = 9; 5$$

Case i: $y = 9$

But $x^2 = y$

$$x^2 = 9$$

$$x = \pm 3$$

Case ii: $y = 5$

But $x^2 = y$

$$x^2 = 5$$

$$x = \pm\sqrt{5}$$

$\therefore$ Roots are: $\pm 3, \pm\sqrt{5}$

EXERCISE 3.4

Concept Corner

Partly Factored Polynomial

Quartic polynomial equations of the form $(ax + b)(cx + d)(px + q)(rx + s) + k = 0, k \neq 0$ which can be rewritten in the form $(ax^2 + \beta x + \lambda)(ax^2 + \beta x + \mu) + k = 0$



Video class

1. Solve: (i) $(x-5)(x-7)(x+6)(x+4) = 504$, (ii) $(x-4)(x-7)(x-2)(x+1) = 16$

(i) $(x-5)(x-7)(x+6)(x+4) = 504$

5M

Given: $(x-5)(x-7)(x+6)(x+4) = 504$

$$[(x-5)(x+4)][(x-7)(x+6)] - 504 = 0$$

$$[(x^2 + 4x - 5x - 20)][(x^2 + 6x - 7x - 42) - 504] = 0$$

$$\Rightarrow (x^2 - x - 20)(x^2 - x - 42) - 504 = 0$$

Put $x^2 - x = y$

$$\Rightarrow (y - 20)(y - 42) - 504 = 0$$

$$\Rightarrow y^2 - 42y - 20y + 840 - 504 = 0$$

$$\Rightarrow y^2 - 62y + 336 = 0$$

$$\Rightarrow (y - 56)(y - 6) = 0$$

$$\Rightarrow y = 56, 6$$

Case: i $y = 56$

But $x^2 - x = y$

$$x^2 - x - 56 = 0$$

$$(x+7)(x-8) = 0$$

$$x = -7, 8$$

Case: ii $y = 6$

But $x^2 - x = y$

$$x^2 - x - 6 = 0$$

$$(x-3)(x+2) = 0$$

$$x = 3, -2$$

$\therefore$ The Roots are: **-2, 3, -7, 8.**

(ii) $(x-4)(x-7)(x-2)(x+1) = 16$

Given: $(x-4)(x-7)(x-2)(x+1) = 16$

$$[(x-4)(x-2)][(x-7)(x+1)] - 16 = 0$$

$$\Rightarrow [(x^2 - 2x - 4x + 8)][(x^2 + x - 7x - 7)] - 16 = 0$$

$$\Rightarrow (x^2 - 6x + 8)(x^2 - 6x - 7) - 16 = 0$$

Put $x^2 - 6x = y$

$$\Rightarrow (y + 8)(y - 7) - 16 = 0$$

$$\Rightarrow y^2 - 7y + 8y - 56 - 16 = 0$$

$$\Rightarrow y^2 + y - 72 = 0$$

$$\Rightarrow (y - 8)(y + 9) = 0$$

$$\Rightarrow y = 8, -9$$

Case i: $y = 8$

But $x^2 - 6x = y$

$$x^2 - 6x - 8 = 0$$

$$x = \frac{6 \pm \sqrt{36+32}}{2}$$

$$= \frac{6 \pm \sqrt{68}}{2} = \frac{6 \pm 2\sqrt{17}}{2}$$

$$x = 3 \pm \sqrt{17}$$

Case ii: $y = -9$

But $x^2 - 6x = y$

$$x^2 - 6x + 9 = 0$$

$$(x-3)(x-3) = 0$$

$$x = 3, 3$$

$\therefore$ The Roots are: **3, 3, $3 + \sqrt{17}$, $3 - \sqrt{17}$**

2. Solve: $(2x - 1)(x + 3)(x - 2)(2x + 3) + 20 = 0$

Given: $[(2x - 1)(x + 3)][(x - 2)(2x + 3)] + 20 = 0$

$$[(2x^2 + 6x - x - 3)][(2x^2 - 4x + 3x - 6)] + 20 = 0$$

$$(2x^2 + 5x - 3)(2x^2 - x - 6) + 20 = 0$$

$$4x^4 - 2x^3 - 12x^2 + 10x^3 - 5x^2 - 30x - 6x^2 + 3x + 18 + 20 = 0$$

$$4x^4 + 8x^3 - 23x^2 - 27x + 38 = 0$$

$$\begin{array}{r|rrrrrr} 1 & 4 & 8 & -23 & -27 & 38 \\ & 0 & 4 & 12 & -11 & -38 \\ -2 & 4 & 12 & -11 & -38 & 0 \\ & 0 & -8 & -8 & 38 & \\ \hline & 4 & 4 & -19 & 0 & \end{array}$$

$$(x - 1)(x + 2)(4x^2 + 4x - 19) = 0$$

$$\Rightarrow x - 1 = 0; \quad x + 2 = 0; \quad 4x^2 + 4x - 19 = 0$$

$$\Rightarrow x = 1; \quad x = -2; \quad x = \frac{-4 \pm \sqrt{16 - 4(4)(-19)}}{2(4)}$$

$$= \frac{-4 \pm \sqrt{16 + 304}}{8}$$

$$= \frac{-4 \pm \sqrt{320}}{8}$$

$$= \frac{-4 \pm 8\sqrt{5}}{8}$$

$$x = \frac{-1 \pm 2\sqrt{5}}{2}$$

$$\therefore \text{The roots are: } 1, -2, \frac{-1 + 2\sqrt{5}}{2}, \frac{-1 - 2\sqrt{5}}{2}$$

EXERCISE 3.5



Video class

Concept Corner

Rational Root Theorem :

Let $a_n x^n + \dots + a_1 x + a_0$ with $a_n \neq 0$, and $a_0 \neq 0$, be a polynomial with integer coefficients.

If $\frac{p}{q}$ with $(p, q) = 1$, is a root of the polynomial, then p is a factor of a_0 and q is a factor of a_n .

Definition:

A polynomial $P(x)$ of degree n is said to be a reciprocal polynomial if one of the following Condition is true.

(i) $P(x) = x^n P\left(\frac{1}{x}\right)$ (Called type 1)

(ii) $P(x) = -x^n P\left(\frac{1}{x}\right)$ (Called type 2)

Example of type 1:

$$6x^5 + x^4 - 43x^3 - 43x^2 + x + 6 = 0$$

The co-efficients from the begining are equal to the co-efficients from the end.

Example of type 2:

$$6x^5 - 41x^4 + 97x^3 - 97x^2 + 41x - 6 = 0$$

The co-efficients from the begining are equal in magnitude to the co-efficients from the end. But in opposite sign.

Theorem:

A polynomial equation $a_n x^n + a_{n-1} x^{n-1} + a_{n-2} x^{n-2} + \dots + a_2 x^2 + a_1 x + a_0 = 0$, ($a_n \neq 0$) is a reciprocal equation if and only if, one of the following two statements is true:

- (i) $a_n = a_0$, $a_{n-1} = a_1$, $a_{n-2} = a_2, \dots$
 (ii) $a_n = -a_0$, $a_{n-1} = -a_1$, $a_{n-2} = -a_2, \dots$

Note:

1. A reciprocal equation cannot have 0 as a solution.
2. The coefficients and the solutions are not restricted to be real.
3. Type I and odd degree, -1 is a root.
4. Type II and odd degree, $+1$ is a root.
5. Type II and even degree, $+1$ and -1 are root.

1. Solve the following equations (i) $\sin^2 x - 5 \sin x + 4 = 0$ (ii) $12x^3 + 8x = 29x^2 - 4$ 3M

(i) $\sin^2 x - 5 \sin x + 4 = 0$

Given: $\sin^2 x - 5 \sin x + 4 = 0$

Let $\sin x = y$

$$y^2 - 5y + 4 = 0$$

$$(y - 1)(y - 4) = 0$$

$$y = 1, 4$$

$$\sin x = 1 \Rightarrow x = \frac{\pi}{2}$$

$$x = n\pi + (-1)^n \frac{\pi}{2}, n \in \mathbb{Z}$$

$$\sin x = 4$$

no solution for $\sin x = 4$

(ii) $12x^3 + 8x = 29x^2 - 4$

$$\Rightarrow 12x^3 + 8x - 29x^2 + 4 = 0$$

$$\Rightarrow 12x^3 - 29x^2 + 8x + 4 = 0$$

Let $p(x) = 12x^3 - 29x^2 + 8x + 4$

$$p(2) = 12(2)^3 - 29(2)^2 + 8(2) + 4$$

$$= 12 \times 8 - 29 \times 4 + 16 + 4$$

$$= 96 - 116 + 20$$

$$= 116 - 116$$

$$= 0$$

2 is a root of the polynomial.

$$\begin{array}{r|rrrr} 2 & 12 & -29 & 8 & 4 \\ & 0 & 24 & -10 & -4 \\ \hline & 12 & -5 & -2 & 0 \end{array}$$

$$\therefore (x - 2)(12x^2 - 5x - 2) = 0$$

$$\Rightarrow (x - 2) \left(x - \frac{2}{3}\right) \left(x + \frac{1}{4}\right) = 0$$

$$\therefore x = 2, \frac{2}{3}, -\frac{1}{4}$$

$$\therefore \text{Solution: } 2, -\frac{1}{4}, \frac{2}{3}$$

2. Examine for the rational roots of

(i) $2x^3 - x^2 - 1 = 0$ (ii) $x^8 - 3x + 1 = 0$.

(i) $2x^3 - x^2 - 1 = 0$

3M

Given: $2x^3 - x^2 - 1 = 0$

Let $a_n = 2$; $a_0 = -1$

Let $\frac{p}{q}$ be a root of the polynomial. p must divide $-1 \Rightarrow 1, -1$ are the values of p q must divide $2 \Rightarrow 1, -1, 2, -2$ are the values of q

We get $\pm \frac{1}{1}, \pm \frac{1}{2}$

Possible rational roots are $1, -1, \pm \frac{1}{2}$

$p(x) = 2x^3 - x^2 - 1$

$p(1) = 2(1)^3 - 1 - 1 = 0$

Others $p(-1), p\left(\pm \frac{1}{2}\right)$ are $\neq 0$

 $\therefore x = 1$ is a only rational root.

(ii) $x^8 - 3x + 1 = 0$

2M

Given: $x^8 - 3x + 1 = 0$

Let $a_n = 1$; $a_0 = 1$

Let $\frac{p}{q}$ be a root of the polynomial. p must divide $1 \Rightarrow 1, -1$ are the values of p q must divide $1 \Rightarrow 1, -1$ are the values of q

We get $\pm \frac{1}{1}$

$p(x) = x^8 - 3x + 1 = 0$

$p(1) = (1)^8 - 3(1) + 1$

$= -1 \neq 0$

$p(-1) = (-1)^8 - 3(-1) + 1$

$= 5 \neq 0$

 $x = 1, -1$ are not a roots $\therefore$ It has no rational roots.

3. Solve: $8x^{\frac{3}{2n}} - 8x^{-\frac{3}{2n}} = 63$

3M

Let $y = x^{\frac{3}{2n}}$ (1)

The equation becomes $8y - \frac{8}{y} = 63$

$8y^2 - 8 = 63y$

$8y^2 - 63y - 8 = 0$

$y = 8$ (or) $8y + 1 = 0$

$y = 8$ (or) $y = -\frac{1}{8}$

Sub in (1) $x^{\frac{3}{2n}} = 8$ (or) $x^{\frac{3}{2n}} = -\frac{1}{8}$

$x^{\frac{3}{2n}} = 2^3$ (or) $x^{\frac{3}{2n}} = \left(-\frac{1}{2}\right)^3$

$x^{\frac{1}{2n}} = 2$ (or) $x^{\frac{1}{2n}} = -\frac{1}{2}$

$x = 2^{2n}$ (or) $x = \frac{1}{2^{2n}}$

$x = 4^n$

($x = \frac{1}{4^n}$ not satisfy the given equation)Solution $x = 4^n$

4. Solve: $2\sqrt{\frac{x}{a}} + 3\sqrt{\frac{a}{x}} = \frac{b}{a} + \frac{6a}{b}$

3M

Let $y = \sqrt{\frac{x}{a}}$ (1)

$2y + \frac{3}{y} = \frac{b}{a} + \frac{6a}{b}$

$\frac{2y^2 + 3}{y} = \frac{b^2 + 6a^2}{ab}$

$2aby^2 + 3ab = b^2y + 6a^2y$

$2aby^2 - b^2y - 6a^2y + 3ab = 0$

$by(2ay - b) - 3a(2ay - b) = 0$

$(2ay - b)(by - 3a) = 0$

$2ay - b = 0$ | $by - 3a = 0$

$y = \frac{b}{2a}$ | $y = \frac{3a}{b}$

Sub in (1) $\sqrt{\frac{x}{a}} = \frac{b}{2a}$ (or) $\sqrt{\frac{x}{a}} = \frac{3a}{b}$

$\frac{x}{a} = \frac{b^2}{4a^2}$ (or) $\frac{x}{a} = \frac{9a^2}{b^2}$

$x = \frac{b^2}{4a}$ (or) $x = \frac{9a^3}{b^2}$

5. Solve the equations (i) $6x^4 - 35x^3 + 62x^2 - 35x + 6 = 0$ (ii) $x^4 + 3x^3 - 3x - 1 = 0$

(i) $6x^4 - 35x^3 + 62x^2 - 35x + 6 = 0$ PTA-1

This is a reciprocal equation.

Type I even degree.

Divide by x^2

$$6x^2 - 35x + 62 - \frac{35}{x} + \frac{6}{x^2} = 0$$

$$\Rightarrow 6\left(x^2 + \frac{1}{x^2}\right) - 35\left(x + \frac{1}{x}\right) + 62 = 0$$

Put $x + \frac{1}{x} = y, x^2 + \frac{1}{x^2} = y^2 - 2$

$$\Rightarrow 6(y^2 - 2) - 35y + 62 = 0$$

$$\Rightarrow 6y^2 - 12 - 35y + 62 = 0$$

$$\Rightarrow 6y^2 - 35y + 50 = 0$$

$$\Rightarrow \left(y - \frac{10}{3}\right)\left(y - \frac{5}{2}\right) = 0$$

$$y = \frac{10}{3}; y = \frac{5}{2}$$

Case: 1

$$y = \frac{10}{3}$$

But $x + \frac{1}{x} = \frac{10}{3}$

$$\frac{x^2+1}{x} = \frac{10}{3}$$

$$3x^2 + 3 = 10x$$

$$3x^2 - 10x + 3 = 0$$

$$\left(x - \frac{1}{3}\right)(x - 3) = 0$$

$$x = \frac{1}{3}, 3$$

Case: 2

$$y = \frac{5}{2}$$

But $x + \frac{1}{x} = \frac{5}{2}$

$$\frac{x^2+1}{x} = \frac{5}{2}$$

$$2x^2 + 2 = 5x$$

$$2x^2 - 5x + 2 = 0$$

$$\left(x - \frac{1}{2}\right)(x - 2) = 0$$

$$x = \frac{1}{2}, 2$$

$\therefore$ The roots are $2, 3, \frac{1}{2}, \frac{1}{3}$

(ii) $x^4 + 3x^3 - 3x - 1 = 0$

3M

Given: $x^4 + 3x^3 - 3x - 1 = 0$

This is a reciprocal equation.

Type II even.

$\therefore 1$ & -1 are roots of the given equation.

$$\begin{array}{r|rrrrr} 1 & 1 & 3 & 0 & -3 & -1 \\ & 0 & 1 & 4 & 4 & 1 \\ -1 & 1 & 4 & 4 & 1 & 0 \\ & 0 & -1 & -3 & -1 & \\ & 1 & 3 & 1 & 0 & \end{array}$$

$\therefore (x - 1)(x + 1)(x^2 + 3x + 1)$ are the factors.

$$\begin{array}{l|l} x - 1 = 0 & x + 1 = 0 \\ x = 1 & x = -1 \\ \hline & x^2 + 3x + 1 = 0 \\ & x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a} \\ & = \frac{-3 \pm \sqrt{9 - 4}}{2} \\ & x = \frac{-3 \pm \sqrt{5}}{2} \end{array}$$

$\therefore$ Solution: $x = 1, -1, \frac{-3 + \sqrt{5}}{2}, \frac{-3 - \sqrt{5}}{2}$.

6. Find all real numbers satisfying $4^x - 3(2^{x+2}) + 2^5 = 0$.

MDL

3M

$$4^x - 3(2^{x+2}) + 2^5 = 0$$

$$\Rightarrow (2^2)^x - 3(2^x \times 2^2) + 32 = 0$$

$$\Rightarrow (2^x)^2 - 12 \times 2^x + 32 = 0$$

Let $y = 2^x$ (1)

$$\Rightarrow y^2 - 12y + 32 = 0$$

$$(y - 8)(y - 4) = 0$$

$$y = 8 \text{ (or)} y = 4$$

Sub in (1)

$$2^x = 8$$

$$2^x = 4$$

$$2^x = 2^3$$

$$2^x = 2^2$$

$$x = 2, 3$$

7. Solve the equation $6x^4 - 5x^3 - 38x^2 - 5x + 6 = 0$ if it is known that $\frac{1}{3}$ is a solution.

5M

Given: $6x^4 - 5x^3 - 38x^2 - 5x + 6 = 0$

It is a reciprocal equation in type I.

Also given that $\frac{1}{3}$ is a solution.

$\Rightarrow 3$ is also an another solution.

$\therefore (3x - 1)(x - 3)$ is a factor.

$\Rightarrow 3x^2 - 9x - x + 3$ is a factor.

$\Rightarrow 3x^2 - 10x + 3$ is a factor.

$3x^2 - 10x + 3$	$2x^2 + 5x + 2$
	$6x^4 - 5x^3 - 38x^2 - 5x + 6$
	$6x^4 - 20x^3 + 6x^2$
	(-) (+) (-)
	$0 \quad 15x^3 - 44x^2 - 5x$
	$15x^3 - 50x^2 + 15x$
	(-) (+) (-)
	$0 \quad 6x^2 - 20x + 6$
	$6x^2 - 20x + 6$
	(-) (+) (-)
	0

$$2x^2 + 5x + 2 = 0$$

$$\Rightarrow \left(x + \frac{1}{2}\right)(x + 2) = 0 \Rightarrow x = -\frac{1}{2}, -2.$$

$\therefore$ The roots are $3, \frac{1}{3}, -2, -\frac{1}{2}$.

EXERCISE 3.6

Concept Corner

Theorem 3.7 (Descartes Rule)

If p is the number of positive zeros of a polynomial $P(x)$ with real coefficients and s is the number of sign changes in coefficients of $P(x)$, then $s - p$ is a non-negative even integer.

Note:

The rules are

- (i) The equation $P(x) = 0$ cannot have more number of $+ve$ roots than the number of changes of sign in the terms of the polynomial $P(x)$.
- (ii) The equation $P(x) = 0$ cannot have more number of $-ve$ roots than the number of changes of sign in the terms of the polynomial $P(-x)$.



Video class

1. Discuss the maximum possible number of positive and negative roots of the polynomial equation $9x^9 - 4x^8 + 4x^7 - 3x^6 + 2x^5 + x^3 + 7x^2 + 7x + 2 = 0$. 2M

Given: $P(x) = 9x^9 - 4x^8 + 4x^7 - 3x^6 + 2x^5 + x^3 + 7x^2 + 7x + 2 = 0$

$$= + \quad - \quad + \quad - \quad + \quad + \quad + \quad + \quad +$$

Number of sign changes of $P(x) = 4$

$\therefore P(x)$ has at most 4 positive real roots.

$$P(-x) = -9x^9 - 4x^8 - 4x^7 - 3x^6 - 2x^5 - x^3 + 7x^2 - 7x + 2 = 0$$

$$= - \quad - \quad - \quad - \quad - \quad - \quad + \quad - \quad +$$

Number of sign changes of $P(x) = 3$

$\therefore P(x)$ has at most 3 negative real roots.

$\therefore P(x)$ has at most 4 positive and at most 3 negative roots.

2. Discuss the maximum possible number of positive and negative zeros of the polynomials $x^2 - 5x + 6$ and $x^2 - 5x + 16$. Also draw rough sketch of the graphs. 5M

Given: $x^2 - 5x + 6$; $x^2 - 5x + 16$

$$P(x) = x^2 - 5x + 6 \quad Q(x) = x^2 - 5x + 16$$

$$= + \quad - \quad + \quad \quad \quad = + \quad - \quad +$$

Number of sign changes of $P(x) = 2, Q(x) = 2$

$P(x), Q(x)$ has at most two positive real roots.

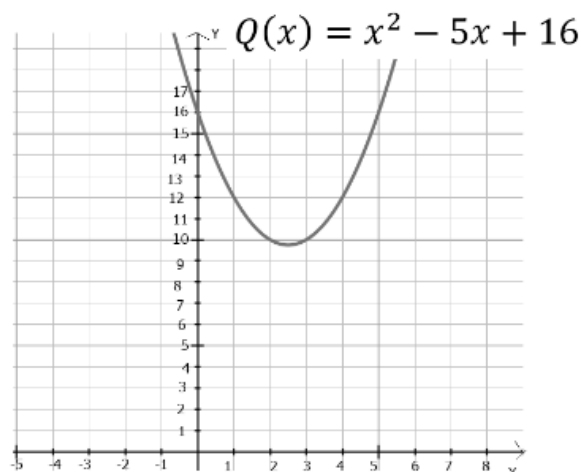
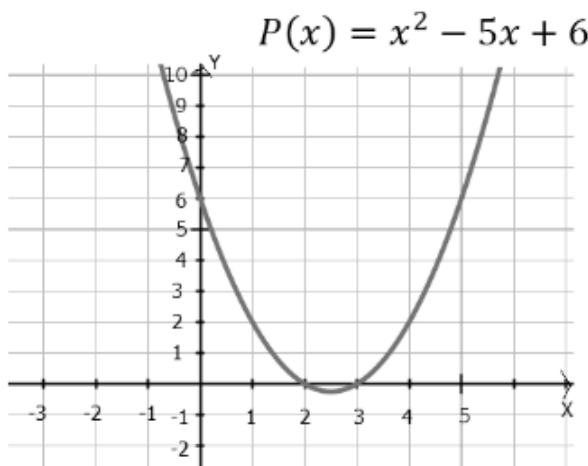
$$P(-x) = x^2 + 5x + 6 \quad Q(-x) = x^2 + 5x + 16$$

$$= + \quad + \quad + \quad \quad \quad = + \quad + \quad +$$

Number of sign changes of $(-x) = 0$, $Q(-x)$ is 0.

Which has no negative real roots.

$\therefore$ The given equation has at most two positive roots.



3. Show that equation $x^9 - 5x^5 + 4x^4 + 2x^2 + 1 = 0$ has atleast 6 imaginary solutions.

Given: $P(x) = x^9 - 5x^5 + 4x^4 + 2x^2 + 1 = 0$

PTA-2

2M

$$= + \quad \overset{\text{arc}}{-} \quad + \quad + \quad +$$

Number of sign changes of $p(x) = 2$

$P(x)$ has atmost two positive real roots.

$$P(-x) = (-x)^9 - 5(-x)^5 + 4(-x)^4 + 2(-x)^2 + 1 = -x^9 + 5x^5 + 4x^4 + 2x^2 + 1$$

$$= - \quad \overset{\text{arc}}{+} \quad + \quad + \quad +$$

$P(x)$ have only one negative real roots.

$\therefore P(x)$ has atmost 2 positive, one negative roots.

Total roots = 9, Remaining = $9 - 3 = 6$

$\therefore P(x)$ has atleast 6 unreal complex roots

4. Determine the number of positive and negative roots of the equation $x^9 - 5x^8 - 14x^7 = 0$

Given: $P(x) = x^9 - 5x^8 - 14x^7$

2M

$$= + \quad \overset{\text{arc}}{-} \quad -$$

One positive real root.

$$P(-x) = -x^9 - 5x^8 + 14x^7$$

$$= - \quad - \quad \overset{\text{arc}}{+}$$

One negative real root.

$\therefore P(x)$ have only one positive real root and one negative real root.

$\therefore$ Other roots are

$$P(x) = x^7(x^2 - 5x - 14) \Rightarrow P(0) = 0^7(x^2 - 5x - 14) = 0$$

$x = 0$ is the real root of order 7

5. Find the exact number of real zeros and imaginary of the polynomial

$$x^9 + 9x^7 + 7x^5 + 5x^3 + 3x$$

2M

Given: $P(x) = x^9 + 9x^7 + 7x^5 + 5x^3 + 3x$

$$= + \quad + \quad + \quad + \quad +$$

Number of sign changes of $P(x) = 0$

$P(x)$ have no positive real root.

$$P(-x) = (-x)^9 + 9(-x)^7 + 7(-x)^5 + 5(-x)^3 + 3(-x)$$

$$= -x^9 - 9x^7 - 7x^5 - 5x^3 - 3x$$

$$= - \quad - \quad - \quad - \quad -$$

Number of sign changes of $P(-x) = 0$

$P(x)$ have no negative real root.

$\therefore P(x)$ have no positive and negative real roots. Also $P(0) = 0$

$x = 0$ is only real root

8 remaining roots are unreal complex roots.

EXERCISE 3.7



Online Test

Choose the correct or the most suitable answer from the given four alternatives :

1. A zero of $x^3 + 64$ is

- (1) 0 (2) 4 (3) $4i$ (4) -4

$$x^3 + 64 = 0$$

$$\Rightarrow x^3 = -64 \Rightarrow x^3 = (-4)^3 \Rightarrow x = -4$$

JUL-21, 22, 23

2. If f and g are polynomials of degrees m and n respectively, and if $h(x) = (f \circ g)(x)$, then the degree of h is

- (1) mn (2) $m + n$ (3) m^n (4) n^m

PTA-5

Explanation with an Example: Let $f(x) = x^3$; $g(x) = x^2$

$$\text{Now } h(x) = (f \circ g)(x) = f[g(x)] = f[x^2] = (x^2)^3 = x^6$$

$$\text{If } m = 3, n = 2 \Rightarrow mn = 6$$

3. A polynomial equation in x of degree n always has

PTA-1,4, APR-20

- (1) n distinct roots (2) n real roots (3) n complex roots (4) at most one root

A polynomial of degree n always has n roots in $\mathbb{C}$ by the Fundamental Theorem of Algebra.

4. If α, β , and γ are the zeros of $x^3 + px^2 + qx + r$, then $\sum \frac{1}{\alpha}$ is

APR-24, 22

- (1) $-\frac{q}{r}$ (2) $-\frac{p}{r}$ (3) $\frac{q}{r}$ (4) $-\frac{q}{p}$

$$\text{Given: } x^3 + px^2 + qx + r = 0$$

Let α, β & γ be the roots

$$\sum 1 = \alpha + \beta + \gamma = -\frac{b}{a} = -p$$

$$\sum 2 = \alpha\beta + \beta\gamma + \gamma\alpha = \frac{c}{a} = q$$

$$\sum 3 = \alpha\beta\gamma = -\frac{d}{a} = -r$$

To find $\sum \frac{1}{\alpha}$

$$\sum \frac{1}{\alpha} = \frac{1}{\alpha} + \frac{1}{\beta} + \frac{1}{\gamma} = \frac{\alpha\beta + \beta\gamma + \gamma\alpha}{\alpha\beta\gamma} = \frac{q}{-r} = -\frac{q}{r}$$

$$\text{Where } a = 1, b = p, \\ c = q, d = r.$$

5. According to the rational root theorem, which number is not possible rational zero of $4x^7 + 2x^4 - 10x^3 - 5$?

- (1) -1 (2) $\frac{5}{4}$ (3) $\frac{4}{5}$ (4) 5

The given polynomial is $P(x) = 4x^7 + 2x^4 - 10x^3 - 5$

$$a_n = 4, a_0 = -5$$

If $\frac{p}{q}$ is a zero of $P(x)$ where $(p, q) = 1$, then p divides 5 and q divides 4 .

p divides $5 \Rightarrow$ The only numbers dividing 5 are ± 1 and ± 5

q divides $4 \Rightarrow$ The only numbers dividing 4 are $\pm 1, \pm 2, \pm 4$.

The possible $\frac{p}{q}$ are $\pm \frac{1}{1}, \pm \frac{5}{2}, \pm \frac{5}{4}$

The rational number $\frac{4}{5}$ cannot be a rational zero of $P(x)$.

6. The polynomial $x^3 - kx^2 + 9x$ has three real zeros if and only if, k satisfies

JUL-20, 24

- (1) $|k| \leq 6$ (2) $k = 0$ (3) $|k| > 6$ (4) $|k| \geq 6$

Given: $x^3 - kx^2 + 9x$

Divide by x

$$\Rightarrow x^2 - kx + 9$$

Have real roots,

where $a = 1, b = -k, c = 9$.

$$\Rightarrow \Delta = b^2 - 4ac \geq 0$$

$$\Rightarrow k^2 - 4(1)9 \geq 0$$

$$\Rightarrow k^2 \geq 36$$

$$\Rightarrow k \geq \pm 6$$

$$\Rightarrow |k| \geq 6$$

7. The number of real numbers in $[0, 2\pi]$ satisfying $\sin^4 x - 2\sin^2 x + 1$ is

APR-25, 23

- (1) 2 (2) 4 (3) 1 (4) ∞

Given: $\sin^4 x - 2\sin^2 x + 1$

Let $\sin^2 x = y$

$$\Rightarrow y^2 - 2y + 1 = 0$$

$$\Rightarrow (y - 1)^2 = 0$$

$$\Rightarrow y = 1$$

But $\sin^2 x = 1 \Rightarrow \sin x = \pm 1$

$$\Rightarrow \sin x = 1$$

$$\Rightarrow x = \frac{\pi}{2}$$

$$\frac{\pi}{2}, \frac{3\pi}{2} \in [0, 2\pi]$$

$\therefore$ Real numbers is 2

$$\sin x = -1$$

$$x = \frac{3\pi}{2}$$

8. If $x^3 + 12x^2 + 10ax + 1999$ definitely has a positive zero, if and only if

- (1) $a \geq 0$ (2) $a > 0$ (3) $a < 0$ (4) $a \leq 0$

Given: $= x^3 + 12x^2 + 10ax + 1999$

$$= + + + +$$

All are positive. $\therefore$ There is no positive real roots. If a is negative value

(ie) $a < 0$, we have

$$= + + - +$$

If a is negative, $p(x)$ has at most 2 positive zeros

Now we have positive real roots if $a < 0$

9. The polynomial $x^3 + 2x + 3$ has

MDL, PTA-3

- (1) one negative and two imaginary zeros (2) one positive and two imaginary zeros
(3) three real zeros (4) no zeros

Given: $P(x) = x^3 + 2x + 3$

$$= + + +$$

No positive real root

$\therefore$ This equation have one negative and 2 imaginary zeros

$P(-x) = -x^3 - 2x + 3$

$$= - - +$$

One negative real root

10. The number of positive zeros of the polynomial $\sum_{r=0}^n nC_r (-1)^r x^r$ is

APR-23, JUL-23

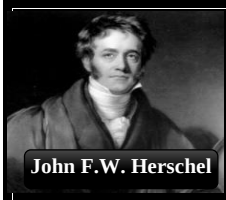
- (1) 0 (2) n (3) $< n$ (4) r

Let $P(x) = \sum_{r=0}^n nC_r (-1)^r x^r$

$$= nC_0 - nC_1 x + nC_2 x^2 + \dots + (-1)^n x^n$$

The number of change in sign in $P(x) = n$

$\therefore$ Number of positive zeros $\leq n$



Chapter - 4

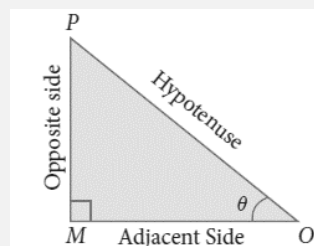
Inverse Trigonometric Functions

EXERCISE 4.1



Concept corner

Trigonometric Ratios

Let $0^\circ < \theta < 90^\circ$ 

$$\sin \theta = \frac{\text{Opposite side}}{\text{Hypotenuse}} = \frac{MP}{OP}$$

$$\operatorname{cosec} \theta = \frac{1}{\sin \theta}$$

$$\cos \theta = \frac{\text{Adjacent side}}{\text{Hypotenuse}} = \frac{OM}{OP}$$

$$\sec \theta = \frac{1}{\cos \theta}$$

$$\tan \theta = \frac{\sin \theta}{\cos \theta} \text{ (or) } \frac{1}{\cot \theta}$$

$$\cot \theta = \frac{\cos \theta}{\sin \theta} \text{ (or) } \frac{1}{\tan \theta}$$

θ	0°	30°	45°	60°	90°
$\sin \theta$	0	$\frac{1}{2}$	$\frac{1}{\sqrt{2}}$	$\frac{\sqrt{3}}{2}$	1
$\cos \theta$	1	$\frac{\sqrt{3}}{2}$	$\frac{1}{\sqrt{2}}$	$\frac{1}{2}$	0
$\tan \theta$	0	$\frac{1}{\sqrt{3}}$	1	$\sqrt{3}$	∞
$\operatorname{cosec} \theta$	∞	2	$\sqrt{2}$	$\frac{2}{\sqrt{3}}$	1
$\sec \theta$	1	$\frac{2}{\sqrt{3}}$	$\sqrt{2}$	2	∞
$\cot \theta$	∞	$\sqrt{3}$	1	$\frac{1}{\sqrt{3}}$	0

Identities : 1) $\sin^2 \theta + \cos^2 \theta = 1$ 2) $1 + \tan^2 \theta = \sec^2 \theta$ 3) $1 + \cot^2 \theta = \operatorname{cosec}^2 \theta$

Signs of trigonometric function:

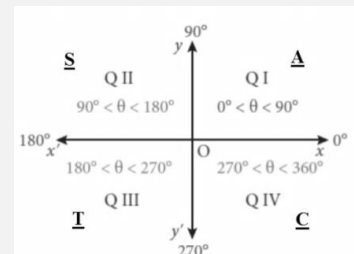
For your memorize: All Silver Tea Cups

A = All are +ve ($\sin \theta, \tan \theta, \cos \theta, \sec \theta, \operatorname{cosec} \theta, \cot \theta$)

S = $\sin \theta, \operatorname{cosec} \theta$ are +ve. others are -ve

T = $\tan \theta, \cot \theta$ are +ve. others are -ve

C = $\cos \theta, \sec \theta$ are +ve. others are -ve

**Some Fundamental Concepts****Even and odd function:**

A function is said to be **even** if $f(-x) = f(x)$, for all x , $-x$ are domain in f . **Ex:** $\cos \theta, \sec \theta, x^2, \dots$

A function is said to be **odd** if $f(-x) = -f(x)$ for all x **Ex:** $\sin \theta, \operatorname{cosec} \theta, \tan \theta, \cot \theta, x^3, \dots$

Domain and Range of trigonometric function:

Trigonometric function	$\sin x$	$\cos x$	$\tan x$	$\operatorname{cosec} x$	$\sec x$	$\cot x$
Domain	$\mathbb{R}$	$\mathbb{R}$	$\mathbb{R} \setminus \{(2n+1)\frac{\pi}{2}, n \in \mathbb{Z}\}$	$\mathbb{R} \setminus \{(n\pi, n \in \mathbb{Z})\}$	$\mathbb{R} \setminus \{(2n+1)\frac{\pi}{2}, n \in \mathbb{Z}\}$	$\mathbb{R} \setminus \{(n\pi, n \in \mathbb{Z})\}$
Range	$[-1, 1]$	$[-1, 1]$	$\mathbb{R}$	$\mathbb{R} \setminus (-1, 1)$	$\mathbb{R} \setminus (-1, 1)$	$\mathbb{R}$

Amplitude and period of a graph:

- The amplitude is the maximum distance of the graph from the x -axis.
- The amplitude of a function is the height from the x -axis to its maximum and minimum.
- The period is the distance required for the function to complete one full cycle.

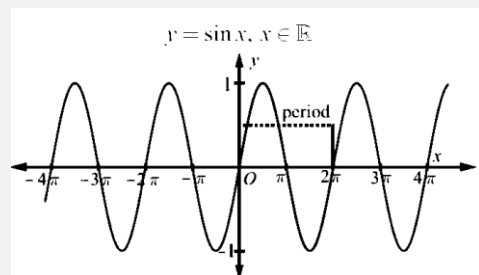
Formulae: $y = a \sin bx$ (i) Amplitude = $|a|$ (ii) Period = $\frac{2\pi}{|b|}$

Note: The graph of an odd function is symmetric with respect to the origin.

The graph of an even function is symmetric about the y -axis.

Sine function and Inverse Sine Function:(i) Domain: $\mathbb{R}$ (ii) Range $[-1, 1]$ (iii) Sine function $y = \sin x$ and Inverse Sine functions $y = \sin^{-1} x$ or $y = \arcsin(x)$

x (in radian)	0	$\frac{\pi}{6}$	$\frac{\pi}{4}$	$\frac{\pi}{3}$	$\frac{\pi}{2}$	π	$\frac{3\pi}{2}$	2π
$y = \sin x$	0	$\frac{1}{2}$	$\frac{1}{\sqrt{2}}$	$\frac{\sqrt{3}}{2}$	1	0	-1	0

**Properties of the sine functions:**

1. $y = \sin x$ is periodic with period 2π .
2. It's amplitude is 1.
3. The *sine* function is continuous function.
4. It is an odd function. The graph is symmetric with respect to the origin.
5. Maximum value of *sine* function is 1 & minimum value of *sin* function is -1 .

$$\therefore -1 \leq \sin x \leq 1, \forall x \in \mathbb{R}.$$

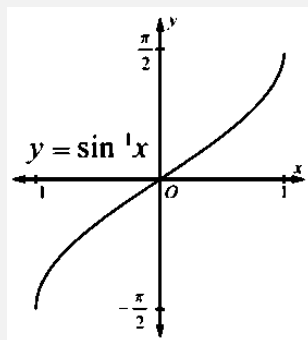
The Inverse sine function:

$\sin^{-1}: [-1, 1] \rightarrow \left[-\frac{\pi}{2}, \frac{\pi}{2}\right]$ is defined by $\sin^{-1} x = y$ if and only if $\sin y = x$ and $y \in \left[-\frac{\pi}{2}, \frac{\pi}{2}\right]$

Some results: $y = \sin^{-1} x$

The restricted domain $\left[-\frac{\pi}{2}, \frac{\pi}{2}\right]$ is called the **principal domain** of sine function and the values of $y = \sin^{-1} x$, $-1 \leq x \leq 1$ are known as **principal values** of the function $y = \sin^{-1} x$

1. $y = \sin^{-1} x$ if and only if $\sin y = x$ for $-1 \leq x \leq 1$ and $-\frac{\pi}{2} \leq y \leq \frac{\pi}{2}$
2. $\sin(\sin^{-1} x) = x$ if $|x| \leq 1$ and has no sense if $|x| > 1$
3. $\sin^{-1}(\sin x) = x$ if $-\frac{\pi}{2} \leq x \leq \frac{\pi}{2}$. Note that $\sin^{-1}(\sin 2\pi) = 0 \neq 2\pi$
4. $\sin^{-1}(\sin x) = \pi - x$ if $\frac{\pi}{2} \leq x \leq \frac{3\pi}{2}$ Note that $-\frac{\pi}{2} \leq \pi - x \leq \frac{\pi}{2}$
5. $y = \sin^{-1} x$ is an odd function.

Graph of $\sin^{-1} x$ (i) Domain $[-1, 1]$ (ii) Range $\left[-\frac{\pi}{2}, \frac{\pi}{2}\right]$ **Important points:**

1. The graph of $y = \sin^{-1} x$ is the reflecting the portion of entire graph of $y = \sin x$ in the interval $\left[-\frac{\pi}{2}, \frac{\pi}{2}\right]$ about the line $y = x$
2. It passes through origin.
3. It is symmetric with respect to the origin & $y = \sin^{-1} x$ is an odd function.

1. Find all the values of x such that

2M

(i) $-10\pi \leq x \leq 10\pi$ and $\sin x = 0$

(ii) $-3\pi \leq x \leq 3\pi$ and $\sin x = -1$

(i) Given: $\sin x = 0, -10\pi \leq x \leq 10\pi$

$$x = \sin^{-1}(0)$$

$$x = 0, \pm\pi, \pm2\pi, \pm3\pi, \dots, \pm10\pi$$

$$x = n\pi, n = 0, \pm1, \pm2, \dots, \pm10 \quad [\text{General solution of } \sin \theta = 0 \text{ is } \theta = n\pi, n \in \mathbb{Z}]$$

(ii) $\sin x = -1; -3\pi \leq x \leq 3\pi$

$$\sin x = \sin\left(-\frac{\pi}{2}\right)$$

$$x = n\pi + (-1)^n\left(-\frac{\pi}{2}\right) = n\pi - (-1)^n\frac{\pi}{2}, n \in \mathbb{Z}$$

General solution of $\sin \theta = \sin \alpha$ is
 $\Rightarrow \theta = n\pi + (-1)^n\alpha, n \in \mathbb{Z}$

Put $n = 0 \Rightarrow x = -\frac{\pi}{2}$; $n = 1 \Rightarrow x = \frac{3\pi}{2}$; $n = -1 \Rightarrow x = -\frac{3\pi}{2}$

Generalise this values $x = (4n - 1)\frac{\pi}{2}, n = 0, \pm1$

2. Find the period and amplitude of

2M

(i) $y = \sin 7x$

(ii) $y = -\sin\left(\frac{1}{3}x\right)$

(iii) $y = 4 \sin(-2x)$

Formula		
Let $y = a \sin bx$	(i) Amplitude = $ a $	(ii) Period = $\frac{2\pi}{ b }$
Let $y = a \cos bx$	(i) Amplitude = $ a $	(ii) Period = $\frac{2\pi}{ b }$
Let $y = a \tan bx$	(i) Amplitude = $ a $	(ii) Period = $\frac{\pi}{ b }$

(i) $y = \sin 7x$

$$a = 1; b = 7$$

$$\text{Amplitude} = |\bar{a}| = |1| = 1$$

$$\text{Period} = \frac{2\pi}{|b|} = \frac{2\pi}{7}$$

(ii) $y = -\sin\left(\frac{1}{3}x\right)$

$$a = -1; b = \frac{1}{3}$$

$$\text{Amplitude} = |a| = |-1| = 1$$

$$\text{Period} = \frac{2\pi}{|b|} = \frac{2\pi}{\frac{1}{3}} = 2\pi \cdot 3 = 6\pi$$

(iii) $y = 4 \sin(-2x)$

$$a = 4; b = -2$$

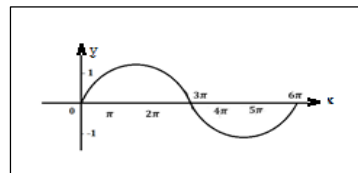
$$\therefore \text{Amplitude} = |\bar{a}| = |4| = 4$$

$$\text{Period} = \frac{2\pi}{|b|} = \frac{2\pi}{2} = \pi$$

3. Sketch the graph of $y = \sin\left(\frac{1}{3}x\right)$ for $0 \leq x < 6\pi$

$$a = 1; b = \frac{1}{3}$$

$$\text{Amplitude} = |a| = 1; \quad \text{Period} = \frac{2\pi}{|b|} = 2\pi \cdot 3 = 6\pi$$



3M

4. Find the value of (i) $\sin^{-1}\left(\sin\left(\frac{2\pi}{3}\right)\right)$

(ii) $\left(\sin^{-1}\left(\sin\frac{5\pi}{4}\right)\right)$

PTA-3, APR-20

2M

(i) $\sin^{-1}\left(\sin\left(\frac{2\pi}{3}\right)\right)$

$$= \pi - \frac{2\pi}{3} \quad \because \frac{2\pi}{3} \in \left[\frac{\pi}{2}, \frac{3\pi}{2}\right]$$

$$= \frac{3\pi - 2\pi}{3}$$

$$= \frac{\pi}{3} \in \left[-\frac{\pi}{2}, \frac{\pi}{2}\right]$$

(ii) $\sin^{-1}\left(\sin\frac{5\pi}{4}\right)$

$$= \pi - \frac{5\pi}{4}$$

$$\because \frac{5\pi}{4} \in \left[\frac{\pi}{2}, \frac{3\pi}{2}\right]$$

$$= \frac{4\pi - 5\pi}{4}$$

$$= -\frac{\pi}{4} \in \left[-\frac{\pi}{2}, \frac{\pi}{2}\right]$$

5. For what value of x does $\sin x = \sin^{-1} x$?

JUL-21

2M

$$\text{Given } \sin x = \sin^{-1} x$$

$$\text{Put } x = 0 \Rightarrow \sin 0 = \sin^{-1} 0$$

$$\therefore x = 0$$

6. Find the domain of the following

3M

$$(i) f(x) = \sin^{-1} \left(\frac{x^2+1}{2x} \right)$$

$$(ii) g(x) = 2 \sin^{-1}(2x - 1) - \frac{\pi}{4}$$

$$(i) f(x) = \sin^{-1} \left(\frac{x^2+1}{2x} \right)$$

$$\text{W.K.T. the domain of } \sin^{-1} x \text{ is } [-1, 1] \Rightarrow -1 \leq \frac{x^2+1}{2x} \leq 1$$

$$\Rightarrow -1 \leq \frac{x^2+1}{2x}$$

$$\text{Multiply by } x, \quad -1 \leq \frac{(x^2+1)x}{2x^2}$$

$$\Rightarrow -2x^2 \leq x^3 + x$$

$$\Rightarrow x^3 + 2x^2 + x \geq 0$$

$$\Rightarrow x(x^2 + 2x + 1) \geq 0$$

$$x(x+1)^2 \geq 0$$

$$x \in [0, \infty) \cup \{-1\}$$

$$\frac{x^2+1}{2x} \leq 1$$

$$\frac{x(x^2+1)}{2x^2} \leq 1$$

$$x^3 + x \leq 2x^2$$

$$x^3 - 2x^2 + x \leq 0$$

$$x(x^2 - 2x + 1) \leq 0$$

$$\Rightarrow x(x-1)^2 \leq 0$$

$$\Rightarrow x \in (-\infty, 0] \cup \{1\}$$

Domain is intersection of these sets $x \in \{-1, 1\}$

$$(ii) \text{ Given } g(x) = 2 \sin^{-1}(2x - 1) - \frac{\pi}{4}$$

$$\text{W.K.T the domain of } \sin^{-1} x \text{ is } [-1, 1] \Rightarrow -1 \leq 2x - 1 \leq 1$$

$$\Rightarrow -1 + 1 \leq 2x \leq 1 + 1$$

$$\Rightarrow 0 \leq 2x \leq 2$$

$$\Rightarrow 0 \leq x \leq 1$$

$$x \in [0, 1]$$

7. Find the value of $\sin^{-1}(\sin \frac{5\pi}{9} \cos \frac{\pi}{9} + \cos \frac{5\pi}{9} \sin \frac{\pi}{9})$

JUL-22, 23

$$\sin^{-1}(\sin \frac{5\pi}{9} \cos \frac{\pi}{9} + \cos \frac{5\pi}{9} \sin \frac{\pi}{9}) = \sin^{-1} \left(\sin \left(\frac{5\pi}{9} + \frac{\pi}{9} \right) \right)$$

$$= \sin^{-1} \left(\sin \left(\frac{6\pi}{9} \right) \right)$$

$$= \sin^{-1} \left(\sin \left(\frac{2\pi}{3} \right) \right)$$

$$= \pi - \frac{2\pi}{3}$$

$$= \frac{3\pi - 2\pi}{3}$$

$$= \frac{\pi}{3} \in \left[\frac{-\pi}{2}, \frac{\pi}{2} \right]$$

$$\therefore \sin(A+B) = \sin A \cos B + \cos A \sin B$$

$$A = \frac{5\pi}{9}, B = \frac{\pi}{9}$$

$$\therefore \frac{2\pi}{3} \in \left[\frac{\pi}{2}, \frac{3\pi}{2} \right]$$

EXERCISE 4.2



Video class

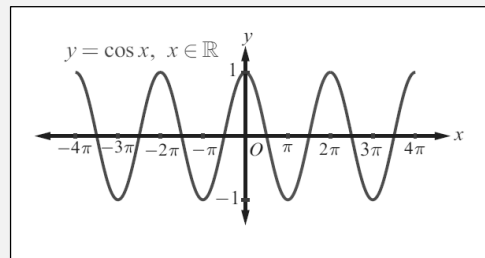
Concept Corner

The cosine and Inverse cosine function:

- (i) Domain: $\mathbb{R}$
 (ii) Range $[-1, 1]$
 (iii) cosine function $y = \cos x$ and

Inverse cosine function $y = \cos^{-1} x$ or $y = \arccos(x)$

x (in radian)	0	$\frac{\pi}{6}$	$\frac{\pi}{4}$	$\frac{\pi}{3}$	$\frac{\pi}{2}$	π	$\frac{3\pi}{2}$	2π
$y = \cos x$	1	$\frac{\sqrt{3}}{2}$	$\frac{1}{\sqrt{2}}$	$\frac{1}{2}$	0	-1	0	1



Properties:

- The cosine function is continuous.
- It is an even function. The graph is symmetric about y-axis.
- The maximum value of cosine function is 1 & minimum value of cosine function is -1.
 (ie) $-1 \leq \cos x \leq 1 \quad \forall x \in \mathbb{R}$

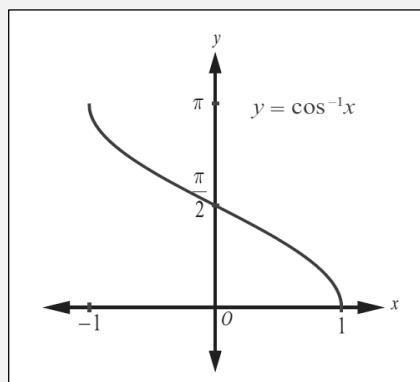
Inverse cosine function

 $\cos^{-1}: [-1, 1] \rightarrow [0, \pi]$ is defined by $\cos^{-1} x = y$ if and only if $\cos y = x$ and $y \in [0, \pi]$ Results: $y = \cos^{-1} x$ The restricted domain $[0, \pi]$ is called the **principal domain** of cosine function and the values of $y = \cos^{-1} x$, $-1 \leq x \leq 1$ are known as **principal values** of the function $y = \cos^{-1} x$

- $y = \cos^{-1} x$ if and only if $x = \cos y$ for $-1 \leq x \leq 1$ and $0 \leq y \leq \pi$
- $(\cos(\cos^{-1} x)) = x$ if $|x| \leq 1$ and has no sense if $|x| > 1$
- $\cos^{-1}(\cos(x)) = x$ if $0 \leq x \leq \pi$, the range of $\cos^{-1} x$. Note that $\cos^{-1}(\cos 3\pi) = \pi$

Graph of the inverse cosine function:

- (i) Domain $[-1, 1]$
 (ii) Range $[0, \pi]$



Note:

- The graph of $y = \cos^{-1} x$ is obtained from the graph $y = \cos x$ by interchanging x & y axes.
- In $y = \cos^{-1} x$, x intercept is 1 & y intercept is $\frac{\pi}{2}$
- The graph is not symmetric w.r.to either origin or y axis.
 $\therefore y = \cos^{-1} x$ is neither even nor odd function.

1. Find all values of x such that

(i) $-6\pi \leq x \leq 6\pi$ and $\cos x = 0$

2M

Given $\cos x = 0$ & $-6\pi \leq x \leq 6\pi$

$$\cos x = 0 = \cos \frac{\pi}{2}$$

$$x = \cos^{-1}(0)$$

$$x = (2n + 1) \frac{\pi}{2}, n = 0, \pm 1, \pm 2, \pm 3, \pm 4, \pm 5, -6$$

General solution of

(i) $\cos \theta = 0$ is $\theta = (2n + 1) \frac{\pi}{2}, n \in \mathbb{Z}$

(ii) $\cos \theta = \cos \alpha$ is $\theta = 2n\pi \pm \alpha, n \in \mathbb{Z}$

(ii) $-5\pi \leq x \leq 5\pi$ and $\cos x = -1$

Given $\cos x = -1$ & $-5\pi \leq x \leq 5\pi$

$$\cos x = -1$$

$$x = \cos^{-1}(-1) = (2n + 1)\pi, n = 0, \pm 1, \pm 2, -3$$

2. State the reason for $\cos^{-1}\left(\cos\left(-\frac{\pi}{6}\right)\right) \neq -\frac{\pi}{6}$

PTA-5

2M

The principal domain of $\cos^{-1}(x)$ is $[0, \pi]$

Here $\cos^{-1}\left(\cos\left(-\frac{\pi}{6}\right)\right) = \frac{\pi}{6} \notin [0, \pi]$

$$\therefore \cos^{-1}\left(\cos\left(-\frac{\pi}{6}\right)\right) \neq -\frac{\pi}{6}$$

3. Is $\cos^{-1}(-x) = \pi - \cos^{-1}(x)$ true? Justify your answer.

2M

Let $\cos^{-1}(-x) = \theta \Rightarrow -x = \cos \theta$

$$\Rightarrow x = -\cos \theta$$

$$\Rightarrow x = \cos(\pi - \theta)$$

$$\Rightarrow \cos^{-1} x = \pi - \theta$$

$$\Rightarrow \theta = \pi - \cos^{-1}(x)$$

$$\Rightarrow \cos^{-1}(-x) = \pi - \cos^{-1}(x) \text{ is true.}$$

4. Find the principal value of $\cos^{-1}\left(\frac{1}{2}\right)$

Let $\cos^{-1}\left(\frac{1}{2}\right) = y$

$$\cos y = \frac{1}{2}$$

$$\cos y = \cos \frac{\pi}{3}$$

$$y = \frac{\pi}{3} \in [0, \pi]$$

$$\therefore \text{The principal value of } \cos^{-1}\left(\frac{1}{2}\right) = \frac{\pi}{3}$$

5. Find the value of (i) $2 \cos^{-1}\left(\frac{1}{2}\right) + \sin^{-1}\left(\frac{1}{2}\right)$

(ii) $\cos^{-1}\left(\frac{1}{2}\right) + \sin^{-1}(-1)$

2M

(iii) $\cos^{-1}\left(\cos \frac{\pi}{7} \cos \frac{\pi}{17} - \sin \frac{\pi}{7} \sin \frac{\pi}{17}\right)$

$$\begin{aligned} \text{(i) } 2 \cos^{-1}\left(\frac{1}{2}\right) + \sin^{-1}\left(\frac{1}{2}\right) &= 2 \frac{\pi}{3} + \frac{\pi}{6} \\ &= \frac{2\pi}{3} + \frac{\pi}{6} \\ &= \frac{4\pi + \pi}{6} \end{aligned}$$

$$\therefore 2 \cos^{-1}\left(\frac{1}{2}\right) + \sin^{-1}\left(\frac{1}{2}\right) = \frac{5\pi}{6}$$

$$\text{(ii) } \cos^{-1}\left(\frac{1}{2}\right) + \sin^{-1}(-1)$$

$$\begin{aligned} \cos^{-1}\left(\frac{1}{2}\right) + \sin^{-1}(-1) &= \cos^{-1}\left(\frac{1}{2}\right) + \sin^{-1}(-1) \\ &= \frac{\pi}{3} - \frac{\pi}{2} = \frac{2\pi - 3\pi}{6} \end{aligned}$$

$$\therefore \cos^{-1}\left(\frac{1}{2}\right) + \sin^{-1}(-1) = \frac{-\pi}{6}$$

$$\text{(iii) } \cos^{-1}\left(\cos \frac{\pi}{7} \cos \frac{\pi}{17} - \sin \frac{\pi}{7} \sin \frac{\pi}{17}\right) = \cos^{-1}\left(\cos\left(\frac{\pi}{7} + \frac{\pi}{17}\right)\right)$$

$$\cos A \cos B - \sin A \sin B = \cos(A + B)$$

$$= \cos^{-1}\left(\cos \frac{17\pi + 7\pi}{119}\right) = \frac{24\pi}{119}$$

$$\therefore \cos^{-1}\left(\cos \frac{\pi}{7} \cos \frac{\pi}{17} - \sin \frac{\pi}{7} \sin \frac{\pi}{17}\right) = \frac{24\pi}{119} \quad \because \frac{24\pi}{119} \in [0, \pi]$$

6. Find the domain of (i) $f(x) = \sin^{-1}\left(\frac{|x|-2}{3}\right) + \cos^{-1}\left(\frac{1-|x|}{4}\right)$ (ii) $g(x) = \sin^{-1}x + \cos^{-1}x$. 3M

(i) Given $f(x) = \sin^{-1}\left(\frac{|x|-2}{3}\right) + \cos^{-1}\left(\frac{1-|x|}{4}\right)$

The Domain of $\sin^{-1}x$ & $\cos^{-1}x$ is $[-1, 1]$

$$\begin{aligned} \Rightarrow -1 &\leq \frac{|x|-2}{3} \leq 1 & \text{and} & \quad -1 \leq \frac{1-|x|}{4} \leq 1 \\ \Rightarrow -3 &\leq |x|-2 \leq 3 & \text{and} & \quad -4 \leq 1-|x| \leq 4 \\ \Rightarrow -3+2 &\leq |x| \leq 3+2 & \text{and} & \quad -4-1 \leq -|x| \leq 3 \\ \Rightarrow -1 &\leq |x| \leq 5 & \text{and} & \quad -5 \leq -|x| \leq 3 \\ \Rightarrow -1 &\leq |x| \leq 5 & \text{and} & \quad 5 \geq |x| \geq -3 \\ \Rightarrow -1 &\leq |x| \leq 5 & \text{and} & \quad -3 \leq |x| \leq 5 \\ & \Rightarrow |x| \leq 5 \Rightarrow -5 \leq x \leq 5 \\ & \Rightarrow x \in [-5, 5] \end{aligned}$$

(ii) $g(x) = \sin^{-1}x + \cos^{-1}x$

The domain of $\sin^{-1}(x)$ is $[-1, 1]$

The domain of $\cos^{-1}(x)$ is $[-1, 1]$

The domain of $g(x)$ is $[-1, 1]$

7. For what value of x , the inequality $\frac{\pi}{2} < \cos^{-1}(3x-1) < \pi$ holds? APR-23 3M

W.K.T The value of $\cos^{-1}x$ lies in II quadrant.

$$\therefore x \in (-1, 0)$$

$$\text{Here } \Rightarrow -1 < 3x - 1 < 0$$

$$\Rightarrow -1 + 1 < 3x < 0 + 1$$

$$\Rightarrow 0 < 3x < 1$$

$$\Rightarrow 0 < x < \frac{1}{3}$$

$$\Rightarrow x \in \left(0, \frac{1}{3}\right)$$

8. Find the value of (i) $\cos\left[\cos^{-1}\left(\frac{4}{5}\right) + \sin^{-1}\left(\frac{4}{5}\right)\right]$ (ii) $\cos^{-1}\left(\cos\left(\frac{4\pi}{3}\right)\right) + \cos^{-1}\left(\cos\left(\frac{5\pi}{4}\right)\right)$. 3M

(i) $\cos\left[\cos^{-1}\left(\frac{4}{5}\right) + \sin^{-1}\left(\frac{4}{5}\right)\right]$

$$\cos\left[\cos^{-1}\left(\frac{4}{5}\right) + \sin^{-1}\left(\frac{4}{5}\right)\right] = \cos\left(\frac{\pi}{2}\right) = 0$$

$$\cos^{-1}x + \sin^{-1}x = \frac{\pi}{2} \quad \therefore \cos\left(\frac{\pi}{2}\right) = 0$$

(ii) $\cos^{-1}\left(\cos\left(\frac{4\pi}{3}\right)\right) + \cos^{-1}\left(\cos\frac{5\pi}{4}\right)$ JUL-21

$$\begin{aligned} \cos^{-1}\left(\cos\left(\frac{4\pi}{3}\right)\right) + \cos^{-1}\left(\cos\frac{5\pi}{4}\right) &= \cos^{-1}\left(\cos\left(2\pi - \frac{2\pi}{3}\right)\right) + \cos^{-1}\left(\cos\left(2\pi - \frac{3\pi}{4}\right)\right) \\ &= \cos^{-1}\left(\cos\left(\frac{2\pi}{3}\right)\right) + \cos^{-1}\left(\cos\frac{3\pi}{4}\right) \quad \left(\because \frac{3\pi}{4} \in (0, \pi)\right) \\ &= \frac{2\pi}{3} + \frac{3\pi}{4} \\ &= \frac{8\pi+9\pi}{12} \\ &= \frac{17\pi}{12} \end{aligned}$$

EXERCISE 4.3



Video class

Concept Corner

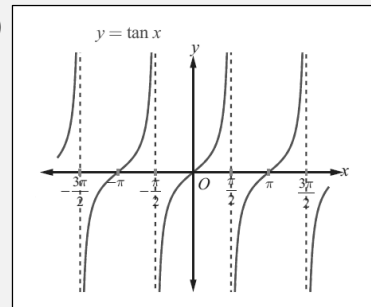
The Tangent Function: $y = \tan x = \frac{\sin x}{\cos x}$

$$\text{Domain } \left\{ x : x \in \mathbb{R}, x \neq \frac{\pi}{2} + k\pi, k \in \mathbb{Z} \right\} = \bigcup_{k=-\infty}^{\infty} \left(\frac{2k+1}{2}\pi, \frac{2k+3}{2}\pi \right)$$

$$\text{Range} = (-\infty, \infty) ; \quad \text{Period} = \pi$$

The graph of tangent function:

x (in radian)	$-\frac{\pi}{3}$	$-\frac{\pi}{4}$	$-\frac{\pi}{6}$	0	$\frac{\pi}{6}$	$\frac{\pi}{4}$	$\frac{\pi}{3}$
$y = \tan x$	$-\sqrt{3}$	-1	$-\frac{\sqrt{3}}{3}$	0	$\frac{\sqrt{3}}{3}$	1	$\sqrt{3}$



Properties:

1. The graph is not continuous. That is it has discontinuity points at $x = (2n+1)\frac{\pi}{2}, n \in \mathbb{Z}$.
2. The partial graph is symmetric about the origin for $-\frac{\pi}{2} < x < \frac{\pi}{2}$
3. It has infinitely many vertical asymptotes $x = (2n+1)\frac{\pi}{2}, n \in \mathbb{Z}$.
4. It has neither maximum nor minimum.

The inverse of Tangent function

$$\text{Domain} = \mathbb{R} ; \quad \text{Range} = \left(-\frac{\pi}{2}, \frac{\pi}{2} \right)$$

Results: $y = \tan^{-1} x$

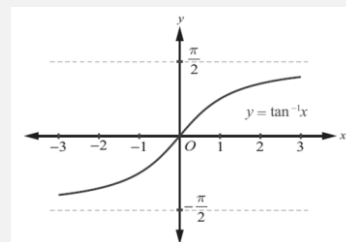
The restricted domain $\left(-\frac{\pi}{2}, \frac{\pi}{2} \right)$ is called the **principal domain** of tangent function and the values of $y = \tan^{-1} x, x \in \mathbb{R}$ are known as **principal values** of the function $y = \tan^{-1} x$

Properties: 1. $y = \tan^{-1} x$ iff $x = \tan y$ for $x \in \mathbb{R}$ and $-\frac{\pi}{2} < y < \frac{\pi}{2}$

2. $\tan(\tan^{-1} x) = x$ for all x is $\mathbb{R}$ and $y = \tan^{-1} x$ is an odd function.

3. $\tan^{-1}(\tan x) = x$ iff $-\frac{\pi}{2} < x < \frac{\pi}{2}$ $\tan^{-1}(\tan \pi) = 0$ and not $= \pi$

Graph of the inverse tangent function:



Note:

1. The inverse tangent function is strictly increasing and continuous on the domain $(-\infty, \infty)$.
2. The graph $y = \tan^{-1} x$ passes through the origin.
3. The graph is symmetric with respect to origin and hence $y = \tan^{-1} x$ is an odd function.

1. Find the domain of the following functions: (i) $\tan^{-1}(\sqrt{9-x^2})$ (ii) $\frac{1}{2}\tan^{-1}(1-x^2) - \frac{\pi}{4}$ 3M

(i) $\tan^{-1}(\sqrt{9-x^2})$

The domain of $\tan^{-1} x$ is $(-\infty, \infty)$

$$\Rightarrow \sqrt{9-x^2} \text{ is always real}$$

$$\Rightarrow 9-x^2 \geq 0 \quad (\text{or}) \quad 9 \geq x^2$$

$$(\text{or}) \quad x^2 \leq 9 \Rightarrow |x| \leq 3 \Rightarrow x \in [-3, 3]$$

(ii) $\frac{1}{2}\tan^{-1}(1-x^2) - \frac{\pi}{4}$

The domain of $\tan^{-1} x$ is $(-\infty, \infty)$

$$-\infty < 1-x^2 < \infty$$

$$-\infty < x < \infty$$

$$\Rightarrow x \in \mathbb{R}$$

2. Find the value of (i) $\tan^{-1}\left(\tan\frac{5\pi}{4}\right)$ (JUL-24) (ii) $\tan^{-1}\left(\tan\left(\frac{-\pi}{6}\right)\right)$ (PTA-1) (3M)

$$\begin{array}{l|l} \text{(i) } \tan^{-1}\left(\tan\frac{5\pi}{4}\right) = \tan^{-1}\left(\tan\left(\frac{5\pi}{4} - \pi\right)\right) = \tan^{-1}\left(\tan\left(\frac{\pi}{4}\right)\right) & \text{(ii) } \tan^{-1}\left(\tan\left(\frac{-\pi}{6}\right)\right) \\ \therefore \tan^{-1}\left(\tan\frac{5\pi}{4}\right) = \frac{\pi}{4} \in \left(\frac{-\pi}{2}, \frac{\pi}{2}\right) & = \frac{-\pi}{6} \in \left(\frac{-\pi}{2}, \frac{\pi}{2}\right) \end{array}$$

3. Find the value of (i) $\tan\left(\tan^{-1}\left(\frac{7\pi}{4}\right)\right)$ (ii) $\tan(\tan^{-1}(1947))$ (iii) $\tan(\tan^{-1}(-0.2021))$

$$\begin{array}{l|l|l} \text{(i) } \tan\left(\tan^{-1}\left(\frac{7\pi}{4}\right)\right) & \text{(ii) } \tan(\tan^{-1}(1947)) & \text{(iii) } \tan(\tan^{-1}(-0.2021)) \\ \text{W.K.T } \tan(\tan^{-1}x) = x & \tan(\tan^{-1}(1947)) = \mathbf{1947} & \tan(\tan^{-1}(-0.2021)) \\ \tan\left(\tan^{-1}\left(\frac{7\pi}{4}\right)\right) = \frac{7\pi}{4} & & = \mathbf{-0.2021} \end{array}$$

4. Find the value of (i) $\tan\left[\cos^{-1}\left(\frac{1}{2}\right) - \sin^{-1}\left(\frac{-1}{2}\right)\right]$ (ii) $\sin\left[\tan^{-1}\left(\frac{1}{2}\right) - \cos^{-1}\left(\frac{4}{5}\right)\right]$ (iii) $\cos\left(\sin^{-1}\left(\frac{4}{5}\right) - \tan^{-1}\left(\frac{3}{4}\right)\right)$

(3M)

$$\begin{array}{l} \text{(i) } \tan\left[\cos^{-1}\left(\frac{1}{2}\right) - \sin^{-1}\left(\frac{-1}{2}\right)\right] = \tan\left[\frac{\pi}{3} - \left(\frac{-\pi}{6}\right)\right] \\ = \tan\left[\frac{\pi}{3} + \frac{\pi}{6}\right] = \tan\left(\frac{2\pi + \pi}{6}\right) = \tan\left(\frac{3\pi}{6}\right) = \tan\left(\frac{\pi}{2}\right) \end{array}$$

$$\therefore \tan\left[\cos^{-1}\left(\frac{1}{2}\right) - \sin^{-1}\frac{-1}{2}\right] = \infty$$

$$\text{(ii) } \sin\left[\tan^{-1}\left(\frac{1}{2}\right) - \cos^{-1}\left(\frac{4}{5}\right)\right]$$

$$\text{Let } x = \tan^{-1}\frac{1}{2}$$

$$\Rightarrow \tan x = \frac{1}{2}$$

$$\therefore \sin x = \frac{1}{\sqrt{5}}$$

$$\cos x = \frac{2}{\sqrt{5}}$$

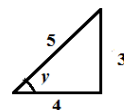
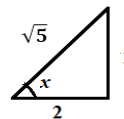
$$\text{Let } y = \cos^{-1}\left(\frac{4}{5}\right)$$

$$\cos y = \frac{4}{5}$$

$$\therefore \sin y = \frac{3}{5}$$

$$\tan y = \frac{3}{4}$$

By Pythagoras theorem,
 $z^2 = 1^2 + 2^2 \Rightarrow z = \sqrt{5}$



$$z^2 = 5^2 - 4^2 \Rightarrow z = 3$$

$$\therefore \sin\left[\tan^{-1}\left(\frac{1}{2}\right) - \cos^{-1}\left(\frac{4}{5}\right)\right] = \sin(x - y)$$

$$= \sin x \cos y - \cos x \sin y$$

$$= \frac{1}{\sqrt{5}} \cdot \frac{4}{5} - \frac{2}{\sqrt{5}} \cdot \frac{3}{5} = \frac{4}{5\sqrt{5}} - \frac{6}{5\sqrt{5}} = \frac{-2}{5\sqrt{5}} = \frac{-2\sqrt{5}}{25}$$

$$\text{(iii) } \cos\left(\sin^{-1}\left(\frac{4}{5}\right) - \tan^{-1}\left(\frac{3}{4}\right)\right)$$

$$\text{Let } x = \sin^{-1}\frac{4}{5}$$

$$\Rightarrow \sin x = \frac{4}{5}$$

$$\Rightarrow \cos x = \frac{3}{5}$$

$$y = \tan^{-1}\frac{3}{4} \Rightarrow \tan y = \frac{3}{4}$$

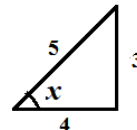
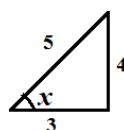
$$\sin y = \frac{3}{5} \Rightarrow y = \sin^{-1}\frac{3}{5}$$

$$\cos y = \frac{4}{5} \Rightarrow y = \cos^{-1}\frac{4}{5}$$

$$\cos\left(\sin^{-1}\left(\frac{4}{5}\right) - \tan^{-1}\left(\frac{3}{4}\right)\right) = \cos(x - y)$$

$$= \cos x \cos y + \sin x \sin y = \frac{3}{5} \cdot \frac{4}{5} + \frac{4}{5} \cdot \frac{3}{5}$$

$$= \frac{12}{25} + \frac{12}{25} = \frac{24}{25}$$



EXERCISE 4.4



Video class

Concept Corner

Functions

Cosecant function: ($\operatorname{cosec} x$)

- (i) Domain : $R \setminus \{n\pi : n \in Z\}$
- (ii) Range $(-\infty, -1] \cup [1, \infty)$.
- (iii) It is an odd function.

The Inverse cosecant function: ($\operatorname{cosec}^{-1} x$)

- (i) Domain : $(-\infty, -1] \cup [1, \infty)$
- (ii) Range : $[-\frac{\pi}{2}, 0) \cup (0, \frac{\pi}{2}]$

Secant function: ($\sec x$)

- (i) Domain : $R \setminus \{(2n+1)\frac{\pi}{2} : n \in Z\}$
- (ii) Range : $(-\infty, -1] \cup [1, \infty)$
- (iii) It is an even function

Inverse secant function : ($\sec^{-1} x$)

- (i) Domain : $R \setminus (-1, 1)$
- (ii) Range : $[0, \pi] \setminus \{\frac{\pi}{2}\}$

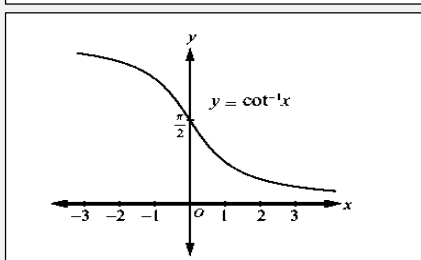
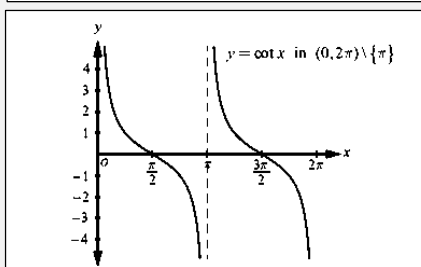
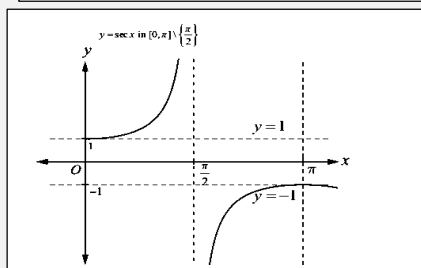
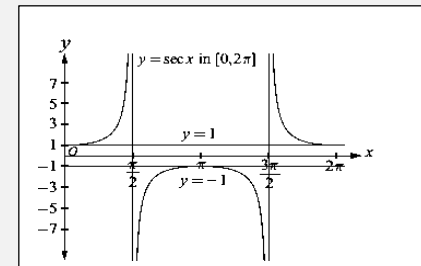
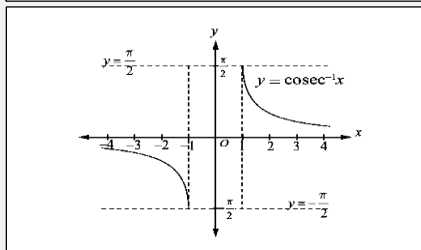
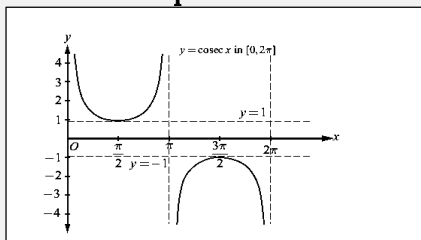
Cotangent function: ($\cot x$)

- (i) Domain : $R \setminus \{n\pi : n \in Z\}$
- (ii) Range : $(-\infty, \infty)$
- (iii) It is an odd function.
- (iv) It is periodic with period π

Inverse of cotangent function ($\cot^{-1} x$)

- (i) Domain : $(-\infty, \infty)$
- (ii) Range : $(0, \pi)$

Graph



Principal Domain and Range:

TRIGONOMETRIC FUNCTION		
FUNCTION	PRINCIPAL DOMAIN	RANGE
$\sin$	$\left[-\frac{\pi}{2}, \frac{\pi}{2}\right]$	$[-1, 1]$
$\cos$	$[0, \pi]$	$[-1, 1]$
$\tan$	$\left(-\frac{\pi}{2}, \frac{\pi}{2}\right)$	R
cosec	$\left[-\frac{\pi}{2}, \frac{\pi}{2}\right] \setminus \{0\}$	$R \setminus (-1, 1)$
$\sec$	$[0, \pi] \setminus \left\{\frac{\pi}{2}\right\}$	$R \setminus (-1, 1)$
$\cot$	$(0, \pi)$	R

INVERSE TRIGONOMETRIC FUNCTION		
FUNCTION	PRINCIPAL DOMAIN	RANGE
$\sin^{-1}$	$[-1, 1]$	$\left[-\frac{\pi}{2}, \frac{\pi}{2}\right]$
$\cos^{-1}$	$[-1, 1]$	$[0, \pi]$
$\tan^{-1}$	R	$\left(-\frac{\pi}{2}, \frac{\pi}{2}\right)$
$\operatorname{cosec}^{-1}$	$R \setminus (-1, 1)$	$\left[-\frac{\pi}{2}, \frac{\pi}{2}\right] \setminus \{0\}$
$\sec^{-1}$	$R \setminus (-1, 1)$	$[0, \pi] \setminus \left\{\frac{\pi}{2}\right\}$
$\cot^{-1}$	R	$(0, \pi)$

1. Find the principal value of (i) $\sec^{-1}\left(\frac{2}{\sqrt{3}}\right)$ (ii) $\cot^{-1}(\sqrt{3})$ (iii) $\operatorname{cosec}^{-1}(-\sqrt{2})$

2M

(i) Let $y = \sec^{-1}\left(\frac{2}{\sqrt{3}}\right)$

$$\sec y = \frac{2}{\sqrt{3}}$$

$$\sec y = \sec \frac{\pi}{6} \Rightarrow y = \frac{\pi}{6}$$

$$\therefore \sec^{-1}\left(\frac{2}{\sqrt{3}}\right) = \frac{\pi}{6}$$

(ii) Let $x = \cot^{-1}(\sqrt{3})$

$$\cot x = \sqrt{3}$$

$$\cot x = \cot \frac{\pi}{6}$$

$$\Rightarrow x = \frac{\pi}{6}$$

$$\therefore \cot^{-1} \sqrt{3} = \frac{\pi}{6}$$

(iii) $\alpha = \operatorname{cosec}^{-1}(-\sqrt{2})$

$$\operatorname{cosec} \alpha = -\sqrt{2}$$

$$\operatorname{cosec} \alpha = -\operatorname{cosec} \frac{\pi}{4}$$

$$\Rightarrow \operatorname{cosec} \alpha = \operatorname{cosec} \left(-\frac{\pi}{4}\right)$$

$$\Rightarrow \alpha = -\frac{\pi}{4}$$

2. Find the value of (i) $\tan^{-1}(\sqrt{3}) - \sec^{-1}(-2)$ (ii) $\sin^{-1}(-1) + \cos^{-1}\left(\frac{1}{2}\right) + \cot^{-1}(2)$
 (iii) $\cot^{-1}(1) + \sin^{-1}\left(\frac{-\sqrt{3}}{2}\right) - \sec^{-1}(-\sqrt{2})$

2M

(i) $\tan^{-1}(\sqrt{3}) - \sec^{-1}(-2)$

$$\tan^{-1}(\sqrt{3}) - \sec^{-1}(-2) = \frac{\pi}{3} - (\pi - \sec^{-1} 2)$$

$$= \frac{\pi}{3} - \pi + \sec^{-1} 2$$

$$= \frac{\pi}{3} - \pi + \frac{\pi}{3} = \frac{\pi}{3} - \frac{2\pi}{3} = \frac{\pi - 2\pi}{3} = \frac{-\pi}{3}$$

$$\therefore \sec^{-1}(-x) = \pi - \sec^{-1} x$$

(ii) $\sin^{-1}(-1) + \cos^{-1}\left(\frac{1}{2}\right) + \cot^{-1}(2)$

3M

Let $\alpha = \sin^{-1}(-1)$

$$\sin \alpha = -1$$

$$\sin \alpha = -\sin \frac{\pi}{2}$$

$$\sin \alpha = \sin \left(-\frac{\pi}{2}\right)$$

$$\therefore \alpha = \frac{-\pi}{2}$$

$$\therefore \sin^{-1}(-1) = \frac{-\pi}{2}$$

$\beta = \cos^{-1}\left(\frac{1}{2}\right)$

$$\cos \beta = \frac{1}{2}$$

$$\cos \beta = \cos \frac{\pi}{3}$$

$$\beta = \frac{\pi}{3}$$

$$\therefore \cos^{-1}\left(\frac{1}{2}\right) = \frac{\pi}{3}$$

$\gamma = \cot^{-1}(2)$

$$\cot \gamma = 2$$

$$\gamma = \cot^{-1}(2)$$

$$\begin{aligned} \therefore \text{The Given: } \sin^{-1}(-1) + \cos^{-1}\left(\frac{1}{2}\right) + \cot^{-1}(2) &= \frac{-\pi}{2} + \frac{\pi}{3} + \cot^{-1}(2) \\ &= \frac{-3\pi + 2\pi}{6} + \cot^{-1}(2) \\ &= \frac{-\pi}{6} + \cot^{-1}(2) \\ &= \cot^{-1}(2) - \frac{\pi}{6} \end{aligned}$$

$$(iii) \cot^{-1}(1) + \sin^{-1}\left(\frac{-\sqrt{3}}{2}\right) - \sec^{-1}(-\sqrt{2})$$

$$\text{Let } \alpha = \cot^{-1}(1)$$

$$\cot \alpha = 1$$

$$\cot \alpha = \cot \frac{\pi}{4}$$

$$\alpha = \frac{\pi}{4}$$

$$\therefore \cot^{-1}(1) = \frac{\pi}{4}$$

$$\beta = \sin^{-1}\left(\frac{-\sqrt{3}}{2}\right)$$

$$\sin \beta = \frac{-\sqrt{3}}{2}$$

$$\sin \beta = \sin\left(\frac{-\pi}{3}\right)$$

$$\therefore \sin^{-1}\left(\frac{-\sqrt{3}}{2}\right) = \frac{-\pi}{3}$$

$$\gamma = \sec^{-1}(-\sqrt{2})$$

$$\sec \gamma = -\sqrt{2}$$

$$\sec \gamma = -\sec \frac{\pi}{4}$$

$$= \sec\left(\pi - \frac{\pi}{4}\right) = \sec \frac{3\pi}{4}$$

$$\therefore \sec^{-1}(-\sqrt{2}) = \frac{3\pi}{4}$$

$$\begin{aligned} \text{Given: } \cot^{-1}(1) + \sin^{-1}\left(\frac{-\sqrt{3}}{2}\right) - \sec^{-1}(-\sqrt{2}) &= \frac{\pi}{4} - \frac{\pi}{3} - \frac{3\pi}{4} \\ &= \frac{\pi}{4} - \frac{3\pi}{4} - \frac{\pi}{3} \\ &= \frac{-2\pi}{4} - \frac{\pi}{3} \\ &= \frac{-\pi}{2} - \frac{\pi}{3} = \frac{-3\pi - 2\pi}{6} = \frac{-5\pi}{6} \end{aligned}$$

EXERCISE 4.5



Video class

Concept Corner

Properties of Inverse Trigonometric function

Property: I

$$(i) \sin^{-1}(\sin \theta) = \theta \text{ if } \theta \in \left[-\frac{\pi}{2}, \frac{\pi}{2}\right]$$

$$(ii) \cos^{-1}(\cos \theta) = \theta, \text{ if } \theta \in [0, \pi]$$

$$(iii) \tan^{-1}(\tan \theta) = \theta, \text{ if } \theta \in \left(-\frac{\pi}{2}, \frac{\pi}{2}\right)$$

$$(iv) \operatorname{cosec}^{-1}(\operatorname{cosec} \theta) = \theta, \text{ if } \theta \in \left[-\frac{\pi}{2}, \frac{\pi}{2}\right] \setminus \{0\}$$

$$(v) \sec^{-1}(\sec \theta) = \theta, \text{ if } \theta \in [0, \pi] \setminus \left\{\frac{\pi}{2}\right\}$$

$$(vi) \cot^{-1}(\cot \theta) = \theta, \text{ if } \theta \in (0, \pi)$$

Property: II

$$(i) \sin(\sin^{-1} x) = x, \text{ if } x \in [-1, 1]$$

$$(ii) \cos(\cos^{-1} x) = x; \text{ if } x \in [-1, 1]$$

$$(iii) \tan(\tan^{-1} x) = x; \text{ if } x \in \mathbb{R}$$

$$(iv) \operatorname{cosec}(\operatorname{cosec}^{-1} x) = x; \text{ if } x \in \mathbb{R} \setminus (-1, 1)$$

$$(v) \sec(\sec^{-1} x) = x; \text{ if } x \in \mathbb{R} \setminus (-1, 1)$$

$$(vi) \cot(\cot^{-1} x) = x; \text{ if } x \in \mathbb{R}$$

Property: III (Reciprocal Inverse Identities)

$$(i) \sin^{-1}\left(\frac{1}{x}\right) = \operatorname{cosec}^{-1} x, \text{ if } x \in \mathbb{R} \setminus (-1, 1)$$

$$(ii) \cos^{-1}\left(\frac{1}{x}\right) = \sec^{-1} x, \text{ if } x \in \mathbb{R} \setminus (-1, 1)$$

$$(iii) \tan^{-1}\left(\frac{1}{x}\right) = \begin{cases} \cot^{-1} x, & \text{if } x > 0 \\ -\pi + \cot^{-1} x & \text{if } x < 0 \end{cases}$$

Property: IV (Reflection Identities)

$$(i) \sin^{-1}(-x) = -\sin^{-1} x, \text{ if } x \in [-1, 1]$$

$$(ii) \cos^{-1}(-x) = \pi - \cos^{-1} x; \text{ if } x \in [-1, 1]$$

$$(ii) \tan^{-1}(-x) = -\tan^{-1} x, \text{ if } x \in \mathbb{R}$$

$$(iii) \operatorname{cosec}^{-1}(-x) = -\operatorname{cosec}^{-1}(x); \text{ if } |x| \geq 1 \text{ or } x \in \mathbb{R} \setminus (-1, 1)$$

$$(v) \sec^{-1}(-x) = \pi - \sec^{-1}(x); \text{ if } |x| \geq 1 \text{ or } x \in \mathbb{R} \setminus (-1, 1)$$

$$(vi) \cot^{-1}(-x) = \pi - \cot^{-1}(x); \text{ if } x \in \mathbb{R}$$

Property: V (Cofunction Inverse Identities)

- (i) $\sin^{-1}(x) + \cos^{-1}(x) = \frac{\pi}{2}, x \in [-1, 1]$
 (ii) $\tan^{-1}(x) + \cot^{-1}(x) = \frac{\pi}{2}; x \in R$
 (iii) $\operatorname{cosec}^{-1}(x) + \sec^{-1}(x) = \frac{\pi}{2}, x \in R \setminus (-1, 1) \text{ or } |x| \geq 1$

Property: VI

- (i) $\sin^{-1}(x) + \sin^{-1}(y) = \sin^{-1}[x\sqrt{1-y^2} + y\sqrt{1-x^2}]$ where either $x^2 + y^2 \leq 1$ or $xy < 0$
 (ii) $\sin^{-1}(x) - \sin^{-1}(y) = \sin^{-1}[x\sqrt{1-y^2} - y\sqrt{1-x^2}]$ where either $x^2 + y^2 \leq 1$ or $xy > 0$
 (iii) $\cos^{-1}(x) - \cos^{-1}(y) = \cos^{-1}[xy + \sqrt{1-x^2}\sqrt{1-y^2}]$, if $x \leq y$
 (iv) $\cos^{-1}(x) + \cos^{-1}(y) = \cos^{-1}[xy - \sqrt{1-x^2}\sqrt{1-y^2}]$, if $x + y \geq 0$
 (v) $\tan^{-1}(x) + \tan^{-1}(y) = \tan^{-1}\left(\frac{x+y}{1-xy}\right)$, if $xy < 1$,
 (vi) $\tan^{-1}(x) - \tan^{-1}(y) = \tan^{-1}\left(\frac{x-y}{1+xy}\right)$, if $xy > -1$

Property: VII

- (i) $2 \tan^{-1} x = \tan^{-1}\left(\frac{2x}{1-x^2}\right), |x| < 1$
 (ii) $2 \tan^{-1} x = \cos^{-1}\left(\frac{1-x^2}{1+x^2}\right), x \geq 0$
 (iii) $2 \tan^{-1} x = \sin^{-1}\left(\frac{2x}{1+x^2}\right), |x| \leq 1$

Property: VIII

- (i) $\sin^{-1}(2x\sqrt{1-x^2}) = 2 \sin^{-1} x$ if $|x| \leq \frac{1}{\sqrt{2}}$ (or) $-\frac{1}{\sqrt{2}} \leq x \leq \frac{1}{\sqrt{2}}$
 (ii) $\sin^{-1}(2x\sqrt{1-x^2}) = 2 \cos^{-1}(x)$ if $\frac{1}{\sqrt{2}} \leq x \leq 1$

Property: IX

- (i) $\sin^{-1}(x) = \cos^{-1}\sqrt{1-x^2}$ if $0 \leq x \leq 1$ (ii) $\sin^{-1}(x) = -\cos^{-1}\sqrt{1-x^2}$; if $-1 \leq x < 0$
 (iii) $\sin^{-1} x = \tan^{-1}\left(\frac{x}{\sqrt{1-x^2}}\right)$ if $-1 < x < 1$ (iv) $\cos^{-1} x = \sin^{-1}\sqrt{1-x^2}$; if $0 \leq x \leq 1$
 (v) $\cos^{-1} x = \pi - \sin^{-1}\sqrt{1-x^2}$ if $-1 \leq x < 0$ (vi) $\tan^{-1} x = \sin^{-1}\left(\frac{x}{\sqrt{1+x^2}}\right) = \cos^{-1}\left(\frac{1}{\sqrt{1+x^2}}\right)$ if $x > 0$

Property: X

- (i) $3 \sin^{-1} x = \sin^{-1}(3x - 4x^3), x \in \left[-\frac{1}{2}, \frac{1}{2}\right]$ (ii) $3 \cos^{-1} x = \cos^{-1}(4x^3 - 3x), x \in \left[\frac{1}{2}, 1\right]$

Note: Toppers மாணவர்கள் பாடத்திலுள்ள Proof-யும் சேர்த்து படித்துக் கொள்வது நல்லது.

1. Find the value, if it exists. If not, give the reason for non-existence.

2M

(i) $\sin^{-1}(\cos \pi)$	(ii) $\tan^{-1}\left(\sin\left(\frac{-5\pi}{2}\right)\right)$	(iii) $\sin^{-1}(\sin 5)$
(i) $\sin^{-1}(\cos \pi)$ $= \sin^{-1}(-1)$ $= -\sin^{-1}(1)$ $= \frac{-\pi}{2}$	(ii) $\tan^{-1}\left(\sin\left(\frac{-5\pi}{2}\right)\right)$ $= \tan^{-1}\left(-\sin\frac{5\pi}{2}\right)$ $= \tan^{-1}\left(-\sin\left(2\pi + \frac{\pi}{2}\right)\right)$ $= \tan^{-1}\left(-\sin 2\pi - \sin\frac{\pi}{2}\right)$ $= \tan^{-1}\left(0 - \sin\frac{\pi}{2}\right) = \tan^{-1}(-1)$ $= -\tan^{-1}(1) = \frac{-\pi}{4}$	(iii) $\sin^{-1}(\sin 5)$ $5 \notin \left[-\frac{\pi}{2}, \frac{\pi}{2}\right]$ $= \sin^{-1}[\sin(2\pi + (5 - 2\pi))]$ $= \sin^{-1}[\sin(5 - 2\pi)]$ $= 5 - 2\pi$ $\because 5 - 2\pi \in \left[-\frac{\pi}{2}, \frac{\pi}{2}\right]$

2. Find the value of the expression in terms of x with the help of a reference triangle.

3M

(i) $\sin(\cos^{-1}(1-x))$ (ii) $\cos(\tan^{-1}(3x-1))$ (iii) $\tan\left(\sin^{-1}\left(x+\frac{1}{2}\right)\right)$

(i) $\sin(\cos^{-1}(1-x))$

Let $\cos^{-1}(1-x) = y$

$\Rightarrow \cos y = 1-x$

W.K.T $\sin y = \sqrt{1-\cos^2 y}$

$= \sqrt{1-(1-x)^2}$

$= \sqrt{1-(1+x^2-2x)}$

$= \sqrt{1-1-x^2+2x}$

$\sin y = \sqrt{2x-x^2}$

$\therefore \sin(\cos^{-1}(1-x)) = \sqrt{2x-x^2}$

(ii) $\cos(\tan^{-1}(3x-1))$

Let $\tan^{-1}(3x-1) = y$

$\Rightarrow \tan y = 3x-1$

W.K.T $\sec y = \sqrt{1+\tan^2 y}$

$= \sqrt{1+(3x-1)^2}$

$= \sqrt{1+9x^2+1-6x}$

$\sec y = \sqrt{9x^2-6x+2}$

$\cos(\tan^{-1}(3x-1)) = \cos y = \frac{1}{\sec y} = \frac{1}{\sqrt{9x^2-6x+2}}$

(iii) $\tan\left(\sin^{-1}\left(x+\frac{1}{2}\right)\right)$

Let $\sin^{-1}\left(x+\frac{1}{2}\right) = y$

$\Rightarrow \sin y = x+\frac{1}{2}$

$= \frac{2x+1}{2} = \frac{O.S}{H}$

Given : $\tan\left(\sin^{-1}\left(x+\frac{1}{2}\right)\right)$

$= \tan y = \frac{O.S}{A.S}$

$= \frac{2x+1}{\sqrt{3-4x^2-4x}}$

$A.S = \sqrt{2^2 - (2x+1)^2}$

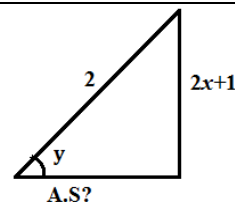
$= \sqrt{4 - (4x^2 + 1 + 4x)}$

$= \sqrt{4 - 4x^2 - 1 - 4x}$

$= \sqrt{3 - 4x^2 - 4x}$

$\tan y = \frac{O.S}{A.S}$

$= \frac{2x+1}{\sqrt{3-4x^2-4x}}$



3. Find the value of (i) $\sin^{-1}\left(\cos\left(\sin^{-1}\left(\frac{\sqrt{3}}{2}\right)\right)\right)$

(ii) $\cot\left(\sin^{-1}\frac{3}{5} + \sin^{-1}\frac{4}{5}\right)$

3M

(iii) $\tan\left(\sin^{-1}\frac{3}{5} + \cot^{-1}\frac{3}{2}\right)$

(i) $\sin^{-1}\left(\cos\left(\sin^{-1}\left(\frac{\sqrt{3}}{2}\right)\right)\right) = \sin^{-1}\left(\cos\left(\frac{\pi}{3}\right)\right)$

$= \sin^{-1}\left(\frac{1}{2}\right) = \frac{\pi}{6}$

(ii) $\cot\left(\sin^{-1}\frac{3}{5} + \sin^{-1}\frac{4}{5}\right) = \cot\left[\sin^{-1}\frac{3}{5}\sqrt{1-\left(\frac{4}{5}\right)^2} + \frac{4}{5}\sqrt{1-\left(\frac{3}{5}\right)^2}\right]$

$= \cot\left[\sin^{-1}\frac{3}{5}\sqrt{1-\frac{16}{25}} + \frac{4}{5}\sqrt{1-\frac{9}{25}}\right]$

$= \cot\left[\sin^{-1}\frac{3}{5}\sqrt{\frac{25-16}{25}} + \frac{4}{5}\sqrt{\frac{25-9}{25}}\right]$

$= \cot\left[\sin^{-1}\left(\frac{3}{5} \cdot \frac{3}{5} + \frac{4}{5} \cdot \frac{4}{5}\right)\right]$

$= \cot\left[\sin^{-1}\left(\frac{9}{25} + \frac{16}{25}\right)\right] = \cot\left[\sin^{-1}\left(\frac{25}{25}\right)\right] = \cot[\sin^{-1}(1)] = \cot\frac{\pi}{2} = 0$

Formula

$[\sin^{-1}x + \sin^{-1}y]$

$= \sin^{-1}[x\sqrt{1-y^2} + y\sqrt{1-x^2}]$

$$\begin{aligned}
 \text{(iii)} \quad & \tan \left(\sin^{-1} \left(\frac{3}{5} \right) + \cot^{-1} \left(\frac{3}{2} \right) \right) \\
 &= \tan \left[\tan^{-1} \frac{3}{4} + \tan^{-1} \frac{2}{3} \right] \\
 &= \tan \left[\tan^{-1} \frac{\frac{3}{4} + \frac{2}{3}}{1 - \frac{3}{4} \cdot \frac{2}{3}} \right] \\
 &= \tan \left[\tan^{-1} \frac{\frac{9+8}{12}}{\frac{12-6}{12}} \right] \\
 &= \tan \left[\tan^{-1} \frac{17}{6} \right] \\
 &= \frac{17}{6}
 \end{aligned}$$

$$\begin{aligned}
 \text{By } \sin^{-1} \left(\frac{3}{5} \right) &= \theta \\
 \sin \theta &= \frac{3}{5} = \frac{O.S}{H} \\
 \tan \theta &= \frac{O.S}{A.S} = \frac{3}{4} \\
 \Rightarrow \theta &= \tan^{-1} \left(\frac{3}{4} \right)
 \end{aligned}$$

$$\begin{aligned}
 \cot^{-1} \left(\frac{3}{2} \right) &= \theta \\
 \cot \theta &= \frac{3}{2} = \frac{A.S}{O.S} \\
 \tan \theta &= \frac{O.S}{A.S} = \frac{2}{3} \\
 \Rightarrow \theta &= \tan^{-1} \left(\frac{2}{3} \right)
 \end{aligned}$$

$$\therefore \left[\tan^{-1} x + \tan^{-1} y = \tan^{-1} \left(\frac{x+y}{1-xy} \right) \right]$$

4. Prove that (i) $\tan^{-1} \frac{2}{11} + \tan^{-1} \frac{7}{24} = \tan^{-1} \frac{1}{2}$ (ii) $\sin^{-1} \frac{3}{5} - \cos^{-1} \frac{12}{13} = \sin^{-1} \frac{16}{65}$

(i) $\tan^{-1} \frac{2}{11} + \tan^{-1} \frac{7}{24} = \tan^{-1} \frac{1}{2}$ 2M

LHS = $\tan^{-1} \frac{2}{11} + \tan^{-1} \frac{7}{24}$ APR-25

$$\therefore [\tan^{-1} x + \tan^{-1} y] = \tan^{-1} \left(\frac{x+y}{1-xy} \right)$$

$$= \tan^{-1} \left[\frac{\frac{2}{11} + \frac{7}{24}}{1 - \frac{2}{11} \cdot \frac{7}{24}} \right]$$

$$= \tan^{-1} \left[\frac{\frac{48+77}{264}}{\frac{264-14}{264}} \right]$$

$$= \tan^{-1} \left[\frac{125}{250} \right]$$

$$= \tan^{-1} \frac{1}{2} = RHS$$

$$\therefore LHS = RHS$$

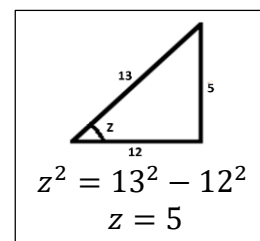
(ii) $\sin^{-1} \frac{3}{5} - \cos^{-1} \frac{12}{13} = \sin^{-1} \frac{16}{65}$ 3M

Let $\cos^{-1} \frac{12}{13} = x$

$$\cos x = \frac{12}{13}$$

$$\sin x = \frac{5}{13}$$

$$\Rightarrow x = \sin^{-1} \left(\frac{5}{13} \right)$$



(ie) $\cos^{-1} \left(\frac{12}{13} \right) = \sin^{-1} \left(\frac{5}{13} \right)$ (1)

LHS = $\sin^{-1} \frac{3}{5} - \cos^{-1} \frac{12}{13} = \sin^{-1} \frac{3}{5} - \sin^{-1} \frac{5}{13}$ use (1)

$$= \sin^{-1} \left[\frac{3}{5} \sqrt{1 - \left(\frac{5}{13} \right)^2} - \frac{5}{13} \sqrt{1 - \left(\frac{3}{5} \right)^2} \right]$$

$$= \sin^{-1} \left[\frac{3}{5} \cdot \frac{12}{13} - \frac{5}{13} \cdot \frac{4}{5} \right]$$

$$= \sin^{-1} \left[\frac{36}{65} - \frac{20}{65} \right]$$

$$= \sin^{-1} \left(\frac{36}{65} - \frac{20}{65} \right)$$

$$= \sin^{-1} \left(\frac{16}{65} \right) \quad LHS = RHS$$

5. Prove that $\tan^{-1} x + \tan^{-1} y + \tan^{-1} z = \tan^{-1} \left[\frac{x+y+z-xyz}{1-xy-yz-zx} \right]$

LHS = $\tan^{-1} x + \tan^{-1} y + \tan^{-1} z$ 3M

$$= \tan^{-1} \frac{x+y}{1-xy} + \tan^{-1} z$$

$$= \tan^{-1} \left[\frac{\frac{x+y}{1-xy} + z}{1 - \left(\frac{x+y}{1-xy} \right) z} \right]$$

$$= \tan^{-1} \left[\frac{\frac{x+y+z-xyz}{1-xy}}{\frac{1-xy-yz-zx}{1-xy}} \right] = \tan^{-1} \left[\frac{x+y+z-xyz}{1-xy-yz-zx} \right] \quad LHS = RHS$$

6. If $\tan^{-1} x + \tan^{-1} y + \tan^{-1} z = \pi$, show that $x + y + z = xyz$

JUL-24

2M

Given: $\tan^{-1} x + \tan^{-1} y + \tan^{-1} z = \pi$

$$\Rightarrow \tan^{-1} \left[\frac{x+y+z-xyz}{1-xy-yz-zx} \right] = \pi$$

$$\Rightarrow \frac{x+y+z-xyz}{1-xy-yz-zx} = \tan \pi$$

$$\Rightarrow \frac{x+y+z-xyz}{1-xy-yz-zx} = \frac{0}{1}$$

$$\Rightarrow x + y + z - xyz = 0$$

$$\Rightarrow x + y + z = xyz$$

$$\therefore \tan^{-1} x + \tan^{-1} y + \tan^{-1} z = \tan^{-1} \left[\frac{x + y + z - xyz}{1 - xy - yz - zx} \right]$$

7. Prove that

$$\tan^{-1} x + \tan^{-1} \frac{2x}{1-x^2} = \tan^{-1} \frac{3x-x^3}{1-3x^2}, |x| < \frac{1}{\sqrt{3}}$$

$$\text{Let } x = x ; \quad y = \frac{2x}{1-x^2}$$

3M

$$= \tan^{-1} \left(\frac{x + \frac{2x}{1-x^2}}{1 - x \left(\frac{2x}{1-x^2} \right)} \right)$$

$$= \tan^{-1} \left(\frac{\frac{x(1-x^2)+2x}{1-x^2}}{1 - \left(\frac{2x^2}{1-x^2} \right)} \right)$$

$$= \tan^{-1} \left(\frac{\frac{x(1-x^2)+2x}{1-x^2}}{\frac{1-x^2-2x^2}{1-x^2}} \right)$$

$$= \tan^{-1} \left(\frac{\frac{x-x^3+2x}{1-x^2}}{\frac{1-x^2-2x^2}{1-x^2}} \right)$$

$$= \tan^{-1} \left(\frac{-x^3+3x}{1-3x^2} \right) = \tan^{-1} \frac{3x-x^3}{1-3x^2}$$

8. Simplify: $\tan^{-1} \frac{x}{y} - \tan^{-1} \frac{x-y}{x+y}$

3M

$$\tan^{-1} \frac{x}{y} - \tan^{-1} \frac{x-y}{x+y} = \tan^{-1} \left[\frac{\frac{x}{y} - \frac{x-y}{x+y}}{1 + \frac{x}{y} \left(\frac{x-y}{x+y} \right)} \right]$$

$$= \tan^{-1} \left[\frac{\frac{x(x+y) - y(x-y)}{y(x+y)}}{\frac{y(x+y) + x(x-y)}{y(x+y)}} \right]$$

$$= \tan^{-1} \left[\frac{x^2 + xy - xy + y^2}{xy + y^2 + x^2 - xy} \right]$$

$$= \tan^{-1} \left(\frac{x^2 + y^2}{x^2 + y^2} \right)$$

$$= \tan^{-1}(1)$$

$$\tan^{-1} \frac{x}{y} - \tan^{-1} \frac{x-y}{x+y} = \frac{\pi}{4}$$

9. Solve: (i) $\sin^{-1} \frac{5}{x} + \sin^{-1} \frac{12}{x} = \frac{\pi}{2}$

(ii) $2 \tan^{-1} x = \cos^{-1} \frac{1-a^2}{1+a^2} - \cos^{-1} \frac{1-b^2}{1+b^2}, a > 0, b > 0$

(iii) $2 \tan^{-1}(\cos x) = \tan^{-1}(2 \operatorname{cosec} x)$

(iv) $\cot^{-1} x - \cot^{-1}(x+2) = \frac{\pi}{12}, x > 0$

(i) $\sin^{-1} \frac{5}{x} + \sin^{-1} \frac{12}{x} = \frac{\pi}{2}$

3M

$$\sin^{-1} \frac{5}{x} + \sin^{-1} \frac{12}{x} = \frac{\pi}{2}$$

$$\sin^{-1} \frac{5}{x} + \sin^{-1} \frac{12}{x} = \sin^{-1}(1)$$

$$\sin^{-1}(1) - \sin^{-1} \frac{5}{x} = \sin^{-1} \frac{12}{x}$$

$$\{ \because [\sin^{-1} x + \sin^{-1} y] = \sin^{-1} [x\sqrt{1-y^2} + y\sqrt{1-x^2}] \}$$

$$\sin^{-1} \left[1\sqrt{1 - \frac{25}{x^2}} - \frac{5}{x}\sqrt{1-1} \right] = \sin^{-1} \frac{12}{x}$$

$$\frac{\sqrt{x^2-25}}{x} = \frac{12}{x}$$

$$\sqrt{x^2-25} = 12$$

$$x^2 - 25 = 12^2$$

$$x^2 = 169$$

$$x = \pm 13$$

$$x = 13$$

(ii) $2 \tan^{-1} x = \cos^{-1} \frac{1-a^2}{1+a^2} - \cos^{-1} \frac{1-b^2}{1+b^2},$

3M

$a > 0, b > 0$

$$2 \tan^{-1} x = \cos^{-1} \frac{1-a^2}{1+a^2} - \cos^{-1} \frac{1-b^2}{1+b^2}$$

$$\therefore 2 \tan^{-1} x = \cos^{-1} \left(\frac{1-x^2}{1+x^2} \right), x \geq 0$$

$$2 \tan^{-1} x = 2 \tan^{-1} a - 2 \tan^{-1} b$$

$$2 \tan^{-1} x = 2 (\tan^{-1} a - \tan^{-1} b)$$

$$\Rightarrow \tan^{-1} x = \tan^{-1} a - \tan^{-1} b$$

$$\tan^{-1} x = \tan^{-1} \left(\frac{a-b}{1+ab} \right)$$

$$x = \frac{a-b}{1+ab}$$

(iii) $2 \tan^{-1}(\cos x) = \tan^{-1}(2 \operatorname{cosec} x)$ **5M**

$$\Rightarrow \tan^{-1}(\cos x) + \tan^{-1}(\cos x)$$

$$\Rightarrow \tan^{-1}(2 \operatorname{cosec} x)$$

$$\tan^{-1} \frac{\cos x + \cos x}{1 - \cos^2 x} = \tan^{-1}(2 \operatorname{cosec} x)$$

$$\frac{2 \cos x}{\sin^2 x} = 2 \operatorname{cosec} x$$

$$\frac{\cos x}{\sin^2 x} = \frac{1}{\sin x}$$

$$\Rightarrow \sin^2 x = \sin x \cos x$$

$$\Rightarrow \sin^2 x - \sin x \cos x = 0$$

$$\Rightarrow \sin x(\sin x - \cos x) = 0$$

$$\sin x = 0$$

$$x = \sin^{-1}(0)$$

$$x = n\pi, n \in \mathbb{Z}$$

$$\sin x - \cos x = 0$$

$$\sin x = \cos x$$

$$\frac{\sin x}{\cos x} = 1$$

$$\tan x = 1$$

$$x = \tan^{-1}(1)$$

$$\therefore x = n\pi + \frac{\pi}{4}, n \in \mathbb{Z}$$

$$\therefore x = n\pi, n\pi + \frac{\pi}{4}, n \in \mathbb{Z}$$

(iv) $\cot^{-1} x - \cot^{-1}(x+2) = \frac{\pi}{12}, x > 0$ **5M**

$$\Rightarrow \tan^{-1}\left(\frac{1}{x}\right) - \tan^{-1}\left(\frac{1}{x+2}\right) = \frac{\pi}{12}$$

$$\Rightarrow \tan^{-1} \frac{\frac{1}{x} - \frac{1}{x+2}}{1 + \frac{1}{x} \cdot \frac{1}{x+2}} = \frac{\pi}{12}$$

$$\therefore \tan \frac{\pi}{12} = 2 - \sqrt{3}$$

$$\Rightarrow \frac{\frac{x+2-x}{x(x+2)}}{\frac{x(x+2)+1}{x(x+2)}} = \tan \frac{\pi}{12}$$

$$\frac{2}{x^2+2x+1} = 2 - \sqrt{3}$$

$$\frac{x^2+2x+1}{2} = \frac{1}{2-\sqrt{3}}$$

$$(x+1)^2 = \frac{2}{2-\sqrt{3}}$$

$$= \frac{2}{2-\sqrt{3}} \times \frac{2+\sqrt{3}}{2+\sqrt{3}} = \frac{4+2\sqrt{3}}{4-3}$$

$$(x+1)^2 = \frac{4+2\sqrt{3}}{1} = 1 + 3 + 2\sqrt{3}$$

$$= 1^2 + (\sqrt{3})^2 + 2\sqrt{3}$$

$$(x+1)^2 = (1 + \sqrt{3})^2$$

$$\Rightarrow x+1 = 1 + \sqrt{3}$$

$$\text{Solution is } \Rightarrow x = \sqrt{3}$$

10. Find the number of solutions of the equation**5M**

$$\tan^{-1}(x-1) + \tan^{-1} x + \tan^{-1}(x+1) = \tan^{-1}(3x)$$

$$\text{Given: } \tan^{-1}(x-1) + \tan^{-1} x + \tan^{-1}(x+1) = \tan^{-1} 3x$$

$$\tan^{-1}(x-1) + \tan^{-1}(x+1) + \tan^{-1}(x) = \tan^{-1} 3x$$

$$\tan^{-1} \frac{x-1+x+1}{1-(x-1)(x+1)} + \tan^{-1} x = \tan^{-1} 3x$$

$$\tan^{-1} \frac{2x}{1-x^2+1} + \tan^{-1} x = \tan^{-1} 3x$$

$$\tan^{-1} \frac{\frac{2x}{2-x^2} + x}{1 - \frac{2x}{2-x^2} \cdot x} = \tan^{-1} 3x$$

$$\tan^{-1} \frac{\frac{2x+(2-x^2)x}{2-x^2}}{\frac{2-x^2-2x^2}{2-x^2}} = \tan^{-1} 3x$$

$$\frac{2x+2x-x^3}{2-3x^2} = \frac{3x}{1}$$

$$4x - x^3 = 6x - 9x^3$$

$$4x - x^3 - 6x + 9x^3 = 0$$

$$8x^3 - 2x = 0$$

$$\Rightarrow 2x(4x^2 - 1) = 0$$

$$x = 0, \pm \frac{1}{2}$$

This is a cubic equation. It has **3** solution.

EXERCISE 4.6



Choose the correct or the most suitable answer from the given four alternatives.

1. The value of $\sin^{-1}(\cos x)$, $0 \leq x \leq \pi$ is

JUL-24

- (1) $\pi - x$ (2) $x - \frac{\pi}{2}$ (3) $\frac{\pi}{2} - x$ (4) $x - \pi$

$$\begin{aligned}\sin^{-1}(\cos x) &= \sin^{-1}\left(\sin\left(\frac{\pi}{2} - x\right)\right) \\ &= \frac{\pi}{2} - x\end{aligned}$$

2. If $\sin^{-1} x + \sin^{-1} y = \frac{2\pi}{3}$; then $\cos^{-1} x + \cos^{-1} y$ is equal to

PTA-1, APR-20

- (1) $\frac{2\pi}{3}$ (2) $\frac{\pi}{3}$ (3) $\frac{\pi}{6}$ (4) π

$$\sin^{-1} x + \sin^{-1} y + \cos^{-1} x + \cos^{-1} y = \frac{\pi}{2} + \frac{\pi}{2} = \frac{2\pi}{2} = \pi \quad \because [\sin^{-1}(x) + \cos^{-1}(x) = \frac{\pi}{2}]$$

$$\text{Given: } \sin^{-1} x + \sin^{-1} y = \frac{2\pi}{3}$$

$$\sin^{-1} x + \sin^{-1} y + \cos^{-1} x + \cos^{-1} y = \pi$$

$$\Rightarrow \cos^{-1} x + \cos^{-1} y = \frac{3\pi - 2\pi}{3}$$

$$\cos^{-1} x + \cos^{-1} y = \frac{\pi}{3}$$

3. $\sin^{-1} \frac{3}{5} - \cos^{-1} \frac{12}{13} + \sec^{-1} \frac{5}{3} - \operatorname{cosec}^{-1} \frac{13}{12}$ is equal to

- (1) 2π (2) π (3) 0 (4) $\tan^{-1} \frac{12}{65}$

$$\begin{aligned}\sin^{-1} \frac{3}{5} - \cos^{-1} \frac{12}{13} + \sec^{-1} \frac{5}{3} - \operatorname{cosec}^{-1} \frac{13}{12} &= \sin^{-1} \frac{3}{5} - \cos^{-1} \frac{12}{13} + \cos^{-1} \frac{3}{5} - \sin^{-1} \frac{12}{13} \\ &= \sin^{-1} \frac{3}{5} + \cos^{-1} \frac{3}{5} - \left(\cos^{-1} \frac{12}{13} + \sin^{-1} \frac{12}{13}\right) \\ &= \frac{\pi}{2} - \frac{\pi}{2} = 0\end{aligned}$$

4. If $\sin^{-1} x = 2 \sin^{-1} \alpha$ has a solution, then

- (1) $|\alpha| \leq \frac{1}{\sqrt{2}}$ (2) $|\alpha| \geq \frac{1}{\sqrt{2}}$ (3) $|\alpha| < \frac{1}{\sqrt{2}}$ (4) $|\alpha| > \frac{1}{\sqrt{2}}$

$$\text{Given } \sin^{-1} x = 2 \sin^{-1} \alpha$$

$$\Rightarrow \frac{-\pi}{2} \leq 2 \sin^{-1} \alpha \leq \frac{\pi}{2}$$

$$\Rightarrow \frac{-\pi}{4} \leq \sin^{-1} \alpha \leq \frac{\pi}{4}$$

$$\Rightarrow -\sin \frac{\pi}{4} \leq \alpha \leq \sin \frac{\pi}{4}$$

$$\Rightarrow -\frac{1}{\sqrt{2}} \leq \alpha \leq \frac{1}{\sqrt{2}}$$

$$\Rightarrow |\alpha| \leq \frac{1}{\sqrt{2}}$$

5. $\sin^{-1}(\cos x) = \frac{\pi}{2} - x$ is valid for

PTA-4, JUL-20

- (1) $-\pi \leq x \leq 0$ (2) $0 \leq x \leq \pi$ (3) $-\frac{\pi}{2} \leq x \leq \frac{\pi}{2}$ (4) $-\frac{\pi}{4} \leq x \leq \frac{3\pi}{4}$

Given $\sin^{-1}(\cos x) = \frac{\pi}{2} - x$

$$\Rightarrow \frac{-\pi}{2} \leq \frac{\pi}{2} - x \leq \frac{\pi}{2}$$

$$\Rightarrow -\frac{\pi}{2} - \frac{\pi}{2} \leq -x \leq \frac{\pi}{2} - \frac{\pi}{2}$$

$$\Rightarrow -\pi \leq -x \leq 0$$

$$\Rightarrow \pi \geq x \geq 0$$

$$\Rightarrow 0 \leq x \leq \pi$$

6. If $\sin^{-1} x + \sin^{-1} y + \sin^{-1} z = \frac{3\pi}{2}$, the value of $x^{2017} + y^{2018} + z^{2019} - \frac{9}{x^{101} + y^{101} + z^{101}}$ is

- (1) 0 (2) 1 (3) 2 (4) 3

Given: $\sin^{-1} x + \sin^{-1} y + \sin^{-1} z = \frac{3\pi}{2} \Rightarrow x = y = z = 1$

$$\begin{aligned} \therefore x^{2017} + y^{2018} + z^{2019} - \frac{9}{x^{101} + y^{101} + z^{101}} \\ = 1^{2017} + 1^{2018} + 1^{2019} - \frac{9}{1^{101} + 1^{101} + 1^{101}} \\ = 1 + 1 + 1 - \frac{9}{1+1+1} \\ = 3 - 3 \\ = 0 \end{aligned}$$

7. If $\cot^{-1} x = \frac{2\pi}{5}$ for some $x \in \mathbb{R}$, the value of $\tan^{-1} x$ is

JUL-20, 23

- (1) $-\frac{\pi}{10}$ (2) $\frac{\pi}{5}$ (3) $\frac{\pi}{10}$ (4) $-\frac{\pi}{5}$

W.K.T. $\tan^{-1} x + \cot^{-1} x = \frac{\pi}{2}$
 $\Rightarrow \tan^{-1} x = \frac{\pi}{2} - \cot^{-1} x$
 $= \frac{\pi}{2} - \frac{2\pi}{5}$
 $= \frac{5\pi - 4\pi}{10}$
 $= \frac{\pi}{10}$

8. The domain of the function defined by $f(x) = \sin^{-1} \sqrt{x-1}$ is

APR-25, JUL-24, MDL

- (1) [1, 2] (2) [-1, 1] (3) [0, 1] (4) [-1, 0]

Given $f(x) = \sin^{-1} \sqrt{x-1}$

$$[\because \sqrt{x-1} \geq 0]$$

$$\Rightarrow -1 \leq \sqrt{x-1} \leq 1$$

$$\Rightarrow 0 \leq \sqrt{x-1} \leq 1$$

$$\Rightarrow 0 \leq x-1 \leq 1$$

$$\Rightarrow 0+1 \leq x \leq 1+1$$

$$\Rightarrow 1 \leq x \leq 2$$

$$\Rightarrow x \in [1, 2]$$

9. If $x = \frac{1}{5}$, the value of $\cos(\cos^{-1} x + 2 \sin^{-1} x)$ is

(1) $-\sqrt{\frac{24}{25}}$

(2) $\sqrt{\frac{24}{25}}$

(3) $\frac{1}{5}$

(4) $-\frac{1}{5}$

$$\begin{aligned}\cos(\cos^{-1} x + 2 \sin^{-1} x) &= \cos(\cos^{-1} x + \sin^{-1} x + \sin^{-1} x) \\ &= \cos\left(\frac{\pi}{2} + \sin^{-1} x\right) \\ &= -\sin(\sin^{-1} x) = -x = -\frac{1}{5}\end{aligned}$$

10. $\tan^{-1}\left(\frac{1}{4}\right) + \tan^{-1}\left(\frac{2}{9}\right)$ is equal to

PTA-3, APR-20

(1) $\frac{1}{2} \cos^{-1}\left(\frac{3}{5}\right)$

(2) $\frac{1}{2} \sin^{-1}\left(\frac{3}{5}\right)$

(3) $\frac{1}{2} \tan^{-1}\left(\frac{3}{5}\right)$

(4) $\tan^{-1}\left(\frac{1}{2}\right)$

$$\begin{aligned}\tan^{-1}\left(\frac{1}{4}\right) + \tan^{-1}\left(\frac{2}{9}\right) &= \tan^{-1}\left[\frac{\frac{1}{4} + \frac{2}{9}}{1 - \frac{1}{4} \cdot \frac{2}{9}}\right] \\ &= \tan^{-1}\left[\frac{\frac{9+8}{36}}{\frac{36-2}{36}}\right] \\ &= \tan^{-1}\left[\frac{17}{\frac{36}{34}}\right] \\ &= \tan^{-1}\left(\frac{1}{2}\right)\end{aligned}$$

11. If the function $f(x) = \sin^{-1}(x^2 - 3)$, then x belongs to

(1) $[-1, 1]$

(2) $[\sqrt{2}, 2]$

(3) $[-2, -\sqrt{2}] \cup [\sqrt{2}, 2]$

(4) $[-2, -\sqrt{2}]$

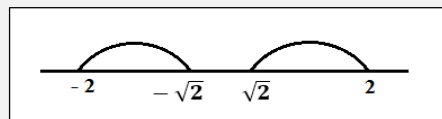
$$-1 \leq x^2 - 3 \leq 1$$

$$-1 + 3 \leq x^2 \leq 1 + 3$$

$$2 \leq x^2 \leq 4$$

$$\sqrt{2} \leq |x| \leq 2$$

$$x \in [-2, -\sqrt{2}] \cup [\sqrt{2}, 2]$$



12. If $\cot^{-1} 2$ and $\cot^{-1} 3$ are two angles of a triangle, then the third angle is

PTA-2

(1) $\frac{\pi}{4}$

(2) $\frac{3\pi}{4}$

(3) $\frac{\pi}{6}$

(4) $\frac{\pi}{3}$

Given $A + B + C = \pi$

$$\tan^{-1}\left(\frac{1}{2}\right) + \tan^{-1}\left(\frac{1}{3}\right) + C = \pi$$

$$\tan^{-1}\left(\frac{\frac{1}{2} + \frac{1}{3}}{1 - \frac{1}{2} \cdot \frac{1}{3}}\right) + C = \pi$$

$$\tan^{-1}\left(\frac{\frac{3+2}{6}}{\frac{6-1}{6}}\right) + C = \pi$$

$$\tan^{-1}\left(\frac{5}{5}\right) + C = \pi$$

$$\frac{\pi}{4} + C = \pi$$

$$C = \pi - \frac{\pi}{4} = \frac{4\pi - \pi}{4} = \frac{3\pi}{4}$$

13. $\sin^{-1}\left(\tan\frac{\pi}{4}\right) - \sin^{-1}\left(\sqrt{\frac{3}{x}}\right) = \frac{\pi}{6}$. Then x is a root of the equation

(1) $x^2 - x - 6 = 0$

(2) $x^2 - x - 12 = 0$

(3) $x^2 + x - 12 = 0$

(4) $x^2 + x - 6 = 0$

Given $\sin^{-1}\left(\tan\frac{\pi}{4}\right) - \sin^{-1}\left(\sqrt{\frac{3}{x}}\right) = \frac{\pi}{6}$

$$\sin^{-1}(1) - \sin^{-1}\left(\sqrt{\frac{3}{x}}\right) = \frac{\pi}{6}$$

$$\frac{\pi}{2} - \frac{\pi}{6} = \sin^{-1}\left(\sqrt{\frac{3}{x}}\right)$$

$$\frac{\pi}{3} = \sin^{-1}\left(\sqrt{\frac{3}{x}}\right)$$

$$\sin \frac{\pi}{3} = \sqrt{\frac{3}{x}}$$

$$\frac{\sqrt{3}}{2} = \frac{\sqrt{3}}{\sqrt{x}}$$

$$\sqrt{x} = 2$$

$$x = 4$$

$$x = 4 \text{ in } x^2 - x - 12 = 0$$

$$(4)^2 - 4 - 12 = 0 \text{ i.e., } x^2 - x - 12 = 0.$$

14. $\sin^{-1}(2 \cos^2 x - 1) + \cos^{-1}(1 - 2 \sin^2 x) =$

PTA-6

(1) $\frac{\pi}{2}$

(2) $\frac{\pi}{3}$

(3) $\frac{\pi}{4}$

(4) $\frac{\pi}{6}$

$$\begin{aligned} \sin^{-1}(2 \cos^2 x - 1) + \cos^{-1}(1 - 2 \sin^2 x) &= \sin^{-1}(\cos 2x) + \cos^{-1}(\cos 2x) \\ &= \frac{\pi}{2} \end{aligned}$$

15. If $\cot^{-1}(\sqrt{\sin \alpha}) + \tan^{-1}(\sqrt{\sin \alpha}) = u$, then $\cos 2u$ is equal to

PTA-4

(1) $\tan^2 \alpha$

(2) 0

(3) -1

(4) $\tan 2\alpha$

$$\cot^{-1}(\sqrt{\sin \alpha}) + \tan^{-1}(\sqrt{\sin \alpha}) = \frac{\pi}{2} = u$$

$$\therefore \cos 2u = \cos 2 \cdot \frac{\pi}{2} = \cos \pi$$

$$\cos 2u = -1$$

16. If $|x| \leq 1$, then $2 \tan^{-1} x - \sin^{-1} \frac{2x}{1+x^2}$ is equal to

(1) $\tan^{-1} x$

(2) $\sin^{-1} x$

(3) 0

(4) π

$$2 \tan^{-1} x - \sin^{-1} \frac{2x}{1+x^2} = 2 \tan^{-1} x - 2 \tan^{-1} x$$

$$2 \tan^{-1} x - \sin^{-1} \frac{2x}{1+x^2} = 0$$

17. The equation $\tan^{-1} x - \cot^{-1} x = \tan^{-1} \left(\frac{1}{\sqrt{3}} \right)$ has

(1) no solution

(2) unique solution

(3) two solutions

(4) infinite number of solutions

$$\begin{aligned}\tan^{-1} x - \cot^{-1} x &= \tan^{-1} \left(\frac{1}{\sqrt{3}} \right) \\ \tan^{-1} x - \left(\frac{\pi}{2} - \tan^{-1} x \right) &= \frac{\pi}{6} \\ 2 \tan^{-1} x &= \frac{\pi}{6} + \frac{\pi}{2} \\ 2 \tan^{-1} x &= \frac{\pi + 3\pi}{6} = \frac{4\pi}{6} = \frac{2\pi}{3} \\ \tan^{-1} x &= \frac{\pi}{3} \\ x &= \tan \frac{\pi}{3} = \sqrt{3} \quad \therefore \text{it is unique solution.}\end{aligned}$$

18. If $\sin^{-1} x + \cot^{-1} \left(\frac{1}{2} \right) = \frac{\pi}{2}$, then x is equal to

APR-24, JUL-23

(1) $\frac{1}{2}$

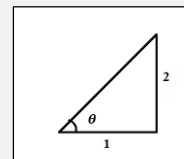
(2) $\frac{1}{\sqrt{5}}$

(3) $\frac{2}{\sqrt{5}}$

(4) $\frac{\sqrt{3}}{2}$

$$\begin{aligned}\sin^{-1} x + \cot^{-1} \frac{1}{2} &= \frac{\pi}{2} \\ \sin^{-1} x + \cos^{-1} \frac{1}{\sqrt{5}} &= \frac{\pi}{2} \\ \Rightarrow x &= \frac{1}{\sqrt{5}}\end{aligned}$$

$$\begin{aligned}\cot x &= \frac{A.S}{O.S} = \frac{1}{2} \\ \cos x &= \frac{A.S}{H} = \frac{1}{\sqrt{5}}\end{aligned}$$



19. If $\sin^{-1} \frac{x}{5} + \operatorname{cosec}^{-1} \frac{5}{4} = \frac{\pi}{2}$, then the value of x is

PTA-5

(1) 4

(2) 5

(3) 2

(4) 3

$$\begin{aligned}\sin^{-1} \frac{x}{5} + \operatorname{cosec}^{-1} \frac{5}{4} &= \frac{\pi}{2} \\ \sin^{-1} \frac{x}{5} + \sin^{-1} \frac{4}{5} &= \frac{\pi}{2} \\ \sin^{-1} \frac{x}{5} + \sin^{-1} \frac{4}{5} &= \frac{\pi}{2} \\ \sin^{-1} \frac{x}{5} + \cos^{-1} \frac{3}{5} &= \frac{\pi}{2} \\ \text{Compare with } \sin^{-1} A + \cos^{-1} A &= \frac{\pi}{2} \\ \frac{x}{5} &= \frac{3}{5} \Rightarrow x = 3\end{aligned}$$

$$\therefore \sin^{-1} \frac{4}{5} = \cos^{-1} \frac{3}{5}$$

20. $\sin(\tan^{-1} x)$, $|x| < 1$ is equal to

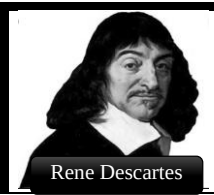
(1) $\frac{x}{\sqrt{1-x^2}}$

(2) $\frac{1}{\sqrt{1-x^2}}$

(3) $\frac{1}{\sqrt{1+x^2}}$

(4) $\frac{x}{\sqrt{1+x^2}}$

$$\begin{aligned}\tan^{-1} x = \theta &\Rightarrow \tan \theta = x & \therefore 1 + \tan^2 \theta &= \sec^2 \theta \\ 1 + x^2 &= \sec^2 \theta \\ \sec \theta &= \sqrt{1 + x^2} \Rightarrow \frac{1}{\cos \theta} = \sqrt{1 + x^2} \Rightarrow \cos \theta = \frac{1}{\sqrt{1+x^2}} \\ \therefore \sin \theta &= \sqrt{1 - \cos^2 \theta} \\ &= \sqrt{1 - \frac{1}{1+x^2}} = \sqrt{\frac{1+x^2-1}{1+x^2}} = \sqrt{\frac{x^2}{1+x^2}} = \frac{x}{\sqrt{1+x^2}} \\ \sin(\tan^{-1} x) &= \frac{x}{\sqrt{1+x^2}}\end{aligned}$$



Rene Descartes

Chapter - 5

Two Dimensional Analytical Geometry-II

EXERCISE 5.1



Video class

Concept Corner

- A **circle** is the locus of a moving point P in a plane which moves such that its distance from a fixed point C in the plane is always a constant.

$$CP = r = \text{constant.}$$

The fixed point C is called the **centre** and the constant distance is called **radius** of the circle.

- **Equation of the circle:**

(1) The equation of the circle with centre at the origin $(0,0)$ & having radius ' r ' is $x^2 + y^2 = r^2$

(2) The equation of the circle with centre $C(h, k)$ & having radius r is $(x - h)^2 + (y - k)^2 = r^2$

- **General Equation of the circle:**

$x^2 + y^2 + 2gx + 2fy + c = 0$ is the general equation of the circle.

This can be written as $(x - (-g))^2 + (y - (-f))^2 = (\sqrt{g^2 + f^2 - c})^2$

$\therefore$ centre of the circle = $C(-g, -f)$

Radius of the circle = $r = \sqrt{g^2 + f^2 - c}$

- If the equation of the circle is of the form $ax^2 + ay^2 + 2gx + 2fy + c = 0$

$$x^2 + y^2 + 2\frac{g}{a}x + 2\frac{f}{a}y + \frac{c}{a} = 0$$

$$\therefore \text{Centre} = C\left(\frac{-g}{a}; \frac{-f}{a}\right); \text{Radius} = r = \sqrt{\frac{g^2}{a} + \frac{f^2}{a} - \frac{c}{a}}$$

- **Condition for the circle:**

The equation $ax^2 + by^2 + 2gx + 2fy + c = 0$

Represent a circle if (1) $a = b$ (coefficient of x^2 = coefficient of y^2)

(2) $h = 0$ (coefficient of xy = 0)

- **Theorem:**

1. The circle passing through the points of intersection of the line $lx + my + n = 0$ and the circle $x^2 + y^2 + 2gx + 2fy + c = 0$ is the circle of the form

$$x^2 + y^2 + 2gx + 2fy + c + \lambda(lx + my + n) = 0, \lambda \in \mathbb{R}^1$$

2. The equation of a circle with (x_1, y_1) and (x_2, y_2) as extremities of one of the diameters of the circle is $(x - x_1)(x - x_2) + (y - y_1)(y - y_2) = 0$.

3. The position of a point $P(x_1, y_1)$ with respect to a given circle $x^2 + y^2 + 2gx + 2fy + c = 0$ in the plane containing the circle is outside or on or inside the circle according as

$$S = x_1^2 + y_1^2 + 2gx_1 + 2fy_1 + c \text{ is } \begin{cases} > 0 \\ = 0 \\ < 0 \end{cases} \text{ or,}$$

- **Parametric Equations:** $x = r \cos \theta$; $y = r \sin \theta$ are the parametric Equation of the circle $x^2 + y^2 = r^2$ where ' θ ' is the parameter.

$\therefore$ Any point on the circle can be taken as $P(r \cos \theta, r \sin \theta)$.

➤ **Tangent and Normals:**

Tangent of a circle is a line which touches the circle at only one point and normal is a line perpendicular to the tangent and passing through the point of contact.

Equation of tangent (x, y) of the circle $xx_1 + yy_1 + g(x + x_1) + f(y + y_1) + c = 0$

Remark:

(1) Equation of tangent drawn at (x_1, y_1) of the circle $x^2 + y^2 = a^2$ is $xx_1 + yy_1 = a^2$

(2) Equation of normal drawn at (x_1, y_1) to the circle $x^2 + y^2 = a^2$ is $xy_1 - yx_1 = 0$

(3) The normal passes through the centre of the circle.

➤ **Condition for tangent:**

(1) The condition for the line $y = mx + c$ to be a tangent to the circle $x^2 + y^2 = a^2$

$$c^2 = a^2(1 + m^2) \quad (\text{or}) \quad c = \pm a\sqrt{1 + m^2}$$

(2) Point of contact = $P\left(\frac{-am}{\sqrt{1+m^2}}; \frac{a}{\sqrt{1+m^2}}\right)$ or $\left(\frac{am}{\sqrt{1+m^2}}; \frac{-a}{\sqrt{1+m^2}}\right)$

Note: The equation of tangent at P to a circle is $y = mx \pm a\sqrt{1 + m^2}$

➤ **Theorem:** From any point outside the circle $x^2 + y^2 = a^2$ two tangents can be drawn.

(1) If (x_1, y_1) is a point outside the circle then both the tangents are real.

(2) If (x_1, y_1) is a point inside the circle then both the tangents are imaginary.

(3) If (x_1, y_1) is a point on the circle then both the tangents coincide.

1. Obtain the equation of the circles with radius 5 cm and touching x -axis at the origin in general form.

Radius = $r = 5$; Centre = $C(0, \pm 5) = (h, k)$

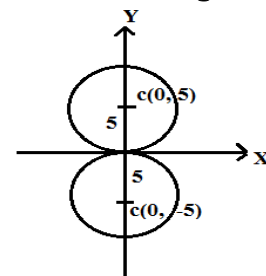
Equation of the circle $(x - h)^2 + (y - k)^2 = r^2$

$$(x - 0)^2 + (y \pm 5)^2 = 5^2$$

$$x^2 + y^2 - 10y \pm 25 = 25$$

$$x^2 + y^2 \pm 10y = 0$$

$\therefore$ Equation of the required circle is $x^2 + y^2 \pm 10y = 0$



2M

2. Find the equation of the circle with centre $(2, -1)$ and passing through the point $(3, 6)$ in standard form.

Centre = $C(2, -1) = (h, k)$

Radius = $CP = \sqrt{(3 - 2)^2 + (6 + 1)^2}$

$$= \sqrt{(1)^2 + (7)^2}$$

$$= \sqrt{1 + 49} = \sqrt{50}$$

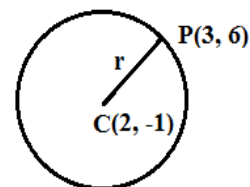
Equation of the circle $(x - h)^2 + (y - k)^2 = r^2$

$$(x - 2)^2 + (y + 1)^2 = (\sqrt{50})^2$$

$$(x - 2)^2 + (y + 1)^2 = 50$$

$$x^2 - 4x + 4 + y^2 + 2y + 1 = 50$$

$$x^2 + y^2 - 4x + 2y - 45 = 0 \Rightarrow (x - 2)^2 + (y + 1)^2 = 50$$



2M

3. Find the equation of circles that touch both the axes & passes through $(-4, -2)$ in general form.

A circle touch both axes has centre $c(h, h)$ and radius h .

$$\text{Radius} = CP = h \quad CP^2 = h^2$$

3M

$$(x - h)^2 + (y - k)^2 = r^2$$

$$(h + 4)^2 + (h + 2)^2 = h^2$$

$$h^2 + 8h + 16 + h^2 + 4h + 4 = h^2$$

$$h^2 + 12h + 20 = 0$$

$$(h + 2)(h + 10) = 0$$

$$h = -2, -10$$

$\therefore$ The centre are $c(-2, -2)$ (or) $c(-10, -10)$

Equation of the circle with centre $c(-2, -2)$ and radius $r = 2$ is

$$(x + 2)^2 + (y + 2)^2 = (2)^2$$

$$x^2 + 4x + 4 + y^2 + 4y + 4 = 4$$

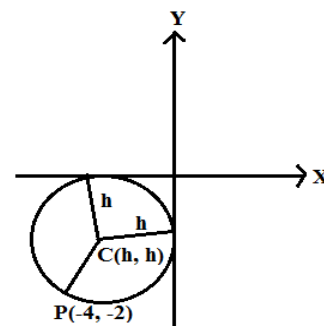
$$x^2 + y^2 + 4x + 4y + 4 = 0$$

Equation of the circle with centre $C(-10, -10)$ and $r = 10$ is

$$(x + 10)^2 + (y + 10)^2 = (10)^2$$

$$x^2 + 20x + 100 + y^2 + 20y + 100 = 100$$

$$x^2 + y^2 + 20x + 20y + 100 = 0$$



4. Find the equation of the circle with centre $(2, 3)$ and passing through the intersection of the lines

$$3x - 2y - 1 = 0 \text{ and } 4x + y - 27 = 0$$

3M

$$3x - 2y = 1 \dots\dots\dots (1)$$

$$4x + y = 27 \dots\dots\dots (2)$$

$$(1) \times 1 \Rightarrow 3x - 2y = 1$$

$$(2) \times 2 \Rightarrow 8x + 2y = 54$$

$$\begin{array}{r} 11x \quad \quad = 55 \end{array}$$

$$x = 5$$

$$(2) \Rightarrow 20 + y = 27$$

$$y = 7$$

$$(x, y) = (5, 7)$$

$$\text{Centre} = c(2, 3) = (h, k)$$

$$\text{Radius} = r = CP = \sqrt{(3)^2 + (4)^2} = \sqrt{9 + 16} = \sqrt{25}$$

$$r = 5$$

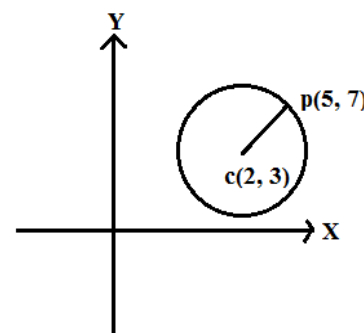
Equation of the circle is

$$(x - h)^2 + (y - k)^2 = r^2$$

$$(x - 2)^2 + (y - 3)^2 = 5^2$$

$$x^2 - 4x + 4 + y^2 - 6y + 9 = 25$$

$$x^2 + y^2 - 4x - 6y - 12 = 0$$



5. Obtain the equation of the circle for which (3, 4) and (2, -7) are the ends of a diameter. (2M)

Equation of the circle having (3, 4) and (2, -7) as ends of a diameter is

JUL-24,PTA-3

$$(x - x_1)(x - x_2) + (y - y_1)(y - y_2) = 0$$

$$(x - 3)(x - 2) + (y - 4)(y + 7) = 0$$

$$x^2 - 5x + 6 + y^2 + 3y - 28 = 0$$

$$x^2 + y^2 - 5x + 3y - 22 = 0$$

$$(x_1, y_1) = (3, 4)$$

$$(x_2, y_2) = (2, -7)$$

6. Find the equation of the circle through the points (1, 0), (-1, 0), and (0, 1) (PTA-5 JUL-23)

Equation of the required circle is

$$x^2 + y^2 + 2gx + 2fy + c = 0 \dots\dots\dots (*)$$

$$\because \text{it passes through } (1, 0) \quad 2g + c = -1 \dots\dots (1)$$

$$\because \text{it passes through } (-1, 0) \quad -2g + c = -1 \dots (2)$$

$$\because \text{it passes through } (0, 1) \quad 2f + c = -1, \dots\dots (3)$$

$$(1) + (2) \Rightarrow 2c = -2$$

$$c = -1$$

$$(2) \Rightarrow 2g - 1 = -1$$

$$2g = 0 \Rightarrow g = 0$$

$$(3) \Rightarrow 2f - 1 = -1$$

$$2f = 0 \Rightarrow f = 0$$

$$(*) \Rightarrow \text{Equation of the circle } x^2 + y^2 - 1 = 0$$

$$x^2 + y^2 = 1$$

7. A circle of area 9π square units has two of its diameters along the lines $x + y = 5$ & $x - y = 1$ Find the equation of the circle. (3M)

$$\text{Area} = \pi r^2 = 9\pi$$

$$r^2 = 9 \Rightarrow r = 3$$

$$x + y = 5 \dots\dots (1)$$

$$x - y = 1 \dots\dots (2)$$

$$2x = 6 \Rightarrow x = 3$$

$$3 + y = 5$$

$$y = 2$$

$$\therefore \text{Centre} = C(h, k) = (3, 2)$$

Equation of the circle is

$$(x - h)^2 + (y - k)^2 = r^2$$

$$(x - 3)^2 + (y - 2)^2 = 3^2$$

$$x^2 - 6x + 9 + y^2 - 4y + 4 = 9$$

$$x^2 + y^2 - 6x - 4y + 4 = 0$$

8. If $y = 2\sqrt{2}x + c$ is a tangent to the circle $x^2 + y^2 = 16$, find the value of c . (2M)

Condition for the line $y = mx + c$ to be a tangent to the circle $x^2 + y^2 = a^2$ is $c^2 = a^2(1 + m^2)$

$$m = 2\sqrt{2} \quad a^2 = 16 \quad c = ?$$

$$c^2 = 16(1 + 8) = 16 \times 9$$

$$c = \pm 4 \times 3 = \pm 12$$

9. Find the equation of the tangent and normal to the circle $x^2 + y^2 - 6x + 6y - 8 = 0$ at (2, 2) (3M)

$$\text{Equation of tangent at } (2, 2) \text{ is } xx_1 + yy_1 - 6 \times \frac{1}{2}(x + x_1) + 6 \times \frac{1}{2}(y + y_1) - 8 = 0$$

$$xx_1 + yy_1 - 3(x + x_1) + 3(y + y_1) - 8 = 0$$

$$x(2) + y(2) - 3(x + 2) + 3(y + 2) - 8 = 0$$

$$2x + 2y - 3x - 6 + 3y + 6 - 8 = 0$$

$$-x + 5y - 8 = 0$$

$$x - 5y + 8 = 0$$

$\therefore$ Normal is perpendicular to tangent; Equation of normal is $5x + y + k = 0$

$$\because \text{it passes through } (2, 2) \Rightarrow 10 + 2 + k = 0$$

$$k = -12$$

$$\text{Equation of the normal is } 5x + y - 12 = 0$$

10. Determine whether the points $(-2, 1)$, $(0, 0)$ and $(-4, -3)$ lie outside, on or inside the circle

$$x^2 + y^2 - 5x + 2y - 5 = 0$$

$$(1) (-2, 1) \Rightarrow x_1^2 + y_1^2 - 5x_1 + 2y_1 - 5 = 4 + 1 + 10 + 2 - 5 = 17 - 5 = 12 > 0$$

$\therefore (-2, 1)$ lies out side of the given circle

$$(2) (0, 0) \Rightarrow x_1^2 + y_1^2 - 5x_1 + y_1 - 5 = 0 + 0 - 0 + 0 - 5 = -5 < 0$$

$\therefore (0, 0)$ lies inside of the given circle

$$(3) (-4, -3) \Rightarrow x_1^2 + y_1^2 - 5x_1 + 2y_1 - 5 = 16 + 9 + 20 - 6 - 5 = 45 - 11 = 34 > 0$$

$\therefore (-4, -3)$ lies outside of the given circle.

3M

11. Find centre and radius of the following circles.

(i) $x^2 + (y + 2)^2 = 0$

(iii) $x^2 + y^2 - x + 2y - 3 = 0$

(i) $x^2 + (y + 2)^2 = 0$

$$(x - 0)^2 + [y - (-2)]^2 = 0^2$$

$$h = 0, k = -2 \quad r = 0$$

$$\therefore \text{centre} = (h, k) = C(0, -2)$$

$$\text{Radius} = r = 0$$

2M

(iii) $x^2 + y^2 - x + 2y - 3 = 0$

$$2g = -1 \Rightarrow g = \frac{-1}{2}$$

$$2f = 2 \Rightarrow f = 1$$

$$c = -3$$

$$\text{centre} = C(-g, -f) = C\left(\frac{1}{2}, -1\right)$$

$$\text{Radius} = r = \sqrt{g^2 + f^2 - c}$$

$$= \sqrt{\frac{1}{4} + 1 + 3}$$

$$= \sqrt{\frac{1}{4} + 4}$$

$$= \sqrt{\frac{17}{4}} = \frac{\sqrt{17}}{2}$$

$$r = \frac{\sqrt{17}}{2}$$

3M

(ii) $x^2 + y^2 + 6x - 4y + 4 = 0$

(iv) $2x^2 + 2y^2 - 6x + 4y + 2 = 0$

(ii) $x^2 + y^2 + 6x - 4y + 4 = 0$

$$2g = 6 \Rightarrow g = 3$$

$$2f = -4 \Rightarrow f = -2$$

$$c = 4$$

$$\text{centre} = C(-g, -f) = C(-3, 2)$$

$$\text{Radius} = r = \sqrt{g^2 + f^2 - c}$$

$$= \sqrt{9 + 4 - 4} = \sqrt{9} = 3$$

JUL-22

3M

(iv) $2x^2 + 2y^2 - 6x + 4y + 2 = 0$

$$x^2 + y^2 - 3x + 2y + 1 = 0$$

PTA-6

(\div 2)

$$2g = -3 \Rightarrow g = \frac{-3}{2}$$

$$2f = 2 \Rightarrow f = 1$$

$$c = 1$$

$$\text{centre} = C(-g, -f) = C\left(\frac{3}{2}, -1\right)$$

$$\text{Radius} = r$$

$$= \sqrt{g^2 + f^2 - c}$$

$$= \sqrt{\frac{9}{4} + 1 - 1} = \sqrt{\frac{9}{4}} = \frac{3}{2}$$

$$r = \frac{3}{2}$$

2M

12. If the equation $3x^2 + (3 - p)xy + qy^2 - 2px = 8pq$ represents a circle, find p and q . Also determine the centre and radius of the circle.

PTA-1

3M

Condition for circle, (1) co-efficient of x^2 = co-efficient of y^2

(2) co-efficient of $xy = 0$

$$3 - p = 0 \Rightarrow p = 3$$

$$3 = q \Rightarrow q = 3$$

$\therefore$ Equation of the circle $3x^2 + 3y^2 - 6x = 8(3)(3)$

$$3x^2 + 3y^2 - 6x - 72 = 0$$

$$x^2 + y^2 - 2x - 24 = 0$$

$$2g = -2 \Rightarrow g = -1 \quad ; \quad 2f = 0 \Rightarrow f = 0$$

$$c = -24$$

$$\therefore \text{centre} = C(-g, -f) = C(1, 0)$$

$$\text{Radius} = r = \sqrt{g^2 + f^2 - c} = \sqrt{1 + 0 + 24} = \sqrt{25} = 5$$

EXERCISE 5.2



Video class

Concept Corner

Conics

A conic is the locus of a point which moves in a plane, so that its distance from a fixed point bears a constant ratio to its distance from a fixed line not containing the fixed point.

The fixed point is called **focus**, the fixed line is called **directrix** and the constant ratio is called **eccentricity**, which is denoted by e .

(i) If this constant $e = 1$ then the conic is called a parabola

(ii) If this constant $e < 1$ then the conic is called an ellipse

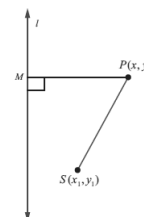
(iii) If this constant $e > 1$ then the conic is called a hyperbola

General Equation of a Conics:

Let $S(x_1, y_1)$ be the focus, l the directrix and e be the eccentricity. Let $P(x, y)$ be the moving point.

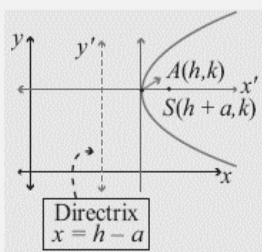
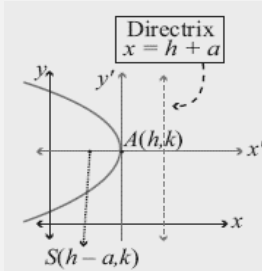
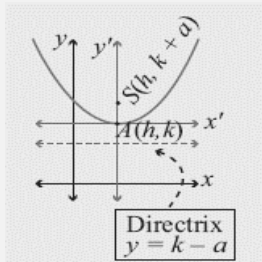
By definition of Conic, $\frac{SP}{PM} = \text{constant} = e$ where $SP = \sqrt{(x - x_1)^2 + (y - y_1)^2}$

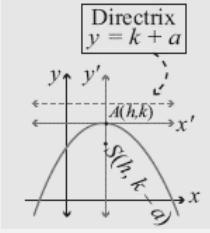
$PM = \text{Perpendicular distance from } P(x, y) \text{ to the line} = \left| \frac{lx + my + n}{\sqrt{l^2 + m^2}} \right|$



Parabola :

Since, $e = 1$, for a parabola, we note that the parabola is the locus of points in a plane that are equidistant from both the directrix and the focus.

Equation	Graph	Vertices	Focus	Axis of symmetry	Equation of directrix	Length of latus rectum
$(y - k)^2 = 4a(x - h)$		(h, k)	$(h + a, 0 + k)$	$y = k$	$x = h - a$	$4a$
$(y - k)^2 = -4a(x - h)$		(h, k)	$(h - a, 0 + k)$	$y = k$	$x = h + a$	$4a$
$(x - h)^2 = 4a(y - k)$		(h, k)	$(0 + h, a + k)$	$x = h$	$y = k - a$	$4a$

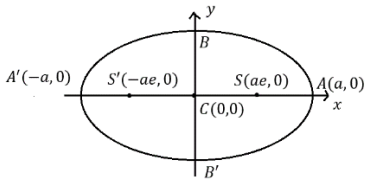
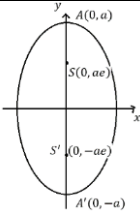
$(x-h)^2 = -4a(y-k)$		(h, k)	$(0+h, -a+k)$	$x = h$	$y = k + a$	$4a$
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Definition :

- (1) The line perpendicular to the directrix and passing through the focus is known as the **Axis** of the parabola.
- (2) The intersection point of the axis with the curve is called **vertex** of the parabola.
- (3) Any chord of the parabola, through its focus is called **focal chord** of the parabola.
- (4) The length of the focal chord perpendicular to the axis is called **latus rectum** of the parabola.

Ellipse: Ellipse is the locus of a point which moves such that $\frac{SP}{PM} = e < 1$, $\therefore ae = c$

Two Types of Ellipse

Equation	$\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1, a > b$	$\frac{x^2}{b^2} + \frac{y^2}{a^2} = 1, a > b$
Major axis along	x - axis	y - axis
Centre	$C(0,0)$	$C(0,0)$
Vertices	$A(a, 0), A'(-a, 0)$	$A(0, a), A'(0, -a)$
Foci	$S(c, 0), S'(-c, 0)$	$S(0, c), S'(0, -c)$
Equation of directrices	$x = \pm \frac{a}{e}$	$y = \pm \frac{a}{e}$
Diagram		

- The line segment AA' is called the **major axis** of the ellipse and is of length $2a$.
- The line segment BB' is called the **minor axis** of the ellipse and is of length $2b$.

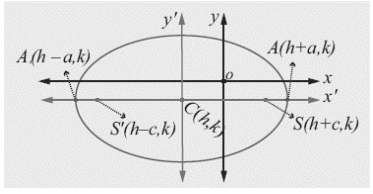
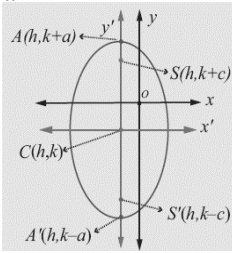
Types of ellipses with centre at (h, k)

(a) **Major axis parallel to the x -axis** $\frac{(x-h)^2}{a^2} + \frac{(y-k)^2}{b^2} = 1, a > b$

The length of the major axis is $2a$. The length of the minor axis is $2b$. The coordinates of the vertices are $(h+a, k)$ and $(h-a, k)$ and the coordinates of the foci are $(h+c, k)$ and $(h-c, k)$ where $c^2 = a^2 - b^2$

(b) **Major axis parallel to the y -axis** $\frac{(x-h)^2}{b^2} + \frac{(y-k)^2}{a^2} = 1, a > b$

The length of the major axis is $2a$. The length of the minor axis is $2b$. The coordinates of the vertices are $(h, k+a)$ and $(h, k-a)$, and the coordinates of the foci are $(h, k+c)$ and $(h, k-c)$, where $c^2 = a^2 - b^2$

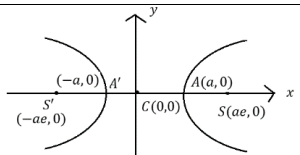
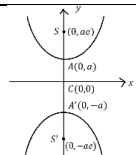
Equation	Centre	Major axis	Vertices	Foci
$\frac{(x-h)^2}{a^2} + \frac{(y-k)^2}{b^2} = 1, a^2 > b^2$  (a) Major axis parallel to the x-axis Foci are c units right and c units left of centre, where $c^2 = a^2 - b^2$	(h, k)	Parallel to the x-axis	$(h - a, k)$ $(h + a, k)$	$(h - c, k)$ $(h + c, k)$
$\frac{(x-h)^2}{b^2} + \frac{(y-k)^2}{a^2} = 1, a^2 > b^2$  (a) Major axis parallel to the y-axis Foci are c units right and c units left of centre, where $c^2 = a^2 - b^2$	(h, k)	Parallel to the y-axis	$(h, k - a)$ $(h, k + a)$	$(h, k - c)$ $(h, k + c)$

➤ The sum of the focal distances of any point on the ellipse is equal to length of the major axis.

Hyperbola :

Hyperbola is the locus of a point which moves such that its distance from its focus is greater than its distance from its directrix, bearing a constant ratio $e (e > 1)$. $\frac{SP}{PM} > 1 \quad \therefore ae = c$

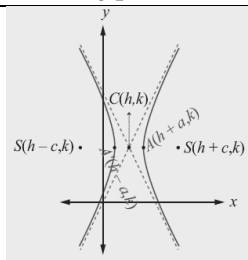
Two Types of Hyperbola:

Equation	$\frac{x^2}{a^2} - \frac{y^2}{b^2} = 1$	$\frac{y^2}{a^2} - \frac{x^2}{b^2} = 1$
Transverse axis along	x axis	y axis
Centre	$C(0,0)$	$C(0,0)$
Vertices	$A(a, 0), A'(-a, 0)$	$A(0, a), A'(0, -a)$
Foci	$S(c, 0), S'(-c, 0)$	$S(0, c), S'(0, -c)$
Equation of directrices	$x = \pm \frac{a}{e}$	$y = \pm \frac{a}{e}$
Diagram		

- The line segment AA' is the **transverse axis** length $2a$.
- The line segment BB' is the **conjugate axis** of length $2b$.
- The line segment $CA =$ the line segment CA' **semi transverse axis** $= a$ and the line segment $CB =$ the line segment $CB' =$ **semi conjugate axis** $= b$.

Asymptotes : Let $P(x, y)$ be a point on the curve defined by $y = f(x)$, which moves further and further away from the origin such that the distance between P and some fixed line tends to zero. This fixed line is called an asymptote.

Note that the hyperbolas admit asymptotes while parabolas and ellipses do not.

Types of Hyperbola with centre at (h, k) 

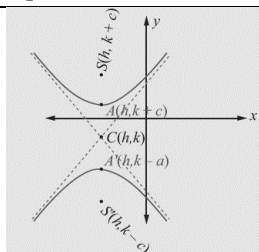
(a) transverse axis parallel to the x-axis

(a) transverse axis parallel to the x-axis

The equation of a hyperbola with centre $C(h, k)$ and transverse axis parallel to the x-axis is given by $\frac{(x-h)^2}{a^2} - \frac{(y-k)^2}{b^2} = 1$.

The coordinates of the vertices are $A(h + a, k)$ and $A'(h - a, k)$. The coordinates of the foci are $S(h + c, k)$ and $S'(h - c, k)$ where $c^2 = a^2 + b^2$.

The equations of directrices are $x = h \pm \frac{a}{e}$



(b) transverse axis parallel to the y-axis

(b) transverse axis parallel to the y-axis

The equation of a hyperbola with centre $C(h, k)$ and transverse axis parallel to the y-axis is given by $\frac{(y-k)^2}{a^2} - \frac{(x-h)^2}{b^2} = 1$.

The coordinates of the vertices are $A(h, k + a)$ and $A'(h, k - a)$. The coordinates of the foci are $S(h, k + c)$ and $S'(h, k - c)$ where $c^2 = a^2 + b^2$.

The equations of directrices are $y = k \pm \frac{a}{e}$

1. Find the equation of the parabola in each of the cases given below:

(i) focus $(4, 0)$ and directrix $x = -4$ (ii) passes through $(2, -3)$ and symmetric about y-axis.

(iii) vertex $(1, -2)$ and focus $(4, -2)$ (iv) end points of latus rectum $(4, -8)$ and $(4, 8)$ 2M

(i) Focus = $S(4, 0) = (a, 0)$

Vertex = $V(0, 0) = (h, k)$

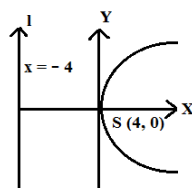
$$a = 4$$

Equation of parabola with $V(0, 0)$ and open rightward is

$$(y - k)^2 = 4a(x - h)$$

$$(y - 0)^2 = 4(4)(x - 0)$$

$$y^2 = 16x$$



JUL-21

(iii) vertex = $V(1, -2) = (h, k)$

Focus = $S(4, -2)$

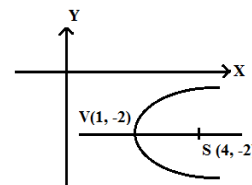
$$VS = a = \sqrt{(3)^2 + 0}$$

$$= \sqrt{9} = 3$$

Equation of the parabola with $V(1, -2)$ & open rightward is $(y - k)^2 = 4a(x - h)$

$$(y + 2)^2 = 4(3)(x - 1)$$

$$(y + 2)^2 = 12(x - 1)$$



(ii) Vertex = $V(0, 0) = (h, k)$

Equation of parabola with $V(0, 0)$ and open downward is

$$(x - h)^2 = -4a(y - k)$$

$$(x - 0)^2 = -4a(y - 0)$$

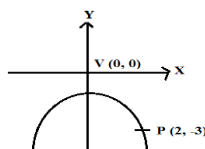
$$x^2 = -4ay$$

∴ it passes through $(2, -3)$

$$4 = -4a(-3) \Rightarrow 1 = 3a \Rightarrow a = \frac{1}{3}$$

∴ Equation of the parabola $x^2 = -4\left(\frac{1}{3}\right)y$

$$3x^2 = -4y$$



(iv) End of latus rectum

$$= (4, 8) (4, -8) = L(a, 2a); L^1(a, -2a)$$

$$a = 4$$

Focus = $S(4, 0)$

∴ vertex is at $(0, 0) = (h, k)$

Equation of parabola with $V(0, 0)$ and open rightward is $(y - k)^2 = 4a(x - h)$

$$(y - 0)^2 = 4(4)(x - 0)$$

$$y^2 = 16x$$

PTA-2

2. Find the equation of the ellipse in each of the cases given below:

3M

(i) foci $(\pm 3, 0)$, $e = \frac{1}{2}$

JUL-24

foci = $S(\pm 3, 0) = (\pm c, 0)$

centre = $C(0, 0)$

$$c = 3$$

$$e = \frac{1}{2}$$

$$a\left(\frac{1}{2}\right) = 3$$

$$a = 6 \Rightarrow a^2 = 36$$

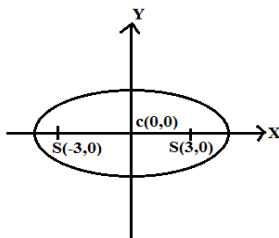
$$b^2 = a^2 - (c)^2 \\ = 36 - 9 = 27$$

Equation of an ellipse with $C(0, 0)$ and having major axis along x -axis is

$$\frac{(x-h)^2}{a^2} + \frac{(y-k)^2}{b^2} = 1$$

$$\frac{(x-0)^2}{36} + \frac{(y-0)^2}{27} = 1$$

$$\frac{x^2}{36} + \frac{y^2}{27} = 1$$



(ii) foci $(0, \pm 4)$ and end points of major axis are $(0, \pm 5)$

Vertices = $A(0, \pm 5) = A(0, \pm a)$

$$a = 5 \Rightarrow a^2 = 25$$

Foci = $S(0, \pm 4)$

$$= S(0, \pm c)$$

Centre = $C(0, 0)$

$$= (h, k)$$

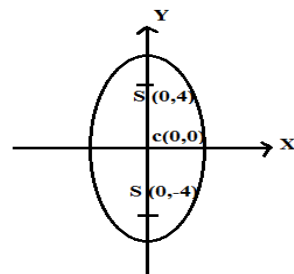
$$b^2 = a^2 - (c)^2$$

$$= 25 - 16 = 9$$

Equation of an ellipse with $C(0, 0)$ & having major axis along y -axis is $\frac{(x-h)^2}{b^2} + \frac{(y-k)^2}{a^2} = 1$

$$\frac{(x-0)^2}{9} + \frac{(y-0)^2}{25} = 1$$

$$\frac{x^2}{9} + \frac{y^2}{25} = 1$$



(iii) Length of latus rectum 8, eccentricity $\frac{3}{5}$, center $(0, 0)$ & major axis on x -axis.
centre = $C(0, 0)$

Length of latus rectum = $\frac{2b^2}{a} = 8$

$$2b^2 = 8a$$

$$b^2 = 4a$$

$$a^2(1 - e^2) = 4a$$

$$a^2\left(1 - \frac{9}{25}\right) = 4a$$

$$a^2\left(\frac{16}{25}\right) = 4a$$

$$16a^2 = 100a$$

$$4a = 25$$

$$a = \frac{25}{4}$$

$$\Rightarrow a^2 = \frac{625}{16}$$

$$b^2 = 4\left(\frac{25}{4}\right) = 25$$

Equation of the ellipse is $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$

$$\frac{x^2}{\frac{625}{16}} + \frac{y^2}{25} = 1$$

$$\frac{16x^2}{625} + \frac{y^2}{25} = 1$$

(iv) length of latus rectum 4, distance between foci $4\sqrt{2}$, center $(0, 0)$ & major axis as y -axis.
centre = $C(0, 0)$

Length of latus rectum = $\frac{2b^2}{a} = 4$

$$2b^2 = 4a$$

$$b^2 = 2a$$

Distance between foci = $SS^1 = 4\sqrt{2}$

$$2c = 4\sqrt{2}$$

$$c = 2\sqrt{2}$$

$$b^2 = 2a$$

$$a^2 - (c)^2 = 2a$$

$$a^2 - 8 = 2a$$

$$a^2 - 2a - 8 = 0$$

$$(a - 4)(a + 2) = 0$$

$$a = 4 \quad a = -2 \text{ absurd}$$

$$a^2 = 16; \quad b^2 = 2(4) = 8$$

Equation of the ellipse with $C(0, 0)$ and having major axis along y -axis

$$\frac{x^2}{b^2} + \frac{y^2}{a^2} = 1$$

$$\frac{x^2}{8} + \frac{y^2}{16} = 1$$

3. Find the equation of the hyperbola in each of the cases given below:

3M

(i) foci $(\pm 2, 0)$,

$$\text{eccentricity} = \frac{3}{2}$$

$$\text{Foci} = S(\pm 2, 0) = (\pm c, 0)$$

$$c = 2$$

$$e = \frac{3}{2}$$

$$a\left(\frac{3}{2}\right) = 2$$

$$a = \frac{4}{3}$$

$$\Rightarrow a^2 = \frac{16}{9}$$

$$\Rightarrow b^2 = (c)^2 - a^2$$

$$= 4 - \frac{16}{9}$$

$$= \frac{36-16}{9}$$

$$b^2 = \frac{20}{9}$$

Equation of the hyperbola with $c(0,0)$ and having transverse axis along x -axis is

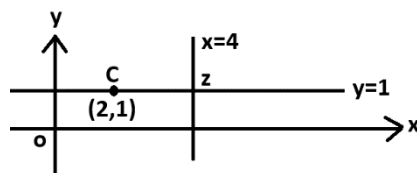
$$\frac{x^2}{a^2} - \frac{y^2}{b^2} = 1$$

$$\frac{x^2}{\frac{16}{9}} - \frac{y^2}{\frac{20}{9}} = 1$$

$$\frac{9x^2}{16} - \frac{9y^2}{20} = 1$$

(ii) Centre $(2, 1)$, one of the foci $(8, 1)$ and corresponding directrix $x = 4$

$$\text{centre} = c(2, 1) = (h, k)$$



$$\text{Foci} = S(8, 1)$$

$$CS = c = \sqrt{(6)^2 + 0} = \sqrt{36}$$

$$c = 6 \dots\dots\dots (1)$$

$$CZ = \frac{a}{e} = 4 - 2$$

$$\frac{\frac{a}{e}}{\frac{3}{2}} = 2 \dots\dots\dots (2)$$

$$(1) \times (2)$$

$$(c) \left(\frac{a}{e}\right) = (6) (2)$$

$$a^2 = 12$$

$$b^2 = (c)^2 - a^2$$

$$= 36 - 12$$

$$b^2 = 24$$

Equation of the hyperbola with $c(2, 1)$ and having transverse axis parallel to x

$$\text{axis is } \frac{(x-h)^2}{a^2} - \frac{(y-k)^2}{b^2} = 1$$

$$\frac{(x-2)^2}{12} - \frac{(y-1)^2}{24} = 1$$

(iii) passing through $(5, -2)$ and length of the transverse axis along x -axis & of length 8 units.

$$\text{centre} = c(0, 0)$$

$$AA' = 2a = 8$$

$$a = 4$$

$$\Rightarrow a^2 = 16$$

Equation of the hyperbola is

$$\frac{x^2}{a^2} - \frac{y^2}{b^2} = 1$$

$\therefore$ it passes through $(5, -2)$,

$$\frac{25}{16} - \frac{4}{b^2} = 1$$

$$\frac{25}{16} - 1 = \frac{4}{b^2}$$

$$\frac{9}{16} = \frac{4}{b^2}$$

$$b^2 = \frac{4 \times 16}{9}$$

$$b^2 = \frac{64}{9}$$

Equation of the hyperbola with $c(0,0)$ and having transverse axis along x -axis is

$$\frac{x^2}{a^2} - \frac{y^2}{b^2} = 1$$

$$\frac{x^2}{16} - \frac{9y^2}{64} = 1$$

4. Find the vertex, focus, equation of directrix and length of the latus rectum of the following:

(i) $y^2 = 16x$

3M

(ii) $x^2 = 24y$

3M

(iii) $y^2 = -8x$

3M

(iv) $x^2 - 2x + 8y + 17 = 0$

5M

(v) $y^2 - 4y - 8x + 12 = 0$

5M

Equation	Vertex	Focus	Equation of directrix	Length of latus rectum
(i) $y^2 = 16x$ $4a = 16$ $\Rightarrow a = 4$	$V(0, 0)$	$S(a, 0) = S(4, 0)$	$x = -a$ $x = -4$ $x + 4 = 0$	$4a = 16$
(i) $x^2 = 24y$ $4a = 24$ $\Rightarrow a = 6$	$V(0, 0)$	$S(0, a) = S(0, 6)$	$y = -a$ $y = -6$ $y + 6 = 0$	$4a = 24$
(iii) $y^2 = -8x$ $4a = 8$ $\Rightarrow a = 2$	$V(0, 0)$	$S(-a, 0) = S(-2, 0)$	$x = a$ $x = 2$ $x - 2 = 0$	$4a = 8$

(iv) $x^2 - 2x = -8y - 17$

$$(x - 1)^2 - 1 = -8y - 17$$

$$(x - 1)^2 = -8y - 16 = -8(y + 2)$$

$$(x - 1)^2 = -4(2)(y - (-2))$$

The above equation is of the form $(x - h)^2 = -4a(y - k)$

$$h = 1, \quad k = -2, \quad a = 2$$

Equation	Vertex	Focus	Equation of directrix	Length of latus rectum
$x^2 - 2x + 8y + 17 = 0$	$V(1, -2)$	$S(h, -a + k)$ $= S(1, -2 - 2) = S(1, -4)$	$y = a + k$ $= 2 - 2 = 0$	$4a = 8$

(v) $y^2 - 4y = 8x - 12$

$$(y - 2)^2 - 4 = 8x - 12 = 8x - 8$$

$$(y - 2)^2 = 8(x - 1) = 4(2)(x - 1)$$

The above equation is of the form $(y - k)^2 = 4a(x - h)$

$$h = 1, \quad k = 2, \quad a = 2$$

APR-24

Equation	Vertex	Focus	Equation of directrix	Length of latus rectum
$y^2 - 4y - 8x + 12 = 0$	$V(1, 2)$	$S(a + h, \quad k)$ $= (2 + 1, 2) = S(3, 2)$	$x = -a + h$ $= -2 + 1 = -1$	$4a = 8$

5. Identify the type of conic and find centre, foci, vertices, and directrices of each of the following:

(i) $\frac{x^2}{25} + \frac{y^2}{9} = 1$

(ii) $\frac{x^2}{3} + \frac{y^2}{10} = 1$

(iii) $\frac{x^2}{25} - \frac{y^2}{144} = 1$

(iv) $\frac{y^2}{16} - \frac{x^2}{9} = 1$

5M

(i) $\frac{x^2}{25} + \frac{y^2}{9} = 1$

JUL-22

The above equation is the equation of an ellipse having major axis along x-axis.

$$\begin{array}{l}
 a^2 = 25 \Rightarrow a = 5 \\
 b^2 = 9 \\
 e = \sqrt{1 - \frac{b^2}{a^2}} = \frac{16}{25}
 \end{array}
 \quad
 \begin{array}{l}
 e = \frac{4}{5} \\
 ae = 5 \left(\frac{4}{5}\right) = 4 \\
 \frac{a}{e} = \frac{5}{\frac{4}{5}} = \frac{25}{4}
 \end{array}$$

Centre	$C(0, 0) = (h, k)$
Vertices	$A(a, 0) = A(5, 0)$ $A'(-a, 0) = A'(-5, 0)$
Foci	$S(ae, 0) = S(4, 0)$ $S'(-ae, 0) = S'(-4, 0)$
Equation of directrix	$x = +\frac{a}{e} = \frac{25}{4}$ $x = -\frac{a}{e} = -\frac{25}{4}$

(ii) $\frac{x^2}{3} + \frac{y^2}{10} = 1$

The above equation is the equation of an ellipse having major along y-axis

$$\begin{array}{l}
 a^2 = 10 \Rightarrow a = \sqrt{10} \\
 b^2 = 3 \\
 e = \sqrt{1 - \frac{b^2}{a^2}} = \frac{7}{10}
 \end{array}
 \quad
 \begin{array}{l}
 e = \frac{\sqrt{7}}{\sqrt{10}} \\
 ae = \sqrt{7} \\
 \frac{a}{e} = \frac{10}{\sqrt{7}}
 \end{array}$$

Centre	$C(0, 0) = (h, k)$
Vertices	$A(0, a) = A(0, \sqrt{10})$ $A'(0, -a) = A'(0, -\sqrt{10})$
Foci	$F(0, ae) = F(0, \sqrt{7})$ $F'(0, -ae) = F'(0, -\sqrt{7})$
Equation of directrix	$y = +\frac{a}{e} = \frac{10}{\sqrt{7}} ; y = -\frac{a}{e} = -\frac{10}{\sqrt{7}}$

$$(iii) \frac{x^2}{25} - \frac{y^2}{144} = 1$$

The above equation is the equation of the hyperbola having transverse axis along x-axis.

$$\begin{aligned} a^2 &= 25 \Rightarrow a = 5 & e &= \frac{13}{5} \\ b^2 &= 144 & c &= ae = 13 \\ e &= \sqrt{1 + \frac{b^2}{a^2}} = \frac{169}{25} & \frac{a}{e} &= \frac{5}{\frac{13}{5}} = \frac{25}{13} \end{aligned}$$

Centre	$C(0, 0) = (h, k)$
Vertices	$A(a, 0) = A(5, 0)$ $A'(-a, 0) = A'(-5, 0)$
Foci	$S(c, 0) = S(13, 0)$ $S'(-c, 0) = S'(-13, 0)$
Equation of directrix	$x = +\frac{a}{e} = \frac{25}{13}$ $x = -\frac{a}{e} = -\frac{25}{13}$

$$(iv) \frac{y^2}{16} - \frac{x^2}{9} = 1$$

The above equation is the equation of the hyperbola having transverse axis along y-axis.

compare with $\frac{y^2}{a^2} - \frac{x^2}{b^2} = 1$

$$\begin{aligned} a^2 &= 16 \Rightarrow a = 4 & e &= \frac{5}{4} \\ b^2 &= 9 & c &= ae = 5 \\ e &= \sqrt{1 + \frac{b^2}{a^2}} = \frac{25}{16} & \frac{a}{e} &= \frac{16}{5} \end{aligned}$$

Centre	$C(0, 0) = (h, k)$
Vertices	$A(0, a) = A(0, 4)$ $A'(0, -a) = A'(0, -4)$
Foci	$F(0, c) = F(0, 5)$ $F'(0, -c) = F'(0, -5)$
Equation of directrix	$y = +\frac{a}{e} = \frac{16}{5}$; $y = -\frac{a}{e} = -\frac{16}{5}$

6. Prove that the length of the latus rectum of the hyperbola $\frac{x^2}{a^2} - \frac{y^2}{b^2} = 1$ is $\frac{2b^2}{a}$

PTA-2

3M

The equation of the hyperbola is $\frac{x^2}{a^2} - \frac{y^2}{b^2} = 1$ (1)

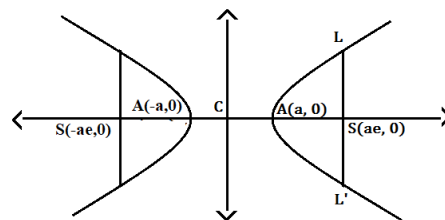
L and L' be the end points of the latus rectum. Then coordinates of L are (ae, SL)

But $L(ae, SL)$ is a point on the hyperbola.

$$\frac{a^2 e^2}{a^2} - \frac{SL^2}{b^2} = 1 \Rightarrow e^2 - \frac{SL^2}{b^2} = 1$$

$$e^2 - 1 = \frac{SL^2}{b^2}$$

$$SL^2 = b^2(e^2 - 1) \text{ (2)}$$



We know $b^2 = a^2(e^2 - 1) \Rightarrow e^2 - 1 = \frac{b^2}{a^2}$

Substituting for $e^2 - 1 = \frac{b^2}{a^2}$, in equation (2) we get $SL^2 = b^2 \times \frac{b^2}{a^2} = \frac{b^4}{a^2} \Rightarrow SL = \frac{b^2}{a}$

$$\therefore LL' = 2 \times SL = \frac{2b^2}{a}. \quad \text{Hence the length of the latus rectum} = \frac{2b^2}{a}$$

7. Show that the absolute value of difference of the focal distances of any point P on the hyperbola is the length of its transverse axis.

3M

Let $P(x, y)$ be any point on the hyperbola $\frac{x^2}{a^2} - \frac{y^2}{b^2} = 1$

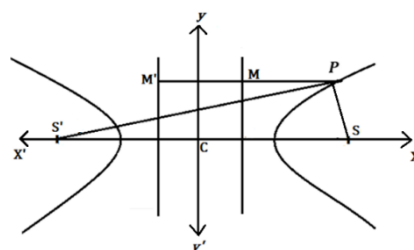
By the definition of hyperbola, $\frac{SP}{PM} = e$ and $\frac{S'P}{PM'} = e$

$$SP = e PM \text{ and } S'P = e PM'$$

$$SP - S'P = e[PM - PM'] = e(MM')$$

$$= e \cdot \frac{2a}{e} = 2a$$

$$S'P - SP = 2a = \text{Length of transverse axis}$$



8. Identify the type of conic and find centre, foci, vertices, and directrices of each of the following:

(i) $\frac{(x-3)^2}{225} + \frac{(y-4)^2}{289} = 1$ (ii) $\frac{(x+1)^2}{100} + \frac{(y-2)^2}{64} = 1$ (iii) $\frac{(x+3)^2}{225} - \frac{(y-4)^2}{64} = 1$ (iv) $\frac{(y-2)^2}{25} - \frac{(x+1)^2}{16} = 1$
 (v) $18x^2 + 12y^2 - 144x + 48y + 120 = 0$ (vi) $9x^2 - y^2 - 36x - 6y + 18 = 0$

(i) $\frac{(x-3)^2}{225} + \frac{(y-4)^2}{289} = 1$

3M

The above equation is the equation of an ellipse having major axis parallel to y-axis.

$$\begin{aligned} h &= 3; \quad k = 4 \\ a^2 &= 289 \\ a &= 17 \\ b^2 &= 225 \end{aligned}$$

$$\begin{aligned} e^2 &= \frac{a^2 - b^2}{a^2} = \frac{64}{289} \Rightarrow e = \frac{8}{17} \\ c &= ae = 8 \\ \frac{a}{e} &= \frac{17}{\frac{8}{17}} = \frac{289}{8} \end{aligned}$$

Centre	$C(3, 4)$	
Vertices	$A(h, a + k) = A(3, 17 + 4) = A(3, 21)$ $(h, -a + k) = A'(3, -17 + 4) = A'(3, -13)$	
Foci	$S(h, k + c) = S(3, 4 + 8) = S(3, 12)$ $S'(h, k - c) = S'(3, 4 - 8) = S'(3, -4)$	
Equation of Directrices	$y - k = +\frac{a}{e}$ $y - 4 = \frac{289}{8}$ $y = \frac{289}{8} + 4 = \frac{289+32}{8} = \frac{321}{8}$	$y - k = -\frac{a}{e}$ $y - 4 = \frac{-289}{8}$ $y = \frac{-289}{8} + 4 = \frac{-289+32}{8} = \frac{-257}{8}$

(ii) $\frac{(x+1)^2}{100} + \frac{(y-2)^2}{64} = 1$

3M

The above equation is the equation of an ellipse having major axis parallel to x-axis.

$$\begin{aligned} h &= -1, k = 2 \\ a^2 &= 100 \\ a &= 10 \\ b^2 &= 64 \end{aligned}$$

$$\begin{aligned} e^2 &= \frac{a^2 - b^2}{a^2} = \frac{36}{100} \Rightarrow e = \frac{6}{10} = \frac{3}{5} \\ c &= ae = 10 \times \frac{3}{5} = 6 \\ \frac{a}{e} &= \frac{10}{\frac{3}{5}} = \frac{50}{3} \end{aligned}$$

Centre	$C(-1, 2)$	
Vertices	$A(a + h, k) = A(10 - 1, 2) = A(9, 2)$ $A'(-a + h, k) = A'(-10 - 1, 2) = A'(-11, 2)$	
Foci	$S(c + h, k) = S(6 - 1, 2) = S(5, 2)$ $S'(-c + h, k) = S'(-6 - 1, 2) = S'(-7, 2)$	
Equation of Directrices	$x - h = +\frac{a}{e}$ $x + 1 = \frac{50}{3}$ $x = \frac{50}{3} - 1$ $x = \frac{47}{3}$	$x - h = -\frac{a}{e}$ $x + 1 = \frac{-50}{3}$ $x = \frac{-50}{3} - 1$ $x = \frac{-53}{3}$

$$(iii) \frac{(x+3)^2}{225} - \frac{(y-4)^2}{64} = 1$$

3M

The above equation is the equation of the hyperbola having transverse axis parallel to x - axis, compare with $\frac{(x-h)^2}{a^2} - \frac{(y-k)^2}{b^2} = 1$

$$\begin{array}{l|l} h = -3 & e^2 = \frac{a^2+b^2}{a^2} = \frac{289}{225} \\ k = 4 & e = \frac{17}{15} \\ a^2 = 225 & c = ae = 17 \\ \Rightarrow a = 15 & \frac{a}{e} = \frac{15}{\frac{17}{15}} = \frac{225}{17} \\ b^2 = 64 & \end{array}$$

Centre	$C(-3, 4)$	
Vertices	$A(a + h, k) = A(15 - 3, 4)$ $= A(12, 4)$	$A'(-a + h, k) = A'(-15 - 3, 4)$ $= A'(-18, 4)$
Foci	$S(c + h, k) = S(17 - 3, 4)$ $= S(14, 4)$	$S'(-c + h, k) = S'(-17 - 3, 4)$ $= S'(-20, 4)$
Equation of Directrices	$x - h = +\frac{a}{e}$ $x + 3 = \frac{225}{17}$ $x = \frac{225}{17} - 3$ $= \frac{225-51}{17}$ $x = \frac{174}{17}$	$x - h = -\frac{a}{e}$ $x + 3 = \frac{-225}{17}$ $x = \frac{-225}{17} - 3$ $= \frac{-225-51}{17}$ $x = \frac{-276}{17}$

$$(iv) \frac{(y-2)^2}{25} - \frac{(x+1)^2}{16} = 1$$

3M

The above equation is the equation of the hyperbola having transverse axis parallel to y -axis, compare with $\frac{(y-k)^2}{a^2} - \frac{(x-h)^2}{b^2} = 1$

$$\begin{array}{l|l} h = -1 & e^2 = \frac{a^2+b^2}{a^2} = \frac{41}{25} \\ k = 2 & e = \frac{\sqrt{41}}{5} \\ a^2 = 25 & c = ae = \sqrt{41} \\ \Rightarrow a = 5 & \frac{a}{e} = \frac{5}{\frac{\sqrt{41}}{5}} = \frac{25}{\sqrt{41}} \\ b^2 = 16 & \end{array}$$

Centre	$C(-1, 2)$	
Vertices	$A(h, a + k) = A(-1, 5 + 2)$ $= A(-1, 7)$	$A'(h, -a + k) = A'(-1, -5 + 2)$ $= A'(-1, -3)$
Foci	$S(h, c + k) = S(-1, \sqrt{41} + 2)$ $= S(-1, 2 + \sqrt{41})$	$S'(h, -c + k) = S'(-1, -\sqrt{41} + 2)$ $= S'(-1, 2 - \sqrt{41})$
Equation of Directrices	$y - k = +\frac{a}{e}$ $y - 2 = \frac{25}{\sqrt{41}}$ $y = \frac{25}{\sqrt{41}} + 2$	$y - k = -\frac{a}{e}$ $y - 2 = -\frac{25}{\sqrt{41}}$ $y = -\frac{25}{\sqrt{41}} + 2$

$$(v) 18x^2 + 12y^2 - 144x + 48y + 120 = 0$$

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5M

$$(18x^2 - 144x) + (12y^2 + 48y) = -120$$

$$18(x^2 - 8x) + 12(y^2 + 4y) = -120$$

$$18[(x - 4)^2 - 16] + 12[(y + 2)^2 - 4] = -120$$

$$18(x - 4)^2 - 288 + 12(y + 2)^2 - 48 = -120$$

$$18(x - 4)^2 + 12(y + 2)^2 = -120 + 336 = 216$$

$$\frac{(x-4)^2}{12} + \frac{(y+2)^2}{18} = 1$$

The above equation is the equation of an ellipse having major axis parallel to y -axis.

$$h = 4; \quad k = -2$$

$$a^2 = 18 = 9 \times 2$$

$$a = 3\sqrt{2}$$

$$b^2 = 12$$

$$e^2 = \frac{a^2 - b^2}{a^2} = \frac{6}{18} = \frac{1}{3}$$

$$e = \frac{1}{\sqrt{3}}$$

$$c = ae = 3\sqrt{2} \times \frac{1}{\sqrt{3}}$$

$$= \sqrt{3}\sqrt{2} = \sqrt{6}$$

$$\frac{a}{e} = \frac{3\sqrt{2}}{\frac{1}{\sqrt{3}}} = 3\sqrt{2} \times \sqrt{3} = 3\sqrt{6}$$

Centre	$C(4, -2)$	
Vertices	$A(h, a + k) = A(4, 3\sqrt{2} - 2)$	$A'(h, a + k) = A'(4, -3\sqrt{2} - 2)$
Foci	$S(h, c + k) = S(4, \sqrt{6} - 2)$	$S'(h, -c + k) = S'(4, -\sqrt{6} - 2)$
Equation of Directrices	$y - k = +\frac{a}{e}$ $y + 2 = 3\sqrt{6}$ $y = -2 + 3\sqrt{6}$	$y - k = +\frac{a}{e} \Rightarrow y + 2 = -3\sqrt{6}$ $y + 2 = -3\sqrt{6}$ $y = -2 - 3\sqrt{6}$

$$(vi) (9x^2 - 36x) - y^2 - 6y = -18$$

5M

$$9(x^2 - 4x) - (y^2 + 6y) = -18$$

$$9[(x - 2)^2 - 4] - [(y + 3)^2 - 9] = -18$$

$$9(x - 2)^2 - 36 - (y + 3)^2 + 9 = -18$$

$$9(x - 2)^2 - (y + 3)^2 = -18 + 27 = 9$$

$$\frac{(x-2)^2}{1} - \frac{(y+3)^2}{9} = 1$$

The above equation is the equation of the hyperbola having transverse axis parallel to x - axis.

$$h = 2; \quad k = -3$$

$$a^2 = 1$$

$$\Rightarrow a = 1$$

$$b^2 = 9$$

$$e^2 = \frac{a^2 + b^2}{a^2} = \frac{10}{1}$$

$$e = c = \sqrt{10}$$

$$\frac{a}{e} = \frac{1}{\sqrt{10}}$$

Centre	$C(2, -3)$	
Vertices	$A(a + h, k) = A(3, -3)$	$A'(-a + h, k) = A'(1, -3)$
Foci	$S(c + h, k) = S(\sqrt{10} + 2, -3)$ $= S(2 + \sqrt{10}, -3)$	$S'(-c + h, k) = S'(-\sqrt{10} + 2, -3)$ $= S'(2 - \sqrt{10}, -3)$
Equation of Directrices	$x - h = +\frac{a}{e}$ $x - 2 = \frac{1}{\sqrt{10}}$ $x = 2 + \frac{1}{\sqrt{10}}$	$x - h = -\frac{a}{e}$ $x - 2 = \frac{-1}{\sqrt{10}}$ $x = 2 - \frac{1}{\sqrt{10}}$

EXERCISE 5.3



Video class

Concept Corner

Identifying the conics from the general equation of the conic

$$Ax^2 + Bxy + Cy^2 + Dx + Ey + F = 0.$$

The graph of the second degree equation is one of a circle, parabola, an ellipse, a hyperbola, a point, an empty set, a single line or a pair of lines. When,

- (1) $A = C = 1, B = 0, D = -2h, E = -2k, F = h^2 + k^2 - r^2$ the general equation reduces to $(x - h)^2 + (y - k)^2 = r^2$ which is a circle.
- (2) $B = 0$ and either A or $C = 0$, the general equation yields a parabola under study, at this level.
- (3) $A \neq C$ and A and C are of the same sign, the general equation yields an ellipse.
- (4) $A \neq C$ and A and C are of opposite signs, the general equation yields a hyperbola
- (5) $A = C$ and $B = D = E = F = 0$, the general equation yields a point $x^2 + y^2 = 0$.
- (6) $A = C = F$ and $B = D = E = 0$, the general equation yields an empty set $x^2 + y^2 + 1 = 0$, as there is no real solution.
- (7) $A \neq 0$ or $C \neq 0$ and others are zeros, the general equation yield coordinate axes.
- (8) $A = -C$ and rests are zero, the general equation yields a pair of lines $x^2 - y^2 = 0$.

Identify the type of conic section for each of the equations.

1.	$2x^2 - y^2 = 7$	$\because A, C$ have opposite sign; this equation represent a hyperbola
2.	$3x^2 + 3y^2 - 4x + 3y + 10 = 0$	$\because A = C; B = 0$ this equation represent a circle.
3.	$3x^2 + 2y^2 = 14$	$\because A \neq C, A, C$ have same sign; this equation represent an Ellipse .
4.	$x^2 + y^2 + x - y = 0$	$\because A = C; B = 0$; This equation represent a circle .
5.	$11x^2 - 25y^2 - 44x + 50y - 256 = 0$	$\because A, C$ have opposite sign; this equation represent a Hyperbola
6.	$y^2 + 4x + 3y + 4 = 0$.	$\because A = 0$ and it has only y^2 $\therefore$ Equation represent a parabola

EXERCISE 5.4

Concept Corner

Parametric form of Conics

Parametric forms : (i) circle $x^2 + y^2 = a^2$ (ii) parabola $y^2 = 4ax$
 (iii) Ellipse $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$ (iv) Hyperbola $\frac{x^2}{a^2} - \frac{y^2}{b^2} = 1$



Video class

Conics	Parametric equations	Parameter	Range of parameter	Any point on the conic
Circle	$x = a \cos \theta; y = a \sin \theta$	θ	$0 \leq \theta \leq 2\pi$	' θ ' or $(a \cos \theta, a \sin \theta)$
Parabola	$x = at^2; y = 2at$	t	$-\infty < t < \infty$	' t ' or $(at^2, 2at)$
Ellipse	$x = a \cos \theta; y = b \sin \theta$	θ	$0 \leq \theta \leq 2\pi$	' θ ' or $(a \cos \theta, b \sin \theta)$
Hyperbola	$x = a \sec \theta; y = b \tan \theta$	θ	$-\pi \leq \theta \leq \pi$ Except $\theta = \pm \frac{\pi}{2}$	' θ ' or $(a \sec \theta, b \tan \theta)$

Equation of tangent and normal to the parabola $y^2 = 4ax$

- (i) Equation of tangent in cartesian form: $yy_1 = 2a(x + x_1)$
 (ii) Equation of tangent in parametric form: $yt = x + at^2$
 (iii) Equation of normal in Cartesian form: $xy_1 + 2ay = x_1y_1 + 2ay_1$
 (iv) Equation of normal in parametric form: $y + xt = at^3 + 2at$

➤ Three normal can be drawn to a parabola $y^2 = 4ax$ from a given point, one of which is always real.

Equations of tangent and normal to Ellipse and Hyperbola :

(1) Equation of the tangent to the ellipse $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$

- (i) at (x_1, y_1) is $\frac{xx_1}{a^2} + \frac{yy_1}{b^2} = 1$ cartesian form
 (ii) at $'\theta'$ is $\frac{x \cos \theta}{a} + \frac{y \sin \theta}{b} = 1$. parametric form

(2) Equation of the normal to the ellipse $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$

- (i) at (x_1, y_1) is $\frac{a^2x}{x_1} - \frac{b^2y}{y_1} = a^2 - b^2$ cartesian form
 (ii) at $'\theta'$ is $\frac{ax}{\cos \theta} - \frac{by}{\sin \theta} = a^2 - b^2$ parametric form

(3) Equation of the tangent to the hyperbola $\frac{x^2}{a^2} - \frac{y^2}{b^2} = 1$

- (i) at (x_1, y_1) is $\frac{xx_1}{a^2} - \frac{yy_1}{b^2} = 1$ cartesian form
 (ii) at $'\theta'$ is $\frac{x \sec \theta}{a} - \frac{y \tan \theta}{b} = 1$ parametric form

(4) Equation of the normal to the hyperbola $\frac{x^2}{a^2} - \frac{y^2}{b^2} = 1$

- (i) at (x_1, y_1) is $\frac{a^2x}{x_1} + \frac{b^2y}{y_1} = a^2 + b^2$ cartesian form
 (ii) at $'\theta'$ is $\frac{ax}{\sec \theta} + \frac{by}{\tan \theta} = a^2 + b^2$ parametric form.

Condition for the line $y = mx + c$ to be a tangent to the conic sections

(i) Parabola $y^2 = 4ax \Rightarrow c = \frac{a}{m}$,

The point of contact is $\left(\frac{a}{m^2}, \frac{2a}{m}\right)$ and the equation of tangent to parabola is $y = mx + \frac{a}{m}$.

(ii) Ellipse $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$:

Condition for line $y = mx + c$ to be the tangent to the ellipse $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$ is $c^2 = a^2m^2 + b^2$ with the point of contact is $\left(-\frac{a^2m}{c}, \frac{b^2}{c}\right)$, and the equation of tangent is $y = mx \pm \sqrt{a^2m^2 + b^2}$.

(iii) Hyperbola $\frac{x^2}{a^2} - \frac{y^2}{b^2} = 1$:

Condition for line $y = mx + c$ to be the tangent to the hyperbola $\frac{x^2}{a^2} - \frac{y^2}{b^2} = 1$ is $c^2 = a^2m^2 - b^2$, with the point of contact is $\left(-\frac{a^2m}{c}, -\frac{b^2}{c}\right)$, and the equation of tangent is $y = mx \pm \sqrt{a^2m^2 - b^2}$.

1. Find the equations of the two tangents that can be drawn from (5, 2) to the ellipse

$$2x^2 + 7y^2 = 14$$

5M

$$\frac{x^2}{7} + \frac{y^2}{2} = 1$$

$$a^2 = 7; \quad b^2 = 2$$

Equation of tangents drawn from (5, 2) is

$$y = mx \pm \sqrt{a^2m^2 + b^2}$$

$$y - mx = \pm \sqrt{7m^2 + 2}$$

$\therefore$ it passes through (5, 2)

$$2 - 5m = \pm \sqrt{7m^2 + 2}$$

$$(2 - 5m)^2 = 7m^2 + 2$$

$$4 + 25m^2 - 20m = 7m^2 + 2$$

$$18m^2 - 20m + 2 = 0$$

$$9m^2 - 10m + 1 = 0$$

$$(9m - 1)(m - 1) = 0$$

$$m = \frac{1}{9} \quad ; \quad 1$$

when $m = \frac{1}{9}$, Equation of tangent at (5, 2) is

$$y - 2 = \frac{1}{9}(x - 5)$$

$$9y - 18 = x - 5$$

$$x - 9y + 13 = 0$$

when $m = 1$, Equation of tangent is

$$y - 2 = 1(x - 5)$$

$$y - 2 = x - 5$$

$$x - y - 3 = 0$$

$\therefore$ Equation of tangents are

$$x - 9y + 13 = 0 \text{ and } x - y - 3 = 0$$

2. Find the equations of tangents to the hyperbola $\frac{x^2}{16} - \frac{y^2}{64} = 1$ which are parallel to $10x - 3y + 9 = 0$

5M

$$\text{slope of line} = m_1 = \frac{-a}{b} = \frac{-10}{-3} = \frac{10}{3}$$

$\therefore$ tangent parallel to line

$$\text{slope of tangent} = m = m_1 = \frac{10}{3}$$

Equation of tangents to the hyperbola

$$a^2 = 16; \quad b^2 = 64$$

$$y = mx \pm \sqrt{a^2m^2 - b^2}$$

$$= mx \pm \sqrt{16m^2 - 64}$$

$$y = \frac{10}{3}x \pm \sqrt{16 \times \frac{100}{9} - 64}$$

$$y = \frac{10}{3}x \pm \sqrt{\frac{1600 - 576}{9}}$$

$$= \frac{10}{3}x \pm \sqrt{\frac{1024}{9}}$$

$$= \frac{10}{3}x \pm \frac{32}{3}$$

$$3y = 10x \pm 32$$

$$10x - 3y \pm 32 = 0$$

Equations are

$$10x - 3y + 32 = 0 \text{ \& } 10x - 3y - 32 = 0$$

3. Show that the line $x - y + 4 = 0$ is a tangent to the ellipse $x^2 + 3y^2 = 12$. Also, find the coordinates of the point of contact.

5M

Line: $y = x + 4$

$$m = 1$$

$$c = 4$$

Ellipse: $\frac{x^2}{12} + \frac{y^2}{4} = 1$

$$a^2 = 12$$

$$b^2 = 4$$

Condition for tangent is $c^2 = a^2m^2 + b^2$

$$a^2m^2 + b^2 = 12(1) + 4$$

$$= 16$$

$$= c^2$$

$\therefore$ The given line is a tangent to the given ellipse.

$$\text{Point of contact} = P\left(\frac{-a^2m}{c}, \frac{b^2}{c}\right)$$

$$= P\left(\frac{-12(1)}{4}, \frac{4}{4}\right)$$

$$= P(-3, 1)$$

4. Find the equation of the tangent to the parabola $y^2 = 16x$ perpendicular to $2x + 2y + 3 = 0$

$$y^2 = 16x$$

$$4a = 16$$

$$a = 4$$

$$\text{Equation of tangents to the parabola } y = mx + \frac{a}{m} = mx + \frac{4}{m}$$

$$\text{Slope of line} = m_1 = \frac{-a}{b} = \frac{-2}{2} = -1$$

$\therefore$ Perpendicular line to tangent

$$\text{Slope of tangent } m = \frac{-1}{m_1} = \frac{-1}{-1} = 1, m = 1$$

$$\text{Equation of tangents are } y = (1)x + \frac{4}{1}$$

$$y = x + 4$$

$$x - y + 4 = 0$$

5. Find the equation of the tangent at $t = 2$ to the parabola $y^2 = 8x$. (Hint: use parametric form)

$$y^2 = 8x$$

$$4a = 8$$

$$a = 2$$

$$\text{Given: } t = 2 \Rightarrow (x_1, y_1) = (at^2; 2at) = (2(4); 2(2)(2)) = (8, 8)$$

$$\text{Equation of tangent at } (8, 8) \text{ is } yy_1 = 8 \times \frac{1}{2} (x + x_1)$$

$$y(8) = 4(x + 8)$$

$$2y = x + 8$$

$$x - 2y + 8 = 0$$

6. Find the equations of the tangent and normal to hyperbola $12x^2 - 9y^2 = 108$ at $\theta = \frac{\pi}{3}$ (Hint: use parametric form)

$$12x^2 - 9y^2 = 108$$

$$\frac{x^2}{9} - \frac{y^2}{12} = 1$$

$$a^2 = 9 \Rightarrow a = 3 \text{ \& } b^2 = 12 \Rightarrow b = 2\sqrt{3}$$

$$\text{Given: } \theta = \frac{\pi}{3}$$

$$(x_1, y_1) = (a \sec \theta; b \tan \theta)$$

$$= \left(3 \sec \frac{\pi}{3}; 2\sqrt{3} \tan \frac{\pi}{3} \right) = (3 \times 2; 2\sqrt{3} \times \sqrt{3}) = (6, 6)$$

$$\text{Equation of tangent is } 12xx_1 - 9yy_1 = 108$$

$$12x(6) - 9y(6) = 108$$

$$\div 6, \quad 12x - 9y = 18$$

$$\div 3, \quad 4x - 3y = 6$$

$$4x - 3y - 6 = 0$$

$$\text{Equation of normal is } 3x + 4y + k = 0$$

$$\therefore \text{ it passes through } (6, 6) \Rightarrow 18 + 24 + k = 0 \Rightarrow k = -42$$

$$\text{Equation of normal is } 3x + 4y - 42 = 0$$

7. Prove that the point of intersection of the tangents at t_1 and t_2 on the parabola $y^2 = 4ax$ is $[at_1t_2, a(t_1 + t_2)]$

JUL-23

3M

Equation of tangent at t_1 to the parabola $yt_1 = x + at_1^2$ (1)

Equation of tangent at t_2 is $yt_2 = x + at_2^2$ (2)

$$(1) - (2) \Rightarrow y(t_1 - t_2) = a(t_1^2 - t_2^2)$$

$$y(t_1 - t_2) = a(t_1 + t_2)(t_1 - t_2)$$

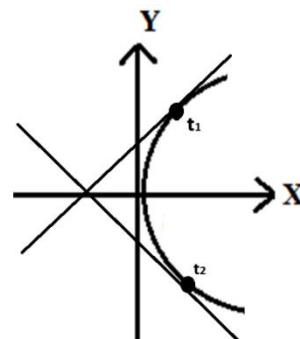
$$y = a(t_1 + t_2)$$

Put y in (1) $a(t_1 + t_2)t_1 = x + at_1^2$

$$at_1^2 + at_1t_2 = x + at_1^2$$

$$x = at_1t_2$$

Point of intersection = $[at_1t_2, a(t_1 + t_2)]$



8. If the normal at the point t_1 on the parabola $y^2 = 4ax$ meets the parabola again at the point t_2 then prove that $t_2 = -\left(t_1 + \frac{2}{t_1}\right)$

3M

Equation of normal at $P(t_1)$ on the parabola is $y + xt_1 = 2at_1 + at_1^3$

$\therefore$ it meets again at $Q(t_2)$ ie $Q(at_2^2; 2at_2)$

$$2at_2 + at_2^2t_1 = 2at_1 + at_1^3$$

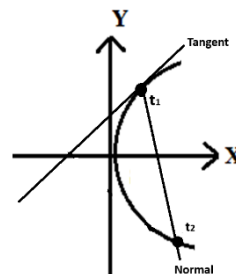
$$t_1t_2^2 - t_1^3 = 2t_1 - 2t_2$$

$$t_1(t_2^2 - t_1^2) = 2(t_1 - t_2)$$

$$t_1(t_2 + t_1)(t_2 - t_1) = -2(t_2 - t_1)$$

$$t_1(t_2 + t_1) = -2$$

$$t_2 + t_1 = -\frac{2}{t_1} \Rightarrow t_2 = -t_1 - \frac{2}{t_1} \Rightarrow t_2 = -\left(t_1 + \frac{2}{t_1}\right)$$



Video class

EXERCISE 5.5

Concept Corner

Reflective property of parabola:

The light or sound or radio waves originating at a parabola's focus are reflected parallel to the parabola's axis and conversely the rays arriving parallel to the axis are directed towards the focus.

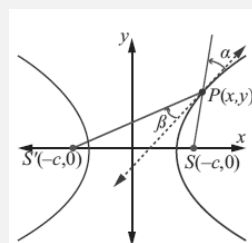
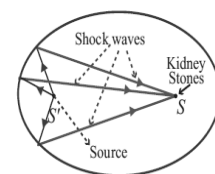
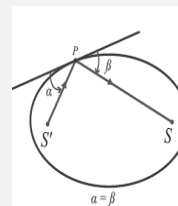
Reflective property of an Ellipse:

The lines from the foci to a point on an ellipse make equal angles with the tangent line at that point. The light or sound or radio waves emitted from one focus hits any point P on the ellipse is received at the other focus.

Reflective Property of a Hyperbola

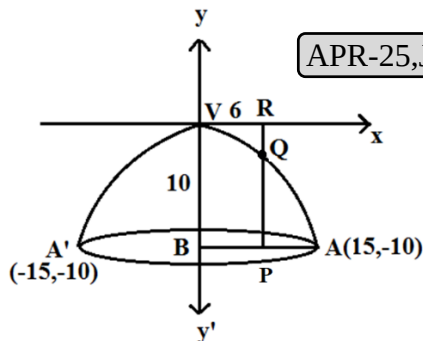
The lines from the foci to a point on a hyperbola make equal angles with the tangent line at that point.

The light or sound or radio waves directed from one focus is received at the other focus as in the case ellipse used in spotting location of ships sailing in deep sea.



1. A bridge has a parabolic arch that is $10m$ high in the centre and $30m$ wide at the bottom. Find the height of the arch $6m$ from the centre, on either sides.

5M



Equation of the parabola $x^2 = -4ay$

With $AA' = 30$

$$AB = BA' = 15$$

Height = $VB = 10$

$\therefore A(15, -10), A'(-15, -10)$

$\therefore A$ lies on the parabola, we have

$$(15)^2 = -4a(-10)$$

$$225 = 40a$$

$$45 = 8a \Rightarrow a = \frac{45}{8}$$

Equation of the parabola

$$x^2 = -4\left(\frac{45}{8}\right)y$$

$$x^2 = -\frac{45}{2}y$$

Given $VR = 6m$. Let $PR = y_1$

$\therefore$ The coordinate of $R = R(6, -y_1)$

$\therefore R$ lies on the parabola, we have

$$(6)^2 = -\frac{45}{2}(-y_1)$$

$$36 = \frac{45}{2}y_1$$

$$4 = \frac{5}{2}y_1 \Rightarrow y_1 = \frac{8}{5}$$

Required height = $RQ - PR$

$$= 10 - \frac{8}{5}$$

$$= \frac{50-8}{5}$$

$$= \frac{42}{5}$$

$$PQ = 8.4m$$

2. A tunnel through a mountain for a four lane highway is to have an elliptical opening. The total width of the highway (not the opening) is to be $16m$, and the height at the edge of the road must be sufficient for a truck $4m$ high to clear if the highest point of the opening is to be $5m$ approximately. How wide must the opening be?

5M

Equation of an ellipse $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$

$$b = 5 \Rightarrow b^2 = 25$$

$$\frac{x^2}{a^2} + \frac{y^2}{5^2} = 1 \dots\dots(1)$$

$(8,4)$ is a point on the Ellipse

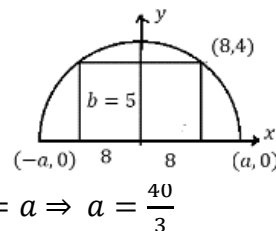
$$\frac{8^2}{a^2} + \frac{4^2}{5^2} = 1$$

$$\frac{8^2}{a^2} = 1 - \frac{16}{25}$$

$$\frac{8^2}{a^2} = \frac{9}{25} = \frac{3^2}{5^2}$$

$$\frac{8}{a} = \frac{3}{5} \Rightarrow \frac{8 \times 5}{3} = a \Rightarrow a = \frac{40}{3}$$

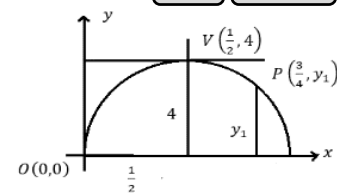
Width of the opening = $2a = \frac{80}{3} = 26.6m$



3. At a water fountain, water attains a maximum height of $4m$ at horizontal distance of $0.5m$ from its origin. If the path of water is a parabola, find the height of water at a horizontal distance of $0.75m$ from the point of origin.

5M

PTA-1



Vertex $V\left(\frac{1}{2}, 4\right)$

Equation of parabola $(x - h)^2 = -4a(y - k)$

$$\left(x - \frac{1}{2}\right)^2 = -4a(y - 4) \dots\dots(1)$$

$(0,0)$ Parabola (1) $\Rightarrow \left(0 - \frac{1}{2}\right)^2 = -4a(0 - 4)$

$$\frac{1}{4} = 4a(4) \Rightarrow 4a = \frac{1}{16}$$

$$\left(x - \frac{1}{2}\right)^2 = -\frac{1}{16}(y - 4) \dots\dots\dots(2)$$

$\left(\frac{3}{4}, y_1\right)$ parabola $\left(\frac{3}{4} - \frac{1}{2}\right)^2 = -\frac{1}{16}(y_1 - 4)$

$$\left(\frac{1}{4}\right)^2 = -\frac{1}{16}(y_1 - 4)$$

$$-\frac{16}{16} = (y_1 - 4)$$

$$4 - 1 = y_1$$

$$y_1 = 3 \Rightarrow \text{Height} = 3m$$

4. An engineer designs a satellite dish with a parabolic cross section. The dish is 5m wide at the opening, and the focus is placed 1.2m from the vertex

(a) Position a coordinate system with the origin at the vertex and the x -axis on the parabola's axis of symmetry and find an equation of the parabola.

5M

(b) Find the depth of the satellite dish at the vertex.

(a) Equation of the parabola

$$y^2 = 4ax.$$

$$\text{Here } a = 1.2$$

$\therefore$ The equation of the parabola

$$y^2 = 4(1.2)x$$

$$y^2 = 4.8x$$

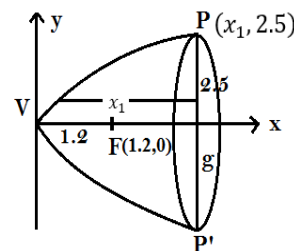
(b) $P(x_1, 2.5)$ lies on the parabola

$$(2.5)^2 = 4.8x_1$$

$$6.25 = 4.8x_1$$

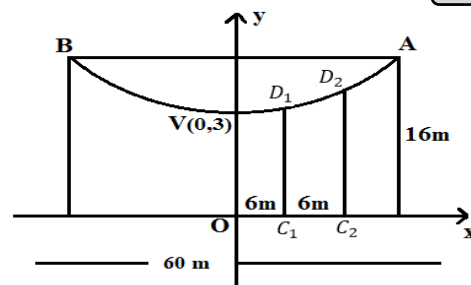
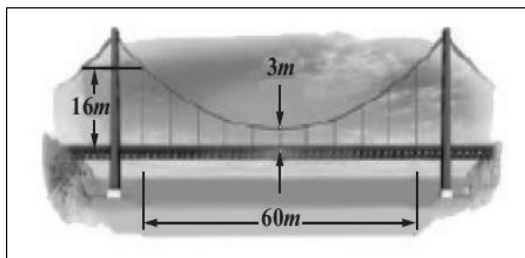
$$x_1 = \frac{6.25}{4.8} = 1.3$$

$\therefore$ Required depth = 1.3m



5. Parabolic cable of a 60m portion of the roadbed of a suspension bridge are positioned as shown below. Vertical Cables are to be spaced every 6m along this portion of the roadbed. Calculate the lengths of first two of these vertical cables from the vertex.

5M



It is given that vertex = $v(0,3)$ and open upward.

$\therefore$ Equation of the parabola

$$(x - h)^2 = 4a(y - k)$$

$$(x - 0)^2 = 4a(y - 3)$$

$$x^2 = 4a(y - 3)$$

It is given the coordinate of A is $A(30,16)$

$\therefore$ A lies on the parabola; we have

$$(30)^2 = 4a(16 - 3)$$

$$900 = 4a(13)$$

$$\Rightarrow 4a = \frac{900}{13}$$

$\therefore$ Equation of the parabola is

$$x^2 = \left(\frac{900}{13}\right)(y - 3)$$

Here $OC_1 = 6m$ Let $C_1D_1 = y_1$

$\therefore$ Coordinate of D_1 is $D_1(6, y_1)$

$\therefore D_1$ lies on the parabola where

$$6^2 = \left(\frac{900}{13}\right)(y_1 - 3)$$

$$\frac{36 \times 13}{900} = y_1 - 3$$

$$0.53 = y_1 - 3$$

$$y_1 = 3 + 0.53 = 3.53$$

The height of first vertical cables $C_1D_1 = 3.53m$

The second vertical cables is C_2D_2 .

Here $OC_2 = 6 + 6 = 12m$

$$C_2D_2 = y_2m$$

$\therefore$ The coordinate of D_2 is $(12, y_2)$

$\therefore D_2$ lies on the parabola $(12)^2 = \left(\frac{900}{13}\right)(y_2 - 3)$

$$144 = \left(\frac{900}{13}\right)(y_2 - 3)$$

$$\frac{144 \times 13}{900} = y_2 - 3$$

$$y_2 - 3 = 2.08$$

$$y_2 = 3 + 2.08$$

$$y_2 = 5.08$$

$\therefore$ the length of second vertical cable is

$$C_2D_2 = 5.08m$$

6. Cross section of a Nuclear cooling tower is in the shape of a hyperbola with equation $\frac{x^2}{30^2} - \frac{y^2}{44^2} = 1$. The tower is $150m$ tall and the distance from the top of the tower to the centre of the hyperbola is half the distance from the base of the tower to the centre of the hyperbola. Find the diameter of the top and base of the tower.

5M

Equation of hyperbola is $\frac{x^2}{30^2} - \frac{y^2}{44^2} = 1$

Height of tower $DC = 150m$

Let the distance of the top of the tower from the centre of the hyperbola $OC = a$

$\therefore$ Distance of bottom of the tower from the centre of the hyperbola $= 150 - a$

Given : Distance of top of tower $= \frac{1}{2}$ (Distance of bottom of tower)

$$a = \frac{1}{2} (150 - a)$$

$$2a = 150 - a$$

$$3a = 150$$

$$\Rightarrow a = 50$$

$\therefore A(x_1, 50)$ lies on the hyperbola

$$\frac{x_1^2}{30^2} - \frac{50^2}{44^2} = 1$$

$$\frac{x_1^2}{30^2} = 1 + \frac{50^2}{44^2}$$

$$x_1^2 = 30^2 \left(\frac{44^2 + 50^2}{44^2} \right)$$

$$x_1 = \frac{30}{44} \sqrt{44^2 + 50^2}$$

$$= \frac{15}{22} \sqrt{1936 + 2500} = \frac{15}{22} \sqrt{4436} = \frac{15}{22} \times 66.60 = 45.41m$$

Diameter of the top $= 45.41 \times 2 = 90.82m$

$B(x_2, -100)$ is lies on the hyperbola

$$\therefore \frac{x_2^2}{30^2} - \frac{100^2}{44^2} = 1$$

$$\frac{x_2^2}{30^2} = 1 + \frac{100^2}{44^2}$$

$$x_2^2 = 30^2 \left(\frac{44^2 + 100^2}{44^2} \right)$$

$$x_2 = \frac{30}{44} \sqrt{44^2 + 100^2}$$

$$= \frac{15}{22} \sqrt{1936 + 10,000}$$

$$= \frac{15}{22} \sqrt{11,936}$$

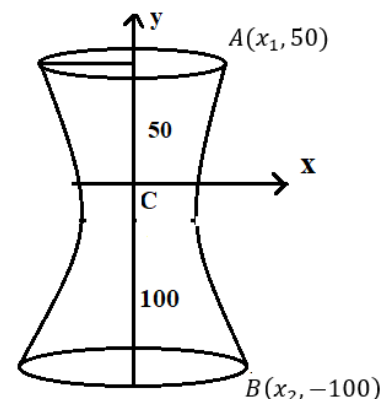
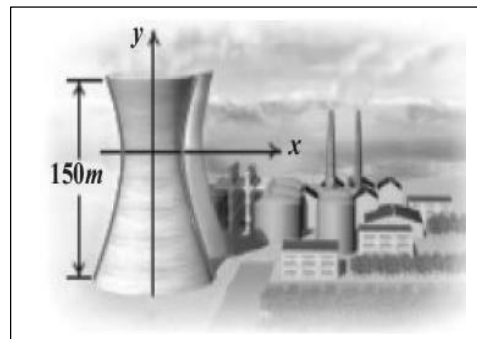
$$= \frac{15}{22} \times 109.25$$

$$x_2 = 74.49m$$

Diameter $= 2x_2 = 148.91$

$\therefore$ Diameter of the top of tower $= 90.82m$

$\therefore$ Diameter of the bottom of tower $= 148.91m$



7. A rod of length $1.2m$ moves with its ends always touching the coordinate axes. The locus of a point P on the rod, which is $0.3m$ from the end in contact with x -axis is an ellipse. Find the eccentricity.

MDL, JUL-20

5M

Let $p(x, y)$ is a point on the rod. $AB = 1.2$

Given $AP = 0.3 \quad \therefore PB = 0.9$

Let $\angle PAQ = \theta \quad \therefore \angle BPR = \theta$

In right angle $\Delta^{le} AQP \quad \sin \theta = \frac{y}{0.3}$

In right angle $\Delta^{le} BRP \quad \cos \theta = \frac{x}{0.9}$

$$\cos^2 \theta + \sin^2 \theta = 1$$

$$\left(\frac{x}{0.9}\right)^2 + \left(\frac{y}{0.3}\right)^2 = 1$$

$$\frac{x^2}{0.81} + \frac{y^2}{0.09} = 1$$

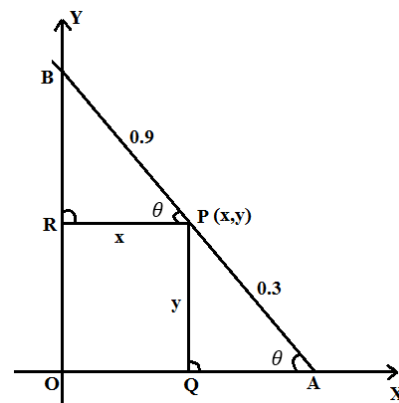
$$a^2 = 0.81 ; b^2 = 0.09$$

$$e^2 = \frac{a^2 - b^2}{a^2}$$

$$= \frac{0.81 - 0.09}{0.81}$$

$$= \frac{0.72}{0.81} = \frac{8}{9} = \frac{4 \times 2}{9}$$

$$e = \frac{2\sqrt{2}}{3}$$



8. Assume that water issuing from the end of a horizontal pipe, $7.5m$ above the ground, describes a parabolic path. The vertex of the parabolic path is at the end of the pipe. At a position $2.5m$ below the line of the pipe, the flow of water has curved outward $3m$ beyond the vertical line through the end of the pipe. How far beyond this vertical line will the water strike the ground?

Equation of the parabola $x^2 = -4ay$

APR-24,20

5M

The coordinates of P is $P(3, -2.5)$

$\therefore P$ lies on the parabola, we have

$$(3)^2 = -4a(-2.5)$$

$$9 = 10a$$

$$a = \frac{9}{10}$$

$\therefore$ Equation of parabola $x^2 = -4\left(\frac{9}{10}\right)y$

Also $VQ' = 7.5$ Let $P'Q' = x_1$

$\therefore$ The coordinate of $P' = P'(x_1; -7.5)$

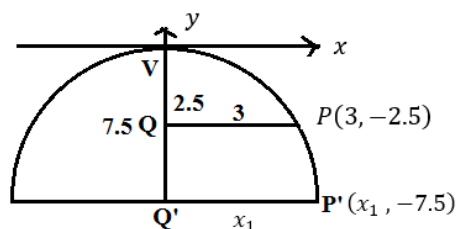
$\therefore P'$ lies on the parabola ; we have

$$x_1^2 = -4\left(\frac{9}{10}\right)(-7.5)$$

$$= \frac{9}{10} \times 30 = 9 \times 3 = 27$$

$$x_1 = 3\sqrt{3}$$

Required distance = $x_1 = 3\sqrt{3}m$



9. On lighting a rocket cracker it gets projected in a parabolic path and reaches a maximum height of $4m$ when it is $6m$ away from the point of projection. Finally it reaches the ground $12m$ away from the starting point. Find the angle of projection.

JUL-21 5M

Equation of the parabola $x^2 = -4ay$(1)

$A(6, -4)$ is a point on the parabola

$$(1) \Rightarrow (6)^2 = -4a(-4)$$

$$36 = 16a$$

$$9 = 4a$$

$$\Rightarrow a = \frac{9}{4}$$

Equation of parabola is $x^2 = -9y$

Differentiate with respect to x

$$2x = -9 \frac{dy}{dx}$$

$$\frac{dy}{dx} = -\frac{2x}{9}$$

$$\text{Slope} = \left(\frac{dy}{dx} \right)_{(-6, -4)}$$

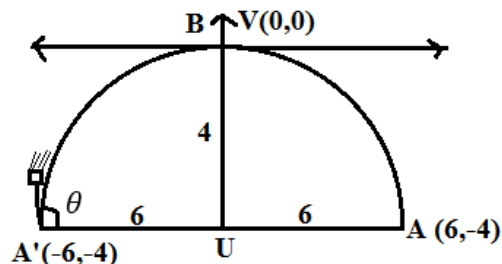
$$= \frac{-2(-6)}{9}$$

$$= \frac{12}{9}$$

$$\tan \theta = \frac{4}{3}$$

$$\theta = \tan^{-1} \left(\frac{4}{3} \right)$$

$$\text{Angle of projection} = \tan^{-1} \left(\frac{4}{3} \right)$$



10. Points A and B are $10km$ apart and it is determined from the sound of an explosion heard at those points at different times that the location of the explosion is $6km$ closer to A than B . Show that the location of the explosion is restricted to a particular curve and find an equation of it.

Let A, B be the Foci of the hyperbola

It is given $PB - PA = 2a = 6$

5M

$$a = 3$$

$$\Rightarrow a^2 = 9$$

$$AB = 2ae = 10$$

$$\Rightarrow ae = 5$$

$$b^2 = (ae)^2 - a^2$$

$$= (5)^2 - 3^2$$

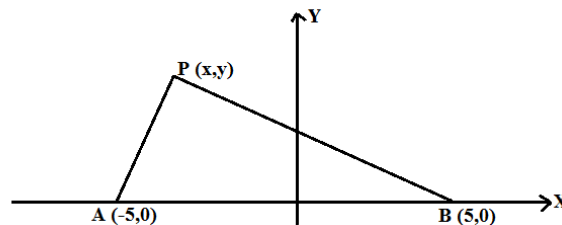
$$= 25 - 9$$

$$b^2 = 16$$

Equation of the curve (hyperbola)

$$\frac{x^2}{a^2} - \frac{y^2}{b^2} = 1$$

$$\frac{x^2}{9} - \frac{y^2}{16} = 1$$



EXERCISE 5.6



Choose the correct or the most suitable answer from the given four alternatives

1. The equation of the circle passing through (1, 5) and (4, 1) and touching y-axis is $x^2 + y^2 - 5x - 6y + 9 + \lambda(4x + 3y - 19) = 0$ where λ is equal to

(1) $0, -\frac{40}{9}$ (2) 0 (3) $\frac{40}{9}$ (4) $-\frac{40}{9}$

$x^2 + y^2 - 5x - 6y + 9 + \lambda(4x + 3y - 19) = 0$ $x^2 + y^2 + 2x\left(2\lambda - \frac{5}{2}\right) + 2y\left(\frac{3\lambda}{2} - 3\right) + 9 - 19\lambda = 0$ Centre y -axis distance = radius $2\lambda - \frac{5}{2} = \sqrt{\left(2\lambda - \frac{5}{2}\right)^2 + \left(\frac{3\lambda}{2} - 3\right)^2 + 19\lambda - 9}$ $\left(2\lambda - \frac{5}{2}\right)^2 = \left(2\lambda - \frac{5}{2}\right)^2 + \left(\frac{3\lambda}{2} - 3\right)^2 + 19\lambda - 9$	$0 = \frac{9\lambda^2}{4} + 9 - 9\lambda + 19\lambda - 9$ $\frac{9\lambda^2}{4} + 10\lambda = 0$ $\lambda\left(\frac{9\lambda + 40}{4}\right) = 0$ $\lambda = 0$ $9\lambda = -40$ $\lambda = -\frac{40}{9}$
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2. The eccentricity of the hyperbola whose latus rectum is 8 and conjugate axis is equal to half the distance between the foci is

PTA-5

(1) $\frac{4}{3}$ (2) $\frac{4}{\sqrt{3}}$ (3) $\frac{2}{\sqrt{3}}$ (4) $\frac{3}{2}$

$\frac{2b^2}{a} = 8$ $b^2 = \frac{8a}{2}$ $b^2 = 4a$ $2b = \frac{1}{2}(2ae)$ $2b = ae$ $a^2 e^2 = 4b^2$ $= 4(4a)$ $= 16a$	$e^2 = \frac{16a}{a^2} = \frac{16}{a}$ $b^2 = a^2 e^2 - a^2$ $4a = 16a - a^2$ $a^2 - 16a + 4a = 0$ $a^2 - 12a = 0$ $a(a - 12) = 0$ $a = 12$ $e^2 = \frac{16}{12} = \frac{4}{3}$ $e = \frac{2}{\sqrt{3}}$
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3. The circle $x^2 + y^2 = 4x + 8y + 5$ intersects the line $3x - 4y = m$ at two distinct points if

PTA-1

(1) $15 < m < 65$ (2) $35 < m < 85$ (3) $-85 < m < -35$ (4) $-35 < m < 15$

$x^2 + y^2 = 4x + 8y + 5$ Centre = $(-g, -f) = (2, 4)$, Radius = $\sqrt{4 + 16 + 5} = 5$ $\text{Distance} = \left \frac{3x_1 - 4y_1 - m}{\sqrt{(3)^2 + (-4)^2}} \right < 5$ $= \left \frac{3(2) - 4(4) - m}{\sqrt{9 + 16}} \right < 5$ $= 6 - 16 - m < 25$ $= 10 + m < 25$ $-25 < 10 + m < 25$ $-35 < m < 15$
--

4. The length of the diameter of the circle which touches the x -axis at the point $(1,0)$ and passes through the point $(2,3)$

(1) $\frac{6}{5}$

(2) $\frac{5}{3}$

(3) $\frac{10}{3}$

(4) $\frac{3}{5}$

 $P(x, y)$ Centre = $(1, a)$

Radius = $\sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$

$$a = \sqrt{(x - 1)^2 + (y - a)^2}$$

It passes through $(2,3)$

$$a^2 = (2 - 1)^2 + (3 - a)^2$$

$$a^2 = 1 + 9 - 6a + a^2$$

$$6a = 10$$

$$a = \frac{10}{6} = \frac{5}{3}$$

Diameter = $2a = \frac{10}{3}$

5. The radius of the circle $3x^2 + by^2 + 4bx - 6by + b^2 = 0$ is

APR-20, JUL-23, 24

(1) 1

(2) 3

(3) $\sqrt{10}$

(4) $\sqrt{11}$

Condition for circle

Coefficient of x^2 = Coefficient of y^2

$$3 = b$$

 $\therefore$ Equation of the circle

$$3x^2 + 3y^2 + 12x - 18y + 9 = 0$$

$$x^2 + y^2 + 4x - 6y + 3 = 0$$

$$g = 2, f = -3, c = 3$$

Radius = $r = \sqrt{g^2 + f^2 - c} = \sqrt{4 + 9 - 3} = \sqrt{13 - 3} = \sqrt{10}$

6. The centre of the circle inscribed in a square formed by the lines $x^2 - 8x - 12 = 0$ and $y^2 - 14y + 45 = 0$ is

(1) $(4, 7)$

(2) $(7, 4)$

(3) $(9, 4)$

(4) $(4, 9)$

$$x^2 - 8x - 12 = 0$$

$$x^2 - 8x + 16 - 16 - 12 = 0$$

$$(x - 4)^2 = 28$$

$$x - 4 = \pm 2\sqrt{7}$$

$$x = 4 \pm 2\sqrt{7}$$

$$y^2 - 14y + 45 = 0$$

$$(y - 9)(y - 5) = 0$$

$$y = 5, 9$$

$$(4 - 2\sqrt{7}, 5) (4 + 2\sqrt{7}, 9)$$

Centre = $\left(\frac{4+2\sqrt{7}+4-2\sqrt{7}}{2}, \frac{5+9}{2}\right) = (4, 7)$

7. The equation of the normal to the circle $x^2 + y^2 - 2x - 2y + 1 = 0$ which is parallel to the line $2x + 4y = 3$ is

(1) $x + 2y = 3$ (2) $x + 2y + 3 = 0$ (3) $2x + 4y + 3 = 0$ (4) $x - 2y + 3 = 0$

$$x^2 + y^2 - 2x - 2y + 1 = 0$$

$$\text{Centre} = (-g, -f) = (1, 1)$$

$$\text{Normal equation } 2x + 4y = k$$

$$\text{It passes } (1, 1)$$

$$2 + 4 = k$$

$$k = 6$$

$$2x + 4y = 6$$

$$x + 2y = 3$$

8. If $P(x, y)$ be any point on $16x^2 + 25y^2 = 400$ with foci $F_1(3, 0)$ and $F_2(-3, 0)$ then $PF_1 + PF_2$ is

(1) 8

(2) 6

(3) 10

(4) 12

APR-24, PTA-2,
JUL-21

$$\frac{x^2}{25} + \frac{y^2}{16} = 1$$

$$F_1P + F_2P = 2a$$

$$a^2 = 25$$

$$\Rightarrow a = 5$$

$$F_1P + F_2P = 2(5)$$

$$F_1P + F_2P = 10$$

9. The radius of the circle passing through the point $(6, 2)$ two of whose diameter are $x + y = 6$ and $x + 2y = 4$ is

(1) 10

(2) $2\sqrt{5}$

(3) 6

(4) 4

$$x + y = 6$$

$$x + 2y = 4$$

$$y = -2, \quad x = 8$$

$$\text{Centre} = C(8, -2), \text{ Point } (6, 2)$$

$$\text{Radius} = \sqrt{(2)^2 + (-4)^2} = \sqrt{4 + 16} = \sqrt{20} = \sqrt{4 \times 5} = 2\sqrt{5}$$

10. The area of quadrilateral formed with foci of the hyperbolas $\frac{x^2}{a^2} - \frac{y^2}{b^2} = 1$ and $\frac{x^2}{a^2} - \frac{y^2}{b^2} = -1$ is

(1) $4(a^2 + b^2)$

(2) $2(a^2 + b^2)$

(3) $a^2 + b^2$

(4) $\frac{1}{2}(a^2 + b^2)$

PTA-6

$$\frac{x^2}{a^2} - \frac{y^2}{b^2} = 1$$

$$A(ae, 0), C(-ae, 0) \quad e = \sqrt{\frac{a^2 + b^2}{a^2}}$$

$$\frac{x^2}{a^2} - \frac{y^2}{b^2} = -1$$

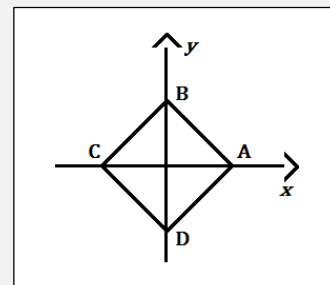
$$\frac{y^2}{b^2} - \frac{x^2}{a^2} = 1$$

$$B(0, be), D(0, -be) \quad e = \sqrt{\frac{a^2 + b^2}{b^2}}$$

$$\text{Area of quadrilateral ABCD} = \frac{1}{2} d_1 \times d_2$$

$$= \frac{1}{2} 2ae \times 2be$$

$$= a \frac{\sqrt{a^2 + b^2}}{a} \times 2b \frac{\sqrt{a^2 + b^2}}{b} = 2(a^2 + b^2)$$



11. If the normals of the parabola $y^2 = 4x$ drawn at the end points of its latus rectum are tangents to the circle $(x - 3)^2 + (y + 2)^2 = r^2$, then the value of r^2 is

(1) 2

(2) 3

(3) 1

(4) 4

$y^2 = 4ax$ $a = 1$ Length of latus rectum $4a = 4$ End Points are $(1, 2), (1, -2)$ $(\because (a, 2a), (a, -2a))$ Tangent equation at $(1, 2)$ $yy_1 = 4\left(\frac{x + x_1}{2}\right)$ $yy_1 = 2(x + x_1)$ $y = (x + 1)$	$x - y + 1 = 0$ Normal equation $x + y + k = 0$ It passes $(1, 2)$ $1 + 2 + k = 0$ $k = -3$ Normal equation $x + y - 3 = 0$, centre : $(3, -2)$ Radius = $\left \frac{3-2-3}{\sqrt{1^2+1^2}}\right = \frac{2}{\sqrt{2}} = \sqrt{2}$ $r = \sqrt{2}; r^2 = 2$
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12. If $x + y = k$ is a normal to the parabola $y^2 = 12x$, then the value of k is JUL-24, MDL, PTA-3

(1) 3

(2) -1

(3) 1

(4) 9

Condition for $y = mx + c$ to be a Normal to $y^2 = 4ax$ is $c = -2am - am^3$ $4a = 12$ $a = 3$ $x + y = k$ $y = -x + k$ $m = -1; c = k$ $k = -2(3)(-1) - 3(-1)^3$ $= 6 + 3 = 9$	
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13. The ellipse $E_1: \frac{x^2}{9} + \frac{y^2}{4} = 1$ is inscribed in a rectangle R whose sides are parallel to the coordinate axes. Another ellipse E_2 passing through the point $(0, 4)$ circumscribes the rectangle R . The eccentricity of the ellipse is

(1) $\frac{\sqrt{2}}{2}$ (2) $\frac{\sqrt{3}}{2}$ (3) $\frac{1}{2}$ (4) $\frac{3}{4}$

Equation of ellipse $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$ It passes $(0, 4)$ $\frac{(0)^2}{b^2} + \frac{(4)^2}{a^2} = 1$ $\frac{16}{a^2} = 1$ $a^2 = 16$	$\frac{x^2}{b^2} + \frac{y^2}{16} = 1$ It passes $(3, 2)$ $\frac{9}{b^2} + \frac{4}{16} = 1$ $\frac{9}{b^2} = 1 - \frac{1}{4}$ $\frac{9}{b^2} = \frac{3}{4}$	$b^2 = \frac{36}{3}$ $b^2 = 12$ $e = \sqrt{\frac{a^2 - b^2}{a^2}}$ $= \sqrt{\frac{4}{16}} = \frac{1}{2}$
--	--	---

14. Tangents are drawn to the hyperbola $\frac{x^2}{9} - \frac{y^2}{4} = 1$ parallel to the straight line $2x - y = 1$. One of the points of contact of tangents on the hyperbola is

- (1) $\left(\frac{9}{2\sqrt{2}}, \frac{-1}{\sqrt{2}}\right)$ (2) $\left(\frac{-9}{2\sqrt{2}}, \frac{1}{\sqrt{2}}\right)$ (3) $\left(\frac{9}{2\sqrt{2}}, \frac{1}{\sqrt{2}}\right)$ (4) $(3\sqrt{3}, -2\sqrt{2})$

$$a^2 = 9, b^2 = 4, m = 2$$

$$c = \pm\sqrt{a^2m^2 - b^2}$$

$$= \pm\sqrt{9 \times 4 - 4}$$

$$= \pm\sqrt{32}$$

Take $c = -4\sqrt{2}$

$$\text{Point of contact} = \left(\frac{-a^2m}{c}, \frac{-b^2}{c}\right) = \left(\frac{-9 \times 2}{-4\sqrt{2}}, \frac{-4}{-4\sqrt{2}}\right) = \left(\frac{9}{2\sqrt{2}}, \frac{1}{\sqrt{2}}\right)$$

15. The equation of the circle passing through the foci of the ellipse $\frac{x^2}{16} + \frac{y^2}{9} = 1$ having centre at (0,3) is

JUL-22

(1) $x^2 + y^2 - 6y - 7 = 0$

(2) $x^2 + y^2 - 6y + 7 = 0$

(3) $x^2 + y^2 - 6y - 5 = 0$

(4) $x^2 + y^2 - 6y + 5 = 0$

$$a^2 = 16 \Rightarrow a = 4$$

$$b^2 = 9$$

$$e^2 = \frac{7}{16} \Rightarrow e = \frac{\sqrt{7}}{4}$$

$$ae = \sqrt{7}$$

$$\text{Foci} = P(\pm\sqrt{7}, 0)$$

$$\text{Centre} = c(0, 3)$$

$$\therefore \text{Equation of the circle } (x - 0)^2 + (y - 3)^2 = 4^2$$

$$x^2 + y^2 - 6y + 9 = 16$$

$$x^2 + y^2 - 6y - 7 = 0$$

$$\text{Radius} = CP = r = \sqrt{(\sqrt{7})^2 + 3^2}$$

$$= \sqrt{7 + 9}$$

$$= \sqrt{16}$$

$$r = 4$$

16. Let C be the circle with centre at (1, 1) and radius = 1. If T is the circle centered at (0, y) passing through the origin and touching the circle C externally, then the radius of T is equal to

(1) $\frac{\sqrt{3}}{\sqrt{2}}$

(2) $\frac{\sqrt{3}}{2}$

(3) $\frac{1}{2}$

(4) $\frac{1}{4}$

$$\text{let } r_1 = 1 \text{ \& } r_2 = y$$

$$c_1c_2 = r_1 + r_2$$

$$c_1(1, 1) \quad c_2(0, y)$$

$$\sqrt{(1 - 0)^2 + (1 - y)^2} = 1 + y$$

$$1 + 1^2 + y^2 - 2y = (1 + y)^2$$

$$1 + 1^2 + y^2 - 2y = 1 + y^2 + 2y$$

$$1 = 2y + 2y$$

$$4y = 1$$

$$y = \frac{1}{4}$$

17. Consider an ellipse whose centre is of the origin and its major axis is along x -axis. If its eccentricity is $\frac{3}{5}$ and the distance between its foci is 6, then the area of the quadrilateral inscribed in the ellipse with diagonals as major and minor axis of the ellipse is

(1) 8 (2) 32 (3) 80 (4) 40

$2ae = 6$ $ae = 3$ $e = \frac{3}{5}$ $a\left(\frac{3}{5}\right) = 3$ $a = 5$	$b^2 = a^2 - a^2e^2 = 25 - 9 = 16 = 4$ Area of the quadrilateral $= \frac{1}{2}d_1 \times d_2$ $= \frac{1}{2}(2a) \times (2b)$ $= 5 \times 2 \times 4$ $= 40 \text{ sq. units}$
--	---

18. Area of the greatest rectangle inscribed in the ellipse $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$ is

PTA-3, APR-23

(1) $2ab$ (2) ab (3) $\sqrt{ab}$ (4) $\frac{a}{b}$

$x = a \cos \theta$ and $y = b \sin \theta$ Area $= l \times b$ $= 2a \cos \theta \times 2b \sin \theta$ $A = 2ab \sin 2\theta$ $A' = 4ab \cos 2\theta$	$A' = 0$ $\cos 2\theta = 0$ $\theta = 45^\circ$ $A = 2ab \sin 2(45^\circ)$ $A = 2ab$
---	--

19. An ellipse has OB as semi minor axes, F and F' its foci and the angle FBF' is a right angle. Then the eccentricity of the ellipse is

APR-25, PTA-6

(1) $\frac{1}{\sqrt{2}}$ (2) $\frac{1}{2}$ (3) $\frac{1}{4}$ (4) $\frac{1}{\sqrt{3}}$

$FF'^2 = BF^2 + BF'^2$ $(2ae)^2 = \left(\sqrt{(ae-0)^2 + (0-b)^2}\right)^2 + \left(\sqrt{(-ae-0)^2 + (0-b)^2}\right)^2$ $4a^2e^2 = a^2e^2 + b^2 + a^2e^2 + b^2$ $4a^2e^2 - 2a^2e^2 = 2b^2$ $2b^2 = 2a^2e^2$ $b^2 = a^2e^2$	$a^2 - a^2e^2 = a^2e^2$ $a^2 = 2a^2e^2$	$e^2 = \frac{1}{2} \Rightarrow e = \frac{1}{\sqrt{2}}$
---	--	--

20. The eccentricity of the ellipse $(x-3)^2 + (y-4)^2 = \frac{y^2}{9}$ is

(1) $\frac{\sqrt{3}}{2}$ (2) $\frac{1}{3}$ (3) $\frac{1}{3\sqrt{2}}$ (4) $\frac{1}{\sqrt{3}}$

$FP^2 = e^2 PM^2$ $(x-3)^2 + (y-4)^2 = \left(\frac{1}{3}\right)^2 \left(\frac{0+y+0}{\sqrt{0+1}}\right)^2$ $= \left(\frac{1}{3}\right)^2 \left(\frac{lx+my+n}{\sqrt{l^2+m^2}}\right)^2$ $\Rightarrow e = \frac{1}{3}$
--

21. If the two tangents drawn from a point P to the parabola $y^2 = 4x$ are at right angles then the locus of P is

PTA-2

- (1) $2x + 1 = 0$ (2) $x = -1$ (3) $2x - 1 = 0$ (4) $x = 1$

The locus of point of intersection of perpendicular tangent to the parabola $y^2 = 4ax$ is the Directrix $x = -a$

$$4a = 4 \Rightarrow a = 1 \quad \therefore x = -1$$

22. The circle passing through $(1, -2)$ and touching the axis of x at $(3, 0)$ passing through the point

MDL

- (1) $(-5, 2)$ (2) $(2, -5)$ (3) $(5, -2)$ (4) $(-2, 5)$

$$(x - 3)^2 + (y - k)^2 = k^2$$

It passes $(1, -2)$

$$(1 - 3)^2 + (-2 - k)^2 = k^2$$

$$4 + 4 + 4k + k^2 = k^2$$

$$4k = -8$$

$$k = -2$$

$$(x - 3)^2 + (y + 2)^2 = 4$$

Lies on $(5, -2)$

$$2^2 + 0^2 = 4$$

$$4 = 4$$

23. The locus of a point whose distance from $(-2, 0)$ is $\frac{2}{3}$ times its distance from the line $x = \frac{-9}{2}$ is

PTA-2, 4

- (1) a parabola (2) a hyperbola (3) an ellipse (4) a circle

$$\text{Given } SP = \frac{2}{3} PM$$

$$\Rightarrow e = \frac{2}{3} < 1$$

$\therefore$ This is an ellipse.

24. The values of m for which the line $y = mx + 2\sqrt{5}$ touches the hyperbola $16x^2 - 9y^2 = 144$ are the roots of $x^2 - (a + b)x - 4 = 0$, then the value of $(a + b)$ is

- (1) 2 (2) 4 (3) 0 (4) -2

$$\text{Line } y = mx + 2\sqrt{5}$$

$$m = ? \quad c = 2\sqrt{5}$$

$$a^2 = 9 \quad b^2 = 16$$

$$\frac{x^2}{9} - \frac{y^2}{16} = 1$$

$$C^2 = a^2 m^2 - b^2$$

$$4 \times 5 = 9m^2 - 16$$

$$m^2 = 4$$

$$m = \pm 2$$

$$\text{Sum of the roots} = 2 - 2$$

$$\frac{(a+b)}{1} = 0$$

$$a + b = 0$$

25. If the coordinates at one end of a diameter of the circle $x^2 + y^2 - 8x - 4y + c = 0$ are $(11, 2)$ the coordinates of the other end are

JUL-20

- (1) $(-5, 2)$ (2) $(-3, 2)$ (3) $(5, -2)$ (4) $(-2, 5)$

$$\text{Centre} = c(4, 2) = \left(\frac{x_1 + x_2}{2}, \frac{y_1 + y_2}{2} \right)$$

$$\text{One end} = A(x_1, y_1) = (11, 2)$$

$$\text{Other end} = B(x_2, y_2) = ?$$

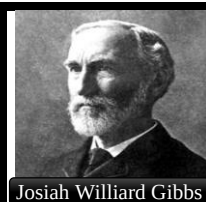
$$\frac{x_1 + x_2}{2} = 4$$

$$\frac{y_1 + y_2}{2} = 2$$

$$11 + x_2 = 8 \quad y_1 + y_2 = 4$$

$$x_2 = -3 \quad 2 + y_2 = 4 \quad y_2 = 2$$

$$\text{The other end} = B(-3, 2)$$



Josiah Willard Gibbs

Chapter - 6

Applications of Vector Algebra

EXERCISE 6.1



Video class

Concept Corner

Geometric introduction to vectors

A vector $\vec{v}$ is represented as a directed straight line segment in a 3-d space R^3 , with an initial point $A = (a_1, a_2, a_3) \in R^3$ & $B = (b_1, b_2, b_3) \in R^3$ & it is denoted by $\overrightarrow{AB}$.

$\overrightarrow{OP}$ is the position vector of the point P ; $\overrightarrow{AB}$ is the zero vector of the point P.

Length of the vector $\overrightarrow{AB} = \sqrt{(b_1 - a_1)^2 + (b_2 - a_2)^2 + (b_3 - a_3)^2}$

In particular, if a vector is the position vector $\vec{b}$ of (b_1, b_2, b_3) , then its length is $\sqrt{b_1^2 + b_2^2 + b_3^2}$

A vector having length 1 is called a unit vector. The notation $\hat{u}$.

Vector Addition : $\vec{a} + \vec{b} = (a_1 + b_1)\hat{i} + (a_2 + b_2)\hat{j} + (a_3 + b_3)\hat{k}$

Scalar multiplication : $\alpha \vec{a} = (\alpha a_1)\hat{i} + (\alpha a_2)\hat{j} + (\alpha a_3)\hat{k}$

where, $\vec{a} = a_1\hat{i} + a_2\hat{j} + a_3\hat{k}$, $\vec{b} = b_1\hat{i} + b_2\hat{j} + b_3\hat{k}$ and $\alpha \in R$

Scalar and vector product for two vectors

Scalar (or) Dot product:

Given two vectors $\vec{a} \cdot \vec{b}$ is calculated by $\vec{a} \cdot \vec{b} = a_1b_1 + a_2b_2 + a_3b_3$

Geometrical interpretation:

$\vec{a}, \vec{b}$ are two vectors and θ is the angle between $\vec{a}, \vec{b}$ then the scalar

(or) dot product of $\vec{a}, \vec{b}$ is defined as $\vec{a} \cdot \vec{b} = |\vec{a}||\vec{b}| \cos \theta$

Property:

1) $\vec{a} \cdot \vec{b} = \vec{b} \cdot \vec{a}$

2) condition for $\vec{a}$ and $\vec{b}$ to be perpendicular is $\vec{a} \cdot \vec{b} = 0$

3) condition for $\vec{a}$ and $\vec{b}$ to be parallel is $\vec{b} = m\vec{a}$ (or) $\vec{a} = n\vec{b}$

4) Angle between $\vec{a}, \vec{b}$ is $\cos \theta = \frac{\vec{a} \cdot \vec{b}}{|\vec{a}||\vec{b}|}$

5) The square of $\vec{a}$ is given by $(\vec{a})^2 = \vec{a} \cdot \vec{a} = a^2 = |\vec{a}|^2 \Rightarrow \therefore (\vec{a})^2 = |\vec{a}|^2$

$$\begin{aligned} 6) \text{ a) } (\vec{a} + \vec{b})^2 &= (\vec{a} + \vec{b}) \cdot (\vec{a} + \vec{b}) \\ &= \vec{a} \cdot \vec{a} + \vec{a} \cdot \vec{b} + \vec{b} \cdot \vec{a} + \vec{b} \cdot \vec{b} \\ &= a^2 + b^2 + \vec{a} \cdot \vec{b} + \vec{a} \cdot \vec{b} \\ &= a^2 + b^2 + 2\vec{a} \cdot \vec{b} \end{aligned}$$

$$(\vec{a} + \vec{b})^2 = |\vec{a} + \vec{b}|^2 = a^2 + b^2 + 2\vec{a} \cdot \vec{b}$$

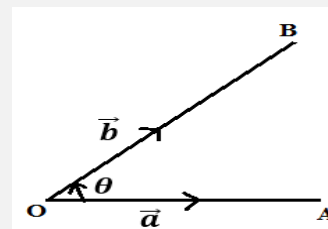
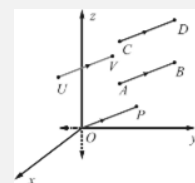
Similarly,

$$(\vec{a} - \vec{b})^2 = |\vec{a} - \vec{b}|^2 = a^2 + b^2 - 2\vec{a} \cdot \vec{b}$$

$$\text{b) } (\vec{a} + \vec{b})^2 + (\vec{a} - \vec{b})^2 = 2(a^2 + b^2)$$

$$(\vec{a} + \vec{b})^2 - (\vec{a} - \vec{b})^2 = 4\vec{a} \cdot \vec{b}$$

$$\begin{aligned} \text{c) } (\vec{a} + \vec{b} + \vec{c})^2 &= |\vec{a} + \vec{b} + \vec{c}|^2 \\ &= a^2 + b^2 + c^2 + 2(\vec{a} \cdot \vec{b} + \vec{b} \cdot \vec{c} + \vec{c} \cdot \vec{a}) \end{aligned}$$



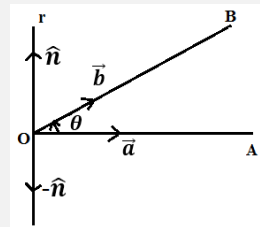
Vector (or) Cross product:

$$\vec{a} = a_1\vec{i} + a_2\vec{j} + a_3\vec{k}; \quad \vec{b} = b_1\vec{i} + b_2\vec{j} + b_3\vec{k}, \text{ then } \vec{a} \times \vec{b} = \begin{vmatrix} \vec{i} & \vec{j} & \vec{k} \\ a_1 & a_2 & a_3 \\ b_1 & b_2 & b_3 \end{vmatrix}$$

Geometrical interpretation:

$\vec{a}, \vec{b}$ are two vector and θ is the angle between $\vec{a}, \vec{b}$. $\hat{n}$ is the unit vector perpendicular to $\vec{a}, \vec{b}$. Then the vector (or) cross product of $\vec{a}, \vec{b}$ is defined as $\vec{a} \times \vec{b} = |\vec{a}||\vec{b}| \sin \theta \hat{n}$

$$\vec{a} \times \vec{b} = ab \sin \theta \hat{n} \Rightarrow |\vec{a} \times \vec{b}| = ab \sin \theta$$

**Property:**

1) $\vec{a} \times \vec{b} \neq \vec{b} \times \vec{a}$ but $\vec{a} \times \vec{b} = -\vec{b} \times \vec{a}$

2) a) The unit vectors perpendicular to both $\vec{a}$ and $\vec{b}$ is $\hat{n} = \pm \frac{(\vec{a} \times \vec{b})}{|\vec{a} \times \vec{b}|}$

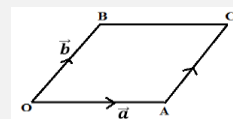
b) The vector magnitude λ which is perpendicular to both $\vec{a}$ and $\vec{b}$ is $\lambda \hat{n} = \pm \frac{\lambda(\vec{a} \times \vec{b})}{|\vec{a} \times \vec{b}|}$

c) The unit vector along the direction of $\vec{a}$ (or) parallel to $\vec{a}$ is given by $\hat{a} = \pm \frac{\vec{a}}{|\vec{a}|}$

3) The $\sin \theta$ angle between $\vec{a}$ and $\vec{b}$ is $\sin \theta = \frac{|\vec{a} \times \vec{b}|}{|\vec{a}||\vec{b}|}$

4) The condition for $\vec{a}, \vec{b}$ to be parallel is $\vec{a} \times \vec{b} = \vec{0}$ In particular $\vec{a} \times \vec{a} = \vec{0}$ [$\because \theta = 0$]

If $\vec{a} = a_1\vec{i} + a_2\vec{j} + a_3\vec{k}; \quad \vec{b} = b_1\vec{i} + b_2\vec{j} + b_3\vec{k}$ are parallel; then $\frac{a_1}{b_1} = \frac{a_2}{b_2} = \frac{a_3}{b_3}$ is their likely components are proportional.

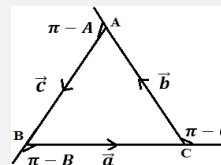
**Geometrical meaning for $\vec{a} \times \vec{b}$:**

a) If $\vec{a}, \vec{b}$ represents to the adjacent sides of a parallelogram, then

1) vector area of parallelogram $= \vec{a} \times \vec{b}$ 2) Area of parallelogram $= |\vec{a} \times \vec{b}|$

b) If $\vec{a}, \vec{b}$ represent any two sides of a triangle then

1) vector area of triangle $= \frac{1}{2}(\vec{a} \times \vec{b})$ 2) Area of triangle $= \frac{1}{2}|\vec{a} \times \vec{b}|$

**Formula:**

If $\overrightarrow{BC} = \vec{a}, \quad \overrightarrow{CA} = \vec{b}, \quad \overrightarrow{AB} = \vec{c}$ then $\vec{a} + \vec{b} + \vec{c} = \vec{0}$

Angle between $\vec{a}, \vec{b}$ is $\pi - C$; Angle between $\vec{b}, \vec{c}$ is $\pi - A$; Angle between $\vec{c}, \vec{a}$ is $\pi - B$

Work done by the force:

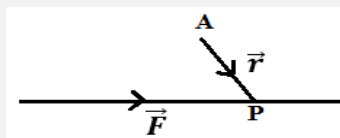
If $\vec{d}$ is the displacement & $\vec{F}$ is the force; then the work done by the force $w = \vec{F} \cdot \vec{d}$

Moment (or) Torque:

If a Force $\vec{F}$ is applied on a particle at a point with position vector $\vec{r}$;

then the moment (or) Torque of $\vec{F}$ is defined as $\vec{\tau}(\text{or}) \vec{m} = \vec{r} \times \vec{F}$

$\vec{r} = \overrightarrow{AP} = \overrightarrow{OP} - \overrightarrow{OA}$ = position vector of the through point - position vector of about point



The torque is also called as Rotational force.

1. Prove by a vector method that if a line is drawn from the centre of a circle to the midpoint of a chord, then the line is perpendicular to the chord. 3M

Let the centre of the circle is $O (\vec{0})$ and AB is the chord.

Let $\vec{OA} = \vec{a}$, $\vec{OB} = \vec{b}$.

Let M is the mid point of chord AB .

$$\therefore \vec{OM} = \frac{\vec{a} + \vec{b}}{2}$$

$$\vec{AB} = \vec{OB} - \vec{OA} \Rightarrow \vec{AB} = \vec{b} - \vec{a}$$

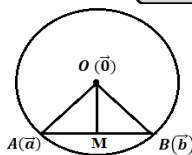
Here Radius = $OA = OB$

$$a = b$$

$$\begin{aligned} \text{Now } \vec{OM} \cdot \vec{AB} &= \frac{1}{2}(\vec{b} + \vec{a}) \cdot (\vec{b} - \vec{a}) \\ &= \frac{1}{2}(b^2 - a^2) \\ &= \frac{1}{2}(a^2 - a^2) \quad [\because a = b] \\ &= 0 \end{aligned}$$

$$\therefore \vec{OM} \perp \vec{AB}$$

A line is drawn from the centre of a circle to the midpoint of a chord, then the line is perpendicular to the chord.

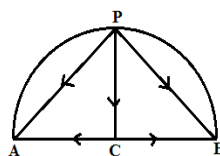


3. Prove by vector method that an angle in a semi-circle is a right angle. 3M

Let AB is a diameter of semi circle with centre at C .

$$\therefore AC = CB$$

Let P is any point on the semi-circle.



$$\therefore \text{Radius} = AC = BC = PC$$

$$\vec{PA} = \vec{PC} + \vec{CA}$$

$$\vec{PB} = \vec{PC} + \vec{CB} = \vec{PC} - \vec{CA}$$

$$\vec{PA} \cdot \vec{PB} = (\vec{PC} + \vec{CA}) \cdot (\vec{PC} - \vec{CA})$$

$$= PC^2 - CA^2$$

$$= PC^2 - PC^2 \quad [\because AC = CP]$$

$$= 0$$

$$\therefore \vec{PA} \perp \vec{PB}$$

$$\therefore \angle APB = 90^\circ$$

An angle in a semi-circle is a right angle.

2. Prove by vector method that the median to the base of an isosceles triangle is perpendicular to the base. 3M

In $\triangle OAB$,

$$\vec{OA} = \vec{a}; \vec{OB} = \vec{b};$$

$$|\vec{a}| = |\vec{b}|$$

C is the midpoint of AB

$$\vec{OC} = \frac{\vec{a} + \vec{b}}{2}$$

$$\vec{AB} = \vec{OB} - \vec{OA} = \vec{b} - \vec{a}$$

$$\vec{OC} \cdot \vec{AB} = \left(\frac{\vec{a} + \vec{b}}{2}\right) \cdot (\vec{b} - \vec{a})$$

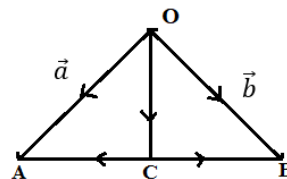
$$= \frac{1}{2}(-\vec{a}^2 + \vec{b}^2)$$

$$= \frac{1}{2}(0)$$

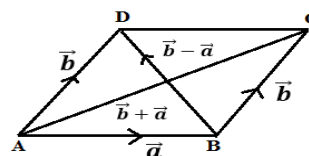
$$\vec{OC} \cdot \vec{AB} = 0$$

$$\therefore \vec{OC} \perp \vec{AB}$$

The median to the base of an isosceles triangle is perpendicular to the base.



4. Prove by vector method that the diagonals of a rhombus bisect each other at right angles. 3M



Let $ABCD$ is a Rhombus

$$\text{Let } \vec{AB} = \vec{a}, \text{ and } \vec{BC} = \vec{b}$$

$\therefore$ a rhombus is a parallelogram, the diagonals are bisect each other.

$$\vec{AC} = \vec{AB} + \vec{BC} = \vec{a} + \vec{b} = \vec{b} + \vec{a}$$

$$\begin{aligned} \vec{BD} &= \vec{BA} + \vec{AD} = -\vec{AB} + \vec{AD} \\ &= -\vec{a} + \vec{b} = \vec{b} - \vec{a} \end{aligned}$$

Here $AB = BC$

$$|\vec{a}| = |\vec{b}| \Rightarrow a = b$$

$$\vec{AC} \cdot \vec{BD} = (\vec{b} + \vec{a}) \cdot (\vec{b} - \vec{a})$$

$$= b^2 - a^2 = a^2 - a^2 = 0$$

$$\therefore \vec{AC} \perp \vec{BD}$$

The diagonals of a rhombus bisect each other at right angles.

5. Using vector method, prove that if the diagonals of a parallelogram are equal, then it is a rectangle. 3M

Let $ABCD$ is a parallelogram.

$$\text{In } \triangle ABC, \quad \vec{AC} = \vec{AB} + \vec{BC}$$

$$(\vec{AC})^2 = (\vec{AB} + \vec{BC})^2$$

$$AC^2 = AB^2 + BC^2 + 2 \vec{AB} \cdot \vec{BC} \dots\dots (1)$$

$$\begin{aligned} \text{In } \triangle BAD, \quad \vec{BD} &= \vec{BA} + \vec{AD} = -\vec{AB} + \vec{AD} \\ &= \vec{AD} - \vec{AB} \end{aligned}$$

$$(\vec{BD})^2 = (\vec{AD} - \vec{AB})^2$$

$$BD^2 = AD^2 + AB^2 - 2 \vec{AD} \cdot \vec{AB}$$

$$= AB^2 + AD^2 - 2 \vec{AB} \cdot \vec{AD}$$

$$= AB^2 + BC^2 - 2 \vec{AB} \cdot \vec{BC} [\because \vec{BC} = \vec{AD}]$$

$$\text{Given } AC = BD$$

$$AC^2 = BD^2$$

$$AB^2 + BC^2 + 2 \vec{AB} \cdot \vec{BC}$$

$$= AB^2 + BC^2 - 2 \vec{AB} \cdot \vec{BC}$$

$$4 \vec{AB} \cdot \vec{BC} = 0$$

$$\vec{AB} \cdot \vec{BC} = 0$$

$$\vec{AB} \perp \vec{BC}, \angle ABC = 90^\circ$$

$\therefore ABCD$ is a rectangle.

6. Prove by vector method that the area of the quadrilateral $ABCD$ having diagonals AC and BD is $\frac{1}{2} |\vec{AC} \times \vec{BD}|$ 3M

Vector area of the quadrilateral $ABCD$

$$= \text{vector area of } \triangle ABC + \text{vector area of } \triangle ACD$$

$$= \frac{1}{2} \vec{AB} \times \vec{AC} + \frac{1}{2} \vec{AC} \times \vec{AD}$$

$$= -\frac{1}{2} \vec{AC} \times \vec{AB} + \frac{1}{2} \vec{AC} \times \vec{AD}$$

Vector area of the quadrilateral $ABCD$

$$= \frac{1}{2} (\vec{AC} \times \vec{AD} - \vec{AC} \times \vec{AB})$$

$$= \frac{1}{2} \vec{AC} \times (\vec{AD} - \vec{AB})$$

$$= \frac{1}{2} \vec{AC} \times (\vec{BA} + \vec{AD})$$

$$= \frac{1}{2} \vec{AC} \times \vec{BD}$$

$$\therefore \text{Area of quadrilateral } ABCD = \frac{1}{2} |\vec{AC} \times \vec{BD}|$$

7. Prove by vector method that the parallelograms on the same base and between the same parallels are equal in area.

Let $ABCD$ & $ABDE$ are two parallelograms on the same base AB . 3M

vector area of parallelogram

$$ABCD = \vec{AB} \times \vec{AD}$$

vector area of parallelogram

$$ABDE = \vec{AB} \times \vec{AE}$$

$$= \vec{AB} \times (\vec{AD} + \vec{DE})$$

$$\because \vec{AB}; \vec{DE} \text{ are parallel } \vec{DE} = t \vec{AB}$$

$$= \vec{AB} \times (\vec{AD} + t \vec{AB})$$

$$= \vec{AB} \times \vec{AD} + t(\vec{0})$$

$$= \vec{AB} \times \vec{AD} + 0$$

$$[\because \vec{a} \times \vec{a} = \vec{0}]$$

vector area of parallelogram $ABDE$

$$= \text{vector area of parallelogram } ABCD$$

Hence the proof.

8. If G is the centroid of a $\triangle ABC$, prove that (area of $\triangle GAB$) = (area of $\triangle GBC$)

$$= (\text{area of } \triangle GCA) = \frac{1}{3} (\text{area of } \triangle ABC) \quad \text{3M}$$

Let A be the origin point.

G is the centroid of $\triangle ABC$.

$$\vec{AG} = \frac{\vec{AA} + \vec{AB} + \vec{AC}}{3}$$

$$= \frac{\vec{AB} + \vec{AC}}{3}$$

$$[\because \vec{AA} = \vec{0}]$$

$$\text{Area of } \triangle ABC = \frac{1}{2} |\vec{AB} \times \vec{AC}|$$

$$\text{Now Area of } \triangle GAB = \frac{1}{2} |\vec{AB} \times \vec{AG}|$$

$$= \frac{1}{2} \left| \vec{AB} \times \frac{1}{3} (\vec{AB} + \vec{AC}) \right|$$

$$= \frac{1}{3} \times \frac{1}{2} |\vec{AB} \times \vec{AB} + \vec{AB} \times \vec{AC}|$$

$$= \frac{1}{3} \times \left(\frac{1}{2} |\vec{AB} \times \vec{AC}| \right)$$

$$\therefore \vec{AB} \times \vec{AB} = \vec{0}$$

$$\text{Area of } \triangle GAB = \frac{1}{3} (\text{Area of } \triangle ABC) \dots\dots\dots (1)$$

Similarly

$$\text{Area of } \triangle GBC = \text{Area of } \triangle GCA$$

$$= \frac{1}{3} (\text{Area of } \triangle ABC)$$

$$\text{Area of } \triangle GAB = \text{Area of } \triangle GBC = \text{Area of } \triangle GCA$$

$$= \frac{1}{3} (\text{Area of } \triangle ABC)$$

$\therefore$ Hence proved.

9. Using vector method, prove that $\cos(\alpha - \beta) = \cos \alpha \cos \beta + \sin \alpha \sin \beta$

Let $\vec{OA} = \vec{a}$, $\vec{OB} = \vec{b}$ are two unit vectors.

JUL-20, 21, 22, 24 APR-23, 25

5M

Let $\angle AOL = \alpha$; $\angle BOL = \beta$

$\therefore \angle AOB = \alpha - \beta$

The coordinate of A, B

are $A(\cos \alpha, \sin \alpha)$; $B(\cos \beta, \sin \beta)$

$$\vec{OA} = \cos \alpha \hat{i} + \sin \alpha \hat{j}$$

$$\vec{OB} = \cos \beta \hat{i} + \sin \beta \hat{j}$$

$$\vec{OB} \cdot \vec{OA} = |\vec{OB}| |\vec{OA}| \cos(\alpha - \beta)$$

$$= 1 \times 1 \times \cos(\alpha - \beta)$$

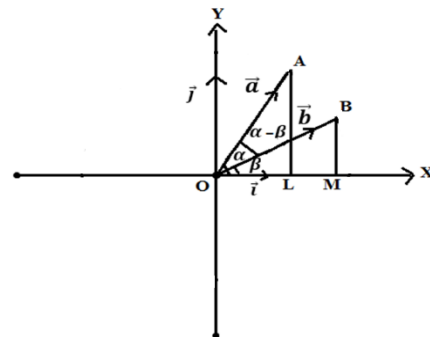
$$= \cos(\alpha - \beta) \quad \dots\dots\dots (1)$$

$$\vec{OB} \cdot \vec{OA} = (\cos \beta \hat{i} + \sin \beta \hat{j}) \cdot (\cos \alpha \hat{i} + \sin \alpha \hat{j})$$

$$= \cos \alpha \cos \beta + \sin \alpha \sin \beta \quad \dots\dots\dots (2)$$

From (1), (2)

$$\cos(\alpha - \beta) = \cos \alpha \cos \beta + \sin \alpha \sin \beta$$

10. Prove by vector method that $\sin(\alpha + \beta) = \sin \alpha \cos \beta + \cos \alpha \sin \beta$

MDL, APR-22

5M

Let $\vec{OA} = \vec{a}$; $\vec{OB} = \vec{b}$ are the unit vectors.

Let $\angle AOL = \alpha$; $\angle BOM = \beta$.

$\therefore \angle AOB = \alpha + \beta$.

The coordinate of A, B are

$A(\cos \alpha, -\sin \alpha)$; $B(\cos \beta, \sin \beta)$

$$\vec{OA} = \cos \alpha \hat{i} - \sin \alpha \hat{j}$$

$$\vec{OB} = \cos \beta \hat{i} + \sin \beta \hat{j}$$

$$\vec{OA} \times \vec{OB} = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ \cos \alpha & -\sin \alpha & 0 \\ \cos \beta & \sin \beta & 0 \end{vmatrix}$$

$$= \hat{i}(0 - 0) + \hat{j}(0 - 0) + \hat{k}(\cos \alpha \sin \beta + \sin \alpha \cos \beta)$$

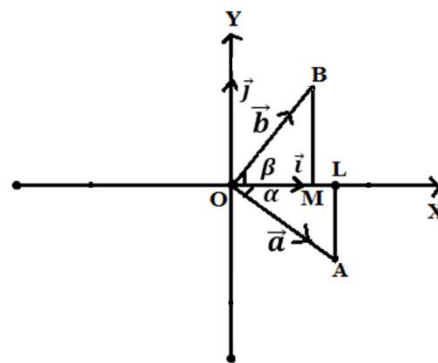
$$= \hat{k}(\cos \alpha \sin \beta + \sin \alpha \cos \beta) \quad \dots\dots\dots (1)$$

$$\vec{OA} \times \vec{OB} = |\vec{OA}| |\vec{OB}| \sin(\alpha + \beta) \vec{k}$$

$$= 1 \times 1 \times \sin(\alpha + \beta) \vec{k}$$

$$= \sin(\alpha + \beta) \vec{k} \quad \dots\dots\dots (2)$$

From (1), (2) $\sin(\alpha + \beta) = \sin \alpha \cos \beta + \cos \alpha \sin \beta$ Hence Proved.



$$[\because |\vec{OA}| = |\vec{OB}| = 1]$$

11. A particle acted on by the constant forces $8\hat{i} + 2\hat{j} - 6\hat{k}$ and $6\hat{i} + 2\hat{j} - 2\hat{k}$ is displaced from the point $(1, 2, 3)$ to the point $(5, 4, 1)$. Find the total work done by the forces. JUL-22 3M

$$\text{Total force} = \vec{F}$$

$$= 8\hat{i} + 2\hat{j} - 6\hat{k} + 6\hat{i} + 2\hat{j} - 2\hat{k}$$

$$= 14\hat{i} + 4\hat{j} - 8\hat{k}$$

$$\vec{OA} = \hat{i} + 2\hat{j} + 3\hat{k}; \quad \vec{OB} = 5\hat{i} + 4\hat{j} + \hat{k}.$$

$$\vec{d} = \vec{OB} - \vec{OA} = 4\hat{i} + 2\hat{j} - 2\hat{k}$$

$$\text{Work done} = W = \vec{F} \cdot \vec{d}$$

$$= (14\hat{i} + 4\hat{j} - 8\hat{k}) \cdot (4\hat{i} + 2\hat{j} - 2\hat{k})$$

$$= 56 + 8 + 16 = \mathbf{80 \text{ units}}$$

13. Find the magnitude and direction cosines of the torque of a force represented by $3\hat{i} + 4\hat{j} - 5\hat{k}$ about the point with position vector $2\hat{i} - 3\hat{j} + 4\hat{k}$ acting through a point whose position vector is $4\hat{i} + 2\hat{j} - 3\hat{k}$. 3M

$$\vec{F} = 3\hat{i} + 4\hat{j} - 5\hat{k}$$

$$\text{Position vector of through point}$$

$$= \vec{OP} = 4\hat{i} + 2\hat{j} - 3\hat{k}$$

$$\text{Position vector of about point}$$

$$= \vec{OA} = 2\hat{i} - 3\hat{j} + 4\hat{k}$$

$$\vec{r} = \vec{OP} - \vec{OA}$$

$$\vec{r} = 2\hat{i} + 5\hat{j} - 7\hat{k}$$

$$\text{Moment (Torque)} = \vec{m} = \vec{r} \times \vec{F}$$

$$\vec{r} \times \vec{F} = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ 2 & 5 & -7 \\ 3 & 4 & -5 \end{vmatrix}$$

$$= \hat{i}(3) - \hat{j}(11) + \hat{k}(-7)$$

$$= 3\hat{i} - 11\hat{j} - 7\hat{k}$$

$$\text{magnitude} = |\vec{r} \times \vec{F}|$$

$$= \sqrt{9 + 121 + 49} = \sqrt{179}$$

$$\text{Direction Cosines} = \left(\frac{x}{r}; \frac{y}{r}; \frac{z}{r} \right)$$

$$= \left(\frac{3}{\sqrt{179}}; \frac{-11}{\sqrt{179}}; \frac{-7}{\sqrt{179}} \right)$$

12. Force of magnitudes $5\sqrt{2}$ and $10\sqrt{2}$ units acting in the directions $3\hat{i} + 4\hat{j} + 5\hat{k}$ and $10\hat{i} + 6\hat{j} - 8\hat{k}$, respectively, act on a particle which is displaced from the point with position vector $4\hat{i} - 3\hat{j} - 2\hat{k}$ to the point with position vector $6\hat{i} + \hat{j} - 3\hat{k}$. Find the work done by the forces. 3M

$$\text{Unit force} = \frac{3\hat{i} + 4\hat{j} + 5\hat{k}}{\sqrt{9+16+25}}$$

$$= \frac{3\hat{i} + 4\hat{j} + 5\hat{k}}{\sqrt{50}}$$

$$= \frac{1}{5\sqrt{2}}(3\hat{i} + 4\hat{j} + 5\hat{k})$$

$$5\sqrt{2} \text{ unit force} = \vec{F}_1$$

$$= 5\sqrt{2} \times \frac{1}{5\sqrt{2}}(3\hat{i} + 4\hat{j} + 5\hat{k})$$

$$= 3\hat{i} + 4\hat{j} + 5\hat{k}$$

$$\text{unit force} = \frac{10\hat{i} + 6\hat{j} - 8\hat{k}}{\sqrt{100+36+64}}$$

$$= \frac{10\hat{i} + 6\hat{j} - 8\hat{k}}{\sqrt{200}}$$

$$= \frac{1}{10\sqrt{2}}(10\hat{i} + 6\hat{j} - 8\hat{k})$$

$$10\sqrt{2} \text{ unit force} = \vec{F}_2$$

$$= 10\sqrt{2} \times \frac{1}{10\sqrt{2}}(10\hat{i} + 6\hat{j} - 8\hat{k})$$

$$= 10\hat{i} + 6\hat{j} - 8\hat{k}$$

$$\text{Total force} = \vec{F}_1 + \vec{F}_2$$

$$= 3\hat{i} + 4\hat{j} + 5\hat{k} + 10\hat{i} + 6\hat{j} - 8\hat{k}$$

$$= 13\hat{i} + 10\hat{j} - 3\hat{k}$$

$$\vec{OA} = 4\hat{i} - 3\hat{j} - 2\hat{k};$$

$$\vec{OB} = 6\hat{i} + \hat{j} - 3\hat{k}$$

$$\vec{d} = \vec{OB} - \vec{OA} = 2\hat{i} + 4\hat{j} - \hat{k}$$

$$\text{work done} = W = \vec{F} \cdot \vec{d}$$

$$= (13\hat{i} + 10\hat{j} - 3\hat{k}) \cdot (2\hat{i} + 4\hat{j} - \hat{k})$$

$$= 26 + 40 + 3$$

$$= \mathbf{69 \text{ units}}$$

14. Find the torque of the resultant of the three forces represented by $-3\hat{i} + 6\hat{j} - 3\hat{k}$, $4\hat{i} - 10\hat{j} + 12\hat{k}$ and $4\hat{i} + 7\hat{j}$ acting at the point with position vector $8\hat{i} - 6\hat{j} - 4\hat{k}$, about the point with position vector $18\hat{i} + 3\hat{j} - 9\hat{k}$. 3M

$$\text{Total force} = \vec{F} = -3\hat{i} + 6\hat{j} - 3\hat{k} + 4\hat{i} - 10\hat{j} + 12\hat{k} + 4\hat{i} + 7\hat{j} = 5\hat{i} + 3\hat{j} + 9\hat{k}$$

$$\text{position vector of through the point} = \vec{OP} = 8\hat{i} - 6\hat{j} - 4\hat{k}$$

$$\text{position vector about the point} = \vec{OA} = 18\hat{i} + 3\hat{j} - 9\hat{k}$$

$$\vec{r} = \vec{OP} - \vec{OA} = -10\hat{i} - 9\hat{j} + 5\hat{k}$$

$$\text{Torque} = \vec{\tau} = \vec{r} \times \vec{F}$$

$$\begin{aligned} \vec{r} \times \vec{F} &= \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ -10 & -9 & 5 \\ 5 & 3 & 9 \end{vmatrix} \\ &= \hat{i}(-81 - 15) - \hat{j}(-90 - 25) + \hat{k}(-30 + 45) \\ &= \hat{i}(-96) - \hat{j}(-115) + \hat{k}(15) \\ \vec{r} \times \vec{F} &= -96\hat{i} + 115\hat{j} + 15\hat{k} \end{aligned}$$

EXERCISE 6.2

Concept Corner

Scalar triple product (or) Box product (or) mixed product:

$\vec{a} \cdot (\vec{b} \times \vec{c})$ (or) $(\vec{a} \times \vec{b}) \cdot \vec{c}$ is defined as scalar triple product of $\vec{a}$, $\vec{b}$, $\vec{c}$ and it is denoted by $[\vec{a} \ \vec{b} \ \vec{c}]$.

$$\therefore [\vec{a} \ \vec{b} \ \vec{c}] = \vec{a} \cdot (\vec{b} \times \vec{c})$$

$$\text{Similarly, } [\vec{b} \ \vec{c} \ \vec{a}] = \vec{b} \cdot (\vec{c} \times \vec{a})$$

$$[\vec{c} \ \vec{a} \ \vec{b}] = \vec{c} \cdot (\vec{a} \times \vec{b})$$

Calculating Formula:

If $\vec{a} = a_1\hat{i} + a_2\hat{j} + a_3\hat{k}$; $\vec{b} = b_1\hat{i} + b_2\hat{j} + b_3\hat{k}$; $\vec{c} = c_1\hat{i} + c_2\hat{j} + c_3\hat{k}$ then

$$[\vec{a} \ \vec{b} \ \vec{c}] = \vec{a} \cdot (\vec{b} \times \vec{c}) = \begin{vmatrix} a_1 & a_2 & a_3 \\ b_1 & b_2 & b_3 \\ c_1 & c_2 & c_3 \end{vmatrix}$$

Property of Scalar triple product:

$$1) \vec{a} \cdot (\vec{b} \times \vec{c}) = (\vec{a} \times \vec{b}) \cdot \vec{c} \text{ i.e., dot and cross are interchangeable.}$$

$$2) \text{ If } \vec{a}, \vec{b}, \vec{c} \text{ are in right handed system (or) anti clockwise then}$$

$$\vec{a} \cdot (\vec{b} \times \vec{c}) = \vec{b} \cdot (\vec{c} \times \vec{a}) = \vec{c} \cdot (\vec{a} \times \vec{b})$$

$$[\vec{a} \ \vec{b} \ \vec{c}] = [\vec{b} \ \vec{c} \ \vec{a}] = [\vec{c} \ \vec{a} \ \vec{b}]$$

$$3) \text{ If } \vec{a}, \vec{b}, \vec{c} \text{ are in left handed system (or) clockwise; then}$$

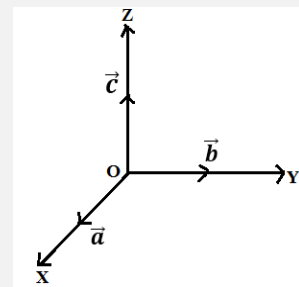
$$[\vec{a} \ \vec{c} \ \vec{b}] = -[\vec{a} \ \vec{b} \ \vec{c}] \quad \text{and} \quad [\vec{c} \ \vec{b} \ \vec{a}] = -[\vec{b} \ \vec{c} \ \vec{a}]$$

$$4) \text{ If any two vectors of Scalar triple product are same then its value is zero}$$

$$[\vec{a} \ \vec{a} \ \vec{c}] = 0; [\vec{a} \ \vec{b} \ \vec{b}] = 0$$



Video class



5) If any two vectors of Scalar triple product are parallel; then its value is zero

Let $\vec{a}$ $\vec{b}$ are parallel, $\vec{b} = \lambda \vec{a}$

$$\begin{aligned} [\vec{a} \vec{b} \vec{c}] &= [\vec{a} \lambda \vec{a} \vec{c}] \\ &= \lambda [\vec{a} \vec{a} \vec{c}] \\ &= \lambda (0) \end{aligned}$$

$$\therefore [\vec{a} \vec{b} \vec{c}] = 0$$

Formula:

- 1) The condition for $\vec{a}$, $\vec{b}$, $\vec{c}$ to be coplanar is $[\vec{a} \vec{b} \vec{c}] = 0$
- 2) The other condition for coplanar of $\vec{a}$, $\vec{b}$, $\vec{c}$ is $\vec{c} = x\vec{a} + y\vec{b}$ x, y are scalars.
- 3) The volume of the parallelopiped having $\vec{a}$; $\vec{b}$; $\vec{c}$ as co-terminus edges is $V = [\vec{a} \vec{b} \vec{c}]$

1. If $\vec{a} = \hat{i} - 2\hat{j} + 3\hat{k}$; $\vec{b} = 2\hat{i} + \hat{j} - 2\hat{k}$; $\vec{c} = 3\hat{i} + 2\hat{j} + \hat{k}$ Find $\vec{a} \cdot (\vec{b} \times \vec{c})$

2M

$$\begin{aligned} \vec{a} \cdot (\vec{b} \times \vec{c}) &= \begin{vmatrix} 1 & -2 & 3 \\ 2 & 1 & -2 \\ 3 & 2 & 1 \end{vmatrix} \\ &= 1(5) + 2(8) + 3(1) = 5 + 16 + 3 = 16 + 8 = 24 \end{aligned}$$

2. Find the volume of the parallelopiped whose coterminous edges are represented by the vectors $-6\hat{i} + 14\hat{j} + 10\hat{k}$, $14\hat{i} - 10\hat{j} - 6\hat{k}$ and $2\hat{i} + 4\hat{j} - 2\hat{k}$

2M

Let $\vec{a} = -6\hat{i} + 14\hat{j} + 10\hat{k}$; $\vec{b} = 14\hat{i} - 10\hat{j} - 6\hat{k}$; $\vec{c} = 2\hat{i} + 4\hat{j} - 2\hat{k}$

volume of parallelepiped $= V = [\vec{a} \vec{b} \vec{c}]$

$$\begin{aligned} &= \begin{vmatrix} -6 & 14 & 10 \\ 14 & -10 & -6 \\ 2 & 4 & -2 \end{vmatrix} \\ &= -6(44) - 14(-16) + 10(76) \\ &= -40 + 760 \\ &= 720 \text{ Cubic units} \end{aligned}$$

3. The volume of the parallelopiped whose coterminous edges are $7\hat{i} + \lambda\hat{j} - 3\hat{k}$, $\hat{i} + 2\hat{j} - \hat{k}$, $-3\hat{i} + 7\hat{j} + 5\hat{k}$ is 90 cubic units. Find the value of λ .

JUL-24

2M

Let $\vec{a} = 7\hat{i} + \lambda\hat{j} - 3\hat{k}$; $\vec{b} = \hat{i} + 2\hat{j} - \hat{k}$; $\vec{c} = -3\hat{i} + 7\hat{j} + 5\hat{k}$

Given Volume $= [\vec{a} \vec{b} \vec{c}] = 90$

$$\begin{vmatrix} 7 & \lambda & -3 \\ 1 & 2 & -1 \\ -3 & 7 & 5 \end{vmatrix} = 90$$

$$7(17) - \lambda(2) - 3(13) = 90$$

$$119 - 2\lambda - 39 = 90$$

$$80 - 2\lambda = 90$$

$$80 - 2\lambda = 90$$

$$-2\lambda = 90 - 80 \Rightarrow 10$$

$$\lambda = -5$$

$$80 - 2\lambda = 90$$

$$2\lambda = 90 + 80$$

$$2\lambda = 170$$

$$\lambda = 85$$

4. If $\vec{a}$, $\vec{b}$, $\vec{c}$ are three non-coplanar vectors represented by concurrent edges of a parallelepiped of volume 4 cubic units, find the value of $(\vec{a} + \vec{b}) \cdot (\vec{b} \times \vec{c}) + (\vec{b} + \vec{c}) \cdot (\vec{c} \times \vec{a}) + (\vec{c} + \vec{a}) \cdot (\vec{a} \times \vec{b})$

$$\text{Given Volume} = [\vec{a} \ \vec{b} \ \vec{c}] = |[\vec{a} \ \vec{b} \ \vec{c}]| = \pm 4$$

3M

$$\begin{aligned} & (\vec{a} + \vec{b}) \cdot (\vec{b} \times \vec{c}) + (\vec{b} + \vec{c}) \cdot (\vec{c} \times \vec{a}) + (\vec{c} + \vec{a}) \cdot (\vec{a} \times \vec{b}) \\ &= \vec{a} \cdot (\vec{b} \times \vec{c}) + \vec{b} \cdot (\vec{b} \times \vec{c}) + \vec{b} \cdot (\vec{c} \times \vec{a}) + \vec{c} \cdot (\vec{c} \times \vec{a}) + \vec{c} \cdot (\vec{a} \times \vec{b}) + \vec{a} \cdot (\vec{a} \times \vec{b}) \\ &= [\vec{a} \ \vec{b} \ \vec{c}] + 0 + [\vec{b} \ \vec{c} \ \vec{a}] + 0 + [\vec{c} \ \vec{a} \ \vec{b}] + 0 \\ &= [\vec{a} \ \vec{b} \ \vec{c}] + [\vec{a} \ \vec{b} \ \vec{c}] + [\vec{a} \ \vec{b} \ \vec{c}] \\ &= 3[\vec{a} \ \vec{b} \ \vec{c}] \\ &= 3(\pm 4) \\ &= \pm 12 \end{aligned}$$

5. Find the altitude of a parallelepiped determined by the vectors $\vec{a} = -2\hat{i} + 5\hat{j} + 3\hat{k}$, $\vec{b} = \hat{i} + 3\hat{j} - 2\hat{k}$ and $\vec{c} = -3\hat{i} + \hat{j} + 4\hat{k}$ if the base is taken as the parallelogram determined by $\vec{b}$ and $\vec{c}$

$$\text{volume} = [\vec{a} \ \vec{b} \ \vec{c}] = (\text{Area of base}) (\text{Height})$$

$$[\vec{a} \ \vec{b} \ \vec{c}] = |\vec{b} \times \vec{c}| (\text{Height})$$

$$\text{Height} = \frac{[\vec{a} \ \vec{b} \ \vec{c}]}{|\vec{b} \times \vec{c}|}$$

$$[\vec{a} \ \vec{b} \ \vec{c}] = \begin{vmatrix} -2 & 5 & 3 \\ 1 & 3 & -2 \\ -3 & 1 & 4 \end{vmatrix}$$

$$= -2(14) - 5(-2) + 3(10)$$

$$= -28 + 10 + 30$$

$$= 40 - 28$$

$$= 12$$

$$\vec{b} \times \vec{c} = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ 1 & 3 & -2 \\ -3 & 1 & 4 \end{vmatrix}$$

PTA-2

3M

$$= \hat{i}(14) - \hat{j}(-2) + \hat{k}(10)$$

$$= 14\hat{i} + 2\hat{j} + 10\hat{k}$$

$$|\vec{b} \times \vec{c}| = \sqrt{196 + 4 + 100} = \sqrt{300}$$

$$= \sqrt{100 \times 3} = 10\sqrt{3}$$

$$\text{Height} = \frac{12}{10\sqrt{3}} = \frac{6}{5\sqrt{3}} = \frac{2 \times 3}{5\sqrt{3}} = \frac{2 \times \sqrt{3} \times \sqrt{3}}{5\sqrt{3}}$$

$$\text{Height} = \frac{2\sqrt{3}}{5} \text{ units}$$

6. Determine whether the three vectors $2\hat{i} + 3\hat{j} + \hat{k}$, $\hat{i} - 2\hat{j} + 2\hat{k}$ and $3\hat{i} + \hat{j} + 3\hat{k}$ are coplanar.

$$\text{Let } \vec{a} = 2\hat{i} + 3\hat{j} + \hat{k}; \quad \vec{b} = \hat{i} - 2\hat{j} + 2\hat{k}; \quad \vec{c} = 3\hat{i} + \hat{j} + 3\hat{k}$$

JUL-21

2M

$$[\vec{a} \ \vec{b} \ \vec{c}] = \begin{vmatrix} 2 & 3 & 1 \\ 1 & -2 & 2 \\ 3 & 1 & 3 \end{vmatrix}$$

$$= 2(-8) - 3(-3) + 1(7)$$

$$= -16 + 9 + 7$$

$$= 16 - 16$$

$$= 0$$

$\therefore$ The given vectors are coplanar.

7. Let $\vec{a} = \hat{i} + \hat{j} + \hat{k}$, $\vec{b} = \hat{i}$ and $\vec{c} = c_1\hat{i} + c_2\hat{j} + c_3\hat{k}$. If $c_1 = 1$ and $c_2 = 2$, find c_3 such that $\vec{a}$, $\vec{b}$ and $\vec{c}$ are coplanar.

3M

$$\vec{a} = \hat{i} + \hat{j} + \hat{k}, \quad \vec{b} = \hat{i}$$

$$c_1 = 1; c_2 = 2 \quad \therefore \vec{c} = c_1\hat{i} + c_2\hat{j} + c_3\hat{k} = (1)\hat{i} + (2)\hat{j} + c_3\hat{k} \\ = \hat{i} + 2\hat{j} + c_3\hat{k}$$

$$\therefore \vec{a}; \vec{b}; \vec{c} \text{ are coplanar } [\vec{a} \ \vec{b} \ \vec{c}] = 0$$

$$\begin{vmatrix} 1 & 1 & 1 \\ 1 & 0 & 0 \\ 1 & 2 & c_3 \end{vmatrix} = 0$$

expand along R_3

$$1(0) - 2(-1) + c_3(-1) = 0$$

$$2 - c_3 = 0$$

$$c_3 = 2$$

8. If $\vec{a} = \hat{i} - \hat{k}$, $\vec{b} = x\hat{i} + \hat{j} + (1 - x)\hat{k}$, $\vec{c} = y\hat{i} + x\hat{j} + (1 + x - y)\hat{k}$, show that $[\vec{a}, \vec{b}, \vec{c}]$ depends on neither x nor y .

3M

$$\text{Let } \vec{a} = \hat{i} - \hat{k}$$

$$\vec{b} = x\hat{i} + \hat{j} + (1 - x)\hat{k}$$

$$\vec{c} = y\hat{i} + x\hat{j} + (1 + x - y)\hat{k}$$

$$[\vec{a} \ \vec{b} \ \vec{c}] = \begin{vmatrix} 1 & 0 & -1 \\ x & 1 & 1 - x \\ y & x & 1 + x - y \end{vmatrix} \\ = 1(1 + x - y - x + x^2) - 0 + (-1)(x^2 - y) \\ = 1(1 - y + x^2) - 1(x^2 - y) \\ = 1 - y + x^2 - x^2 + y$$

$$[\vec{a} \ \vec{b} \ \vec{c}] = 1$$

$$\therefore [\vec{a} \ \vec{b} \ \vec{c}] \text{ depends on neither } x \text{ nor } y.$$

9. If the vectors $a\hat{i} + a\hat{j} + c\hat{k}$, $\hat{i} + \hat{k}$ and $c\hat{i} + c\hat{j} + b\hat{k}$ are coplanar, prove that c is the geometric mean of a and b

3M

$$\text{Let } \vec{x} = a\hat{i} + a\hat{j} + c\hat{k}, \quad \vec{y} = \hat{i} + \hat{k} \quad \vec{z} = c\hat{i} + c\hat{j} + b\hat{k}$$

$$\text{They are coplanar} \Rightarrow [\vec{x} \ \vec{y} \ \vec{z}] = 0$$

$$\begin{vmatrix} a & a & c \\ 1 & 0 & 1 \\ c & c & b \end{vmatrix} = 0$$

$$a(-c) - a(b - c) + c(c) = 0$$

$$-ac - ab + ac + c^2 = 0$$

$$c^2 = ab$$

$$c = \sqrt{ab}$$

$$\therefore c \text{ is the geometric mean of } a \text{ and } b.$$

10. Let $\vec{a}$, $\vec{b}$, $\vec{c}$ be three non-zero vectors such that $\vec{c}$ is a unit vector perpendicular to both $\vec{a}$ and $\vec{b}$. If the angle between $\vec{a}$ and $\vec{b}$ is $\frac{\pi}{6}$, show that $[\vec{a} \ \vec{b} \ \vec{c}]^2 = \frac{1}{4} |\vec{a}|^2 |\vec{b}|^2$

3M

Given $\vec{c}$ is perpendicular to both $\vec{a}$, $\vec{b}$

But $\vec{a} \times \vec{b}$ is perpendicular both $\vec{a}$, $\vec{b}$

$$\vec{c} = \vec{a} \times \vec{b} \Rightarrow |\vec{c}| = |\vec{a} \times \vec{b}| \Rightarrow |\vec{c}| = 1$$

$$[\vec{a} \ \vec{b} \ \vec{c}] = (\vec{a} \times \vec{b}) \cdot \vec{c}$$

$$|[\vec{a} \ \vec{b} \ \vec{c}]| = |\vec{a}| |\vec{b} \times \vec{c}|$$

$$= |\vec{a}| |\vec{b}| |\vec{c}| \sin\left(\frac{\pi}{6}\right) = |\vec{a}| |\vec{b}| |\vec{c}| \left(\frac{1}{2}\right)$$

Squaring on both sides, $|[\vec{a} \ \vec{b} \ \vec{c}]|^2 = \left(|\vec{a}| |\vec{b}| |\vec{c}| \left(\frac{1}{2}\right)\right)^2$

$$[\vec{a} \ \vec{b} \ \vec{c}]^2 = |\vec{a}|^2 |\vec{b}|^2 |\vec{c}|^2 \left(\frac{1}{4}\right) \quad \because |\vec{c}| = 1$$

$$[\vec{a} \ \vec{b} \ \vec{c}]^2 = \frac{1}{4} |\vec{a}|^2 |\vec{b}|^2$$

$$\text{L.H.S} = \text{R.H.S}$$

EXERCISE 6.3



Video class

Concept Corner

Vector triple product:

$\vec{a} \times (\vec{b} \times \vec{c})$ (or) $(\vec{a} \times \vec{b}) \times \vec{c}$ is defined as the vector triple product of $\vec{a}$; $\vec{b}$; $\vec{c}$

$$\vec{a} \times (\vec{b} \times \vec{c}) = (\vec{a} \cdot \vec{c})\vec{b} - (\vec{a} \cdot \vec{b})\vec{c}$$

$$(\vec{a} \times \vec{b}) \times \vec{c} = (\vec{a} \cdot \vec{c})\vec{b} - (\vec{b} \cdot \vec{c})\vec{a}$$

Vector triple product is not associative. $\vec{a} \times (\vec{b} \times \vec{c}) \neq (\vec{a} \times \vec{b}) \times \vec{c}$

Note: (1) $\vec{a} \times (\vec{b} \times \vec{c})$ is coplanar with $\vec{b}$; $\vec{c}$ (2) $(\vec{a} \times \vec{b}) \times \vec{c}$ is coplanar with $\vec{a}$; $\vec{b}$

Jacobi's Identity: $\vec{a} \times (\vec{b} \times \vec{c}) + \vec{b} \times (\vec{c} \times \vec{a}) + \vec{c} \times (\vec{a} \times \vec{b}) = \vec{0}$

Scalar product of four vectors: (Lagrange's Identity):

$(\vec{a} \times \vec{b}) \cdot (\vec{c} \times \vec{d})$ is defined as scalar product of four vectors $\vec{a}$, $\vec{b}$, $\vec{c}$, $\vec{d}$.

$$(\vec{a} \times \vec{b}) \cdot (\vec{c} \times \vec{d}) = \begin{vmatrix} \vec{a} \cdot \vec{c} & \vec{a} \cdot \vec{d} \\ \vec{b} \cdot \vec{c} & \vec{b} \cdot \vec{d} \end{vmatrix}$$

Vector product of four vectors:

$(\vec{a} \times \vec{b}) \times (\vec{c} \times \vec{d})$ is defined as vector product of four vectors $\vec{a}$, $\vec{b}$, $\vec{c}$, $\vec{d}$.

$$(1) (\vec{a} \times \vec{b}) \times (\vec{c} \times \vec{d}) = [\vec{a} \ \vec{c} \ \vec{d}]\vec{b} - [\vec{b} \ \vec{c} \ \vec{d}]\vec{a}$$

$$(2) (\vec{a} \times \vec{b}) \times (\vec{c} \times \vec{d}) = [\vec{a} \ \vec{b} \ \vec{d}]\vec{c} - [\vec{a} \ \vec{b} \ \vec{c}]\vec{d}$$

Formula:

(1) $(\vec{a} \times \vec{b}) \times (\vec{c} \times \vec{d})$ is coplanar with $\vec{a}$, $\vec{b}$ and also $(\vec{a} \times \vec{b}) \times (\vec{c} \times \vec{d})$ is coplanar with $\vec{c}$, $\vec{d}$

(2) The condition for $\vec{a}$, $\vec{b}$, $\vec{c}$, $\vec{d}$ to be coplanar $(\vec{a} \times \vec{b}) \times (\vec{c} \times \vec{d}) = \vec{0}$

1. If $\vec{a} = \hat{i} - 2\hat{j} + 3\hat{k}$; $\vec{b} = 2\hat{i} + \hat{j} - 2\hat{k}$; $\vec{c} = 3\hat{i} + 2\hat{j} + \hat{k}$, find (i) $(\vec{a} \times \vec{b}) \times \vec{c}$ (ii) $\vec{a} \times (\vec{b} \times \vec{c})$

Given: $\vec{a} = \hat{i} - 2\hat{j} + 3\hat{k}$; $\vec{b} = 2\hat{i} + \hat{j} - 2\hat{k}$; $\vec{c} = 3\hat{i} + 2\hat{j} + \hat{k}$.

3M

(i) $(\vec{a} \times \vec{b}) \times \vec{c}$

$$\begin{aligned}\vec{a} \times \vec{b} &= \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ 1 & -2 & 3 \\ 2 & 1 & -2 \end{vmatrix} \\ &= \hat{i}(1) - \hat{j}(-8) + \hat{k}(5) \\ &= \vec{i} + 8\vec{j} + 5\vec{k} \\ (\vec{a} \times \vec{b}) \times \vec{c} &= \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ 1 & 8 & 5 \\ 3 & 2 & 1 \end{vmatrix} \\ &= \hat{i}(-2) - \hat{j}(-14) + \hat{k}(-22) \\ &= -2\hat{i} + 14\hat{j} - 22\hat{k}\end{aligned}$$

(ii) $\vec{a} \times (\vec{b} \times \vec{c})$

$$\begin{aligned}\vec{b} \times \vec{c} &= \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ 2 & 1 & -2 \\ 3 & 2 & 1 \end{vmatrix} \\ &= \hat{i}(5) - \hat{j}(8) + \hat{k}(1) \\ &= 5\vec{i} - 8\vec{j} + \vec{k} \\ \vec{a} \times (\vec{b} \times \vec{c}) &= \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ 1 & -2 & 3 \\ 5 & -8 & 1 \end{vmatrix} \\ &= \hat{i}(22) - \hat{j}(-14) + \hat{k}(2) \\ &= 22\hat{i} + 14\hat{j} + 2\hat{k}\end{aligned}$$

2. For any vector $\vec{a}$, prove that $\hat{i} \times (\vec{a} \times \hat{i}) + \hat{j} \times (\vec{a} \times \hat{j}) + \hat{k} \times (\vec{a} \times \hat{k}) = 2\vec{a}$

3M

$$\begin{aligned}\text{L.H.S} &= \hat{i} \times (\vec{a} \times \hat{i}) + \hat{j} \times (\vec{a} \times \hat{j}) + \hat{k} \times (\vec{a} \times \hat{k}) \\ &= (\hat{i} \cdot \hat{i})\vec{a} - (\hat{i} \cdot \vec{a})\hat{i} + (\hat{j} \cdot \hat{j})\vec{a} - (\hat{j} \cdot \vec{a})\hat{j} + (\hat{k} \cdot \hat{k})\vec{a} - (\hat{k} \cdot \vec{a})\hat{k} \\ &= (1)\vec{a} + (1)\vec{a} + (1)\vec{a} - [(\hat{i} \cdot \vec{a})\hat{i} + (\hat{j} \cdot \vec{a})\hat{j} + (\hat{k} \cdot \vec{a})\hat{k}] \\ &= 3\vec{a} - [(\hat{i} \cdot \vec{a})\hat{i} + (\hat{j} \cdot \vec{a})\hat{j} + (\hat{k} \cdot \vec{a})\hat{k}] \quad [\because \hat{i} \cdot \hat{i} = 1, \hat{j} \cdot \hat{j} = 1, \hat{k} \cdot \hat{k} = 1]\end{aligned}$$

Let $\vec{a} = x\hat{i} + y\hat{j} + z\hat{k}$

$$\hat{i} \cdot \vec{a} = (1)x = x$$

$$\hat{j} \cdot \vec{a} = (1)y = y$$

$$\hat{k} \cdot \vec{a} = (1)z = z$$

$$\text{L.H.S} = 3\vec{a} - [x\hat{i} + y\hat{j} + z\hat{k}]$$

$$\text{L.H.S} = 3\vec{a} - \vec{a} = 2\vec{a} = \text{R.H.S}$$

3. Prove that $[\vec{a} - \vec{b}, \vec{b} - \vec{c}, \vec{c} - \vec{a}] = 0$ (or) prove $\vec{a} - \vec{b}$, $\vec{b} - \vec{c}$, $\vec{c} - \vec{a}$ are coplanar.

3M

$$\begin{aligned}\text{L.H.S} &= [\vec{a} - \vec{b}, \vec{b} - \vec{c}, \vec{c} - \vec{a}] \\ &= (\vec{a} - \vec{b}) \cdot \{(\vec{b} - \vec{c}) \times (\vec{c} - \vec{a})\} \\ &= (\vec{a} - \vec{b}) \cdot \{\vec{b} \times \vec{c} - \vec{b} \times \vec{a} - \vec{c} \times \vec{c} + \vec{c} \times \vec{a}\} \\ &= (\vec{a} - \vec{b}) \cdot \{\vec{b} \times \vec{c} - \vec{b} \times \vec{a} + \vec{c} \times \vec{a}\} \quad [\because \vec{c} \times \vec{c} = \vec{0}] \\ &= \vec{a} \cdot (\vec{b} \times \vec{c}) - \vec{a} \cdot (\vec{b} \times \vec{a}) + \vec{a} \cdot (\vec{c} \times \vec{a}) - \vec{b} \cdot (\vec{b} \times \vec{c}) + \vec{b} \cdot (\vec{b} \times \vec{a}) - \vec{b} \cdot (\vec{c} \times \vec{a}) \\ &= [\vec{a} \vec{b} \vec{c}] - 0 + 0 - 0 + 0 - [\vec{b} \vec{c} \vec{a}]\end{aligned}$$

$$\text{L.H.S} = [\vec{a} \vec{b} \vec{c}] - [\vec{a} \vec{b} \vec{c}] = 0 = \text{R.H.S}$$

$$\therefore [\vec{a} - \vec{b}, \vec{b} - \vec{c}, \vec{c} - \vec{a}] = 0$$

similarly we can prove $[\vec{b} - \vec{c}, \vec{c} - \vec{a}, \vec{a} - \vec{b}] = 0$, $[\vec{c} - \vec{a}, \vec{a} - \vec{b}, \vec{b} - \vec{c}] = 0$.

APR-24

4. If $\vec{a} = 2\hat{i} + 3\hat{j} - \hat{k}$, $\vec{b} = 3\hat{i} + 5\hat{j} + 2\hat{k}$, $\vec{c} = -\hat{i} - 2\hat{j} + 3\hat{k}$, verify that

5M

(i) $(\vec{a} \times \vec{b}) \times \vec{c} = (\vec{a} \cdot \vec{c})\vec{b} - (\vec{b} \cdot \vec{c})\vec{a}$ (JUL-23) (ii) $\vec{a} \times (\vec{b} \times \vec{c}) = (\vec{a} \cdot \vec{c})\vec{b} - (\vec{a} \cdot \vec{b})\vec{c}$

$$\vec{a} = 2\hat{i} + 3\hat{j} - \hat{k}, \quad \vec{b} = 3\hat{i} + 5\hat{j} + 2\hat{k}, \quad \vec{c} = -\hat{i} - 2\hat{j} + 3\hat{k}$$

$$(i) \quad \vec{a} \times \vec{b} = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ 2 & 3 & -1 \\ 3 & 5 & 2 \end{vmatrix}$$

$$= \hat{i}(11) - \hat{j}(7) + \hat{k}(1) = 11\hat{i} - 7\hat{j} + \hat{k}$$

$$\text{L.H.S} = (\vec{a} \times \vec{b}) \times \vec{c}$$

$$= \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ 11 & -7 & 1 \\ -1 & -2 & 3 \end{vmatrix}$$

$$= \hat{i}(-19) - \hat{j}(34) + \hat{k}(-29)$$

$$= -19\hat{i} - 34\hat{j} - 29\hat{k} \quad \dots\dots\dots (1)$$

$$\vec{a} \cdot \vec{c} = (2\hat{i} + 3\hat{j} - \hat{k}) \cdot (-\hat{i} - 2\hat{j} + 3\hat{k})$$

$$\vec{a} \cdot \vec{c} = -2 - 6 - 3 = -11$$

$$\vec{b} \cdot \vec{c} = (3\hat{i} + 5\hat{j} + 2\hat{k}) \cdot (-\hat{i} - 2\hat{j} + 3\hat{k})$$

$$= -3 - 10 + 6 = -7$$

$$\text{R.H.S} = (-11)\vec{b} - (-7)\vec{a}$$

$$= (-11)(3\hat{i} + 5\hat{j} + 2\hat{k}) - (-7)(2\hat{i} + 3\hat{j} - \hat{k})$$

$$= -33\hat{i} - 55\hat{j} - 22\hat{k} + 14\hat{i} + 21\hat{j} - 7\hat{k}$$

$$= -19\hat{i} - 34\hat{j} - 29\hat{k} \quad \dots\dots\dots (2)$$

$$\text{From (1), (2)} \quad (\vec{a} \times \vec{b}) \times \vec{c} = (\vec{a} \cdot \vec{c})\vec{b} - (\vec{b} \cdot \vec{c})\vec{a}$$

$$(ii) \quad \vec{b} \times \vec{c} = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ 3 & 5 & 2 \\ -1 & -2 & 3 \end{vmatrix}$$

$$= \hat{i}(19) - \hat{j}(11) + \hat{k}(-1)$$

$$= 19\hat{i} - 11\hat{j} - \hat{k}$$

$$\vec{a} \times (\vec{b} \times \vec{c}) = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ 2 & 3 & -1 \\ 19 & -11 & -1 \end{vmatrix}$$

$$= \hat{i}(-14) - \hat{j}(17) + \hat{k}(-79) \quad \dots\dots\dots (3)$$

$$\vec{a} \cdot \vec{c} = -2 - 6 - 3 = -11$$

$$\vec{a} \cdot \vec{b} = 6 + 15 - 2 = 19$$

$$\text{R.H.S} = (-11)\vec{b} - (19)\vec{c}$$

$$= (-11)(3\hat{i} + 5\hat{j} + 2\hat{k}) -$$

$$(19)(-\hat{i} - 2\hat{j} + 3\hat{k})$$

$$= -33\hat{i} - 55\hat{j} - 22\hat{k} + 19\hat{i} + 38\hat{j} - 57\hat{k}$$

$$= -14\hat{i} - 17\hat{j} - 79\hat{k} \quad \dots\dots\dots (4)$$

From (3), (4)

$$\vec{a} \times (\vec{b} \times \vec{c}) = (\vec{a} \cdot \vec{c})\vec{b} - (\vec{a} \cdot \vec{b})\vec{c}$$

5. $\vec{a} = 2\hat{i} + 3\hat{j} - \hat{k}$, $\vec{b} = -\hat{i} + 2\hat{j} - 4\hat{k}$, $\vec{c} = \hat{i} + \hat{j} + \hat{k}$ then, find the value of $(\vec{a} \times \vec{b}) \cdot (\vec{a} \times \vec{c})$

$$(\vec{a} \times \vec{b}) \cdot (\vec{a} \times \vec{c})$$

3M

$$\vec{a} \times \vec{b} = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ 2 & 3 & -1 \\ -1 & 2 & -4 \end{vmatrix}$$

$$= \hat{i}(-10) - \hat{j}(-9) + \hat{k}(7)$$

$$= -10\hat{i} + 9\hat{j} + 7\hat{k}$$

$$(\vec{a} \times \vec{b}) \cdot (\vec{a} \times \vec{c}) = (-10)(4) + (9)(-3) + (7)(-1)$$

$$= -40 - 27 - 7 = -74$$

$$\vec{a} \times \vec{c} = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ 2 & 3 & -1 \\ 1 & 1 & 1 \end{vmatrix}$$

$$= \hat{i}(4) - \hat{j}(3) + \hat{k}(-1)$$

$$= 4\hat{i} - 3\hat{j} - \hat{k}$$

6. If $\vec{a}$, $\vec{b}$, $\vec{c}$, $\vec{d}$ are coplanar vectors, show that $(\vec{a} \times \vec{b}) \times (\vec{c} \times \vec{d}) = \vec{0}$

3M

$$(\vec{a} \times \vec{b}) \times (\vec{c} \times \vec{d}) = [\vec{a} \ \vec{b} \ \vec{d}]\vec{c} - [\vec{a} \ \vec{b} \ \vec{c}]\vec{d}$$

$$= (0)\vec{c} - (0)\vec{d}$$

$$\therefore (\vec{a} \times \vec{b}) \times (\vec{c} \times \vec{d}) = \vec{0}$$

$$\because \vec{a}, \vec{b}, \vec{c}, \vec{d} \text{ are coplanar.} \\ [\vec{a} \ \vec{b} \ \vec{d}] = 0; [\vec{a} \ \vec{b} \ \vec{c}] = 0$$

7. If $\vec{a} = \hat{i} + 2\hat{j} + 3\hat{k}$, $\vec{b} = 2\hat{i} - \hat{j} + \hat{k}$, $\vec{c} = 3\hat{i} + 2\hat{j} + \hat{k}$ and $\vec{a} \times (\vec{b} \times \vec{c}) = l\vec{a} + m\vec{b} + n\vec{c}$, find the values of l, m, n .

PTA-5

3M

$$\text{Given } \vec{a} \times (\vec{b} \times \vec{c}) = l\vec{a} + m\vec{b} + n\vec{c}$$

$$(\vec{a} \cdot \vec{c})\vec{b} - (\vec{a} \cdot \vec{b})\vec{c} = l\vec{a} + m\vec{b} + n\vec{c}$$

$$\text{i.e. } (0)\vec{a} + (\vec{a} \cdot \vec{c})\vec{b} + (-(\vec{a} \cdot \vec{b}))\vec{c} = l\vec{a} + m\vec{b} + n\vec{c}$$

Equating likely components; we get $l = 0$

$$m = \vec{a} \cdot \vec{c} = 3 + 4 + 3 = 10$$

$$n = -(\vec{a} \cdot \vec{b}) = -(2 - 2 + 3) = -3$$

8. If $\hat{a}, \hat{b}, \hat{c}$ are three unit vectors such that $\hat{b}$ and $\hat{c}$ are non-parallel and $\hat{a} \times (\hat{b} \times \hat{c}) = \frac{1}{2} \hat{b}$, find the angle between $\hat{a}$ and $\hat{c}$.

MDL, JUL-20

2M

$$\vec{a} \times (\vec{b} \times \vec{c}) = \frac{1}{2} \vec{b}$$

$$(\vec{a} \cdot \vec{c})\vec{b} - (\vec{a} \cdot \vec{b})\vec{c} = \frac{1}{2} \vec{b} - (0)\vec{c}$$

Equating the likely components, we get

$$\vec{a} \cdot \vec{c} = \frac{1}{2}$$

$$\text{ac } \cos \theta = \frac{1}{2}$$

$$(1)(1) \cos \theta = \frac{1}{2}$$

$$\cos \theta = \frac{1}{2}$$

$$\theta = \frac{\pi}{3}$$

$$\Rightarrow \cos \theta = \cos \frac{\pi}{3}$$

$$\vec{a} \cdot \vec{b} = 0$$

$$\vec{a} \perp \vec{b}$$

The angle between $\hat{a}$ and $\hat{c}$ is $\theta = \frac{\pi}{3}$

EXERCISE 6.4



Video class

Concept Corner

A Point on the Straight line:

I a) **Parametric Form:** The Vector Equation of a straight line passing through a fixed point A with position vector $\vec{a}$ and parallel to $\vec{b}$ is $\vec{r} = \vec{a} + t\vec{b}$ $t \in R$.

b) **Non-Parametric Form:** The Vector Equation of a line passing through the point $A(\vec{a})$ and parallel to $\vec{b}$ is $(\vec{r} - \vec{a}) \times \vec{b} = 0$

c) **Cartesian Form:** The Cartesian Equation of a line passing through point $A(x_1, y_1, z_1)$ and parallel to $\vec{b} = (b_1, b_2, b_3)$ is $\frac{x-x_1}{b_1} = \frac{y-y_1}{b_2} = \frac{z-z_1}{b_3}$

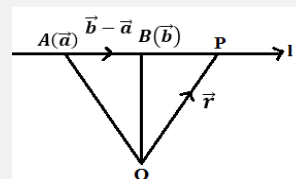
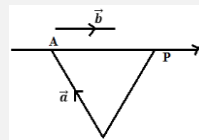
Straight line passing through two given points:

II a) **Parametric Form:** The Vector Equation of a straight line passing through two given points $A(\vec{a})$ & $B(\vec{b})$ is $\vec{r} = \vec{a} + t(\vec{b} - \vec{a})$ (or) $\vec{r} = (1-t)\vec{a} + t\vec{b}$ $t \in R$

b) **Non-Parametric Form:** The Vector Equation of a line passing through two given points $A(\vec{a})$ and $B(\vec{b})$ is $(\vec{r} - \vec{a}) \times (\vec{b} - \vec{a}) = 0$

c) **Cartesian Form:** The Cartesian Equation of a line passing through two points

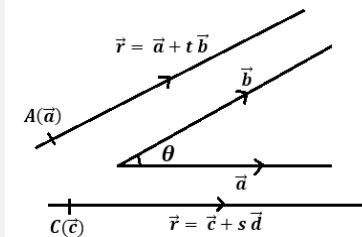
$$A(x_1, y_1, z_1) \text{ and } B(x_2, y_2, z_2) \text{ is } \frac{x-x_1}{x_2-x_1} = \frac{y-y_1}{y_2-y_1} = \frac{z-z_1}{z_2-z_1}$$



III Angle between two lines:

Angle between two lines $\vec{r} = \vec{a} + t\vec{b}$; $\vec{r} = \vec{c} + s\vec{d}$ is same as the angle between their parallel vectors $\vec{b}, \vec{d}$

$$\therefore \text{Angle between two lines is } \cos \theta = \frac{|\vec{b} \cdot \vec{d}|}{|\vec{b}| |\vec{d}|}$$



Note: (1) condition for two lines to be perpendicular is $\vec{b} \cdot \vec{d} = 0$

(2) condition for two lines to be parallel is $\vec{b} = m\vec{d}$ $\vec{d} = n\vec{b}$ m, n are scalars

b) Cartesian Form: If two lines are given in Cartesian form

$$\frac{x-x_1}{b_1} = \frac{y-y_1}{b_2} = \frac{z-z_1}{b_3} \quad \text{and} \quad \frac{x-x_2}{d_1} = \frac{y-y_2}{d_2} = \frac{z-z_2}{d_3}$$

1. Find the non-parametric form of vector equation and Cartesian equations of the straight line passing through the point with position vector $4\hat{i} + 3\hat{j} - 7\hat{k}$ and parallel to the vector $2\hat{i} - 6\hat{j} + 7\hat{k}$

$$\vec{a} = 4\hat{i} + 3\hat{j} - 7\hat{k}; \quad \vec{b} = 2\hat{i} - 6\hat{j} + 7\hat{k}$$

2M

The non-parametric form of the line $[\vec{r} - \vec{a}] \times \vec{b} = 0$

$$[\vec{r} - (4\hat{i} + 3\hat{j} - 7\hat{k})] \times (2\hat{i} - 6\hat{j} + 7\hat{k}) = 0$$

The Cartesian equation of the line $\frac{x-x_1}{b_1} = \frac{y-y_1}{b_2} = \frac{z-z_1}{b_3}$

$$(x_1, y_1, z_1) = (4, 3, -7); \quad (b_1, b_2, b_3) = (2, -6, 7)$$

$$\frac{x-4}{2} = \frac{y-3}{-6} = \frac{z+7}{7}$$

2. Find the parametric form of vector equation and Cartesian equations of the straight line passing through the point $(-2, 3, 4)$ and parallel to the straight line $\frac{x-1}{-4} = \frac{y+3}{5} = \frac{8-z}{6}$

2M

$$\text{Line } \frac{x-1}{-4} = \frac{y+3}{5} = \frac{8-z}{6}$$

$$\frac{x-1}{-4} = \frac{y+3}{5} = \frac{z-8}{-6}$$

The required line passes through the point $(-2, 3, 4)$ and parallel to $\vec{b} = (-4, 5, -6)$

$$\vec{a} = -2\hat{i} + 3\hat{j} + 4\hat{k}; \quad \vec{b} = -4\hat{i} + 5\hat{j} - 6\hat{k}$$

Vector Equation in parametric form $\vec{r} = \vec{a} + t\vec{b}$

$$\vec{r} = (-2\hat{i} + 3\hat{j} + 4\hat{k}) + t(-4\hat{i} + 5\hat{j} - 6\hat{k})$$

Cartesian Equation of the line $\frac{x-x_1}{b_1} = \frac{y-y_1}{b_2} = \frac{z-z_1}{b_3}$

$$(x_1, y_1, z_1) = (-2, 3, 4); \quad (b_1, b_2, b_3) = (-4, 5, -6)$$

$$\frac{x+2}{-4} = \frac{y-3}{5} = \frac{z-4}{-6}$$

3. Find the points where the straight line passes through (6, 7, 4) and (8, 4, 9) cuts the xz and yz planes. 5M

Equation of line joining (6, 7, 4) and (8, 4, 9)

$$\frac{x-x_1}{x_2-x_1} = \frac{y-y_1}{y_2-y_1} = \frac{z-z_1}{z_2-z_1}$$

$$\frac{x-6}{2} = \frac{y-7}{-3} = \frac{z-4}{5} = t$$

Any point on this line is $P(2t + 6, -3t + 7, 5t + 4)$

$\therefore$ The line cuts xz - plane put $y = 0$

$$-3t + 7 = 0$$

$$3t = 7$$

$$t = \frac{7}{3}$$

$$\text{Required point} = P\left(\frac{14}{3} + 6, 0, \frac{35}{3} + 4\right) = P\left(\frac{32}{3}, 0, \frac{47}{3}\right)$$

$\therefore$ The line cuts yz -plane

$$\text{put } x = 0, \quad 2t + 6 = 0$$

$$2t = -6$$

$$t = -3$$

$$\text{Required point} = Q(0, 9 + 7, -15 + 4) = Q(0, 16, -11)$$

4. Find the direction cosines of the straight line passes through the points (5, 6, 7) and (7, 9, 13). Also, find the parametric form of vector equation and Cartesian equations of the straight line passing through two given points. 3M

(1) Direction Cosines of line = Direction Cosines of parallel vector

$$\vec{a} = 5\hat{i} + 6\hat{j} + 7\hat{k}; \quad \vec{b} = 7\hat{i} + 9\hat{j} + 13\hat{k}$$

$$\vec{b} - \vec{a} = 2\hat{i} + 3\hat{j} + 6\hat{k}$$

$$r = \sqrt{4 + 9 + 36} = \sqrt{49} = 7$$

$$x = 2; \quad y = 3; \quad z = 6$$

$$\text{Direction Cosines of the line} = \left(\frac{x}{r}, \frac{y}{r}, \frac{z}{r}\right) = \left(\frac{2}{7}, \frac{3}{7}, \frac{6}{7}\right)$$

(2) vector equation of line in parametric form

$$\vec{r} = \vec{a} + t(\vec{b} - \vec{a})$$

$$\vec{r} = (5\vec{i} + 6\vec{j} + 7\vec{k}) + t(2\vec{i} + 3\vec{j} + 6\vec{k})$$

(3) Cartesian equation of line is

$$\frac{x-x_1}{x_2-x_1} = \frac{y-y_1}{y_2-y_1} = \frac{z-z_1}{z_2-z_1}$$

$$\frac{x-5}{7-5} = \frac{y-6}{9-6} = \frac{z-7}{13-7}$$

$$\frac{x-5}{2} = \frac{y-6}{3} = \frac{z-7}{6}$$

5. Find the acute angle between the following lines.

2M

(i) $\vec{r} = (4\hat{i} - \hat{j}) + t(\hat{i} + 2\hat{j} - 2\hat{k})$, $\vec{r} = (\hat{i} - 2\hat{j} + 4\hat{k}) + s(-\hat{i} - 2\hat{j} + 2\hat{k})$

(ii) $\frac{x+4}{3} = \frac{y-7}{4} = \frac{z+5}{5}$, $\vec{r} = 4\hat{k} + t(2\hat{i} + \hat{j} + \hat{k})$

(iii) $2x = 3y = -z$ and $6x = -y = -4z$

(i) Angle between the lines

$$\cos \theta = \frac{|\vec{b} \cdot \vec{d}|}{|\vec{b}||\vec{d}|}$$

$$\vec{b} = \hat{i} + 2\hat{j} - 2\hat{k}$$

$$\vec{d} = -\hat{i} - 2\hat{j} + 2\hat{k}$$

$$\vec{b} \cdot \vec{d} = -1 - 4 - 4 = -9$$

$$|\vec{b}| = \sqrt{1 + 4 + 4} = \sqrt{9} = 3$$

$$|\vec{d}| = \sqrt{1 + 4 + 4} = \sqrt{9} = 3$$

$$\cos \theta = \frac{|-9|}{3 \times 3} = \frac{9}{9}$$

$$\cos \theta = 1$$

$$\cos \theta = \cos 0$$

$$\theta = 0$$

(ii) $\vec{b} = 3\hat{i} + 4\hat{j} + 5\hat{k}$

$$\vec{d} = 2\hat{i} + \hat{j} + \hat{k}$$

$$\cos \theta = \frac{|\vec{b} \cdot \vec{d}|}{|\vec{b}||\vec{d}|}$$

$$\vec{b} \cdot \vec{d} = 6 + 4 + 5 = 15$$

$$|\vec{b}| = \sqrt{9 + 16 + 25} = \sqrt{50}$$

$$= \sqrt{25 \times 2} = 5\sqrt{2}$$

$$|\vec{d}| = \sqrt{4 + 1 + 1} = \sqrt{6}$$

$$\cos \theta = \frac{15}{5\sqrt{2} \times \sqrt{6}}$$

$$= \frac{3}{\sqrt{2} \times \sqrt{2} \times \sqrt{3}} = \frac{\sqrt{3} \times \sqrt{3}}{2\sqrt{3}}$$

$$\cos \theta = \frac{\sqrt{3}}{2} = \cos \frac{\pi}{6}$$

$$\theta = \frac{\pi}{6}$$

(iii) $\frac{x}{\frac{1}{2}} = \frac{y}{\frac{1}{3}} = \frac{z}{-1}$

$$\frac{x}{\frac{1}{2}} = \frac{y}{-1} = \frac{z}{-1}$$

i.e) $\vec{b} = \frac{1}{2}\hat{i} + \frac{1}{3}\hat{j} - \hat{k}$

$$\vec{d} = \frac{1}{6}\hat{i} - \hat{j} - \frac{1}{4}\hat{k}$$

$$\vec{b} \cdot \vec{d} = \left(\frac{1}{2}\right)\left(\frac{1}{6}\right) + \left(\frac{1}{3}\right)(-1) + (-1)\left(-\frac{1}{4}\right)$$

$$= \frac{1}{12} - \frac{1}{3} + \frac{1}{4}$$

$$= \frac{1-4+3}{12}$$

$$\vec{b} \cdot \vec{d} = 0$$

$$\Rightarrow \vec{b} \perp \vec{d}$$

$$\Rightarrow \theta = \frac{\pi}{2}$$

PTA-4

6. The vertices of $\triangle ABC$ are $A(7, 2, 1)$, $B(6, 0, 3)$, and $C(4, 2, 4)$. Find $\angle ABC$

3M

$$\vec{OA} = 7\hat{i} + 2\hat{j} + \hat{k}; \quad \vec{OB} = 6\hat{i} + 0\hat{j} + 3\hat{k}; \quad \vec{OC} = 4\hat{i} + 2\hat{j} + 4\hat{k}.$$

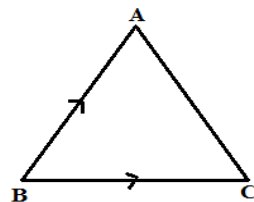
$$\vec{BA} = \vec{OA} - \vec{OB} = \hat{i} + 2\hat{j} - 2\hat{k}$$

$$\vec{BC} = \vec{OC} - \vec{OB} = -2\hat{i} + 2\hat{j} + \hat{k}$$

$$\vec{BA} \cdot \vec{BC} = (1)(-2) + (2)(2) + (-2)(1) = -2 + 4 - 2 = 0$$

$$\vec{BA} \perp \vec{BC}$$

$$\therefore \angle ABC = \frac{\pi}{2}$$



7. If the straight line joining the points $(2, 1, 4)$ and $(a-1, 4, -1)$ is parallel to the line joining the points $(0, 2, b-1)$ and $(5, 3, -2)$, find the values of a and b

3M

$$\vec{OA} = 2\hat{i} + \hat{j} + 4\hat{k}$$

$$\vec{OB} = (a-1)\hat{i} + 4\hat{j} - \hat{k}$$

$$\vec{AB} = \vec{OB} - \vec{OA} = (a-3)\hat{i} + 3\hat{j} - 5\hat{k}$$

$$\vec{OC} = 0\hat{i} + 2\hat{j} + (b-1)\hat{k}$$

$$\vec{OD} = 5\hat{i} + 3\hat{j} - 2\hat{k}$$

$$\vec{CD} = \vec{OD} - \vec{OC} = 5\hat{i} + \hat{j} + (-1-b)\hat{k}$$

$$\therefore \vec{AB}; \vec{CD} \text{ are parallel} \quad \frac{b_1}{a_1} = \frac{b_2}{a_2} = \frac{b_3}{a_3}$$

$$\frac{5}{a-3} = \frac{1}{3} = \frac{-b-1}{-5}$$

$$\frac{5}{a-3} = \frac{1}{3}$$

$$15 = a - 3$$

$$a = 18$$

$$\frac{1}{3} = \frac{-b-1}{-5}$$

$$\frac{-5}{3} = -b - 1$$

$$b = \frac{5}{3} - 1 = \frac{2}{3}$$

8. If the straight lines $\frac{x-5}{5m+2} = \frac{2-y}{5} = \frac{1-z}{-1}$ and $x = \frac{2y+1}{4m} = \frac{1-z}{-3}$ are perpendicular to each other, find the value of m . 3M

$$\begin{aligned}\frac{x-5}{5m+2} &= \frac{2-y}{5} = \frac{1-z}{-1} \\ \frac{x-5}{5m+2} &= \frac{y-2}{-5} = \frac{z-1}{1} \\ \vec{b} &= (5m+2)\hat{i} - 5\hat{j} + \hat{k}\end{aligned}$$

$$\begin{aligned}x &= \frac{2y+1}{4m} = \frac{1-z}{-3} \\ \frac{x-0}{1} &= \frac{y+\frac{1}{2}}{2m} = \frac{z-1}{3} \\ \vec{d} &= \hat{i} + (2m)\hat{j} + 3\hat{k}\end{aligned}$$

$$\therefore \vec{b} \perp \vec{d} \Rightarrow \vec{b} \cdot \vec{d} = 0$$

$$(5m+2)(1) + (-5)(2m) + (1)(3) = 0$$

$$5m + 2 - 10m + 3 = 0$$

$$-5m + 5 = 0$$

$$-m + 1 = 0$$

$$m = 1$$

9. Show that the points (2, 3, 4), (-1, 4, 5) and (8, 1, 2) are collinear. 2M

Equation of line joining (2, 3, 4) (-1, 4, 5) is $\frac{x-2}{-3} = \frac{y-3}{1} = \frac{z-4}{1} = t$

Any point on the line $P(-3t+2, t+3, t+4)$

Put $t = -2$

$$P(6+2, -2+3, -2+4) = P(8, 1, 2)$$

$\therefore$ The given points are collinear.

EXERCISE 6.5



Video class

Concept Corner

Point of intersection of two straight lines:

$$\frac{x-x_1}{a_1} = \frac{y-y_1}{a_2} = \frac{z-z_1}{a_3} \quad \text{and} \quad \frac{x-x_2}{b_1} = \frac{y-y_2}{b_2} = \frac{z-z_2}{b_3}$$

$$\text{Form: } (x_1 + sa_1, y_1 + sa_2, z_1 + sa_3) \text{ and } (x_2 + tb_1, y_2 + tb_2, z_2 + tb_3)$$

Distance between parallel lines and skew lines:

To find the point of intersection of given lines, solve the given equation.

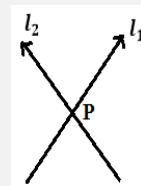
Intersecting lines:

Intersecting lines are the lines which meet at a point.

Parallel lines:



Parallel lines are the lines which do not meet at a point.

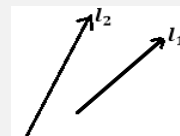


Coplanar:

Coplanar lines are the lines which lie in the same plane.

Skew lines:

Skew lines are the lines which are neither parallel nor intersect.



Important Formulas:

(1) The shortest distance between the two parallel lines $\vec{r} = \vec{a} + s\vec{b}$ and $\vec{r} = \vec{c} + t\vec{b}$ is given by

$$d = \frac{|(\vec{c}-\vec{a}) \times \vec{b}|}{|\vec{b}|}; \quad |\vec{b}| \neq 0$$

(2) The shortest distance between the skew lines $\vec{r} = \vec{a} + s\vec{b}$ and $\vec{r} = \vec{c} + t\vec{d}$ is

$$\delta = \frac{|(\vec{c}-\vec{a}) \cdot (\vec{b} \times \vec{d})|}{|\vec{b} \times \vec{d}|} = \frac{|[\vec{c}-\vec{a} \ \vec{b} \ \vec{d}]|}{|\vec{b} \times \vec{d}|}; \quad |\vec{b} \times \vec{d}| \neq 0$$

(3) The condition for the lines $\vec{r} = \vec{a} + s\vec{b}$; $\vec{r} = \vec{c} + t\vec{d}$ to be skew lines is $(\vec{c} - \vec{a}) \cdot (\vec{b} \times \vec{d}) \neq 0$

$$\text{i.e. } [\vec{c} - \vec{a} \ \vec{b} \ \vec{d}] \neq 0$$

(4) The condition for the lines $\vec{r} = \vec{a} + s\vec{b}$ and $\vec{r} = \vec{c} + t\vec{d}$ to intersect is $(\vec{c} - \vec{a}) \cdot (\vec{b} \times \vec{d}) = 0$

$$\text{i.e. } [\vec{c} - \vec{a} \ \vec{b} \ \vec{d}] = 0 \quad (\text{or}) \quad \begin{vmatrix} x_2 - x_1 & y_2 - y_1 & z_2 - z_1 \\ b_1 & b_2 & b_3 \\ d_1 & d_2 & d_3 \end{vmatrix} = 0$$

1. Find the parametric form of vector equation and Cartesian equations of a straight line passing through (5, 2, 8) and is perpendicular to the straight lines $\vec{r} = (\hat{i} + \hat{j} - \hat{k}) + s(2\hat{i} - 2\hat{j} + \hat{k})$ and $\vec{r} = (2\hat{i} - \hat{j} - 3\hat{k}) + t(\hat{i} + 2\hat{j} + 2\hat{k})$

3M

$$\vec{r} = (\hat{i} + \hat{j} - \hat{k}) + s(2\hat{i} - 2\hat{j} + \hat{k})$$

$$\vec{b} = 2\hat{i} - 2\hat{j} + \hat{k}; \quad \vec{d} = \hat{i} + 2\hat{j} + 2\hat{k}$$

$$\vec{b} \times \vec{d} = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ 2 & -2 & 1 \\ 1 & 2 & 2 \end{vmatrix}$$

$$= \hat{i}(-6) - \hat{j}(3) + \hat{k}(6)$$

$$= -3(2\hat{i} + \hat{j} - 2\hat{k})$$

The required line passes through A(5, 2, 8) and parallel to $\vec{b} \times \vec{d}$.

Vector Equation in parametric form.

$$\vec{r} = (5\hat{i} + 2\hat{j} + 8\hat{k}) + s(-3)(2\hat{i} + \hat{j} - 2\hat{k})$$

$$\vec{r} = (5\hat{i} + 2\hat{j} + 8\hat{k}) + t(2\hat{i} + \hat{j} - 2\hat{k}), t \in \mathbb{R}$$

$$\text{Equation of the line } \frac{x-x_1}{b_1} = \frac{y-y_1}{b_2} = \frac{z-z_1}{b_3}$$

$$(x_1, y_1, z_1) = (5, 2, 8) \quad \&$$

$$(b_1, b_2, b_3) = (2, 1, -2)$$

$$\frac{x-5}{2} = \frac{y-2}{1} = \frac{z-8}{-2}$$

2. Show that the lines

$$\vec{r} = (6\hat{i} + \hat{j} + 2\hat{k}) + s(\hat{i} + 2\hat{j} - 3\hat{k}) \text{ and}$$

$$\vec{r} = (3\hat{i} + 2\hat{j} - 2\hat{k}) + t(2\hat{i} + 4\hat{j} - 5\hat{k})$$

are skew lines and hence find the shortest distance between them.

$$\vec{a} = 6\hat{i} + \hat{j} + 2\hat{k}; \quad \vec{b} = \hat{i} + 2\hat{j} - 3\hat{k}$$

$$\vec{c} = 3\hat{i} + 2\hat{j} - 2\hat{k}; \quad \vec{d} = 2\hat{i} + 4\hat{j} - 5\hat{k}$$

$$\vec{c} - \vec{a} = -3\hat{i} + \hat{j} - 4\hat{k}$$

3M

condition for skew lines

$$(\vec{c} - \vec{a}) \cdot (\vec{b} \times \vec{d}) \neq 0$$

$$\text{i.e. } [\vec{c} - \vec{a} \ \vec{b} \ \vec{d}] \neq 0$$

$$[\vec{c} - \vec{a} \ \vec{b} \ \vec{d}] = \begin{vmatrix} -3 & 1 & -4 \\ 1 & 2 & -3 \\ 2 & 4 & -5 \end{vmatrix}$$

$$= -3(2) - 1(1) - 4(0)$$

$$= -6 - 1 = -7 \neq 0$$

The given lines are skew lines

$$\text{Distance } \delta = \frac{|[\vec{c}-\vec{a} \ \vec{b} \ \vec{d}]|}{|\vec{b} \times \vec{d}|}$$

$$|[\vec{c} - \vec{a} \ \vec{b} \ \vec{d}]| = |-7| = 7$$

$$\vec{b} \times \vec{d} = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ 1 & 2 & -3 \\ 2 & 4 & -5 \end{vmatrix}$$

$$= \hat{i}(2) - \hat{j}(1) + \hat{k}(0) = 2\hat{i} - \hat{j}$$

$$|\vec{b} \times \vec{d}| = \sqrt{4 + 1 + 0} = \sqrt{5}$$

$$D = \frac{7}{\sqrt{5}} \text{ units}$$

3. If the two lines $\frac{x-1}{2} = \frac{y+1}{3} = \frac{z-1}{4}$ and $\frac{x-3}{1} = \frac{y-m}{2} = z$ intersect at a point, find the value of m .

3M

$$\frac{x-1}{2} = \frac{y-(-1)}{3} = \frac{z-1}{4} \text{ and } \frac{x-3}{1} = \frac{y-m}{2} = \frac{z-0}{1}$$

$$\vec{a} = \hat{i} - \hat{j} + \hat{k}; \quad \vec{b} = 2\hat{i} + 3\hat{j} + 4\hat{k}; \quad \vec{c} = 3\hat{i} + m\hat{j} + 0\hat{k}; \quad \vec{d} = \hat{i} + 2\hat{j} + \hat{k}$$

Condition for two lines are intersect $[\vec{c} - \vec{a} \quad \vec{b} \quad \vec{d}] = 0$

$$\begin{vmatrix} 3-1 & m+1 & 0-1 \\ 2 & 3 & 4 \\ 1 & 2 & 1 \end{vmatrix} = 0$$

$$\begin{vmatrix} 2 & m+1 & -1 \\ 2 & 3 & 4 \\ 1 & 2 & 1 \end{vmatrix} = 0$$

$$2(3-8) - (m+1)(2-4) - 1(4-3) = 0$$

$$2(-5) + 2(m+1) - 1(1) = 0$$

$$2m - 9 = 0$$

$$m = \frac{9}{2}$$

4. Show that the lines $\frac{x-3}{3} = \frac{y-3}{-1}, z-1=0$ and $\frac{x-6}{2} = \frac{y-2}{3}, y-2=0$ intersect. Also find the point of intersection.

5M

Let the equation of the first line $\frac{x-3}{3} = \frac{y-3}{-1} = \frac{z-1}{0} = \lambda$

Let $P(3\lambda + 3, -\lambda + 3, 1)$ be the any point on this line.

Let the equation of the second line $\frac{x-6}{2} = \frac{y-2}{0} = \frac{z-1}{3} = \mu$

Let $Q(2\mu + 6, 2, 3\mu + 1)$ be the any point on this line.

$$\vec{a} = 3\hat{i} + 3\hat{j} - \hat{k}, \quad \vec{b} = 3\hat{i} - \hat{j},$$

$$\vec{c} = 6\hat{i} + 2\hat{j} - \hat{k}, \quad \vec{d} = 2\hat{i} + 3\hat{k},$$

$$[\vec{c} - \vec{a} \quad \vec{b} \quad \vec{d}] = \begin{vmatrix} 3 & -1 & 0 \\ 3 & -1 & 0 \\ 2 & 0 & 3 \end{vmatrix}$$

$$= 0 \quad [\because R_1 \equiv R_2]$$

Hence, Given two lines are intersect each other.

$$P(3\lambda + 3, -\lambda + 3, 1) = Q(2\mu + 6, 2, 3\mu + 1)$$

$$-\lambda + 3 = 2 \quad \left| \quad 1 = 3\mu + 1 \right.$$

$$-\lambda = -1 \quad \left| \quad 3\mu = 0 \right.$$

$$\lambda = 1 \quad \mu = 0$$

$$\text{Point of intersection} = P(3 + 3, -1 + 3, 1) = \mathbf{P(6, 2, 1)}$$

5. Show that the straight lines $x + 1 = 2y = -12z$ & $x = y + 2 = 6z - 6$ are skew and hence find the shortest distance between them.

3M

$$x + 1 = 2y = -12z$$

$$\frac{x+1}{1} = \frac{y}{\frac{1}{2}} = \frac{z}{\frac{-1}{12}}$$

$$\frac{x-(-1)}{1} = \frac{y-0}{\frac{1}{2}} = \frac{z-0}{\frac{-1}{12}}$$

$$\vec{a} = -\hat{i}$$

$$\vec{b} = \hat{i} + \frac{1}{2}\hat{j} - \frac{1}{12}\hat{k}$$

$$x = y + 2 = 6z - 6$$

$$= 6(z - 1)$$

$$\frac{x-0}{1} = \frac{y-(-2)}{1} = \frac{z-1}{\frac{1}{6}}$$

$$\vec{c} = -2\hat{j} + \hat{k} \quad \vec{d} = \hat{i} + \hat{j} + \frac{1}{6}\hat{k}$$

$$\vec{c} - \vec{a} = \hat{i} - 2\hat{j} + \hat{k}$$

$$[\vec{c} - \vec{a} \quad \vec{b} \quad \vec{d}] = \begin{vmatrix} 1 & -2 & 1 \\ 1 & \frac{1}{2} & \frac{-1}{12} \\ 1 & 1 & \frac{1}{6} \end{vmatrix}$$

$$= 1\left(\frac{1}{12} + \frac{1}{12}\right) + 2\left(\frac{1}{6} + \frac{1}{12}\right) + 1\left(1 - \frac{1}{2}\right)$$

$$= 1\left(\frac{1}{6}\right) + 2\left(\frac{3}{12}\right) + 1\left(\frac{1}{2}\right)$$

$$= \frac{1}{6} + 1 = \frac{7}{6} \neq 0$$

∴ The given lines are skew lines

$$\text{Distance} = D = \frac{|\vec{c} - \vec{a} \cdot \vec{b} \cdot \vec{d}|}{|\vec{b} \times \vec{d}|}$$

$$|[\vec{c} - \vec{a} \quad \vec{b} \quad \vec{d}]| = \left|\frac{7}{6}\right| = \frac{7}{6}$$

$$\vec{b} \times \vec{d} = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ 1 & \frac{1}{2} & \frac{-1}{12} \\ 1 & 1 & \frac{1}{6} \end{vmatrix}$$

$$= \hat{i}\left(\frac{1}{6}\right) - \hat{j}\left(\frac{3}{12}\right) + \hat{k}\left(\frac{1}{2}\right)$$

$$= \hat{i}\left(\frac{1}{6}\right) + \hat{j}\left(\frac{-1}{4}\right) + \hat{k}\left(\frac{1}{2}\right)$$

$$|\vec{b} \times \vec{d}| = \sqrt{\frac{1}{36} + \frac{1}{16} + \frac{1}{4}} = \sqrt{\frac{4+9+36}{144}}$$

$$= \sqrt{\frac{49}{144}} = \frac{7}{12}$$

$$D = \frac{\frac{7}{6}}{\frac{7}{12}} = \frac{7}{6} \times \frac{12}{7} = 2 \text{ units}$$

6. Find the parametric form of vector equation of the straight line passing through $(-1, 2, 1)$ and parallel to the straight line $\vec{r} = (2\hat{i} + 3\hat{j} - \hat{k}) + t(\hat{i} - 2\hat{j} + \hat{k})$ and hence find the shortest distance between the lines.

vector equation of the required line

3M

$$\vec{r} = (-\hat{i} + 2\hat{j} + \hat{k}) + s(\hat{i} - 2\hat{j} + \hat{k})$$

$$\vec{r} = (2\hat{i} + 3\hat{j} - \hat{k}) + t(\hat{i} - 2\hat{j} + \hat{k})$$

$$\vec{a} = -\hat{i} + 2\hat{j} + \hat{k} \quad \vec{c} = 2\hat{i} + 3\hat{j} - \hat{k}$$

$$\vec{c} - \vec{a} = 3\hat{i} + \hat{j} - 2\hat{k} \quad \vec{b} = \hat{i} - 2\hat{j} + \hat{k}$$

$$|\vec{b}| = \sqrt{1 + 4 + 1} = \sqrt{6}$$

$$\text{Distance between parallel lines} = D = \frac{|(\vec{c} - \vec{a}) \times \vec{b}|}{|\vec{b}|}$$

$$(\vec{c} - \vec{a}) \times \vec{b} = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ 3 & 1 & -2 \\ 1 & -2 & 1 \end{vmatrix}$$

$$= \hat{i}(-3) - \hat{j}(5) + \hat{k}(-7) = -3\hat{i} - 5\hat{j} - 7\hat{k}$$

$$|(\vec{c} - \vec{a}) \times \vec{b}| = \sqrt{9 + 25 + 49} = \sqrt{83}$$

$$D = \frac{\sqrt{83}}{\sqrt{6}} \text{ units}$$

7. Find the foot of the perpendicular drawn from the point $(5, 4, 2)$ to the line $\frac{x+1}{2} = \frac{y-3}{3} = \frac{z-1}{-1}$

Also, find the equation of the perpendicular.

Let F is the foot of perpendicular from $D(5, 4, 2)$

$$\text{Line: } \frac{x+1}{2} = \frac{y-3}{3} = \frac{z-1}{-1} = t$$

5M

Any point on the line $F = (2t - 1, 3t + 3, -t + 1)$

$$\overrightarrow{DF} = \overrightarrow{OF} - \overrightarrow{OD}$$

PTA-2, JUL-23

$$= (2t - 1)\hat{i} + (3t + 3)\hat{j} + (-t + 1)\hat{k} - (5\hat{i} + 4\hat{j} + 2\hat{k})$$

$$= (2t - 1)\hat{i} + (3t + 3)\hat{j} + (-t + 1)\hat{k} - 5\hat{i} - 4\hat{j} - 2\hat{k}$$

$$= (2t - 6)\hat{i} + (3t - 1)\hat{j} + (-t - 1)\hat{k}$$

$$\vec{b} = 2\hat{i} + 3\hat{j} - \hat{k}$$

$$\therefore \overrightarrow{DF} \perp \vec{b} \Rightarrow \overrightarrow{DF} \cdot \vec{b} = 0$$

$$(2t - 6)(2) + (3t - 1)(3) + (-t - 1)(-1) = 0$$

$$4t - 12 + 9t - 3 + t + 1 = 0$$

$$14t - 14 = 0$$

$$t - 1 = 0 \Rightarrow t = 1$$

Foot of perpendicular = $F(2 - 1, 3 + 3, -1 + 1)$

$$= F(1, 6, 0)$$

Equation of perpendicular joining $D(5, 4, 2)$

$$\text{and } F(1, 6, 0) \text{ is } \frac{x-5}{-4} = \frac{y-4}{2} = \frac{z-2}{-2}$$

EXERCISE 6.6



Video class

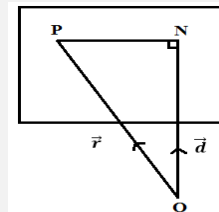
Concept Corner

Different forms of Equations of a plane

I. Normal Form:

- a) The Vector Equation of the plane at a distance p from the origin and perpendicular to the unit normal vector $\hat{d}$ is $\vec{r} \cdot \hat{d} = p$
- b) If $\vec{d} = l\vec{i} + m\vec{j} + n\vec{k}$ and $\vec{r} = x\vec{i} + y\vec{j} + z\vec{k}$ then $lx + my + nz = p$ is the equation of the plane.
- c) If $\vec{d}$ is normal but not a unit vector, then vector Equation is $\vec{r} \cdot \vec{d} = q$
- d) If $\vec{d} = a\vec{i} + b\vec{j} + c\vec{k}$ is normal but not a unit vector; then Equation of the plane $ax + by + cz = q$
- e) $\vec{r} \cdot \hat{d} = q \quad \vec{r} \cdot \frac{\vec{d}}{|\vec{d}|} = \frac{q}{|\vec{d}|} \quad \text{unit normal vector} = \hat{d} = \frac{\vec{d}}{|\vec{d}|}$

Length of perpendicular from the origin to the plane $\vec{r} \cdot \vec{d} = q$ is $\frac{q}{|\vec{d}|}$



II. Point and Normal form:

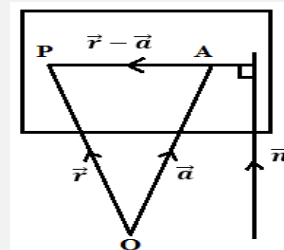
- a) The vector equation of the plane passing through the point $A(\vec{a})$ and normal ($\vec{n}$) to the plane is $(\vec{r} - \vec{a}) \cdot \vec{n} = 0$ (or) $\vec{r} \cdot \vec{n} = \vec{a} \cdot \vec{n}$
- b) If $\vec{r} = x\vec{i} + y\vec{j} + z\vec{k}$

$$\vec{a} = x_1\vec{i} + y_1\vec{j} + z_1\vec{k}$$

$$a(x - x_1) + b(y - y_1) + c(z - z_1) = 0$$

$$\vec{r} - \vec{a} = (x - x_1)\vec{i} + (y - y_1)\vec{j} + (z - z_1)\vec{k} \text{ and } \vec{n} = a\vec{i} + b\vec{j} + c\vec{k}$$

$$(\vec{r} - \vec{a}) \cdot \vec{n} = 0$$



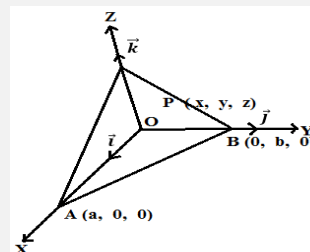
III. Intercept Form

The Equation of the plane in the intercept form is $\frac{x}{a} + \frac{y}{b} + \frac{z}{c} = 1$

This plane passes through $A(a, 0, 0)$ $B(0, b, 0)$ and $C(0, 0, c)$

Important formula:

- (1) If $\vec{r} \cdot (a\vec{i} + b\vec{j} + c\vec{k}) = q$ is the vector equation of the plane; then the Cartesian equation is $ax + by + cz = q$
- (2) If $ax + by + cz = q$ is the equation of the plane; then vector equation is $\vec{r} \cdot (a\vec{i} + b\vec{j} + c\vec{k}) = q$



1. Find a parametric form of vector equation of a plane which is at a distance of 7 units from the origin having 3, -4, 5 as direction ratios of a normal to it.

APR-23

2M

$$p = 7 \quad \vec{d} = 3\hat{i} - 4\hat{j} + 5\hat{k}$$

$$|\vec{d}| = \sqrt{9 + 16 + 25} = \sqrt{50} = \sqrt{25 \times 2} = 5\sqrt{2}$$

$$\hat{d} = \frac{\vec{d}}{|\vec{d}|}$$

$$\hat{d} = \frac{3\hat{i} - 4\hat{j} + 5\hat{k}}{5\sqrt{2}}$$

vector equation of the plane in the normal form is $\vec{r} \cdot \hat{d} = p$

$$\vec{r} \cdot \left(\frac{3\hat{i} - 4\hat{j} + 5\hat{k}}{5\sqrt{2}} \right) = 7$$

2. Find the direction cosines of the normal to the plane $12x + 3y - 4z = 65$. Also, find the non-parametric form of vector equation of a plane and the length of the perpendicular to the plane from the origin.

Equation of the plane $12x + 3y - 4z = 65$

Vector equation of the plane $\vec{r} \cdot (12\hat{i} + 3\hat{j} - 4\hat{k}) = 65$

2M

$$(1) \quad \vec{d} = 12\hat{i} + 3\hat{j} - 4\hat{k} \quad q = 65$$

$$|\vec{d}| = \sqrt{144 + 9 + 16} = \sqrt{169} = 13$$

$$\text{Direction cosines of normal} = \left(\frac{12}{13}, \frac{3}{13}, \frac{-4}{13}\right)$$

$$(2) \quad \hat{d} = \frac{\vec{d}}{|\vec{d}|} = \frac{1}{13}(12\hat{i} + 3\hat{j} - 4\hat{k})$$

Vector equation of the plane is $p = \frac{q}{|\vec{d}|} = \frac{65}{13} = 5$

$$\vec{r} \cdot \hat{d} = p$$

$$\vec{r} \cdot \left(\frac{12\hat{i} + 3\hat{j} - 4\hat{k}}{13}\right) = 5$$

$$(3) \text{ Length of perpendicular from the origin} = \frac{q}{|\vec{d}|} = \frac{65}{13} = 5$$

3. Find the vector and Cartesian equations of the plane passing through the point with position vector $2\hat{i} + 6\hat{j} + 3\hat{k}$ and normal to the vector $\hat{i} + 3\hat{j} + 5\hat{k}$.

2M

vector equation of the plane

JUL-24

$$\vec{r} \cdot \vec{n} = \vec{a} \cdot \vec{n}$$

$$\vec{a} = 2\hat{i} + 6\hat{j} + 3\hat{k};$$

$$\vec{n} = \hat{i} + 3\hat{j} + 5\hat{k}$$

$$\begin{aligned} \vec{r} \cdot (\hat{i} + 3\hat{j} + 5\hat{k}) &= (2\hat{i} + 6\hat{j} + 3\hat{k}) \cdot (\hat{i} + 3\hat{j} + 5\hat{k}) \\ &= 2 + 18 + 15 \end{aligned}$$

$$\vec{r} \cdot (\hat{i} + 3\hat{j} + 5\hat{k}) = 35$$

Cartesian equation of the plane

Put $\vec{r} = x\hat{i} + y\hat{j} + z\hat{k}$

$$(x\hat{i} + y\hat{j} + z\hat{k}) \cdot (\hat{i} + 3\hat{j} + 5\hat{k}) = 35$$

$$x + 3y + 5z = 35$$

4. A plane passes through the point $(-1, 1, 2)$ and the normal to the plane of magnitude $3\sqrt{3}$ makes equal acute angles with the coordinate axes. Find the equation of the plane.

2M

Given magnitude $= 3\sqrt{3}$

$$\vec{a} = -\hat{i} + \hat{j} + 2\hat{k}$$

Here $\alpha = \beta = \gamma$

$$\cos^2 \alpha + \cos^2 \beta + \cos^2 \gamma = 1$$

$$\cos^2 \alpha + \cos^2 \alpha + \cos^2 \alpha = 1$$

$$3 \cos^2 \alpha = 1$$

$$\cos^2 \alpha = \frac{1}{3} \Rightarrow \cos \alpha = \frac{1}{\sqrt{3}}$$

$$\vec{n} = r(l\hat{i} + m\hat{j} + n\hat{k})$$

$$= 3\sqrt{3} \left(\frac{1}{\sqrt{3}}\hat{i} + \frac{1}{\sqrt{3}}\hat{j} + \frac{1}{\sqrt{3}}\hat{k}\right) = 3\hat{i} + 3\hat{j} + 3\hat{k}$$

Vector equation of the plane $\vec{r} \cdot \vec{n} = \vec{a} \cdot \vec{n}$

$$\vec{r} \cdot (3\hat{i} + 3\hat{j} + 3\hat{k}) = (-\hat{i} + \hat{j} + 2\hat{k}) \cdot (3\hat{i} + 3\hat{j} + 3\hat{k})$$

$$\vec{r} \cdot (3\hat{i} + 3\hat{j} + 3\hat{k}) = -3 + 3 + 6$$

$$\vec{r} \cdot (3\hat{i} + 3\hat{j} + 3\hat{k}) = 6$$

$$\vec{r} \cdot (\hat{i} + \hat{j} + \hat{k}) = 2$$

Cartesian equation of the plane

$$(x\hat{i} + y\hat{j} + z\hat{k}) \cdot (3\hat{i} + 3\hat{j} + 3\hat{k}) = 6$$

$$3x + 3y + 3z = 6$$

$$x + y + z = 2$$

EXERCISE 6.10



Choose the correct or the most suitable answer from the given four alternatives.

1. If $\vec{a}$ and $\vec{b}$ are parallel vectors, then $[\vec{a}, \vec{c}, \vec{b}]$ is equal to

APR-23

- (1) 2 (2) -1 (3) 1 (4) 0

$$\because \vec{a} \text{ parallel to } \vec{b}, \vec{b} = \lambda \vec{a}$$

$$\begin{aligned} [\vec{a} \vec{c} \vec{b}] &= [\vec{a} \vec{c} \lambda \vec{a}] \\ &= \lambda [\vec{a} \vec{c} \vec{a}] = \lambda(0) \\ &= 0 \end{aligned}$$

2. If a vector $\vec{a}$ lies in the plane of $\vec{\beta}$ and $\vec{\gamma}$, then

APR-24, JUL-22

- (1) $[\vec{a}, \vec{\beta}, \vec{\gamma}] = 1$ (2) $[\vec{a}, \vec{\beta}, \vec{\gamma}] = -1$ (3) $[\vec{a}, \vec{\beta}, \vec{\gamma}] = 0$ (4) $[\vec{a}, \vec{\beta}, \vec{\gamma}] = 2$

$$\because \vec{a}, \vec{\beta}, \vec{\gamma} \text{ are coplanar} \quad \therefore [\vec{a} \vec{\beta} \vec{\gamma}] = 0$$

3. If $\vec{a} \cdot \vec{b} = \vec{b} \cdot \vec{c} = \vec{c} \cdot \vec{a} = 0$, then the value of $[\vec{a}, \vec{b}, \vec{c}]$ is

JUL-23

- (1) $|\vec{a}||\vec{b}||\vec{c}|$ (2) $\frac{1}{3}|\vec{a}||\vec{b}||\vec{c}|$ (3) 1 (4) -1

$$\text{Here } \vec{a}, \vec{b}, \vec{c} \text{ are mutually perpendicular}$$

$$\therefore [\vec{a} \vec{b} \vec{c}] = abc = |\vec{a}||\vec{b}||\vec{c}|$$

4. If $\vec{a}, \vec{b}, \vec{c}$ are three unit vectors such that $\vec{a}$ is perpendicular to $\vec{b}$, and is parallel to $\vec{c}$ then $\vec{a} \times (\vec{b} \times \vec{c})$ is equal to

- (1) $\vec{a}$ (2) $\vec{b}$ (3) $\vec{c}$ (4) $\vec{0}$

$$\begin{aligned} \because \vec{a} \perp \vec{b} &\Rightarrow \vec{a} \cdot \vec{b} = 0 \\ \vec{a} \parallel \vec{c} &\Rightarrow \vec{a} \cdot \vec{c} = |\vec{a}||\vec{c}| \cos 0 \quad [\because \theta = 0] \\ &= 1 \times 1 \times 1 \\ &= 1 \end{aligned}$$

$$\begin{aligned} \therefore \vec{a} \times (\vec{b} \times \vec{c}) &= (\vec{a} \cdot \vec{c}) \vec{b} - (\vec{a} \cdot \vec{b}) \vec{c} \\ &= (1)\vec{b} - (0) \vec{c} \\ &= \vec{b} - 0 = \vec{b} \end{aligned}$$

5. If $[\vec{a}, \vec{b}, \vec{c}] = 1$, then the value of $\frac{\vec{a} \cdot (\vec{b} \times \vec{c})}{(\vec{c} \times \vec{a}) \cdot \vec{b}} + \frac{\vec{b} \cdot (\vec{c} \times \vec{a})}{(\vec{a} \times \vec{b}) \cdot \vec{c}} + \frac{\vec{c} \cdot (\vec{a} \times \vec{b})}{(\vec{b} \times \vec{c}) \cdot \vec{a}}$ is

PTA-6

- (1) 1 (2) -1 (3) 2 (4) 3

$$\begin{aligned} \frac{[\vec{a} \vec{b} \vec{c}]}{[\vec{c} \vec{a} \vec{b}]} + \frac{[\vec{b} \vec{c} \vec{a}]}{[\vec{a} \vec{b} \vec{c}]} + \frac{[\vec{c} \vec{a} \vec{b}]}{[\vec{b} \vec{c} \vec{a}]} &= \frac{[\vec{a} \vec{b} \vec{c}]}{[\vec{a} \vec{b} \vec{c}]} + \frac{[\vec{a} \vec{b} \vec{c}]}{[\vec{a} \vec{b} \vec{c}]} + \frac{[\vec{a} \vec{b} \vec{c}]}{-[\vec{a} \vec{b} \vec{c}]} \\ &= 1 + 1 - 1 = 1 \end{aligned}$$

$$[\because [\vec{a} \vec{b} \vec{c}] = [\vec{b} \vec{c} \vec{a}] = [\vec{c} \vec{a} \vec{b}]]$$

6. The volume of the parallelepiped with its edges represented by the vectors $\hat{i} + \hat{j}$, $\hat{i} + 2\hat{j}$, $\hat{i} + \hat{j} + \pi\hat{k}$ is

APR-25, MDL, JUL-21

- (1) $\frac{\pi}{2}$ (2) $\frac{\pi}{3}$ (3) π (4) $\frac{\pi}{4}$

$$\begin{aligned} V &= [\vec{a} \vec{b} \vec{c}] \\ &= \begin{vmatrix} 1 & 1 & 0 \\ 1 & 2 & 0 \\ 1 & 1 & \pi \end{vmatrix} = \pi(1) = \pi \end{aligned}$$

7. If $\vec{a}$ and $\vec{b}$ are unit vectors such that $[\vec{a}, \vec{b}, \vec{a} \times \vec{b}] = \frac{1}{4}$, then the angle between $\vec{a}$ and $\vec{b}$ is

- (1) $\frac{\pi}{6}$ (2) $\frac{\pi}{4}$ (3) $\frac{\pi}{3}$ (4) $\frac{\pi}{2}$

$[\vec{a} \ \vec{b} \ \vec{a} \times \vec{b}] = \frac{1}{4}$ $(\vec{a} \times \vec{b}) \cdot (\vec{a} \times \vec{b}) = \frac{1}{4}$ $\Rightarrow (\vec{a} \times \vec{b})^2 = \frac{1}{4}$ $\text{i.e. } \vec{a} \times \vec{b} ^2 = \frac{1}{4} \Rightarrow \therefore \vec{a} \times \vec{b} = \frac{1}{2}$	$ab \sin \theta = \frac{1}{2}$ $(1)(1) \sin \theta = \frac{1}{2}$ $\sin \theta = \sin \frac{\pi}{6}$ $\theta = \frac{\pi}{6}$
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8. If $\vec{a} = \hat{i} + \hat{j} + \hat{k}$, $\vec{b} = \hat{i} + \hat{j}$, $\vec{c} = \hat{i}$ and $(\vec{a} \times \vec{b}) \times \vec{c} = \lambda \vec{a} + \mu \vec{b}$, then the value of $\lambda + \mu$ is

- (1) 0 (2) 1 (3) 6 (4) 3 PTA-3, JUL-20, 24

$(\vec{a} \times \vec{b}) \times \vec{c} = \lambda \vec{a} + \mu \vec{b}$ $(\vec{a} \cdot \vec{c}) \vec{b} - (\vec{b} \cdot \vec{c}) \vec{a} = \lambda \vec{a} + \mu \vec{b}$ $\mu = \vec{a} \cdot \vec{c} = 1; \lambda = -\vec{b} \cdot \vec{c} = -1$ $\therefore \lambda + \mu = -1 + 1 = 0$
--

9. If $\vec{a}, \vec{b}, \vec{c}$ are non-coplanar, non-zero vectors such that $[\vec{a}, \vec{b}, \vec{c}] = 3$, then $\{[\vec{a} \times \vec{b}, \vec{b} \times \vec{c}, \vec{c} \times \vec{a}]\}^2$ is equal to

- (1) 81 (2) 9 (3) 27 (4) 18

$[\vec{a} \ \vec{b} \ \vec{c}] = 3; \{[\vec{a} \times \vec{b} \ \vec{b} \times \vec{c} \ \vec{c} \times \vec{a}]\}^2 = \{[\vec{a} \ \vec{b} \ \vec{c}]^2\}^2 = [\vec{a} \ \vec{b} \ \vec{c}]^4 = 3^4 = 81$
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10. If $\vec{a}, \vec{b}, \vec{c}$ are three non-coplanar unit vectors such that $\vec{a} \times (\vec{b} \times \vec{c}) = \frac{\vec{b} + \vec{c}}{\sqrt{2}}$, then the angle between $\vec{a}$ and $\vec{b}$ is

- (1) $\frac{\pi}{2}$ (2) $\frac{3\pi}{4}$ (3) $\frac{\pi}{4}$ (4) π

$\vec{a} \times (\vec{b} \times \vec{c}) = \frac{\vec{b} + \vec{c}}{\sqrt{2}}$ $(\vec{a} \cdot \vec{c}) \vec{b} - (\vec{a} \cdot \vec{b}) \vec{c} = \frac{\vec{b}}{\sqrt{2}} + \frac{\vec{c}}{\sqrt{2}}$ $-(\vec{a} \cdot \vec{b}) = \frac{1}{\sqrt{2}} \Rightarrow \vec{a} \cdot \vec{b} = -\frac{1}{\sqrt{2}}$	$ab \cos \theta = -\frac{1}{\sqrt{2}}$ $(1)(1) \cos \theta = -\frac{1}{\sqrt{2}} = \cos \left(\pi - \frac{\pi}{4} \right)$ $\cos \theta = \cos \frac{3\pi}{4} \Rightarrow \theta = \frac{3\pi}{4}$
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11. If the volume of the parallelepiped with $\vec{a} \times \vec{b}$, $\vec{b} \times \vec{c}$, $\vec{c} \times \vec{a}$ as coterminous edges is 8 cubic units, then the volume of the parallelepiped with $(\vec{a} \times \vec{b}) \times (\vec{b} \times \vec{c})$, $(\vec{b} \times \vec{c}) \times (\vec{c} \times \vec{a})$ and $(\vec{c} \times \vec{a}) \times (\vec{a} \times \vec{b})$ as coterminous edges is,

- (1) 8 cubic units (2) 512 cubic units (3) 64 cubic units (4) 24 cubic units PTA-5

Given $V = [\vec{a} \times \vec{b} \ \vec{b} \times \vec{c} \ \vec{c} \times \vec{a}] = 8$ $[\vec{a} \ \vec{b} \ \vec{c}]^2 = 8$ $(\vec{a} \times \vec{b}) \times (\vec{b} \times \vec{c}) = [\vec{a} \ \vec{b} \ \vec{c}] \vec{b}$ $(\vec{b} \times \vec{c}) \times (\vec{c} \times \vec{a}) = [\vec{a} \ \vec{b} \ \vec{c}] \vec{c}$ $(\vec{c} \times \vec{a}) \times (\vec{a} \times \vec{b}) = [\vec{a} \ \vec{b} \ \vec{c}] \vec{a}$	$V = [(\vec{a} \ \vec{b} \ \vec{c}) \vec{b} \ (\vec{a} \ \vec{b} \ \vec{c}) \vec{c} \ (\vec{a} \ \vec{b} \ \vec{c}) \vec{a}]$ $= [\vec{a} \ \vec{b} \ \vec{c}]^3 [\vec{b} \ \vec{c} \ \vec{a}] = [\vec{a} \ \vec{b} \ \vec{c}]^3 [\vec{a} \ \vec{b} \ \vec{c}]$ $= [\vec{a} \ \vec{b} \ \vec{c}]^4$ $= \{[\vec{a} \ \vec{b} \ \vec{c}]^2\}^2$ $= 8^2 = 64$
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12. Consider the vectors $\vec{a}, \vec{b}, \vec{c}, \vec{d}$ such that $(\vec{a} \times \vec{b}) \times (\vec{c} \times \vec{d}) = \vec{0}$. Let P_1 and P_2 be the planes determined by the pairs of vectors $\vec{a}, \vec{b}$ and $\vec{c}, \vec{d}$ respectively. Then the angle between P_1 and P_2 is
 (1) 0° (2) 45° (3) 60° (4) 90°

$$\because P_1, P_2 \text{ lies in same plane} \quad (\vec{a} \times \vec{b}) \times (\vec{c} \times \vec{d}) = 0$$

$$\Rightarrow \vec{a} \times \vec{b} \text{ parallel } \vec{c} \times \vec{d} \quad \therefore \text{angle between } P_1 \text{ and } P_2 \text{ is } \theta = 0^\circ$$

13. If $\vec{a} \times (\vec{b} \times \vec{c}) = (\vec{a} \times \vec{b}) \times \vec{c}$, where $\vec{a}, \vec{b}, \vec{c}$ are any three vectors such that $\vec{b} \cdot \vec{c} \neq 0$ and $\vec{a} \cdot \vec{b} \neq 0$, then $\vec{a}$ and $\vec{c}$ are
 (1) perpendicular (2) parallel
 (3) inclined at an angle $\frac{\pi}{3}$ (4) inclined at an angle $\frac{\pi}{6}$

$$\vec{a} \times (\vec{b} \times \vec{c}) = (\vec{a} \times \vec{b}) \times \vec{c} \Rightarrow (\vec{a} \cdot \vec{c})\vec{b} - (\vec{a} \cdot \vec{b})\vec{c} = (\vec{a} \cdot \vec{c})\vec{b} - (\vec{b} \cdot \vec{c})\vec{a} \Rightarrow \vec{c} = \frac{\vec{b} \cdot \vec{c}}{\vec{a} \cdot \vec{b}} \vec{a}$$

$$\Rightarrow \vec{c} = \lambda \vec{a} \quad \therefore \vec{a}, \vec{c} \text{ are parallel}$$

14. If $\vec{a} = 2\hat{i} + 3\hat{j} - \hat{k}$, $\vec{b} = \hat{i} + 2\hat{j} - 5\hat{k}$, $\vec{c} = 3\hat{i} + 5\hat{j} - \hat{k}$, then a vector perpendicular to $\vec{a}$ and lies in the plane containing $\vec{b}$ and $\vec{c}$ is
 (1) $-17\hat{i} + 21\hat{j} - 97\hat{k}$ (2) $17\hat{i} + 21\hat{j} - 123\hat{k}$
 (3) $-17\hat{i} - 21\hat{j} + 97\hat{k}$ (4) $-17\hat{i} - 21\hat{j} - 97\hat{k}$

$$\vec{a} \cdot \vec{d} = (2\hat{i} + 3\hat{j} - \hat{k}) \cdot (-17\hat{i} - 21\hat{j} - 97\hat{k}) = -34 - 63 + 97 = 0$$

$$\therefore \vec{a} \perp \vec{d}, \text{ Also } [\vec{d} \vec{b} \vec{c}] = 0$$

$$\therefore \text{Required vector is } -17\hat{i} - 21\hat{j} - 97\hat{k}$$

15. The angle between the lines $\frac{x-2}{3} = \frac{y+1}{-2}, z = 2$ and $\frac{x-1}{1} = \frac{2y+3}{3} = \frac{z+5}{2}$ is
 (1) $\frac{\pi}{6}$ (2) $\frac{\pi}{4}$ (3) $\frac{\pi}{3}$ (4) $\frac{\pi}{2}$

APR-25, PTA-5

$$\frac{x-2}{3} = \frac{y+1}{-2}, \frac{z-2}{0}, \frac{x-1}{1} = \frac{y+\frac{3}{2}}{\frac{3}{2}} = \frac{z+5}{2}$$

$$\vec{n}_1 = 3\hat{i} - 2\hat{j}, \quad \vec{n}_2 = \hat{i} + \frac{3}{2}\hat{j} + 2\hat{k}$$

$$\vec{n}_1 \cdot \vec{n}_2 = 3(1) + (-2)\frac{3}{2} + (0)(2) = 3 - 3 + 0 = 0$$

$$\vec{n}_1 \perp \vec{n}_2 \quad \Rightarrow \theta = 90^\circ = \frac{\pi}{2}$$

16. If the line $\frac{x-2}{3} = \frac{y-1}{-5} = \frac{z+2}{2}$ lies in the plane $x + 3y - \alpha z + \beta = 0$, then (α, β) is
 (1) $(-5, 5)$ (2) $(-6, 7)$ (3) $(5, -5)$ (4) $(6, -7)$

MDL

$$\vec{b} = 3\hat{i} - 5\hat{j} + 2\hat{k}, \quad \vec{n} = \hat{i} + 3\hat{j} - \alpha\hat{k}$$

$$\vec{b} \cdot \vec{n} = 0$$

$$3 - 15 - 2\alpha = 0$$

$$-2\alpha = 12$$

$$\alpha = -6$$

$$(2, 1, -2) \text{ lies in } x + 3y - \alpha z + \beta = 0$$

$$2 + 3 - 12 + \beta = 0$$

$$\beta = 7$$

$$(\alpha, \beta) = (-6, 7)$$

17. The angle between the line $\vec{r} = (\hat{i} + 2\hat{j} - 3\hat{k}) + t(2\hat{i} + \hat{j} - 2\hat{k})$ and the plane $\vec{r} \cdot (\hat{i} + \hat{j}) + 4 = 0$ is

- (1) 0° (2) 30° (3) 45° (4) 90° APR-24, PTA-2

$$\vec{b} = 2\hat{i} + \hat{j} - 2\hat{k}; \quad \vec{n} = \hat{i} + \hat{j}$$

$$\vec{b} \cdot \vec{n} = 2 + 1 = 3, \quad |\vec{b}| = 3 \quad |\vec{n}| = \sqrt{2}$$

$$\sin \theta = \frac{3}{3 \times \sqrt{2}} = \frac{1}{\sqrt{2}} = \sin \frac{\pi}{4} \Rightarrow \theta = \frac{\pi}{4} = 45^\circ$$

18. The coordinates of the point where the line $\vec{r} = (6\hat{i} - \hat{j} - 3\hat{k}) + t(-\hat{i} + 4\hat{k})$ meets the plane $\vec{r} \cdot (\hat{i} + \hat{j} - \hat{k}) = 3$ are JUL-20

- (1) (2, 1, 0) (2) (7, -1, -7) (3) (1, 2, -6) (4) (5, -1, 1)

$$\text{Line } \frac{x-6}{-1} = \frac{y+1}{0} = \frac{z+3}{4} = \lambda$$

$$\text{any point } (-\lambda + 6, -1, 4\lambda - 3) \dots\dots\dots (1)$$

$$\text{This lies on Plane } x + y - z = 3$$

$$\Rightarrow -\lambda + 6 - 1 - 4\lambda + 3 - 3 = 0$$

$$-5\lambda = -5$$

$$\lambda = 1$$

$$(1) \Rightarrow \text{Point} = (-1 + 6, -1, 4 - 3) = (5, -1, 1)$$

19. Distance from the origin to the plane $3x - 6y + 2z + 7 = 0$ is

PTA-2, JUL-22, 24, APR-23

- (1) 0 (2) 1 (3) 2 (4) 3

$$D = \frac{|d|}{\sqrt{a^2+b^2+c^2}} = \frac{|7|}{\sqrt{9+36+4}} = \frac{7}{7} = 1$$

20. The distance between the planes $x + 2y + 3z + 7 = 0$ and $2x + 4y + 6z + 7 = 0$ is

- (1) $\frac{\sqrt{7}}{2\sqrt{2}}$ (2) $\frac{7}{2}$ (3) $\frac{\sqrt{7}}{2}$ (4) $\frac{7}{2\sqrt{2}}$ APR-20

$$x + 2y + 3z + 7 = 0; \quad x + 2y + 3z + \frac{7}{2} = 0$$

$$D = \frac{|d_1 - d_2|}{\sqrt{a^2+b^2+c^2}} = \frac{|7 - \frac{7}{2}|}{\sqrt{1+4+9}} = \frac{|\frac{7}{2}|}{\sqrt{14}} = \frac{\sqrt{7}\sqrt{7}}{2\sqrt{2}\sqrt{7}} = \frac{\sqrt{7}}{2\sqrt{2}}$$

21. If the direction cosines of a line are $\frac{1}{c}, \frac{1}{c}, \frac{1}{c}$, then

- (1) $c = \pm 3$ (2) $c = \pm \sqrt{3}$ (3) $c > 0$ (4) $0 < c < 1$

$$\text{W.K.T } \cos^2 \alpha + \cos^2 \beta + \cos^2 \gamma = 1$$

$$\frac{1}{c^2} + \frac{1}{c^2} + \frac{1}{c^2} = 1$$

$$\frac{3}{c^2} = 1 \Rightarrow c^2 = 3$$

$$c = \pm \sqrt{3}$$

22. The vector equation $\vec{r} = (\hat{i} - 2\hat{j} - \hat{k}) + t(6\hat{j} - \hat{k})$ represents a straight line passing through the points

(1) $(0, 6, -1)$ and $(1, -2, -1)$

(2) $(0, 6, -1)$ and $(-1, -4, -2)$

(3) $(1, -2, -1)$ and $(1, 4, -2)$

(4) $(1, -2, -1)$ and $(0, -6, 1)$

Cartesian form $\frac{x-1}{0} = \frac{y+2}{6} = \frac{z+1}{-1} = \lambda$

any point $(1, 6\lambda - 2, -\lambda - 1)$

Put $\lambda = 0$ $(1, -2, -1)$

$\lambda = 1$ $(1, 4, -2)$

23. If the distance of the point $(1, 1, 1)$ from the origin is half of its distance from the plane $x + y + z + k = 0$, then the values of k are

PTA-3

(1) ± 3

(2) ± 6

(3) $-3, 9$

(4) $3, -9$

Distance between $(0, 0, 0)$ and $(1, 1, 1) = \frac{1}{2}$ (Distance of the plane $x + y + z + k = 0$ from $(1, 1, 1)$)

$$\sqrt{1+1+1} = \frac{1}{2} \left| \frac{x_1 + y_1 + z_1 + k}{\sqrt{1+1+1}} \right|$$

$$\sqrt{3} = \frac{1}{2} \left| \frac{1+1+1+k}{\sqrt{3}} \right|$$

$$2\sqrt{3} = \left| \frac{3+k}{\sqrt{3}} \right|$$

$$= \pm \frac{(3+k)}{\sqrt{3}}$$

$$\pm 2\sqrt{3} \times \sqrt{3} = 3 + k$$

$$\pm 6 = 3 + k$$

$$6 = 3 + k$$

$$3 = k$$

$$-6 = 3 + k$$

$$-6 - 3 = k$$

$$-9 = k$$

$$\therefore k = 3, -9$$

24. If the planes $\vec{r} \cdot (2\hat{i} - \lambda\hat{j} + \hat{k}) = 3$ and $\vec{r} \cdot (4\hat{i} + \hat{j} - \mu\hat{k}) = 5$ are parallel, then the value of λ and μ are

PTA-4, JUL-21

(1) $\frac{1}{2}, -2$

(2) $-\frac{1}{2}, 2$

(3) $-\frac{1}{2}, -2$

(4) $\frac{1}{2}, 2$

$\therefore$ the planes are parallel

$$\vec{n}_1 = 2\hat{i} - \lambda\hat{j} - \hat{k}$$

$$\vec{n}_2 = 4\hat{i} + \hat{j} - \mu\hat{k} \text{ are parallel}$$

$$\frac{4}{2} = \frac{1}{-\lambda} = \frac{-\mu}{1}$$

$$2 = \frac{-1}{\lambda} = -\mu$$

$$2 = \frac{-1}{\lambda}$$

$$\lambda = \frac{-1}{2}$$

$$2 = -\mu$$

$$\mu = -2$$

25. If the length of the perpendicular from the origin to the plane $2x + 3y + \lambda z = 1, \lambda > 0$ is $\frac{1}{5}$, then the value of λ is

(1) $2\sqrt{3}$

(2) $3\sqrt{2}$

(3) 0

(4) 1

$$2x + 3y + \lambda z - 1 = 0$$

Given $\left| \frac{-1}{\sqrt{4+9+\lambda^2}} \right| = \frac{1}{5}$

$$\frac{1}{\sqrt{13+\lambda^2}} = \frac{1}{5}$$

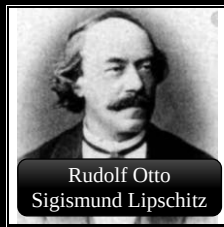
$$5 = \sqrt{13 + \lambda^2}$$

$$25 = 13 + \lambda^2$$

$$\lambda^2 = 12$$

$$\lambda^2 = 4 \times 3$$

$$\Rightarrow \lambda = \pm 2\sqrt{3}$$



Chapter - 7

Applications of Differential Calculus

EXERCISE 7.1



Concept Corner

Average and instantaneous rate of change Slope or Gradient of a curve:

1. For the function $y = f(x)$

(i) Points $(x, f(x))$ and $(x + h, f(x + h))$ slope of line $\frac{f(x+h) - f(x)}{h}$

Taking the limit $h \rightarrow 0$ as $\lim_{h \rightarrow 0} \frac{f(x+h) - f(x)}{h} = f'(x)$

(ii) The average rate of change in an interval $[a, b] = \frac{f(b) - f(a)}{b - a}$

(iii) The instantaneous rate of change of with respect to $x = f'(x) = \frac{dy}{dx}$

2. Derivative as rate of change

(i) The rate of change of y with respect to $x = \frac{dy}{dx}$.

(ii) The rate of change of the distance (s) with respect to time (t) $= \frac{ds}{dt}$

3. Velocity and acceleration

If $s = f(t)$ is a position of an object as a function of time t , then

(i) Velocity $= v(t) = \frac{ds}{dt}$

(ii) Acceleration $= a(t) = \frac{dv}{dt} = \frac{d^2s}{dt^2}$

(iii) Speed = absolute value of velocity $= |v(t)| = \left| \frac{ds}{dt} \right|$

Note :

- ❖ Initial velocity means velocity at $t = 0$
- ❖ Initial acceleration means acceleration at $t = 0$
- ❖ If the motion is upward at the maximum height, velocity $v = 0$
- ❖ If the motion is horizontal velocity $v = 0$ when the particle come to rest.

1. A particle moves along a straight line in such a way that after t seconds its distance from the origin is $s = 2t^2 + 3t$ metres. (i) Find the average velocity between $t = 3$ and $t = 6$ seconds. (ii) Find the instantaneous velocities at $t = 3$ and $t = 6$ seconds.

3M

$$s(t) = 2t^2 + 3t$$

(i) When $t = 3\text{sec}$

$$s(3) = 2(3)^2 + 3(3) = 18 + 9 = 27$$

When $t = 6\text{sec}$

$$s(6) = 2(6)^2 + 3(6) = 72 + 18 = 90$$

$$\begin{aligned} \text{Average velocity} &= \frac{s(6) - s(3)}{6 - 3} \\ &= \frac{90 - 27}{3} = \frac{63}{3} = 21\text{m/sec} \end{aligned}$$

(ii) $s(t) = 2t^2 + 3t$ and $s'(t) = 4t + 3$

Instantaneous velocity at $t = 3\text{sec} = s'(3)$

$$= 4(3) + 3$$

$$= 12 + 3$$

$$= 15\text{m/sec}$$

Instantaneous velocity at $t = 6\text{sec} = s'(6)$

$$= 4(6) + 3$$

$$= 27\text{m/sec}$$

2. A camera is accidentally knocked off an edge of a cliff 400ft high. The camera falls a distance of $s = 16t^2$ in t seconds. (i) How long does the camera fall before it hits the ground? (ii) What is the average velocity with which the camera falls during the last 2 seconds? (iii) What is the instantaneous velocity of the camera when it hits the ground? JUL-21 5M

(i) $s = 16t^2$ When $s = 400$ $16t^2 = 400$ $t = \frac{400}{16} = 25$ $t = \pm 5$ Since, time is always +ve, $t = 5\text{sec}$ $\therefore$ The camera falls 5 seconds before it hits the ground.	(ii) Required average velocity = $\frac{s(5)-s(3)}{5-3}$ $= \frac{16(5^2)-16(3^2)}{2}$ $= \frac{400-144}{2}$ $= \frac{256}{2}$ = 128 ft/sec	(iii) $s(t) = 16t^2$ $s'(t) = 32t$ When $t = 5\text{sec}$ $s'(5) = 32(5)$ $= 160 \text{ ft/sec}$ $\therefore$ The instantaneous velocity of the camera when it hits the ground = 160 ft/sec
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3. A particle moves along a line according to the law $s(t) = 2t^3 - 9t^2 + 12t - 4$, where $t \geq 0$ (i) At what times the particle changes direction? (ii) Find the total distance travelled by the particle in the first 4 seconds. (iii) Find the particle's acceleration each time the velocity is zero. JUL-23 5M

(i) $s(t) = 2t^3 - 9t^2 + 12t - 4$

$$s'(t) = 6t^2 - 18t + 12$$

The particle changes direction when $s'(t) = 0$

$$\Rightarrow 6t^2 - 18t + 12 = 0$$

$$\Rightarrow t^2 - 3t + 2 = 0 \quad [\div 6]$$

$$\Rightarrow (t-1)(t-2) = 0$$

$$t = 1 \text{ or } t = 2$$

$\therefore$ The particle changes direction at

$t = 1 \text{ sec}$ & $t = 2 \text{ seconds}$.

(iii) $s(t) = 2t^3 - 9t^2 + 12t - 4$

$$s'(t) = 6t^2 - 18t + 12$$

$$s''(t) = 12t - 18$$

When $s'(t) = 0$, $6t^2 - 18t + 12 = 0$

$$\Rightarrow t^2 - 3t + 2 = 0 \quad [\div 6]$$

$$(t-1)(t-2) = 0,$$

$$t = 1 \text{ or } t = 2$$

Velocity is zero when, $t = 1 \text{ sec}$ & $t = 2 \text{ sec}$

At $t = 1 \text{ sec}$

$$s''(t) = s''(1) = 12(1) - 18 = -6\text{m/sec}^2$$

At $t = 2 \text{ sec}$

$$s''(t) = s''(2) = 12(2) - 18 = 24 - 18 = 6\text{m/sec}^2$$

(ii) At $t = 0$,

$$s(0) = 0 - 0 + 0 - 4 = -4$$

At $t = 1$,

$$s(1) = 2 - 9 + 12 - 4$$

$$= 14 - 13$$

$$= 1$$

At $t = 2$,

$$s(2) = 2(2^3) - 9(2^2) + 12(2) - 4$$

$$= 16 - 36 + 24 - 4$$

$$= 40 - 40$$

$$= 0$$

At $t = 4$,

$$s(4) = 2(4^3) - 9(4^2) + 12(4) - 4$$

$$= 128 - 144 + 48 - 4$$

$$s(4) = 176 - 148$$

$$= 28$$

Total distance travelled in the first 4 seconds

$$= |s(1) - s(0)| + |s(2) - s(1)| + |s(4) - s(2)|$$

$$= |1 - (-4)| + |0 - 1| + |28 - 0|$$

$$= 5 + 1 + 28$$

$$= \mathbf{34m}$$

4. If the volume of a cube of side length x is $v = x^3$. Find the rate of change of the volume with respect to x when $x = 5$ units.

$$v = x^3$$

2M

$$\frac{dv}{dx} = 3x^2$$

$$\text{When } x = 5, \frac{dv}{dx} = 3(5^2)$$

$$= 3 \times 25 = 75$$

$\therefore$ The rate of change of volume w. r. to x when $x = 5$ units is 75 cubic units.

5. If the mass $m(x)$ (in kilograms) of a thin rod of length x (in metres) is given by, $m(x) = \sqrt{3x}$ then what is the rate of change of mass with respect to the length When it is $x = 3$ and $x = 27$ metres.

2M

$$m(x) = \sqrt{3x} = \sqrt{3} \sqrt{x}$$

MDL

$$m'(x) = \sqrt{3} \frac{1}{2\sqrt{x}} = \frac{\sqrt{3}}{2\sqrt{x}}$$

(i) When $x = 3$,

$$m'(3) = \frac{\sqrt{3}}{2\sqrt{3}} = \frac{1}{2} \text{ kg/m}$$

(ii) When $x = 27$

$$m'(27) = \frac{\sqrt{3}}{2\sqrt{27}} = \frac{\sqrt{3}}{2 \times 3\sqrt{3}} = \frac{1}{6} \text{ kg/m}$$

6. A stone is dropped into a pond causing ripples in the form of concentric circles. The radius r of the outer ripple is increasing at a constant rate at 2cm per second. When the radius is 5cm find the rate of changing of the total area of the disturbed water?

2M

$$\text{Area of the disturbed water} = A = \pi r^2$$

The rate of change of the area with respect to $t = \frac{dA}{dt} = 2\pi r \frac{dr}{dt}$

$$\text{When } r = 5 \text{ and } \frac{dr}{dt} = 2 \text{ cm/sec}$$

$$\frac{dA}{dt} = 2\pi \times 5 \times 2$$

$$= 20\pi \text{ cm}^2/\text{sec}$$

7. A beacon makes one revolution every 10 seconds. It is located on a ship which is anchored 5km from a straight shore line. How fast is the beam moving along the shore line when it makes an angle of 45° with the shore?

BC = Straight line shore line

A = Location of the beacon on a ship

Let $\angle BAC = \theta$ be the angle of revolution of the beacon at any time t .

Let $BC = x \text{ km}$

Given that the beacon makes one revolution every 10 seconds

(ie) Number of revolution made by the beacon in 10sec = 2π

Number of revolution made by the

$$\text{beacon in 1sec} = \frac{2\pi}{10} = \frac{\pi}{5}$$

$$\therefore \frac{d\theta}{dt} = \frac{\pi}{5}$$

$$\text{In } \triangle ABC, \tan \theta = \frac{x}{5}$$

$$\Rightarrow x = 5 \tan \theta$$

$$\frac{dx}{dt} = 5 \sec^2 \theta \frac{d\theta}{dt}$$

When $\theta = 45^\circ$

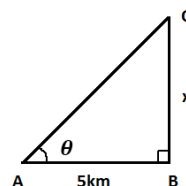
$$\frac{dx}{dt} = 5 \sec^2(45^\circ) \times \frac{\pi}{5}$$

$$= \frac{1}{\left(\frac{1}{\sqrt{2}}\right)^2} \times \pi$$

$$= (\sqrt{2})^2 \times \pi$$

$$= 2\pi \text{ km/sec}$$

$\therefore$ The beam is moving along the shore line at the rate of $2\pi \text{ km/sec}$ when $\theta = 45^\circ$



8. A conical water tank with vertex down of 12 metres height has a radius of 5 metres at the top. If water flows into the tank at a rate 10 cubic m/min , how fast is the depth of the water increases when the water is 8 metres deep?

Let r , h and v be the radius, height and volume at any time 't' minutes.

$$V = \frac{1}{3}\pi r^2 h \dots\dots\dots (1)$$

5M

Since $\triangle OAB \sim \triangle OCD$,

$$\frac{r}{5} = \frac{h}{12}$$

$$\Rightarrow r = \frac{5h}{12}$$

$$(1) \Rightarrow v = \frac{1}{3}\pi \left(\frac{5h}{12}\right)^2 h$$

$$v = \frac{\pi}{3} \left(\frac{25h^2}{144}\right) h$$

$$v = \frac{25\pi}{3 \times 144} h^3$$

$$\frac{dv}{dt} = \frac{25\pi}{3 \times 144} \times 3h^2 \frac{dh}{dt}$$

$$\frac{dv}{dt} = \frac{25\pi}{144} \times h^2 \frac{dh}{dt}$$

When $h = 8\text{m}$ and $\frac{dv}{dt} = 10\text{m}^3/\text{min}$

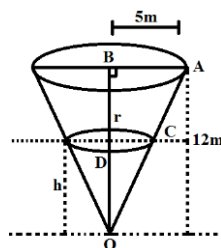
$$10 = \frac{25\pi}{144} \times 8 \times 8 \times \frac{dh}{dt}$$

$$10 = \frac{100\pi}{9} \times \frac{dh}{dt}$$

$$\Rightarrow \frac{dh}{dt} = 10 \times \frac{9}{100\pi}$$

$$= \frac{9}{10\pi}$$

$\therefore$ The depth of water increases at the rate of $\frac{9}{10\pi} \text{ m/min}$



9. A ladder 17 metre long is leaning against the wall. The base of the ladder is pulled away from the wall at a rate of 5 m/sec. when the base of the ladder is 8 metres from the wall,

(i) How fast is the top of the ladder moving down the wall?

(ii) At what rate, the area of the triangle formed by the ladder, wall and the floor, is changing?

5M

- (i) $AB = \text{Length of the ladder} = 17\text{m}$

At any time $t \text{ sec}$, let $AC = x$ and $BC = y$

In $\triangle ABC$, by Pythagoras theorem

$$AC^2 + BC^2 = AB^2$$

$$\Rightarrow x^2 + y^2 = 289$$

When $x = 8\text{m}$

$$8^2 + y^2 = 289$$

$$y^2 = 289 - 64 = 225$$

$$y = 15\text{m}$$

$$x^2 + y^2 = 289$$

Differentiating w. r to x , we get

$$2x \frac{dx}{dt} + 2y \frac{dy}{dt} = 0$$

$$\div \text{ by } 2 \Rightarrow x \frac{dx}{dt} + y \frac{dy}{dt} = 0$$

When $x = 8$, $\frac{dx}{dt} = 5\text{m/s}$ and $y = 15\text{m}$

$$8(5) + 15 \frac{dy}{dt} = 0$$

$$\frac{dy}{dt} = -\frac{40}{15} = -\frac{8}{3}$$

$\therefore$ The top of the ladder moving down the wall at the rate of $-\frac{8}{3} \text{ m/sec}$

- (ii) Area of the $\triangle ABC$ at any time $t \text{ sec}$

$$A = \frac{1}{2} \times \text{base} \times \text{height}$$

$$A = \frac{1}{2} xy$$

$$\frac{dA}{dt} = \frac{1}{2} \left[x \frac{dy}{dt} + y \frac{dx}{dt} \right]$$

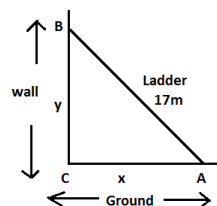
$$= \frac{1}{2} \left[8 \left(-\frac{8}{3} \right) + 5(15) \right]$$

$$= \frac{1}{2} \left[-\frac{64}{3} + 75 \right] = \frac{1}{2} \left[\frac{-64 + 225}{3} \right]$$

$$= \frac{1}{2} \left[\frac{161}{3} \right] = \frac{161}{6}$$

$$= 26.83 \text{ m}^2/\text{sec}$$

$\therefore$ Area of the triangle is changing at the rate of $26.83 \text{ m}^2/\text{sec}$.



10. A police jeep, approaching an orthogonal intersection from the northern direction, is chasing a speeding car that has turned and moving straight east. When the jeep is 0.6km north of the intersection and the car is 0.8km to the east. The police determine with a radar that the distance between them and the car is increasing at 20km/hr. if the jeep is moving at 60km/hr at the instant of measurement, what is the speed of the car?

APR-20

5M

At any time t hrs, let

A = position of the car in the eastern direction

B = position of the car in the northern direction

C = point of intersection of two roads.

AC = x km ; BC = y km and AB = z km

Speed of the jeep = $\frac{dy}{dt} = -60$ km/hr

The rate of change of the distance between the jeep and the car = $\frac{dz}{dt} = 20$ km/hr.

Speed of the car = $\frac{dx}{dt}$ km/hr

By the Pythagore's Theorem,

$$x^2 + y^2 = z^2$$

When $x = 0.8$ and $y = 0.6$

$$0.8^2 + 0.6^2 = z^2$$

$$z^2 = 0.64 + 0.36$$

$$z^2 = 1$$

$$\therefore z = 1 \text{ km}$$

$$x^2 + y^2 = z^2$$

Diff. w. r. to x , we get

$$2x \frac{dx}{dt} + 2y \frac{dy}{dt} = 2z \frac{dz}{dt}$$

Divide by 2

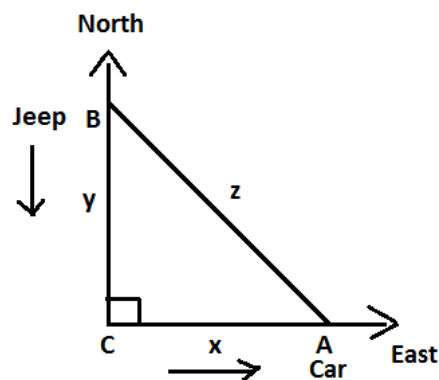
$$\Rightarrow x \frac{dx}{dt} + y \frac{dy}{dt} = z \frac{dz}{dt}$$

$$(0.8) \frac{dx}{dt} + (0.6)(-60) = 20(1)$$

$$(0.8) \frac{dx}{dt} = 20 + 36 = 56$$

$$\frac{dx}{dt} = \frac{56}{0.8} = \frac{560}{8} = 70 \text{ km/hr}$$

$\therefore$ Speed of the car = **70 km/hr.**



EXERCISE 7.2



Video class

Concept Corner

Equation of Tangent and Normal

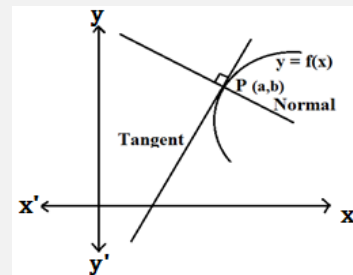
1. If $y = f(x)$ represents a curve,

(a) Slope of the tangent at $(a, b) = \left(\frac{dy}{dx}\right)_{(a,b)}$

(b) Slope of the normal at $(a, b) = -\frac{1}{\left(\frac{dy}{dx}\right)_{(a,b)}}$

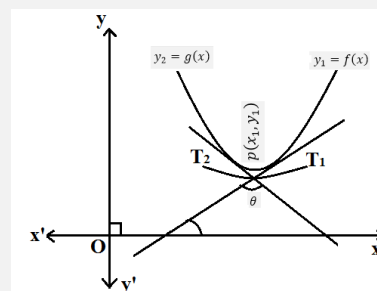
(c) Equation of the tangent at (a, b) is $(y - b) = \left(\frac{dy}{dx}\right)_{(a,b)}(x - a)$

(d) Equation of the normal at (a, b) is $(y - b) = -\frac{1}{\left(\frac{dy}{dx}\right)_{(a,b)}}(x - a)$

2. In the adjoining figure, two curves $y_1 = f(x)$ and $y_2 = g(x)$ intersect at $p(x_1, y_1)$ PT_1 = Tangent to the curve $y_1 = f(x)$ PT_2 = Tangent to the curve $y_2 = g(x)$

(a) Angle between two curves

$$y_1 = f(x) \text{ and } y_2 = g(x) \left\} = \tan \theta = \left| \frac{m_1 - m_2}{1 + m_1 m_2} \right|, \text{ Where}$$

 m_1 = Slope of the tangent to the curve $y_1 = f(x)$ at $p(x_1, y_1)$ m_2 = Slope of the tangent to the curve $y_2 = g(x)$ at $p(x_1, y_1)$ (b) If the two curves are parallel at $p(x_1, y_1)$ then $m_1 = m_2$ (c) If the two curves cut orthogonally at $p(x_1, y_1)$ then $m_1 m_2 = -1$ 

1. Find the slope of the tangent to the following curves at the respective given points.

(i) $y = x^4 + 2x^2 - x$ at $x = 1$, 2M

(ii) $x = a \cos^3 t, y = b \sin^3 t$ at $t = \frac{\pi}{2}$

(i) $y = x^4 + 2x^2 - x$

$$\frac{dy}{dx} = 4x^3 + 4x - 1$$

Slope of the tangent (at $x = 1$) = $\left(\frac{dy}{dx}\right)_{(x=1)}$

$$= 4(1)^3 + 4(1) - 1$$

$$= 8 - 1 = 7$$

(ii) $x = a \cos^3 t$

$$\frac{dx}{dt} = 3a \cos^2 t (-\sin t) = -3a \cos^2 t \sin t$$

$$y = b \sin^3 t \Rightarrow \frac{dy}{dt} = 3b \sin^2 t \cos t$$

$$\frac{dy}{dx} = \frac{\frac{dy}{dt}}{\frac{dx}{dt}} = \frac{3b \sin^2 t \cos t}{-3a \cos^2 t \sin t} = -\frac{b \sin t}{a \cos t}$$

$$\frac{dy}{dx} = -\frac{b}{a} \tan t$$

Slope of the tangent (at $t = \frac{\pi}{2}$) = $-\frac{b}{a} \tan \frac{\pi}{2}$

$$= -\frac{b}{a}(\infty) = \infty$$

2. Find the point on the curve $y = x^2 - 5x + 4$ at which the tangent is parallel to the line $3x + y = 7$. JUL-21

Given: $y = x^2 - 5x + 4$ (1)

$$\frac{dy}{dx} = 2x - 5$$

Slope of the line $3x + y = 7$

$$y = -3x + 7$$

$$m = \frac{-3}{1} = -3$$

Since the tangent to the curve is parallel to the line $3x + y = 7$

$$2x - 5 = -3$$

$$2x - 5 + 3 = 0$$

$$2x - 2 = 0$$

$$2x = 2 \Rightarrow x = 1$$

(1) $\Rightarrow y = 1^2 - 5(1) + 4 = 5 - 5 = 0$

 $\therefore$ Required point is $(1, 0)$

3. Find the points on the curve $y = x^3 - 6x^2 + x + 3$ where the normal is parallel to the line $x + y = 1729$.

Given:

PTA-1

$$y = x^3 - 6x^2 + x + 3 \dots (1)$$

3M

$$\frac{dy}{dx} = 3x^2 - 12x + 1$$

$$\text{Slope of the line } x + y = 1729$$

$$y = -x + 1729$$

$$m_1 = -\frac{1}{1} = -1$$

$$m_1 \times m_2 = -1$$

$$-1 \times m_2 = -1$$

$$m_2 = 1$$

Since the normal to the curve is parallel to the line $x + y = 1729$

$$3x^2 - 12x + 1 = 1$$

$$3x^2 - 12x = 0$$

$$3x(x - 4) = 0$$

$$3x = 0 \text{ (or) } x - 4 = 0$$

$$x = 0 \text{ (or) } x = 4$$

When $x = 0$,

$$(1) \Rightarrow y = 0 - 0 + 0 + 3 = 3$$

When $x = 4$

$$(1) \Rightarrow y = 4^3 - 6(4)^2 + 4 + 3$$

$$= 64 - 96 + 4 + 3$$

$$= 71 - 96$$

$$= -25$$

$\therefore$ Required points are **(0, 3)** and **(4, -25)**

4. Find the points on the curve $y^2 - 4xy = x^2 + 5$ for which the tangent is horizontal.

3M

The tangent is horizontal is $m = 0$

$$y^2 - 4xy = x^2 + 5 \dots (1)$$

Differentiating w. r. to x , we get

$$2yy' - 4[xy' + y(1)] = 2x$$

$\div$ by 2

$$\Rightarrow yy' - 2[xy' + y] = x$$

$$yy' - 2xy' - 2y = x$$

$$(y - 2x)y' = x + 2y$$

$$\therefore y' = \frac{x+2y}{y-2x}$$

$$m = 0$$

$$\frac{x+2y}{y-2x} = 0$$

$$\Rightarrow x + 2y = 0$$

$$\therefore x = -2y \dots (2)$$

$$(1) \Rightarrow y - 4(-2y)y = (-2y)^2 + 5$$

$$y^2 + 8y^2 = 4y^2 + 5$$

$$9y^2 - 4y^2 = 5$$

$$5y^2 = 5$$

$$\therefore y^2 = 1$$

$$y = \pm 1$$

When $y = 1$

$$(2) \Rightarrow x = -2(1)$$

$$x = -2$$

when $y = -1$

$$(2) \Rightarrow x = -2(-1)$$

$$x = 2$$

$\therefore$ Required points are **(-2, 1)** and **(2, -1)**.

5. Find the tangent and normal to the following curves at the given points on the curve.

- (i) $y = x^2 - x^4$ at **(1, 0)**

2M

JUL-22

$$y = x^2 - x^4$$

$$\frac{dy}{dx} = 2x - 4x^3$$

$$m = \text{Slope of the tangent at } (1,0) = 2(1) - 4(1)^3 = 2 - 4 = -2$$

$$\text{Equation of the tangent at } (1,0) \quad y - y_1 = m(x - x_1)$$

$$y - 0 = -2(x - 1) \Rightarrow y = -2x + 2$$

$$2x + y - 2 = 0 \Rightarrow 2x + y = 2$$

$$\text{Equation of the normal at } (1,0), \quad y - y_1 = -\frac{1}{m}(x - x_1)$$

$$y - 0 = -\frac{1}{-2}(x - 1) \Rightarrow 2y = x - 1$$

$$x - 2y - 1 = 0 \Rightarrow x - 2y = 1$$

(ii) $y = x^4 + 2e^x$ at $(0, 2)$

3M

$$y = x^4 + 2e^x$$

$$\frac{dy}{dx} = 4x^3 + 2e^x$$

$$m = \left. \begin{array}{l} \text{Slope of the tangent} \\ \text{at } (0, 2) \end{array} \right\} = 4(0)^3 + 2e^0$$

$$= 0 + 2(1) = 2$$

$$m = 2$$

Equation of the tangent at $(0, 2)$

$$y - y_1 = m(x - x_1)$$

$$y - 2 = 2(x - 0)$$

$$y - 2 = 2x$$

$$2x - y + 2 = 0 \Rightarrow 2x - y = -2$$

Equation of the normal at $(0, 2)$

$$y - y_1 = -\frac{1}{m}(x - x_1)$$

$$y - 2 = -\frac{1}{2}(x - 0)$$

$$2y - 4 = -x$$

$$x + 2y - 4 = 0 \Rightarrow x + 2y = 4$$

(iii) $y = x \sin x$ at $\left(\frac{\pi}{2}, \frac{\pi}{2}\right)$

3M

$$y = x \sin x$$

$$\frac{dy}{dx} = x \cos x + \sin x(1)$$

$$m = \left. \begin{array}{l} \text{Slope of the tangent} \\ \text{at } \left(\frac{\pi}{2}, \frac{\pi}{2}\right) \end{array} \right\} = \frac{\pi}{2} \cos \frac{\pi}{2} + \sin \frac{\pi}{2}$$

$$= 0 + 1$$

$$m = 1$$

Equation of the tangent at $\left(\frac{\pi}{2}, \frac{\pi}{2}\right)$

$$y - y_1 = m(x - x_1)$$

$$y - \frac{\pi}{2} = 1 \left(x - \frac{\pi}{2}\right)$$

$$\frac{2y - \pi}{2} = \frac{2x - \pi}{2}$$

$$2y - \pi = 2x - \pi$$

$$\Rightarrow y = x$$

$$x - y = 0$$

Equation of the normal at $\left(\frac{\pi}{2}, \frac{\pi}{2}\right)$

$$y - y_1 = -\frac{1}{m}(x - x_1)$$

$$y - \frac{\pi}{2} = -\frac{1}{1} \left(x - \frac{\pi}{2}\right)$$

$$y - \frac{\pi}{2} = -x + \frac{\pi}{2}$$

$$x + y - \frac{\pi}{2} - \frac{\pi}{2} = 0$$

$$x + y - \pi = 0 \Rightarrow x + y = \pi$$

(iv) $x = \cos t, y = 2\sin^2 t$ at $t = \frac{\pi}{3}$

5M

$$x = \cos t \quad y = 2\sin^2 t$$

$$\frac{dx}{dt} = -\sin t \quad \frac{dy}{dt} = 4\sin t \cos t$$

$$\frac{dy}{dx} = \frac{\frac{dy}{dt}}{\frac{dx}{dt}} = \frac{4\sin t \cos t}{-\sin t} = -4\cos t$$

$$m = \left. \begin{array}{l} \text{Slope of the tangent} \\ \text{at } t = \frac{\pi}{3} \end{array} \right\} = -4\cos \frac{\pi}{3}$$

$$= -4\cos \frac{\pi}{3} = -4\left(\frac{1}{2}\right) = -2$$

$$m = -2$$

$$\text{At } t = \frac{\pi}{3}, x = \cos \frac{\pi}{3} = \frac{1}{2}$$

$$y = 2\sin^2 \frac{\pi}{3} = 2\left(\frac{\sqrt{3}}{2}\right)^2 = 2 \times \frac{3}{4} = \frac{3}{2}$$

Equation of the tangent at $t = \frac{\pi}{3}$ at $\left(\frac{1}{2}, \frac{3}{2}\right)$

$$y - y_1 = m(x - x_1)$$

$$y - \frac{3}{2} = -2 \left(x - \frac{1}{2}\right)$$

$$\frac{2y - 3}{2} = -2 \left[\frac{2x - 1}{2}\right]$$

$$2y - 3 = -4x + 2$$

$$4x + 2y = 2 + 3$$

$$4x + 2y = 5$$

Equation of the normal at $\left(\frac{1}{2}, \frac{3}{2}\right)$

$$y - y_1 = \frac{1}{m}(x - x_1)$$

$$y - \frac{3}{2} = -\frac{1}{-2} \left(x - \frac{1}{2}\right)$$

$$\frac{2y - 3}{2} = \frac{1}{2} \left(\frac{2x - 1}{2}\right)$$

$$2(2y - 3) = 2x - 1$$

$$4y - 6 = 2x - 1$$

$$1 - 6 = 2x - 4y$$

$$2x - 4y = -5$$

6. Find the equations of the tangents to the curve $y = 1 + x^3$ for which the tangent is orthogonal with the line $x + 12y = 12$. 5M

Slope of the tangent $x + 12y = 12$

$$y = -\frac{1}{12}x + 1$$

$$m_2 = -\frac{1}{12}$$

$$y = 1 + x^3$$

$$y' = m_1 = 3x^2 \dots\dots\dots (1)$$

The tangent to the curve is

orthogonal with the line $x + 12y = 12$

$$\therefore m_1 m_2 = -1$$

$$(3x^2) \left(-\frac{1}{12} \right) = -1$$

$$3x^2 = 12$$

$$x^2 = 4$$

$$x = \pm 2$$

At $x = 2$	At $x_1 = -2$
(1) $\Rightarrow y_1 = 1 + 2^3$	(1) $\Rightarrow y = 1 + (-2)^3$
$y = 1 + 8$	$= 1 - 8 = -7$
$y_1 = 9$	$y_1 = -7$

$\therefore$ Tangential points are (2,9) and (-2, -7)

$m_1 =$ Slope at (2, 9) $= 3(2)^2 = 12$

Equation of the tangent at (2, 9)

$$y - y_1 = m_1(x - x_1)$$

$$y - 9 = 12(x - 2)$$

$$y - 9 = 12x - 24$$

$$24 - 9 = 12x - y$$

$$12x - y = 15$$

$m =$ Slope at (-2, -7) $= 3(-2)^2 = 12$

Equation of the tangent at (-2, -7)

$$y - y_1 = m(x - x_1)$$

$$y - (-7) = 12(x - (-2))$$

$$y + 7 = 12(x + 2)$$

$$y + 7 = 12x + 24$$

$$7 - 24 = 12x - y$$

$$12x - y = -17$$

7. Find the equations of the tangent to the curve $y = \frac{x+1}{x-1}$ which are parallel to line $x + 2y = 6$. 5M

Let $p(x_1, y_1)$ be the tangential points

$$\therefore y_1 = \frac{x_1+1}{x_1-1} \dots\dots\dots (1)$$

$$y = \frac{x+1}{x-1}$$

$$\frac{dy}{dx} = \frac{(x-1)(1)-(x+1)(1)}{(x-1)^2} = \frac{x-1-x-1}{(x-1)^2}$$

$$\frac{dy}{dx} = \frac{-2}{(x-1)^2}$$

$$m_1 = \left. \begin{array}{l} \text{Slope of the tangent} \\ \text{at } (x_1, y_1) \end{array} \right\} = \frac{-2}{(x_1-1)^2}$$

$$m_2 = \text{Slope of the line } x + 2y = 6 \text{ is } = -\frac{1}{2}$$

Given: Tangent to the given curve is parallel to the line $x + 2y = 6$

$$\therefore m_1 = m_2$$

$$\frac{-2}{(x_1-1)^2} = -\frac{1}{2}$$

$$4 = (x_1 - 1)^2$$

$$\Rightarrow x_1 - 1 = \pm 2$$

$$x_1 = 1 \pm 2 \quad x_1 = 3 \text{ or } x_1 = -1$$

$$\text{when } x_1 = 3 \quad (1) \Rightarrow y_1 = \frac{3+1}{3-1} = \frac{4}{2} = 2$$

$$\text{when } x_1 = -1 \quad (1) \Rightarrow y_1 = \frac{-1+1}{-1-1} = 0$$

$\therefore$ Tangential points are (3,2) and (-1,0)

$$m = \text{Slope} = -\frac{1}{2}$$

Equation of the tangent at (3,2)

$$y - y_1 = m(x - x_1)$$

$$y - 2 = -\frac{1}{2}(x - 3)$$

$$2y - 4 = -x + 3$$

$$x + 2y = 7$$

Equation of the tangent at (-1, 0)

$$y - y_1 = m(x - x_1)$$

$$y - 0 = -\frac{1}{2}(x + 1)$$

$$2y = -x - 1$$

$$x + 2y = -1$$

8. Find the equation of tangent and normal to the curve given by $x = 7 \cos t$ & $y = 2 \sin t$, $t \in R$ at any point on the curve.

5M

$$x = 7 \cos t \quad y = 2 \sin t$$

APR-25

$$\frac{dx}{dt} = -7 \sin t \quad \frac{dy}{dt} = 2 \cos t$$

$$m = \text{Slope of the tangent at } t \text{ is } = \frac{dy}{dx} = \frac{\frac{dy}{dt}}{\frac{dx}{dt}}$$

$$= \frac{2 \cos t}{-7 \sin t}$$

$$m = -\frac{2 \cos t}{7 \sin t}$$

Equation of the tangent at any point

$$y - y_1 = m(x - x_1)$$

$$y - 2 \sin t = -\frac{2 \cos t}{7 \sin t}(x - 7 \cos t)$$

$$7 \sin t [y - 2 \sin t] = -2 \cos t (x - 7 \cos t)$$

$$(7 \sin t)y - 14 \sin^2 t = (-2 \cos t)x + 14 \cos^2 t$$

$$(2 \cos t)x + (7 \sin t)y = 14(\sin^2 t + \cos^2 t)$$

$$(2 \cos t)x + (7 \sin t)y = 14$$

Equation of the normal at any points

$$y - y_1 = \frac{-1}{m}(x - x_1)$$

$$y - 2 \sin t = -\frac{1}{\left(\frac{-2 \cos t}{7 \sin t}\right)}(x - 7 \cos t)$$

$$y - 2 \sin t = \frac{7 \sin t}{2 \cos t}(x - 7 \cos t)$$

$$(2 \cos t)y - 4 \sin t \cos t = (7 \sin t)x - 49 \sin t \cos t$$

$$49 \sin t \cos t - 4 \sin t \cos t = (7 \sin t)x - (2 \cos t)y$$

$$(7 \sin t)x - (2 \cos t)y = 45 \sin t \cos t$$

9. Find the angle between the rectangular hyperbola $xy = 2$ and the parabola $x^2 + 4y = 0$

5M

Equation of the R. H: $xy = 2$

$$y = \frac{2}{x} \dots\dots\dots(1)$$

JUL-24

Equation of the parabola: $x^2 + 4y = 0$

$$x^2 + 4\left(\frac{2}{x}\right) = 0 \text{ [Using (1)]}$$

$$x^3 + 8 = 0$$

$$x^3 = -8 = (-2)^3$$

$$\therefore x = -2$$

Sub: $x = -2$ in (1)

$$y = \frac{2}{-2} = -1$$

$$y = -1$$

Point of intersection = $(-2, -1)$

$$(1) \Rightarrow y = \frac{2}{x}$$

$$\frac{dy}{dx} = -\frac{2}{x^2}$$

$$m_1 = \text{Slope of the tangent at } (-2, -1) \left\} = \frac{-2}{(-2)^2} = -\frac{2}{4} = -\frac{1}{2}\right.$$

$$m_1 = -\frac{1}{2}$$

Parabola $x^2 + 4y = 0$

Diff: w. r. to x , we get

$$2x + 4 \frac{dy}{dx} = 0$$

$$\frac{dy}{dx} = \frac{-2x}{4}$$

$$\frac{dy}{dx} = -\frac{x}{2}$$

$$m_2 = \text{Slope of the tangent at } (-2, -1) \left\} = -\left(\frac{-2}{2}\right) = 1\right.$$

Let θ be the angle between the two curves.

$$\tan \theta = \left| \frac{m_1 - m_2}{1 + m_1 m_2} \right|$$

$$= \left| \frac{-\frac{1}{2} - 1}{1 + \left(-\frac{1}{2}\right)(1)} \right|$$

$$= \left| \frac{-3/2}{-1/2} \right|$$

$$\tan \theta = 3$$

$$\theta = \tan^{-1}(3)$$

10. Show that the two curves $x^2 - y^2 = r^2$ are $xy = c^2$ where c, r are constants, cut orthogonally.

Equation of the two curves: $x^2 - y^2 = r^2$ and $xy = c^2$

3M

Let $p(x_1, y_1)$ be the point of intersection of the two curves

Then $x_1^2 - y_1^2 = r^2$ (1) and $x_1 y_1 = c^2$ (2)

Curve - I $x^2 - y^2 = r^2$

Differentiating w. r. to x , we get $2x - 2y \frac{dy}{dx} = 0$

divide by 2 $\Rightarrow x - y \frac{dy}{dx} = 0$

$$x = y \frac{dy}{dx}$$

$$\frac{dy}{dx} = \frac{x}{y}$$

$m_1 = \text{Slope of the tangent at } (x_1, y_1) = \frac{x_1}{y_1}$

Curve - II

$$xy = c^2$$

$$y = \frac{c^2}{x}$$

$$\frac{dy}{dx} = -\frac{c^2}{x^2}$$

$m_2 = \text{Slope of the tangent at } (x_1, y_1)$

$$= -\frac{c^2}{x_1^2}$$

$$m_1 m_2 = \left(\frac{x_1}{y_1}\right) \left(-\frac{c^2}{x_1^2}\right) = -\frac{c^2}{x_1 y_1} = -\frac{c^2}{c^2} = -1$$

$\therefore$ Given two curve cut orthogonally.

EXERCISE 7.3



Video class

Concept Corner

Mean Value Theorem

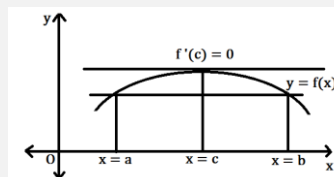
1. Intermediate value theorem

If f is continuous on a closed interval $[a, b]$, and c is any number between $f(a)$ and $f(b)$ inclusive, then there is at least one number $x \in [a, b]$ such that $f(x) = c$.

2. Rolle's Theorem

Let $f(x)$ be continuous on a closed interval $[a, b]$ and differentiable on the open interval (a, b) . If $f(a) = f(b)$, then there is atleast one point $c \in (a, b)$ where $f'(c) = 0$

Geometrical meaning: If the tangents is moving along the curve starting at $x = a$ towards $x = b$ then there exists $c \in (a, b)$ at which the tangent is parallel to the x -axis.



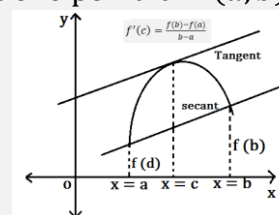
3. Lagrange's mean value theorem

Let $f(x)$ be continuous in a closed interval $[a, b]$ and differentiable in the open interval (a, b) (where $f(a), f(b)$ are not necessarily equal). Then there exists at least one point $c \in (a, b)$ such that $f'(c) = \frac{f(b) - f(a)}{b - a}$.

Remark: If $f(a) = f(b)$ then Lagrange's Mean Value Theorem gives the Rolle's theorem. It is also known as rotated Rolle's Theorem.

Geometrical meaning:

The instantaneous rate of change at some interior point is equal to the average rate of change over the entire interval. The secant to the curve $y = f(x)$ between $x = a$ and $x = b$ is parallel to a tangent line of the curve at some point $c \in (a, b)$.



4. Theorem

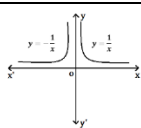
If $f(x)$ is continuous in $[a, b]$ and differentiable in (a, b) and if $f'(x) > 0, \forall x \in (a, b)$ and if $f'(x) < 0, \forall x \in (a, b)$, then for $x_1, x_2 \in [a, b]$, such that $x_1 < x_2$ we have $f(x_1) < f(x_2)$.

1. Explain why Rolle's theorem is not applicable to the following functions in the respective intervals.

(i) $f(x) = \left| \frac{1}{x} \right|, x \in [-1, 1]$ 3M

$f(x)$ is not continuous at $x = 0$

Thus, Rolle's Theorem is not applicable.



(ii) $f(x) = \tan x, x \in [0, \pi]$

At $x = \frac{\pi}{2}, f(x) = \tan \frac{\pi}{2} = \infty$

$\therefore f(x)$ is discontinuous at $x = \frac{\pi}{2}$

$\therefore f(x)$ is not continuous on $[0, \pi]$

Thus, Rolle's Theorem is not applicable.

(iii) $f(x) = x - 2 \log x, x \in [2, 7]$

$f(x)$ is continuous on $[2, 7]$

$f(x)$ is differentiable on $(2, 7)$

$f(2) = 2 - 2 \log 2$

$f(7) = 7 - 2 \log 7$

Since $f(2) \neq f(7)$

Rolle's Theorem is not applicable.

2. Using the Rolle's theorem, determine the values of x at which the tangent is parallel to the x -axis for the following functions. 3M

(i) $f(x) = x^2 - x, x \in [0, 1]$ (ii) $f(x) = \frac{x^2 - 2x}{x + 2}, x \in [-1, 6]$ (iii) $f(x) = \sqrt{x} - \frac{x}{3}, x \in [0, 9]$

(i) $f(x) = x^2 - x$

$f'(x) = 2x - 1$

(1) $f(x)$ is continuous in $[0, 1]$

(2) $f(x)$ is differentiable in $(0, 1)$

(3) $f(0) = f(1)$

By Rolle's Theorem there exist $c \in (0, 1)$ such that $f'(c) = 0$

$\Rightarrow 2c - 1 = 0 \Rightarrow c = \frac{1}{2} \in (0, 1)$

$\therefore$ At $x = \frac{1}{2}$

the tangent is parallel to the x -axis.

(iii) $f(x) = \sqrt{x} - \frac{x}{3}, x \in [0, 9]$

$f'(x) = \frac{1}{2\sqrt{x}} - \frac{1}{3}(1) = \frac{1}{2\sqrt{x}} - \frac{1}{3}$

(1) $f(x)$ is continuous in $[0, 9]$

(2) $f(x)$ is differentiable in $(0, 9)$

(3) $f(0) = \sqrt{0} - \frac{0}{3} = 0$

$f(9) = \sqrt{9} - \frac{9}{3} = 3 - 3 = 0$

$f(0) = f(9)$

By Rolle's Theorem, there exist $c \in (0, 9)$ such that $f'(c) = 0$

$\Rightarrow \frac{1}{2\sqrt{c}} - \frac{1}{3} = 0 \Rightarrow \frac{1}{2\sqrt{c}} = \frac{1}{3}$

$\Rightarrow 3 = 2\sqrt{c} \Rightarrow \sqrt{c} = \frac{3}{2}$

$c = \frac{9}{4}$ (Squaring)

$\therefore c = \frac{9}{4} \in (0, 9)$

$\therefore$ At $x = \frac{9}{4}$, the tangent is parallel to x -axis.

(ii) $f(x) = \frac{x^2 - 2x}{x + 2}$

$f'(x) = \frac{(x+2)(2x-2) - (x^2-2x)(1)}{(x+2)^2}$

$= \frac{2x^2 - 2x + 4x - 4 - x^2 + 2x}{(x+2)^2}$

$f'(x) = \frac{x^2 + 4x - 4}{(x+2)^2}$

(1) $f(x)$ is continuous in $[-1, 6]$

(2) $f(x)$ is differentiable in $(-1, 6)$

(3) $f(-1) = \frac{(-1)^2 - 2(-1)}{-1+2}$

$= \frac{1+2}{1} = 3$

$f(6) = \frac{(6)^2 - 2(6)}{6+2} = \frac{36-12}{8} = \frac{24}{8} = 3$

$\therefore f(-1) = f(6)$

By Rolle's Theorem, there exist $c \in (-1, 6)$ such that $f'(c) = 0$

$\frac{c^2 + 4c - 4}{(c+2)^2} = 0$

$\Rightarrow c^2 + 4c - 4 = 0$

$c = \frac{-4 \pm \sqrt{16 - 4(1)(-4)}}{2 \times 1}$

$c = \frac{-4 \pm \sqrt{32}}{2} = \frac{-4 \pm 4\sqrt{2}}{2}$

$= -2 \pm 2\sqrt{2}$

$c = -2 + 2\sqrt{2} \in (-1, 6)$

$\therefore$ At $x = -2 + 2\sqrt{2}$,

The tangent is parallel to x -axis.

3. Explain why Lagrange's mean value theorem is not applicable to the following functions in the respective intervals. (i) $f(x) = \frac{x+1}{x}$, $x \in [-1, 2]$ (ii) $f(x) = |3x + 1|$, $x \in [-1, 3]$.

(i) $f(x) = \frac{x+1}{x}$

At $x = 0$, $f(0) = \frac{0+1}{0} = \infty$

∴ $f(x)$ is not continuous at $x = 0$

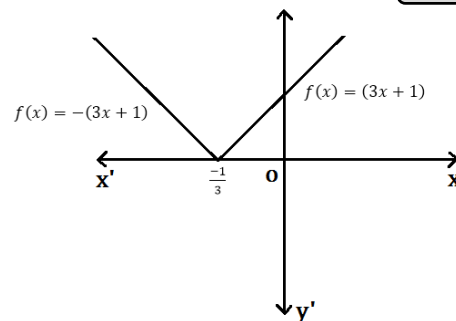
∴ Lagrange's mean value theorem is not applicable.

(ii) $f(x) = |3x + 1| = \left|x - \left(-\frac{1}{3}\right)\right|$

(1) $f(x)$ is continuous in $[-1, 3]$

(2) $f(x)$ is not differentiable at $-\frac{1}{3} \in (-1, 3)$

Thus, Lagrange's mean value theorem is not applicable.



3M

4. Using the Lagrange's mean value theorem determine the values of x at which the tangent is parallel to the secant line at the end points of the given interval.

3M

(i) $f(x) = x^3 - 3x + 2$, $x \in [-2, 2]$

(i) $f(x) = x^3 - 3x + 2$

$f'(x) = 3x^2 - 3$

(1) $f(x)$ is continuous in $[-2, 2]$

(2) $f(x)$ is differentiable in $(-2, 2)$

By Lagrange's mean value theorem, there exist $c \in (-2, 2)$ such that

$$f'(c) = \frac{f(2) - f(-2)}{2 - (-2)}$$

$$3c^2 - 3 = \frac{[2^3 - 3(2) + 2] - [(-2)^3 - 3(-2) + 2]}{4}$$

$$3c^2 - 3 = \frac{(8 - 6 + 2) - (-8 + 6 + 2)}{4}$$

$$= \frac{4 - 0}{4} = 1$$

$$3c^2 = 1 + 3 = 4$$

$$c^2 = \frac{4}{3}$$

$$c = \pm \frac{2}{\sqrt{3}} \in (-2, 2)$$

∴ Tangents at $x = \pm \frac{2}{\sqrt{3}}$ are parallel to the secant line at the end points of the given interval.

(ii) $f(x) = (x - 2)(x - 7)$, $x \in [3, 11]$

(ii) $f(x) = (x - 2)(x - 7)$

$$= x^2 - 9x + 14$$

$$f'(x) = 2x - 9$$

(1) $f(x)$ is continuous in $[3, 11]$

(2) $f(x)$ is differentiable in $(3, 11)$

$$f(3) = 3^2 - 9(3) + 14$$

$$= 9 - 27 + 14 = -4$$

$$f(11) = 11^2 - 9(11) + 14$$

$$= 121 - 99 + 14$$

$$= 135 - 99 = 36$$

By LMV theorem, there exist $c \in (3, 11)$ such that

$$f'(c) = \frac{f(11) - f(3)}{11 - 3}$$

$$2c - 9 = \frac{36 - (-4)}{8}$$

$$2c - 9 = \frac{40}{8} = 5$$

$$2c = 5 + 9 = 14$$

$$c = 7 \in (3, 11)$$

∴ Tangent at $x = 7$ is parallel to the secant line at end points of the given interval.

5. Show that the value in the conclusion of the mean value theorem for (i) $f(x) = \frac{1}{x}$ on a closed interval of positive numbers $[a, b]$ is $\sqrt{ab}$ (ii) $f(x) = Ax^2 + Bx + C$ on any interval $[a, b]$ is $\frac{a+b}{2}$ 3M PTA-2

(i) $f(x) = \frac{1}{x}$, $[a, b]$ and a, b are +ve numbers

$$f'(x) = -\frac{1}{x^2}$$

(1) $f(x)$ is continuous in $[a, b]$

(2) $f(x)$ is differentiable in (a, b)

By LMVT, there exist $c \in (a, b)$ such that

$$f'(c) = \frac{f(b)-f(a)}{b-a}$$

$$-\frac{1}{c^2} = \frac{\left(\frac{1}{b} - \frac{1}{a}\right)}{b-a}$$

$$-\frac{1}{c^2} = \frac{a-b}{ab} \times \frac{1}{b-a}$$

$$-\frac{1}{c^2} = -\frac{(b-a)}{ab} \times \frac{1}{(b-a)}$$

$$\frac{1}{c^2} = \frac{1}{ab} \Rightarrow c^2 = ab$$

$$\therefore c = \pm\sqrt{ab} \quad c = \sqrt{ab} \in (a, b)$$

(ii) $f(x) = Ax^2 + Bx + C$, $[a, b]$

$$f'(x) = 2Ax + B$$

(1) $f(x)$ is continuous in $[a, b]$

(2) $f(x)$ is differentiable in (a, b)

By LMVT, there exist $c \in (a, b)$ such that

$$f'(x) = \frac{f(b)-f(a)}{b-a}$$

$$2Ac + B = \frac{(Ab^2+Bb+C)-(Aa^2+Ba+C)}{b-a}$$

$$= \frac{A(b^2-a^2)+B(b-a)+C-C}{b-a}$$

$$= \frac{A(b-a)(b+a)+B(b-a)}{b-a}$$

$$2Ac + B = A(b+a) + B$$

$$c = \frac{(b+a)A}{2A}$$

$$c = \frac{a+b}{2} \in (a, b)$$

6. A race car driver is in kilometer stone 20. If his speed never exceeds 150 km/hr, what is the maximum kilometer he can reach in the next two hours. 5M

Let $f(t)$ be the distance covered by the car driver at any time $t \in [0, 2]$.

Then, speed of the car $= f'(t)$

Given: $f(0) = 20$ and $f'(t) \leq 150$

(i) $f(t)$ is continuous in $[0, 2]$

(ii) $f(t)$ is differentiable in $(0, 2)$

By LMVT, there exist $t \in (0, 2)$ such that

$$f'(t) = \frac{f(2)-f(0)}{2-0} \\ \Rightarrow \frac{f(2)-20}{2} \leq 150$$

$$f(2) \leq 300 + 20 \quad \therefore f(2) \leq 320$$

$\therefore$ The maximum distance can cover in the next two hours = **320km**

7. Suppose that for a function $f(x)$, $f'(x) \leq 1$ for all $1 \leq x \leq 4$. Show that $f(4) - f(1) \leq 3$

Given: $f'(x) \leq 1$ for all $x \in [1, 4]$ 3M

(i) $f(x)$ is continuous in $[1, 4]$

(ii) $f(x)$ is differentiable in $(1, 4)$

By LMVT, there exist $x \in (1, 4)$ such that

$$f'(x) = \frac{f(4)-f(1)}{4-1} \\ \Rightarrow \frac{f(4)-f(1)}{3} \leq 1$$

$$\therefore f(4) - f(1) \leq 3$$

8. Does there exist a differentiable function $f(x)$ such that $f(0) = -1$, $f(2) = 4$ and $f'(x) \leq 2$ for all x . Justify your answer.

Given: $f(0) = -1$, $f(2) = 4$ and 3M

$$f'(x) \leq 2 \text{ for all } x$$

$$f'(x) = \frac{f(2)-f(0)}{2-0}$$

$$\Rightarrow \frac{4-(-1)}{2} \leq 2 \Rightarrow \frac{5}{2} \leq 2$$

$$2.5 \leq 2 \text{ which is not possible}$$

$\Rightarrow f'(x)$ cannot be 2.5 at any point in $(0, 2)$

No, the differentiable function does not exist for the given conditions.

9. Show that there lies a point on the curve $f(x) = x(x+3)e^{-\frac{\pi}{2}}$, $-3 \leq x \leq 0$ where tangent drawn is parallel to the x -axis. 5M

$$f(x) = (x^2 + 3x)e^{-\frac{\pi}{2}}$$

$$f'(x) = (2x + 3)e^{-\frac{\pi}{2}}$$

- (i) $f(x)$ is continuous in $[-3, 0]$
 (ii) $f(x)$ is differentiable in $(-3, 0)$
 (iii) $f(-3) = -3(-3+3)e^{-\frac{\pi}{2}} = 0$

$$f(0) = (0)(0+3)e^{-\frac{\pi}{2}} = 0$$

$$f(-3) = f(0)$$

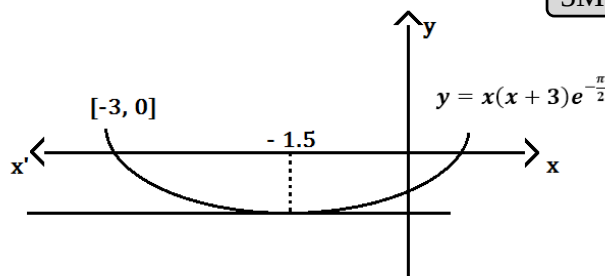
By Rolle's Theorem there exist $c \in (-3, 0)$ such that $f'(c) = 0$

Tangent at $x = c$ to the curve is parallel to x -axis $(2c + 3)e^{-\frac{\pi}{2}} = 0$

$$\Rightarrow 2c + 3 = 0$$

$$c = -\frac{3}{2} = -1.5 \in (-3, 0)$$

$\therefore$ The tangent drawn at $x = -1.5$ to the curve is parallel to the x -axis.



10. Using mean value theorem prove that for, $a > 0, b > 0, |e^{-a} - e^{-b}| < |a - b|$. 5M

$$f(x) = e^{-x}, [a, b] \text{ where } a > 0, b > 0$$

$$f'(x) = -e^{-x}$$

- (i) $f(x)$ is continuous in $[a, b]$ (ii) $f(x)$ is differentiable in (a, b)

By LMVT, there exist $c \in (a, b)$ such that $f'(c) = \frac{f(b) - f(a)}{b - a}$

$$-e^{-c} = \frac{e^{-b} - e^{-a}}{b - a} \Rightarrow -\frac{[e^{-a} - e^{-b}]}{-[a - b]} = -e^{-c}$$

$$\frac{|e^{-a} - e^{-b}|}{|a - b|} = |-e^{-c}| < 1$$

$$\therefore |e^{-a} - e^{-b}| < |a - b|$$

Hence proved.

EXERCISE 7.4

Concept Corner

Series Expansions

1. **Taylor's Series:** Let $f(x)$ be a functions infinitely differentiable at $x = a$. Then $f(x)$ can be expanded as a series, in an interval $(a - \varepsilon, a + \varepsilon)$, $\varepsilon > 0$ of the form

$$\begin{aligned} f(x) &= \sum_{n=0}^{\infty} \frac{f^{(n)}(a)}{n!} (x - a)^n \\ &= f(a) + \frac{f'(a)}{1!} (x - a) + \dots + \frac{f^n(a)}{n!} (x - a)^n + \dots \end{aligned}$$

2. **Maclaurin's Series:** If $a = 0$, the expansion takes the form

$$\begin{aligned} f(x) &= \sum_{n=0}^{\infty} \frac{f^{(n)}(0)}{n!} (x)^n \\ f(x) &= f(0) + \frac{f'(0)}{1!} x + \frac{f''(0)}{2!} x^2 + \dots + \frac{f^n(0)}{n!} x^n + \dots \end{aligned}$$



Video class

1. Write the Maclaurin series expansion of the following functions:

(i) e^x

(ii) $\sin x$

(iii) $\cos x$

(iv) $\log(1-x), -1 \leq x < 1$

(v) $\tan^{-1}(x), -1 \leq x \leq 1$

(vi) $\cos^2 x$

(i) $f(x) = e^x$ Maclaurin's series: $f(x) = f(0) + \frac{f'(0)}{1!}x + \frac{f''(0)}{2!}x^2 + \dots$$e^x = 1 + \frac{1}{1!}x + \frac{1}{2!}x^2 + \frac{1}{3!}x^3 + \dots$$\therefore e^x = 1 + \frac{x}{1!} + \frac{x^2}{2!} + \frac{x^3}{3!} + \dots$	3M <table><tr><th>Function and its derivatives</th><th>e^x and its derivatives</th><th>Value at $x = 0$</th></tr><tr><td>$f(x)$</td><td>e^x</td><td>$e^0 = 1$</td></tr><tr><td>$f'(x)$</td><td>e^x</td><td>1</td></tr><tr><td>$f''(x)$</td><td>e^x</td><td>1</td></tr><tr><td>$f'''(x)$</td><td>e^x</td><td>1</td></tr><tr><td>$f^{iv}(x)$</td><td>e^x</td><td>1</td></tr><tr><td>$f^v(x)$</td><td>e^x</td><td>1</td></tr></table>	Function and its derivatives	e^x and its derivatives	Value at $x = 0$	$f(x)$	e^x	$e^0 = 1$	$f'(x)$	e^x	1	$f''(x)$	e^x	1	$f'''(x)$	e^x	1	$f^{iv}(x)$	e^x	1	$f^v(x)$	e^x	1
Function and its derivatives	e^x and its derivatives	Value at $x = 0$																				
$f(x)$	e^x	$e^0 = 1$																				
$f'(x)$	e^x	1																				
$f''(x)$	e^x	1																				
$f'''(x)$	e^x	1																				
$f^{iv}(x)$	e^x	1																				
$f^v(x)$	e^x	1																				
(ii) $f(x) = \sin x$ Maclaurin's Series: $f(x) = f(0) + \frac{f'(0)}{1!}x + \frac{f''(0)}{2!}x^2 + \dots$$\sin x = 0 + \frac{1}{1!}x + 0 - \frac{1}{3!}x^3 + 0 + \frac{1}{5!}x^5 + \dots$$\sin x = \frac{x}{1!} - \frac{x^3}{3!} + \frac{x^5}{5!} - \frac{x^7}{7!} + \dots$	3M <table><tr><th>Functions and its derivatives</th><th>$\sin x$ and its derivatives</th><th>Value at $x = 0$</th></tr><tr><td>$f(x)$</td><td>$\sin x$</td><td>$\sin 0 = 0$</td></tr><tr><td>$f'(x)$</td><td>$\cos x$</td><td>$\cos 0 = 1$</td></tr><tr><td>$f''(x)$</td><td>$-\sin x$</td><td>0</td></tr><tr><td>$f'''(x)$</td><td>$-\cos x$</td><td>-1</td></tr><tr><td>$f^{iv}(x)$</td><td>$\sin x$</td><td>0</td></tr><tr><td>$f^v(x)$</td><td>$\cos x$</td><td>1</td></tr></table>	Functions and its derivatives	$\sin x$ and its derivatives	Value at $x = 0$	$f(x)$	$\sin x$	$\sin 0 = 0$	$f'(x)$	$\cos x$	$\cos 0 = 1$	$f''(x)$	$-\sin x$	0	$f'''(x)$	$-\cos x$	-1	$f^{iv}(x)$	$\sin x$	0	$f^v(x)$	$\cos x$	1
Functions and its derivatives	$\sin x$ and its derivatives	Value at $x = 0$																				
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(iii) $f(x) = \cos x$ Maclaurin's Series: $f(x) = f(0) + \frac{f'(0)}{1!}x + \frac{f''(0)}{2!}x^2 + \dots$$\cos x = 1 + 0 - \frac{1}{2!}x^2 + 0 + \frac{1}{4!}x^4 + 0 \dots$$\cos x = 1 - \frac{x^2}{2!} + \frac{x^4}{4!} - \frac{x^6}{6!} + \dots$	3M <table><tr><th>Functions and its derivatives</th><th>$\cos x$ and its derivatives</th><th>Value at $x = 0$</th></tr><tr><td>$f(x)$</td><td>$\cos x$</td><td>$\cos 0 = 1$</td></tr><tr><td>$f'(x)$</td><td>$-\sin x$</td><td>0</td></tr><tr><td>$f''(x)$</td><td>$-\cos x$</td><td>-1</td></tr><tr><td>$f'''(x)$</td><td>$\sin x$</td><td>0</td></tr><tr><td>$f^{iv}(x)$</td><td>$\cos x$</td><td>1</td></tr><tr><td>$f^v(x)$</td><td>$-\sin x$</td><td>0</td></tr></table>	Functions and its derivatives	$\cos x$ and its derivatives	Value at $x = 0$	$f(x)$	$\cos x$	$\cos 0 = 1$	$f'(x)$	$-\sin x$	0	$f''(x)$	$-\cos x$	-1	$f'''(x)$	$\sin x$	0	$f^{iv}(x)$	$\cos x$	1	$f^v(x)$	$-\sin x$	0
Functions and its derivatives	$\cos x$ and its derivatives	Value at $x = 0$																				
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$f'''(x)$	$\sin x$	0																				
$f^{iv}(x)$	$\cos x$	1																				
$f^v(x)$	$-\sin x$	0																				
(iv) $f(x) = \log(1 - x), -1 \leq x < 1$ Maclaurin's Series: $f(x) = f(0) + \frac{f'(0)}{1!}x + \frac{f''(0)}{2!}x^2 + \dots$$\log(1 - x) = 0 - \frac{1}{1!}x - \frac{1}{2!}x^2 - \frac{2}{1 \times 2 \times 3}x^3$$- \frac{6}{1 \times 2 \times 3 \times 4}x^4 - \frac{24}{1 \times 2 \times 3 \times 4 \times 5}x^5 \dots$$= -\frac{x}{1} - \frac{x^2}{2} - \frac{x^3}{3} - \frac{x^4}{4} - \frac{x^5}{5} \dots$$= -\frac{x}{1} - \frac{x^2}{2} - \frac{x^3}{3} - \frac{x^4}{4} - \frac{x^5}{5} \dots$$\therefore \log(1 - x) = -\left[x + \frac{x^2}{2} + \frac{x^3}{3} + \frac{x^4}{4} + \frac{x^5}{5} + \dots\right]$	5M <table><tr><th>Functions and its derivatives</th><th>$\log(1 - x)$ and its derivatives</th><th>Value at $x = 0$</th></tr><tr><td>$f(x)$</td><td>$\log(1 - x)$</td><td>$\log 1 = 0$</td></tr><tr><td>$f'(x)$</td><td>$\frac{-1}{1-x}$</td><td>$\frac{-1}{1-0} = -1$</td></tr><tr><td>$f''(x)$</td><td>$\frac{-1}{(1-x)^2}$</td><td>$\frac{-1}{(1-0)^2} = -1$</td></tr><tr><td>$f'''(x)$</td><td>$\frac{-2}{(1-x)^3}$</td><td>$-\frac{2}{(1-0)^3} = -2$</td></tr><tr><td>$f^{iv}(x)$</td><td>$\frac{-6}{(1-x)^4}$</td><td>$\frac{-6}{(1-0)^4} = -6$</td></tr><tr><td>$f^v(x)$</td><td>$\frac{-24}{(1-x)^5}$</td><td>$\frac{-24}{(1-0)^5} = -24$</td></tr></table>	Functions and its derivatives	$\log(1 - x)$ and its derivatives	Value at $x = 0$	$f(x)$	$\log(1 - x)$	$\log 1 = 0$	$f'(x)$	$\frac{-1}{1-x}$	$\frac{-1}{1-0} = -1$	$f''(x)$	$\frac{-1}{(1-x)^2}$	$\frac{-1}{(1-0)^2} = -1$	$f'''(x)$	$\frac{-2}{(1-x)^3}$	$-\frac{2}{(1-0)^3} = -2$	$f^{iv}(x)$	$\frac{-6}{(1-x)^4}$	$\frac{-6}{(1-0)^4} = -6$	$f^v(x)$	$\frac{-24}{(1-x)^5}$	$\frac{-24}{(1-0)^5} = -24$
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$f'''(x)$	$\frac{-2}{(1-x)^3}$	$-\frac{2}{(1-0)^3} = -2$																				
$f^{iv}(x)$	$\frac{-6}{(1-x)^4}$	$\frac{-6}{(1-0)^4} = -6$																				
$f^v(x)$	$\frac{-24}{(1-x)^5}$	$\frac{-24}{(1-0)^5} = -24$																				

(v) $\tan^{-1}(x), -1 \leq x \leq 1$

5M

Maclaurin Series: $f(x) = f(0) + \frac{f'(0)}{1!}x + \frac{f''(0)}{2!}x^2 + \dots$

$$\tan^{-1}(x) = 0 + \frac{1}{1!}x + \frac{0}{2!}x^2 - \frac{2}{1 \times 2 \times 3}x^3 + 0 + \frac{24}{1 \times 2 \times 3 \times 4 \times 5}x^5 + \dots$$

$$\tan^{-1}(x) = x - \frac{x^3}{3} + \frac{x^5}{5} - \frac{x^7}{7} + \dots$$

Functions and its derivatives	$\tan^{-1}x$ and its derivatives	Value at $x = 0$
$f(x)$	$\tan^{-1}x$	$\tan^{-1}(0) = 0$
$f'(x)$	$\frac{1}{1+x^2} = (1+x^2)^{-1} = 1 - x^2 + x^4 - x^6 + \dots$	$\frac{1}{1+0} = 1$
$f''(x)$	$-2x + 4x^3 - 6x^5 + \dots$	0
$f'''(x)$	$-2 + 12x^2 - 30x^4 + \dots$	-2
$f^{iv}(x)$	$24x - 120x^3 + \dots$	0
$f^v(x)$	$24 - 360x^2 + \dots$	24

(vi) $\cos^2 x$

5M

Maclaurin's Series: $f(x) = f(0) + \frac{f'(0)}{1!}x + \frac{f''(0)}{2!}x^2 + \dots$

$$\cos^2 x = 1 + 0 - \frac{2}{2!}x^2 + 0 + \frac{8}{4!}x^4 + 0 + \dots$$

$$\cos^2 x = 1 - \frac{2}{2!}x^2 + \frac{2^3}{4!}x^4 - \frac{2^5}{6!}x^6 + \dots$$

Functions and its derivatives	$\cos^2 x$ and its derivatives	Value at $x = 0$
$f(x)$	$\cos^2 x$	$\cos^2(0) = 1$
$f'(x)$	$2 \cos x (-\sin x) = -\sin 2x$	0
$f''(x)$	$-2 \cos 2x$	$-2 \cos(0) = -2$
$f'''(x)$	$-2(-\sin 2x)(2) = 4 \sin 2x$	0
$f^{iv}(x)$	$4 \cos 2x(2) = 8 \cos 2x$	$8 \cos(0) = 8$
$f^v(x)$	$8(-\sin 2x)(2) = -16 \sin 2x$	0

2. Write down the Taylor series expansion, of the function $\log x$ about $x = 1$ upto three non-zero terms for $x > 0$

3M

Functions and its derivatives	$\log x$ and its derivatives	Value at $x = 1$
$f(x)$	$\log x$	$\log 1 = 0$
$f'(x)$	$\frac{1}{x}$	$\frac{1}{1} = 1$
$f''(x)$	$-\frac{1}{x^2}$	$-\frac{1}{1^2} = -1$
$f'''(x)$	$\frac{2}{x^3}$	$\frac{2}{1^3} = 2$
$f^{iv}(x)$	$-\frac{6}{x^4}$	$-\frac{6}{1^4} = -6$
$f^v(x)$	$\frac{24}{x^5}$	$\frac{24}{1^5} = 24$

Taylor's Series: $f(x) = f(a) + \frac{f'(a)}{1!}(x-a) + \frac{f''(a)}{2!}(x-a)^2 + \dots$

$$\log x = f(1) + \frac{f'(1)}{1!}(x-1) + \frac{f''(1)}{2!}(x-1)^2 + \dots$$

$$\log x = 0 + \frac{1}{1!}(x-1) - \frac{1}{2!}(x-1)^2 + \frac{2}{1 \times 2 \times 3}(x-1)^3 - \frac{6}{1 \times 2 \times 3 \times 4}(x-1)^4 + \frac{24}{1 \times 2 \times 3 \times 4 \times 5}(x-1)^5 + \dots$$

$$\log x = (x-1) - \frac{1}{2}(x-1)^2 + \frac{1}{3}(x-1)^3 - \frac{1}{4}(x-1)^4 + \frac{1}{5}(x-1)^5 + \dots$$

3. Expand $\sin x$ in ascending powers $x - \frac{\pi}{4}$ upto three non-zero terms.

3M

Function and its derivatives	$\sin x$ and its derivatives	Value at $x = \frac{\pi}{4}$
$f(x)$	$\sin x$	$\sin \frac{\pi}{4} = \frac{1}{\sqrt{2}}$
$f'(x)$	$\cos x$	$\cos \frac{\pi}{4} = \frac{1}{\sqrt{2}}$
$f''(x)$	$-\sin x$	$-\sin \frac{\pi}{4} = -\frac{1}{\sqrt{2}}$
$f'''(x)$	$-\cos x$	$-\cos \frac{\pi}{4} = -\frac{1}{\sqrt{2}}$

Taylor's series:

$$f(x) = f(a) + \frac{f'(a)}{1!}(x-a) + \frac{f''(a)}{2!}(x-a)^2 + \dots$$

$$\sin x = f\left(\frac{\pi}{4}\right) + \frac{f'\left(\frac{\pi}{4}\right)}{1!}\left(x - \frac{\pi}{4}\right) + \frac{f''\left(\frac{\pi}{4}\right)}{2!}\left(x - \frac{\pi}{4}\right)^2 + \frac{f'''\left(\frac{\pi}{4}\right)}{3!}\left(x - \frac{\pi}{4}\right)^3 + \dots$$

$$= \frac{1}{\sqrt{2}} + \frac{\frac{1}{\sqrt{2}}}{1!}\left(x - \frac{\pi}{4}\right) - \frac{\frac{1}{\sqrt{2}}}{2!}\left(x - \frac{\pi}{4}\right)^2 - \frac{\frac{1}{\sqrt{2}}}{3!}\left(x - \frac{\pi}{4}\right)^3 + \dots$$

$$= \frac{1}{\sqrt{2}} \left[1 + \frac{1}{1!}\left(x - \frac{\pi}{4}\right) - \frac{1}{2!}\left(x - \frac{\pi}{4}\right)^2 - \frac{1}{3!}\left(x - \frac{\pi}{4}\right)^3 + \dots \right]$$

$$= \frac{1}{\sqrt{2}} \times \frac{\sqrt{2}}{\sqrt{2}} \left[1 + \frac{1}{1!}\left(x - \frac{\pi}{4}\right) - \frac{1}{2!}\left(x - \frac{\pi}{4}\right)^2 - \frac{1}{3!}\left(x - \frac{\pi}{4}\right)^3 + \dots \right]$$

$$\sin x = \frac{\sqrt{2}}{2} \left[1 + \frac{1}{1!}\left(x - \frac{\pi}{4}\right) - \frac{1}{2!}\left(x - \frac{\pi}{4}\right)^2 - \frac{1}{3!}\left(x - \frac{\pi}{4}\right)^3 + \dots \right]$$

4. Expand the polynomial $f(x) = x^2 - 3x + 2$ in powers of $x - 1$

3M

Function and its derivatives	$x^2 - 3x + 2$ and its derivatives	Value at $x = 1$
$f(x)$	$x^2 - 3x + 2$	$1^2 - 3(1) + 2 = 0$
$f'(x)$	$2x - 3$	$2(1) - 3 = -1$
$f''(x)$	2	2

Taylor's series:

$$f(x) = f(a) + \frac{f'(a)}{1!}(x-a) + \frac{f''(a)}{2!}(x-a)^2 + \dots$$

$$x^2 - 3x + 2 = f(1) + \frac{f'(1)}{1!}(x-1) + \frac{f''(1)}{2!}(x-1)^2 + \dots$$

$$= 0 - \frac{1}{1}(x-1) + \frac{2}{2}(x-1)^2 + 0$$

$$f(x) = -(x-1) + (x-1)^2$$

EXERCISE 7.5



Video class

Concept Corner

Indeterminate Forms

1. The l' Hospital's Rule:

Suppose $f(x)$ and $g(x)$ are differentiable functions and $g'(x) \neq 0$ with

$$(i) \lim_{x \rightarrow a} f(x) = 0 = \lim_{x \rightarrow a} g(x)$$

$$\text{Then } \lim_{x \rightarrow a} \frac{f(x)}{g(x)} = \lim_{x \rightarrow a} \frac{f'(x)}{g'(x)}$$

$$(ii) \lim_{x \rightarrow a} f(x) = \pm\infty = \lim_{x \rightarrow a} g(x)$$

$$\text{Then } \lim_{x \rightarrow a} \frac{f(x)}{g(x)} = \lim_{x \rightarrow a} \frac{f'(x)}{g'(x)}$$

2. Indeterminate forms:

$$\frac{0}{0}, \frac{\infty}{\infty}, 0 \times \infty, \infty - \infty$$

$$0^0, 1^\infty \text{ and } \infty^0$$

3. **Theorem:** Let $\lim_{x \rightarrow a} g(x)$ exist and let it be L and let $f(x)$ be a continuous function at $x = L$.

$$\text{Then } \lim_{x \rightarrow a} f[g(x)] = f\left(\lim_{x \rightarrow a} g(x)\right)$$

Evaluate the following limits, if necessary use L' Hospital Rule:

2M

$$1. \lim_{x \rightarrow 0} \frac{1 - \cos x}{x^2}$$

2M

$$\lim_{x \rightarrow 0} \frac{1 - \cos x}{x^2} = \frac{1 - \cos 0}{0^2} = \frac{1 - 1}{0} = \frac{0}{0}$$

(Indeterminate form)

Let us apply L' Hospital's rule

$$\lim_{x \rightarrow 0} \frac{1 - \cos x}{x^2} = \lim_{x \rightarrow 0} \frac{0 + \sin x}{2x} \left(\frac{0}{0} \text{ form} \right)$$

Apply again L' Hospital's rule

$$= \lim_{x \rightarrow 0} \frac{\cos x}{2} = \frac{\cos 0}{2} = \frac{1}{2}$$

$$2. \lim_{x \rightarrow \infty} \frac{2x^2 - 3}{x^2 - 5x + 3}$$

APR-22

3M

$$\lim_{x \rightarrow \infty} \frac{2x^2 - 3}{x^2 - 5x + 3} = \frac{\infty}{\infty} \text{ (Indeterminate form)}$$

Let us apply L' Hospital rule

$$\lim_{x \rightarrow \infty} \frac{2x^2 - 3}{x^2 - 5x + 3} = \lim_{x \rightarrow \infty} \frac{4x}{2x - 5} \left(\frac{\infty}{\infty} \text{ form} \right)$$

Apply again L' Hospital's rule

$$\begin{aligned} &= \lim_{x \rightarrow \infty} \frac{4}{2} \\ &= 2 \end{aligned}$$

$$3. \lim_{x \rightarrow \infty} \frac{x}{\log x}$$

$$\lim_{x \rightarrow \infty} \frac{x}{\log x} = \frac{\infty}{\log \infty} = \frac{\infty}{\infty} \text{ (Indeterminate form)}$$

Let us apply L' Hospital's rules

$$\begin{aligned} \lim_{x \rightarrow \infty} \frac{x}{\log x} &= \lim_{x \rightarrow \infty} \frac{1}{\frac{1}{x}} \\ &= \lim_{x \rightarrow \infty} x = \infty \end{aligned}$$

$$4. \lim_{x \rightarrow \frac{\pi}{2}} \frac{\sec x}{\tan x}$$

2M

$$\lim_{x \rightarrow \frac{\pi}{2}} \frac{\sec x}{\tan x} = \frac{\sec \frac{\pi}{2}}{\tan \frac{\pi}{2}} = \frac{\infty}{\infty} \text{ (Indeterminate form)}$$

Let us apply L' Hospital's rules

$$\begin{aligned} \lim_{x \rightarrow \frac{\pi}{2}} \frac{\sec x}{\tan x} &= \lim_{x \rightarrow \frac{\pi}{2}} \frac{\sec x \tan x}{\sec^2 x} \\ &= \lim_{x \rightarrow \frac{\pi}{2}} \frac{\sin x}{\cos x} \\ &= \lim_{x \rightarrow \frac{\pi}{2}} \sin x \\ &= \sin \frac{\pi}{2} = 1 \end{aligned}$$

5. $\lim_{x \rightarrow \infty} e^{-x} \sqrt{x}$

3M

$$\lim_{x \rightarrow \infty} e^{-x} \sqrt{x} = \lim_{x \rightarrow \infty} \frac{\sqrt{x}}{e^x} = \frac{\infty}{\infty}$$

(Indeterminate form)

Let us apply L' Hospital's rule

$$\begin{aligned} \lim_{x \rightarrow \infty} \frac{\sqrt{x}}{e^x} &= \lim_{x \rightarrow \infty} \frac{1}{2\sqrt{x}} \times \frac{1}{e^x} \\ &= \frac{1}{\infty} \times \frac{1}{e^\infty} = \frac{1}{\infty} = 0 \end{aligned}$$

6. $\lim_{x \rightarrow 0} \left(\frac{1}{\sin x} - \frac{1}{x} \right)$

3M

$$\lim_{x \rightarrow 0} \left(\frac{1}{\sin x} - \frac{1}{x} \right) = \frac{1}{\sin 0} - \frac{1}{0} = \infty - \infty$$

(Indeterminate form)

$$\frac{1}{\sin x} - \frac{1}{x} = \frac{x - \sin x}{x \sin x}$$

Let us apply L' Hospital's rule

$$\begin{aligned} \lim_{x \rightarrow 0} \left(\frac{1}{\sin x} - \frac{1}{x} \right) &= \lim_{x \rightarrow 0} \left(\frac{x - \sin x}{x \sin x} \right) \\ &= \lim_{x \rightarrow 0} \frac{1 - \cos x}{x \cos x + \sin x} \\ &= \frac{1 - \cos 0}{0 + \sin 0} = \frac{1 - 1}{0} = \frac{0}{0} \end{aligned}$$

(Indeterminate form)

Apply again L' Hospitals rule

$$\begin{aligned} \lim_{x \rightarrow 0} \left(\frac{1}{\sin x} - \frac{1}{x} \right) &= \lim_{x \rightarrow 0} \frac{\sin x}{-x \sin x + \cos x + \cos x} \\ &= \frac{\sin 0}{0 + \cos 0 + \cos 0} = \frac{0}{0 + 1 + 1} = 0 \end{aligned}$$

7. $\lim_{x \rightarrow 1^+} \left(\frac{2}{x^2 - 1} - \frac{x}{x - 1} \right)$

3M

$$\begin{aligned} \lim_{x \rightarrow 1^+} \left(\frac{2}{x^2 - 1} - \frac{x}{x - 1} \right) &= \lim_{x \rightarrow 1^+} \left(\frac{2}{(x+1)(x-1)} - \frac{x}{x-1} \right) \\ &= \lim_{x \rightarrow 1^+} \left(\frac{2 - x(x+1)}{(x+1)(x-1)} \right) \\ &= \lim_{x \rightarrow 1} \frac{2 - x^2 - x}{x^2 - 1} = \frac{2 - 1 - 1}{1 - 1} = \frac{0}{0} \end{aligned}$$

(Indeterminate form)

Let us apply L' Hospital's rule

$$\begin{aligned} \lim_{x \rightarrow 1^+} \left(\frac{2}{x^2 - 1} - \frac{x}{x - 1} \right) &= \lim_{x \rightarrow 1^+} \left(\frac{2 - x^2 - x}{x^2 - 1} \right) \\ &= \lim_{x \rightarrow 1^+} \frac{-2x - 1}{2x} \\ &= \frac{-2(1) - 1}{2(1)} = -\frac{3}{2} \end{aligned}$$

8. $\lim_{x \rightarrow 0^+} x^x$

3M

$$\lim_{x \rightarrow 0^+} x^x = 0^0 \text{ form}$$

$$\text{Let } y = x^x$$

$$\log y = \log x^x$$

$$\log y = x \log x$$

$$\lim_{x \rightarrow 0^+} \log y = \lim_{x \rightarrow 0^+} \frac{\log x}{\left(\frac{1}{x}\right)} \left(\frac{\infty}{\infty} \text{ form} \right)$$

Let us apply L' Hospital's rule

$$\begin{aligned} \log \left(\lim_{x \rightarrow 0^+} y \right) &= \lim_{x \rightarrow 0^+} \frac{\frac{1}{x}}{-\frac{1}{x^2}} \\ &= \lim_{x \rightarrow 0^+} \left(\frac{1}{x} \right) \left(-\frac{x^2}{1} \right) \\ &= \lim_{x \rightarrow 0^+} (-x) = 0 \end{aligned}$$

$$\log_e \left(\lim_{x \rightarrow 0^+} y \right) = 0$$

$$\Rightarrow \lim_{x \rightarrow 0^+} x^x = e^0 = 1$$

9. $\lim_{x \rightarrow \infty} \left(1 + \frac{1}{x} \right)^x$

3M

$$\lim_{x \rightarrow \infty} \left(1 + \frac{1}{x} \right)^x = \left(1 + \frac{1}{\infty} \right)^\infty$$

$$\text{Let } y = \left(1 + \frac{1}{x} \right)^x$$

$$\log y = \log \left(1 + \frac{1}{x} \right)^x$$

$$\lim_{x \rightarrow \infty} (\log y) = \lim_{x \rightarrow \infty} \frac{\log \left(1 + \frac{1}{x} \right)}{\frac{1}{x}}$$

Let us apply L' Hospital's rule

$$\begin{aligned} \log_e \left(\lim_{x \rightarrow \infty} y \right) &= \lim_{x \rightarrow \infty} \frac{\frac{1}{1 + \frac{1}{x}} \left(-\frac{1}{x^2} \right)}{\left(-\frac{1}{x^2} \right)} \\ &= \lim_{x \rightarrow \infty} \frac{1}{1 + \frac{1}{x}} \end{aligned}$$

$$\log_e \left(\lim_{x \rightarrow \infty} y \right) = \frac{1}{1 + \frac{1}{\infty}}$$

$$= \frac{1}{1 + 0} = 1$$

$$\lim_{x \rightarrow \infty} y = e^1$$

$$\therefore \lim_{x \rightarrow \infty} \left(1 + \frac{1}{x} \right)^x = e$$

10. $\lim_{x \rightarrow \frac{\pi}{2}} (\sin x)^{\tan x}$

JUL-23

5M

$$\lim_{x \rightarrow \frac{\pi}{2}} (\sin x)^{\tan x} = \left(\sin \frac{\pi}{2} \right)^{\tan \frac{\pi}{2}}$$

Let $y = (\sin x)^{\tan x}$ (Indeterminate form)

$$\begin{aligned} \log y &= \log(\sin x)^{\tan x} \\ &= \tan x \log(\sin x) \\ &= \frac{\log \sin x}{\frac{1}{\tan x}} = \frac{\log \sin x}{\cot x} \end{aligned}$$

$$\lim_{x \rightarrow \frac{\pi}{2}} \log y = \lim_{x \rightarrow \frac{\pi}{2}} \frac{\log \sin x}{\cot x} \left(\frac{0}{0} \text{ form} \right)$$

Let us apply L' Hospital's rule

$$\begin{aligned} \log_e \left(\lim_{x \rightarrow \frac{\pi}{2}} y \right) &= \lim_{x \rightarrow \frac{\pi}{2}} \frac{\frac{1}{\sin x} \cdot \cos x}{-\operatorname{cosec}^2 x} \\ &= \lim_{x \rightarrow \frac{\pi}{2}} \frac{\frac{\cos x}{\sin x}}{-\frac{1}{\sin^2 x}} \\ &= \lim_{x \rightarrow \frac{\pi}{2}} \frac{\cos x}{\sin x} \left(\frac{-\sin^2 x}{1} \right) \\ &= \cos \frac{\pi}{2} \left(-\sin \frac{\pi}{2} \right) \end{aligned}$$

$$\log_e \left(\lim_{x \rightarrow \frac{\pi}{2}} y \right) = 0$$

$$\lim_{x \rightarrow \frac{\pi}{2}} y = e^0 = 1$$

$$\therefore \lim_{x \rightarrow \frac{\pi}{2}} (\sin x)^{\tan x} = 1$$

11. $\lim_{x \rightarrow 0^+} (\cos x)^{\frac{1}{x^2}}$

5M

$$\lim_{x \rightarrow 0^+} (\cos x)^{\frac{1}{x^2}} = (\cos 0)^{\frac{1}{0}} = 1^\infty$$

$$\text{Let } y = (\cos x)^{\frac{1}{x^2}}$$

$$\log y = \log(\cos x)^{\frac{1}{x^2}}$$

$$\log y = \frac{1}{x^2} \log \cos x$$

$$\lim_{x \rightarrow 0^+} \log y = \lim_{x \rightarrow 0^+} \frac{\log \cos x}{x^2}$$

Let us apply L' Hospital's rule

$$\begin{aligned} \log \left(\lim_{x \rightarrow 0^+} y \right) &= \lim_{x \rightarrow 0^+} \frac{\frac{1}{\cos x} \cdot (-\sin x)}{2x} \\ &= \lim_{x \rightarrow 0^+} \left(\frac{-1}{2 \cos x} \right) \left(\frac{\sin x}{x} \right) \\ &= \frac{-1}{2 \cos 0} \left(\lim_{x \rightarrow 0^+} \frac{\sin x}{x} \right) \\ &= \frac{1}{2(1)} (1) \end{aligned}$$

$$\log_e \left(\lim_{x \rightarrow 0^+} y \right) = -\frac{1}{2}$$

$$\lim_{x \rightarrow 0^+} y = e^{-\frac{1}{2}}$$

$$\lim_{x \rightarrow 0^+} (\cos x)^{\frac{1}{x^2}} = \frac{1}{e^{\frac{1}{2}}}$$

$$\lim_{x \rightarrow 0^+} (\cos x)^{\frac{1}{x^2}} = \frac{1}{\sqrt{e}}$$

12. If an initial amount A_0 of money is invested at an interest rate r compounded n times a year, the value of the investment after t years is $A = A_0 \left(1 + \frac{r}{n} \right)^{nt}$. If the interest is compounded continuously, (that is $n \rightarrow \infty$), show that the amount after t years is $A = A_0 e^{rt}$

3M

Method I:

$$\begin{aligned} \lim_{n \rightarrow \infty} A &= \lim_{n \rightarrow \infty} A_0 \left(1 + \frac{r}{n} \right)^{nt} \\ &= A_0 \left[\lim_{n \rightarrow \infty} \left(1 + \frac{r}{n} \right)^n \right]^t = A_0 (e^r)^t \end{aligned}$$

$$\lim_{n \rightarrow \infty} A = A_0 e^{rt}$$

$\therefore$ The amount after t years is $A = A_0 e^{rt}$

Method II:

$$\lim_{n \rightarrow \infty} A = \lim_{n \rightarrow \infty} A_0 \left(1 + \frac{r}{n} \right)^{nt} = A_0 \left(1 + \frac{r}{\infty} \right)^\infty$$

$$\text{Let } y = \left(1 + \frac{r}{n} \right)^{nt}$$

$$\lim_{n \rightarrow \infty} \log y = \lim_{n \rightarrow \infty} \frac{t \log \left(1 + \frac{r}{n} \right)}{\frac{1}{n}} \left(\frac{0}{0} \text{ form} \right)$$

Let us apply L' Hospital's rule:

PTA-6

$$\begin{aligned} \log \left(\lim_{n \rightarrow \infty} y \right) &= \lim_{n \rightarrow \infty} \frac{t \cdot \frac{1}{1 + \frac{r}{n}} \left(-\frac{r}{n^2} \right)}{\left(-\frac{1}{n^2} \right)} = \lim_{n \rightarrow \infty} rt \left(\frac{1}{1 + \frac{r}{n}} \right) \\ &= rt \left(\frac{1}{1 + \frac{r}{\infty}} \right) = rt \left(\frac{1}{1 + 0} \right) = rt \end{aligned}$$

$$\lim_{n \rightarrow \infty} y = e^{rt} \quad \lim_{n \rightarrow \infty} \left(1 + \frac{r}{n} \right)^{nt} = e^{rt}$$

$$\therefore \lim_{n \rightarrow \infty} A_0 \left(1 + \frac{r}{n} \right)^{nt} = A_0 e^{rt}$$

$$\Rightarrow \lim_{n \rightarrow \infty} A = A_0 e^{rt}$$

$\therefore$ The amount after t years is $A = A_0 e^{rt}$

EXERCISE 7.10



1. The volume of a sphere is increasing in volume at the rate of $3\pi \text{ cm}^3/\text{sec}$.
The rate of change of its radius when radius is $\frac{1}{2} \text{ cm}$

(1) 3 cm/s (2) 2 cm/s (3) 1 cm/s (4) $\frac{1}{2} \text{ cm/s}$

Given: $\frac{dv}{dt} = 3\pi \text{ cm}^3/\text{sec}$

Volume of a sphere, $V = \frac{4}{3}\pi r^3$

$$\frac{dv}{dt} = \frac{4}{3}\pi(3r^2) \frac{dr}{dt}$$

$$3\pi = 4\pi \left(\frac{1}{2}\right)^2 \frac{dr}{dt}$$

$$3 = 4 \times \frac{1}{4} \times \frac{dr}{dt}$$

$$\therefore \frac{dr}{dt} = 3 \text{ cm/sec}$$

2. A balloon rises straight up at 10 m/s . An observer is 40 m away from the spot where the balloon left the ground. The rate of change of the balloon's angle of elevation in radian per second when the balloon is 30 metres above the ground.

(1) $\frac{3}{25} \text{ radians/sec}$ (2) $\frac{4}{25} \text{ radian/sec}$ (3) $\frac{1}{5} \text{ radians/sec}$ (4) $\frac{1}{3} \text{ radians/sec}$

$$\tan \theta = \frac{h}{40}$$

$$\Rightarrow h = 40 \tan \theta$$

$$\frac{dh}{dt} = 40 \sec^2 \theta \cdot \frac{d\theta}{dt} \quad \dots\dots\dots (1)$$

$$\frac{d\theta}{dt} = \frac{1}{40} \cos^2 \theta \cdot \frac{dh}{dt}$$

$$\begin{aligned} \text{When } h = 30 \text{ m } \frac{d\theta}{dt} &= \frac{1}{40} \left(\frac{40}{50}\right)^2 (10) \\ &= \frac{4}{25} \text{ radian / sec} \end{aligned}$$

3. The position of a particle moving along a horizontal line of any time t is given by $s(t) = 3t^2 - 2t - 8$. The time at which the particle is at rest is

(1) $t = 0$ (2) $t = \frac{1}{3}$ (3) $t = 1$ (4) $t = 3$

$$s(t) = 3t^2 - 2t - 8$$

$$\text{Velocity} = s'(t) = 6t - 2$$

$$\text{The particle is at rest when } s'(t) = 0$$

$$6t - 2 = 0$$

$$t = \frac{2}{6} = \frac{1}{3}$$

4. A stone is thrown up vertically. The height it reaches at time t seconds is given by $x = 80t - 16t^2$. The stone reaches the maximum height in time t seconds is given by

(1) 2 (2) 2.5 (3) 3 (4) 3.5

$$x = 80t - 16t^2$$

$$\text{Velocity, } v = \frac{dx}{dt} = 80 - 32t$$

The stone reaches the maximum height when velocity = 0

$$\therefore \frac{dx}{dt} = 80 - 32t = 0$$

$$80 = 32t$$

$$\therefore t = \frac{80}{32} = \frac{5}{2} = 2.5 \text{ second}$$

JUL-21
APR-20
PTA-4

JUL-22

APR-25

5. The point on the curve $6y = x^3 + 2$ at which y-coordinate changes 8 times as fast as x-coordinate is

(1) (4, 11) (2) (4, -11) (3) (-4, 11) (4) (-4, -11)

Given: y-coordinate changes 8 times as fast as x-coordinate $(ie) \frac{dy}{dt} = 8 \left(\frac{dx}{dt} \right)$ Equation of the curve by $6y = x^3 + 2 \dots\dots\dots (1)$ Diff. w.r.to t $6 \frac{dy}{dt} = 3x^2 \frac{dx}{dt}$ $2 \times 8 \frac{dx}{dt} = x^2 \frac{dx}{dt}$ $x^2 = 16 \Rightarrow x = \pm 4$	When $x = 4$ $(1) \Rightarrow 6y = 4^3 + 2$ $6y = 66$ $y = 11$	when $x = -4$ $6y = (-4)^3 + 2$ $6y = -64 + 2$ $6y = -62$ $y = -\frac{62}{6}$
The point on the curve: (4, 11) or $(-4, -\frac{62}{6})$		

6. The abscissa of the point on the curve $f(x) = \sqrt{8-2x}$ at which the slope of the tangent is -0.25 ?

(1) -8 (2) -4 (3) -2 (4) 0 PTA-4, APR-23

$f(x) = \sqrt{8-2x}$ $f'(x) = \frac{-2}{2\sqrt{8-2x}} = \frac{-1}{\sqrt{8-2x}}$ Given: Slope of the tangent = -0.25 $\frac{-1}{\sqrt{8-2x}} = -0.25$ $\frac{-1}{\sqrt{8-2x}} = \frac{1}{4} \Rightarrow \sqrt{8-2x} = 4$	Squaring on both sides, we get $8-2x = 16$ $\Rightarrow 2x = -8$ $\Rightarrow x = -4$
---	--

7. The slope of the line normal to the curve $f(x) = 2 \cos 4x$ at $x = \frac{\pi}{12}$ is

PTA-6

(1) $-4\sqrt{3}$ (2) -4 (3) $\frac{\sqrt{3}}{12}$ (4) $4\sqrt{3}$

$f(x) = 2 \cos 4x$ $f'(x) = -8 \sin 4x$ $f' \left(\frac{\pi}{12} \right) = -8 \sin \frac{\pi}{3}$ $= -8 \times \frac{\sqrt{3}}{2} = -4\sqrt{3}$ $\therefore$ Slope of the normal = $-\frac{1}{m} = \frac{1}{4\sqrt{3}}$	$\times$ and $\div$ by $\sqrt{3}$ $= \frac{\sqrt{3}}{4 \times \sqrt{3} \times \sqrt{3}} = \frac{\sqrt{3}}{12}$
---	--

8. The tangent to the curve $y^2 - xy + 9 = 0$ is vertical when

(1) $y = 0$ (2) $y = \pm\sqrt{3}$ (3) $y = \frac{1}{2}$ (4) $y = \pm 3$

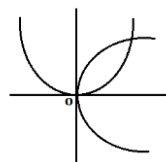
$y^2 - xy + 9 = 0$ Diff. w.r.to x, we get $2y \frac{dy}{dx} - \left[x \frac{dy}{dx} + y \right] = 0$ $2y \frac{dy}{dx} - x \frac{dy}{dx} - y = 0$ $\frac{dy}{dx} (2y - x) = y$ $\frac{dy}{dx} = \frac{y}{2y-x}$	Given: The tangent to the curve $y^2 - xy + 9 = 0$ is vertical (ie) $\frac{dy}{dx} = \infty \Rightarrow \frac{y}{2y-x} = \frac{1}{0}$ $\Rightarrow 0 = 2y - x \Rightarrow x = 2y$ $\therefore y^2 - xy + 9 = 0 \Rightarrow y^2 - (2y)y + 9 = 0$ $\Rightarrow -y^2 = -9$ $y^2 = 9 \Rightarrow y = \pm 3$
---	---

9. Angle between $y^2 = x$ and $x^2 = y$ at the origin is

PTA-2, JUL-20, 22, APR-23

- (1) $\tan^{-1} \frac{3}{4}$ (2) $\tan^{-1} \left(\frac{4}{3}\right)$ (3) $\frac{\pi}{2}$ (4) $\frac{\pi}{4}$

Tangent at origin is x axis and y axis,
angle between them is $\frac{\pi}{2}$



10. The value of the limit $\lim_{x \rightarrow 0} \left(\cot x - \frac{1}{x} \right)$ is

JUL-24

- (1) 0 (2) 1 (3) 2 (4) ∞

$$\begin{aligned} \lim_{x \rightarrow 0} \left[\cot x - \frac{1}{x} \right] &= \lim_{x \rightarrow 0} \left[\frac{\cos x}{\sin x} - \frac{1}{x} \right] \\ &= \lim_{x \rightarrow 0} \left[\frac{x \cos x - \sin x}{x \sin x} \right] \left(\frac{0}{0} \text{ form} \right) \end{aligned}$$

Applying L' Hospital's Rule, we get

$$= \lim_{x \rightarrow 0} \left[\frac{x(-\sin x) + \cos x(1) - \cos x}{x \cos x + \sin x(1)} \right]$$

$$\begin{aligned} \lim_{x \rightarrow 0} \left[\cot x - \frac{1}{x} \right] &= \lim_{x \rightarrow 0} \left[\frac{-x \sin x}{x \cos x + \sin x} \right] \left(\frac{0}{0} \text{ form} \right) \\ \text{Applying again L' Hospital's Rule, we get} \\ &= \lim_{x \rightarrow 0} \left[\frac{-x \cos x - \sin x}{-x \sin x + \cos x + \cos x} \right] \\ &= \frac{0}{0+1+1} = 0 \end{aligned}$$

11. The function $\sin^4 x + \cos^4 x$ is increasing in the interval.

MDL

- (1) $\left[\frac{5\pi}{8}, \frac{3\pi}{4} \right]$ (2) $\left[\frac{\pi}{2}, \frac{5\pi}{8} \right]$ (3) $\left[\frac{\pi}{4}, \frac{\pi}{2} \right]$ (4) $\left[0, \frac{\pi}{4} \right]$

$$\begin{aligned} f(x) &= \sin^4 x + \cos^4 x \\ f'(x) &= 4 \sin^3 x \cos x + 4 \cos^3 x (-\sin x) \\ &= -4 \sin x \cos x [\cos^2 x - \sin^2 x] \\ &= -2(2 \sin x \cos x (\cos^2 x - \sin^2 x)) \\ &= -2 \sin 2x \cos 2x = -\sin 4x \end{aligned}$$

$$\begin{aligned} f'(x) &= 0 \Rightarrow \sin 4x = 0 \\ 4x &= 0, \pi, 2\pi, \dots \Rightarrow x = 0, \frac{\pi}{4}, \frac{\pi}{2}, \dots \\ f'(x) &\geq 0 \text{ in the interval } \left[\frac{\pi}{4}, \frac{\pi}{2} \right] \end{aligned}$$

12. The number given by the Rolle's theorem for the function $f(x) = x^3 - 3x^2, x \in [0, 3]$

APR-24,
PTA-6,
JUL-20

- (1) 1 (2) $\sqrt{2}$ (3) $\frac{3}{2}$ (4) 2

$$\begin{aligned} f(x) &= x^3 - 3x^2 \\ f'(x) &= 3x^2 - 6x \end{aligned}$$

By the Rolle's theorem there must exist
 $c \in (0, 3)$ such that $f'(c) = 0$

$$\begin{aligned} &\Rightarrow 3c^2 - 6c = 0 \\ \text{Divide by 3} &\Rightarrow c^2 - 2c = 0 \\ c(c - 2) &= 0 \\ c &= 0 \text{ (or) } c = 2 \in (0, 3) \end{aligned}$$

13. The number given by the Mean value theorem for the function $f(x) = \frac{1}{x}, x \in [1, 9]$ is

PTA-3, JUL-23

- (1) 2 (2) 2.5 (3) 3 (4) 3.5

$$\begin{aligned} f(x) &= \frac{1}{x} \\ f'(x) &= -\frac{1}{x^2} \end{aligned}$$

By the mean value theorem, there exists
 $c \in (1, 9)$ such that $f'(c) = \frac{f(9) - f(1)}{9 - 1}$

$$-\frac{1}{c^2} = \frac{\frac{1}{9} - 1}{8}$$

$$\begin{aligned} -\frac{1}{c^2} &= -\frac{8}{9} \times \frac{1}{8} \\ c^2 &= 9 \\ c &= \pm 3 \\ c &= 3 \in (1, 9) \end{aligned}$$

14. The minimum value of the function $f(x) = |3 - x| + 9$ is

PTA-4, APR-22

(1) 0

(2) 3

(3) 6

(4) 9

$$f(x) = |3 - x| + 9$$

$$f'(x) = \frac{3-x}{|3-x|} (-1)$$

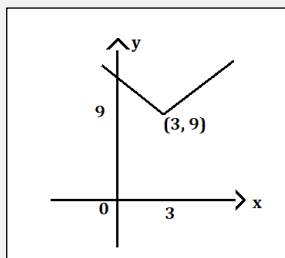
$$\left[\because \frac{d|x|}{dx} = \frac{x}{|x|} \right]$$

$$f'(x) = \frac{x-3}{|3-x|}$$

$$f'(x) = 0$$

$$\Rightarrow x - 3 = 0$$

$$\Rightarrow x = 3$$



$$f'(x) < 0 \text{ in } (-\infty, 3) \text{ and}$$

$$f'(x) > 0 \text{ in } (3, \infty)$$

Since $f'(x)$ changes from negative to positive when passing through $x = 3$,

$f(x)$ is minimum at $x = 3$

Minimum value = $f(3)$

$$= |3 - 3| + 9$$

Minimum value = 9

15. The maximum slope of the tangent to the curve $y = e^x \sin x$, $x \in [0, 2\pi]$ is at

(1) $x = \frac{\pi}{4}$ (2) $x = \frac{\pi}{2}$ (3) $x = \pi$ (4) $x = \frac{3\pi}{2}$

$$y = e^x \sin x$$

$$m = \text{Slope of the tangent} = \frac{dy}{dx}$$

$$= e^x \cos x + \sin x (e^x)$$

$$m = e^x [\sin x + \cos x]$$

$$\frac{dm}{dx} = e^x [\cos x - \sin x] + e^x [\sin x + \cos x]$$

$$\frac{dm}{dx} = 2e^x \cos x$$

$$\frac{d^2m}{dx^2} = 2e^x [\cos x - \sin x]$$

$$\frac{dm}{dx} = 0$$

$$\Rightarrow 2e^x \cos x = 0$$

$$\cos x = 0$$

$$\Rightarrow x = \frac{\pi}{2} \in [0, 2\pi]$$

$$\text{At } x = \frac{\pi}{2}, \frac{d^2m}{dx^2} = 2e^{\frac{\pi}{2}} \left[\cos \frac{\pi}{2} - \sin \frac{\pi}{2} \right]$$

$$= 2e^{\frac{\pi}{2}} [0 - 1]$$

$$= -2e^{\frac{\pi}{2}}$$

$$= -ve$$

$\therefore$ The maximum slope of the tangent to the curve is $x = \frac{\pi}{2}$.

16. The maximum value of the function $f(x) = x^2 e^{-2x}$, $x > 0$ is

APR-23

(1) $\frac{1}{e}$ (2) $\frac{1}{2e}$ (3) $\frac{1}{e^2}$ (4) $\frac{4}{e^4}$

$$f(x) = x^2 e^{-2x}$$

$$f'(x) = x^2 [-2e^{-2x}] + e^{-2x} (2x)$$

$$f'(x) = 2xe^{-2x} [1 - x]$$

$$f'(x) = 0 \Rightarrow x = 0 \text{ (or) } x = 1$$

$$f'(x) < 0 \text{ in } (-\infty, 0)$$

$$f'(x) > 0 \text{ in } (0, 1)$$

$$f'(x) < 0 \text{ in } (1, \infty)$$

Since $f'(x)$ changes from +ve to -ve when passing through $x = 1$, $f(x)$ is maximum at $x = 1$

$$\text{Maximum value} = f(1) = (1)^2 e^{-2(1)}$$

$$= e^{-2} = \frac{1}{e^2}$$

17. One of the closest points on the curve $x^2 - y^2 = 4$ to the point (6,0) is

- (1) (2,0) (2) $(\sqrt{5}, 1)$ (3) $(3, \sqrt{5})$ (4) $(\sqrt{13}, -\sqrt{3})$

Let (x, y) be a point on the curve $x^2 - y^2 = 4$ $y^2 = x^2 - 4 \quad \dots\dots\dots (1)$ The distance between the points (6,0) and $(x, y) = d = \sqrt{(x-6)^2 + (y-0)^2}$ $d^2 = (x-6)^2 + y^2 = (x-6)^2 + (x^2 - 4)$ Let $d^2 = f(x)$ $f(x) = x^2 - 12x + 36 + x^2 - 4$ $f(x) = 2x^2 - 12x + 32$ $f'(x) = 4x - 12$	$f''(x) = 4$ $f'(x) = 0 \Rightarrow 4x - 12 = 0$ $x = 3$ At $x = 3, f''(x) = 4 = +ve$ $\therefore f(x) = d^2$ is minimum at $x = 3$ $(1) \Rightarrow y^2 = 3^2 - 4 = 5$ $y = \pm\sqrt{5}$ $\therefore$ The closest point on the given curve to the point (6,0) is $(3, \sqrt{5})$
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18. The maximum product of two positive numbers, when their sum of the squares is 200, is

- (1) 100 (2) $25\sqrt{7}$ (3) 28 (4) $24\sqrt{14}$

Let x and y be the two +ve numbers Given: $x^2 + y^2 = 200$ $y^2 = 200 - x^2 \quad \dots\dots\dots (1)$ Product, $p = xy$ $p^2 = x^2 y^2 = x^2 (200 - x^2)$ Let $f(x) = p^2 = 200x^2 - x^4$ $f'(x) = 400x - 4x^3$ $f''(x) = 400 - 12x^2$ $f'(x) = 0$ $\Rightarrow 400x - 4x^3 = 0 \quad [\text{Divide by } 4]$	$\Rightarrow 100x - x^3 = 0$ $x(100 - x^2) = 0$ $x = 0 \text{ (or) } x^2 = 100$ $x = 10 \quad [\because x \text{ is a +ve numbers}]$ At $x = 10, f''(10)$ $= 400 - 12(10^2) = -ve$ $\therefore f(x) = p^2$ is maximum at $x = 10$ $(1) \Rightarrow y^2 = 200 - 10^2 = 200 - 100 = 100$ $y = 10 \quad [\because y \text{ is +ve integer}]$ $\therefore$ Maximum product $= xy = 10 \times 10 = 100$
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19. The curve $y = ax^4 + bx^2$ with $ab > 0$

- (1) has no horizontal tangent (2) is concave up
(3) is concave down (4) has no points of inflection

MDL, PTA-2, JUL-23

$y = ax^4 + bx^2$ $\frac{dy}{dx} = 4ax^3 + 2bx$ $\frac{d^2y}{dx^2} = 12ax^2 + 2b$ $\frac{d^2y}{dx^2} = 0$ $\Rightarrow 12ax^2 + 2b = 0$	$x^2 = \frac{-2b}{12a} = \frac{-b}{6a}$ $x = \sqrt{\frac{-b}{6a}}$ [Since $ab > 0$ $\frac{-b}{6a}$ is $-ve$ for all a, b] x is an imaginary number $\therefore$ The curve has no points of inflection.
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20. The point of inflection of the curve $y = (x-1)^3$ is

- (1) (0,0) (2) (0,1) (3) (1,0) (4) (1,1)

JUL-24, PTA-5, APR-25,22

$y = (x-1)^3$ $\frac{dy}{dx} = 3(x-1)^2$ $\frac{d^2y}{dx^2} = 6(x-1) = 0$ $\Rightarrow 6(x-1) = 0 \Rightarrow x-1 = 0 \Rightarrow x = 1$	$\frac{d^2y}{dx^2} < 0 \text{ in } (-\infty, 1)$ $\frac{d^2y}{dx^2} > 0 \text{ in } (1, \infty)$ At $x = 1, y = (1-1)^3 = 0$ $\therefore$ The point of inflection is (1,0)
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Godfried W Leibniz

Chapter - 8

Differentials and Partial Derivatives

EXERCISE 8.1



Video class

Concept Corner

Linear approximation:

Let $f: (a, b) \rightarrow R$ be a differentiable function and $x_0 \in (a, b)$ we define the linear approximation L of f at x_0 by

$$L(x) = f(x_0) + f'(x_0)(x - x_0), \quad \forall x \in (a, b)$$

Absolute error, relative error, percentage error

$$f(x_0 + h) - f(x_0) \approx f'(x_0)h$$

Absolute error = |Actual value - approximate value|

$$\text{Relative error} = \frac{\text{Absolute error}}{\text{Actual value}}$$

Percentage error = Relative error $\times 100$

1. Let $f(x) = \sqrt[3]{x}$. Find the linear approximation at $x = 27$. Use the linear approximation to approximate $\sqrt[3]{27.2}$

3M

$$f(x) = \sqrt[3]{x} = x^{\frac{1}{3}} \quad x_0 = 27 \quad \Delta x = 0.2$$

$$\begin{aligned} f(x_0) &= f(27) \\ &= \sqrt[3]{27} \\ &= 3 \end{aligned}$$

$$\begin{aligned} f'(x) &= \frac{1}{3} x^{\frac{1}{3}-1} \\ &= \frac{1}{3} x^{-\frac{2}{3}} \\ &= \frac{1}{3 x^{\frac{2}{3}}} \end{aligned}$$

$$\begin{aligned} f'(27) &= \frac{1}{3} (27)^{-\frac{2}{3}} \\ &= \frac{1}{3} \times \frac{1}{27} \\ &= \frac{1}{81} \end{aligned}$$

Linear approximation

$$\begin{aligned} L(x) &= f(x_0) + f'(x_0)(x - x_0) \\ &= 3 + \frac{1}{81}(x - 27) \\ &= 3 + \frac{x}{81} - 1 \end{aligned}$$

$$L(x) = \frac{x}{81} + 2$$

This is the required linear approximation

Put $x = 27.2$

$$f(27.2) \approx L(27.2)$$

$$\approx \frac{27.2}{81} + 2$$

$$= 1.0074 + 2$$

$$\sqrt[3]{27.2} \approx 3.0074$$

2. Use the linear approximation to find approximate values of (i) $(123)^{\frac{2}{3}}$ (ii) $\sqrt[4]{15}$ (iii) $\sqrt[3]{26}$

3M

(i) $(123)^{\frac{2}{3}}$

APR-23

$$\text{Let } f(x) = x^{\frac{2}{3}}$$

$$x_0 = 125 ; \Delta x = -2$$

$$f(x_0) = (125)^{\frac{2}{3}}$$

$$= (5^3)^{\frac{2}{3}}$$

$$= 5^2$$

$$= 25$$

$$f'(x) = \frac{2}{3}x^{\frac{2}{3}-1}$$

$$= \frac{2}{3} \frac{x^{\frac{2}{3}}}{x}$$

$$f'(x_0) = f'(125)$$

$$= \frac{\frac{2}{3}(125)^{\frac{2}{3}}}{125}$$

$$= \frac{2(25)}{3(125)}$$

$$= \frac{2}{15}$$

$\therefore$ Linear approximation

$$L(x) = f(x_0) + f'(x_0)(x - x_0)$$

$$L(x) = 25 + \frac{2}{15}(x - 125)$$

$$= 25 + \frac{2x}{15} - 2 \times \frac{125}{15}$$

$$= \frac{2x}{15} + 25 - \frac{50}{3}$$

$$= \frac{2x}{15} + \frac{75-50}{3}$$

$$L(x) = \frac{2x}{15} + \frac{25}{3}$$

$$\text{Put } x = 123$$

$$f(123) = (123)^{\frac{2}{3}}$$

$$\cong L(123)$$

$$\cong \frac{2(123)}{15} + \frac{25}{3}$$

$$= \frac{82}{5} + \frac{25}{3}$$

$$= \frac{246+125}{15}$$

$$= \frac{371}{15}$$

$$(123)^{\frac{2}{3}} \cong \mathbf{24.733}$$

(ii) $\sqrt[4]{15}$

$$\text{Let } f(x) = x^{\frac{1}{4}}$$

$$x_0 = 16 ; \Delta x = -1$$

$$f(x_0) = f(16)$$

$$= 16^{\frac{1}{4}}$$

$$= (2^4)^{\frac{1}{4}}$$

$$= 2$$

$$f'(x) = \frac{1}{4}x^{\frac{1}{4}-1}$$

$$= \frac{1}{4} \frac{x^{\frac{1}{4}}}{x^1}$$

$$f'(16) = \frac{1}{4} \frac{16^{\frac{1}{4}}}{16}$$

$$= \frac{2}{64}$$

$$= \frac{1}{32}$$

Linear approximation

$$L(x) = f(x_0) + f'(x_0)(x - x_0)$$

$$= 2 + \frac{1}{32}(x - 16)$$

$$= 2 + \frac{x}{32} - \frac{16}{32}$$

$$= \frac{x}{32} + 2 - \frac{1}{2}$$

$$L(x) = \frac{x}{32} + \frac{3}{2}$$

$$\text{Put } x = 15$$

$$f(15) = \sqrt[4]{15}$$

$$\cong L(15)$$

$$\cong \frac{15}{32} + \frac{3}{2}$$

$$= \frac{15+48}{32}$$

$$= \frac{63}{32}$$

$$\sqrt[4]{15} \cong \mathbf{1.968}$$

(iii) $\sqrt[3]{26}$

$$\text{Let } f(x) = x^{\frac{1}{3}}$$

$$x_0 = 27 ; \Delta x = -1$$

$$f(x) = \sqrt[3]{x}$$

$$= x^{\frac{1}{3}}$$

$$f(x_0) = \sqrt[3]{27}$$

$$= 3$$

$$f'(x) = \frac{1}{3}x^{\frac{1}{3}-1}$$

$$= \frac{1}{3} \frac{x^{\frac{1}{3}}}{x}$$

$$f'(x_0) = f'(27)$$

$$= \frac{1}{3} \frac{(27)^{\frac{1}{3}}}{27}$$

$$= \frac{1}{3} \frac{3}{27}$$

$$= \frac{1}{27}$$

Linear approximation

$$L(x) = f(x_0) + f'(x_0)(x - x_0)$$

$$= 3 + \frac{1}{27}(x - 27)$$

$$= 3 + \frac{x}{27} - 1$$

$$L(x) = \frac{x}{27} + 2$$

$$\text{Put } x = 26$$

$$f(26) = \sqrt[3]{26}$$

$$\cong L(26)$$

$$\cong \frac{26}{27} + 2$$

$$\cong 0.9629 + 2$$

$$\sqrt[3]{26} \approx \mathbf{2.963}$$

3. Find a linear approximation for the following functions at the indicated points.

(i) $f(x) = x^3 - 5x + 12$, $x_0 = 2$

$$f(x_0) = 2^3 - 5(2) + 12$$

$$= 8 - 10 + 12$$

$$f(x_0) = 10$$

$$f'(x) = 3x^2 - 5$$

$$f'(x_0) = f'(2)$$

$$= 3(2)^2 - 5 = 12 - 5 = 7$$

$$L(x) = f(x_0) + f'(x_0)(x - x_0)$$

$$L(x) = 10 + 7(x - 2) = 10 + 7x - 14$$

$$= 7x - 4$$

(ii) $g(x) = \sqrt{x^2 + 9}$, $x_0 = -4$

$$g(x) = \sqrt{x^2 + 9}$$

$$g(x_0) = \sqrt{(-4)^2 + 9} = \sqrt{16 + 9} = \sqrt{25} = 5$$

$$g'(x) = \frac{1}{2\sqrt{x^2 + 9}} 2x$$

$$g'(x_0) = g'(-4) = \frac{-4}{\sqrt{16+9}} = -\frac{4}{5}$$

Linear approximation

$$L(x) = g(x_0) + g'(x_0)(x - x_0)$$

$$= 5 + \left(-\frac{4}{5}\right)(x + 4)$$

$$= 5 - \frac{4x}{5} - \frac{16}{5}$$

$$= \frac{25-16}{5} - \frac{4x}{5} = \frac{9}{5} - \frac{4x}{5}$$

$$L(x) = \frac{9-4x}{5}$$

(iii) $h(x) = \frac{x}{x+1}$, $x_0 = 1$

$$h(x_0) = h(1) = \frac{1}{1+1} = \frac{1}{2}$$

$$h'(x) = \frac{(x+1)(1) - x(1)}{(x+1)^2} \quad \left| \quad h'(x_0) = h'(1) \right.$$

$$= \frac{x+1-x}{(x+1)^2} \quad \left| \quad = \frac{1}{(1+1)^2} = \frac{1}{2^2} = \frac{1}{4} \right.$$

Linear approximation

$$L(x) = h(x_0) + h'(x_0)(x - x_0)$$

$$= \frac{1}{2} + \frac{1}{4}(x - 1) = \frac{1}{2} + \frac{x}{4} - \frac{1}{4}$$

$$= \frac{x}{4} + \frac{1}{2} - \frac{1}{4} = \frac{x+2-1}{4}$$

$$L(x) = \frac{x+1}{4}$$

4. The radius of a circular plate is measured as 12.65 cm instead of the actual length 12.5 cm. find the following in calculating change in the area of the circular plate:

(i) Absolute error (ii) Relative error

(iii) Percentage error

$$r = 12.65 \quad \Delta r = 12.65 - 12.5 = 0.15$$

$$\text{Area } A(r) = \pi r^2 \Rightarrow dA = 2\pi r dr$$

$$\Delta A \cong dA = 2\pi r dr$$

$$= 2\pi(12.65)(0.15) = 3.795\pi \text{ cm}^2$$

$$\text{Actual error} = A(12.65) - A(12.5)$$

$$= \pi(12.65)^2 - \pi(12.5)^2$$

$$= 160.0225\pi - 156.25\pi$$

$$= 3.7725\pi \text{ cm}^2$$

$$\text{Absolute Error} = \text{Actual error} - \text{Approximate error}$$

$$= 3.795\pi - 3.7725\pi$$

$$= 0.0225\pi \text{ cm}^2$$

$$\text{Relative error} = \frac{\text{Absolute error}}{\text{Actual value of Error}}$$

$$= \frac{0.0225\pi}{3.775\pi} = 0.00596$$

$$= 0.00596 \cong 0.006$$

$$\text{Percentage Error} = 0.006 \times 100 = 0.6\%$$

5. A sphere is made of ice having radius 10 cm. Its radius decreases from 10 cm to 9.8 cm. Find approximations for the following:

(i) change in the volume

(ii) change in the surface area

$$\text{Actual radius } r = 10 ; \text{ Approximate } r = 9.8$$

$$\text{Difference } h = 10 - 9.8 = 0.2$$

$$\text{Volume of Sphere } V = \frac{4}{3} \pi r^3$$

$$V'(r) = \frac{4}{3} \times 3\pi r^2 = 4\pi r^2$$

(i) Change in the volume

$$V(10) - V(9.8) \cong V'(10)h$$

$$\cong 4\pi(10)^2 0.2 \cong 80\pi \text{ cm}^3$$

(ii) Change in SA

$$\text{SA of sphere } S = 4\pi r^2 ; \quad S' = 8\pi r$$

$$\text{Change in S.A} = S(10) - S(9.8)$$

$$\cong S'(r)h$$

$$\cong 8\pi(10)(0.2)$$

$$\cong 16\pi \text{ cm}^2$$

Surface area decreased by $16\pi \text{ cm}^2$

6. The time T , taken for a complete oscillation of a single pendulum with length l , is given by the equation $T = 2\pi\sqrt{\frac{l}{g}}$, where g is a constant. Find the approximate percentage error in the calculated value of T corresponding to an error of 2 percent in the value of l . 3M

$$T = 2\pi\sqrt{\frac{l}{g}}$$

$$\begin{aligned}\text{Taking log on both sides, } \log T &= \log 2\pi \left(\frac{l}{g}\right)^{1/2} \\ &= \log 2\pi + \log \left(\frac{l}{g}\right)^{1/2} \\ &= \log 2\pi + \frac{1}{2} \log \left(\frac{l}{g}\right) \\ \log T &= \log 2\pi + \frac{1}{2} [\log l - \log g]\end{aligned}$$

$$\begin{aligned}\text{Diff. w.r.to } l, \frac{1}{T} \frac{dT}{dl} &= 0 + \frac{1}{2} \left[\frac{1}{l} - 0\right] \\ \frac{dT}{T} &= \frac{dl}{2l} \quad \dots\dots\dots(1)\end{aligned}$$

$$\begin{aligned}\text{Given that } dl &= 2\% \text{ of } l = \frac{2l}{100} \\ (1) \Rightarrow \frac{dT}{T} &= \frac{2l/100}{2l} \\ \frac{dT}{T} &= \frac{1}{100}\end{aligned}$$

$$\text{Relative error } \frac{\Delta T}{T} \cong \frac{dT}{T} = \frac{1}{100}$$

$$\begin{aligned}\text{Percentage error} &= \text{Relative error} \times 100 \\ &= \frac{1}{100} \times 100 = 1\%\end{aligned}$$

7. Show that the percentage error in the n^{th} root of a number is approximately $\frac{1}{n}$ times the percentage error in the number.

Let the number be x , JUL-24,23 3M

Let its n^{th} root $x^{1/n} = y$

$$y = x^{1/n}$$

Taking log on both sides, $\log y = \log x^{1/n}$

$$\log y = \frac{1}{n} \log x$$

Diff.w.r.to x ,

$$\frac{1}{y} \frac{dy}{dx} = \frac{1}{n} \frac{1}{x}$$

$$\frac{dy}{y} = \frac{1}{n} \frac{dx}{x}$$

Multiply by 100,

$$\frac{dy}{y} \times 100 = \frac{1}{n} \left(\frac{dx}{x} \times 100\right)$$

$$\begin{aligned}\frac{\Delta y}{y} \times 100 &\cong \frac{dy}{y} \times 100 \\ &= \frac{1}{n} \left(\frac{dx}{x} \times 100\right)\end{aligned}$$

Percentage error of $y \cong \frac{1}{n}$ (% error of x)

EXERCISE 8.2

Concept Corner

Differentials : Let $f: (a, b) \rightarrow R$ be a differentiable function for $x \in (a, b)$ and Δx the increment given to x , we define the differential of f by $df = f'(x)\Delta x$

Properties of differentials :

1. If $f = c$ then $df = 0$
2. If $f(x) = x$ then $df = 1 dx$
3. If f is differentiable, $c \in R$, then $d(cf) = cf'(x)dx$
4. If f, g are differentiable, then $d(f + g) = df + dg = f'(x)dx + g'(x)dx$
5. If f, g are differentiable, then $d(fg) = fdg + gdf = (f(x)g'(x) + f'(x)g(x))dx$
6. If f, g are differentiable, then $d\left(\frac{f}{g}\right) = \frac{gdf - fdg}{g^2} = \frac{g(x)f'(x) - f(x)g'(x)}{g^2(x)} dx$, where $g(x) \neq 0$
7. If f, g are differentiable and $h = f \circ g$ is defined, then $dh = f'(g(x))g'(x)dx$
8. If $h(x) = e^{f(x)}$ then $dh = e^{f(x)}f'(x)dx$
9. If $f(x) > 0, \forall x$ & $g(x) = \log f(x)$ then $dg = \frac{f'(x)}{f(x)} dx$.



Video class

1. Find differential dy for each of the following functions:

2M

(i) $y = \frac{(1-2x)^3}{3-4x}$

$f = (1-2x)^3$
 $g = 3-4x$

$$d\left(\frac{f}{g}\right) = \frac{gdf - fdg}{g^2}$$

$$\frac{dy}{dx} = \frac{(3-4x)3(1-2x)^2(-2) - (1-2x)^3(0-4)}{(3-4x)^2}$$

$$= (1-2x)^2 \frac{[-6(3-4x) + 4(1-2x)]}{(3-4x)^2}$$

$$= (1-2x)^2 \frac{[-18 + 24x + 4 - 8x]}{(3-4x)^2}$$

$$dy = \frac{(1-2x)^2(16x-14)}{(3-4x)^2} dx = \frac{2(1-2x)^2(8x-7)}{(3-4x)^2} dx$$

(ii) $y = (3 + \sin(2x))^{\frac{2}{3}}$

$$\frac{dy}{dx} = \frac{2}{3} (3 + \sin 2x)^{\frac{2}{3}-1} (0 + \cos 2x (2))$$

$$= \frac{4}{3} (3 + \sin 2x)^{-\frac{1}{3}} \cos 2x$$

$$dy = \frac{4 \cos 2x}{3(3 + \sin 2x)^{\frac{1}{3}}} dx$$

$$dy = \frac{4}{3} \frac{\cos 2x}{(3 + \sin 2x)^{\frac{1}{3}}} dx$$

(iii) $y = e^{x^2-5x+7} \cos(x^2 - 1)$

$$d(fg) = f dg + g df$$

$$\frac{dy}{dx} = e^{x^2-5x+7} (\cos(x^2 - 1))' + (\cos(x^2 - 1)) (e^{x^2-5x+7})'$$

$$= e^{x^2-5x+7} (-\sin(x^2 - 1) (2x) + \cos(x^2 - 1) e^{x^2-5x+7} (2x - 5))$$

$$\frac{dy}{dx} = e^{x^2-5x+7} [-2x \sin(x^2 - 1) + (2x - 5) \cos(x^2 - 1)]$$

$$dy = e^{x^2-5x+7} [(2x - 5) \cos(x^2 - 1) - 2x \sin(x^2 - 1)] dx$$

2. Find df for $f(x) = x^2 + 3x$ and evaluate it for

2M

(i) $x = 2$ and $dx = 0.1$

$$f(x) = x^2 + 3x$$

APR-24, 20, 22

$$\frac{df}{dx} = 2x + 3$$

$$df = (2x + 3)dx$$

$$df = (2(2) + 3)(0.1) = (4 + 3)(0.1) = 0.7$$

(ii) $x = 3$ and $dx = 0.02$

JUL-22

$$f(x) = x^2 + 3x$$

$$df = (2x + 3)dx$$

$$df = (2(3) + 3)(0.02)$$

$$= (6 + 3)(0.02) = 0.18$$

3. Find Δf and df for the function f for the indicated values of x , Δx and compare

3M

(i) $f(x) = x^3 - 2x^2$; $x = 2$, $\Delta x = dx = 0.5$ (ii) $f(x) = x^2 + 2x + 3$; $x = -0.5$, $\Delta x = dx = 0.1$

(i) $f(x) = x^3 - 2x^2$; $x = 2$, $\Delta x = dx = 0.5$ To find df

$$f(x) = x^3 - 2x^2 \quad x = 2,$$

$$\frac{df}{dx} = 3x^2 - 4x \quad dx = 0.5$$

$$df = (3x^2 - 4x)dx$$

$$df = (3(2)^2 - 4(2))(0.5)$$

$$= (12 - 8)0.5 = 2.0$$

To find $\Delta f = f(x + \Delta x) - f(x)$

$$= f(2 + 0.5) - f(2)$$

$$= f(2.5) - f(2) = f\left(\frac{5}{2}\right) - f(2)$$

$$= \left(\frac{5}{2}\right)^3 - 2\left(\frac{5}{2}\right)^2 - (2^3 - 2(2)^2)$$

$$= \frac{125}{8} - 2\left(\frac{25}{4}\right) - (8 - 8) = \frac{125-100}{8} = \frac{25}{8}$$

$$\Delta f = 3.125$$

To find df

$$f(x) = x^2 + 2x + 3 \quad x = (-0.5)$$

$$\frac{df}{dx} = 2x + 2 \quad dx = 0.1$$

$$df = (2x + 2)dx$$

$$= (2(-0.5) + 2)(0.1) = (-1 + 2)0.1 = 0.1$$

To find $\Delta f = f(x + \Delta x) - f(x)$

$$= f(-0.5 + 0.1) - f(-0.5)$$

$$= f(-0.4) - f(-0.5)$$

$$= ((-0.4)^2 + 2(-0.4) + 3) - ((-0.5)^2 + 2(-0.5) + 3)$$

$$= (0.16 - 0.8 + 3) - (0.25 - 1 + 3)$$

$$= 3.16 - 0.8 - (2 + 0.25) = 3.16 - 3.05$$

$$\Delta f = 0.11$$

4. Assuming $\log_{10} e = 0.4343$, find an approximate value of $\log_{10} 1003$.

$$\text{Let } y = f(x) = \log_{10} x$$

3M

$$\begin{aligned} f(1000) &= \log_{10} 1000 \\ &= \log_{10} 10^3 \\ &= 3 \log_{10} 10 \\ &= 3 \end{aligned}$$

$$x = 1000, \Delta x = 3$$

$$x + \Delta x = 1003$$

$$\Delta x \cong dx = 3$$

$$y = \log_{10} x$$

Diff.w.r.to x ,

$$\frac{dy}{dx} = \frac{1}{x} \log_{10} e$$

$$dy = \frac{0.4343}{x} dx$$

$$dy = \frac{0.4343}{1000} \times 3 = \frac{1.3029}{1000}$$

$$dy = 0.001303$$

$$f(x + \Delta x) \cong y + dy$$

$$f(1000 + 3) \cong f(1000) + 0.001303$$

$$\log_{10} 1003 \cong 3 + 0.001303$$

$$\cong \mathbf{3.0013029}$$

6. An egg of a particular bird is very nearly spherical. If the radius to the inside of the shell is 5mm and radius to the outside of the shell is 5.3mm find the volume of the shell approximately.

2M

$$\begin{aligned} r_1 &= 5 \text{ mm}, r_2 = 5.3 \text{ mm}, r = 5, \\ dr &= 0.3 \text{ mm} \end{aligned}$$

$$\text{Volume of sphere} = \frac{4}{3} \pi r^3$$

$$V(r) = \frac{4}{3} \pi r^3$$

$$V'(r) = \frac{4}{3} \pi r^2 (3) dr$$

$$V'(r) = 4 \pi r^2 dr$$

$$V(5.3) - V(5.0) \cong V'(r) dr$$

$$= 4 \pi (5)^2 (0.3) \cong 30 \pi$$

$$\text{Volume of shell} \cong \mathbf{30 \pi \text{ mm}^3}$$

5. The trunk of a tree has diameter 30cm. During the following year, the circumference grew 6 cm

(i) Approximately, how much did the tree's diameter grow?

3M

(ii) What is the percentage increase in area of the tree's cross-section.

$$\text{Actual diameter } 2r = 30$$

$$\text{Actual circumference} = 2\pi r_1 = 30\pi \quad C_A = 30\pi$$

$$\text{After increasing circumference} = 30\pi + 6 \quad C_I = 30\pi + 6$$

$$2\pi r_2 = 30\pi + 6$$

(i) Increasing diameter

$$\text{Difference of circumference} = 6$$

$$2\pi r_2 - 2\pi r_1 = 6$$

$$(2r_2 - 2r_1)\pi = 6$$

$$2r_2 - 2r_1 = \frac{6}{\pi}$$

$$r_2 - r_1 = \frac{3}{\pi}$$

$$r = 15$$

$$\text{Increasing in diameter} = \frac{6}{\pi} \text{ cm}$$

$$dr = \frac{3}{\pi}$$

$$\text{ii) Percentage increasing} = \frac{\text{Increasing area}}{\text{actual area}} \times 100$$

$$A = \pi r^2$$

$$r = 15$$

$$\frac{dA}{dr} = 2\pi r$$

$$dr = \frac{3}{\pi}$$

$$dA = 2\pi r dr$$

$$dA = 2\pi (15) \frac{3}{\pi} = 90$$

$$\begin{aligned} \text{(i)} \Rightarrow \text{Percentage of increasing} &= \frac{90}{\pi \times 15 \times 15} \times 100 \\ &= \frac{40}{\pi} \% \end{aligned}$$

7. Assume that the cross section of the artery of human is circular. A drug is given to a patient to dilate his arteries. If the radius of an artery is increased from 2 mm to 2.1 mm, how much is cross-sectional area increased approximately?

3M

Artery cross section is like circle.

$$\text{Area of circle } A = \pi r^2$$

APR-22

$$\frac{dA}{dr} = 2\pi r$$

$$r = 2$$

$$dA = 2\pi r dr \quad dr = 2.1 - 2$$

$$dA = 2\pi (2) 0.1 = 0.1$$

$$dA = 0.4 \pi$$

$$A(2.1) - A(2.0) \cong dA$$

$$\text{Area increased} \cong \mathbf{0.4 \pi \text{ mm}^2}$$

8. In a newly developed city, it is estimated that the voting population (in thousands) will increase according to $V(t) = 30 + 12t^2 - t^3$, $0 \leq t \leq 8$ where t is the time in years. Find the approximate change in voters for the time change from 4 to $4\frac{1}{6}$ year. 2M

$$V(t) = 30 + 12t^2 - t^3$$

$$V'(t) = 0 + 24t - 3t^2$$

$$\frac{dV}{dt} = 24t - 3t^2$$

$$dV = (24t - 3t^2)dt$$

$$dV = (24(4) - 3(4^2))\frac{1}{6}$$

$$dV = (96 - 48)\frac{1}{6} = (48)\frac{1}{6}$$

$$dV = 8000$$

$$V\left(4\frac{1}{6}\right) - V(4) \cong 8000$$

Change in voters $\cong 8000$

9. The relation between the number of words y a person learns in x hours is given by $y = 52\sqrt{x}$, $0 \leq x \leq 9$. What is the approximate number of words learned when x changes from 3M

(i) 1 to 1.1 hour? (ii) 4 to 4.1 hour?

$$y = 52\sqrt{x}, \quad 0 \leq x \leq 9$$

(i) $x = 1$; $dx = \Delta x = 1.1 - 1 = 0.1$

$$y = 52\sqrt{x}$$

$$\frac{dy}{dx} = 52 \cdot \frac{1}{2\sqrt{x}}$$

$$dy = 26 \cdot \frac{1}{\sqrt{x}} dx$$

$$dy = 26 \cdot \frac{1}{\sqrt{1}} (0.1) = 2.6$$

Change in words learn $= 2.6 \cong 3$ words.

(ii) $x = 4$ $dx = \Delta x = 4.1 - 4 = 0.1$

$$dy = 26 \cdot \frac{1}{\sqrt{4}} dx$$

$$dy = 26 \cdot \frac{1}{2} (0.1) = \frac{26}{2} (0.1)$$

$$dy = 1.3$$

No. of word learned $\cong 1$ word

10. A circular plate expands uniformly under the influence of heat. If its radius increases from 10.5 cm to 10.75 cm , then find an approximate change in the area and the approximate percentage change in the area.

Given radius 10.5 cm

PTA-4, JUL-22

$$r = 10.5$$

$$\Delta r = 10.75 - 10.5 = 0.25$$

3M

Area of circular plate $A = \pi r^2$

$$\frac{dA}{dr} = 2\pi r$$

$$dA = 2\pi r dr$$

$$dA = 2\pi(10.5)0.25$$

$$= 21\pi \times 0.25 = 5.25\pi$$

Approximate change in Area

$$A(10.75) - A(10.5) \cong dA$$

$$\cong 5.25\pi \text{ cm}^2$$

Approximate percentage of change $= \frac{dA}{A} \times 100$

$$= \frac{5.25}{\pi(10.5)^2} \times 100 = \frac{525}{10.5}$$

$$= \frac{10}{2.1} = \frac{100}{21} = 4.76\%$$

11. A coat of paint of thickness 0.2 cm is applied to the faces of a cube whose edge is 10 cm . Use the differentials to find approximately how many cubic centimeters of paint is used to paint this cube. Also calculate the exact amount of paint used to paint this cube.

Edge of a cube $= 10 \text{ cm}$

3M

$$a = 10$$

Thickness of paint $= 0.2 \text{ cm}$

$$da = 0.2$$

Volume of cube $V = a^3$

$$\frac{dV}{da} = 3a^2$$

$$dV = 3a^2 da$$

$$dV = 3(10)^2 0.2$$

$$dV = 300 \times 0.2 = 60 \text{ cm}^3$$

Volume of paint $= V(10.2) - V(10)$

$$\cong dV \cong 60 \text{ cm}^3$$

Exact volume of paint $= V(10.2) - V(10)$

$$= (10.2)^3 - 10^3$$

$$= 1061.2 - 1000$$

$$= 61.2 \text{ cm}^3$$

EXERCISE 8.3



Video class

Concept Corner

Limits and continuity of functions of two variables

Limits of a function:

Suppose that $A = \{(x, y) / a < x < b, c < y < d\} \subset R^2, F: A \rightarrow R$ we say F has a limit L at (u, v) if. For every neighbourhood. $(L - \varepsilon, L + \varepsilon), \varepsilon > 0$ of L .

There exists a δ - neighbourhood. $B_\delta(u, v) \subset A$ Such that $(x, y) \in B_\delta(u, v) \setminus \{(u, v)\}, \delta > 0 \Rightarrow f(x) \in (L - \varepsilon, L + \varepsilon)$

We denote this by $\lim_{(x,y) \rightarrow (u,v)} F(x, y) = L$

Continuity :

Suppose that $A = \{(x, y) / a < x < b, c < y < d\} \subset R^2, F: A \rightarrow R$. We say that F is continuous at (u, v) if the following hold :

If (1) F is defined at (u, v) (2) $\lim_{(x,y) \rightarrow (u,v)} F(x, y) = L$ exists (3) $L = F(u, v)$

1. Evaluate $\lim_{(x,y) \rightarrow (1,2)} g(x, y)$, if the limit exists,

2M

where $g(x, y) = \frac{3x^2 - xy}{x^2 + y^2 + 3}$.

$$\begin{aligned} \lim_{(x,y) \rightarrow (1,2)} g(x, y) &= \lim_{(x,y) \rightarrow (1,2)} \frac{3x^2 - xy}{x^2 + y^2 + 3} \\ &= \frac{3(1)^2 - 1(2)}{1^2 + 2^2 + 3} = \frac{3-2}{1+4+3} = \frac{1}{8} \end{aligned}$$

$$\lim_{(x,y) \rightarrow (1,2)} g(x, y) = \frac{1}{8}$$

2. Evaluate $\lim_{(x,y) \rightarrow (0,0)} \cos\left(\frac{x^3 + y^2}{x + y + 2}\right)$, If the limit exists.

2M

$$\begin{aligned} \lim_{(x,y) \rightarrow (0,0)} \cos\left(\frac{x^3 + y^2}{x + y + 2}\right) &= \cos\left(\frac{0+0}{0+0+2}\right) \\ &= \cos(0) \end{aligned}$$

$$\lim_{(x,y) \rightarrow (0,0)} \cos\left(\frac{x^3 + y^2}{x + y + 2}\right) = 1$$

3. Let $f(x, y) = \frac{y^2 - xy}{\sqrt{x} - \sqrt{y}}$ for $(x, y) \neq (0, 0)$.

Show that $\lim_{(x,y) \rightarrow (0,0)} f(x, y) = 0$.

3M

To prove that $\lim_{(x,y) \rightarrow (0,0)} f(x, y) = 0$

$$\begin{aligned} \lim_{(x,y) \rightarrow (0,0)} f(x, y) &= \lim_{(x,y) \rightarrow (0,0)} \frac{y^2 - xy}{\sqrt{x} - \sqrt{y}} \times \frac{\sqrt{x} + \sqrt{y}}{\sqrt{x} + \sqrt{y}} \\ &= \lim_{(x,y) \rightarrow (0,0)} \frac{y(y-x)(\sqrt{x} + \sqrt{y})}{x - y} \\ &= \lim_{(x,y) \rightarrow (0,0)} -y(\sqrt{x} + \sqrt{y}) \\ &= 0(0 + 0) \end{aligned}$$

$$\lim_{(x,y) \rightarrow (0,0)} f(x, y) = 0$$

4. Evaluate $\lim_{(x,y) \rightarrow (0,0)} \cos\left(\frac{e^x \sin y}{y}\right)$, if the limits exists.

2M

$$\begin{aligned} \lim_{(x,y) \rightarrow (0,0)} \cos\left(\frac{e^x \sin y}{y}\right) &= \cos\left(\lim_{(x,y) \rightarrow (0,0)} \frac{e^x \sin y}{y}\right) \\ &= \cos\left(\lim_{(x,y) \rightarrow (0,0)} e^x \cdot \lim_{(x,y) \rightarrow (0,0)} \frac{\sin y}{y}\right) \\ &= \cos(e^0 \cdot 1) \end{aligned}$$

$$\lim_{(x,y) \rightarrow (0,0)} \cos\left(\frac{e^x \sin y}{y}\right) = \cos(1)$$

∴ By Composite function theorem

$$\lim_{x \rightarrow a} g(f(x)) = g(\lim_{x \rightarrow a} f(x))$$

5. Let $g(x, y) = \frac{x^2 y}{x^4 + y^2}$ for $(x, y) \neq (0, 0)$ and $f(0, 0) = 0$.

3M

(i) Show that $\lim_{(x,y) \rightarrow (0,0)} g(x, y) = 0$ along every line $y = mx, m \in \mathbb{R}$.

(ii) Show that $\lim_{(x,y) \rightarrow (0,0)} g(x, y) = \frac{k}{1+k^2}$ along every parabola $y = kx^2, k \in \mathbb{R} \setminus \{0\}$.

(i) Move along $y = mx$

$$\begin{aligned} \lim_{(x,y) \rightarrow (0,0)} g(x, y) &= \lim_{(x,y) \rightarrow (0,0)} \frac{x^2 y}{x^4 + y^2} \\ &= \lim_{(x,y) \rightarrow (0,0)} \frac{x^2 mx}{x^4 + m^2 x^2} \\ &= \lim_{(x,y) \rightarrow (0,0)} \frac{x^2 mx}{x^2(x^2 + m^2)} \\ &= \frac{0}{0+m^2} \\ \lim_{(x,y) \rightarrow (0,0)} g(x, y) &= 0 \end{aligned}$$

(ii) along every parabola $y = kx^2$

$$\begin{aligned} \lim_{(x,y) \rightarrow (0,0)} g(x, y) &= \lim_{(x,y) \rightarrow (0,0)} \frac{x^2 y}{x^4 + y^2} \\ &= \lim_{(x,y) \rightarrow (0,0)} \frac{x^2 kx^2}{x^4 + (kx^2)^2} \\ &= \lim_{(x,y) \rightarrow (0,0)} \frac{x^4 k}{x^4(1+k^2)} \\ &= \lim_{(x,y) \rightarrow (0,0)} \frac{k}{(1+k^2)} \\ &= \frac{k}{(1+k^2)} \end{aligned}$$

6. Show that $f(x, y) = \frac{x^2 - y^2}{y^2 + 1}$ is continuous at every $(x, y) \in \mathbb{R}^2$.

3M

$$f(x, y) = \frac{x^2 - y^2}{y^2 + 1}$$

Let $(a, b) \in \mathbb{R}^2$ be an arbitrary point.

(i) $f(a, b) = \frac{a^2 - b^2}{b^2 + 1}$ is defined. For $\forall (a, b) \in \mathbb{R}^2$

(ii) $\lim_{(x,y) \rightarrow (a,b)} f(x, y) = \lim_{(x,y) \rightarrow (a,b)} \frac{x^2 - y^2}{y^2 + 1} = \frac{a^2 - b^2}{b^2 + 1} = L$

(iii) From (i) and (ii)

$$\lim_{(x,y) \rightarrow (a,b)} f(x, y) = L = f(a, b) = \frac{a^2 - b^2}{b^2 + 1}$$

$\therefore f$ satisfies all 3 conditions of continuity of f at (a, b) .

$\therefore f$ is continuous at every point on $\mathbb{R}^2$.

7. Let $g(x, y) = \frac{e^y \sin x}{x}$, for $x \neq 0$ and $g(0, 0) = 1$. Show that g is continuous at $(0, 0)$.

3M

(i) Given $g(0, 0) = 1$,

g is defined at $(0, 0)$

(ii) $\lim_{(x,y) \rightarrow (0,0)} g(x, y) = \lim_{(x,y) \rightarrow (0,0)} e^y \frac{\sin x}{x}$

$$\begin{aligned} &= \lim_{(x,y) \rightarrow (0,0)} e^y \lim_{(x,y) \rightarrow (0,0)} \frac{\sin x}{x} \\ &= 1.1 \\ &= 1 \end{aligned}$$

Limit exist at $(0, 0)$

(iii) $\lim_{(x,y) \rightarrow (0,0)} g(x, y) = 1 = g(0, 0)$

$\therefore g$ is continuous at $(0, 0)$

EXERCISE 8.4

Concept Corner

Partial derivatives

Let $A = \left\{ x, y / \begin{matrix} a < x < b \\ c < y < d \end{matrix} \right\} \subset \mathbb{R}^2$ $F: A \rightarrow \mathbb{R}$ and $(x_0, y_0) \in A$

(i) We say that F has a partial derivatives with respect to x at $(x_0, y_0) \in A$. If

$\lim_{h \rightarrow 0} \frac{F(x_0 + h, y_0) - F(x_0, y_0)}{h}$ exists. In this case, the limit value is denoted by $\frac{\partial F}{\partial x}(x_0, y_0)$

(ii) F has a partial derivatives w.r to y at $(x_0, y_0) \in A$. If $\lim_{k \rightarrow 0} \frac{F(x_0, y_0 + k) - F(x_0, y_0)}{k}$ exists. In this case, the limit value is denoted by $= \frac{\partial F}{\partial y}(x_0, y_0)$



Video class

Clairaut's Theorem:

Suppose that $A = \{(x, y)/a < x < b, c < y < d\} \subset \mathbb{R}^2$, $F: A \rightarrow \mathbb{R}$ if f_{xy} and f_{yx} exist in A are continuous in A , then $f_{xy} = f_{yx}$ in A .

Laplace's equation

Let $A = \{(x, y)/a < x < b, c < y < d\} \subset \mathbb{R}^2$, A function $u: A \rightarrow \mathbb{R}^2$ is said to be harmonic in A . if it satisfies $\frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2} = 0, \forall (x, y) \in A$. This equation is called Laplace's equation.

1. Find the partial derivations of the following functions at the indicated points.

(i) $f(x, y) = 3x^2 - 2xy + y^2 + 5x + 2, (2, -5)$ (ii) $g(x, y) = 3x^2 + y^2 + 5x + 2, (1, -2)$

(iii) $h(x, y, z) = x \sin(xy) + z^2x, \left(2, \frac{\pi}{4}, 1\right)$ (iv) $G(x, y) = e^{x+3y} \log(x^2 + y^2), (-1, 1)$

(i) $f(x, y) = 3x^2 - 2xy + y^2 + 5x + 2, (2, -5)$

Diff partially w. r to x

2M

$$\frac{\partial f}{\partial x} = 6x - 2y + 0 + 5 + 0$$

$$\frac{\partial f}{\partial x(2, -5)} = 6(2) - 2(-5) + 5$$

$$= 12 + 10 + 5 = 27$$

Diff partially w. r to y

$$\frac{\partial f}{\partial y} = 0 - 2x + 2y + 0 + 0$$

$$\frac{\partial f}{\partial y(2, -5)} = -2(2) + 2(-5)$$

$$= -4 - 10 = -14$$

(ii) $g(x, y) = 3x^2 + y^2 + 5x + 2$

2M

Diff partially w. r to x

$$\frac{\partial g}{\partial x} = 6x + 0 + 5 + 0$$

$$\frac{\partial g}{\partial x(1, -2)} = 6(1) + 5 = 11$$

Differentiate partially w. r to y

$$\frac{\partial g}{\partial y} = 0 + 2y + 0$$

$$\frac{\partial g}{\partial y(1, -2)} = 2(-2) = -4$$

(iii) $h(x, y, z) = x \sin(xy) + z^2x, \left(2, \frac{\pi}{4}, 1\right)$

Diff partially w. r to x

3M

$$\frac{\partial h}{\partial x} = x(\cos xy)(y) + \sin xy(1) + z^2$$

$$\frac{\partial h}{\partial x}\left(2, \frac{\pi}{4}, 1\right) = 2\left(\cos \frac{2\pi}{4}\right)\frac{\pi}{4} + \sin \frac{2\pi}{4} + 1^2$$

$$= 2(0)\left(\frac{\pi}{4}\right) + 1 + 1$$

$$= 1 + 1 = 2$$

Diff partially w.r.to y

$$\frac{\partial h}{\partial y} = x \cos xy(x) + 0$$

$$\frac{\partial h}{\partial y}\left(2, \frac{\pi}{4}, 1\right) = 2\left(\cos \frac{2\pi}{4}\right)2$$

$$= 0$$

Diff partially w. r to z

$$\frac{\partial h}{\partial z} = 0 + 2zx$$

$$\frac{\partial h}{\partial z}\left(2, \frac{\pi}{4}, 1\right) = 2(2)$$

$$= 4$$

$$(iv) G(x, y) = e^{x+3y} \log(x^2 + y^2) \quad (-1, 1)$$

Diff partially w. r to x

3M

$$\frac{\partial G}{\partial x} = e^{x+3y} (\log x^2 + y^2)' +$$

$$\log(x^2 + y^2)(e^{x+3y})'$$

$$= e^{x+3y} \frac{1}{x^2 + y^2} 2x + \log(x^2 + y^2) e^{x+3y}$$

$$\frac{\partial G}{\partial x} = e^{x+3y} \left(\frac{2x}{x^2 + y^2} + \log(x^2 + y^2) \right)$$

$$\frac{\partial G}{\partial x}_{(-1,1)} = e^{-1+3(1)} \left(\frac{2(-1)}{(-1)^2 + 1^2} + \log((-1)^2 + (1)^2) \right)$$

$$= e^2 \left(\frac{-2}{2} + \log(1 + 1) \right)$$

$$= e^2 ((\log 2) - 1)$$

Diff partially w. r to y

$$\frac{\partial G}{\partial y} = e^{x+3y} (\log(x^2 + y^2))' +$$

$$+ \log(x^2 + y^2)(e^{x+3y})'$$

$$= e^{x+3y} \left(\frac{1}{x^2 + y^2} 2y + \log(x^2 + y^2) e^{x+3y} (3) \right)$$

$$\frac{\partial G}{\partial y} = e^{x+3y} \left(\frac{2y}{x^2 + y^2} + 3 \log(x^2 + y^2) \right)$$

$$\frac{\partial G}{\partial y}_{(-1,1)} = e^{-1+3} \left(\frac{2}{(-1)^2 + 1} + 3 \log((-1)^2 + 1^2) \right)$$

$$= e^2 \left(\frac{2}{2} + 3 \log(1 + 1) \right)$$

$$= e^2 (1 + 3 \log 2)$$

$$= e^2 (1 + \log 2^3)$$

$$\frac{\partial G}{\partial y}_{(-1,1)} = e^2 (1 + \log 8)$$

2. For each of the following functions find the

f_x, f_y and show that $f_{xy} = f_{yx}$

$$(i) f(x, y) = \frac{3x}{y + \sin x} \quad (ii) f(x, y) = \tan^{-1} \left(\frac{x}{y} \right)$$

$$(iii) f(x, y) = \cos(x^2 - 3xy)$$

$$(i) f(x, y) = \frac{3x}{y + \sin x}$$

PTA-3

5M

$$f_x = \frac{(y + \sin x)(3x)' - 3x(y + \sin x)'}{(y + \sin x)^2}$$

$$= \frac{(y + \sin x)(3) - 3x(\cos x)}{(y + \sin x)^2}$$

$$f_x = \frac{3y + 3\sin x - 3x\cos x}{(y + \sin x)^2}$$

$$f_y = \frac{(y + \sin x)(3x)' - 3x(y + \sin x)'}{(y + \sin x)^2}$$

$$= \frac{(y + \sin x)(0) - 3x(1)}{(y + \sin x)^2}$$

$$f_y = \frac{-3x}{(y + \sin x)^2}$$

$$f_{xy} = \frac{\partial}{\partial x} \left(\frac{\partial f}{\partial y} \right)$$

$$= \frac{\partial}{\partial x} \left(\frac{-3x}{(y + \sin x)^2} \right)$$

$$= \frac{(y + \sin x)^2 (-3x)' + 3x((y + \sin x)^2)'}{(y + \sin x)^4}$$

$$f_{xy}$$

$$= \frac{(y + \sin x)(-3) + 3x(2(y + \sin x)(0 + \cos x))}{(y + \sin x)^4}$$

$$f_{xy} = \frac{3(y + \sin x)(-1(y + \sin x) + 2x \cos x)}{(y + \sin x)^4}$$

$$f_{xy} = \frac{3(2x \cos x - y - \sin x)}{(y + \sin x)^3} \dots \dots \dots (1)$$

$$f_{yx} = \frac{\partial}{\partial y} (f_x)$$

$$= \frac{\partial}{\partial y} \left(\frac{3y + 3\sin x - 3x\cos x}{(y + \sin x)^2} \right)$$

$$f_{yx} = \frac{(y + \sin x)^2 (3y + 3\sin x - 3x\cos x)' - (3y + 3\sin x - 3x\cos x)((y + \sin x)^2)'}{(y + \sin x)^4}$$

$$= \frac{(y + \sin x)^2 (3 + 0 + 0) - (3y + 3\sin x - 3x\cos x) \times 2(y + \sin x)}{(y + \sin x)^4}$$

$$= \frac{3[y + \sin x - 2y - 2\sin x + 2x\cos x]}{(y + \sin x)^3}$$

$$f_{yx} = \frac{3(2x\cos x - y - \sin x)}{(y + \sin x)^3} \dots \dots \dots (2)$$

From (1) & (2), $f_{xy} = f_{yx}$

(ii) $f(x, y) = \tan^{-1} \left(\frac{x}{y} \right)$

5M

$$f_x = \frac{\partial f}{\partial x} = \frac{1}{1 + \left(\frac{x}{y}\right)^2} \left(\frac{1}{y} \right)$$

$$= \frac{1}{\frac{y^2 + x^2}{y^2}} \left(\frac{1}{y} \right)$$

$$= \frac{y^2}{y^2 + x^2} \frac{1}{y}$$

$$f_x = \frac{y}{x^2 + y^2}$$

$$f_y = \frac{1}{1 + \left(\frac{x}{y}\right)^2} \left(\frac{-x}{y^2} \right)$$

$$= \frac{1}{\frac{y^2 + x^2}{y^2}} \left(\frac{-x}{y^2} \right)$$

$$= \frac{y^2}{y^2 + x^2} \left(\frac{-x}{y^2} \right)$$

$$f_y = \frac{-x}{x^2 + y^2}$$

$$f_{xy} = \frac{\partial}{\partial x} (f_y)$$

$$f_{xy} = \frac{(x^2 + y^2)(-x)' - (-x)(x^2 + y^2)'}{(x^2 + y^2)^2}$$

$$= \frac{(x^2 + y^2)(-1) + x(2x)}{(x^2 + y^2)^2}$$

$$= \frac{-x^2 - y^2 + 2x^2}{(x^2 + y^2)^2}$$

$$f_{xy} = \frac{x^2 - y^2}{(x^2 + y^2)^2} \quad \dots\dots\dots (1)$$

$$f_{yx} = \frac{\partial}{\partial y} (f_x) = \frac{\partial}{\partial y} \left(\frac{y}{x^2 + y^2} \right)$$

$$= \frac{(x^2 + y^2)(y)' - y(x^2 + y^2)'}{(x^2 + y^2)^2}$$

$$= \frac{(x^2 + y^2)(1) - y(2y)}{(x^2 + y^2)^2}$$

$$= \frac{x^2 + y^2 - 2y^2}{(x^2 + y^2)^2}$$

$$f_{yx} = \frac{x^2 - y^2}{(x^2 + y^2)^2} \quad \dots\dots\dots (2)$$

From (1) & (2)

$$f_{xy} = f_{yx}$$

(iii) $f(x, y) = \cos(x^2 - 3xy)$

3M

$$f_x = \frac{\partial f}{\partial x} = -\sin(x^2 - 3xy)(2x - 3y)$$

$$f_x = -(2x - 3y) \sin(x^2 - 3xy)$$

$$f_y = \frac{\partial f}{\partial y} = -\sin(x^2 - 3xy)(0 - 3x)$$

$$f_y = 3x \sin(x^2 - 3xy)$$

$$f_{xy} = \frac{\partial}{\partial x} (f_y)$$

$$= \frac{\partial}{\partial x} (3x \cdot \sin(x^2 - 3xy))$$

$$= 3x(\sin(x^2 - 3xy))' + \sin(x^2 - 3xy)(3x)'$$

$$= 3x \cos(x^2 - 3xy)(2x - 3y) + \sin(x^2 - 3xy)(3)$$

$$f_{xy} = 3[(2x^2 - 3xy) \cos(x^2 - 3xy) + \sin(x^2 - 3xy)] \quad \dots\dots\dots (1)$$

$$f_{yx} = \frac{\partial}{\partial y} (f_x)$$

$$= \frac{\partial}{\partial y} (-(2x - 3y) \sin(x^2 - 3xy))$$

$$= -(2x - 3y)(\sin(x^2 - 3xy))' + \sin(x^2 - 3xy)(-(2x - 3y))$$

$$= -(2x - 3y) \cos(x^2 - 3xy)(0 - 3x) + \sin(x^2 - 3xy)(-(0 - 3))$$

$$f_{yx} = 3[(2x^2 - 3xy) \cos(x^2 - 3xy) + \sin(x^2 - 3xy)] \quad \dots\dots\dots (2)$$

From (1) & (2),

$$f_{xy} = f_{yx}$$

3. If $U(x, y, z) = \frac{x^2+y^2}{xy} + 3z^2y$, find $\frac{\partial U}{\partial x}$, $\frac{\partial U}{\partial y}$ and $\frac{\partial U}{\partial z}$

$$U(x, y, z) = \frac{x^2}{xy} + \frac{y^2}{xy} + 3z^2y \quad (3M)$$

$$U(x, y, z) = \frac{x}{y} + \frac{y}{x} + 3z^2y$$

$$\frac{\partial U}{\partial x} = \frac{1}{y} + y \left(\frac{-1}{x^2} \right) + 0 = \frac{1}{y} - \frac{y}{x^2} = \frac{x^2 - y^2}{yx^2}$$

$$\frac{\partial U}{\partial y} = x \left(\frac{-1}{y^2} \right) + \frac{1}{x} + 3z^2 = \frac{-x^2 + y^2}{xy^2} + 3z^2$$

$$\frac{\partial U}{\partial z} = 0 + 0 + 3y(2z) = 6yz$$

4. If $U(x, y, z) = \log(x^3 + y^3 + z^3)$, find

$$\frac{\partial U}{\partial x} + \frac{\partial U}{\partial y} + \frac{\partial U}{\partial z}. \quad \text{PTA-3, APR-20, JUL-23}$$

$$U = \log(x^3 + y^3 + z^3) \quad (3M)$$

$$\frac{\partial U}{\partial x} = \frac{1}{x^3 + y^3 + z^3} (3x^2 + 0 + 0)$$

$$\frac{\partial U}{\partial x} = \frac{3x^2}{x^3 + y^3 + z^3}$$

Similarly we get

$$\frac{\partial U}{\partial y} = \frac{3y^2}{x^3 + y^3 + z^3}, \quad \frac{\partial U}{\partial z} = \frac{3z^2}{x^3 + y^3 + z^3}$$

$$\frac{\partial U}{\partial x} + \frac{\partial U}{\partial y} + \frac{\partial U}{\partial z}$$

$$= \frac{3x^2}{x^3 + y^3 + z^3} + \frac{3y^2}{x^3 + y^3 + z^3} + \frac{3z^2}{x^3 + y^3 + z^3}$$

$$= \frac{3x^2 + 3y^2 + 3z^2}{x^3 + y^3 + z^3}$$

$$= \frac{3(x^2 + y^2 + z^2)}{(x^3 + y^3 + z^3)}$$

5. For each of the following functions find the

g_{xy} , g_{xx} , g_{yy} and g_{yx}

(i) $g(x, y) = xe^y + 3x^2y$

(3M)

(ii) $g(x, y) = \log(5x + 3y)$

(iii) $g(x, y) = x^2 + 3xy - 7y + \cos(5x)$

(i) $g(x, y) = xe^y + 3x^2y$

$$g_x = (1)e^y + 3y(2x) = e^y + 6xy$$

$$g_y = xe^y + 3x^2$$

$$g_{xx} = 0 + 6y = 6y$$

$$g_{xy} = \frac{\partial}{\partial x}(g_y) = e^y + 6x$$

$$g_{yy} = \frac{\partial}{\partial y}(g_y) = xe^y + 0 = xe^y$$

$$g_{yx} = \frac{\partial}{\partial y}(g_x) = e^y + 6x$$

(ii) $g(x, y) = \log(5x + 3y)$

(3M)

$$g_x = \frac{1}{5x+3y} (5)$$

$$g_{xx} = 5 \left(\frac{-1}{(5x+3y)^2} \right) (5)$$

$$g_{xx} = \frac{-25}{(5x+3y)^2}$$

$$g_y = \frac{1}{5x+3y} (3)$$

$$g_{yy} = \frac{\partial}{\partial y}(g_y)$$

$$= 3 \left(-\frac{1}{(5x+3y)^2} \right) (3)$$

$$= \frac{-9}{(5x+3y)^2}$$

$$g_{yx} = \frac{\partial}{\partial x}(g_x)$$

$$= 5 \left(-\frac{1}{(5x+3y)^2} \right) (3)$$

$$= \frac{-15}{(5x+3y)^2}$$

$$g_{xy} = \frac{\partial}{\partial x}(g_y)$$

$$= 3 \left(-\frac{1}{(5x+3y)^2} \right) (5)$$

$$= \frac{-15}{(5x+3y)^2}$$

(iii) $g(x, y) = x^2 + 3xy - 7y + \cos(5x)$

$$g_x = 2x + 3y - 0 - \sin 5x (5)$$

$$g_y = 0 + 3x - 7 + 0$$

$$g_{xy} = \frac{\partial}{\partial x}(g_y)$$

$$= 3(1)$$

$$= 3$$

$$g_{xx} = \frac{\partial}{\partial x}(g_x)$$

$$= 2 + 0 - 5 \cos 5x (5)$$

$$= 2 - 25 \cos 5x$$

$$g_{yy} = \frac{\partial}{\partial y}(g_y)$$

$$= 0$$

$$g_{yx} = \frac{\partial}{\partial y}(g_x)$$

$$= 3 - 0$$

$$= 3$$

6. Let $w(x, y, z) = \frac{1}{\sqrt{x^2+y^2+z^2}}$, $(x, y, z) \neq (0, 0, 0)$ Show that $\frac{\partial^2 w}{\partial x^2} + \frac{\partial^2 w}{\partial y^2} + \frac{\partial^2 w}{\partial z^2} = 0$.

3M

$$w = \frac{1}{(x^2+y^2+z^2)^{\frac{1}{2}}}$$

PTA-4

$$\frac{\partial w}{\partial x} = -\frac{1}{2}(x^2 + y^2 + z^2)^{-\frac{1}{2}-1} (2x) = -x(x^2 + y^2 + z^2)^{-\frac{3}{2}}$$

$$\begin{aligned}\frac{\partial^2 w}{\partial x^2} &= -x \left((x^2 + y^2 + z^2)^{-\frac{3}{2}} \right)' + (x^2 + y^2 + z^2)^{-\frac{3}{2}}(-x)' \\ &= -x \left(-\frac{3}{2}(x^2 + y^2 + z^2)^{-\frac{3}{2}-1} 2x \right) + (x^2 + y^2 + z^2)^{-\frac{3}{2}}(-1) \\ &= \frac{3x^2}{(x^2+y^2+z^2)^{\frac{5}{2}}} - \frac{1}{(x^2+y^2+z^2)^{\frac{3}{2}}} = \frac{3x^2 - (x^2+y^2+z^2)}{(x^2+y^2+z^2)^{\frac{5}{2}}} = \frac{3x^2 - x^2 - y^2 - z^2}{(x^2+y^2+z^2)^{\frac{5}{2}}}\end{aligned}$$

$$\frac{\partial^2 w}{\partial x^2} = \frac{2x^2 - y^2 - z^2}{(x^2+y^2+z^2)^{\frac{5}{2}}} \dots\dots\dots (1)$$

$$\text{Similarly } \frac{\partial^2 w}{\partial y^2} = \frac{2y^2 - x^2 - z^2}{(x^2+y^2+z^2)^{\frac{5}{2}}} \dots\dots\dots (2) \quad \& \quad \frac{\partial^2 w}{\partial z^2} = \frac{2z^2 - x^2 - y^2}{(x^2+y^2+z^2)^{\frac{5}{2}}} \dots\dots\dots (3)$$

$$\text{Add (1) (2) \& (3) } \frac{\partial^2 w}{\partial x^2} + \frac{\partial^2 w}{\partial y^2} + \frac{\partial^2 w}{\partial z^2} = \frac{2x^2 - y^2 - z^2 + 2y^2 - x^2 - z^2 + 2z^2 - x^2 - y^2}{(x^2+y^2+z^2)^{\frac{5}{2}}} = \frac{0}{(x^2+y^2+z^2)^{\frac{5}{2}}} = 0$$

7. If $V(x, y) = e^x(x \cos y - y \sin y)$, then prove that $\frac{\partial^2 V}{\partial x^2} + \frac{\partial^2 V}{\partial y^2} = 0$.

5M

$$V(x, y) = e^x(x \cos y - y \sin y)$$

$$\frac{\partial V}{\partial x} = e^x(x \cos y - y \sin y)' + (x \cos y - y \sin y)(e^x)' = e^x(\cos y - 0) + (x \cos y - y \sin y)e^x$$

$$\frac{\partial V}{\partial x} = e^x(\cos y + x \cos y - y \sin y)$$

$$\frac{\partial^2 V}{\partial x^2} = e^x(\cos y + x \cos y - y \sin y)' + (\cos y + x \cos y - y \sin y)(e^x)'$$

$$= e^x(0 + \cos y + 0) + (\cos y + x \cos y - y \sin y)e^x = e^x(\cos y + \cos y + x \cos y - y \sin y)$$

$$\frac{\partial^2 V}{\partial x^2} = e^x(2 \cos y - y \sin y + x \cos y) \dots\dots\dots (1)$$

$$\frac{\partial V}{\partial y} = e^x(x(-\sin y) - (y \cos y + \sin y(1))) = e^x(-x \sin y - y \cos y - \sin y)$$

$$\frac{\partial^2 V}{\partial y^2} = e^x(-x \cos y - (y(-\sin y) + \cos y(1)) - \cos y) = e^x(-x \cos y + y \sin y - \cos y - \cos y)$$

$$\frac{\partial^2 V}{\partial y^2} = e^x(-2 \cos y + y \sin y - x \cos y) \dots\dots\dots (2)$$

$$\text{Add (1) (2) } \frac{\partial^2 V}{\partial x^2} + \frac{\partial^2 V}{\partial y^2} = e^x(2 \cos y - y \sin y + x \cos y) + e^x(-2 \cos y + y \sin y - x \cos y)$$

$$\frac{\partial^2 V}{\partial x^2} + \frac{\partial^2 V}{\partial y^2} = e^x[2 \cos y - y \sin y + x \cos y - 2 \cos y + y \sin y - x \cos y] = e^x(0) = 0$$

8. If $w(x, y) = xy + \sin(xy)$, then prove

that $\frac{\partial^2 w}{\partial y \partial x} = \frac{\partial^2 w}{\partial x \partial y}$

5M

$$\frac{\partial w}{\partial x} = y + \cos xy(y)$$

$$\frac{\partial w}{\partial x} = y(1 + \cos xy)$$

$$\frac{\partial^2 w}{\partial y \partial x} = \frac{\partial}{\partial y} \left(\frac{\partial w}{\partial x} \right)$$

$$= \frac{\partial}{\partial y} (y(1 + \cos xy))$$

$$= y(0 - \sin xy(x) + (1 + \cos xy)(1))$$

$$\frac{\partial^2 w}{\partial y \partial x} = 1 - xy \sin xy + \cos xy \quad \dots (1)$$

$$\frac{\partial w}{\partial y} = x + \cos xy(x) = x(1 + \cos xy)$$

$$\frac{\partial^2 w}{\partial x \partial y} = \frac{\partial}{\partial x} (x(1 + \cos xy))$$

$$= x(0 - \sin xy(y)) + (1 + \cos xy)(1)$$

$$\frac{\partial^2 w}{\partial x \partial y} = 1 - xy \sin xy + \cos xy \quad \dots (2)$$

From (1) & (2)

$$\frac{\partial^2 w}{\partial y \partial x} = \frac{\partial^2 w}{\partial x \partial y} \quad \text{Hence proved.}$$

9. If $v(x, y, z) = x^3 + y^3 + z^3 + 3xyz$,

show that $\frac{\partial^2 v}{\partial y \partial z} = \frac{\partial^2 v}{\partial z \partial y}$

3M

$$\frac{\partial v}{\partial z} = 0 + 0 + 3z^2 + 3xy(1)$$

$$\frac{\partial v}{\partial z} = 3z^2 + 3xy$$

$$\frac{\partial^2 v}{\partial y \partial z} = \frac{\partial}{\partial y} \left(\frac{\partial v}{\partial z} \right) = \frac{\partial}{\partial y} (3z^2 + 3xy)$$

$$= 0 + 3x$$

$$\frac{\partial^2 v}{\partial y \partial z} = 3x \quad \dots (1)$$

$$\frac{\partial v}{\partial y} = 0 + 3y^2 + 0 + 3xz$$

$$\frac{\partial v}{\partial y} = 3y^2 + 3xz$$

$$\frac{\partial^2 v}{\partial z \partial y} = \frac{\partial}{\partial z} \left(\frac{\partial v}{\partial y} \right)$$

$$= \frac{\partial}{\partial z} (3y^2 + 3xz)$$

$$= 0 + 3x$$

$$\frac{\partial^2 v}{\partial z \partial y} = 3x \quad \dots (2)$$

From (1) & (2) $\frac{\partial^2 v}{\partial y \partial z} = \frac{\partial^2 v}{\partial z \partial y}$

10. A firm produces two types of calculators each week, x number of type A and y number of type B. The weekly revenue and cost functions (in rupees) are

$$R(x, y) = 80x + 90y + 0.04xy - 0.05x^2 - 0.05y^2$$

$$\text{and } C(x, y) = 8x + 6y + 2000 \text{ respectively.}$$

(i) Find the profit function $P(x, y)$

3M

(ii) Find $\frac{\partial P}{\partial x}(1200, 1800)$ and $\frac{\partial P}{\partial y}(1200, 1800)$

and interpret these results.

$$R(x, y) = 80x + 90y + 0.04xy - 0.05x^2 - 0.05y^2$$

$$\text{and } C(x, y) = 8x + 6y + 2000$$

(i) Profit = Revenue - cost

$$P(x, y) = R(x, y) - C(x, y)$$

$$= 80x + 90y + 0.04xy - 0.05x^2 - 0.05y^2 - (8x + 6y + 2000)$$

$$P(x, y) = 72x + 84y + 0.04xy - 0.05x^2 - 0.05y^2 - 2000$$

$$(ii) \frac{\partial P}{\partial x} = 72 + 0 + 0.04y - 0.05(2x) - 0 - 0$$

$$\frac{\partial P}{\partial x} = 72 + 0.04y - 0.1x$$

$$\frac{\partial P}{\partial x}(1200, 1800) = 72 + 0.04(1800) - 0.1(1200)$$

$$= 72 + 72 - 120$$

$$= 144 - 120 = 24$$

$$\frac{\partial P}{\partial y} = 0 + 84 + 0.04x - 0 - 0.05(2y) - 0$$

$$\frac{\partial P}{\partial y} = 84 + 0.04x - 0.1y$$

$$\frac{\partial P}{\partial y}(1200, 1800) = 84 + 0.04(1200) - 0.1(1800)$$

$$= 84 + 48 - 180$$

$$= 132 - 180 = -48$$

At (1200, 1800) $\frac{\partial P}{\partial x}$ is +ve and $\frac{\partial P}{\partial y}$ is -ve

$\therefore$ Keeping y constant and increase x values then we get profit increasing.

EXERCISE 8.5



Video class

Concept Corner

Linear approximation and differential of a function of several variable

❖ $A = \{(x, y) \mid a < x < b, c < y < d\} \subset \mathbb{R}^2$, $F: A \rightarrow \mathbb{R}$ and $(x_0, y_0) \in A$ The linear approximation of F at $(x_0, y_0) \in A$ is defined to be

$$L(x, y) = F(x_0, y_0) + \frac{\partial F}{\partial x}(x_0, y_0)(x - x_0) + \frac{\partial F}{\partial y}(x_0, y_0)(y - y_0)$$

❖ The differential of F is $dF = \frac{\partial F}{\partial x}(x, y) dx + \frac{\partial F}{\partial y}(x, y) dy$ where $dx = \Delta x, dy = \Delta y$ ❖ The linear approximation of F at $(x_0, y_0, z_0) \in A$ is defined to be

$$L(x, y, z) = F(x_0, y_0, z_0) + \frac{\partial F}{\partial x}(x_0, y_0, z_0)(x - x_0) + \frac{\partial F}{\partial y}(x_0, y_0, z_0)(y - y_0) + \frac{\partial F}{\partial z}(x_0, y_0, z_0)(z - z_0)$$

❖ The differential of F is

$$dF = \frac{\partial F}{\partial x}(x, y, z) dx + \frac{\partial F}{\partial y}(x, y, z) dy + \frac{\partial F}{\partial z}(x, y, z) dz \text{ where } dx = \Delta x, dy = \Delta y, dz = \Delta z$$

1. If $w(x, y) = x^3 - 3xy + 2y^2$, $x, y \in \mathbb{R}$, find the linear approximation for w at $(1, -1)$.

3M

$$w = x^3 - 3x + 2y^2$$

$$w(1, -1) = 1 - 3(1)(-1) + 2(-1)^2 = 1 + 3 + 2 = 6$$

$$\frac{\partial w}{\partial x} = 3x^2 - 3y$$

$$\frac{\partial w}{\partial y} = 0 - 3x + 4y$$

$$\frac{\partial w}{\partial x}(1, -1) = 3(1) - 3(-1) = 3 + 3 = 6$$

$$\frac{\partial w}{\partial y}(1, -1) = -3(1) + 4(-1) = -3 - 4 = -7$$

linear approximation $L(x, y) = f(x_0, y_0) + \frac{\partial F}{\partial x}(x - x_0) + \frac{\partial F}{\partial y}(y - y_0)$

$$L(x, y) = w(1, -1) + \frac{\partial w}{\partial x}(1, -1)(x - 1) + \frac{\partial w}{\partial y}(1, -1)(y + 1)$$

$$L(x, y) = 6 + 6(x - 1) + (-7)(y + 1)$$

$$L(x, y) = 6 + 6x - 6 - 7y - 7 = 6x - 7y - 7$$

2. Let $z(x, y) = x^2y + 3xy^4$, $x, y \in \mathbb{R}$ Find the linear approximation for z at $(2, -1)$.

3M

$$z(x, y) = x^2y + 3xy^4$$

$$\frac{\partial z}{\partial x} = 2xy + 3y^4$$

$$\frac{\partial z}{\partial y} = x^2 + 12xy^3$$

$$\frac{\partial z}{\partial x}(2, -1) = 2(2)(-1) + 3(-1)^4 = -4 + 3 = -1$$

$$\frac{\partial z}{\partial y}(2, -1) = 2^2 + 12(2)(-1)^3 = 4 - 24 = -20$$

$$z(2, -1) = 2^2(-1) + 3(2)(-1)^4 = -4 + 6 = 2$$

linear approximation $L(x, y) = z(x_0, y_0) + \frac{\partial z}{\partial x}(x - x_0) + \frac{\partial z}{\partial y}(y - y_0)$

$$L(x, y) = z(2, -1) + \frac{\partial z}{\partial x}(x - 2) + \frac{\partial z}{\partial y}(y + 1)$$

$$= 2 + (-1)(x - 2) + (-20)(y + 1) = 2 - x + 2 - 20y - 20 = -x - 20y - 16$$

$$L(x, y) = -(x + 20y + 16)$$

EXERCISE 8.8



Online Test

Choose the correct or the most suitable answer from the given four alternatives :

1. A circular template has a radius of 10 cm. The measurement of radius has an approximate error of 0.02 cm. Then the percentage error in calculating area of this template is
 (1) 0.2% (2) **0.4%** (3) 0.04% (4) 0.08%

$r = 10\text{cm}$ $dr = 0.02\text{cm}$ $\text{Area} = \pi r^2$ $\frac{dA}{dr} = 2\pi r$ $dA = 2\pi r dr$	$\text{Percentage error} = \frac{dA}{A} \times 100 = \frac{2\pi r dr}{\pi r^2} \times 100$ $= \frac{2(10)}{10 \times 10} 0.02 \times 100$ $= 0.04 \times 10$ $= 0.4\%$
--	---

2. The percentage error of fifth root of 31 is approximately how many times the percentage error in 31?

JUL-22

- (1) $\frac{1}{31}$ (2) **$\frac{1}{5}$** (3) 5 (4) 31

W. k. t Percentage error on n th root of a number = $\frac{1}{n}$ times Percentage on that number
 Percentage error of $\sqrt[5]{31} = \frac{1}{5}$ (Percentage error of 31)

3. If $u(x, y) = e^{x^2+y^2}$, then $\frac{\partial u}{\partial x}$ is equal to

PTA-5, APR-25, 20, JUL-23

- (1) $e^{x^2+y^2}$ (2) **$2xu$** (3) x^2u (4) y^2u

$$u(x, y) = e^{x^2+y^2}; \quad \frac{\partial u}{\partial x} = e^{x^2+y^2} (2x + 0) = 2xu$$

4. If $v(x, y) = \log(e^x + e^y)$, then $\frac{\partial v}{\partial x} + \frac{\partial v}{\partial y}$ is equal to

- (1) $e^x + e^y$ (2) $\frac{1}{e^x+e^y}$ (3) 2 (4) **1**

$$v = \log(e^x + e^y)$$

$$\frac{\partial v}{\partial x} = \frac{1}{e^x+e^y} e^x; \quad \frac{\partial v}{\partial y} = \frac{e^y}{e^x+e^y}$$

$$\frac{\partial v}{\partial x} + \frac{\partial v}{\partial y} = \frac{e^x+e^y}{e^x+e^y} = 1$$

5. If $w(x, y) = x^y, x > 0$ then $\frac{\partial w}{\partial x}$ is equal to

PTA-2, JUL-20

- (1) $x^y \log x$ (2) $y \log x$ (3) **yx^{y-1}** (4) $x \log y$

$$w = x^y; \quad \frac{\partial w}{\partial x} = yx^{y-1} \quad (y \text{ is constant})$$

6. If $f(x, y) = e^{xy}$, then $\frac{\partial^2 f}{\partial x \partial y}$ is equal to

PTA-5

- (1) $xy e^{xy}$ (2) **$(1 + xy)e^{xy}$** (3) $(1 + y)e^{xy}$ (4) $(1 + x)(e^{xy})$

$$f = e^{xy}$$

$$\frac{\partial f}{\partial y} = e^{xy}(x); \quad \frac{\partial^2 f}{\partial x \partial y} = x e^{xy}(y) + e^{xy}(1) = xy e^{xy} + 1 e^{xy} = e^{xy}(xy + 1)$$

7. If we measure the side of a cube to be 4 cm with an error of 0.1 cm then the error in our calculation of the volume is

PTA-1, JUL-21

- (1) 0.4 cu. cm (2) 0.45 cu. cm (3) 2 cu. cm **(4) 4.8 cu cm**

$a = 4$ $da = 0.1$ Volume of cube $v = a^3$	$\frac{dv}{da} = 3a^2$ $dv = 3a^2 da$ $dv = 3(4)^2 0.1 = 4.8 \text{ cu. cm}$
---	--

8. The change in the surface area $s = 6x^2$ of a cube when the edge length varies from x_0 to $x_0 + dx$ is

- (1) $12x_0 + dx$ **(2) $12x_0 dx$** (3) $6x_0 dx$ (4) $6x_0 + dx$

$s = 6x^2$ $\frac{ds}{dx} = 12x \Rightarrow ds = 12x dx$ $x = x_0$	$dx = (x_0 + dx) - x_0$ $dx = dx$ Change in surface $ds = 12x_0 dx$
--	---

9. The approximate change in the volume V of a cube of side x metres caused by increasing the side by 1% is

- (1) $0.3x dx \text{ m}^3$ (2) $0.03x \text{ m}^3$ (3) $0.03x^2 \text{ m}^3$ **(4) $0.03x^3 \text{ m}^3$**

Volume of cube $v = x^3$ $dx = 1\%x$ $= \frac{1}{100}x$ $dx = 0.01x$	$\frac{dv}{dx} = 3x^2$ $dv = 3x^2 dx$ $dv = 3x^2 0.01x$ $dv = 0.03x^3 \text{ m}^3$
---	---

10. If $g(x, y) = 3x^2 - 5y + 2y^2$, $x(t) = e^t$ and $y(t) = \cos t$, then $\frac{dg}{dt}$ is equal to

- (1) $6e^{2t} + 5 \sin t - 4 \cos t \sin t$** (2) $6e^{2t} - 5 \sin t + 4 \cos t \sin t$
 (3) $3e^{2t} + 5 \sin t + 4 \cos t \sin t$ (4) $3e^{2t} - 5 \sin t + 4 \cos t \sin t$

$g = 3x^2 - 5y + 2y^2$	
$\frac{\partial g}{\partial x} = 6x$ $x = e^t$ $\frac{dx}{dt} = e^t$	$\frac{\partial g}{\partial y} = -5 + 4y$ $y = \cos t$ $\frac{dy}{dt} = -\sin t$
$\frac{dg}{dt} = \frac{\partial g}{\partial x} \frac{dx}{dt} + \frac{\partial g}{\partial y} \frac{dy}{dt} = 6x(e^t) + (-5 + 4y)(-\sin t) = 6e^t e^t + (-5 + 4 \cos t)(-\sin t)$ $= 6e^{2t} + 5 \sin t - 4 \sin t \cos t$	

11. If $f(x) = \frac{x}{x+1}$, then its differential is given by

APR-24, 22, PTA-4

- (1) $\frac{-1}{(x+1)^2} dx$ **(2) $\frac{1}{(x+1)^2} dx$** (3) $\frac{1}{x+1} dx$ (4) $\frac{-1}{x+1} dx$

$f(x) = \frac{x}{x+1}$ $\frac{df}{dx} = \frac{(x+1)(1) - (x)(1)}{(x+1)^2} = \frac{x+1-x}{(x+1)^2}$ $df = \frac{dx}{(x+1)^2}$
--

12. If $u(x, y) = x^2 + 3xy + y - 2019$, then $\frac{\partial u}{\partial x(4, -5)}$ is equal to

- (1) -4 (2) -3 (3) -7 (4) 13

$$u = x^2 + 3xy + y - 2019$$

$$\frac{\partial u}{\partial x} = 2x + 3y$$

$$\frac{\partial u}{\partial x(4, -5)} = 2(4) + 3(-5) = 8 - 15 = -7$$

13. Linear approximation for $g(x) = \cos x$ at $x = \frac{\pi}{2}$ is

JUL-23

- (1) $x + \frac{\pi}{2}$ (2) $-x + \frac{\pi}{2}$ (3) $x - \frac{\pi}{2}$ (4) $-x - \frac{\pi}{2}$

$$g(x) = \cos x \Rightarrow g\left(\frac{\pi}{2}\right) = \cos \frac{\pi}{2} = 0$$

$$g'(x) = -\sin x \Rightarrow g'\left(\frac{\pi}{2}\right) = -\sin \frac{\pi}{2} = -1$$

$$L(x) = g(x_0) + g'(x_0)(x - x_0) = 0 + (-1)\left(x - \frac{\pi}{2}\right) = -x + \frac{\pi}{2}$$

14. $w(x, y, z) = x^2(y - z) + y^2(z - x) + z^2(x - y)$, then $\frac{\partial w}{\partial x} + \frac{\partial w}{\partial y} + \frac{\partial w}{\partial z}$ is

- (1) $xy + yz + zx$ (2) $x(y + z)$ (3) $y(z + x)$ (4) 0

$$\frac{\partial w}{\partial x} = 2x(y - z) + y^2(0 - 1) + z^2(1 - 0) = 2x(y - z) - (y^2 - z^2)$$

$$\text{Similarly } \frac{\partial w}{\partial y} = 2y(z - x) - (z^2 - x^2) \text{ \& } \frac{\partial w}{\partial z} = 2z(x - y) - (x^2 - y^2)$$

$$\frac{\partial w}{\partial x} + \frac{\partial w}{\partial y} + \frac{\partial w}{\partial z} = 2x(y - z) + 2y(z - x) + 2z(x - y) - (y^2 - z^2 + z^2 - x^2 + x^2 - y^2) \\ = 2[xy - xz + yz - yx - zx - zy] = 0$$

15. If $f(x, y, z) = xy + yz + zx$, then $f_x - f_z$ is equal to

JUL-24, PTA-6

- (1) $z - x$ (2) $y - z$ (3) $x - z$ (4) $y - x$

$$f_x = y + z ; \quad f_z = y + x$$

$$f_x - f_z = y + z - y - x = z - x$$

Creative Questions

1. 1. If $f(x) = \frac{x-1}{x+1}$, then its differential is given by

PTA-3

- (1) $\frac{2}{(x+1)^2} dx$ (2) $-\frac{2}{(x+1)^2} dx$ (3) $\frac{x}{(x+1)^2} dx$ (4) $\frac{-x}{(x+1)^2} dx$

$$f(x) = y = \frac{x-1}{x+1}$$

$$\frac{dy}{dx} = \frac{(x+1)(1) - (x-1)(1)}{(x+1)^2} \Rightarrow \frac{dy}{dx} = \frac{x+1-x+1}{(x+1)^2} \Rightarrow dy = \frac{2}{(x+1)^2} dx$$

2. If $u = (x - y)^2$, then $\frac{\partial u}{\partial x} + \frac{\partial u}{\partial y}$ is

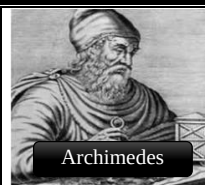
MDL

- (1) 1 (2) -1 (3) 0 (4) 2

$$u = (x - y)^2 = x^2 + y^2 - 2xy$$

$$\frac{\partial u}{\partial x} = 2x - 2y, \quad \frac{\partial u}{\partial y} = 2y - 2x$$

$$\frac{\partial u}{\partial x} + \frac{\partial u}{\partial y} = 2x - 2y + 2y - 2x = 0$$



Archimedes

Chapter - 9

Applications of Integration

EXERCISE 9.1



Video class

Concept Corner

Definite integral as the limit of a sum

Riemann Integral:

$$\sum_{i=1}^n f(\xi_i)(x_i - x_{i-1}) = f(\xi_1)(x_1 - x_0) + f(\xi_2)(x_2 - x_1) + \cdots + f(\xi_n)(x_n - x_{n-1}) \dots \dots \dots (1)$$

is called a **Riemann sum** of $f(x)$ corresponding to the partition $[x_0, x_1], [x_1, x_2], \dots [x_{n-1}, x_n]$ of $[a, b]$. The sum in (1) tends to a finite value, then it is called definite integral of $f(x)$ with respect to x on $[a, b]$. Also it is called **Riemann integral** of $f(x)$ on $[a, b]$.

- (i) $\int_a^b f(x) dx = \lim_{n \rightarrow \infty}$ and $\max(x_i - x_{i-1}) \rightarrow 0$ $\sum_{i=1}^n f(x_{i-1})(x_i - x_{i-1})$ is known as the **left - end rule** for evaluating the Riemann integral.
- (ii) $\int_a^b f(x) dx = \lim_{n \rightarrow \infty}$ and $\max(x_i - x_{i-1}) \rightarrow 0$ $\sum_{i=1}^n f(x_i)(x_i - x_{i-1})$ is known as the **right - end rule** for evaluating the Riemann integral.
- (iii) $\int_a^b f(x) dx = \lim_{n \rightarrow \infty}$ and $\max(x_i - x_{i-1}) \rightarrow 0$ $\sum_{i=1}^n f\left(\frac{x_{i-1} + x_i}{2}\right)(x_i - x_{i-1})$ is known as the **mid-point rule** for evaluating the Riemann integral.

Formulae :

$$1. \int (ax + b)^n dx = \frac{1}{a} \left[\frac{(ax+b)^{n+1}}{n+1} \right] + c$$

$$2. \int \frac{1}{ax+b} dx = \frac{1}{a} \log_e |(ax + b)| + c$$

$$3. \int e^{ax} dx = \frac{e^{ax}}{a} + c$$

$$4. \int \frac{1}{a^2 + x^2} dx = \frac{1}{a} \tan^{-1} \left(\frac{x}{a} \right) + c$$

$$5. \int \frac{1}{a^2 - x^2} dx = \frac{1}{2a} \log_e \left| \frac{a+x}{a-x} \right| + c$$

$$6. \int \frac{1}{x^2 - a^2} dx = \frac{1}{2a} \log_e \left| \frac{x-a}{x+a} \right| + c$$

$$7. \int \frac{1}{\sqrt{a^2 + x^2}} dx = \log_e |x + \sqrt{a^2 + x^2}| + c$$

$$8. \int \frac{1}{\sqrt{a^2 - x^2}} dx = \sin^{-1} \left(\frac{x}{a} \right) + c$$

$$9. \int \frac{1}{\sqrt{x^2 - a^2}} dx = \log_e |x + \sqrt{x^2 - a^2}| + c$$

$$10. \int \sin(ax + b) dx = -\frac{\cos(ax+b)}{a} + c$$

$$11. \int \cos(ax + b) dx = \frac{\sin(ax+b)}{a} + c$$

$$12. \int \tan(ax + b) dx = \frac{1}{a} \log |\sec(ax + b)| + c$$

$$13. \int \cot(ax + b) dx = \frac{1}{a} \log |\sin(ax + b)| + c$$

$$14. \int \sec(ax + b) dx = \frac{1}{a} \log |\sec(ax + b) + \tan(ax + b)| + c$$

$$15. \int \operatorname{cosec}(ax + b) dx = \frac{-1}{a} \log |\operatorname{cosec}(ax + b) - \cot(ax + b)| + c$$

$$16. \int \sqrt{a^2 + x^2} dx = \frac{x \sqrt{a^2 + x^2}}{2} + \frac{a^2}{2} \log_e |x + \sqrt{a^2 + x^2}| + c$$

$$17. \int \sqrt{a^2 - x^2} dx = \frac{x \sqrt{a^2 - x^2}}{2} + \frac{a^2}{2} \sin^{-1} \left(\frac{x}{a} \right) + c$$

1. Find an approximate value of $\int_1^{1.5} x \, dx$ by applying the left – end rule with the partition {1.1, 1.2, 1.3, 1.4, 1.5}. 3M

$$\int_1^{1.5} x \, dx \text{ Given interval } \{1.1, 1.2, 1.3, 1.4, 1.5\} \quad \therefore n = 5$$

$$\text{Let } x_0 = 1; x_1 = 1.1; x_2 = 1.2; x_3 = 1.3; x_4 = 1.4; x_5 = 1.5$$

$$\Delta x = x_2 - x_1 = 1.2 - 1.1 = 0.1$$

$$\text{Let } f(x) = x$$

$$\begin{aligned} \text{The left end rule for Riemann Integral } S &= [f(x_0) + f(x_1) + f(x_2) + f(x_3) + f(x_4)]\Delta x \\ &= (1 + 1.1 + 1.2 + 1.3 + 1.4)0.1 = 6.0 \times 0.1 \end{aligned}$$

$$\int_1^{1.5} x \, dx \simeq S = 0.6$$

2. Find an approximate value of $\int_1^{1.5} x^2 \, dx$ by applying the right-end rule with the partitions {1.1, 1.2, 1.3, 1.4, 1.5}. 3M

$$\int_1^{1.5} x^2 \, dx \text{ Given interval } \{1.1, 1.2, 1.3, 1.4, 1.5\} \quad \therefore n = 5$$

$$\text{Let } x_1 = 1.1; x_2 = 1.2; x_3 = 1.3; x_4 = 1.4; x_5 = 1.5$$

$$\Delta x = x_2 - x_1 = 1.2 - 1.1 = 0.1$$

$$\text{Let } f(x) = x^2$$

$$\begin{aligned} \text{The right end rule for Riemann Integral } S &= [f(x_1) + f(x_2) + f(x_3) + f(x_4) + f(x_5)]\Delta x \\ &= [(f(1.1) + f(1.2) + f(1.3) + f(1.4) + f(1.5))]0.1 \\ &= (1.21 + 1.44 + 1.69 + 1.96 + 2.25)0.1 \\ &= 8.55 \times 0.1 \end{aligned}$$

$$\int_1^{1.5} x^2 \, dx \simeq S = 0.855$$

3. Find an approximate value of $\int_1^{1.5} (2 - x) \, dx$ by applying the mid – point rule with the partition {1.1, 1.2, 1.3, 1.4, 1.5}. 3M

$$\text{Given } \int_1^{1.5} (2 - x) \, dx \text{ interval } \{1.1, 1.2, 1.3, 1.4, 1.5\} \quad \therefore n = 5$$

$$\text{Let } x_0 = 1; x_1 = 1.1; x_2 = 1.2; x_3 = 1.3; x_4 = 1.4; x_5 = 1.5$$

$$\Delta x = x_2 - x_1 = 1.2 - 1.1 = 0.1$$

$$\text{Let } f(x) = 2 - x$$

Now by the mid point rule

$$\begin{aligned} S &= \left[f\left(\frac{x_0+x_1}{2}\right) + f\left(\frac{x_1+x_2}{2}\right) + f\left(\frac{x_2+x_3}{2}\right) + f\left(\frac{x_3+x_4}{2}\right) + f\left(\frac{x_4+x_5}{2}\right) \right] \Delta x \\ &= \left[f\left(\frac{1+1.1}{2}\right) + f\left(\frac{1.1+1.2}{2}\right) + f\left(\frac{1.2+1.3}{2}\right) + f\left(\frac{1.3+1.4}{2}\right) + f\left(\frac{1.4+1.5}{2}\right) \right] 0.1 \\ &= \left[f\left(\frac{2.1}{2}\right) + f\left(\frac{2.3}{2}\right) + f\left(\frac{2.5}{2}\right) + f\left(\frac{2.7}{2}\right) + f\left(\frac{2.9}{2}\right) \right] 0.1 \\ &= [f(1.05) + f(1.15) + f(1.25) + f(1.35) + f(1.45)] 0.1 \\ &= [(2 - 1.05) + (2 - 1.15) + (2 - 1.25) + (2 - 1.35) + (2 - 1.45)](0.1) \\ &= [10 - (1.05 + 1.15 + 1.25 + 1.35 + 1.45)](0.1) \\ &= [10 - 6.25](0.1) = [3.75](0.1) = 0.375 \end{aligned}$$

EXERCISE 9.2



Video class

Concept Corner

Limit Formula to Evaluate $\int_a^b f(x)dx$

$$\int_a^b f(x)dx = \lim_{n \rightarrow \infty} \frac{b-a}{n} \sum_{r=1}^n f\left(a + (b-a)\frac{r}{n}\right)$$

If $a = 0$; $b = 1$ then we get

$$\int_0^1 f(x)dx = \lim_{n \rightarrow \infty} \frac{1}{n} \sum_{r=1}^n f\left(\frac{r}{n}\right)$$

Use it for problems:

$$\sum 1 = n$$

$$\sum r = \frac{n(n+1)}{2}$$

$$\sum r^2 = \frac{n(n+1)(2n+1)}{6}$$

$$\sum r^3 = \left[\frac{n(n+1)}{2}\right]^2$$

1. Evaluate the following integrals as the limits of sum.

5M

$$(i) \int_0^1 (5x + 4)dx$$

$$f(x) = (5x + 4); a = 0; b = 1$$

$$\begin{aligned} \int_0^1 (5x + 4)dx &= \lim_{n \rightarrow \infty} \frac{1}{n} \sum_{r=1}^n f\left(\frac{r}{n}\right) \\ &= \lim_{n \rightarrow \infty} \frac{1}{n} \sum_{r=1}^n \left(5\frac{r}{n} + 4\right) \\ &= \lim_{n \rightarrow \infty} \frac{1}{n} \sum_{r=1}^n \left(\frac{5r}{n} + 4\right) \\ &= \lim_{n \rightarrow \infty} \frac{1}{n} \left(\frac{5}{n} \sum_{r=1}^n r + 4 \sum_{r=1}^n 1\right) \\ &= \lim_{n \rightarrow \infty} \frac{1}{n} \left[\frac{5}{n} \frac{n(n+1)}{2} + 4n\right] \\ &= \lim_{n \rightarrow \infty} \left[\frac{5}{n^2} \frac{n(n+1)}{2} + \frac{4}{n} n\right] \\ &= \lim_{n \rightarrow \infty} \left[\frac{5}{n^2} \frac{n^2(1+\frac{1}{n})}{2} + 4\right] \\ &= 5\left(\frac{1+0}{2}\right) + 4 \\ &= \frac{5+8}{2} \end{aligned}$$

$$\therefore \int_0^1 (5x + 4)dx = \frac{13}{2}$$

$$(ii) \int_1^2 (4x^2 - 1)dx$$

$$f(x) = (4x^2 - 1); a = 1; b = 2$$

$$\therefore \int_a^b f(x)dx = \lim_{n \rightarrow \infty} \sum_{r=1}^n \frac{1}{n} f\left(a + (b-a)\frac{r}{n}\right) \dots\dots (1)$$

$$\begin{aligned} f\left(a + (b-a)\frac{r}{n}\right) &= f\left(1 + (2-1)\frac{r}{n}\right) \\ &= f\left(1 + \frac{r}{n}\right) \\ &= 4\left(1 + \frac{r}{n}\right)^2 - 1 \\ &= 4 + \frac{4r^2}{n^2} + 8\frac{r}{n} - 1 \end{aligned}$$

$$f\left(a + (b-a)\frac{r}{n}\right) = 3 + \frac{4r^2}{n^2} + 8\frac{r}{n}$$

$$\text{From (1), } = \lim_{n \rightarrow \infty} \frac{2-1}{n} \sum_{r=1}^n \left(3 + \frac{4r^2}{n^2} + 8\frac{r}{n}\right)$$

$$\begin{aligned} &= \lim_{n \rightarrow \infty} \frac{1}{n} \sum_{r=1}^n \left[3 + \frac{4r^2}{n^2} + \frac{8r}{n}\right] \\ &= \lim_{n \rightarrow \infty} \sum_{r=1}^n \left[\frac{3}{n} + \frac{4r^2}{n^3} + \frac{8r}{n^2}\right] \\ &= \lim_{n \rightarrow \infty} \left[\frac{3}{n} \sum 1 + \frac{4}{n^3} \sum r^2 + \frac{8}{n^2} \sum r\right] \\ &= \lim_{n \rightarrow \infty} \left[\frac{3}{n} n + \frac{4}{n^3} \frac{n(n+1)(2n+1)}{6} + \frac{8}{n^2} \frac{n(n+1)}{2}\right] \\ &= \lim_{n \rightarrow \infty} \left[3 + \frac{4}{n^3} n^3 \frac{(1+\frac{1}{n})(2+\frac{1}{n})}{6} + \frac{8}{n^2} n^2 \frac{(1+\frac{1}{n})}{2}\right] \\ &= 3 + \frac{2}{3} (1+0)(2+0) + 4(1+0) \\ &= 3 + \frac{4}{3} + 4 = 7 + \frac{4}{3} \\ &= 3 + \frac{4}{3} + 4 = 7 + \frac{4}{3} \end{aligned}$$

$$\therefore \int_1^2 (4x^2 - 1)dx = \frac{25}{3}$$

EXERCISE 9.3



Video class

Concept Corner

Fundamental Theorems of Integral calculus

Theorem 1 (First fundamental theorem):

If $f(x)$ be a continuous function defined on a closed interval $[a, b]$ and $F(x) = \int_a^x f(u)du$,
 $a < x < b$ then $\frac{d}{dx} F(x) = f(x)$ In other words, $F(x)$ is an anti-derivative of $f(x)$.

Theorem 2 (Second fundamental Theorem):

If $f(x)$ be a continuous function defined on a closed interval $[a, b]$ and $F(x)$ is an anti-derivative of $f(x)$, then $\int_a^b f(x)dx = F(b) - F(a)$

Properties:

1. $\int_a^b f(x)dx = \int_a^b f(u)du, a < b$
2. $\int_a^b f(x)dx = -\int_b^a f(x)dx$
3. $\int_a^b f(x)dx = \int_a^c f(x)dx + \int_c^b f(x)dx, \text{ if } a < c < b.$
4. $\int_a^b [\alpha f(x) + \beta g(x)]dx = \alpha \int_a^b f(x)dx + \beta \int_a^b g(x)dx, \alpha, \beta \text{ are constants}$
5. if $x = g(u)$, then $\int_a^b f(x)dx = \int_c^d f(g(u)) \frac{dg(u)}{du} du$ where $g(c) = a$ & $g(d) = b$.
6. $\int_{-a}^a f(x)dx = 2 \int_0^a f(x)dx$, if $f(x)$ is an even function
7. $\int_{-a}^a f(x)dx = 0$, if $f(x)$ is an odd function
8. If $f(2a - x) = f(x)$ then $\int_0^{2a} f(x)dx = 2 \int_0^a f(x)dx$
9. If $f(2a - x) = -f(x)$ then $\int_0^{2a} f(x)dx = 0$
10. $\int_0^a xf(x)dx = \frac{a}{2} \int_0^a f(x)dx$, if $f(a - x) = f(x)$

1. Evaluate the following definite integrals.

(i) $\int_3^4 \frac{dx}{x^2-4}$

JUL-20

2M

$$\int \frac{dx}{x^2-a^2} = \frac{1}{2a} \log \left| \frac{x-a}{x+a} \right| + c$$

$$\begin{aligned} \int_3^4 \frac{dx}{x^2-4} &= \left[\frac{1}{2(2)} \log \left| \frac{x-2}{x+2} \right| \right]_3^4 \\ &= \frac{1}{4} \left[\log \left(\frac{4-2}{4+2} \right) - \log \left(\frac{3-2}{3+2} \right) \right] \\ &= \frac{1}{4} \left[\log \frac{2}{6} - \log \frac{1}{5} \right] \\ &= \frac{1}{4} \left[\log \frac{2}{6} \times \frac{5}{1} \right] \\ &= \frac{1}{4} [\log \frac{5}{3}] \end{aligned}$$

(ii) $\int_{-1}^1 \frac{dx}{x^2+2x+5}$

2M

$$\int \frac{dx}{a^2+x^2} = \frac{1}{a} \tan^{-1} \frac{x}{a} + c$$

$$\begin{aligned} \int_{-1}^1 \frac{dx}{x^2+2x+5} &= \int_{-1}^1 \frac{dx}{x^2+2x+1+4} \\ &= \int_{-1}^1 \frac{dx}{(x+1)^2+2^2} \\ &= \left[\frac{1}{2} \tan^{-1} \frac{x+1}{2} \right]_{-1}^1 \\ &= \frac{1}{2} \left[\tan^{-1} \frac{1+1}{2} - \tan^{-1} \frac{-1+1}{2} \right] \\ &= \frac{1}{2} \left[\tan^{-1} \frac{2}{2} - \tan^{-1}(0) \right] \\ &= \frac{1}{2} \left[\frac{\pi}{4} - 0 \right] \\ &= \frac{\pi}{8} \end{aligned}$$

(iii) $\int_0^1 \sqrt{\frac{1-x}{1+x}} dx$

JUL-21

3M

$$\begin{aligned} \int_0^1 \sqrt{\frac{1-x}{1+x}} dx &= \int_0^1 \sqrt{\frac{(1-x)(1-x)}{(1+x)(1-x)}} dx \\ &= \int_0^1 \frac{(1-x)}{\sqrt{1-x^2}} dx \\ &= \int_0^1 \frac{1}{\sqrt{1-x^2}} dx + \int_0^1 \frac{-x}{\sqrt{1-x^2}} dx \\ &= [\sin^{-1} x]_0^1 + \frac{1}{2} \int_0^1 \frac{-2x}{\sqrt{1-x^2}} dx \\ &= \sin^{-1}(1) - \sin^{-1}(0) + \frac{1}{2} \int_0^1 (1-x^2)^{-\frac{1}{2}} (-2x) dx \\ &= \frac{\pi}{2} - 0 + \frac{1}{2} \left[\frac{(1-x^2)^{-\frac{1}{2}+1}}{-\frac{1}{2}+1} \right]_0^1 \\ &= \frac{\pi}{2} + \frac{1}{2} \left[\frac{(1-x^2)^{\frac{1}{2}}}{\frac{1}{2}} \right]_0^1 = \frac{\pi}{2} + [0 - 1] = \frac{\pi}{2} - 1 \end{aligned}$$

(iv) $\int_0^{\frac{\pi}{2}} e^x \left(\frac{1+\sin x}{1+\cos x} \right) dx$

3M

$$\begin{aligned} \text{I} &= \int_0^{\frac{\pi}{2}} e^x \left(\frac{1+2 \sin \frac{x}{2} \cos \frac{x}{2}}{2 \cos^2 \frac{x}{2}} \right) dx \\ &= \int_0^{\frac{\pi}{2}} \frac{e^x}{2 \cos^2 \frac{x}{2}} dx + \int_0^{\frac{\pi}{2}} e^x \tan \frac{x}{2} dx \\ &= \frac{1}{2} \int_0^{\frac{\pi}{2}} e^x \sec^2 \frac{x}{2} dx + \int_0^{\frac{\pi}{2}} e^x \tan \frac{x}{2} dx \\ &= \frac{1}{2} \left[e^x \tan \frac{x}{2} \right]_0^{\frac{\pi}{2}} - \frac{1}{2} \int_0^{\frac{\pi}{2}} e^x \frac{\tan \frac{x}{2}}{\frac{1}{2}} dx + \int_0^{\frac{\pi}{2}} e^x \tan \frac{x}{2} dx \\ &= \left[e^x \tan \frac{x}{2} \right]_0^{\frac{\pi}{2}} \\ &= e^{\frac{\pi}{2}}(1) \\ \int_0^{\frac{\pi}{2}} e^x \left(\frac{1+\sin x}{1+\cos x} \right) dx &= e^{\frac{\pi}{2}} \end{aligned}$$

$$u = e^x, \\ dv = \sec^2 \frac{x}{2} dx$$

(v) $\int_0^{\frac{\pi}{2}} \sqrt{\cos \theta} \sin^3 \theta d\theta$

3M

$$\begin{aligned} I &= \int_0^{\frac{\pi}{2}} \sqrt{\cos \theta} \sin^3 \theta d\theta \\ &= \int_1^0 \sqrt{t} \sin^3 \theta \frac{-dt}{\sin \theta} \\ &= -\int_1^0 \sqrt{t} \sin^2 \theta dt \\ &= \int_0^1 \sqrt{t} (1 - \cos^2 \theta) dt \\ &= \int_0^1 \sqrt{t} (1 - t^2) dt \\ &= \int_0^1 (t^{\frac{1}{2}} - t^{\frac{5}{2}}) dt \\ &= \left[\frac{t^{\frac{3}{2}}}{\frac{3}{2}} - \frac{t^{\frac{7}{2}}}{\frac{7}{2}} \right]_0^1 \\ &= \left[\frac{2}{3} t^{\frac{3}{2}} - \frac{2}{7} t^{\frac{7}{2}} \right]_0^1 \\ &= \frac{2}{3} - \frac{2}{7} \end{aligned}$$

$$\int_0^{\frac{\pi}{2}} \sqrt{\cos \theta} \sin^3 \theta d\theta = \frac{8}{21}$$

Let $\cos \theta = t$
 $-\sin \theta d\theta = dt$
 $\theta = 0 \Rightarrow t = \cos \theta = 1$
 $\theta = \frac{\pi}{2} \Rightarrow t = \cos \frac{\pi}{2} = 0$

(vi) $\int_0^1 \frac{1-x^2}{(1+x^2)^2} dx$

3M

$$\begin{aligned} I &= \int_0^1 \frac{x^2 (\frac{1}{x^2} - 1)}{x^2 (\frac{1}{x^2} + x)^2} dx \\ &= \int_0^1 \frac{\frac{1}{x^2} - 1}{(\frac{1}{x^2} + x)^2} dx \\ &= \int_{\infty}^2 \frac{-dt}{t^2} \\ &= \left[\frac{1}{t} \right]_{\infty}^2 \\ &= \frac{1}{2} - \frac{1}{\infty} \\ &= \frac{1}{2} - 0 \end{aligned}$$

Let $\frac{1}{x} + x = t$
 $\left(1 - \frac{1}{x^2}\right) dx = dt$
 $\left(\frac{1}{x^2} - 1\right) dx = -dt$
 $x = 0 \Rightarrow t = \infty$
 $x = 1 \Rightarrow t = 2$

$$\int_0^1 \frac{1-x^2}{(1+x^2)^2} dx = \frac{1}{2}$$

2. Evaluate the following integrals using properties of integration.

(i) $\int_{-5}^5 x \cos \left(\frac{e^x - 1}{e^{x+1}} \right) dx$

3M

$$\begin{aligned} f(x) &= x \cos \left(\frac{e^x - 1}{e^{x+1}} \right) \\ f(-x) &= -x \cos \left(\frac{e^{-x} - 1}{e^{-x+1}} \right) \\ &= -x \cos \left(\frac{\frac{1}{e^x} - 1}{\frac{1}{e^x} + 1} \right) = -x \cos \left(\frac{1 - e^x}{1 + e^x} \right) \\ f(-x) &= -x \cos \left[-\left(\frac{e^x - 1}{1 + e^x} \right) \right] \\ f(-x) &= -x \cos \left(\frac{e^x - 1}{1 + e^x} \right) = -f(x) \\ \therefore f(x) &\text{ is an odd function } \quad \mathbf{I = 0} \end{aligned}$$

$\therefore \int_{-a}^a f(x) dx = 0$ if $f(x)$ is an odd function

(iii) $\int_{-\frac{\pi}{4}}^{\frac{\pi}{4}} \sin^2 x dx$

3M

$$\begin{aligned} \int_{-\frac{\pi}{4}}^{\frac{\pi}{4}} \sin^2 x dx &= 2 \int_0^{\frac{\pi}{4}} \sin^2 x dx \\ [\because \sin^2 x \text{ is an even function}] \\ &= 2 \int_0^{\frac{\pi}{4}} \left(\frac{1 - \cos 2x}{2} \right) dx \\ &= 2 \cdot \frac{1}{2} \int_0^{\frac{\pi}{4}} (1 - \cos 2x) dx \\ &= \left[x - \frac{\sin 2x}{2} \right]_0^{\frac{\pi}{4}} \\ &= \left[\frac{\pi}{4} - \frac{\sin 2 \cdot \frac{\pi}{4}}{2} - 0 \right] = \frac{\pi}{4} - \frac{1}{2} \end{aligned}$$

$$\int_{-\frac{\pi}{4}}^{\frac{\pi}{4}} \sin^2 x dx = \frac{\pi - 2}{4}$$

(ii) $\int_{-\frac{\pi}{2}}^{\frac{\pi}{2}} (x^5 + x \cos x + \tan^3 x + 1) dx$

3M

$$\begin{aligned} \int_{-\frac{\pi}{2}}^{\frac{\pi}{2}} (x^5 + x \cos x + \tan^3 x + 1) dx &= \int_{-\frac{\pi}{2}}^{\frac{\pi}{2}} x^5 dx + \int_{-\frac{\pi}{2}}^{\frac{\pi}{2}} x \cos x dx + \int_{-\frac{\pi}{2}}^{\frac{\pi}{2}} \tan^3 x dx + \int_{-\frac{\pi}{2}}^{\frac{\pi}{2}} 1 dx \\ &= 0 + 0 + 0 + 2 \int_0^{\frac{\pi}{2}} dx \quad [\because x^5, x \cos x, \tan^3 x \text{ are odd function}] \\ &= 2 \left[x \right]_0^{\frac{\pi}{2}} \\ &= 2 \left[\frac{\pi}{2} - 0 \right] \end{aligned}$$

$$\int_{-\frac{\pi}{2}}^{\frac{\pi}{2}} (x^5 + x \cos x + \tan^3 x + 1) dx = \pi$$

(iv) $\int_0^{2\pi} x \log \left[\frac{3+\cos x}{3-\cos x} \right] dx$ PTA-4 3M

$$f(x) = X \log \left[\frac{3+\cos x}{3-\cos x} \right]$$

$$f(2\pi - x) = \log \left[\frac{3+\cos(2\pi-x)}{3-\cos(2\pi-x)} \right]$$

$$= \log \left[\frac{3+\cos x}{3-\cos x} \right] = f(x)$$

$$\int_0^a x f(x) dx = \int_0^a \frac{a}{2} f(x) dx$$

$$I = \frac{2\pi}{2} \int_0^{\frac{\pi}{2}} \log \left[\frac{3+\cos x}{3-\cos x} \right] dx$$

$$= 2\pi \int_0^{\pi} \log \left[\frac{3+\cos x}{3-\cos x} \right] dx$$

$$= 2\pi(0) = 0 \quad [\because f(2a-x) = -f(x)]$$

(v) $\int_0^{2\pi} \sin^4 x \cos^3 x dx$ 3M

$$\int_0^{2\pi} \sin^4 x \cos^3 x dx = \int_0^{2\pi} (\sin^2 x)^2 \cos^3 x dx$$

$$= \int_0^{2\pi} (1 - \cos^2 x)^2 \cos^3 x dx$$

$$= \int_0^{2\pi} (1 + \cos^4 x - 2\cos^2 x) \cos^3 x dx$$

$$= 2 \int_0^{\pi} (\cos^3 x + \cos^7 x - 2\cos^5 x) dx \quad \dots (1)$$

$$= 2 \int_0^{\pi} [\cos^3(\pi-x) + \cos^7(\pi-x) - 2\cos^5(\pi-x)] dx$$

$$= 2 \int_0^{\pi} [-\cos^3 x - \cos^7 x + 2\cos^5 x] dx \dots (2)$$

$$(1) + (2) \Rightarrow 2I = 0$$

$$\Rightarrow I = 0$$

(vi) $\int_0^1 |5x-3| dx$ 3M

$$\text{Clearly, } |5x-3| = \begin{cases} -(5x-3) & \text{when } 5x-3 < 0 \\ 5x-3 & \text{when } 5x-3 \geq 0 \end{cases}$$

$$|5x-3| = \begin{cases} -(5x-3) & x < \frac{3}{5} \\ 5x-3 & x \geq \frac{3}{5} \end{cases}$$

$$\therefore I = \int_0^1 |5x-3| dx$$

$$= \int_0^{\frac{3}{5}} -(5x-3) dx + \int_{\frac{3}{5}}^1 (5x-3) dx$$

$$= \left[3x - \frac{5x^2}{2} \right]_0^{\frac{3}{5}} + \left[\frac{5x^2}{2} - 3x \right]_{\frac{3}{5}}^1$$

$$= \left(\frac{9}{5} - \frac{5}{2} \times \frac{9}{25} \right) + \left(\frac{5}{2} - 3 \right) - \left(\frac{9}{10} - \frac{9}{5} \right)$$

$$= \left(\frac{9}{5} - \frac{9}{10} \right) + \left(\frac{-1}{2} + \frac{9}{10} \right)$$

$$I = \frac{13}{10}$$

(vii) $\int_0^{\sin^2 x} \sin^{-1} \sqrt{t} dt + \int_0^{\cos^2 x} \cos^{-1} \sqrt{t} dt,$ 5M
 $x \in \left[0, \frac{\pi}{2} \right]$

$$\text{Let } I_1 = \int_0^{\sin^2 x} \sin^{-1} \sqrt{t} dt$$

$$\sqrt{t} = \sin \theta$$

$$t = \sin^2 \theta$$

$$dt = 2 \sin \theta \cos \theta$$

$$dt = \sin 2\theta d\theta$$

$$t = 0, \sin^2 \theta = 0 \Rightarrow \theta = 0$$

$$t = \sin^2 x, \sin^2 \theta = \sin^2 x \Rightarrow \theta = x$$

$$I_1 = \int_0^x \sin^{-1} \sin \theta \sin 2\theta d\theta$$

$$= \int_0^x \theta \sin 2\theta d\theta$$

t	0	$\sin^2 x$
θ	0	x

Similarly,

$$\text{Let } I_2 = \int_0^{\cos^2 x} \cos^{-1} \sqrt{t} dt$$

$$t = \cos^2 \theta \Rightarrow \sqrt{t} = \cos \theta$$

$$dt = -2 \sin \theta \cos \theta d\theta$$

$$t = 0, \cos^2 \theta = 0 \Rightarrow \theta = \frac{\pi}{2}$$

$$t = \cos^2 x \Rightarrow \theta = x$$

t	0	$\cos^2 x$
θ	$\frac{\pi}{2}$	x

$$I_2 = \int_{\pi/2}^x -\theta \sin 2\theta d\theta$$

$$I = I_1 + I_2$$

$$I = \int_0^x \theta \sin 2\theta d\theta - \int_{\pi/2}^x \theta \sin 2\theta d\theta$$

$$I = \int_0^x \theta \sin 2\theta d\theta + \int_x^{\pi/2} \theta \sin 2\theta d\theta$$

$$I = \int_0^{\pi/2} \theta \sin 2\theta d\theta$$

$$= \left[\theta \left(\frac{-\cos 2\theta}{2} \right) - 1 \left(\frac{-\sin 2\theta}{4} \right) \right]_0^{\pi/2}$$

$$= \left[\frac{\pi}{2} \left(\frac{-\cos \pi}{2} \right) + 0 \right] - [0 + 0]$$

$$= \frac{\pi}{2} \left(\frac{1}{2} \right)$$

$$= \frac{\pi}{4}$$

(viii) $\int_0^1 \frac{\log(1+x)}{1+x^2} dx$

5M

$$\begin{aligned} \text{put } x &= \tan \theta ; dx = \sec^2 \theta \\ x = 0 &\Rightarrow \theta = 0 \\ x = 1 &\Rightarrow \theta = \frac{\pi}{4} \end{aligned}$$

$$\begin{aligned} I &= \int_0^{\frac{\pi}{4}} \frac{\log(1+\tan \theta)}{1+\tan^2 \theta} \sec^2 \theta d\theta \\ &= \int_0^{\frac{\pi}{4}} \log(1+\tan \theta) d\theta \quad \dots\dots\dots (1) \end{aligned}$$

$$\begin{aligned} &= \int_0^{\frac{\pi}{4}} \log(1+\tan(\frac{\pi}{4}-\theta)) d\theta \\ &= \int_0^{\frac{\pi}{4}} \log\left[1+\frac{1-\tan \theta}{1+\tan \theta}\right] d\theta \end{aligned}$$

$$= \int_0^{\frac{\pi}{4}} \log \frac{2}{1+\tan \theta} d\theta$$

$$I = \int_0^{\frac{\pi}{4}} [\log 2 - \log(1+\tan \theta)] d\theta \dots (2)$$

$$\begin{aligned} (1) + (2) \Rightarrow 2I &= \int_0^{\frac{\pi}{4}} \log 2 d\theta \\ &= \log 2 [\theta]_0^{\frac{\pi}{4}} \end{aligned}$$

$$2I = \frac{\pi}{4} \log 2$$

$$I = \frac{\pi}{8} \log 2$$

(ix) $\int_0^{\pi} \frac{x \sin x}{1+\sin x} dx$

5M

$$I = \int_0^{\pi} \frac{x \sin x}{1+\sin x} dx \quad \dots\dots\dots (1)$$

$$= \int_0^{\pi} \frac{(\pi-x) \sin(\pi-x)}{1+\sin(\pi-x)} dx$$

$$= \int_0^{\pi} \frac{(\pi-x) \sin x}{1+\sin x} dx \quad \dots\dots\dots (2)$$

$$(1) + (2) \Rightarrow 2I = \int_0^{\pi} \frac{\pi \sin x}{1+\sin x} dx$$

$$I = \frac{\pi}{2} \int_0^{\pi} \frac{\sin x}{1+\sin x} \frac{(1-\sin x)}{1-\sin x} dx$$

$$= \frac{\pi}{2} \int_0^{\pi} \frac{\sin x - \sin^2 x}{\cos^2 x} dx$$

$$= \frac{\pi}{2} \int_0^{\pi} (\sec x \tan x - \tan^2 x) dx$$

$$= \frac{\pi}{2} \int_0^{\pi} (\sec x \tan x - \sec^2 x + 1) dx$$

$$= \frac{\pi}{2} [\sec x - \tan x + x]_0^{\pi}$$

$$I = \frac{\pi}{2} [\pi - 2]$$

(x) $\int_{\frac{\pi}{8}}^{\frac{3\pi}{8}} \frac{1}{1+\sqrt{\tan x}} dx$

APR-24

5M

$$I = \int_{\frac{\pi}{8}}^{\frac{3\pi}{8}} \frac{1}{1+\sqrt{\tan x}} dx \quad \dots\dots\dots (1)$$

$$= \int_{\frac{\pi}{8}}^{\frac{3\pi}{8}} \frac{1}{1+\sqrt{\tan(\frac{3\pi}{8}+\frac{\pi}{8}-x)}} dx \quad \frac{\frac{3\pi}{8} + \frac{\pi}{8} = \frac{\pi}{2}}$$

$$= \int_{\frac{\pi}{8}}^{\frac{3\pi}{8}} \frac{1}{1+\sqrt{\tan(\frac{\pi}{2}-x)}} dx$$

$$I = \int_{\frac{\pi}{8}}^{\frac{3\pi}{8}} \frac{1}{1+\sqrt{\cot x}} dx \quad \dots\dots\dots (2)$$

$$(1) + (2) \Rightarrow 2I = \int_{\frac{\pi}{8}}^{\frac{3\pi}{8}} \left(\frac{1}{1+\sqrt{\tan x}} + \frac{1}{1+\sqrt{\cot x}} \right) dx$$

$$= \int_{\frac{\pi}{8}}^{\frac{3\pi}{8}} \left(\frac{\sqrt{\sin x}}{\sqrt{\sin x} + \sqrt{\cos x}} + \frac{\sqrt{\cos x}}{\sqrt{\sin x} + \sqrt{\cos x}} \right) dx$$

$$= \int_{\frac{\pi}{8}}^{\frac{3\pi}{8}} \left(\frac{\sqrt{\sin x} + \sqrt{\cos x}}{\sqrt{\sin x} + \sqrt{\cos x}} \right) dx$$

$$2I = \int_{\frac{\pi}{8}}^{\frac{3\pi}{8}} dx$$

$$= [x]_{\frac{\pi}{8}}^{\frac{3\pi}{8}} = \frac{3\pi}{8} - \frac{\pi}{8} = \frac{\pi}{4}$$

$$I = \frac{\pi}{8}$$

(xi) $\int_0^{\pi} x [\sin^2(\sin x) + \cos^2(\cos x)] dx$

5M

$$I = \int_0^{\pi} x [\sin^2(\sin x) + \cos^2(\cos x)] dx$$

$$= \frac{\pi}{2} \int_0^{\pi} [\sin^2(\sin x) + \cos^2(\cos x)] dx$$

$$= \frac{\pi}{2} 2 \int_0^{\frac{\pi}{2}} [\sin^2(\sin x) + \cos^2(\cos x)] dx$$

$$= \pi \int_0^{\frac{\pi}{2}} [\sin^2(\sin x) + \cos^2(\cos x)] dx \dots (1)$$

$$= \pi \int_0^{\frac{\pi}{2}} [\sin^2(\sin(\frac{\pi}{2}-x)) + \cos^2(\cos(\frac{\pi}{2}-x))] dx$$

$$I = \pi \int_0^{\frac{\pi}{2}} [\sin^2(\cos x) + \cos^2(\sin x)] dx \dots (2)$$

$$(1) + (2) \Rightarrow$$

$$I + I = \pi \int_0^{\frac{\pi}{2}} (\sin^2(\cos x) + \cos^2(\sin x) + \sin^2(\sin x) + \cos^2(\cos x)) dx$$

$$2I = \pi \int_0^{\frac{\pi}{2}} (1+1) dx = 2\pi \int_0^{\frac{\pi}{2}} dx$$

$$I = \pi [x]_0^{\frac{\pi}{2}}$$

$$= \pi \left(\frac{\pi}{2} \right)$$

$$I = \frac{\pi^2}{2}$$

EXERCISE 9.4



Video class

Concept Corner

Bernoulli's Formula:

$$\int uv \, dx = uv_{(1)} - u^{(1)}v_{(2)} + u^{(2)}v_{(3)} - u^{(3)}v_{(4)} + \dots$$

This is called Bernoulli's formula for integration of product of two function.

Evaluate the following.

1. $\int_0^1 x^3 e^{-2x} \, dx$

$$u = x^3; \quad v = e^{-2x}$$

By Bernoulli's formula,

$$\int uv \, dx = uv_1 - u^1 v_2 + u^2 v_3 - u^3 v_4 + \dots$$

$$= \left[x^3 \left(\frac{e^{-2x}}{-2} \right) - 3x^2 \left(\frac{e^{-2x}}{4} \right) + 6x \left(\frac{e^{-2x}}{-8} \right) - 6 \left(\frac{e^{-2x}}{16} \right) \right]_0^1$$

$$= \left[x^3 \left(\frac{e^{-2x}}{-2} \right) - 3x^2 \left(\frac{e^{-2x}}{4} \right) + 6x \left(\frac{e^{-2x}}{-8} \right) - 6 \left(\frac{e^{-2x}}{16} \right) - \left(-6 \left(\frac{e^0}{16} \right) \right) \right]_0^1$$

$$= \left(1 \left(\frac{e^{-2}}{-2} \right) - 3 \left(\frac{e^{-2}}{4} \right) + 6 \left(\frac{e^{-2}}{-8} \right) - 6 \left(\frac{e^{-2}}{16} \right) \right) - \left(-6 \frac{e^0}{16} \right) [\because e^0 = 1]$$

$$= -e^{-2} \left[\frac{1}{2} + \frac{3}{4} + \frac{3}{4} + \frac{3}{8} \right] + \frac{3}{8}$$

$$= -e^{-2} \left[\frac{4+6+6+3}{8} \right] + \frac{3}{8}$$

$$= \frac{-19}{8} e^{-2} + \frac{3}{8}$$

$$= \frac{3}{8} - \frac{19}{8} e^{-2}$$

5M

$$dv = e^{-2x} \, dx$$

$$u = x^3 \quad v = \frac{e^{-2x}}{-2}$$

$$u_1 = 3x^2 \quad v_1 = \frac{e^{-2x}}{4}$$

$$u_2 = 6x \quad v_2 = \frac{e^{-2x}}{-8}$$

$$u_3 = 6 \quad v_3 = \frac{e^{-2x}}{16}$$

2. $\int_0^1 \frac{\sin(3 \tan^{-1} x) \tan^{-1} x}{1+x^2} \, dx$

5M

$$t = \tan^{-1} x; \quad dt = \frac{1}{1+x^2} \, dx$$

$$x = 0 \Rightarrow t = 0 \quad \& \quad x = 1 \Rightarrow t = \frac{\pi}{4}$$

$$I = \int_0^{\frac{\pi}{4}} t \sin 3t \, dt$$

$$= \left[t \left(\frac{-\cos 3t}{3} \right) - 1 \left(\frac{-\sin 3t}{3.3} \right) \right]_0^{\frac{\pi}{4}}$$

$$= \left[\frac{-t \cos 3t}{3} + \frac{\sin 3t}{9} \right]_0^{\frac{\pi}{4}}$$

$$= \left[\left(\frac{-\pi}{12} \cos \frac{3\pi}{4} + \frac{\sin \frac{3\pi}{4}}{9} \right) - 0 \right]$$

$$= \frac{-\pi}{12} \left(-\cos \frac{\pi}{4} \right) + \frac{1}{9} \sin \frac{\pi}{4}$$

$$= \frac{\pi}{12} \frac{1}{\sqrt{2}} + \frac{1}{9} \frac{1}{\sqrt{2}}$$

$$I = \frac{1}{\sqrt{2}} \left(\frac{\pi}{12} + \frac{1}{9} \right)$$

$$dv = \sin 3t \, dt$$

$$u = t \quad v = \frac{-\cos 3t}{3}$$

$$u_1 = 1 \quad v_1 = \frac{-\sin 3t}{9}$$

3. $\int_0^{\frac{1}{\sqrt{2}}} \frac{e^{\sin^{-1} x} \sin^{-1} x}{\sqrt{1-x^2}} \, dx$

5M

$$t = \sin^{-1} x$$

$$dt = \frac{1}{\sqrt{1-x^2}} \, dx$$

$$x = 0 \Rightarrow t = 0$$

$$x = \frac{1}{\sqrt{2}} \Rightarrow t = \frac{\pi}{4}$$

$$I = \int_0^{\frac{\pi}{4}} e^t t \, dt$$

$$= [te^t - e^t]_0^{\frac{\pi}{4}}$$

$$= \frac{\pi}{4} e^{\frac{\pi}{4}} - e^{\frac{\pi}{4}} + 1$$

$$I = 1 + e^{\frac{\pi}{4}} \left(\frac{\pi}{4} - 1 \right)$$

$$dv = e^t \, dt$$

$$u = t \quad v = e^t$$

$$u_1 = 1 \quad v_1 = e^t$$

4. $\int_0^{\frac{\pi}{2}} x^2 \cos 2x \, dx$

Taking $u = x^2$; $v = \cos 2x$

$$\begin{aligned} I &= \int_0^{\frac{\pi}{2}} x^2 \cos 2x \, dx \\ &= \left[x^2 \frac{\sin 2x}{2} - 2x \frac{(-\cos 2x)}{2.2} + 2 \frac{(-\sin 2x)}{2.2.2} \right]_0^{\frac{\pi}{2}} \\ &= \left[x^2 \frac{\sin 2x}{2} + 2x \frac{\cos 2x}{4} - 2 \frac{\sin 2x}{8} \right]_0^{\frac{\pi}{2}} \\ &= \left(0 + \frac{2 \cdot \frac{\pi}{2} \cos 2 \cdot \frac{\pi}{2}}{4} - 0 \right) - (0) \\ &= \frac{\pi \cos \pi}{4} \\ &= \frac{\pi}{4} (-1) \\ I &= -\frac{\pi}{4} \end{aligned}$$

$$\begin{aligned} dv &= \cos 2x \, dx \\ u &= x^2 & v &= \frac{\sin 2x}{2} \\ u_1 &= 2x & v_1 &= \frac{-\cos 2x}{4} \\ u_1 &= 2 & v_1 &= \frac{-\sin 2x}{8} \end{aligned}$$

EXERCISE 9.5



Concept Corner

Improper Integrals

- $\int_a^\infty f(x) \, dx = \lim_{t \rightarrow \infty} \int_a^t f(x) \, dx$
- $\int_{-\infty}^a f(x) \, dx = \lim_{t \rightarrow -\infty} \int_t^a f(x) \, dx$
- $\int_{-\infty}^\infty f(x) \, dx = \lim_{t \rightarrow \infty} \int_{-t}^t f(x) \, dx$

They are called improper integral of first kind. If the limit exists, then the improper integrals are said to be convergent.

1. Evaluate the following

PTA-3

3M

(i) $\int_0^{\frac{\pi}{2}} \frac{dx}{1+5 \cos^2 x}$

APR-25

$$I = \int_0^{\frac{\pi}{2}} \frac{dx}{1+5 \cos^2 x}$$

Dividing numerator & denominator by $\cos^2 x$

$$\begin{aligned} I &= \int_0^{\frac{\pi}{2}} \frac{\frac{dx}{\cos^2 x}}{\frac{1}{\cos^2 x} + 5} \\ &= \int_0^{\frac{\pi}{2}} \frac{\sec^2 x}{\sec^2 x + 5} \, dx \\ &= \int_0^{\frac{\pi}{2}} \frac{\sec^2 x}{1 + \tan^2 x + 5} \, dx \\ &= \int_0^{\frac{\pi}{2}} \frac{\sec^2 x}{\tan^2 x + 6} \, dx \\ &= \int_0^\infty \frac{dt}{t^2 + \sqrt{6}^2} = \left[\frac{1}{\sqrt{6}} \tan^{-1} \frac{t}{\sqrt{6}} \right]_0^\infty \\ &= \frac{1}{\sqrt{6}} [\tan^{-1} \infty - \tan^{-1} 0] \\ I &= \frac{\pi}{2\sqrt{6}} \end{aligned}$$

$$\begin{aligned} \text{put } \tan x &= t \\ \sec^2 x \, dx &= dt \\ x = 0 &\Rightarrow t = 0 \\ x = \frac{\pi}{2} &\Rightarrow t = \infty \end{aligned}$$

(ii) $\int_0^{\frac{\pi}{2}} \frac{dx}{5+4 \sin^2 x}$

$$I = \int_0^{\frac{\pi}{2}} \frac{dx}{5+4 \sin^2 x}$$

Dividing numerator & denominator by $\cos^2 x$

$$\begin{aligned} I &= \int_0^{\frac{\pi}{2}} \frac{\sec^2 x}{5 \sec^2 x + 4 \tan^2 x} \, dx \\ &= \int_0^{\frac{\pi}{2}} \frac{\sec^2 x}{5(1 + \tan^2 x) + 4 \tan^2 x} \, dx \\ &= \int_0^{\frac{\pi}{2}} \frac{\sec^2 x}{5 + 9 \tan^2 x} \, dx \\ &= \int_0^\infty \frac{dt}{5 + 9t^2} \\ &= \int_0^\infty \frac{dt}{9 \left[t^2 + \left(\frac{\sqrt{5}}{3} \right)^2 \right]} \\ &= \frac{1}{9} \int_0^\infty \frac{dt}{t^2 + \left(\frac{\sqrt{5}}{3} \right)^2} \\ &= \frac{1}{9} \left[\frac{1}{\sqrt{5}/3} \tan^{-1} \frac{t}{\sqrt{5}/3} \right]_0^\infty = \frac{1}{9} \frac{3}{\sqrt{5}} [\tan^{-1} \frac{3t}{\sqrt{5}}]_0^\infty \\ &= \frac{1}{3\sqrt{5}} [\tan^{-1} \infty - \tan^{-1} 0] \\ I &= \frac{\pi}{6\sqrt{5}} \end{aligned}$$

$$\begin{aligned} \tan x &= t \\ \sec^2 x \, dx &= dt \\ x = 0 &\Rightarrow t = 0 \\ x = \frac{\pi}{2} &\Rightarrow t = \infty \end{aligned}$$

6. A watermelon has an ellipsoid shape which can be obtained by revolving an ellipse with major axis 20 cm and minor-axis 10 cm about its major-axis. Find its volume using integration.

Length of major axis = 20 cm

$$\therefore 2a = 20$$

$$\Rightarrow a = 10$$

Length of

minor axis = 10 cm

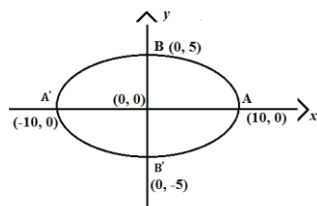
$$\therefore 2b = 10$$

$$\Rightarrow b = 5$$

Equation of ellipse $\frac{x^2}{100} + \frac{y^2}{25} = 1$

$$y^2 = 25 \left[1 - \frac{x^2}{100} \right]$$

$$y^2 = \frac{1}{4} (100 - x^2)$$



$$\therefore \text{Volume } V = \pi \int_a^b y^2 dx$$

5M

$$= \pi \int_{-10}^{10} \frac{1}{4} (100 - x^2) dx$$

$$= \frac{\pi}{4} 2 \int_0^{10} (100 - x^2) dx$$

$$= \frac{\pi}{2} \left[100x - \frac{x^3}{3} \right]_0^{10}$$

$$= \frac{\pi}{2} \left[\left(1000 - \frac{1000}{3} \right) - (0) \right]$$

$$= \frac{\pi}{2} \left[\frac{3000 - 1000}{3} \right]$$

$$= \frac{\pi}{2} \cdot \frac{2000}{3}$$

$$\therefore \text{Volume } V = \frac{1000}{3} \pi \text{ cm}^3$$

EXERCISE 9.10

Choose the correct or the most suitable answer from the given four alternatives

1. The value of $\int_0^{\frac{2}{3}} \frac{dx}{\sqrt{4-9x^2}}$ is

APR-25,20



Online Test

(1) $\frac{\pi}{6}$

(2) $\frac{\pi}{2}$

(3) $\frac{\pi}{4}$

(4) π

$$\int_0^{\frac{2}{3}} \frac{dx}{\sqrt{4-9x^2}} = \int_0^{\frac{2}{3}} \frac{dx}{\sqrt{9\left(\frac{4}{9}-x^2\right)}} = \frac{1}{3} \int_0^{\frac{2}{3}} \frac{dx}{\sqrt{\left(\frac{2}{3}\right)^2 - x^2}}$$

$$= \frac{1}{3} \left[\sin^{-1} \frac{x}{\frac{2}{3}} \right]_0^{\frac{2}{3}} = \frac{1}{3} \left[\sin^{-1} \frac{3x}{2} \right]_0^{\frac{2}{3}} = \frac{1}{3} \left[\sin^{-1} \frac{3}{2} \left(\frac{2}{3} \right) - 0 \right] = \frac{1}{3} [\sin^{-1} 1] = \frac{1}{3} \cdot \frac{\pi}{2}$$

$$\int_0^{\frac{2}{3}} \frac{dx}{\sqrt{4-9x^2}} = \frac{\pi}{6}$$

2. The value of $\int_{-1}^2 |x| dx$ is

JUL-22, 23, 24

(1) $\frac{1}{2}$

(2) $\frac{3}{2}$

(3) $\frac{5}{2}$

(4) $\frac{7}{2}$

$$\int_{-1}^2 |x| dx = \int_{-1}^0 -x dx + \int_0^2 x dx$$

$$= \left[\frac{-x^2}{2} \right]_{-1}^0 + \left[\frac{x^2}{2} \right]_0^2 = -\left(0 - \frac{1}{2} \right) + \left(\frac{4}{2} - 0 \right) = \frac{1}{2} + \frac{4}{2} = \frac{5}{2}$$

3. For any value of $n \in \mathbb{Z}$, $\int_0^\pi e^{\cos^2 x} \cos^3[(2n+1)x] dx$ is

(1) $\frac{\pi}{2}$

(2) π

(3) 0

(4) 2

$$f(x) = e^{\cos^2 x} \cos^3[(2n+1)x]$$

$$f(\pi - x) = e^{\cos^2(\pi - x)} (\cos(2n+1)(\pi - x))^3$$

$$= e^{\cos^2 x} (-\cos(2n+1)x)^3 = -e^{\cos^2 x} \cos^3[(2n+1)x]$$

$$f(\pi - x) = -f(x)$$

$$\therefore \int_0^{2a} f(x) dx = 0 \text{ When } f(2a - x) = -f(x)$$

$$I = 0$$

4. The value of $\int_{-\frac{\pi}{2}}^{\frac{\pi}{2}} \sin^2 x \cos x \, dx$ is

(1) $\frac{3}{2}$ (2) $\frac{1}{2}$ (3) 0 (4) $\frac{2}{3}$

$$\begin{aligned}\int_{-\frac{\pi}{2}}^{\frac{\pi}{2}} \sin^2 x \cos x \, dx &= 2 \int_0^{\frac{\pi}{2}} \sin^2 x \cos x \, dx = 2 \int_0^{\frac{\pi}{2}} \sin^2 x \, d(\sin x) \\ &= 2 \left[\frac{\sin^3 x}{3} \right]_0^{\frac{\pi}{2}} = \frac{2}{3} \left[\left(\sin \frac{\pi}{2} \right)^3 - 0 \right] = \frac{2}{3} (1) = \frac{2}{3}\end{aligned}$$

5. The value of $\int_{-4}^4 \left[\tan^{-1} \left(\frac{x^2}{x^4+1} \right) + \tan^{-1} \left(\frac{x^4+1}{x^2} \right) \right] dx$ is

(1) π (2) 2π (3) 3π (4) 4π

$$\begin{aligned}\int_{-4}^4 \left[\tan^{-1} \left(\frac{x^2}{x^4+1} \right) + \tan^{-1} \left(\frac{x^4+1}{x^2} \right) \right] dx &= 2 \int_0^4 \left[\tan^{-1} \left(\frac{x^2}{x^4+1} \right) + \cot^{-1} \left(\frac{x^2}{x^4+1} \right) \right] dx \\ &= 2 \int_0^4 \frac{\pi}{2} \, dx = \pi \int_0^4 dx \\ &= \pi [x]_0^4 \\ &= 4\pi\end{aligned}$$

6. The value of $\int_{-\frac{\pi}{4}}^{\frac{\pi}{4}} \left(\frac{2x^7-3x^5+7x^3-x+1}{\cos^2 x} \right) dx$ is

(1) 4 (2) 3 (3) 2 (4) 0

$$\begin{aligned}\int_{-\frac{\pi}{4}}^{\frac{\pi}{4}} \left(\frac{2x^7-3x^5+7x^3-x+1}{\cos^2 x} \right) dx &= \int_{-\frac{\pi}{4}}^{\frac{\pi}{4}} \left(\frac{2x^7-3x^5+7x^3-x}{\cos^2 x} \right) dx + \int_{-\frac{\pi}{4}}^{\frac{\pi}{4}} \left(\frac{1}{\cos^2 x} \right) dx \\ &= 0 + \int_{-\frac{\pi}{4}}^{\frac{\pi}{4}} \sec^2 x \, dx = 2 \int_{-\frac{\pi}{4}}^{\frac{\pi}{4}} \sec^2 x \, dx \\ &= 2 [\tan x]_0^{\frac{\pi}{4}} \\ &= 2 \left[\tan \frac{\pi}{4} - 0 \right] = 2\end{aligned}$$

7. If $f(x) = \int_0^x t \cos t \, dt$, then $\frac{df}{dx} =$

PTA-6, JUL-24,20

(1) $\cos x - x \sin x$ (2) $\sin x + x \cos x$ (3) $x \cos x$ (4) $x \sin x$

$$\begin{aligned}f(x) &= \int_0^x t \cos t \, dt \\ &= [t \sin t + \cos t]_0^x \\ &= [(x \sin x + \cos x) - (0 - \cos 0)] \\ f(x) &= x \sin x + \cos x - 1 \\ \frac{df}{dx} &= x \cos x + \sin x (1) - \sin x - 0 = x \cos x\end{aligned}$$

8. The area between the parabola $y^2 = 4x$ and its latus rectum is

JUL-22, APR-24, 23

(1) $\frac{2}{3}$ (2) $\frac{4}{3}$ (3) $\frac{8}{3}$ (4) $\frac{5}{3}$

Compare, $y^2 = 4ax$ and $y^2 = 4x$ so we get, $a = 1$

The area between the parabola and its latus rectum is $A = \frac{8}{3} a^2 = \frac{8}{3} (1)^2 = \frac{8}{3}$

9. The value of $\int_0^1 x(1-x)^{99} dx$ is

APR-22

- (1) $\frac{1}{11000}$ (2) $\frac{1}{10100}$ (3) $\frac{1}{10010}$ (4) $\frac{1}{10001}$

$$\begin{aligned}\int_0^1 x(1-x)^{99} dx &= \int_0^1 x(1-x)^{99} dx = \int_0^1 (x^{99} - x^{100}) dx \\ &= \left[\frac{x^{100}}{100} - \frac{x^{101}}{101} \right]_0^1 \\ &= \frac{1}{100} - \frac{1}{101} = \frac{1}{10100}\end{aligned}$$

$$\begin{aligned}f(x) &= x(1-x)^{99} \\ f(1-x) &= (1-x)x^{99} \\ f(1-x) &= x^{99} - x^{100}\end{aligned}$$

10. The value of $\int_0^\pi \frac{dx}{1+5\cos x}$ is

MDL

- (1) $\frac{\pi}{2}$ (2) π (3) $\frac{3\pi}{2}$ (4) 2π

$$\begin{aligned}I &= \int_0^\pi \frac{dx}{1+5\cos x} \dots\dots\dots (1) \\ f(x) &= \frac{1}{1+5\cos x} \\ f(\pi-x) &= \frac{1}{1+5\cos(\pi-x)} = \frac{1}{1+5\cos x} = \frac{1}{1+\frac{1}{5\cos x}} \\ &= \frac{5\cos x}{1+5\cos x} \\ I &= \int_0^\pi \frac{5\cos x}{1+5\cos x} dx \dots\dots\dots (2)\end{aligned}$$

$$\int_0^{2a} f(x) dx = \int_0^a f(x) dx + \int_0^a f(a-x) dx$$

$$\begin{aligned}\text{Add (1) \& (2) } I + I &= \int_0^\pi \left(\frac{5\cos x + 1}{5\cos x + 1} \right) dx \\ 2I &= \int_0^\pi dx \\ &= [x]_0^\pi \\ I &= \frac{\pi}{2}\end{aligned}$$

11. If $\frac{\Gamma(n+2)}{\Gamma(n)} = 90$ then n is

- (1) 10 (2) 5 (3) 8 (4) 9

$$\begin{aligned}\frac{\Gamma(n+2)}{\Gamma(n)} &= 90 \\ \Rightarrow \frac{(n+1)\Gamma(n+1)}{\Gamma(n)} &= 90 \\ \Rightarrow \frac{(n+1)n\Gamma(n)}{\Gamma(n)} &= 90\end{aligned}$$

$$\begin{aligned}n(n+1) &= 90 \\ &= 9 \times 10 \\ n &= 9\end{aligned}$$

12. The value of $\int_0^{\frac{\pi}{6}} \cos^3 3x dx$ is

- (1) $\frac{2}{3}$ (2) $\frac{2}{9}$ (3) $\frac{1}{9}$ (4) $\frac{1}{3}$

$$\begin{aligned}\int_0^{\frac{\pi}{6}} \cos^3 3x dx &= \int_0^{\frac{\pi}{2}} \cos^3 t \frac{dt}{3} = \frac{1}{3} \int_0^{\frac{\pi}{2}} \cos^3 t dt \\ n &= 3 \text{ (odd number)} \\ &= \frac{1}{3} \cdot \frac{n-1}{n} \cdot \frac{n-3}{n-2} \dots \\ &= \frac{1}{3} \cdot \frac{3-1}{3} = \frac{1}{3} \times \frac{2}{3} = \frac{2}{9}\end{aligned}$$

$$\begin{aligned}\text{Let } 3x &= t \\ 3dx &= dt \\ dx &= \frac{dt}{3} \\ x = 0 &\Rightarrow t = 0 \\ x = \frac{\pi}{6} &\Rightarrow t = \frac{\pi}{2}\end{aligned}$$

13. The value of $\int_0^\pi \sin^4 x dx$ is

APR-20, JUL-23

- (1) $\frac{3\pi}{10}$ (2) $\frac{3\pi}{8}$ (3) $\frac{3\pi}{4}$ (4) $\frac{3\pi}{2}$

$$\begin{aligned}\int_0^\pi \sin^4 x dx &= 2 \int_0^{\frac{\pi}{2}} \sin^4 x dx \\ n &= 4 \text{ (even number)} \\ &= 2 \cdot \frac{n-1}{n} \cdot \frac{n-3}{n-2} \dots \dots \frac{\pi}{2} \\ &= 2 \cdot \frac{4-1}{4} \cdot \frac{4-3}{4-2} \cdot \frac{\pi}{2} = 2 \cdot \frac{3}{4} \cdot \frac{1}{2} \cdot \frac{\pi}{2} = \frac{3\pi}{8}\end{aligned}$$

14. The value of $\int_0^{\infty} e^{-3x} x^2 dx$ is

PTA-1, APR-22

(1) $\frac{7}{27}$

(2) $\frac{5}{27}$

(3) $\frac{4}{27}$

(4) $\frac{2}{27}$

$$\int_0^{\infty} e^{-3x} x^2 dx$$

$$n = 2; a = 3$$

$$\int_0^{\infty} e^{-3x} x^2 dx = \frac{2!}{3^{2+1}} = \frac{2!}{3^3} = \frac{2}{27}$$

$$\int_0^{\infty} e^{-ax} x^n dx = \frac{n!}{a^{n+1}}$$

15. If $\int_0^a \frac{1}{4+x^2} dx = \frac{\pi}{8}$ then a is

PTA-2

(1) 4

(2) 1

(3) 3

(4) 2

$$\int_0^a \frac{1}{4+x^2} dx = \frac{\pi}{8}$$

$$\left[\frac{1}{2} \tan^{-1} \frac{x}{2} \right]_0^a = \frac{\pi}{8}$$

$$\frac{1}{2} \tan^{-1} \frac{a}{2} = \frac{\pi}{8}$$

$$\tan^{-1} \frac{a}{2} = \frac{\pi}{4}$$

$$\frac{a}{2} = \tan \frac{\pi}{4} = 1$$

$$a = 2$$

16. The volume of solid of revolution of the region bounded by $y^2 = x(a-x)$ about x -axis is

(1) πa^3

(2) $\frac{\pi a^3}{4}$

(3) $\frac{\pi a^3}{5}$

(4) $\frac{\pi a^3}{6}$

APR-25, MDL

$$\text{Given } y^2 = x(a-x)$$

$$\text{Required Volume } V = \pi \int_a^b y^2 dx$$

$$= \pi \int_0^a (ax - x^2) dx$$

$$= \pi \left[\frac{ax^2}{2} - \frac{x^3}{3} \right]_0^a$$

$$= \pi \left[\frac{a^3}{2} - \frac{a^3}{3} \right] = \pi a^3 \left[\frac{3-2}{6} \right]$$

$$V = \frac{\pi a^3}{6}$$

$$\begin{aligned} \text{put } y &= 0 \\ \Rightarrow x(x-a) &= 0 \\ x &= 0; x = a \end{aligned}$$

17. If $f(x) = \int_1^x \frac{e^{\sin u}}{u} du$, $x > 1$ & $\int_1^3 \frac{e^{\sin x^2}}{x} dx = \frac{1}{2} [f(a) - f(1)]$, then one of the possible value of a is

(1) 3

(2) 6

(3) 9

(4) 5

By Leibniz theorem,

$$f'(x) = \int_1^x \frac{\partial}{\partial x} \left(\frac{e^{\sin u}}{u} \right) du + \frac{d}{dx} (x) \frac{e^{\sin x}}{x} - \frac{d}{dx} (1) \frac{e^{\sin 1}}{1} = 0 + \frac{e^{\sin x}}{x} - 0 = \frac{e^{\sin x}}{x}$$

$$\text{Let } I = \int_1^3 \frac{e^{\sin x^2}}{x} dx$$

$$= \int_1^9 \frac{1}{2} \frac{e^{\sin t}}{t} dt$$

$$= \frac{1}{2} [f(t)]_1^9$$

$$= \frac{1}{2} [f(9) - f(1)]$$

$$= \frac{1}{2} [f(a) - f(1)]$$

$$\therefore a = 9$$

$$\text{Let } x^2 = t$$

$$2x dx = dt \Rightarrow x dx = \frac{1}{2} dt$$

$$\frac{x dx}{x^2} = \frac{1}{2} \frac{dt}{x^2} \Rightarrow \frac{dx}{x} = \frac{1}{2} \frac{dt}{t}$$

$$x = 1 \Rightarrow t = 1$$

$$x = 3 \Rightarrow t = 3^2 = 9$$

18. The value of $\int_0^1 (\sin^{-1} x)^2 dx$ is

(1) $\frac{\pi^2}{4} - 1$

(2) $\frac{\pi^2}{4} + 2$

(3) $\frac{\pi^2}{4} + 1$

(4) $\frac{\pi^2}{4} - 2$

Let $t = \sin^{-1} x$

$$x = \sin t$$

$$dt = \frac{1}{\sqrt{1-x^2}} dx$$

$$= \frac{1}{\sqrt{1-\sin^2 t}} dx$$

$$\cos t dt = dx$$

$$\therefore \int_0^1 (\sin^{-1} x)^2 dx = \int_0^{\frac{\pi}{2}} t^2 \cos t dt$$

$$= [t^2(\sin t) - 2t(-\cos t) + 2(-\sin t)]_0^{\frac{\pi}{2}}$$

$$= \left[\left(\frac{\pi^2}{4} \sin \frac{\pi}{2} - 0 - 2 \sin \frac{\pi}{2} \right) - 0 \right]$$

$$= \frac{\pi^2}{4} (1) - 2 = \frac{\pi^2}{4} - 2$$

$$x = 0 \Rightarrow t = 0$$

$$x = 1 \Rightarrow t = \frac{\pi}{2}$$

19. The value of $\int_0^a (\sqrt{a^2 - x^2})^3 dx$ is

APR-24

(1) $\frac{\pi a^3}{16}$

(2) $\frac{3\pi a^4}{16}$

(3) $\frac{3\pi a^2}{8}$

(4) $\frac{3\pi a^4}{8}$

Let $x = a \cos \theta$

$$dx = -a \sin \theta d\theta$$

$$\int_{\frac{\pi}{2}}^0 (\sqrt{a^2 - a^2 \cos^2 \theta})^3 (-a \sin \theta d\theta) = \int_0^{\frac{\pi}{2}} (a \sin \theta)^3 (a \sin \theta) d\theta$$

$$= a^4 \int_0^{\frac{\pi}{2}} \sin^4 \theta d\theta$$

$n = 4$ (even number)

$$\int_{\frac{\pi}{2}}^0 (\sqrt{a^2 - a^2 \cos^2 \theta})^3 (-a \sin \theta d\theta) = a^4 \left[\frac{4-1}{4} \cdot \frac{4-3}{4-2} \cdot \frac{\pi}{2} \right]$$

$$= a^4 \cdot \frac{3}{4} \cdot \frac{1}{2} \cdot \frac{\pi}{2}$$

$$= \frac{3\pi a^4}{16}$$

$$x = 0 \Rightarrow a \cos \theta = 0 \quad \theta = \frac{\pi}{2}$$

$$x = a \Rightarrow a \cos \theta = a \quad \theta = 0$$

20. If $\int_0^x f(t) dt = x + \int_x^1 t f(t) dt$, then the value of $f(1)$ is

PTA-3

(1) $\frac{1}{2}$

(2) 2

(3) 1

(4) $\frac{3}{4}$

$$\frac{d}{dx} \left[\int_0^x f(t) dt \right] = \frac{d}{dx} \left[x + \int_x^1 t f(t) dt \right]$$

$$f(x) = 1 + 0 - x f(x)$$

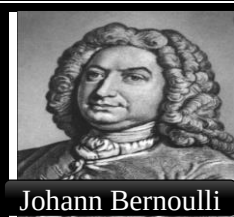
$$f(x) = 1 - x f(x)$$

$$f(x) + x f(x) = 1$$

$$f(x) = \frac{1}{1+x}$$

$$\Rightarrow f(1) = \frac{1}{1+1}$$

$$f(1) = \frac{1}{2}$$



Chapter - 10

Ordinary Differential Equations

EXERCISE 10.1



Concept Corner

Differential Equation, Order and degree

(1) Differential equation:

Any equation contains at least one derivative of an unknown function, either ordinary or partial derivative.

(2) Order of a Differential Equation

Highest order derivative present in the differential equation is called **Order of differential equation**.

(3) Degree of a Differential Equation

The power of the highest order derivative involved in the differential equation (when it is expressing in polynomial form) is called **Degree of a differential equation**.

- If a differential equation is not expressible to polynomial equation form having the highest order derivative as the leading term then that the degree of the differential equation is not defined.
- Degree of the differential equation is always a positive integer.

1. For each of the following differential equations, determine its order, degree (if exists).

2M

(i) $\frac{dy}{dx} + xy = \cot x$

Highest order derivative is $\frac{dy}{dx}$

Order = 1 ;

Degree = 1

This is a polynomial degree of $\frac{dy}{dx}$ is 1.

(ii) $\left(\frac{d^3y}{dx^3}\right)^{\frac{2}{3}} - 3\frac{d^2y}{dx^2} + 5\frac{dy}{dx} + 4 = 0$

$$\left(\frac{d^3y}{dx^3}\right)^{\frac{2}{3}} = 3\frac{d^2y}{dx^2} - 5\frac{dy}{dx} - 4$$

Cubing on both sides

$$\left(\frac{d^3y}{dx^3}\right)^2 = \left(3\frac{d^2y}{dx^2} - 5\frac{dy}{dx} - 4\right)^2 \dots\dots (1)$$

Highest order derivative term $\left(\frac{d^3y}{dx^3}\right)^2$

Order = 3 ;

Degree = 2

(iii) $\left(\frac{d^2y}{dx^2}\right)^2 + \left(\frac{dy}{dx}\right)^2 = x \sin\left(\frac{d^2y}{dx^2}\right)$

Highest order derivative is $\frac{d^2y}{dx^2}$

order = 2

This cannot be expressed as polynomial with $\frac{d^2y}{dx^2}$ as the leading term.

∴ Degree does not exist

(iv) $\sqrt{\frac{dy}{dx}} - 4\frac{dy}{dx} - 7x = 0$

PTA-1

$$\sqrt{\frac{dy}{dx}} = 4\frac{dy}{dx} + 7x$$

Squaring on both sides

$$\frac{dy}{dx} = \left(4\frac{dy}{dx} + 7x\right)^2$$

$$\frac{dy}{dx} = 16\left(\frac{dy}{dx}\right)^2 + 56x\frac{dy}{dx} + 49x^2$$

Highest order derivative is $\frac{dy}{dx}$

Order = 1 ; Degree = 2

$$(v) \quad y \left(\frac{dy}{dx} \right) = \frac{x}{\left(\frac{dy}{dx} \right) + \left(\frac{dy}{dx} \right)^3}$$

PTA-4

Cross multiplying, $y \frac{dy}{dx} \left(\frac{dy}{dx} + \left(\frac{dy}{dx} \right)^3 \right) = x$

$$y \left(\frac{dy}{dx} \right)^2 + y \left(\frac{dy}{dx} \right)^4 = x$$

Highest order derivative is $\frac{dy}{dx}$.

Order = 1; Degree = 4

$$(vi) \quad x^2 \frac{d^2y}{dx^2} + \left(1 + \left(\frac{dy}{dx} \right)^2 \right)^{\frac{1}{2}} = 0$$

$$x^2 \frac{d^2y}{dx^2} = - \left(1 + \left(\frac{dy}{dx} \right)^2 \right)^{\frac{1}{2}}$$

Squaring on both sides, $\left(x^2 \frac{d^2y}{dx^2} \right)^2 = 1 + \left(\frac{dy}{dx} \right)^2$

$$x^4 \left(\frac{d^2y}{dx^2} \right)^2 = 1 + \left(\frac{dy}{dx} \right)^2$$

Highest order derivative $\frac{d^2y}{dx^2}$.

Order = 2; Degree = 2

$$(vii) \quad \left(\frac{d^2y}{dx^2} \right)^3 = \sqrt{1 + \left(\frac{dy}{dx} \right)}$$

Squaring on both sides, $\left(\frac{d^2y}{dx^2} \right)^6 = 1 + \frac{dy}{dx}$

Highest order derivative is $\frac{d^2y}{dx^2}$

Order = 2 ; Degree = 6

$$(viii) \quad \frac{d^2y}{dx^2} = xy + \cos \left(\frac{dy}{dx} \right)$$

Highest order derivative is $\frac{d^2y}{dx^2}$

$\therefore$ order = 2

This cannot be expressed as a polynomial with $\frac{d^2y}{dx^2}$ as its leading term

$\therefore$ Degree is not defined.

Order = 2 ; Degree does not exist

$$(ix) \quad \frac{d^2y}{dx^2} + 5 \frac{dy}{dx} + \int y \, dx = x^3$$

Differentiate with respect to x

$$\frac{d^3y}{dx^3} + 5 \frac{d^2y}{dx^2} + y = 3x^2$$

Order = 3 ; Degree = 1

$$(x) \quad x = e^{xy \left(\frac{dy}{dx} \right)}$$

Highest order derivative is $\frac{dy}{dx}$

Taking log on both sides,

$$\log x = xy \left(\frac{dy}{dx} \right)$$

Order = 1 ; Degree = 1

EXERCISE 10.2

Concept Corner

Classification of differential Equations

a) Ordinary Differential Equation (ODE)

Contains only ordinary derivatives of one or more functions with respect to a single independent variable.

b) Partial Differential Equation (PDE)

Involving only partial derivatives of one or more functions of 2 or more independent variable.

c) Form of a general linear ODE

$$a_n(x)y^{(n)} + a_{n-1}(x)y^{(n-1)} + \dots + a_1(x)y' + a_0y = g(x) \text{ with } a_n(x) \neq 0,$$

$a_0(x), a_1(x), \dots, a_{n-1}(x)$ and $g(x)$ are any function of x .

d) If $g(x) = 0$ in the above form then it is said to be **homogeneous**.

Otherwise, ($g(x) \neq 0$) it is called **non-homogeneous**.



Video class

1. Express each of the following physical statements in the form of differential equation.

2M

- (i) Radium decays at a rate proportional to the amount Q present.

Given that Q be the amount of Radium at any time t .

$$\frac{dQ}{dt} \propto Q$$

$$\frac{dQ}{dt} = kQ \quad \text{Where } k \text{ is a constant.}$$

- (iv) A saving amount pays 8% interest per year, compounded continuously. In addition, the income from another investment is credited to the amount continuously at the rate of ₹ 400 per year.

Let x be the amount at any time t .

Rate of change $\propto$ amount at present

$$\frac{dx}{dt} \propto x \quad k = 8\% = \frac{8}{100}$$

$$\frac{dx}{dt} = \frac{8}{100} x \quad k = 0.08$$

The income is credited to the amount continuously at the rate of ₹ 400 per year.

$$\therefore \frac{dx}{dt} = \frac{2}{25} x + 400$$

- (ii) The population P of a city increases at a rate proportional to the product of population and to the difference between 5,00,000 and the population.

P be the population at time t

Rate of population $\propto P \times (500000 - P)$

$$\frac{dP}{dt} \propto P(500000 - P)$$

$$\frac{dP}{dt} = kP(500000 - P)$$

- (iii) For a certain substance, the rate of change of vapor pressure P with respect to temperature T is proportional to the vapor pressure and inversely proportional to the square of the Temperature.

Given pressure = P temperature = T

Rate of change of P with respect to $T \propto \frac{P}{T^2}$

$$\frac{dP}{dT} \propto \frac{P}{T^2}$$

$$\frac{dP}{dT} = \frac{kP}{T^2}$$

2. Assume that a spherical rain drop evaporates at a rate proportional to its surface area. From a differential equation involving the rate of change of the radius of the rain drop.

2M

Let r be the radius of the spherical rain drop.

Let V be the volume and A be the surface area of it.

PTA-2, JUL-23

Given evaporate rate $\propto$ surface area

$$\frac{dV}{dt} \propto A$$

$$\frac{dV}{dt} = -kA \quad (\because -\text{sign for decreasing})$$

$$\frac{d}{dt} \left(\frac{4}{3} \pi r^3 \right) = -k(4\pi r^2)$$

$$\frac{4\pi}{3} \frac{d}{dt} (r^3) = -4k\pi r^2$$

$$\frac{1}{3} 3r^2 \frac{dr}{dt} = -kr^2$$

$$\frac{dr}{dt} = \frac{-kr^2}{r^2}$$

$$\frac{dr}{dt} = -k$$

$$\begin{aligned} \text{Volume of sphere} &= \frac{4}{3} \pi r^3 \\ \text{SA} &= 4\pi r^2 \end{aligned}$$

EXERCISE 10.3



Concept Corner

Formation of Differential Equations from Geometrical Problems.

To eliminate n arbitrary constants differentiate the equation n times successively, We get the differential equation of order n .

1. Find the differential equation of the family of
 (i) all non-vertical lines in a plane
 (ii) all non-horizontal lines in a plane.

(i) The equation of all non-vertical family of line in a plane is $y = mx + c$, $m \neq 0$ **2M**

Here we have two arbitrary constants m & c

$$y' = \frac{dy}{dx} = m$$

$$y'' = \frac{d^2y}{dx^2} = 0$$

$$\frac{d^2y}{dx^2} = 0 \text{ This is the required D.E}$$

- (ii) The equation of all non-horizontal family of lines in a place is $y = mx + c$, $m \neq 0$

Differentiate with respect to y ,

$$\frac{dy}{dy} = m \left(\frac{dx}{dy} \right) + 0$$

$$1 = m \left(\frac{dx}{dy} \right) + 0$$

$$\frac{1}{m} = \frac{dx}{dy}$$

$$\frac{d^2x}{dy^2} = 0$$

This is the required D.E

2. Form the differential equation of all straight lines touching the circle $x^2 + y^2 = r^2$.

General equation of tangent to the circle $x^2 + y^2 = r^2$ is **2M**

$$y = mx \pm r \sqrt{1 + m^2} \dots\dots\dots (1)$$

m is a only arbitrary constant (r is given radius)

$\therefore$ Differentiate one time with respect to x

$$\frac{dy}{dx} = m \dots\dots\dots (2)$$

$$\text{Sub in (1)} \Rightarrow y = \left(\frac{dy}{dx} \right) x \pm r \sqrt{1 + \left(\frac{dy}{dx} \right)^2}$$

$$\left(y - \frac{dy}{dx} x \right) = \pm r \sqrt{1 + \left(\frac{dy}{dx} \right)^2}$$

Squaring on both sides,

$$\left(y - x \frac{dy}{dx} \right)^2 = r^2 \left(1 + \left(\frac{dy}{dx} \right)^2 \right)$$

3. Find the differential equation of the family of circles passing through the origin and having their centres on the x -axis. **3M**

Equation of circles passing through origin and centre at $(h, 0)$ is

$$\text{Equation of circle centre } (h, k) \text{ radius } r \text{ is } (x - h)^2 + (y - k)^2 = r^2$$

$$(x - h)^2 + (y - k)^2 = a^2$$

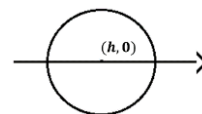
$$(x - h)^2 + y^2 = a^2 \dots\dots\dots (1)$$

a is a only arbitrary constant

Differentiating (1) with respect to x ,

$$2(x - h) + 2y \frac{dy}{dx} = 0 \text{ [Divide by 2]}$$

$$(x - h) + y \frac{dy}{dx} = 0$$



$$x - h = -y \frac{dy}{dx}$$

$$h = x + y \frac{dy}{dx}$$

$$(1) \Rightarrow \left(-y \frac{dy}{dx} \right)^2 + y^2 = \left(x + y \frac{dy}{dx} \right)^2$$

$$y^2 \left(\frac{dy}{dx} \right)^2 + y^2 = x^2 + y^2 \left(\frac{dy}{dx} \right)^2 + 2xy \frac{dy}{dx}$$

$$y^2 = x^2 + 2xy \frac{dy}{dx}$$

$$x^2 + 2xy \frac{dy}{dx} - y^2 = 0$$

This is the required differential equation

$$x^2 + 2xy \frac{dy}{dx} - y^2 = 0.$$

4. Find the differential equation of the family of all the parabolas with latus rectum $4a$ and whose axes are parallel to the x -axis.

Equation of parabola with centre (h, k) axes parallel to x axis is

$$(y - k)^2 = 4a(x - h) \dots\dots\dots (1)$$

Here a is given constant, h, k are the arbitrary constants.

Differentiating with respect to x

$$2(y - k)y' = 4a(1) \text{ [Divide by 2]}$$

$$(y - k)y' = 2a \dots\dots\dots (2)$$

Again differentiating with respect to x

$$(y - k)y'' + y' y' = 0$$

$$(y - k)y'' + (y')^2 = 0$$

$$(y - k)y'' = -(y')^2$$

$$(y - k)y' y'' = -(y')^3 \text{ [Multiply by } y']$$

$$(y - k)y' = \frac{-(y')^3}{y''}$$

$$\text{Sub this in (2)} \Rightarrow \frac{-(y')^3}{y''} = 2a$$

$$-y'^3 = 2ay''$$

$$2ay'' + (y')^3 = 0$$

This is the required differential equation.

$$2ay'' + (y')^3 = 0$$

7. Find the differential equation corresponding to the family of curves represented by the equation

$y = Ae^{8x} + Be^{-8x}$, where A and B are arbitrary constants.

$$\text{Given } y = Ae^{8x} + Be^{-8x}$$

Differentiate y with respect to x .

$$\frac{dy}{dx} = 8Ae^{8x} + Be^{-8x}(-8)$$

Again, differentiate with respect to x

$$\frac{d^2y}{dx^2} = 8Ae^{8x}(8) + Be^{-8x}(-8)(-8)$$

$$= 64Ae^{8x} + 64Be^{-8x}$$

$$= 64(Ae^{8x} + Be^{-8x})$$

$$\frac{d^2y}{dx^2} = 64y$$

5. Find the differential equation of the family of parabolas with vertex at $(0, -1)$ and having axis along the y -axis.

Equation of parabola with vertex (h, k) and axis parallel to y axis is

$$(x - h)^2 = 4a(y - k)$$

$$(x - 0)^2 = 4a(y + 1)$$

$$x^2 = 4a(y + 1) \dots\dots\dots (1)$$

a is the only arbitrary constant.

Differentiating with respect to x ,

$$2x = 4a(1) \frac{dy}{dx} = 4ay'$$

$$\frac{2x}{y'} = 4a$$

$$(1) \Rightarrow x^2 = \frac{2x}{y'} (y + 1)$$

$$\Rightarrow x^2 \times \frac{y'}{x} = 2(y + 1)$$

$$xy' = 2y + 2$$

$$xy' - 2y - 2 = 0.$$

This is the required differential equation.

6. Find the differential equations of the family of all the ellipses having foci on the y -axis and centre at the origin.

Equation of ellipse having foci on y axis and centre at the origin is $\frac{x^2}{b^2} + \frac{y^2}{a^2} = 1 \dots\dots\dots (1)$

a, b are arbitrary constants.

Differentiating with respect to x .

$$\frac{2x}{b^2} + \frac{2yy'}{a^2} = 0 \text{ [Divide by 2]}$$

$$\frac{x}{b^2} + \frac{yy'}{a^2} = 0 \dots\dots\dots (2)$$

Again differentiating with respect to x

$$\frac{1}{b^2} + \frac{yy'' + y'y'}{a^2} = 0 \dots\dots\dots (3)$$

From (1), (2), (3) compare the co-efficients and eliminate $\frac{1}{a^2}, \frac{1}{b^2}$

$$\begin{vmatrix} x^2 & y^2 & 1 \\ x & yy' & 0 \\ 1 & yy'' + y'^2 & 0 \end{vmatrix} = 0$$

$$x^2(0 - 0) - y^2(0 - 0)$$

$$+ 1(xy'' + y'^2 - yy') = 0$$

$$xyy'' + xy'^2 - yy' = 0.$$

This is the required differential equation.

8. Find the differential equation of the curve represented by $xy = ae^x + be^{-x} + x^2$.

PTA-2

3M

$$xy = ae^x + be^{-x} + x^2$$

$$xy - x^2 = ae^x + be^{-x} \dots\dots\dots (1)$$

a, b are arbitrary constants to eliminate a, b

Differentiate with respect to x

$$xy' + yx' - 2x = ae^x + be^{-x} (-1)$$

$$xy' + y - 2x = ae^x - be^{-x}$$

Again, differentiate with respect to x

$$xy'' + y'(1) + y' - 2 = ae^x - be^{-x} (-1)$$

$$xy'' + 2y' - 2 = ae^x + be^{-x}$$

Using (1) we have

$$xy'' + 2y' - 2 = xy - x^2$$

$$xy'' + 2y' + x^2 - xy - 2 = 0$$

EXERCISE 10.4



Video class

Concept Corner

Solution of ODE

(1) solution of DE

A **solution** of a differential equation is an expression for the dependent variable in terms of the independent variable which satisfies the differential equation.

(2) General solution

The solution which contains as many arbitrary constants as the order of the DE is called the general solution.

(3) Particular solution

If we give arbitrary constants takes some particular values in the general solution of D.E. The resulting solution is called **particular solution**

1. Show that each of the following expressions is a solution of the corresponding given differential equation.

(i) $y = 2x^2$; $xy' = 2y$

2M

$$y = 2x^2$$

Differentiate with respect to x ,

$$y' = 2x \quad (2)$$

Multiply by x , $xy' = 2x^2(2)$

$$xy' = 2y$$

$\therefore$ Given function is a solution of the given Differential equation.

(ii) $y = ae^x + be^{-x}$; $y'' - y = 0$

2M

$$y = ae^x + be^{-x}$$

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Differentiate with respect to x two times,

$$y' = ae^x + be^{-x}(-1)$$

$$y'' = ae^x + be^{-x}(-1)(-1)$$

$$y'' = ae^x + be^{-x}$$

$$y'' = y$$

$$y'' - y = 0$$

$\therefore$ The given function is a solution of the Differential equation is given.

2. Find value of m so that the function $y = e^{mx}$ is a solution of the given differential equation. 3M

(i) $y' + 2y = 0$

$$y' + 2y = 0 \quad \dots\dots\dots (1)$$

$$y = e^{mx}$$

Differentiate with respect to x , $y' = e^{mx} (m)$

$$y' = my$$

$$y' - my = 0 \quad \dots\dots (2)$$

Compare (1) & (2) $-m = 2$

$$m = -2$$

(ii) $y'' - 5y' + 6y = 0$

$$(2) \Rightarrow y' - my = 0$$

Again, differentiate with respect to x

$$y'' - my' = 0$$

$$y'' = my'$$

$$y'' = m(my) \quad [\because y' = my]$$

$$y'' = m^2y$$

$$(3) \Rightarrow y'' - 5y' + 6y = 0$$

$$m^2y - 5my + 6y = 0$$

$$y(m^2 - 5m + 6) = 0$$

$$y = 0 \text{ (or) } m^2 - 5m + 6 = 0$$

$$(m - 2)(m - 3) = 0$$

$$m - 2 = 0; m - 3 = 0$$

$$m = 2; m = 3$$

$$m = 2, 3$$

4. Show that $y = e^{-x} + mx + n$ is a solution of the differential equation $e^x \left(\frac{d^2y}{dx^2} \right) - 1 = 0$

$$y = e^{-x} + mx + n$$

Two constants m, n 3M PTA-3

Differentiate with respect to x

$$\frac{dy}{dx} = e^{-x} (-1) + m$$

Again, differentiate with respect to x

$$\frac{d^2y}{dx^2} = e^{-x} (-1)(-1) + 0$$

$$\frac{d^2y}{dx^2} = e^{-x}$$

$$\frac{d^2y}{dx^2} = \frac{1}{e^x}$$

$$e^x \frac{d^2y}{dx^2} = 1$$

$$e^x \frac{d^2y}{dx^2} - 1 = 0$$

$\therefore$ Given function is a solution of the given D.E.

3. The slope of the tangent to the curve at any point is the reciprocal of four times the ordinate at that point. The curve passes through $(2, 5)$. Find the equation of the curve. Let the any point on the curve be $P(x, y)$

$$\text{slope} = \frac{1}{4(\text{ordinate})} \quad \text{3M}$$

$$\frac{dy}{dx} = \frac{1}{4y}$$

$$4ydy = dx$$

Integrating on both sides

$$4 \int y dy = \int dx$$

$$4 \frac{y^2}{2} = x + c$$

$$2y^2 = x + c \quad \dots\dots\dots (1)$$

This curve passing through $(2, 5)$

$$(1) \Rightarrow 2(5)^2 = 2 + c$$

$$50 = 2 + c$$

$$50 - 2 = c$$

$$c = 48$$

$$(1) \Rightarrow 2y^2 = x + 48$$

This is the required equation of the curve.

5. Show that $y = ax + \frac{b}{x}$, $x \neq 0$ is a solution of the differential equation $x^2y'' + xy' - y = 0$. 2M MDL

$$y = ax + \frac{b}{x} \quad \dots\dots\dots (1)$$

Differentiate with respect to x

$$y' = a(1) + b \left(\frac{-1}{x^2} \right) \quad \dots\dots\dots (2)$$

Again, differentiate with respect to x

$$y'' = 0 + \frac{2b}{x^3}$$

$$\frac{2b}{x^3} = y'' \Rightarrow 2b = y''x^3$$

$$b = \frac{x^3y''}{2}$$

$$(2) \Rightarrow y' = a - \frac{b}{x^2}$$

$$y' = a - \frac{x^3y''}{2x^2} = a - \frac{xy''}{2}$$

$$y' + \frac{xy''}{2} = a$$

$$(1) \Rightarrow y = \left(y' + \frac{xy''}{2} \right) x + \frac{x^3y''}{2} \left(\frac{1}{x} \right)$$

$$y = xy' + \frac{x^2y''}{2} + \frac{x^2y''}{2} = xy' + \frac{2x^2y''}{2}$$

$$x^2y'' + xy' - y = 0$$

$\therefore$ The given function is a solution of the given D.E.

6. Show that $y = ae^{-3x} + b$, where a and b are arbitrary constants, is a solution of the differential equation $\frac{d^2y}{dx^2} + 3\frac{dy}{dx} = 0$.

$$y = ae^{-3x} + b$$

5M

Differentiate with respect to x

$$\frac{dy}{dx} = a(-3)e^{-3x} + 0$$

$$\frac{dy}{dx} = -3ae^{-3x} \dots\dots\dots (1)$$

Again, differentiate with respect to x

$$\frac{d^2y}{dx^2} = -3ae^{-3x}(-3)$$

$$\frac{d^2y}{dx^2} = -3\frac{dy}{dx} \quad [\because \text{using (1)}]$$

$$\frac{d^2y}{dx^2} + 3\frac{dy}{dx} = 0$$

$\therefore$ Given function is a solution of given D.E.

8. Show that $y = a \cos bx$ is a solution of the differential equation $\frac{d^2y}{dx^2} + b^2y = 0$ 3M

$$y = a \cos bx$$

Differentiate with respect to x

$$\frac{dy}{dx} = a(-\sin bx) b$$

Again differentiate with respect to x

$$\frac{d^2y}{dx^2} = -ab \cos bx \quad b$$

$$\frac{d^2y}{dx^2} = -b^2 (a \cos bx)$$

$$\frac{d^2y}{dx^2} = -b^2 y$$

$$\frac{d^2y}{dx^2} + b^2y = 0$$

$\therefore$ The given function is a solution of the given D.E.

7. Show that the differential equation representing the family of curves $y^2 = 2a \left(x + a^{\frac{2}{3}}\right)$, where a is a positive parameter, is $\left(y^2 - 2xy \frac{dy}{dx}\right)^3 = 8 \left(y \frac{dy}{dx}\right)^5$.

Given:

5M

$$y^2 = 2a \left(x + a^{\frac{2}{3}}\right) \dots\dots\dots (1)$$

Differentiate with respect to x ,

$$2y \frac{dy}{dx} = 2a(1 + 0)$$

$$\text{Divided by 2,} \quad y \frac{dy}{dx} = a$$

$$(1) \Rightarrow y^2 = 2 \left(y \frac{dy}{dx}\right) \left(x + \left(y \frac{dy}{dx}\right)^{\frac{2}{3}}\right)$$

$$y^2 = 2xy \frac{dy}{dx} + 2 \left(y \frac{dy}{dx}\right) \left(y \frac{dy}{dx}\right)^{\frac{2}{3}}$$

$$y^2 - 2xy \frac{dy}{dx} = 2 \left(y \frac{dy}{dx}\right)^{1+\frac{2}{3}}$$

$$y^2 - 2xy \frac{dy}{dx} = 2 \left(y \frac{dy}{dx}\right)^{\frac{5}{3}}$$

Cubing on both sides,

$$\left(y^2 - 2xy \frac{dy}{dx}\right)^3 = 8 \left(y \frac{dy}{dx}\right)^5$$

$\therefore$ The given family of curve is the solution of the given D.E.

EXERCISE 10.5



Video class

Concept Corner

Solution of first order and first degree D.E

Variables separable method.

A differential equation in first order can be written as $h(y) y' = g(x)$.

Where the left side function of y and right side function of x .

Rewriting this type is called method of variable separable.

An equation of the form $f_1(x)g_1(y)dx + f_2(x)g_2(y)dy = 0$ is called an equation with **variable separable** or simply a **separable equations**.

General Solution of D.E :

$$\int \frac{f_1(x)}{f_2(x)} dx = - \int \frac{g_2(y)}{g_1(y)} dy + C, \text{ where } C \text{ is an arbitrary constant.}$$

1. If F is the constant force generated by the motor of an automobile of mass M , its velocity V is given by $M \frac{dv}{dt} = F - kV$, where k is a constant. Express V in terms of t given that $V = 0$ when $t = 0$. 3M

$$M \frac{dv}{dt} = F - kV$$

$$\frac{dv}{F - kV} = \frac{dt}{M}$$

Integrating on both sides,

$$\int \frac{dv}{F - kV} = \int \frac{1}{M} dt$$

Multiply both sides by $-k$,

$$\int \frac{-k dv}{F - kV} = \frac{-k}{M} \int dt$$

$$\log(F - kV) = \frac{-k}{M} t + c \dots\dots\dots (1)$$

$$\left[\because \int \frac{f'(x)}{f(x)} dx = \log f(x) + c \right]$$

$$\text{When } t = 0 \Rightarrow V = 0$$

$$(1) \Rightarrow \log F = 0 + c$$

$$c = \log F$$

Substitute this in (1)

$$\log(F - kV) = \frac{-k}{M} t + \log F$$

$$\log(F - kV) - \log F = \frac{-kt}{M}$$

$$\log\left(\frac{F - kV}{F}\right) = \frac{-kt}{M}$$

$$\left[\because \log a - \log b = \log \frac{a}{b} \right]$$

$$\text{Taking anti log } \frac{F - kV}{F} = e^{\frac{-kt}{M}}$$

$$\frac{F - kV}{F} = \frac{1}{e^{\frac{kt}{M}}}$$

$$(F - kV) e^{\frac{kt}{M}} = F$$

2. The velocity v , of a parachute falling vertically satisfies the equation $v \frac{dv}{dx} = g \left(1 - \frac{v^2}{k^2}\right)$, where g and k are constants. If v and x are both initially zero, find v in terms of x . 5M

$$v \frac{dv}{dx} = g \left(1 - \frac{v^2}{k^2}\right)$$

$$\frac{v dv}{\left(1 - \frac{v^2}{k^2}\right)} = g dx$$

$$\frac{\frac{-2v}{k^2} dv}{\left(1 - \frac{v^2}{k^2}\right)} = \frac{-2g}{k^2} dx \quad [\times \text{ by } \frac{-2}{k^2}]$$

Integrating on both sides

$$\int \frac{\frac{-2v}{k^2} dv}{\left(1 - \frac{v^2}{k^2}\right)} = \frac{-2g}{k^2} \int dx$$

$$\log\left(1 - \frac{v^2}{k^2}\right) = \frac{-2gx}{k^2} + c \dots\dots\dots (1)$$

$$\therefore \left[\int \frac{f'(x)}{f(x)} dx = \log f(x) + c \right]$$

$$\text{Given } v = 0, x = 0$$

$$(1) \Rightarrow \log(1 - 0) = 0 + c$$

$$\log 1 = c \quad [\log 1 = 0]$$

$$c = 0$$

$$(1) \Rightarrow \log\left(1 - \frac{v^2}{k^2}\right) = \frac{-2gx}{k^2} + 0$$

Taking antilog

$$1 - \frac{v^2}{k^2} = e^{\frac{-2gx}{k^2}}$$

$$1 - e^{\frac{-2gx}{k^2}} = \frac{v^2}{k^2}$$

$$k^2 \left(1 - e^{\frac{-2gx}{k^2}}\right) = v^2$$

$$v^2 = k^2 \left(1 - e^{\frac{-2gx}{k^2}}\right)$$

3. Find the equation of the curve whose slope is $\frac{y-1}{x^2+x}$ and which passes through the point $(1, 0)$ 5M

$$\text{Slope} = \frac{y-1}{x^2+x}$$

$$\frac{dy}{dx} = \frac{y-1}{x^2+x}$$

$$\frac{dy}{y-1} = \frac{1}{x^2+x} dx = \left[\frac{x+1-x}{x(x+1)} \right] = \left(\frac{1}{x} - \frac{1}{x+1} \right) dx$$

$$\text{Integrating on both sides } \int \frac{dy}{y-1} = \int \frac{dx}{x} - \int \frac{dx}{x+1}$$

$$\log(y-1) = \log x - \log(x+1) + \log c = \log \frac{xc}{x+1}$$

$$\text{Taking anti log, } y-1 = \frac{cx}{x+1} \dots\dots\dots (2)$$

$$\text{This curve passing through } (1, 0) \quad (2) \Rightarrow 0 - 1 = \frac{c(1)}{1+1}$$

$$-1 = \frac{c}{2} \Rightarrow -2 = c$$

$$(2) \Rightarrow y-1 = \frac{-2x}{x+1} \Rightarrow y = 1 - \frac{2x}{x+1} = \frac{x+1-2x}{x+1} = \frac{1-x}{1+x}$$

4. Solve the following differential equations:

(i) $\frac{dy}{dx} = \sqrt{\frac{1-y^2}{1-x^2}}$ 2M PTA-6, JUL-21,
APR-22, 25

$$\frac{dy}{\sqrt{1-y^2}} = \frac{dx}{\sqrt{1-x^2}} \quad \because \left[\int \frac{dx}{\sqrt{1-x^2}} = \sin^{-1} x + C \right]$$

Integrating on both sides

$$\int \frac{dy}{\sqrt{1-y^2}} = \int \frac{dx}{\sqrt{1-x^2}}$$

$$\sin^{-1} y = \sin^{-1} x + C$$

(ii) $y dx + (1+x^2) \tan^{-1} x dy = 0$ 3M

$$y dx = -(1+x^2) \tan^{-1} x dy$$

$$\frac{dx}{(1+x^2) \tan^{-1} x} = \frac{-dy}{y} \dots\dots\dots (1)$$

Put $\tan^{-1} x = v$ Differentiate with respect to x

$$\frac{1}{1+x^2} = \frac{dv}{dx}$$

$$\frac{dx}{1+x^2} = dv$$

$$(1) \Rightarrow \frac{dv}{v} = \frac{-dy}{y}$$

Integrating on both sides $\int \frac{dv}{v} = - \int \frac{dy}{y}$

$$\log v + \log y = \log C$$

$$\log vy = \log C$$

Taking anti log, $vy = C$ Replace the v value, $y \tan^{-1} x = C$

(iii) $\sin \frac{dy}{dx} = a$ $y(0) = 1$ 3M

$$\sin \frac{dy}{dx} = a$$

$$\frac{dy}{dx} = \sin^{-1}(a)$$

$$dy = (\sin^{-1} a) dx$$

Integrating on both sides $\int dy = \sin^{-1}(a) \int dx$

$$y = x \sin^{-1}(a) + C \dots\dots (1)$$

When $x = 0$, $y = 1$ (1) $\Rightarrow 1 = \sin^{-1} a(0) + C$

$$1 = C$$

$$(1) \Rightarrow y = (\sin^{-1} a)x + 1$$

$$y - 1 = (\sin^{-1} a)x$$

$$\frac{y-1}{x} = \sin^{-1} a$$

$$\Rightarrow \sin \left(\frac{y-1}{x} \right) = a$$

(iv) $\frac{dy}{dx} = e^{x+y} + x^3 e^y$ JUL-20 3M

$$\frac{dy}{dx} = e^x e^y + x^3 e^y = e^y (e^x + x^3)$$

$$e^{-y} dy = (e^x + x^3) dx$$

Integrating on both sides

$$\int e^{-y} dy = \int e^x dx + \int x^3 dx$$

$$\frac{e^{-y}}{-1} = e^x + \frac{x^4}{4} + C_1$$

$$-e^{-y} = e^x + \frac{x^4}{4} + C_1$$

$$-C_1 = e^x + e^{-y} + \frac{x^4}{4}$$

$$e^x + e^{-y} + \frac{x^4}{4} = C \text{ [where } C = -C_1]$$

(v) $(e^y + 1) \cos x dx + e^y \sin x dy = 0$

$$(e^y + 1) \cos x dx = -e^y \sin x dy$$
 5M

$$\frac{\cos x}{\sin x} dx = \frac{-e^y}{e^y+1} dy$$

JUL-21

Integrating on both sides

$$\int \frac{\cos x}{\sin x} dx = - \int \frac{e^y}{e^y+1} dy$$

$$\log \sin x = -\log(e^y + 1) + \log C$$

$$\log \sin x + \log(e^y + 1) = \log C$$

$$\log \sin x (e^y + 1) = \log C$$

Taking antilog

$$(e^y + 1) \sin x = C$$

(vi) $(y dx - x dy) \cot \left(\frac{x}{y} \right) = ny^2 dx$

Divide by y^2

$$\frac{y dx - x dy}{y^2} \cot \frac{x}{y} = n dx$$

$$\cot \frac{x}{y} d \left(\frac{x}{y} \right) = n dx$$

$$\because \left[d \left(\frac{x}{y} \right) = \frac{y dx - x dy}{y^2} \right]$$

Integrating on both sides

$$\int \cot \frac{x}{y} d \left(\frac{x}{y} \right) = n \int dx$$

$$\because \left[\int \cot x dx = \log \sin x + C \right]$$

$$\log \sin \left(\frac{x}{y} \right) = nx + C$$

Taking antilog

$$\sin \left(\frac{x}{y} \right) = e^{nx+C}$$

(vii) $\frac{dy}{dx} - x\sqrt{25 - x^2} = 0$

3M

$$\frac{dy}{dx} = x\sqrt{25 - x^2}$$

$$dy = x\sqrt{25 - x^2} dx \dots\dots\dots (1)$$

Put $25 - x^2 = t$

Differentiate with respect to x

$$-2x = \frac{dt}{dx}$$

$$x dx = \frac{dt}{-2}$$

$$(1) \Rightarrow dy = \sqrt{t} \left(\frac{dt}{-2} \right)$$

Integrating on both sides

$$2 \int dy = - \int t^{\frac{1}{2}} dt$$

$$2y = -\frac{t^{\frac{3}{2}}}{\frac{3}{2}} + 2C$$

Multiply both sides by $\frac{3}{2}$

$$\frac{3}{2} 2y = -t^{\frac{3}{2}} + \frac{3}{2} 2C$$

$$3y = -t^{\frac{3}{2}} + 3C$$

Replace $t = 25 - x^2$

$$3y = -(25 - x^2)^{\frac{3}{2}} + 3C$$

(viii) $x \cos y \, dy = e^x (x \log x + 1) \, dx$

$$\cos y \, dy = e^x \frac{(x \log x + 1)}{x} \, dx$$

APR-23

$$\cos y \, dy = e^x \left(\log x + \frac{1}{x} \right) \, dx$$

3M

$$\Rightarrow \cos y \, dy = \left(e^x \log x + e^x \frac{1}{x} \right) \, dx$$

$$\cos y \, dy = \log x \, d(e^x) + e^x \, d(\log x)$$

$$\cos y \, dy = d(e^x \log x)$$

$$[\because d(uv) = u \, dv + v \, du]$$

Integrating on both sides

$$\int \cos y \, dy = \int d(e^x \log x)$$

$$\sin y = e^x \log x + C$$

(ix) $\tan y \frac{dy}{dx} = \cos(x + y) + \cos(x - y)$

3M

$$\tan y \frac{dy}{dx} = \cos x \cos y - \sin x \sin y + \cos x \cos y + \sin x \sin y$$

$$\tan y \frac{dy}{dx} = 2 \cos x \cos y$$

$$\frac{\tan y}{\cos y} \, dy = 2 \cos x \, dx$$

$$\sec y \tan y \, dy = 2 \cos x \, dx$$

Integrating on both sides

$$\int \sec y \tan y \, dy = \int 2 \cos x \, dx$$

$$\sec y = 2 \sin x + C$$

(x) $\frac{dy}{dx} = \tan^2(x + y)$

5M

$$\frac{dy}{dx} = \tan^2(x + y) \dots\dots\dots (1)$$

Let $x + y = z$

Differentiate with respect to x ,

$$1 + \frac{dy}{dx} = \frac{dz}{dx}$$

$$\Rightarrow \frac{dy}{dx} = \frac{dz}{dx} - 1$$

$$(1) \Rightarrow \frac{dz}{dx} - 1 = \tan^2 z$$

$$\frac{dz}{dx} = 1 + \tan^2 z = \sec^2 z$$

$$\frac{dz}{\sec^2 z} = dx \Rightarrow \cos^2 z \, dz = dx$$

$$\left(\frac{1 + \cos 2z}{2} \right) dz = dx$$

Integrating on both sides

$$\frac{1}{2} [\int (1 + \cos 2z) \, dz] = \int dx$$

$$\frac{1}{2} \left[z + \frac{1}{2} \sin 2z \right] = x + C$$

$$\frac{1}{2} \left[(x + y) + \frac{1}{2} \sin 2(x + y) \right] = x + C$$

$$\frac{1}{2} \left[(x + y) + \frac{1}{2} 2 \sin(x + y) \cos(x + y) \right] = x + C$$

$$\frac{1}{2} [(x + y) + \sin(x + y) \cos(x + y)] = x + C$$

EXERCISE 10.6



Video class

Concept Corner

Homogeneous Form or Homogeneous Differential Equation

Homogeneous function of degree n

A function $f(x, y)$ is said to be a homogeneous function of degree n in the variables x and y if, $f(tx, ty) = t^n f(x, y)$. This is known as **Euler's homogeneity**.

Homogeneous Differential Equation

An ODE is said to be in homogeneous form, it is written as $\frac{dy}{dx} = g\left(\frac{y}{x}\right)$.

Method to solve

I Put $y = vx$

$$\frac{dy}{dx} = v + x \frac{dv}{dx}$$

Then use variable separable method for (v, x) II Put $x = vy$

$$\frac{dx}{dy} = v + y \frac{dv}{dy}$$

Then use variable separable method for (v, y)

Solve the following differential equations:

1. $\left[x + y \cos\left(\frac{y}{x}\right)\right] dx = x \cos\left(\frac{y}{x}\right) dy$ (5M)

$$\frac{x+y \cos \frac{y}{x}}{x \cos \frac{y}{x}} = \frac{dy}{dx} \dots\dots\dots (1)$$

This is a Homogeneous differential equation.

Put $y = vx$

$$\frac{dy}{dx} = v + x \frac{dv}{dx}$$

$$(1) \Rightarrow \frac{x+vx \cos \frac{vx}{x}}{x \cos \frac{vx}{x}} = v + x \frac{dv}{dx}$$

$$\frac{x(1+v \cos v)}{x \cos v} = v + x \frac{dv}{dx}$$

$$\frac{1+v \cos v}{\cos v} - v = x \frac{dv}{dx}$$

$$\frac{1+v \cos v - v \cos v}{\cos v} = x \frac{dv}{dx}$$

$$\frac{1}{\cos v} = x \frac{dv}{dx}$$

$$\frac{dx}{x} = \cos v \, dv$$

Integrating on both sides

Replace $y = vx$
 $\frac{y}{x} = v$

$$\int \cos v \, dv = \int \frac{dx}{x}$$

$$\sin v = \log|x| + \log|C|$$

$$\sin\left(\frac{y}{x}\right) = \log|Cx|$$

2. $(x^3 + y^3) dy - x^2 y dx = 0$ (5M)

$$(x^3 + y^3) dy = x^2 y dx$$

$$\frac{dx}{dy} = \frac{x^3 + y^3}{x^2 y}$$

$$\frac{dx}{dy} = \frac{x}{y} + \frac{y^2}{x^2} \dots\dots\dots (1)$$

This is a homogeneous D.E

Put $x = vy$

$$\frac{dx}{dy} = v + y \frac{dv}{dy}$$

$$(1) \Rightarrow v + y \frac{dv}{dy} = v + \frac{1}{v^2}$$

$$y \frac{dv}{dy} = \frac{1}{v^2}$$

$$v^2 dv = \frac{dy}{y}$$

Integrating on both sides,

$$\int v^2 \, dv = \int \frac{dy}{y}$$

$$\frac{v^3}{3} = \log y + \log c$$

$$\frac{1}{3} v^3 = \log y + \log c$$

$$\frac{1}{3} \frac{x^3}{y^3} = \log(yc)$$

$$e^{\frac{x^3}{3y^3}} = yc$$

$$y = ce^{\frac{x^3}{3y^3}}$$

$$3. y e^{\frac{x}{y}} dx = \left(x e^{\frac{x}{y}} + y \right) dy$$

5M

$$\frac{dx}{dy} = \frac{x e^{\frac{x}{y}} + y}{y e^{\frac{x}{y}}} \dots\dots\dots(1)$$

This is a Homogeneous D.E

Put $x = vy$

$$\frac{dx}{dy} = v + y \frac{dv}{dy}$$

$$(1) \Rightarrow v + y \frac{dv}{dy} = \frac{vy e^{\frac{vy}{y}} + y}{y e^{\frac{vy}{y}}}$$

$$v + y \frac{dv}{dy} = \frac{y (ve^v + 1)}{y e^v}$$

$$y \frac{dv}{dy} = \frac{ve^v + 1}{e^v} - v = \frac{ve^v + 1 - ve^v}{e^v} = \frac{1}{e^v}$$

$$e^v dv = \frac{dy}{y}$$

Integrating on both sides,

$$\int e^v dv = \int \frac{dy}{y}$$

$$e^v = \log|y| + \log|C| \text{ [Replace } v = \frac{x}{y}]$$

$$e^{\frac{x}{y}} = \log|Cy|$$

$$4. 2xy dx + (x^2 + 2y^2) dy = 0$$

5M

$$2xy dx = -(x^2 + 2y^2) dy$$

$$\frac{dx}{dy} = \frac{-(x^2 + 2y^2)}{2xy} \dots\dots\dots(1)$$

$$\frac{dx}{dy} = -\frac{1}{2} \frac{x}{y} - \frac{y}{x}$$

This is a homogeneous D.E

Put $x = vy$

$$\frac{dx}{dy} = v + y \frac{dv}{dy}$$

$$(1) \Rightarrow v + y \frac{dv}{dy} = -\frac{1}{2} v - \frac{1}{v}$$

$$y \frac{dv}{dy} = -\frac{3}{2} v - \frac{1}{v}$$

$$y \frac{dv}{dy} = \frac{-3v^2 - 2}{2v}$$

$$\frac{2v}{3v^2 + 2} dv = -\frac{dy}{y}$$

Integrating on both sides

$$\frac{1}{3} \int \frac{6v dv}{3v^2 + 2} = - \int \frac{dy}{y}$$

$$\frac{1}{3} \log(3v^2 + 2) = -\log y + \log c$$

$$(3v^2 + 2) = \left(\frac{c}{y}\right)^3$$

$$3 \frac{x^2}{y^2} + 2 = \frac{c}{y^3}$$

$$3x^2y + 2y^3 = C$$

$$5. (y^2 - 2xy) dx = (x^2 - 2xy) dy$$

5M

$$\frac{dy}{dx} = \frac{y^2 - 2xy}{x^2 - 2xy} \dots\dots\dots(1)$$

This is a homogeneous D.E

$$\text{Put } y = vx ; \quad \frac{dy}{dx} = v + x \frac{dv}{dx}$$

$$(1) \Rightarrow v + x \frac{dv}{dx} = \frac{v^2 x^2 - 2xvx}{x^2 - 2xvx}$$

$$v + x \frac{dv}{dx} = \frac{x^2 (v^2 - 2v)}{x^2 (1 - 2v)}$$

$$x \frac{dv}{dx} = \frac{v^2 - 2v}{1 - 2v} - v$$

$$= \frac{v^2 - 2v - v(1 - 2v)}{1 - 2v} = \frac{v^2 - 2v - v + 2v^2}{1 - 2v}$$

$$x \frac{dv}{dx} = \frac{3v^2 - 3v}{1 - 2v} = -3 \frac{v^2 - v}{2v - 1}$$

$$\frac{2v-1}{v^2-v} dv = -3 \frac{dx}{x}$$

Integrating on both sides

$$\int \frac{2v-1}{v^2-v} dv = -3 \int \frac{dx}{x}$$

$$\log|v^2 - v| = -3 \log|x| + \log|c|$$

$$|x^3(v^2 - v)| = |c|$$

$$v^2 - v = \frac{c}{x^3}$$

$$\frac{y^2}{x^2} - \frac{y}{x} = \frac{c}{x^3}$$

$$xy^2 - x^2y = C$$

$$6. x \frac{dy}{dx} = y - x \cos^2 \left(\frac{y}{x} \right)$$

5M

$$x \frac{dy}{dx} = y - x \cos^2 \left(\frac{y}{x} \right) \dots\dots\dots(1)$$

This is a homogeneous D.E.

$$\text{Put } y = vx ; \quad \frac{dy}{dx} = v + x \frac{dv}{dx}$$

$$(1) \Rightarrow x \left(v + x \frac{dv}{dx} \right) = vx - x \cos^2 \frac{vx}{x}$$

$$x^2 \frac{dv}{dx} = -x \cos^2 v$$

$$\frac{dv}{\cos^2 v} = -\frac{dx}{x}$$

Integrating on both sides

$$\int \sec^2 v dv = - \int \frac{dx}{x}$$

$$\tan v = -\log|x| + \log|C| = \log \left| \frac{C}{x} \right|$$

$$\text{Putting } v = \frac{y}{x} ; \tan \frac{y}{x} = \log \left| \frac{C}{x} \right|$$

Taking antilog

$$e^{\tan \left(\frac{y}{x} \right)} = \left| \frac{C}{x} \right|$$

$$x e^{\tan \left(\frac{y}{x} \right)} = \pm C$$

$$\therefore x e^{\tan \left(\frac{y}{x} \right)} = C \quad [\text{where } C = \pm C]$$

EXERCISE 10.9



Choose the correct or the most suitable answer from the given four alternatives.

1. The order & degree of the differential equation $\frac{d^2y}{dx^2} + \left(\frac{dy}{dx}\right)^{\frac{1}{3}} + x^{\frac{1}{4}} = 0$ are respectively.

(1) 2, 3

(2) 3, 3

(3) 2, 6

(4) 2, 4 PTA-6, JUL-24,21

$\frac{d^2y}{dx^2} + x^{\frac{1}{4}} = -\left(\frac{dy}{dx}\right)^{\frac{1}{3}}$ Cubing on both sides $\left(\frac{d^2y}{dx^2} + x^{\frac{1}{4}}\right)^3 = \frac{-dy}{dx}$	Highest order derivative is $\frac{d^2y}{dx^2}$ its degree = 3 order, degree = 2, 3
--	---

2. The differential equation representing the family of curves $y = A \cos(x + B)$, where A and B are parameters, is

JUL-22

(1) $\frac{d^2y}{dx^2} - y = 0$

(2) $\frac{d^2y}{dx^2} + y = 0$

(3) $\frac{d^2y}{dx^2} = 0$

(4) $\frac{d^2x}{dy^2} = 0$

$y = A \cos(x + B)$ Differentiate with respect to x $\frac{dy}{dx} = A(-\sin(x + B))$ again differentiate with respect to x	$\frac{d^2y}{dx^2} = -A \cos(x + B)$ $\frac{d^2y}{dx^2} = -y$ $\frac{d^2y}{dx^2} + y = 0$
---	---

3. The order and degree of the differential equation $\sqrt{\sin x} (dx + dy) = \sqrt{\cos x} (dx - dy)$ is

(1) 1, 2

(2) 2, 2

(3) 1, 1

(4) 2, 1

$\sqrt{\sin x} (dx + dy) = \sqrt{\cos x} (dx - dy)$ $\sqrt{\sin x} dx + \sqrt{\sin x} dy = \sqrt{\cos x} dx - \sqrt{\cos x} dy$ $\sqrt{\sin x} dy + \sqrt{\cos x} dy = \sqrt{\cos x} dx - \sqrt{\sin x} dx$ $dy (\sqrt{\sin x} + \sqrt{\cos x}) = dx (\sqrt{\cos x} - \sqrt{\sin x})$ $\frac{dy}{dx} = \frac{\sqrt{\cos x} - \sqrt{\sin x}}{\sqrt{\sin x} + \sqrt{\cos x}}$ Highest order derivative is $\frac{dy}{dx}$; Order = 1, degree = 1
--

4. The order of the differential equation of all circles with centre at (h, k) and radius ' a ' is

(1) 2

(2) 3

(3) 4

(4) 1 JUL-22, APR-20

Equation of circle with centre (h, k) radius a is $(x - h)^2 + (y - k)^2 = a^2 \dots\dots\dots (1)$ Differentiate with respect to x $2(x - h) + 2(y - k)y' = 0 \quad [\div 2]$ $(x - h) + (y - k)y' = 0 \dots\dots\dots (2)$	Again differentiate with respect to x $0 + (y - k)y'' + y'(y') = 0$ $(y - k)y'' = -y'^2$ $y - k = \frac{-y'^2}{y''}$ To eliminate k we have again differentiate w.r.to x $\therefore \text{Order} = 3$
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5. The differential equation of the family of curves $y = Ae^x + Be^{-x}$, where A and B are arbitrary constants is APR-24, PTA-4

(1) $\frac{d^2y}{dx^2} + y = 0$ (2) $\frac{d^2y}{dx^2} - y = 0$ (3) $\frac{dy}{dx} + y = 0$ (4) $\frac{dy}{dx} - y = 0$

$$y = Ae^x + Be^{-x}$$

Differentiate with respect to x two times

$$\frac{dy}{dx} = Ae^x + Be^{-x}(-1)$$

$$\frac{d^2y}{dx^2} = Ae^x + Be^{-x}(-1)(-1) = Ae^x + Be^{-x} = y \Rightarrow \frac{d^2y}{dx^2} - y = 0$$

6. The general solution of the differential equation $\frac{dy}{dx} = \frac{y}{x}$ is APR-23

(1) $xy = k$ (2) $y = k \log x$ (3) $y = kx$ (4) $\log y = kx$

$$\frac{dy}{dx} = \frac{y}{x} \Rightarrow \frac{dy}{y} = \frac{dx}{x}$$

Integrating on both sides

$$\int \frac{dy}{y} = \int \frac{dx}{x}$$

$$\log y = \log x + \log k = \log kx$$

$$y = kx$$

7. The solution of the differential equation $2x \frac{dy}{dx} - y = 3$ represents PTA-6

(1) straight lines (2) circles (3) parabola (4) ellipse

$$2x \frac{dy}{dx} - y = 3$$

$$2x \frac{dy}{dx} = y + 3$$

$$\frac{dy}{y+3} = \frac{dx}{2x}$$

Integrating on both sides

$$\int \frac{dy}{y+3} = \frac{1}{2} \int \frac{1}{x} dx$$

$$2 \log(y+3) = \log x + \log c$$

$$\log(y+3)^2 = \log cx$$

$$(y+3)^2 = cx \quad \text{This is an equation of parabola.}$$

8. The solution of $\frac{dy}{dx} + p(x)y = 0$ is JUL-21, APR-22

(1) $y = ce^{\int p dx}$ (2) $y = ce^{-\int p dx}$ (3) $x = ce^{-\int p dy}$ (4) $x = ce^{\int p dy}$

$$\frac{dy}{dx} + p(x)y = 0$$

$$Q = 0$$

This is linear in y

$$IF = e^{\int P(x) dx}$$

$$\text{solution is } y(IF) = \int Q(IF) dx + c$$

$$ye^{\int p(x) dx} = c$$

$$y = ce^{-\int p(x) dx}$$

9. The integrating factor of the differential equation $\frac{dy}{dx} + y = \frac{1+y}{x}$ is

(1) $\frac{x}{e^x}$ (2) $\frac{e^x}{x}$ (3) xe^x (4) e^x

$$\frac{dy}{dx} + y = \frac{1+y}{x}$$

$$\frac{dy}{dx} + y = \frac{1}{x} + \frac{y}{x}$$

$$\frac{dy}{dx} + \left(y - \frac{y}{x}\right) = \frac{1}{x}$$

$$P = 1 - \frac{1}{x}$$

$$IF = e^{\int P dx} = e^{\int \left(1 - \frac{1}{x}\right) dx}$$

$$= e^{x - \log x}$$

$$= \frac{e^x}{e^{\log x}} = \frac{e^x}{x}$$

$$IF = \frac{e^x}{x}$$

10. The integrating factor of the differential equation $\frac{dy}{dx} + P(x)y = Q(x)$ is x , then $P(x)$

- (1) x (2) $\frac{x^2}{2}$ (3) $\frac{1}{x}$ (4) $\frac{1}{x^2}$

$$e^{\int P(x)dx} = x$$

$$\int P(x)dx = \log x$$

Differentiate with respect to x , $P(x) = \frac{1}{x}$

11. The degree of the differential equation $y(x) = 1 + \frac{dy}{dx} + \frac{1}{1.2} \left(\frac{dy}{dx}\right)^2 + \frac{1}{1.2.3} \left(\frac{dy}{dx}\right)^3 + \dots$ is **PTA-3**

- (1) 2 (2) 3 (3) 1 (4) 4

$$y(x) = 1 + \frac{y'}{1!} + \frac{y''}{2!} + \frac{y'''}{3!} + \dots$$

$$y(x) = e^{y'}$$

taking log on both sides, $\log y = y'$

$$\frac{dy}{dx} = \log y$$

order = 1

degree = 1

12. If p and q are the order and degree of the differential equation $y \frac{dy}{dx} + x^3 \left(\frac{d^2y}{dx^2}\right) + xy = \cos x$, when

- (1) $p < q$ (2) $p = q$ (3) $p > q$ (4) p exists and q does not exist

order $p = 2$; degree $q = 1 \Rightarrow p > q$

13. The solution of the differential equation $\frac{dy}{dx} + \frac{1}{\sqrt{1-x^2}} = 0$ is **PTA-4**

- (1) $y + \sin^{-1} x = c$ (2) $x + \sin^{-1} y = 0$ (3) $y^2 + 2 \sin^{-1} x = C$ (4) $x^2 + 2 \sin^{-1} y = 0$

$$\frac{dy}{dx} = -\frac{1}{\sqrt{1-x^2}}$$

$$dy = -\frac{dx}{\sqrt{1-x^2}}$$

Integrating on both sides

$$\int dy = -\int \frac{dx}{\sqrt{1-x^2}}$$

$$y = -\sin^{-1} x + C$$

$$y + \sin^{-1} x = C$$

14. The solution of the differential equation $\frac{dy}{dx} = 2xy$ is **JUL-23**

- (1) $y = Ce^{x^2}$ (2) $y = 2x^2 + C$ (3) $y = Ce^{-x^2} + C$ (4) $y = x^2 + C$

$$\frac{dy}{y} = 2x dx$$

Integrating on both sides

$$\int \frac{dy}{y} = 2 \int x dx$$

$$\log y = \frac{2x^2}{2} + \log C$$

$$\log \frac{y}{C} = x^2$$

$$\frac{y}{C} = e^{x^2} \Rightarrow y = Ce^{x^2}$$

15. The general solution of the differential equation $\log \left(\frac{dy}{dx}\right) = x + y$ is **PTA-6**

- (1) $e^x + e^y = C$ (2) $e^x + e^{-y} = C$ (3) $e^{-x} + e^y = C$ (4) $e^{-x} + e^{-y} = C$

$$\log \left(\frac{dy}{dx}\right) = x + y$$

taking antilog

$$\frac{dy}{dx} = e^{x+y} = e^x e^y$$

$$e^{-y} dy = e^x dx$$

Integrating on both sides

$$\int e^{-y} dy = \int e^x dx$$

$$\int e^x dx = \int e^{-y} dy$$

$$e^x + e^{-y} = C$$

16. The solution of $\frac{dy}{dx} = 2^{y-x}$ is

- (1) $2^x + 2^y = C$ (2) $2^x - 2^y = C$ (3) $\frac{1}{2^x} - \frac{1}{2^y} = C$ (4) $x + y = C$

$\frac{dy}{dx} = 2^y 2^{-x}$ $\frac{dy}{2^y} = dx 2^{-x}$ Integrating on both sides $\int 2^{-y} dy = \int 2^{-x} dx$	$-\frac{2^{-y}}{\log 2} = \frac{-2^{-x}}{\log 2} + \frac{C}{\log 2}$ $2^{-y} = 2^{-x} - C$ $\frac{1}{2^y} = \frac{1}{2^x} - C$ $\frac{1}{2^x} - \frac{1}{2^y} = C$
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17. The solution of the differential equation $\frac{dy}{dx} = \frac{y}{x} + \frac{\varphi(\frac{y}{x})}{\varphi'(\frac{y}{x})}$ is

MDL, PTA-1, JUL-20

- (1) $x\varphi\left(\frac{y}{x}\right) = k$ (2) $\varphi\left(\frac{y}{x}\right) = kx$ (3) $y\varphi\left(\frac{y}{x}\right) = k$ (4) $\varphi\left(\frac{y}{x}\right) = ky$

$\frac{dy}{dx} = \frac{y}{x} + \frac{\varphi\left(\frac{y}{x}\right)}{\varphi'\left(\frac{y}{x}\right)}$ This is a Homogeneous D.E. Put $y = vx$ $\frac{y}{x} = v$ $\frac{dy}{dx} = V + x \frac{dv}{dx}$ $v + x \frac{dv}{dx} = v + \frac{\varphi(V)}{\varphi'(V)}$	$\frac{\varphi'(V)}{\varphi(V)} dv = \frac{dx}{x}$ Integrating on both sides $\int \frac{\varphi'(V)}{\varphi(V)} dv = \int \frac{dx}{x}$ $\log \varphi(v) = \log x + \log k$ $\log \varphi\left(\frac{y}{x}\right) = \log kx$ $\varphi\left(\frac{y}{x}\right) = kx$
--	--

18. If $\sin x$ is the integrating factor of the linear differential equation $\frac{dy}{dx} + Py = Q$, then P is

- (1) $\log \sin x$ (2) $\cos x$ (3) $\tan x$ (4) $\cot x$ APR-22, JUL-24

$e^{\int P dx} = \sin x$ taking log on both sides $\int P dx = \log \sin x$ diff. w. r to x $P = \frac{1}{\sin x} \cos x = \cot x$
--

19. The number of arbitrary constants in the general solutions of order n and $n+1$ are respectively

PTA-6

- (1) $n-1, n$ (2) $n, n+1$ (3) $n+1, n+2$ (4) $n+1, n$

n arbitrary constants = n order $n+1$ arbitrary constants = $n+1$ order
--

20. The number of arbitrary constants in the particular solution of a differential equation of the third order is

- (1) 3 (2) 2 (3) 1 (4) 0

Order of the Differential equation is = No. of arbitrary constants Order three $\therefore$ arbitrary constants = 3 but particular solution has no arbitrary constants.
--

21. Integrating factor of the differential equation $\frac{dy}{dx} = \frac{x+y+1}{x+1}$ is

(1) $\frac{1}{x+1}$

(2) $x + 1$

(3) $\frac{1}{\sqrt{x+1}}$

(4) $\sqrt{x+1}$

$$\begin{aligned}\frac{dy}{dx} &= \frac{x+1+y}{x+1} \\ \frac{dy}{dx} &= \frac{x+1}{x+1} + \frac{y}{x+1} \\ \frac{dy}{dx} - \frac{y}{x+1} &= 1\end{aligned}$$

$$P = \frac{-1}{x+1} \quad Q = 1$$

$$IF = e^{\int P dx}$$

$$= e^{-\int \frac{1}{x+1} dx} = e^{-\log x+1} = e^{\log(x+1)^{-1}}$$

$$= (x+1)^{-1} = \frac{1}{x+1}$$

22. The population P in any year t is such that the rate of increase in the population is proportional to the population. Then

PTA-3, JUL-23

(1) $P = Ce^{kt}$

(2) $P = Ce^{-kt}$

(3) $P = Ckt$

(4) $P = C$

$$\frac{dp}{dt} = kP$$

$$\frac{dp}{p} = k dt$$

$$\log P = kt + \log C \Rightarrow \log \frac{P}{C} = kt$$

$$\frac{P}{C} = e^{kt} \Rightarrow P = Ce^{kt}$$

23. P is the amount of certain substance left in after time t . If the rate of evaporation of the substance is proportional to the amount remaining, then

PTA-1

(1) $P = Ce^{kt}$

(2) $P = Ce^{-kt}$

(3) $P = Ckt$

(4) $Pt = C$

$$\frac{dp}{dt} = -kP \Rightarrow P = Ce^{-kt}$$

24. If the solution of the differential equation $\frac{dy}{dx} = \frac{ax+3}{2y+f}$ represents a circle, then the value of a is

PTA-2

(1) 2

(2) -2

(3) 1

(4) -1

$$(2y+f) dy = (ax+3) dx$$

Integrating on both sides

$$\int (2y+f) dy = \int (ax+3) dx$$

$$\frac{2y^2}{2} + fy = \frac{ax^2}{2} + 3x + C = 0$$

$$ax^2 - 2y^2 - 2fy + 6x + 2C = 0$$

This represent a circle only when

$$\text{Co-efficient of } x^2 = \text{co-efficient of } y^2$$

$$a = -2$$

25. The slope at any point of a curve $y = f(x)$ is given by $\frac{dy}{dx} = 3x^2$ and it passes through $(-1, 1)$

APR-25

Then the equation of the curve is

(1) $y = x^3 + 2$

(2) $y = 3x^2 + 4$

(3) $y = 3x^3 + 4$

(4) $y = x^3 + 5$

$$\frac{dy}{dx} = 3x^2$$

$$dy = 3x^2 dx$$

$$\text{Integrating on both sides } \int dy = \int 3x^2 dx$$

$$y = \frac{3x^3}{3} + C = x^3 + C \dots\dots\dots (1)$$

This passes through $(-1, 1)$

$$(1) \Rightarrow 1 = (-1)^3 + C$$

$$1 + 1 = C \Rightarrow C = 2$$

$$(1) \Rightarrow y = x^3 + 2$$



Laplace

Chapter - 11

Probability Distributions

EXERCISE 11.1



Video class

Concept Corner

Random variables

A **random variable** X is a function defined on a sample space S into the real number $\mathbb{R}$ such that the inverse image of points or subset or interval of $\mathbb{R}$ is an event in S , for which probability is assigned.

1. Suppose X is the number of tails occurred when three fair coins are tossed once simultaneously. Find the values of the random variable X and number of points in its inverse images.

APR-22

Sample space

2M

$$S = \{ \text{HHH, HHT, HTH, THH,} \\ \text{HTT, THT, TTH, TTT} \}$$

Given that $n(S) = 8$

X be the random variable denotes no of tails $X = 0, 1, 2, 3$

0 tail $X(\text{HHH}) = 0$

1 tail $X(\text{HHT}) = 1$ $X(\text{HTH}) = 1$

$X(\text{THH}) = 1$

2 tail $X(\text{HTT}) = 2$ $X(\text{THT}) = 2$

$X(\text{TTH}) = 2$

3 tail $X(\text{TTT}) = 3$

$X(\omega)$ denotes the number of tails

$$X(\omega) = \begin{cases} 0 & \text{if } \omega = \text{HHH} \\ 1 & \text{if } \omega = \text{HHT, THH, HTH} \\ 2 & \text{if } \omega = \text{HTT, THT, TTH} \\ 3 & \text{if } \omega = \text{TTT} \end{cases}$$

Values of Random variable	0	1	2	3	Total
Number of points in inverse image	1	3	3	1	8

2. An urn contains 5 mangoes and 4 apples. Three fruits are taken at random. If the number of apples taken is a random variable, then find the values of the random variable and number of points in its inverse images.

Let X be the random variable denotes the number of apples taken $X = 0, 1, 2, 3$

3M

$X(\omega)$ denotes number of apples.

$$X(0) = {}^4C_0 \times {}^5C_3 = 1 \times \frac{5 \times 4}{1 \times 2} = 10$$

$$X(1) = {}^4C_1 \times {}^5C_2 = 4 \times \frac{5 \times 4}{1 \times 2} = 40$$

$$X(2) = {}^4C_2 \times {}^5C_1 = \frac{4 \times 3}{1 \times 2} \times 5 = 30$$

$$X(3) = {}^4C_3 \times {}^5C_0 = 4 \times 1 = 4$$

Value of random variable	0	1	2	3	Total
Number of points in inverse image	10	40	30	4	84

3. Two balls are chosen randomly from an urn containing 6 red and 8 black balls. Suppose that we win ₹ 15 for each red ball selected and we lose ₹ 10 for each black ball selected. If X denotes the winning amount, find the values of X and number of points in its inverse images.

Number of Red balls = 6

3M

Number of Black balls = 8

Let X be the amount won

$$X = -10 \times 2, -10 + 15, 15 + 15$$

$$X = -20, 5, 30$$

$$X(W) = \begin{cases} -20 & \text{if two black balls} \\ 5 & \text{if one black one red} \\ 30 & \text{if two red balls} \end{cases}$$

$$X(-20) = {}^6C_0 \times {}^8C_2 = 1 \times \frac{8 \times 7}{1 \times 2} = 28$$

$$X(5) = {}^6C_1 \times {}^8C_1 = 6 \times 8 = 48$$

$$X(30) = {}^6C_2 \times {}^8C_0 = \frac{5 \times 6}{1 \times 2} \times 1 = 15$$

Value of random variable	-20	5	30	Total
Number of points in inverse image	28	48	15	91

4. A six sided die is marked '2' on one face, '3' on two of its faces, and '4' on remaining three faces. The die is thrown twice. If X denotes the total score in two throws, find the values of the random variable and number of points in its inverse images.

3M

X is assigned to each point (α, β) the sum on the faces.

$$n(S) = 36$$

II \ I	2	3	3	4	4	4
2	4	5	5	6	6	6
3	5	6	6	7	7	7
3	5	6	6	7	7	7
4	6	7	7	8	8	8
4	6	7	7	8	8	8
4	6	7	7	8	8	8

$$X(\alpha, \beta) = \alpha + \beta, \quad X(2, 2) = 4$$

$$X(2, 3) = X(3, 2) = 5$$

$$X(2, 4) = X(3, 3) = X(4, 2) = 6$$

$$X(4, 3) = X(3, 4) = 7, \quad X(4, 4) = 8$$

X takes the values 4, 5, 6, 7, 8

Value of random variable	4	5	6	7	8	Total
Number of points in inverse image	1	4	10	12	9	36

EXERCISE 11.2



Video class

Concept Corner

Discrete random variable :

A random variable X is said to be a **discrete random variable** if the range of X is countable.

Probability Mass function (P.M.F) :

If X is a discrete random variable with discrete values $x_1, x_2, \dots, x_n, \dots$ then the function $f(\cdot)$ or $p(\cdot)$ defined by $f(x_k) = P(X = x_k)$, for $k = 1, 2, \dots, n$.

is called Probability Mass function (PMF)

$$f(x_k) \geq 0 \text{ for } k = 1, 2, \dots, n \text{ and } \sum_k f(x_k) = 1$$

Cumulative Distribution Function (or) Distribution Function

The cumulative distribution function $F(x)$ of a discrete random variable X , taking the values $x_1, x_2, \dots$ such that $x_1 < x_2 < x_3 < \dots$ with P.M.F $f(x_i)$

$$F(x) = P(X \leq x) = \sum_{x_i \leq x} f(x_i), x \in R$$

$$f(x_i) = F(x_i) - F(x_{i-1}), i = 1, 2, 3, \dots$$

1. Three fair coins are tossed simultaneously. Find the probability mass function for number of heads occurred.

APR-24,20

2M

$$S = \{HHH, HHT, HTH, THH, HTT, THT, TTH, TTT\},$$

$$n(S) = 8$$

Let X be random variable denotes number of heads

$$X = 0, 1, 2, 3.$$

Value of random variable	0	1	2	3	Total
Number of points in inverse image	1	3	3	1	8

$$P(X = 0) = \frac{1}{8}, \quad P(X = 1) = \frac{3}{8}, \quad P(X = 2) = \frac{3}{8}, \quad P(X = 3) = \frac{1}{8}$$

Probability mass function is

x	0	1	2	3
$P[X = x]$	$\frac{1}{8}$	$\frac{3}{8}$	$\frac{3}{8}$	$\frac{1}{8}$

2. A six sided die is marked '1' on one face, '3' on two of its faces, and '5' on remaining three faces. The die is thrown twice. If X denotes the total score in two throws, find (i) the probability mass function (ii) the cumulative distribution function (iii) $P(4 \leq X < 10)$ (iv) $P(X \geq 6)$

II \ I	1	3	3	5	5	5
1	2	4	4	6	6	6
3	4	6	6	8	8	8
3	4	6	6	8	8	8
5	6	8	8	10	10	10
5	6	8	8	10	10	10
5	6	8	8	10	10	10

Let X denote the total score in two throws it takes 2, 4, 6, 8, 10.

Value of random variable	2	4	6	8	10	Total
Number of points in inverse image	1	4	10	12	9	36

$$P(X = 2) = \frac{1}{36},$$

APR-24

$$P(X = 4) = \frac{4}{36}$$

$$P(X = 6) = \frac{10}{36}$$

$$P(X = 8) = \frac{12}{36},$$

$$P(X = 10) = \frac{9}{36}$$

- (i) The probability mass function is

3M

x	2	4	6	8	10	Total
$f(x)$	$\frac{1}{36}$	$\frac{4}{36}$	$\frac{10}{36}$	$\frac{12}{36}$	$\frac{9}{36}$	1

- (ii) The cumulative distribution function.

$$F(x) = \begin{cases} 0 & \text{for } x < 2 \\ 1/36 & \text{for } 2 \leq x < 4 \\ 5/36 & \text{for } 4 \leq x < 6 \\ 15/36 & \text{for } 6 \leq x < 8 \\ 27/36 & \text{for } 8 \leq x < 10 \\ 1 & \text{for } 10 \leq x < \infty \end{cases}$$

- (iii) $P(4 \leq X < 10)$

$$= P(X = 4) + P(X = 6) + P(X = 8)$$

$$= \frac{4}{36} + \frac{10}{36} + \frac{12}{36}$$

$$= \frac{26}{36}$$

$$= \frac{13}{18}$$

- (iv) $P(X \geq 6)$

$$= P(X = 6) + P(X = 8) + P(X = 10)$$

$$= \frac{10}{36} + \frac{12}{36} + \frac{9}{36}$$

$$= \frac{31}{36}$$

3. Find the probability mass function and cumulative distribution function of number of girl child in families with 4 children, assuming equal probabilities for boys and girls.

3M

$$n(s) = 2^4 = 16$$

Let X be the random variable denotes number of girls child.

$$X = 0, 1, 2, 3, 4$$

Value of random variable X	0	1	2	3	4
Number of elements in inverse image	4C_0	4C_1	4C_2	4C_3	4C_4
	1	4	6	4	1

$$P(X = 0) = \frac{1}{16} = P(X = 4)$$

$$P(X = 1) = \frac{4}{16} = P(X = 3)$$

$$P(X = 2) = \frac{6}{16} = \frac{3}{8}$$

Probability mass function is

$$f(x) = \begin{cases} \frac{1}{16} & \text{for } x = 0, 4 \\ \frac{4}{16} & \text{for } x = 1, 3 \\ \frac{6}{16} & \text{for } x = 2 \end{cases}$$

X	0	1	2	3	4
f(x)	$\frac{1}{16}$	$\frac{4}{16}$	$\frac{6}{16}$	$\frac{4}{16}$	$\frac{1}{16}$

$$F(x) = P(X \leq x)$$

$$F(0) = P(X \leq 0) = \frac{1}{16}$$

$$F(1) = P(X \leq 1) = \frac{1}{16} + \frac{4}{16} = \frac{5}{16}$$

$$F(2) = P(X \leq 2) = \frac{5}{16} + \frac{6}{16} = \frac{11}{16}$$

$$F(3) = P(X \leq 3) = \frac{11}{16} + \frac{4}{16} = \frac{15}{16}$$

$$F(4) = P(X \leq 4) = \frac{15}{16} + \frac{1}{16} = \frac{16}{16} = 1$$

Cumulative distribution function is

$$F(x) = \begin{cases} 0 & \text{for } x < 0 \\ 1/16 & \text{for } 0 \leq x < 1 \\ 5/16 & \text{for } 1 \leq x < 2 \\ 11/16 & \text{for } 2 \leq x < 3 \\ 15/16 & \text{for } 3 \leq x < 4 \\ 1 & \text{for } 4 \leq x < \infty \end{cases}$$

4. Suppose a discrete random variable can only take the values 0, 1, and 2. The probability mass

function is defined by $f(x) = \begin{cases} \frac{x^2+1}{k}, & \text{for } x = 0, 1, 2 \\ 0 & \text{otherwise} \end{cases}$

MDL, PTA-1

Find (i) the value of k (ii) cumulative distribution function (iii) $P(X \geq 1)$.

5M

Given $f(x)$ is a P.M.F

$$(i) \quad \therefore \sum f(x) = 1$$

$$\frac{0^2+1}{k} + \frac{1^2+1}{k} + \frac{2^2+1}{k} = 1$$

$$\frac{1}{k} + \frac{2}{k} + \frac{5}{k} = 1$$

$$\frac{8}{k} = 1$$

$$\Rightarrow k = 8$$

$$\therefore f(x) = \begin{cases} \frac{x^2+1}{8} & \text{for } x = 0, 1, 2 \\ 0 & \text{otherwise} \end{cases}$$

(or)

x	0	1	2
f(x)	$\frac{1}{8}$	$\frac{2}{8}$	$\frac{5}{8}$

$$(ii) \quad F(x) = P(X \leq x)$$

$$F(0) = P(X = 0) = \frac{1}{8}$$

$$F(1) = P(X = 0) + P(X = 1) = \frac{3}{8}$$

$$F(2) = P(X = 0) + P(X = 1) + P(X = 2)$$

$$F(2) = \frac{8}{8} = 1$$

$$F(x) = \begin{cases} 0 & \text{for } x < 0 \\ \frac{1}{8} & \text{for } 0 \leq x < 1 \\ \frac{3}{8} & \text{for } 1 \leq x < 2 \\ \frac{8}{8} = 1 & \text{for } 2 \leq x < \infty \end{cases}$$

$$(iii) \quad P(X \geq 1) = P(X = 1) + P(X = 2)$$

$$= \frac{2}{8} + \frac{5}{8} = \frac{7}{8}$$

5. The cumulative distribution function of a discrete random variable is given by 3M

$$F(x) = \begin{cases} 0 & -\infty < x < -1 \\ 0.15 & -1 \leq x < 0 \\ 0.35 & 0 \leq x < 1 \\ 0.60 & 1 \leq x < 2 \\ 0.85 & 2 \leq x < 3 \\ 1 & 3 \leq x < \infty \end{cases} \quad \text{PTA-3}$$

Find (i) the probability mass function

(ii) $P(X < 1)$ (iii) $P(X \geq 2)$

$$P(X = -1) = F(-1) = 0.15$$

$$P(X = 0) = F(0) - F(-1) \\ = 0.35 - 0.15 = 0.20$$

$$P(X = 1) = F(1) - F(0) \\ = 0.60 - 0.35 = 0.25$$

$$P(X = 2) = F(2) - F(1) = 0.85 - 0.60 = 0.25$$

$$P(X = 3) = F(3) - F(2) = 1 - 0.85 = 0.15$$

(i) Probability mass function is

x	-1	0	1	2	3
$f(x)$	0.15	0.20	0.25	0.25	0.15

$$\begin{aligned} \text{(ii)} \quad P(X < 1) &= P(X \leq 0) = F(0) + F(-1) \\ &= P(X = -1) + P(X = 0) \\ &= 0.20 + 0.15 \\ &= \mathbf{0.35} \end{aligned}$$

$$\begin{aligned} \text{(iii)} \quad P(X \geq 2) &= P(2 \leq X \leq 3) \\ &= P(X = 2) + P(X = 3) \\ &= 0.25 + 0.15 \\ &= \mathbf{0.40} \end{aligned}$$

7. The cumulative distribution function of a discrete random variable is given by Find

$$F(x) = \begin{cases} 0 & \text{for } -\infty < x < 0 \\ \frac{1}{2} & \text{for } 0 \leq x < 1 \\ \frac{3}{5} & \text{for } 1 \leq x < 2 \\ \frac{4}{5} & \text{for } 2 \leq x < 3 \\ \frac{9}{10} & \text{for } 3 \leq x < 4 \\ 1 & \text{for } 4 \leq x < \infty \end{cases} \quad \begin{matrix} \text{JUL-22} \\ 5M \end{matrix}$$

Find (i) The probability mass function

(ii) $P(X < 3)$ (iii) $P(X \geq 2)$

$$f(0) = P(X = 0) = F(0) = \frac{1}{2}$$

$$f(1) = P(X = 1) = F(1) - F(0) = \frac{3}{5} - \frac{1}{2}$$

$$f(1) = \frac{6-5}{10} = \frac{1}{10}$$

$$f(2) = P(X = 2) = F(2) - F(1) = \frac{4}{5} - \frac{3}{5} = \frac{1}{5}$$

$$f(3) = P(X = 3) = F(3) - F(2) = \frac{9}{10} - \frac{4}{5} = \frac{1}{10}$$

$$f(4) = P(X = 4) = F(4) - F(3) = 1 - \frac{9}{10} = \frac{1}{10}$$

The probability mass function is

x	0	1	2	3	4
$f(x)$	$\frac{1}{2}$	$\frac{1}{10}$	$\frac{1}{5}$	$\frac{1}{10}$	$\frac{1}{10}$

$$\begin{aligned} \text{(ii)} \quad P(X < 3) &= P(X = 0) + P(X = 1) + P(X = 2) \\ &= P(X \leq 2) = F(2) = \frac{4}{5} \end{aligned}$$

$$\begin{aligned} \text{(iii)} \quad P(X \geq 2) &= 1 - P(X < 2) \\ &= 1 - P(X \leq 1) \\ &= 1 - F(1) \\ P(X \geq 2) &= 1 - \frac{3}{5} = \frac{2}{5} \end{aligned}$$

6. A random variable X has the following probability mass function. Find (i) the value of k

(ii) $P(2 \leq X < 5)$ (iii) $P(3 < X)$

x	1	2	3	4	5
$f(x)$	k^2	$2k^2$	$3k^2$	$2k$	$3k$

JUL-22, 21

5M

(i) $f(x)$ is a Probability Mass function

$$\sum f(x) = 1$$

$$k^2 + 2k^2 + 3k^2 + 2k + 3k = 1$$

$$6k^2 + 5k - 1 = 0$$

$$\left(k + \frac{6}{6}\right)\left(k - \frac{1}{6}\right) = 0$$

$$k + 1 = 0, \quad k - \frac{1}{6} = 0$$

$$k = -1, \quad k = \frac{1}{6}$$

x	1	2	3	4	5
$f(x)$	$\frac{1}{36}$	$\frac{2}{36}$	$\frac{3}{36}$	$\frac{2}{6}$	$\frac{3}{6}$

(ii) $P(2 \leq X < 5) = P(X = 2) + P(X = 3) + P(X = 4)$

$$= 2k^2 + 3k^2 + 2k$$

$$= \frac{2}{36} + \frac{3}{36} + \frac{2}{6}$$

$$= \frac{2+3+12}{36} = \frac{17}{36}$$

(iii) $P(3 < X) = P(X = 4) + P(X = 5)$

$$= 2k + 3k = 5k = \frac{5}{6}$$

EXERCISE 11.3

Concept Corner

Continuous random variable : Let S be a sample space and let a random variable $X: S \rightarrow R$ takes on any value in a set I of R . Then X is called a Continuous random variable if $P(X = x) = 0$ for every x in I .

Probability Density function (P.d.f): A non-negative real valued function $f(x)$ is said to be p.d.f. If for each possible outcome x , $x \in [a, b]$ of a continuous random variable X having the property $P(a \leq X \leq b) = \int_a^b f(x) dx$

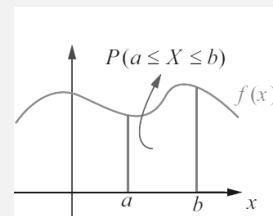
Theorem: $f(x)$ is a p.d.f (i) $f(x) \geq 0$ (ii) $\int_{-\infty}^{\infty} f(x) dx = 1$

Cumulative distribution function: The distribution function or cumulative distribution function $F(x)$ of a continuous random variable X with probability density $f(x)$ is $F(x) = P(X \leq x) = \int_{-\infty}^x f(u) du$ $-\infty < u < \infty$

Properties of distribution function: A continuous random variable X , the cumulative distribution function satisfies the properties.

- (i) $0 \leq F(x) \leq 1$ (ii) $F(x)$ is a real valued non-decreasing. i.e., $x < y$ then, $F(x) \leq F(y)$
 (iii) $F(x)$ is continuous everywhere. (iv) $\lim_{x \rightarrow -\infty} F(x) = F(-\infty) = 0$ and $\lim_{x \rightarrow \infty} F(x) = F(+\infty) = 1$
 (v) $P(X > x) = 1 - P(X \leq x) = 1 - F(x)$ (vi) $P(a < x < b) = F(b) - F(a)$

Probability Density function : $f(x) = \frac{dF(x)}{dx}$ or $F'(x)$



1. The probability density function of X is given by $f(x) = \begin{cases} kx e^{-2x} & \text{for } x > 0 \\ 0 & \text{for } x \leq 0 \end{cases}$

2M

JUL-21

Find the value of k .

$f(x)$ is a P.d.f

$$\therefore \int_{-\infty}^{\infty} f(x) dx = 1$$

$$\int_0^{\infty} kx^1 e^{-2x} dx = 1$$

When $a = 2, n = 1,$

$$k \left[\frac{1!}{2^{1+1}} \right] = 1 \quad k = 2^2, k = 4$$

$$\therefore \int_0^{\infty} x^n e^{-ax} dx = \frac{n!}{a^{n+1}}$$

2. The probability density function of X is $f(x) = \begin{cases} x & 0 < x < 1 \\ 2 - x & 1 \leq x < 2 \\ 0 & \text{otherwise} \end{cases}$

3M

Find (i) $P(0.2 \leq X < 0.6)$ (ii) $P(1.2 \leq X < 1.8)$ (iii) $P(0.5 \leq X < 1.5)$

$$\begin{aligned} \text{(i) } P(0.2 \leq X < 0.6) &= \int_{0.2}^{0.6} f(x) dx \\ &= \int_{0.2}^{0.6} x dx = \left[\frac{x^2}{2} \right]_{0.2}^{0.6} \\ &= \frac{1}{2} [0.6^2 - 0.2^2] \\ &= \frac{1}{2} [0.36 - 0.04] \\ &= \frac{1}{2} [0.32] = \mathbf{0.16} \end{aligned}$$

$$\begin{aligned} \text{(ii) } P(1.2 \leq X < 1.8) &= \int_{1.2}^{1.8} f(x) dx \\ &= \int_{1.2}^{1.8} (2 - x) dx = - \left[\frac{(2-x)^2}{2} \right]_{1.2}^{1.8} \\ &= - \frac{1}{2} [(2 - 1.8)^2 - (2 - 1.2)^2] \\ &= - \frac{1}{2} [0.2^2 - 0.8^2] = - \frac{1}{2} [0.04 - 0.64] \\ &= - \frac{1}{2} [-0.60] = \mathbf{0.30} \end{aligned}$$

$$\begin{aligned} \text{(iii) } P(0.5 \leq X < 1.5) &= \int_{0.5}^{1.5} f(x) dx \\ &= \int_{0.5}^1 f(x) dx + \int_1^{1.5} f(x) dx \\ &= \int_{0.5}^1 x dx + \int_1^{1.5} (2 - x) dx \\ &= \left[\frac{x^2}{2} \right]_{0.5}^1 + \left[\frac{-(2-x)^2}{2} \right]_1^{1.5} \\ &= \frac{1}{2} [(1^2 - 0.5^2) - ((2 - 1.5)^2 - (2 - 1)^2)] \\ &= \frac{1}{2} [(1 - 0.25) - (0.5^2 - 1^2)] \\ &= \frac{1}{2} [0.75 + 0.75] \\ &= \mathbf{0.75} \end{aligned}$$

3. Suppose the amount of milk sold daily at a milk booth is distributed with a minimum of 200 litres and a maximum of 600 litres with probability density function

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5M

$f(x) = \begin{cases} k & 200 \leq x \leq 600 \\ 0 & \text{otherwise} \end{cases}$ Find (i) the value of k (ii) The distribution function (iii) the probability that daily sales will fall between 300 litres and 500 litres?

(i) given $f(x)$ is a p.d.f $\int_{-\infty}^{\infty} f(x) dx = 1$

$$\int_{200}^{600} k dx = 1 \Rightarrow k[x]_{200}^{600} = 1$$

$$k(600 - 200) = 1$$

$$400k = 1$$

$$k = \frac{1}{400}$$

(iii) $P(300 < x < 500) = F(500) - F(300)$

$$= \left(\frac{500}{400} - \frac{1}{2} \right) - \left(\frac{300}{400} - \frac{1}{2} \right)$$

$$= \frac{500-300}{400} - \frac{1}{2} + \frac{1}{2}$$

$$= \frac{200}{400} = \frac{1}{2}$$

(ii) distribution function $F(x) = \int_{-\infty}^x f(t) dt$

$$\begin{aligned} \text{a) when } x \in (-\infty, 200) \quad F(x) &= \int_{-\infty}^x f(t) dt \\ &= \int_{-\infty}^{200} 0 dt = 0 \end{aligned}$$

$$\begin{aligned} \text{b) when } x \in [200, 600] \quad F(x) &= \int_{-\infty}^x f(t) dt \\ &= \int_{-\infty}^{200} f(t) dt + \int_{200}^x f(t) dt \\ &= 0 + \int_{200}^x \frac{1}{400} dt = \frac{1}{400} [t]_{200}^x \\ &= \frac{x}{400} - \frac{200}{400} = \frac{x}{400} - \frac{1}{2} \end{aligned}$$

c) when $x \in (600, \infty)$

$$\begin{aligned} F(x) &= \int_{-\infty}^x f(t) dt \\ &= \int_{-\infty}^{200} f(t) dt + \int_{200}^{600} f(t) dt + \int_{600}^x f(t) dt \\ &= 0 + \int_{200}^{600} \frac{1}{400} dt = \frac{1}{400} [t]_{200}^{600} \\ &= \frac{1}{400} (400) = 1 \end{aligned}$$

$$F(x) = \begin{cases} 0 & x < 200 \\ \frac{x}{400} - \frac{1}{2} & 200 \leq x \leq 600 \\ 1 & x > 600 \end{cases}$$

4. The Probability density function of X is given $f(x) = \begin{cases} k e^{-\frac{x}{3}} & \text{for } x > 0 \\ 0 & \text{for } x \leq 0 \end{cases}$ Find (i) the value of

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k (ii) the distribution function (iii) $P(X < 3)$ (iv) $P(5 \leq X)$ (v) $P(X \leq 4)$.

(i) Given $f(x)$ is a p.d.f

$$\int_{-\infty}^{\infty} f(x) dx = 1$$

$$\Rightarrow \int_{-\infty}^{\infty} k e^{-\frac{x}{3}} dx = 1$$

$$\frac{k}{-\frac{1}{3}} \left[e^{-\frac{x}{3}} \right]_0^{\infty} = 1$$

$$-3k[e^{-\infty} - e^0] = 1$$

$$-3k[0 - 1] = 1$$

$$3k = 1$$

$$k = \frac{1}{3}$$

(ii) distribution function $F(x) = \int_{-\infty}^x f(t) dt$

$$\text{a) } x \in (-\infty, 0] \quad F(x) = \int_{-\infty}^x f(t) dt = 0$$

$$\begin{aligned} \text{b) } x \in (0, \infty) \quad F(x) &= \int_{-\infty}^x f(t) dt \\ &= \int_{-\infty}^0 f(t) dt + \int_0^x f(t) dt \\ &= 0 + \int_0^x \frac{1}{3} e^{-\frac{t}{3}} dt \\ &= \frac{1}{3} \frac{1}{-\frac{1}{3}} \left[e^{-\frac{t}{3}} \right]_0^x = -e^{-\frac{x}{3}} + e^0 \\ &= 1 - e^{-\frac{x}{3}} \end{aligned}$$

$$F(x) = \begin{cases} 0 & \text{for } x \leq 0 \\ 1 - e^{-\frac{x}{3}} & \text{for } x > 0 \end{cases}$$

$$\text{(iii) } P(X < 3) = P(X \leq 3) = F(3) = 1 - e^{-\frac{3}{3}} = 1 - e^{-1}$$

$$\text{(iv) } P(5 \leq X) = P(X \geq 5) = 1 - P(X < 5) = 1 - F(5) = 1 - \left(1 - e^{-\frac{5}{3}} \right) = e^{-\frac{5}{3}}$$

$$\text{(v) } P(X \leq 4) = F(4) = 1 - e^{-\frac{4}{3}}$$

5. If X is the random variable with probability density function $f(x)$ given by,

$$f(x) = \begin{cases} x+1, & -1 \leq x < 0 \\ -x+1, & 0 \leq x < 1 \\ 0, & \text{otherwise} \end{cases}$$

Then find (i) the distribution function $F(x)$

(i) distribution function

a) when $x \in (-\infty, -1)$

$$F(x) = \int_{-\infty}^x f(t) dt = 0$$

b) when $x \in [-1, 0)$

$$\begin{aligned} F(x) &= \int_{-\infty}^x f(t) dt \\ &= \int_{-\infty}^{-1} f(t) dt + \int_{-1}^x f(t) dt \\ &= 0 + \int_{-1}^x (t+1) dt \\ &= \left[\frac{t^2}{2} + t \right]_{-1}^x = \frac{x^2}{2} + x - \frac{1}{2} + 1 \end{aligned}$$

$$F(x) = \frac{x^2}{2} + x + \frac{1}{2}$$

c) when $x \in [0, 1)$ $F(x) = \int_{-\infty}^x f(t) dt$

$$\begin{aligned} &= \int_{-\infty}^{-1} f(t) dt + \int_{-1}^0 f(t) dt + \int_0^x f(t) dt \\ &= 0 + \int_{-1}^0 (t+1) dt + \int_0^x (-t+1) dt \\ &= \left[\frac{t^2}{2} + t \right]_{-1}^0 + \left[-\frac{t^2}{2} + t \right]_0^x \\ &= 0 - \left(\frac{1}{2} - 1 \right) + \left(-\frac{x^2}{2} + x \right) - (0 + 0) \\ &= \frac{1}{2} - \frac{x^2}{2} + x \end{aligned}$$

(ii) $P(-0.5 \leq X \leq 0.5)$

d) when $x \in [1, \infty)$ $F(x) = \int_{-\infty}^x f(t) dt$

$$\begin{aligned} &= \int_{-\infty}^{-1} f(t) dt + \int_{-1}^0 f(t) dt + \\ &\quad \int_0^1 f(t) dt + \int_1^{\infty} f(t) dt \end{aligned}$$

$$= 0 + \frac{1}{2} + \int_0^1 f(-t+1) dt + 0$$

$$\begin{aligned} &= \frac{1}{2} + \left(\frac{-t^2}{2} + t \right)_0^1 = \frac{1}{2} + \left(-\frac{1}{2} + 1 \right) - 0 \\ &= 1 \end{aligned}$$

$$F(x) = \begin{cases} 0 & x \leq -1 \\ \frac{x^2}{2} + x + \frac{1}{2} & -1 \leq x < 0 \\ -\frac{x^2}{2} + x + \frac{1}{2} & 0 \leq x < 1 \\ 1 & 1 \leq x \end{cases}$$

(ii) $P(-0.5 \leq x \leq 0.5) = F(0.5) - F(-0.5)$

$$= F\left(\frac{1}{2}\right) - F\left(-\frac{1}{2}\right)$$

$$= \left(-\frac{\left(\frac{1}{2}\right)^2}{2} + \frac{1}{2} + \frac{1}{2} \right) - \left(\frac{-\left(\frac{1}{2}\right)^2}{2} - \frac{1}{2} + \frac{1}{2} \right)$$

$$= \left(-\frac{1}{8} + 1 \right) - \left(\frac{1}{8} - \frac{1}{2} + \frac{1}{2} \right)$$

$$= 1 - \frac{2}{8} = \frac{6}{8} = \mathbf{0.75}$$

6. If X is the random variable with distribution function $F(x)$ given by, $F(x) = \begin{cases} 0, & -\infty < x < 0 \\ \frac{1}{2}(x^2 + x), & 0 \leq x < 1 \\ 1, & 1 \leq x < \infty \end{cases}$

Then find (i) the probability density function $f(x)$ (ii) $P(0.3 \leq X \leq 0.6)$

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$$f(x) = \frac{d}{dx}(F(x))$$

$$f(x) = \begin{cases} 0 & x < 0 \\ \frac{1}{2}(2x+1) & 0 \leq x < 1 \\ 0 & 1 \leq x \end{cases}$$

$$f(x) = \begin{cases} \frac{1}{2}(2x+1) & 0 \leq x < 1 \\ 0 & \text{otherwise} \end{cases}$$

(ii) $P(0.3 \leq X \leq 0.6) = F(0.6) - F(0.3)$

$$= \frac{1}{2}(0.6^2 + 0.6) - \frac{1}{2}(0.3^2 + 0.3)$$

$$= \frac{1}{2}[0.36 + 0.6 - 0.09 - 0.3]$$

$$P(0.3 \leq X \leq 0.6) = \frac{1}{2}[0.96 - 0.39]$$

$$= \frac{1}{2}[0.57]$$

$$= \mathbf{0.285}$$

EXERCISE 11.4



Video class

Concept Corner

Mathematical Expectation

Mean :

The expected value or mean or Mathematical Expectation of X , denote by $E(X)$ or μ is

$$E(X) = \begin{cases} \sum_x x f(x) & \text{if } X \text{ is discrete} \\ \int_{-\infty}^{\infty} x f(x) dx & \text{if } X \text{ is continuous} \end{cases}$$

Theorem : $E(g(X)) = \begin{cases} \sum_x g(x) f(x) & \text{if } g(x) \text{ is discrete} \\ \int_{-\infty}^{\infty} g(x) f(x) dx & \text{if } g(x) \text{ is continuous} \end{cases}$

Variance : $V(X) = E(X - E(X))^2 = E(X - \mu)^2$

Properties : (1) $E(aX + b) = aE(X) + b$ where a and b are constants.

(2) $V(X) = E(X^2) - (E(X))^2$ (3) $\text{Var}(aX + b) = a^2 \text{Var}(X)$, where a and b are constants.

1. For the random variable X with the given probability mass function as below, find the mean and

variance. (i) $f(x) = \begin{cases} \frac{1}{10} & x = 2, 5 \\ \frac{1}{5} & x = 0, 1, 3, 4 \end{cases}$

(ii) $f(x) = \begin{cases} \frac{4-x}{6} & x = 1, 2, 3 \end{cases}$

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(iii) $f(x) = \begin{cases} 2(x-1) & 1 < x < 2 \\ 0 & \text{otherwise} \end{cases}$

(iv) $f(x) = \begin{cases} \frac{1}{2} e^{-\frac{x}{2}} & \text{for } x > 0 \\ 0 & \text{otherwise} \end{cases}$

(i) $f(x) = \begin{cases} \frac{1}{10} & x = 2, 5 \\ \frac{1}{5} & x = 0, 1, 3, 4 \end{cases}$

x	0	1	2	3	4	5
$f(x)$	$\frac{1}{5}$	$\frac{1}{5}$	$\frac{1}{10}$	$\frac{1}{5}$	$\frac{1}{5}$	$\frac{1}{10}$
$xf(x)$	0	$\frac{1}{5}$	$\frac{2}{10}$	$\frac{3}{5}$	$\frac{4}{5}$	$\frac{5}{10}$
$x^2 f(x)$	0	$\frac{1}{5}$	$\frac{4}{10}$	$\frac{9}{5}$	$\frac{16}{5}$	$\frac{25}{10}$

Mean = $\sum_x x f(x)$

$$= 0 + \frac{1}{5} + \frac{2}{10} + \frac{3}{5} + \frac{4}{5} + \frac{5}{10}$$

$$E(X) = \frac{8}{5} + \frac{7}{10} = \frac{16+7}{10} = \frac{23}{10}$$

Mean = 2.3

$E(X^2) = \sum_x x^2 f(x)$

$$= 0 + \frac{1}{5} + \frac{4}{10} + \frac{9}{5} + \frac{16}{5} + \frac{25}{10}$$

$$= \frac{29}{10} + \frac{26}{5} = \frac{29+52}{10}$$

$$= \frac{81}{10} = 8.1$$

$\text{Var}(X) = E(X^2) - (E(X))^2$

$$= 8.1 - 2.3^2$$

$$= 8.1 - 5.29 = 2.81$$

Mean = 2.3

Variance = 2.81

(ii) $f(x) = \begin{cases} \frac{4-x}{6} & x = 1, 2, 3 \end{cases}$

x	1	2	3
$f(x)$	$\frac{3}{6}$	$\frac{2}{6}$	$\frac{1}{6}$
$xf(x)$	$\frac{3}{6}$	$\frac{4}{6}$	$\frac{3}{6}$
$x^2 f(x)$	$\frac{3}{6}$	$\frac{8}{6}$	$\frac{9}{6}$

$E(X) = \sum x f(x)$

$$= \frac{3}{6} + \frac{4}{6} + \frac{3}{6} = \frac{10}{6} = 1.667$$

mean = 1.67

$E(X^2) = \sum x^2 f(x)$

$$= \frac{3}{6} + \frac{8}{6} + \frac{9}{6} = \frac{20}{6}$$

$\text{Var}(X) = E(X^2) - (E(X))^2$

$$= \frac{20}{6} - \left(\frac{5}{3}\right)^2$$

$$= \frac{20}{6} - \frac{25}{9}$$

$$= \frac{60-50}{18}$$

$$= \frac{10}{18} = 0.56$$

variance = 0.56

Mean = 1.67

Variance = 0.56

$$(iii) f(x) = \begin{cases} 2(x-1) & 1 < x < 2 \\ 0 & \text{otherwise} \end{cases}$$

$$E(X) = \int_1^2 x f(x) dx$$

$$= 2 \int_1^2 x(x-1) dx$$

$$= 2 \int_1^2 (x^2 - x) dx$$

$$= 2 \left[\frac{x^3}{3} - \frac{x^2}{2} \right]_1^2$$

$$= 2 \left[\left(\frac{8}{3} - \frac{4}{2} \right) - \left(\frac{1}{3} - \frac{1}{2} \right) \right]$$

$$= 2 \left[\frac{8}{3} - \frac{4}{2} - \frac{1}{3} + \frac{1}{2} \right]$$

$$= 2 \left[\frac{7}{3} - \frac{3}{2} \right] = 2 \left[\frac{14-9}{6} \right]$$

$$E(X) = \frac{5}{3}$$

$$E(X^2) = \int_1^2 x^2 f(x) dx$$

$$= \int_1^2 x^2 2(x-1) dx$$

$$= 2 \int_1^2 (x^3 - x^2) dx$$

$$= 2 \left[\frac{x^4}{4} - \frac{x^3}{3} \right]_1^2$$

$$= 2 \left[\left(\frac{16}{4} - \frac{8}{3} \right) - \left(\frac{1}{4} - \frac{1}{3} \right) \right]$$

$$= 2 \left[\frac{16}{4} - \frac{8}{3} - \frac{1}{4} + \frac{1}{3} \right] = 2 \left[\frac{45-28}{12} \right]$$

$$E(X^2) = \frac{17}{6}$$

$$\text{var}(X) = E(X^2) - (E(X))^2$$

$$= \frac{17}{6} - \left(\frac{5}{3} \right)^2 = \frac{17}{6} - \frac{25}{9}$$

$$= \frac{51-50}{18} = \frac{1}{18}$$

$$\text{Mean} = \frac{5}{3}; \quad \text{Variance} = \frac{1}{18}$$

$$(iv) f(x) = \begin{cases} \frac{1}{2} e^{-\frac{x}{2}} & \text{for } x > 0 \\ 0 & \text{otherwise} \end{cases}$$

$$E(X) = \int_0^{\infty} x f(x) dx$$

$$= \int_0^{\infty} x \frac{1}{2} e^{-\frac{x}{2}} dx$$

$$= \frac{1}{2} \frac{1!}{\left(\frac{1}{2}\right)^{1+1}} = \frac{1}{2} 2^2$$

$$E(X) = 2$$

$$E(X^2) = \int_0^{\infty} x^2 f(x) dx$$

$$E(X^2) = \int_0^{\infty} x^2 \frac{1}{2} e^{-\frac{x}{2}} dx$$

$$= \frac{1}{2} \frac{2!}{\left(\frac{1}{2}\right)^{2+1}} = \frac{1}{2} 2 \times 2^3 = 8$$

$$\text{Var}(X) = E(X^2) - (E(X))^2$$

$$= 8 - 2^2 = 8 - 4 = 4$$

$$\text{Mean} = 2 \quad \text{Variance} = 4$$

2. Two balls are drawn in succession without replacement from an urn containing four red balls and three black balls. Let X be the possible outcomes drawing red balls. Find the probability mass function and mean for X.

3M

RED	BLACK	TOTAL
4	3	7

Let X be the random variable denotes the number of red balls

$$X = 0, 1, 2$$

$$f(0) = P(X = 0) = \frac{{}^4C_0 \times {}^3C_2}{{}^7C_2}$$

$$= \frac{1 \times 3}{{}^7 \times 6} = \frac{3}{21} = \frac{1}{7}$$

$$f(1) = P(X = 1) = \frac{{}^4C_1 \times {}^3C_1}{{}^7C_2}$$

$$= \frac{4 \times 3}{{}^7 \times 6} = \frac{4 \times 3}{21} = \frac{4}{7}$$

$$f(2) = P(X = 2) = \frac{{}^4C_2 \times {}^3C_0}{{}^7C_2}$$

$$= \frac{6 \times 1}{{}^7 \times 6} = \frac{6 \times 1}{21} = \frac{2}{7}$$

Probability mass function is

x	0	1	2
P(X = x)(or) f(x)	$\frac{1}{7}$	$\frac{4}{7}$	$\frac{2}{7}$

$$E(X) = \sum_x x f(x)$$

$$= 0 \left(\frac{1}{7} \right) + 1 \left(\frac{4}{7} \right) + 2 \left(\frac{2}{7} \right)$$

$$= 0 + \frac{4}{7} + \frac{4}{7}$$

$$\text{Mean} = \frac{8}{7}$$

3. If μ and σ^2 are the mean and variance of the discrete random variable X & $E(X + 3) = 10$ and $E(X + 3)^2 = 116$, find μ and σ^2 .

3M

PTA-1

$$E(X + 3) = 10$$

$$E(X) + 10 - 3 = 7 \quad \mu = 7$$

$$E(X + 3)^2 = 116 \quad E(aX + b) = aE(X) + b$$

$$E(X^2) + 6E(X) + 9 = 116$$

$$E(X^2) + 6(7) + 9 = 116$$

$$E(X^2) + 51 = 116$$

$$E(X^2) = 116 - 51 = 65$$

$$\text{var}(X) = E(X^2) - (E(X))^2 = 65 - 7^2$$

$$= 65 - 49 = 16$$

$$\mu = 7; \quad \sigma^2 = 16$$

4. Four fair coins are tossed once. Find the probability mass function, mean and variance for number of heads occurred.

$$n(s) = 2^4 = 16 \quad \text{PTA-6} \quad \text{3M}$$

Let X be the random variable denotes number of heads $X = 0, 1, 2, 3, 4$

$$P(X = 0) = \frac{{}^4C_0}{16} = \frac{1}{16}$$

$$P(X = 1) = \frac{{}^4C_1}{16} = \frac{4}{16}$$

$$P(X = 2) = \frac{{}^4C_2}{16} = \frac{6}{16}$$

$$P(X = 3) = \frac{{}^4C_3}{16} = \frac{4}{16}$$

$$P(X = 4) = \frac{{}^4C_4}{16} = \frac{1}{16}$$

Probability mass function is

x	0	1	2	3	4
$f(x)$	$\frac{1}{16}$	$\frac{4}{16}$	$\frac{6}{16}$	$\frac{4}{16}$	$\frac{1}{16}$
$xf(x)$	0	$\frac{4}{16}$	$\frac{12}{16}$	$\frac{12}{16}$	$\frac{4}{16}$
$x^2f(x)$	0	$\frac{4}{16}$	$\frac{24}{16}$	$\frac{36}{16}$	$\frac{16}{16}$

$$E(X) = \sum_x xf(x)$$

$$= 0 + \frac{4}{16} + \frac{12}{16} + \frac{12}{16} + \frac{4}{16}$$

$$= \frac{32}{16}$$

$$= 2$$

$$E(X^2) = \sum_x x^2f(x)$$

$$= 0 + \frac{4}{16} + \frac{24}{16} + \frac{36}{16} + \frac{16}{16}$$

$$= \frac{80}{16}$$

$$= 5$$

$$\text{Var}(X) = E(X^2) - (E(X))^2$$

$$= 5 - 2^2$$

$$= 1$$

$$\text{Mean} = 2$$

$$\text{Variance} = 1$$

5. A commuter train arrives punctually at a station every half hour. Each morning a student leaves his house to the train station. Let X denote the amount of time, in minutes, that the student waits for the train from the time he reaches the train station. It is known that the pdf of X is
- $$f(x) = \begin{cases} \frac{1}{30} & 0 < x < 30 \\ 0 & \text{elsewhere} \end{cases}$$
- Obtain and interpret the expected value of the random variable X . 2M

Given X be the random variable denotes the waiting time.

X is continuous on $(0, 30)$.

$$E(X) = \int_{-\infty}^{\infty} xf(x) dx$$

$$= \int_0^{30} x \frac{1}{30} dx$$

$$= \frac{1}{30} \left[\frac{x^2}{2} \right]_0^{30}$$

$$= \frac{1}{30 \times 2} [30^2 - 0]$$

$$= \frac{30 \times 30}{30 \times 2} = 15$$

$$E(X) = 15$$

Expected value of waiting time = 15 minutes

6. The time to failure in thousands of hours of an electronic equipment used in a manufactured computer has the density function
- $$f(x) = \begin{cases} 3e^{-3x} & x > 0 \\ 0 & \text{elsewhere} \end{cases}$$
- Find the expected life of this electronic equipment.

$$E(X) = \int_{-\infty}^{\infty} xf(x) dx \quad \text{PTA-3} \quad \text{2M}$$

$$= \int_0^{\infty} x' 3e^{-3x} dx$$

$$= 3 \frac{1!}{3^{1+1}} = \frac{3}{9} = \frac{1}{3}$$

$$E(X) = \frac{1}{3}$$

$$\text{unuseful life time} = \frac{1}{3} \times 1000$$

$$\text{useful lifetime} = 1 - \frac{1}{3} = \frac{2}{3} \times 1000 = 666.7 \text{ hours}$$

7. The probability density function of the random variable X is given by

$$f(x) = \begin{cases} 16x e^{-4x} & \text{for } x > 0 \\ 0 & \text{for } x \leq 0 \end{cases} \quad \text{Find the mean and variance of X.}$$

PTA-4

3M

$$\begin{aligned} E(X) &= \int_{-\infty}^{\infty} x f(x) dx \\ &= \int_0^{\infty} x 16x e^{-4x} dx \\ &= 16 \int_0^{\infty} x^2 e^{-4x} dx \\ &= 16 \frac{2!}{4^2+1} \\ &= 16 \times \frac{2}{4 \times 4 \times 4} = \frac{1}{2} \end{aligned}$$

$$\text{mean} = \frac{1}{2}$$

$$\begin{aligned} E(X^2) &= \int_{-\infty}^{\infty} x^2 f(x) dx \\ &= 16 \int_0^{\infty} x^3 e^{-4x} dx \\ &= 16 \frac{3!}{4^4} \\ &= 16 \times \frac{6}{16 \times 16} = \frac{3}{8} \end{aligned}$$

$$E(X^2) = \frac{3}{8}$$

$$\begin{aligned} \text{Var}(X) &= E(X^2) - (E(X))^2 \\ &= \frac{3}{8} - \left(\frac{1}{2}\right)^2 \\ &= \frac{3}{8} - \frac{1}{4} = \frac{3-2}{8} = \frac{1}{8} \\ \text{variance} &= \frac{1}{8} \end{aligned}$$

EXERCISE 11.5

Concept Corner

The Bernoulli distribution :

Let X be a random variable follows Bernoulli's trial.

$$X(\text{Success}) = 1, X(\text{Failure}) = 0$$

$$\text{Such that } f(x) = \begin{cases} p & x = 1 \\ q = 1 - p & x = 0 \end{cases} \quad \text{where } 0 < p < 1$$

X is called Bernoulli R.V

$$f(x) \text{ is called Bernoulli's distribution.} \quad \text{Mean} = p, \quad \text{Variance} = pq$$

Binomial distribution: The binomial random variable X, equals number of success with probability p, q = 1 - p for a failure in n-independent trials has a binomial distribution denoted by $X \sim B(n, p)$. The Probability mass function of X is

$$f(x) = {}^nC_x p^x (1-p)^{n-x} \quad x = 0, 1, 2, \dots, n$$

$$f(x) = {}^nC_x p^x q^{n-x}, \quad x = 0, 1, 2, \dots, n, \quad p + q = 1$$

$$\text{Mean } \mu = np$$

$$\text{variance} = \sigma^2 = npq = np(1-p)$$



Video class

1. Compute $P(X = k)$ for the binomial distribution, $B(n, p)$ where

2M

(i) $n = 6, p = \frac{1}{3}, k = 3$

(ii) $n = 10, p = \frac{1}{5}, k = 4$

(iii) $n = 9, p = \frac{1}{2}, k = 7$

(i) $n = 6, p = \frac{1}{3}, k = 3$

$$P(X = x) = n_{C_x} p^x (1 - p)^{n-x}$$

$$P(X = k) = 6_{C_k} \left(\frac{1}{3}\right)^k \left(1 - \frac{1}{3}\right)^{6-k}$$

$$P(X = 3) = 6_{C_3} \left(\frac{1}{3}\right)^3 \left(1 - \frac{1}{3}\right)^{6-3}$$

$$= \frac{6 \times 5 \times 4}{1 \times 2 \times 3} \left(\frac{1}{3}\right)^3 \left(\frac{2}{3}\right)^3 = \frac{20 \times 2^3}{3^6}$$

$$P(X = 3) = \frac{20 \times 8}{9 \times 9 \times 9} = \frac{160}{729}$$

(ii) $n = 10, p = \frac{1}{5}, k = 4$

PTA-6

$$P(X = k) = n_{C_k} p^k (1 - p)^{n-k}$$

$$P(X = 4) = 10_{C_4} \left(\frac{1}{5}\right)^4 \left(\frac{4}{5}\right)^{10-4}$$

$$= \frac{10 \times 9 \times 8 \times 7}{1 \times 2 \times 3 \times 4} \left(\frac{1}{5}\right)^4 \left(\frac{4}{5}\right)^6$$

$$P(X = 4) = 210 \left(\frac{1}{5}\right)^4 \left(\frac{4}{5}\right)^6$$

(iii) $n = 9, p = \frac{1}{2}, k = 7$

$$P(X = k) = n_{C_k} p^k (1 - p)^{n-k}$$

$$P(X = 7) = 9_{C_7} \left(\frac{1}{2}\right)^7 \left(1 - \frac{1}{2}\right)^{9-7}$$

$$P(X = 7) = 9_{C_7} \left(\frac{1}{2}\right)^7 \left(\frac{1}{2}\right)^2$$

2. The probability that Mr.Q hits a target at any trial is $\frac{1}{4}$. Suppose he tries at the target 10 times. Find the probability that he hits the target (i) exactly 4 times (ii) at least one time.

$n = 10$ Let X be the random variable denote number of hits

3M

$$X = 0, 1, 2, \dots, 10.$$

X follows the Binomial distribution.

$$p = \text{probability of success} = \frac{1}{4}$$

$$p = \frac{1}{4},$$

$$q = 1 - p = 1 - \frac{1}{4} = \frac{3}{4}$$

$$f(x) \text{ or } P(X = x) = n_{C_x} p^x q^{n-x}$$

$$P(X = x) \text{ or } f(x) = 10_{C_x} \left(\frac{1}{4}\right)^x \left(\frac{3}{4}\right)^{10-x}$$

(i) exactly 4 times $f(4) = 10_{C_4} \left(\frac{1}{4}\right)^4 \left(\frac{3}{4}\right)^{10-4}$

$$= 10_{C_4} \left(\frac{1}{4}\right)^4 \left(\frac{3}{4}\right)^6$$

(ii) atleast one time $= P(X \geq 1)$

$$= 1 - P(X < 1)$$

$$= 1 - P(X = 0)$$

$$= 1 - 10_{C_0} \left(\frac{1}{4}\right)^0 \left(\frac{3}{4}\right)^{10-0}$$

$$= 1 - \left(\frac{3}{4}\right)^{10}$$

$$= 1 - \frac{3^{10}}{4^{10}}$$

3. Using binomial distribution find the mean and variance of X for the following experiments

(i) A fair coin is tossed 100 times, and X denote the number of heads.

2M

(ii) A fair die is tossed 240 times, and X denote the number of times that four appeared.

(i) A fair coin is tossed 100 times and X denote the number of heads.

$n = 100$ P = Probability of getting head

$$p = \frac{1}{2}$$

$$q = 1 - p = \frac{1}{2}$$

$$\text{Mean} = np = 100 \times \frac{1}{2} = 50$$

$$\text{Variance} = npq = 100 \times \frac{1}{2} \cdot \frac{1}{2} = 25$$

(ii) A fair die is tossed 240 times, and X denote the number of times that four appeared.

$n = 240$ p = Probability of getting 4

$$p = \frac{1}{6} \quad q = 1 - p = 1 - \frac{1}{6} = \frac{5}{6}$$

$$\text{Mean} = np = 240 \times \frac{1}{6} = 40$$

$$\text{Variance} = npq = 240 \times \frac{1}{6} \times \frac{5}{6} = \frac{200}{6} = \frac{100}{3}$$

4. The probability that a certain kind of component will survive a electrical test is $\frac{3}{4}$. Find the probability that exactly 3 of the 5 components tested survive. (2M)

$$n = 5, p = \frac{3}{4}$$

Let X be the random variable denotes number of survive components.

$$X = 0, 1, 2, 3, 4, 5$$

$$X \sim B\left(5, \frac{3}{4}\right)$$

p = Probability of a components survive after test

$$p = \frac{3}{4}, q = 1 - p, q = 1 - \frac{3}{4} = \frac{1}{4}$$

$$P(X = x) = {}_nC_x p^x q^{n-x} \quad x = 0, 1, \dots, 5$$

$$P(X = x) = {}_5C_x \left(\frac{3}{4}\right)^x \left(\frac{1}{4}\right)^{5-x}$$

$$P(\text{exactly 3 survive}) = P(X = 3)$$

$$= {}_5C_3 \left(\frac{3}{4}\right)^3 \left(\frac{1}{4}\right)^{5-3}$$

$$= \frac{5 \times 4}{1 \times 2} \left(\frac{3}{4}\right)^3 \left(\frac{1}{4}\right)^2$$

$$= 10 \frac{3^3}{4^3} \frac{1}{4^2}$$

$$= \frac{270}{1024}$$

5. A retailer purchases a certain kind of electronic device from a manufacturer. The manufacturer indicates that the defective rate of the device is 5%. The inspector of the retailer randomly picks 10 items from a shipment. What is the probability that there will be (i) at least one defective item (ii) exactly two defective items? (3M)

$$n = 10$$

X be the random variable denotes number of defective items

p = Probability of a defective item

$$X \sim B(10, 0.05)$$

$$p = 5\% \Rightarrow p = 0.05; \quad q = 1 - p = 0.95$$

$$P(X = x) = {}_nC_x p^x q^{n-x}$$

$$P(X = x) = {}_{10}C_x (0.05)^x (0.95)^{10-x}$$

$$(i) P(\text{atleast one defective}) = P(X \geq 1)$$

$$= 1 - P(X < 1)$$

$$= 1 - P(X = 0)$$

$$= 1 - {}_{10}C_0 (0.05)^0 (0.95)^{10-0}$$

$$= 1 - (0.95)^{10}$$

$$(ii) P(\text{exactly 2 defective}) = P(X = 2)$$

$$= {}_{10}C_2 (0.05)^2 (0.95)^{10-2}$$

$$= 10C_2 (0.05)^2 (0.95)^8$$

6. If the probability that a fluorescent light has a useful life of at least 600 hours is 0.9, find the probabilities that among 12 such lights (5M)

(i) exactly 10 will have a useful life of at least 600 hours;

(ii) at least 11 will have a useful life of at least 600 hours:

(iii) at least 2 will not have a useful life of at least 600 hours.

$$n = 12 \quad P = \text{Probability of useful life}$$

$$p = 0.9 \quad q = 1 - p = 1 - 0.9 = 0.1$$

X be the random variable denotes useful life of atleast 600 hours of a light.

$$X \sim B(12, 0.09)$$

$$P(X = x) = {}_nC_x p^x q^{n-x} \quad P(X = x) = {}_{12}C_x (0.9)^x (0.1)^{12-x} \quad x = 0, 1, 2, \dots, 12$$

$$(i) P(\text{exactly 10}) = P(X = 10)$$

$$= {}_{12}C_{10} (0.9)^{10} (0.1)^{12-10} = 12C_{10} (0.9)^{10} (0.1)^2$$

$$(ii) P(\text{atleast 11 fluorescent light}) = P(X \geq 11) = P(X = 11) + P(X = 12)$$

$$= {}_{12}C_{11} (0.9)^{11} (0.1)^{12-11} + {}_{12}C_{12} (0.9)^{12} (0.1)^{12-12}$$

$$= 12(0.9)^{11} (0.1) + 1(0.9)^{12} (1)$$

$$= (0.9)^{11} [12 \times 0.1 + 0.9] = (0.9)^{11} [1.2 + 0.9]$$

$$P(X \geq 11) = 2.1(0.9)^{11}$$

$$(iii) P(\text{atleast 2 will not have fluorescent light}) = P(X < 11)$$

$$P(\text{atleast 2 will not}) = 1 - P(X \geq 11) = 1 - (2.1)(0.9)^{11}$$

7. The mean and standard deviation of a binomial variate X are respectively 6 and 2. Find

- (i) the probability mass function (ii) $P(X = 3)$ (iii) $P(X \geq 2)$

5M

$$\text{mean} = 6 \quad S.D = 2 \quad S.D^2 = 4$$

$$np = 6 \dots \dots \dots (1)$$

$$\text{variance} = 4 \quad npq = 4 \dots \dots \dots (2)$$

$$\frac{(2)}{(1)} \Rightarrow \frac{npq}{np} = \frac{4}{6} \Rightarrow q = \frac{2}{3}$$

$$P = 1 - q$$

$$= 1 - \frac{2}{3}$$

$$p = \frac{1}{3}$$

(i) the probability mass function

$$P(X = x) = n_{C_x} p^x q^{n-x}$$

$$(1) \Rightarrow n \frac{1}{3} = 6$$

$$n = 18$$

$$P(X = x) = 18_{C_x} \left(\frac{1}{3}\right)^x \left(\frac{2}{3}\right)^{18-x} \quad x = 0, 1 \dots 18$$

$$(ii) P(X = 3) = 18_{C_3} \left(\frac{1}{3}\right)^3 \left(\frac{2}{3}\right)^{18-3}$$

$$= 18_{C_3} \left(\frac{1}{3}\right)^3 \left(\frac{2}{3}\right)^{15}$$

$$(iii) P(X \geq 2) = 1 - P(X < 2)$$

$$= 1 - [P(X = 0) + P(X = 1)]$$

$$= 1 - \left[18_{C_0} \left(\frac{1}{3}\right)^0 \left(\frac{2}{3}\right)^{18-0} + 18_{C_1} \left(\frac{1}{3}\right)^1 \left(\frac{2}{3}\right)^{18-1} \right]$$

$$= 1 - \left[1(1) \left(\frac{2}{3}\right)^{18} + 18 \frac{1}{3} \left(\frac{2}{3}\right)^{17} \right]$$

$$= 1 - \left(\frac{2}{3}\right)^{17} \left[\frac{2}{3} + 6 \right]$$

$$\Rightarrow P(X \geq 2) = 1 - \frac{20}{3} \left(\frac{2}{3}\right)^{17}$$

8. If $X \sim B(n, p)$ such that

$4P(X = 4) = P(X = 2)$ and $n = 6$. Find the distribution, mean and standard deviation of X . PTA-6, JUL-20 3M

$$n = 6$$

$$4P(X = 4) = P(X = 2)$$

$$4(6_{C_4} (p)^4 (q)^{6-4}) = 6_{C_2} p^2 (q)^{6-2}$$

$$4 \times 6_{C_2} p^4 q^2 = 6_{C_2} p^2 q^4$$

$$4 \frac{p^4}{p^2} = \frac{q^4}{q^2}$$

$$4(p^2) = q^2$$

Taking square root

$$2p = q \Rightarrow 2p = 1 - p$$

$$3p = 1 \Rightarrow p = \frac{1}{3}$$

$$q = 1 - p = 1 - \frac{1}{3} \Rightarrow q = \frac{2}{3}$$

The distribution is

$$P(X = x) = n_{C_x} p^x q^{n-x}$$

$$= 6_{C_x} \left(\frac{1}{3}\right)^x \left(\frac{2}{3}\right)^{6-x} \quad x = 0, 1, \dots 6$$

$$\text{Mean} = np = 6 \times \frac{1}{3} = 2$$

$$\text{Variance} = npq = 6 \times \frac{1}{3} \times \frac{2}{3} = \frac{4}{3}$$

$$\text{Standard deviation} = \sqrt{npq} = \sqrt{\frac{4}{3}} = \frac{2}{\sqrt{3}}$$

9. In a binomial distribution consisting of 5 independent trials, the probability of 1 and 2 successes are 0.4096 and 0.2048 respectively. Find the mean and variance of the random variable.

$$n = 5 \quad \text{given that}$$

3M

$$P(X = 1) = 0.4096$$

$$P(X = 2) = 0.2048$$

$$P(X = x) = n_{C_x} p^x q^{n-x}$$

$$P(X = 1) = 5_{C_1} p^1 q^{5-1} = 0.4096$$

$$\Rightarrow 5pq^4 = 0.4096 \dots \dots \dots (1)$$

$$P(X = 2) = 5_{C_2} p^2 q^{5-2} = 0.2048$$

$$\Rightarrow 10p^2 q^3 = 0.2048 \dots \dots \dots (2)$$

$$(1) \div (2) \Rightarrow$$

$$\frac{5pq^4}{10p^2 q^3} = \frac{0.4096}{0.2048}$$

$$\frac{q}{2p} = 2$$

$$1 - p = 4p$$

$$1 = 5p$$

$$p = \frac{1}{5}$$

$$\text{Sub } p = \frac{1}{5} \text{ in } q$$

$$q = 1 - p = 1 - \frac{1}{5} = \frac{4}{5}$$

$$\text{Mean} = np = 5 \times \frac{1}{5} = 1$$

$$\text{Variance} = npq = 1 \times \frac{4}{5} = \frac{4}{5}$$

EXERCISE 11.6



Online Test

1. Let X be random variable with probability density function $f(x) = \begin{cases} \frac{2}{x^3} & x \geq 1 \\ 0 & x < 1 \end{cases}$

Which of the following statement is correct?

- (1) both mean and variance exist (2) mean exists but variance does not exist
(3) both mean and variance do not exist (4) variance exist but Mean does not exist.

$$\begin{aligned} \text{Mean} &= \int_1^{\infty} x f(x) dx \\ &= \int_1^{\infty} x \times \frac{2}{x^3} dx \\ &= 2 \int_1^{\infty} \frac{1}{x^2} dx \\ &= 2 \int_1^{\infty} x^{-2} dx \\ &= 2 \left[\frac{x^{-1}}{-1} \right]_1^{\infty} = -2 \left[\frac{1}{x} \right]_1^{\infty} \\ &= -2[0 - 1] = 2 \end{aligned}$$

$$\begin{aligned} \text{But } E(x^2) &= \int_1^{\infty} x^2 f(x) dx \\ &= \int_1^{\infty} x^2 \times \frac{2}{x^3} dx \\ &= 2 \int_1^{\infty} \frac{1}{x} dx \\ &= 2[\log x]_1^{\infty} \\ &= 2(\infty - 0) = 2(\infty) = \infty \\ \therefore \text{Variance does not exist.} \end{aligned}$$

2. A rod of length $2l$ is broken into two pieces at random. The probability density function of the shorter of the two pieces is $f(x) = \begin{cases} \frac{1}{l} & 0 < x < l \\ 0 & l \leq x < 2l \end{cases}$ The mean and variance of the shorter of the two pieces are respectively

PTA-1

- (1) $\frac{l}{2}, \frac{l^2}{3}$ (2) $\frac{l}{2}, \frac{l^2}{6}$ (3) $l, \frac{l^2}{12}$ (4) $\frac{l}{2}, \frac{l^2}{12}$

$$\begin{aligned} \text{Mean} = E(x) &= \int_0^l x f(x) dx = \int_0^l x \left(\frac{1}{l} \right) dx = \frac{1}{l} \left(\frac{x^2}{2} \right)_0^l = \frac{1}{l} \times \frac{l^2}{2} = \frac{l}{2} \\ E(x^2) &= \int_0^l x^2 f(x) dx = \int_0^l x^2 \times \frac{1}{l} dx = \frac{1}{l} \left[\frac{x^3}{3} \right]_0^l = \frac{1}{l} \left(\frac{l^3}{3} \right) = \frac{l^2}{3} \\ \text{Variance} &= \frac{l^2}{3} - \left(\frac{l}{2} \right)^2 = \frac{l^2}{3} - \frac{l^2}{4} = \frac{4l^2 - 3l^2}{12} = \frac{l^2}{12} \end{aligned}$$

3. Consider a game where the player tosses a six-sided fair die. If the face that comes up is 6, the player wins ₹ 36, otherwise he loses ₹ k^2 , where is the face that comes up $k = \{1, 2, 3, 4, 5\}$. The expected amount to win at this game in ₹ is

- (1) $\frac{19}{6}$ (2) $-\frac{19}{6}$ (3) $\frac{3}{2}$ (4) $-\frac{3}{2}$

x is the gaining amount.

x takes the value $-1, -4, -9, -16, -25$ and 36 .

x	-1	-4	-9	-16	-25	36
$p(x)$	$\frac{1}{6}$	$\frac{1}{6}$	$\frac{1}{6}$	$\frac{1}{6}$	$\frac{1}{6}$	$\frac{1}{6}$

$$E(x) = \frac{1}{6}(-1 - 4 - 9 - 16 - 25 + 36) = \frac{1}{6}(-55 + 36) = -\frac{19}{6}$$

4. A pair of dice numbered 1, 2, 3, 4, 5, 6 of a six-sided die and 1, 2, 3, 4 of a four-sided die is rolled and the sum is determined. Let the random variable X denote this sum. Then the number of elements in the inverse image of 7 is

JUL-21

- (1) 1 (2) 2 (3) 3 (4) 4

$\therefore$ Sum of 7 is $\{(6, 1), (5, 2), (4, 3), (3, 4)\}$

$\therefore$ Number of inverse of images 7 is 4

5. A random variable X has binomial distribution $n = 25$ and $p = 0.8$ then standard deviation of X is

(1) 6 (2) 4 (3) 3 (4) 2

$$n = 25, p = 0.8, q = 1 - p = 0.2$$

PTA-2, 6, APR-20,24, JUL-23

$$\sigma = \sqrt{npq} = \sqrt{25 \times 0.8 \times 0.2} = \sqrt{20 \times 0.2} = \sqrt{4} = 2$$

6. Let X represent the difference between the number of heads and the number of tails obtained when a coin is tossed n times. Then the possible values of X are

(1) $i + 2n, i = 0, 1, 2 \dots n$ (2) $2i - n, i = 0, 1, 2 \dots n$
 (3) $n - i, i = 0, 1, 2 \dots n$ (4) $2i + 2n, i = 0, 1, 2 \dots n$

Here n is the number of tails and i is the no of heads

No. of tails = $\{0, 1, 2 \dots \dots n\}$

$n = 1$	$X = 0, 1$
$n = 2$	$X = -2, 0, 2$
$n = 3$	$X = -3, -1, 1, 3$

$\therefore$ The possible value of X is $2i - n: i = 1, 2 \dots \dots n$

7. If the function $f(x) = \frac{1}{12}$ for $a < x < b$, represents a probability density function of a continuous random variable X , then which of the following cannot be the value of a and b ?

(1) 0 and 12 (2) 5 and 17 (3) 7 and 19 (4) 16 and 24

$$\because f(x) \text{ is probability density function} = \int_{-\infty}^{\infty} f(x) dx = 1$$

PTA-5, JUL-21

$$\int_a^b \frac{1}{12} dx = 1$$

$$b - a = 12$$

Hence the not possible values are $a = 16, b = 24$

$$b - a = 24 - 16 = 8 \neq 12$$

8. Four buses carrying 160 students from the same school arrive at a football stadium. The buses carry, respectively, 42, 36, 34, and 48 students. One of the students is randomly selected. Let X denote the number of students that were on the bus carrying the randomly selected student. One of the 4 bus drivers is also randomly selected. Let Y denote the number of students on that bus. Then $E[X]$ and $E[Y]$ respectively are

(1) 50, 40 (2) 40, 50 (3) 40.75, 40 (4) 41, 41

$$P(X = 42) = \frac{42}{160}; P(X = 36) = \frac{36}{160}; P(X = 34) = \frac{34}{160}; P(X = 48) = \frac{48}{160}$$

X	42	36	34	48
$P(X)$	$\frac{42}{160}$	$\frac{36}{160}$	$\frac{34}{160}$	$\frac{48}{160}$

$$E(X) = 42 \left(\frac{42}{160} \right) + 36 \left(\frac{36}{160} \right) + 34 \left(\frac{34}{160} \right) + 48 \left(\frac{48}{160} \right) = 40.75$$

Y	42	36	34	48
$P(Y)$	$\frac{1}{4}$	$\frac{1}{4}$	$\frac{1}{4}$	$\frac{1}{4}$

$$E(Y) = \frac{1}{4}(42 + 36 + 34 + 48) = \frac{160}{4} = 40$$

9. Two coins are to be flipped. The first coin will land on heads with probability 0.6, the second with probability 0.5. Assume that the results of the flips are independent, and let X equal the total number of heads that result. The value of $E(X)$ is

(1) 0.11 (2) 1.1 (3) 11 (4) 1

$$P(X = 0) = P(\text{no tails}) = (0.4)(0.5) = 0.20$$

$$P(X = 1) = P(\text{tail head}) = (0.4)(0.5) + (0.6)(0.5) = 0.20 + 0.30 = 0.50$$

$$P(X = 2) = P(\text{two heads}) = (0.6)(0.5) = 0.30$$

X	0	1	2
$P(X)$	0.20	0.50	0.30

$$E(X) = 0 + 1(0.50) + 2(0.30) = 0.50 + 0.60 = 1.10 = 1.1$$

10. On a multiple-choice exam with 3 possible destructives for each of the 5 questions, the probability that a student will get 4 or more correct answers just by guessing is

PTA-5

(1) $\frac{11}{243}$ (2) $\frac{3}{8}$ (3) $\frac{1}{243}$ (4) $\frac{5}{243}$

$$n = 5, p = \frac{1}{3}, q = \frac{2}{3}$$

$$P(X = x) = {}^5C_x \left(\frac{1}{3}\right)^x \left(\frac{2}{3}\right)^{5-x}$$

$$P(X \geq 4) = P(4) + P(5)$$

$$= {}^5C_4 \left(\frac{1}{3}\right)^4 \left(\frac{2}{3}\right)^1 + {}^5C_5 \left(\frac{1}{3}\right)^5 (1)$$

$$P(X \geq 4) = 5 \left(\frac{1}{3}\right)^4 \left(\frac{2}{3}\right) + (1) \left(\frac{1}{3}\right)^5$$

$$P(X \geq 4) = \left(\frac{1}{3}\right)^4 \left(\frac{10}{3} + \frac{1}{3}\right)$$

$$P(X \geq 4) = \frac{1}{81} \times \left(\frac{11}{3}\right)$$

$$P(X \geq 4) = \frac{11}{243}$$

11. If $P(X = 0) = 1 - P(X = 1)$. If $E(X) = 3 \text{ Var}(X)$, then $P(X = 0)$ is

MDL

(1) $\frac{2}{3}$ (2) $\frac{2}{5}$ (3) $\frac{1}{5}$ (4) $\frac{1}{3}$

$$P(X = 0) = q$$

$$P(X = 0) + P(X = 1) = 1$$

$$\therefore X = 0, 1$$

$$p = P(X = 1), q = P(X = 0)$$

$$E(X) = 3 \text{ Var}(X)$$

$$np = 3npq$$

$$q = P(X = 0) = \frac{1}{3}$$

12. If X is a binomial random variable with expected value 6 and variance 2.4, Then $P(X = 5)$ is

(1) $\binom{10}{5} \left(\frac{3}{5}\right)^6 \left(\frac{2}{5}\right)^4$ (2) $\binom{10}{5} \left(\frac{3}{5}\right)^{10}$ (3) $\binom{10}{5} \left(\frac{3}{5}\right)^4 \left(\frac{2}{5}\right)^6$ (4) $\binom{10}{5} \left(\frac{3}{5}\right)^5 \left(\frac{2}{5}\right)^5$

PTA-1

$$E(x) = np = 6$$

$$\text{Variance}(x) = npq = 2.4$$

$$6q = 2.4$$

$$q = \frac{2.4}{6} = \frac{24}{60} = \frac{4}{10} = \frac{2}{5}$$

$$p = \frac{3}{5}$$

$$n \left(\frac{3}{5}\right) = 6$$

$$\Rightarrow 3n = 30$$

$$n = 10$$

$$P(x = 5) = {}^{10}C_5 \left(\frac{3}{5}\right)^5 \left(\frac{2}{5}\right)^5$$

13. The random variable X has the probability density function $f(x) = \begin{cases} ax + b & 0 < x < 1 \\ 0 & \text{otherwise} \end{cases}$ and $E(X) = \frac{7}{12}$, then a and b respectively

PTA-4

- (1) 1 and $\frac{1}{2}$ (2) $\frac{1}{2}$ and 1 (3) 2 and 1 (4) 1 and 2

$$\begin{aligned} E(X) &= \int_0^1 x f(x) dx \\ &= \int_0^1 x (ax + b) dx \\ &= \int_0^1 (ax^2 + bx) dx \\ &= \left[a \frac{x^3}{3} + b \frac{x^2}{2} \right]_0^1 = \frac{a}{3} + \frac{b}{2} = \frac{7}{12} \end{aligned}$$

$$\begin{aligned} \text{Put } a &= 1; b = \frac{1}{2} \\ E(X) &= \frac{1}{3} + \frac{\frac{1}{2}}{2} \\ &= \frac{1}{3} + \frac{1}{4} = \frac{7}{12} \\ \therefore a &= 1, b = \frac{1}{2} \end{aligned}$$

14. Suppose that X takes 0, 1 and 2. If for some constant k, $P(X = i) = k P(X = i - 1)$ for $i = 1, 2$ and $P(X = 0) = \frac{1}{7}$, then the value of k is

PTA-2, APR-23, 25

- (1) 1 (2) 2 (3) 3 (4) 4

$$\begin{aligned} P(X = 1) &= kP(X = 0) = \frac{1}{7}k \\ P(X = 2) &= kP(X = 1) = \frac{1}{7}k^2 \\ \therefore \frac{1}{7} + \frac{1}{7}k + \frac{1}{7}k^2 &= 2 \\ k^2 + k &= 6 \end{aligned}$$

$$\begin{aligned} k^2 + k - 6 &= 0 \\ (k + 3)(k - 2) &= 0 \\ k &= -3, 2 \\ \Rightarrow k &= 2 \end{aligned}$$

15. Which of the following is a discrete random variable?

- I. The number of cars crossing a particular signal in a day.
II. The number of customers in a queue to buy train tickets at a moment.
III. The time taken to complete a telephone call.

- (1) I and II (2) II only (3) III only (4) II and III

16. If $f(x) = \begin{cases} 2x & 0 \leq x \leq a \\ 0 & \text{otherwise} \end{cases}$ is a probability density function of a random variable, then the value of a is

PTA-6, APR-22

- (1) 1 (2) 2 (3) 3 (4) 4

$$\begin{aligned} \text{Total probability} &= 1 \\ \int_0^a 2x dx &= 1 \\ [x^2]_0^a &= 1 \\ a^2 &= 1 \Rightarrow a = 1 \end{aligned}$$

17. The probability mass function of a random variable is defined as:

PTA-3

x	-2	-1	0	1	2
f(x)	k	2k	3k	4k	5k

Then E(X) is equal to:

- (1) $\frac{1}{15}$ (2) $\frac{1}{10}$ (3) $\frac{1}{3}$ (4) $\frac{2}{3}$

$$\begin{aligned} 15k &= 1 \Rightarrow k = \frac{1}{15} \\ E(x) &= -2k - 2k + 0 + 4k + 10k = 10k = 10 \left(\frac{1}{15} \right) = \frac{2}{3} \end{aligned}$$

18. Let X have a Bernoulli distribution with mean 0.4, then the variance of $(2X - 3)$ is

- (1) 0.24 (2) 0.48 (3) 0.6 (4) 0.96

For Bernoulli $E(x) = p = 0.4$

$$q = 1 - 0.4 = 0.6$$

$$\text{Variance}(x) = pq = (0.4)(0.6) = 0.24$$

$$\text{Variance}(2x - 3) = 4 \text{ variance}(x) = 4(0.24) = 0.96$$

19. If in 6 trials, X is a binomial variable which follows the relation $9P(X = 4) = P(X = 2)$, then the probability of success is

- (1) 0.125 (2) 0.25 (3) 0.375 (4) 0.75

APR-25, JUL-24

$$n = 6$$

$$9p(x = 4) = p(x = 2)$$

$$9(6C_4 p^4 q^2) = 6C_2 (p^2 q^4)$$

$$9(p^4 q^2) = p^2 q^4$$

$$9p^2 = q^2 \Rightarrow q = 3p$$

$$p + q = 1$$

$$p + 3p = 1$$

$$4p = 1 \Rightarrow p = \frac{1}{4} = 0.25$$

20. A computer salesperson knows from his past experience that he sells computers to one in every twenty customers who enter the showroom. What is the probability that he will sell a computer to exactly two of the next three customers?

- (1) $\frac{57}{20^3}$ (2) $\frac{57}{20^2}$ (3) $\frac{19^3}{20^3}$ (4) $\frac{57}{20}$

$$p = \frac{1}{20}; q = \frac{19}{20}; n = 3$$

$$p(X = x) = 3C_x \left(\frac{1}{20}\right)^x \left(\frac{19}{20}\right)^{3-x}$$

$$p(x = 2) = 3C_2 \left(\frac{1}{20}\right)^2 \left(\frac{19}{20}\right)^1$$

$$= 3 \times \frac{1}{20^2} \times \frac{19}{20} = \frac{57}{20^3}$$

Creative Questions

1. The value of $\text{Var}(3)$ is

- (1) 0 (2) 3 (3) $\text{Var}(3)$ (4) 9

APR-24

2. If a fair die is thrown once then the probability to get a prime number on the face is

- (1) 0 (2) $\frac{1}{2}$ (3) $\frac{1}{4}$ (4) $\frac{1}{6}$

JUL-22

Prime numbers on dice = 2, 3, 5

Total numbers = 6

$$\text{Required Probability} = \frac{3}{6} = \frac{1}{2}$$

3. A random variable X takes the probability mass function

JUL-22

X	-2	3	1
$P(X = x)$	$\frac{\lambda}{6}$	$\frac{\lambda}{4}$	$\frac{\lambda}{12}$

The value of λ is

- (1) 1 (2) 2 (3) 3 (4) 4

Total probability = 1

$$\frac{\lambda}{6} + \frac{\lambda}{4} + \frac{\lambda}{12} = 1$$

$$\frac{4\lambda + 6\lambda + 2\lambda}{24} = 1 \Rightarrow 12\lambda = 24 \Rightarrow \lambda = 2$$



John von Neumann

Chapter - 12

Discrete Mathematics

EXERCISE 12.1



Video class

Concept Corner

➤ Binary operation (or) closure property:

A Binary operation $*$ on S is $\forall a, b \in S, a * b$ is unique and $a * b \in S$, where S is any non-empty set.

Properties:

1. Commutative property: $a * b = b * a, \forall a, b \in S$

2. Associative Property: $a * (b * c) = (a * b) * c \quad \forall a, b, c \in S$

3. Existence of identity property: $e \in S$ is an identity element $a * e = e * a = a, \forall a \in S$

4. Existence of Inverse Property: $b \in S$ is said to be Inverse element of a

$$a * b = b * a = e, \quad \forall a \in S \text{ i.e., } b = a^{-1}$$

➤ Some binary operations on Boolean Matrices: It is a real matrix whose entries are either 0 or 1.

(1) Join of A and B $A \vee B = [a_{ij}] \vee [b_{ij}] = [a_{ij} \vee b_{ij}] = [c_{ij}]$

$$[c_{ij}] = \begin{cases} 1, & \text{if either } a_{ij} = 1 \text{ or } b_{ij} = 1 \\ 0, & \text{if } a_{ij} = 0 \text{ and } b_{ij} = 0 \end{cases}$$

(2) Meet of A and B $A \wedge B = [a_{ij}] \wedge [b_{ij}] = [a_{ij} \wedge b_{ij}] = [c_{ij}]$

$$[c_{ij}] = \begin{cases} 1, & \text{if } a_{ij} = 1 \text{ and } b_{ij} = 1 \\ 0, & \text{if either } a_{ij} = 0 \text{ or } b_{ij} = 0 \end{cases}$$

Note: $(a \vee b) = \max\{a, b\}; (a \wedge b) = \min\{a, b\}, a, b \in \{0, 1\}$

Properties:

1. Closure: $A, B \in \mathbb{B}, A \vee B = [a_{ij}] \vee [b_{ij}] = [a_{ij} \vee b_{ij}] \in \mathbb{B}$ because, $(a_{ij} \vee b_{ij}) = 0 / 1$

2. Associative Property: $A \vee (B \vee C) = (A \vee B) \vee C, \quad \forall A, B, C \in \mathbb{B}$

3. Existence of identity property: $\forall A \in \mathbb{B}$ null matrix $0 \in \mathbb{B}, A \vee 0 = 0 \vee A = A$

4. Existence of Inverse Property: $A \in \mathbb{B}, A \vee B = B \vee A = 0$

➤ Modular Arithmetic:

(i) Addition modulo: $a, b \in Z_n$ then $a +_n b =$ the remainder of $a + b$ on division by n .

(ii) Multiplication modulo n : $a, b \in Z_n$ then, $a \times_n b$ the remainder of $a \times b$ on division by n .

1. Determine whether $*$ is a binary operation on the sets given below.

2M

(i) $a * b = a \cdot |b|$ on $\mathbb{R}$

Let $a * b \in \mathbb{R},$

$|b| \in \mathbb{R}$

$a \cdot |b| \in \mathbb{R}$

$a * b \in \mathbb{R} \quad \forall a, b \in \mathbb{R}$

Yes,

$*$ is the binary operation on $\mathbb{R}$

(ii) $a * b = \min(a, b)$ on

$A = \{1, 2, 3, 4, 5\}$

Let $a, b \in A$

$\min(a, b) \in A$

$a * b \in A \quad \forall a, b \in A$

Yes,

$*$ is the binary operation on A

Example

$\min(1, 2) = 1 \in A$
 $\min(2, 3) = 2 \in A$

(iii) $(a * b) = a\sqrt{b}$ is binary on $\mathbb{R}$.

Let $a, b \in \mathbb{R}$

Root of $-ve$ numbers not in $\mathbb{R}$

$\therefore a\sqrt{b} \notin \mathbb{R}$

$[\because 2, -2 \in \mathbb{R}, \quad 2\sqrt{-2} \notin \mathbb{R}]$

$a * b \notin \mathbb{R}$

No, $*$ is not binary on $\mathbb{R}$

2. On $\mathbb{Z}$, define $*$ by $(m * n) = m^n + n^m$:

$\forall m, n \in \mathbb{Z}$. Is $*$ binary on $\mathbb{Z}$? (2M)

Let $m, n \in \mathbb{Z}$

Take $m = 2, n = -2$ (PTA-3)

$$m^n + n^m = 2^{-2} + (-2)^2$$

$$= \frac{1}{4} + 4$$

$$= \frac{17}{4} \notin \mathbb{Z}$$

$$m^n + n^m \notin \mathbb{Z}$$

$$\therefore m * n \notin \mathbb{Z}$$

$\therefore$ No, $*$ is not binary on $\mathbb{Z}$

3. Let $*$ be defined on $\mathbb{R}$ by (PTA-2, APR-22)

$$(a * b) = a + b + ab - 7.$$

Is $*$ binary on $\mathbb{R}$? If so, find $3 * \left(\frac{-7}{15}\right)$ (2M)

Let $a, b \in \mathbb{R}$,

Clearly $a, b, ab \in \mathbb{R}$

$$a + b + ab - 7 \in \mathbb{R}$$

$$a * b \in \mathbb{R}$$

$*$ is binary operator on $\mathbb{R}$

$$3 * \left(-\frac{7}{15}\right) = 3 - \frac{7}{15} + 3\left(-\frac{7}{15}\right) - 7$$

$$= \frac{45 - 7 - 21 - 105}{15}$$

$$= \frac{45 - 133}{15}$$

$$= \frac{-88}{15}$$

4. Let $A = \{a + \sqrt{5}b : a, b \in \mathbb{Z}\}$. Check whether the usual multiplication is a binary operation on A . (2M)

$$\text{Let } x = a + \sqrt{5}b,$$

$$y = c + d\sqrt{5}$$

$$x, y \in A, \quad a, b, c, d \in \mathbb{Z}$$

$$xy = (a + \sqrt{5}b)(c + \sqrt{5}d)$$

$$= ac + 5bd + \sqrt{5}ad + \sqrt{5}bc$$

$$= (ac + 5bd) + \sqrt{5}(ad + bc) \in A$$

$$\therefore xy \in A$$

Yes, Multiplication is binary on A . (PTA-5)

5. (i) Define an operation $*$ on $\mathbb{Q}$ as follows

$a * b = \left(\frac{a+b}{2}\right); a, b \in \mathbb{Q}$. Examine the closure, commutative and associative properties satisfied by $*$ on $\mathbb{Q}$ (5M)

1. Closure property (JUL-21)

Given, $a, b \in \mathbb{Q}; a + b \in \mathbb{Q}$

$$\frac{a+b}{2} \in \mathbb{Q}$$

$$a * b \in \mathbb{Q} \quad \forall a, b \in \mathbb{Q}$$

$*$ is closed on $\mathbb{Q}$

2. Commutative property

$$a * b = \frac{a+b}{2} = \frac{b+a}{2} = b * a$$

$$a * b = b * a \quad \forall a, b \in \mathbb{Q}$$

$*$ is commutative on $\mathbb{Q}$

3. Associative property

$$a * (b * c) = a * \left(\frac{b+c}{2}\right)$$

$$= \frac{a + \left(\frac{b+c}{2}\right)}{2} = \frac{2a+b+c}{2} \times \frac{1}{2}$$

$$= \frac{2a+b+c}{4} \dots (1)$$

$$(a * b) * c = \left(\frac{a+b}{2}\right) * c$$

$$= \frac{a+b+2c}{2} \times \frac{1}{2} = \frac{a+b+2c}{4} \dots (2)$$

From (1) and (2), $(a * b) * c \neq a * (b * c)$

$*$ is not associative on $\mathbb{Q}$

(ii) Define an operation $*$ on $\mathbb{Q}$ as follows:

$a * b = \left(\frac{a+b}{2}\right); a, b \in \mathbb{Q}$. Examine the existence of identity and the existence of inverse for the operation $*$ on $\mathbb{Q}$.

(i) Existence of Identity

Let $a \in \mathbb{Q}$, e be the identity element on $\mathbb{Q}$

By definition of $*$, $a * e = \frac{a+e}{2}$

By definition of e , $a * e = a$

$$\frac{a+e}{2} = a$$

$$a + e = 2a$$

$$e = 2a - a = a \quad \forall a \in \mathbb{Q}$$

i.e., every element is a identity element.

This is not possible

$\mathbb{Q}$ has no identity element.

(ii) Existence of Inverse

$*$ has no Identity element cannot defined

$$\text{as } a * a^{-1} = a^{-1} * a = e$$

$\mathbb{Q}$ has no inverse.

6. Fill in the following table so that the binary operation $*$ on $A = \{a, b, c\}$ is commutative.

2M

$*$	a	b	c
a	b	?	?
b	c	b	a
c	a	?	c

(i) From table $b * a = c \Rightarrow a * b = c$

(ii) $c * a = a \Rightarrow a * c = a$

(iii) $b * c = a \Rightarrow c * b = a$

$*$	a	b	c
a	b	c	a
b	c	b	a
c	a	a	c

7. Consider the binary operation $*$ defined on the set $A = \{a, b, c, d\}$ by the following table:

2M

$*$	a	b	c	d
a	a	c	b	d
b	d	a	b	c
c	c	d	a	a
d	d	b	a	c

Is it commutative and associative?

1. Commutative property

$$a * b = c \text{ but } b * a = d \Rightarrow a * b \neq b * a$$

$$a * c = b \text{ but } c * a = c \Rightarrow a * c \neq c * a$$

$$a * b \neq b * a; a * c \neq c * a$$

No, $*$ is not commutative.

2. Associative property

$$(a * b) * c = c * c = a$$

$$a * (b * c) = a * b = c$$

$$(a * b) * c \neq a * (b * c)$$

No, $*$ is not associative

8. Let $A = \begin{pmatrix} 1 & 0 & 1 & 0 \\ 0 & 1 & 0 & 1 \\ 1 & 0 & 0 & 1 \end{pmatrix}$, $B = \begin{pmatrix} 0 & 1 & 0 & 1 \\ 1 & 0 & 1 & 0 \\ 1 & 0 & 0 & 1 \end{pmatrix}$, $C = \begin{pmatrix} 1 & 1 & 0 & 1 \\ 0 & 1 & 1 & 0 \\ 1 & 1 & 1 & 1 \end{pmatrix}$ be any three Boolean

matrices of the same type. Find (i) $A \vee B$ (ii) $A \wedge B$ (iii) $(A \vee B) \wedge C$ (iv) $(A \wedge B) \vee C$

3M

$$(i) A \vee B = \begin{pmatrix} 1 & 0 & 1 & 0 \\ 0 & 1 & 0 & 1 \\ 1 & 0 & 0 & 1 \end{pmatrix} \vee \begin{pmatrix} 0 & 1 & 0 & 1 \\ 1 & 0 & 1 & 0 \\ 1 & 0 & 0 & 1 \end{pmatrix} = \begin{pmatrix} 1 & 1 & 1 & 1 \\ 1 & 1 & 1 & 1 \\ 1 & 0 & 0 & 1 \end{pmatrix}$$

$$(ii) A \wedge B = \begin{pmatrix} 1 & 0 & 1 & 0 \\ 0 & 1 & 0 & 1 \\ 1 & 0 & 0 & 1 \end{pmatrix} \wedge \begin{pmatrix} 0 & 1 & 0 & 1 \\ 1 & 0 & 1 & 0 \\ 1 & 0 & 0 & 1 \end{pmatrix} = \begin{pmatrix} 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 1 & 0 & 0 & 1 \end{pmatrix}$$

$$(iii) (A \vee B) \wedge C = \begin{pmatrix} 1 & 1 & 1 & 1 \\ 1 & 1 & 1 & 1 \\ 1 & 0 & 0 & 1 \end{pmatrix} \wedge \begin{pmatrix} 1 & 1 & 0 & 1 \\ 0 & 1 & 1 & 0 \\ 1 & 1 & 1 & 1 \end{pmatrix} = \begin{pmatrix} 1 & 1 & 0 & 1 \\ 0 & 1 & 1 & 0 \\ 1 & 0 & 0 & 1 \end{pmatrix}$$

$$(iv) (A \wedge B) \vee C = \begin{pmatrix} 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 1 & 0 & 0 & 1 \end{pmatrix} \vee \begin{pmatrix} 1 & 1 & 0 & 1 \\ 0 & 1 & 1 & 0 \\ 1 & 1 & 1 & 1 \end{pmatrix} = \begin{pmatrix} 1 & 1 & 0 & 1 \\ 0 & 1 & 1 & 0 \\ 1 & 1 & 1 & 1 \end{pmatrix}$$

9. (i) Let $M = \left\{ \begin{pmatrix} x & x \\ x & x \end{pmatrix} ; x \in \mathbb{R} - \{0\} \right\}$ and let $*$ be the matrix multiplication. Determine whether M is closed under $*$. If so, examine the commutative & associative properties satisfied by $*$ on M . JUL-23

$$\text{Let } A = \begin{pmatrix} x & x \\ x & x \end{pmatrix}, B = \begin{pmatrix} y & y \\ y & y \end{pmatrix}$$

where, $y \in \mathbb{R} - \{0\}, A, B \in M$ 5M

$$\because x, y \in \mathbb{R} - \{0\}, \\ xy \in \mathbb{R} - \{0\}$$

1) Closure property:

$$A, B \in M$$

$$\begin{aligned} A * B &= \begin{pmatrix} x & x \\ x & x \end{pmatrix} \begin{pmatrix} y & y \\ y & y \end{pmatrix} \\ &= \begin{pmatrix} 2xy & 2xy \\ 2xy & 2xy \end{pmatrix} \in M \end{aligned}$$

$$A, B \in M \Rightarrow A * B \in M$$

$*$ is closed on M

2) Commutative property :

$$A, B \in M$$

$$\begin{aligned} A * B &= \begin{pmatrix} x & x \\ x & x \end{pmatrix} \begin{pmatrix} y & y \\ y & y \end{pmatrix} \\ &= \begin{pmatrix} 2xy & 2xy \\ 2xy & 2xy \end{pmatrix} \\ &= \begin{pmatrix} 2yx & 2yx \\ 2yx & 2yx \end{pmatrix} \\ &= \begin{pmatrix} y & y \\ y & y \end{pmatrix} \begin{pmatrix} x & x \\ x & x \end{pmatrix} \end{aligned}$$

$$A * B = B * A$$

$*$ is commutative on M

3) Associative property :

Matrix multiplication is always associative.

$$A * (B * C) = (A * B) * C \quad \forall A, B, C \in M$$

$*$ is associative on M

9. (ii) Let $M = \left\{ \begin{pmatrix} x & x \\ x & x \end{pmatrix} : x \in \mathbb{R} - \{0\} \right\}$ & let $*$ be the matrix multiplication. Determine whether M is closed under $*$. If so, examine the existence of identity, existence of inverse properties for the operation $*$ on M .

1. Closure property:

$$A, B \in M, \text{ where } A = \begin{pmatrix} x & x \\ x & x \end{pmatrix}, B = \begin{pmatrix} y & y \\ y & y \end{pmatrix} \\ x, y \in \mathbb{R} - \{0\}$$

$$\begin{aligned} A * B &= \begin{pmatrix} x & x \\ x & x \end{pmatrix} \begin{pmatrix} y & y \\ y & y \end{pmatrix} \\ &= \begin{pmatrix} 2xy & 2xy \\ 2xy & 2xy \end{pmatrix} \in M \end{aligned}$$

$$A, B \in M \Rightarrow A * B \in M$$

$*$ is closed on M

2. Existence of Identity property

Let $A \in M, E = \begin{pmatrix} e & e \\ e & e \end{pmatrix}$ be the identity element

$$\begin{aligned} AE &= A \\ \begin{pmatrix} x & x \\ x & x \end{pmatrix} \begin{pmatrix} e & e \\ e & e \end{pmatrix} &= \begin{pmatrix} x & x \\ x & x \end{pmatrix} \\ \begin{pmatrix} 2xe & 2xe \\ 2xe & 2xe \end{pmatrix} &= \begin{pmatrix} x & x \\ x & x \end{pmatrix} \\ 2xe &= x \\ 2e &= 1 \\ e &= \frac{1}{2} \in \mathbb{R} - \{0\} \end{aligned}$$

$$E = \begin{pmatrix} \frac{1}{2} & \frac{1}{2} \\ \frac{1}{2} & \frac{1}{2} \end{pmatrix} \in M, \text{ satisfies } AE = A$$

Similarly, $EA = A, \forall A \in M$

$*$ has Identity element on M

3. Existence of Inverse property

$$A^{-1} = \begin{pmatrix} x^{-1} & x^{-1} \\ x^{-1} & x^{-1} \end{pmatrix} \text{ be the inverse of } A \\ AA^{-1} = E$$

$$\begin{aligned} \begin{pmatrix} x & x \\ x & x \end{pmatrix} \begin{pmatrix} x^{-1} & x^{-1} \\ x^{-1} & x^{-1} \end{pmatrix} &= \begin{pmatrix} 1/2 & 1/2 \\ 1/2 & 1/2 \end{pmatrix} \\ \begin{pmatrix} 2xx^{-1} & 2xx^{-1} \\ 2xx^{-1} & 2xx^{-1} \end{pmatrix} &= \begin{pmatrix} 1/2 & 1/2 \\ 1/2 & 1/2 \end{pmatrix} \\ 2xx^{-1} &= 1/2 \end{aligned}$$

$$x^{-1} = \frac{1}{4x} \in \mathbb{R} - \{0\}$$

$$A^{-1} = \begin{pmatrix} 1/4x & 1/4x \\ 1/4x & 1/4x \end{pmatrix} \in M \text{ is the inverse of } A \text{ in } M$$

$$\text{Similarly, } A^{-1}A = E \quad \forall A \in M$$

$*$ has inverse on M

10. (i) Let A be $Q \setminus \{1\}$, Define $*$ on A by $x * y = x + y - xy$. Is $*$ binary on A ? If so, examine the commutative and associative properties satisfied by $*$ on A .

5M

$$A = Q - \{1\}, x, y \in A \quad x \neq 1, y \neq 1$$

$*$ is defined by $x * y = x + y - xy$

1. Closure property

$$\text{Let } x, y \in A \quad x \neq 1, y \neq 1$$

$$x - 1 \neq 0, y - 1 \neq 0$$

$$(x - 1)(y - 1) \neq 0$$

$$xy - x - y + 1 \neq 0$$

$$1 \neq x + y - xy$$

$$x * y \neq 1 \Rightarrow x * y \in A$$

$$x, y \in A \Rightarrow x * y \in A$$

$*$ is closed on A

2. Commutative property

$$\text{Let } x, y \in A$$

$$x * y = x + y - xy$$

$$= y + x - yx$$

$$x * y = y * x \quad \forall x, y \in A$$

$*$ is commutative on A

3. Associative property

$$(x * y) * z = (x + y - xy) * z$$

$$= (x + y - xy) + z - (x + y - xy)z$$

$$(x * y) * z = x + y + z - xy - xz - yz + xyz \dots\dots\dots(1)$$

$$x * (y * z) = x * (y + z - yz)$$

$$= x + (y + z - yz) - x(y + z - yz)$$

$$x * (y * z) = x + y + z - xy - xz - yz + xyz \dots\dots\dots(2)$$

$$\text{From (1), (2), } (x * y) * z = x * (y * z) \quad \forall x, y, z \in A$$

$*$ is associative on A

- (ii) Examine the existence of identity, existence of inverse properties for the operation $*$ on A

1. Closure property

$$\text{Let } x, y \in A \quad x \neq 1, y \neq 1$$

$$x - 1 \neq 0, y - 1 \neq 0$$

$$(x - 1)(y - 1) \neq 0$$

$$xy - x - y + 1 \neq 0$$

$$1 \neq x + y - xy$$

$$x * y \neq 1$$

$$x, y \in A \Rightarrow x * y \in A$$

$*$ is closed on A

2. Existence of Identity property

Let $x \in A$, e be the identity element.

By definition of $*$, $x * e = x + e - xe$

By definition of e , $x * e = x$

$$x + e - xe = x$$

$$e(1 - x) = 0$$

$$e = \frac{0}{1-x} = 0 \in A$$

Identity element $e = 0 \in A$

$*$ has identity element on A

3. Existence of Inverse property

Let $x \in A$, x^{-1} be the inverse of x

By definition of $*$, $x * x^{-1} = x + x^{-1} - xx^{-1}$

By definition of x^{-1} , $x * x^{-1} = e$

$$x + x^{-1} - xx^{-1} = 0$$

$$x^{-1}(1 - x) = -x$$

$$x^{-1} = \frac{-x}{1-x} \in A$$

$$\text{Inverse of } x \text{ is } x^{-1} = \frac{-x}{1-x} \in A \quad \forall x \in A$$

$*$ has inverse element $\forall x \in A$

Mathematical Logic and Truth Tables

3. Truth table for OR [v]

p	q	$p \vee q$
T	T	T
T	F	T
F	T	T
F	F	F

6. Truth table

Exclusive OR (EOR) $\bar{V}$

p	q	$p \bar{\vee} q$
T	T	F
T	F	T
F	T	T
F	F	F

10. **Duality:** The dual of a statement formula is obtained by replacing

 $\mathbb{T}$ by $\mathbb{F}$ and $\mathbb{F}$ by $\mathbb{T}$

(ii) $p \wedge p = p$

(ii) $p \wedge q = q \wedge p$
$$(ii) \quad p \wedge (q \wedge r) = (p \wedge q) \wedge r$$
$$(ii) \ p \wedge (q \vee r) = (p \wedge q) \vee (p \wedge r)$$

(ii) $p \wedge T = p$ and $p \wedge F = F$

(ii) $\neg T = F$ and $\neg F = T$

$$(ii) \neg(p \vee q) = \neg p \wedge \neg q$$
$$(ii) \quad p \wedge (p \vee q) = p$$

2M

(v) $p \leftrightarrow q$: Jupiter is a planet if and only if India is an Island.

2. Write each of the following sentences in symbolic form using statement variables p and q .

2M

p : 19 is a prime number q : All the angles of a triangle are equal.

(i) 19 is not a prime number and all the angles of a triangle are equal.

19 is not a prime number and all the angles of a triangle are equal $\neg p \wedge q$

$\neg$ p $\wedge$ q

(ii) 19 is a prime number or all the angles of a triangle are not equal

19 is a prime number or all the angles of a triangle are not equal $p \vee \neg q$

p $\vee$ $\neg$ q

(iii) 19 is a prime number and all the angles of a triangle are equal

19 is a prime number and all the angles of a triangle are equal $p \wedge q$

p $\wedge$ q

(iv) 19 is not a prime number

19 is not a prime number $\neg p$

$\neg$ p

2M

3. Determine the truth value of each of the following statements

(i) If $6 + 2 = 5$, then the milk is white

p : $6 + 2 = 5$ F

q : The milk is white T

Symbolic form is $p \rightarrow q$

$F \rightarrow T$

Truth value is T

(ii) China is in Europe or $\sqrt{3}$ is an integer.

p : China is in Europe F

q : $\sqrt{3}$ is an integer F

Symbolic form is $p \vee q$

$F \vee F$

Truth value is F

(iii) It is not true that $5 + 5 = 9$ or Earth is a planet

p : It is true that $5 + 5 = 9$

q : Earth is a planet.

Symbolic form is $\neg p \vee q$

Truth value $T \vee T$

Truth value is T

(iv) 11 is a prime number and all the sides of a rectangle are equal.

p : 11 is a prime number

q : all the sides of a rectangle are equal

Symbolic form is $p \wedge q$

Truth value $T \wedge F$

Truth value is F

4. Which one of the following sentences is a proposition?

2M

(i) $4 + 7 = 12$ (proposition)

(ii) What are you doing? (not a proposition)

(iii) $3^n \leq 81$, $n \in \mathbb{N}$ (proposition)

(iv) Peacock is our national bird (proposition)

(v) How tall this mountain is! (not a proposition)

$\therefore$ (i),(iii),(iv) are propositions.

5. Write the converse, inverse and contrapositive of each of the following implication.

PTA-6

(i) If x and y are numbers such that $x = y$, then $x^2 = y^2$

3M

p : x and y are numbers such that $x = y$

q : $x^2 = y^2$

Given statement symbolic form is $p \rightarrow q$

(1) converse $q \rightarrow p$: If x and y are numbers such that $x^2 = y^2$ then $x = y$

(2) Inverse $\neg p \rightarrow \neg q$: If x and y are numbers such that $x \neq y$ then $x^2 \neq y^2$

(3) contrapositive: $\neg q \rightarrow \neg p$: If x and y are numbers such that $x^2 \neq y^2$ then $x \neq y$

(ii) If a quadrilateral is a square then it is a rectangle

p : a quadrilateral is a square

q : a quadrilateral is a rectangle

Given statement is $p \rightarrow q$

1. converse: $q \rightarrow p$: If a quadrilateral is a rectangle then it is a square

2. Inverse: $\neg p \rightarrow \neg q$: If a quadrilateral is not a square then it is not a rectangle.

3. contrapositive: $\neg q \rightarrow \neg p$: If a quadrilateral is not a rectangle then it is not a square.

6. Construct the truth table for the following statements

3M

(i) $\neg p \wedge \neg q$

p	q	$\neg p$	$\neg q$	$\neg p \wedge \neg q$
T	T	F	F	F
T	F	F	T	F
F	T	T	F	F
F	F	T	T	T

(ii) $\neg(p \wedge \neg q)$

p	q	$\neg q$	$p \wedge \neg q$	$\neg(p \wedge \neg q)$
T	T	F	F	T
T	F	T	T	F
F	T	F	F	T
F	F	T	F	T

(iii) $(p \vee q) \vee \neg q$

p	q	$p \vee q$	$\neg q$	$(p \vee q) \vee (\neg q)$
T	T	T	F	T
T	F	T	T	T
F	T	T	F	T
F	F	F	T	T

(iv) $(\neg p \rightarrow r) \wedge (p \leftrightarrow q)$

p	q	r	$\neg p$	$\neg p \rightarrow r$	$p \leftrightarrow q$	$(\neg p \rightarrow r) \wedge (p \leftrightarrow q)$
T	T	T	F	T	T	T
T	T	F	F	T	T	T
T	F	T	F	T	F	F
T	F	F	F	T	F	F
F	T	T	T	T	F	F
F	T	F	T	F	F	F
F	F	T	T	T	T	T
F	F	F	T	F	T	F

7. Verify whether the following compound propositions are tautologies or contradictions or contingency

(i) $(p \wedge q) \wedge \neg(p \vee q)$

3M

p	q	$p \wedge q$	$p \vee q$	$\neg(p \vee q)$	$(p \wedge q) \wedge \neg(p \vee q)$
T	T	T	T	F	F
T	F	F	T	F	F
F	T	F	T	F	F
F	F	F	F	T	F

Last column contains only F .
This is a contradiction.

(ii) $((p \vee q) \wedge \neg p) \rightarrow q$

3M

p	q	$p \vee q$	$\neg p$	$(p \vee q) \wedge \neg p$	$((p \vee q) \wedge \neg p) \rightarrow q$
T	T	T	F	F	T
T	F	T	F	F	T
F	T	T	T	T	T
F	F	F	T	F	T

Last column contains only T .
This is tautology.

(iii) $(p \rightarrow q) \leftrightarrow (\neg p \rightarrow q)$

PTA-2, JUL-22

5M

p	q	$p \rightarrow q$	$\neg p$	$\neg p \rightarrow q$	$(p \rightarrow q) \leftrightarrow (\neg p \rightarrow q)$
T	T	T	F	T	T
T	F	F	F	T	F
F	T	T	T	T	T
F	F	T	T	F	F

Last column contains T and F .
This is a contingency.

(iv) $((p \rightarrow q) \wedge (q \rightarrow r)) \rightarrow (p \rightarrow r)$

5M

p	q	r	$p \rightarrow q$	$q \rightarrow r$	$(p \rightarrow q) \wedge (q \rightarrow r)$	$p \rightarrow r$	$((p \rightarrow q) \wedge (q \rightarrow r)) \rightarrow (p \rightarrow r)$
T	T	T	T	T	T	T	T
T	T	F	T	F	F	F	T
T	F	T	F	T	F	T	T
T	F	F	F	T	F	F	T
F	T	T	T	T	T	T	T
F	T	F	T	F	F	T	T
F	F	T	T	T	T	T	T
F	F	F	T	T	T	T	T

Last column contains only T , This is a tautology

8. Show that (i) $\neg(p \wedge q) \equiv \neg p \vee \neg q$

3M

Table (1)

$\neg(p \wedge q)$			
p	q	$p \wedge q$	$\neg(p \wedge q)$
T	T	T	F
T	F	F	T
F	T	F	T
F	F	F	T

Table (2)

$\neg p \vee \neg q$				
p	q	$\neg p$	$\neg q$	$\neg p \vee \neg q$
T	T	F	F	F
T	F	F	T	T
F	T	T	F	T
F	F	T	T	T

From table (1) and (2), Last columns are Identical.

$$\therefore \neg(p \wedge q) \equiv \neg p \vee \neg q$$

JUL-21

(ii) $\neg(p \rightarrow q) \equiv p \wedge \neg q$

Table (1)

$$\neg(p \rightarrow q)$$

p	q	$p \rightarrow q$	$\neg(p \rightarrow q)$
T	T	T	F
T	F	F	T
F	T	T	F
F	F	T	F

Table (2)

$$p \wedge \neg q$$

p	q	$\neg q$	$p \wedge \neg q$
T	T	F	F
T	F	T	T
F	T	F	F
F	F	T	F

From table (1) and (2), Last columns are Identical.

$$\therefore \neg(p \rightarrow q) \equiv p \wedge \neg q$$

9. Prove that $q \rightarrow p \equiv \neg p \rightarrow \neg q$

JUL-24

3M

Table (1)

$$q \rightarrow p$$

p	q	$q \rightarrow p$
T	T	T
T	F	T
F	T	F
F	F	T

Table (2)

$$\neg p \rightarrow \neg q$$

p	q	$\neg p$	$\neg q$	$\neg p \rightarrow \neg q$
T	T	F	F	T
T	F	F	T	T
F	T	T	F	F
F	F	T	T	T

From table (1) and (2), Last columns are Identical.

$$\therefore q \rightarrow p \equiv \neg p \rightarrow \neg q$$

10. Show that $p \rightarrow q$ and $q \rightarrow p$ are not equivalent.

APR-23

3M

Table (1)

$$p \rightarrow q$$

p	q	$p \rightarrow q$
T	T	T
T	F	F
F	T	T
F	F	T

Table (2)

$$q \rightarrow p$$

p	q	$q \rightarrow p$
T	T	T
T	F	T
F	T	F
F	F	T

From table (1) and (2), Last columns are not Identical.

$$\therefore p \rightarrow q \not\equiv q \rightarrow p$$

11. Show that $\neg(p \leftrightarrow q) \equiv p \leftrightarrow \neg q$

3M

Table (1)

$$p \leftrightarrow q$$

p	q	$p \leftrightarrow q$	$\neg(p \leftrightarrow q)$
T	T	T	F
T	F	F	T
F	T	F	T
F	F	T	F

Table (2)

$$p \leftrightarrow \neg q$$

p	q	$\neg q$	$p \leftrightarrow \neg q$
T	T	F	F
T	F	T	T
F	T	F	T
F	F	T	F

From table (1) and (2), Last columns are Identical.

$$\therefore \neg(p \leftrightarrow q) \equiv p \leftrightarrow \neg q$$

12. Check whether the statement $p \rightarrow (q \rightarrow p)$ is a tautology or a contradiction without using the truth table.

$$\begin{aligned}
 p \rightarrow (q \rightarrow p) &\equiv p \rightarrow (\neg q \vee p) \\
 &\equiv \neg p \vee (\neg q \vee p) \\
 &\equiv \neg p \vee (p \vee \neg q) & (\because \text{commutative law}) \\
 &\equiv (\neg p \vee p) \vee \neg q & (\because \text{associative law}) \\
 &\equiv \mathbb{T} \vee \neg q \\
 &\equiv \mathbb{T}
 \end{aligned}$$

5M

$p \rightarrow (q \rightarrow p)$ is a Tautology

13. Using truth table check whether the statements $\neg(p \vee q) \vee (\neg p \wedge q)$ & $\neg p$ are logically equivalent.

p	q	$p \vee q$	$\neg(p \vee q)$	$\neg p$	$(\neg p \wedge q)$	$\neg(p \vee q) \vee (\neg p \wedge q)$
T	T	T	F	F	F	F
T	F	T	F	F	F	F
F	T	T	F	T	T	T
F	F	F	T	T	F	T

MDL, PTA-3

3M

From the table last column and the fifth columns are Identical

yes. logically equivalent $\therefore \neg(p \vee q) \vee (\neg p \wedge q) \equiv \neg p$

14. Prove $p \rightarrow (q \rightarrow r) \equiv (p \wedge q) \rightarrow r$ with out using truth table.

5M

$$\begin{aligned}
 p \rightarrow (q \rightarrow r) &\equiv p \rightarrow (\neg q \vee r) \\
 &\equiv \neg p \vee (\neg q \vee r) \\
 &\equiv (\neg q \vee \neg p) \vee r & (\because \text{associative law}) \\
 &\equiv \neg(p \wedge q) \vee r \\
 &\equiv (p \wedge q) \rightarrow r \quad \text{Hence proved}
 \end{aligned}$$

15. Prove that $p \rightarrow (\neg q \vee r) \equiv \neg p \vee (\neg q \vee r)$ using truth table.

APR-23

5M

Table (1) $p \rightarrow (\neg q \vee r)$

p	q	r	$\neg q$	$(\neg q \vee r)$	$p \rightarrow (\neg q \vee r)$
T	T	T	F	T	T
T	T	F	F	F	F
T	F	T	T	T	T
T	F	F	T	T	T
F	T	T	F	T	T
F	T	F	F	F	T
F	F	T	T	T	T
F	F	F	T	T	T

Table (2) $\neg p \vee (\neg q \vee r)$

p	q	r	$\neg p$	$\neg q$	$(\neg q \vee r)$	$\neg p \vee (\neg q \vee r)$
T	T	T	F	F	T	T
T	T	F	F	F	F	F
T	F	T	F	T	T	T
T	F	F	F	T	T	T
F	T	T	T	F	T	T
F	T	F	T	F	F	T
F	F	T	T	T	T	T
F	F	F	T	T	T	T

From table (1) and (2), last columns are identical

$$\therefore p \rightarrow (\neg q \vee r) \equiv \neg p \vee (\neg q \vee r)$$

EXERCISE 12.3



Online Test

Choose the correct or the most suitable answer from the given four alternatives.

1. A binary operation on a set S is a function from

(1) $S \rightarrow S$ (2) $(S \times S) \rightarrow S$ (3) $S \rightarrow (S \times S)$ (4) $(S \times S) \rightarrow (S \times S)$

By definition of binary, $(S \times S) \rightarrow S$

2. Subtraction is not a binary operation in

PTA-6, APR-20, 25

(1) $\mathbb{R}$ (2) $\mathbb{Z}$ (3) $\mathbb{N}$ (4) $\mathbb{Q}$

$2, 3 \in \mathbb{N} \Rightarrow 2 - 3 = -1 \notin \mathbb{N}$
 -1 not binary on $\mathbb{N}$

3. Which one of the following is a binary operation on $\mathbb{N}$?

APR-22, JUL-22

(1) Subtraction (2) **Multiplication** (3) Division (4) All the above

$2, 3 \in \mathbb{N}, 2 \times 3 = 6 \in \mathbb{N} \quad \forall a, b \in \mathbb{N}$
 $a * b \in \mathbb{N}$

4. In the set $\mathbb{R}$ of real numbers '*' is defined as follows. Which one of the following is not a binary operation on $\mathbb{R}$?

PTA-4, JUL-21

(1) $a * b = \min(a, b)$ (2) $a * b = \max(a, b)$
 (3) $a * b = a$ (4) **$a * b = a^b$**

Let $a = 0, b = 0$
 $a * b = a^b = 0^0$
 Indeterminate form $a * b = a^b$
 $*$ is not binary on $\mathbb{R}$

5. The operation $*$ defined by $a * b = \frac{ab}{7}$ is not a binary operation on

PTA-1, JUL-20, APR-23

(1) $\mathbb{Q}^+$ (2) $\mathbb{Z}$ (3) $\mathbb{R}$ (4) $\mathbb{C}$

$a, b \in \mathbb{Z} \Rightarrow a * b \notin \mathbb{Z}$ For example, $4, 3 \in \mathbb{Z} \Rightarrow 4 * 3 = \frac{4 \times 3}{7} = \frac{12}{7} \notin \mathbb{Z}$

6. In the set $\mathbb{Q}$ define $a \odot b = a + b + ab$. For what value of y , $3 \odot (y \odot 5) = 7$?

(1) $y = \frac{2}{3}$ (2) **$y = -\frac{2}{3}$** (3) $y = \frac{-3}{2}$ (4) $y = 4$

$3 \odot (y \odot 5) = 7$
 $3 + (y \odot 5) + 3(y \odot 5) = 7$
 $3 + y + 5 + 5y + 3(y + 5 + 5y) = 7$
 $8 + 6y + 3(6y + 5) = 7$
 $8 + 6y + 18y + 15 = 7$
 $24y = 7 - 23 = -16$
 $y = -\frac{16}{24} = -\frac{2}{3}$

7. If $a * b = \sqrt{a^2 + b^2}$ on the real numbers then $*$ is

PTA-4

- (1) commutative but not associative (2) associative but not commutative
(3) **both commutative and associative** (4) neither commutative nor associative

Commutative: $a, b \in R$ $a * b = \sqrt{a^2 + b^2} = \sqrt{b^2 + a^2} = b * a$ $\forall a, b \in R$ $*$ is commutative	Associative: $a, b, c \in R$ $a * (b * c) = a * (\sqrt{b^2 + c^2}) = \sqrt{a^2 + b^2 + c^2}$ $(a * b) * c = \sqrt{a^2 + b^2} * c = \sqrt{a^2 + b^2 + c^2}$ $*$ is associative
Both commutative and associative.	

8. Which one of the following statements has the truth value T ?

PTA-2

- (1) $\sin x$ is an even function
(2) Every square matrix is non-singular
(3) The product of complex number and its conjugate is purely imaginary
(4) **$\sqrt{5}$ is an irrational number**

9. Which one of the following statements has the truth value F ?

- (1) Chennai is in India or $\sqrt{2}$ is an integer
(2) Chennai is in India or $\sqrt{2}$ is an irrational number
(3) **Chennai is in China or $\sqrt{2}$ is an integer**
(4) Chennai is in China or $\sqrt{2}$ is an irrational number

Chennai is in India = T	$\sqrt{2}$ is an integer = F
Chennai is in China = F	$\sqrt{2}$ is an irrational number = T
(1) $T \vee F = T$	(2) $T \vee T = T$ (3) $F \vee F = F$ (4) $F \vee T = T$

10. If a compound statement involves 3 simple statements, then the number of rows in the truth table is

JUL-24, 23

- (1) 9 (2) **8** (3) 6 (4) 3

$2^3 = 8$

11. Which one is the inverse of the statement $(p \vee q) \rightarrow (p \wedge q)$?

PTA-2, 3

- (1) $(p \wedge q) \rightarrow (p \vee q)$ (2) $\neg(p \vee q) \rightarrow (p \wedge q)$
(3) $(\neg p \vee \neg q) \rightarrow (\neg p \wedge \neg q)$ (4) **$(\neg p \wedge \neg q) \rightarrow (\neg p \vee \neg q)$**

Inverse of $p \rightarrow q$ is $\neg p \rightarrow \neg q$ Similarly $(p \vee q) \rightarrow (p \wedge q)$ $\neg(p \vee q) \rightarrow \neg(p \wedge q)$

12. Which one is the contrapositive of the statement $(p \vee q) \rightarrow r$?

MDL

- (1) $\neg r \rightarrow (\neg p \wedge \neg q)$ (2) $\neg r \rightarrow (p \vee q)$
(3) $r \rightarrow (p \wedge q)$ (4) $p \rightarrow (q \vee r)$

Contrapositive of $p \rightarrow q$ is $\neg q \rightarrow \neg p$ Similarly $\neg r \rightarrow \neg(p \vee q)$; $\neg r \rightarrow (\neg p \wedge \neg q)$

13. The truth table for $(p \wedge q) \vee \neg q$ is given below

p	q	$(p \wedge q) \vee (\neg q)$
T	T	(a)
T	F	(b)
F	T	(c)
F	F	(d)

Which one of the following is true?

(a) (b) (c) (d)

(1) T T T T

(2) T F T T

(3) T T F T

(4) T F F F

p	q	$p \wedge q$	$\neg q$	$(p \wedge q) \vee (\neg q)$
T	T	T	F	T
T	F	F	T	T
F	T	F	F	F
F	F	F	T	T

14. In the last column of the truth table for $\neg(p \vee \neg q)$ the number of final outcomes of the truth value F are

PTA-5

- (1) 1 (2) 2 (3) 3 (4) 4

p	q	$\neg q$	$(p \vee \neg q)$	$\neg(p \vee (\neg q))$
T	T	F	T	F
T	F	T	T	F
F	T	F	F	T
F	F	T	T	F

15. Which one of the following is incorrect? For any two propositions p and q , we have

- (1) $\neg(p \vee q) \equiv \neg p \wedge \neg q$ (2) $\neg(p \wedge q) \equiv \neg p \vee \neg q$
- (3) $\neg(p \vee q) \equiv \neg p \vee \neg q$ (4) $\neg(\neg p) \equiv p$

16.

p	q	$(p \wedge q) \rightarrow (\neg p)$
T	T	(a)
T	F	(b)
F	T	(c)
F	F	(d)

Which one of the following is correct for the truth value of $(p \wedge q) \rightarrow (\neg p)$?

- (a) (b) (c) (d)
- (1) T T T T
- (2) F T T T
- (3) F F T T
- (4) T T T F

p	q	$p \wedge q$	$\neg p$	$(p \wedge q) \rightarrow (\neg p)$
T	T	T	F	F
T	F	F	F	T
F	T	F	T	T
F	F	F	T	T

17. The dual of $\neg(p \vee q) \vee [p \vee (p \wedge \neg r)]$ is ____.

- (1) $\neg(p \wedge q) \wedge [p \vee (p \wedge \neg r)]$ (2) $(p \wedge q) \wedge [p \wedge (p \vee \neg r)]$
- (3) $\neg(p \wedge q) \wedge [p \wedge (p \wedge r)]$ (4) $\neg(p \wedge q) \wedge [p \wedge (p \vee \neg r)]$

Replace $\wedge$ by $\vee$

$\vee$ by $\wedge$ we get the dual of a statement $\therefore \neg(p \wedge q) \wedge [p \wedge (p \vee \neg r)]$

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